



# IEEE Standard for System, Software, and Hardware Verification and Validation

IEEE Computer Society

Developed by the  
Software and Systems Engineering Standards Committee

IEEE Std 1012™-2024

**STANDARDS**

# **IEEE Standard for System, Software, and Hardware Verification and Validation**

Developed by the

**Software and Systems Engineering Standards Committee**  
of the  
**IEEE Computer Society**

Approved 12 November 2024

**IEEE SA Standards Board**

**Abstract:** The Verification and Validation (V&V) processes are used to determine whether the development products of a given activity conform to the requirements of that activity and whether the product satisfies its intended use and user needs. V&V life cycle process requirements are specified for different integrity levels. The scope of V&V processes encompasses systems, software, and hardware, and it includes their interfaces. Systems, software, and hardware being developed, maintained, or reused [legacy, commercial off-the-shelf (COTS), non-developmental items] can be the system of interest to which this standard is applied. The term software also includes firmware and microcode, and each of the terms system, software, and hardware includes related information or documentation. V&V processes include the analysis, evaluation, review, inspection, assessment, and testing of products.

**Keywords:** acceptance testing, adaptive, Agile, AI, architecture evaluation, artificial intelligence, component testing, concept documentation evaluation, COTS, criticality, criticality analysis, design evaluation, disposal plan evaluation, environmental Verification and Validation (V&V) factors, field programmable gate array, firmware, FPGA, hardware life cycle, hardware V&V, hardware Verification and Validation, hazard analysis, IEEE 1012™, implementation evaluation, independent V&V, integration testing, integrity level, interface analysis, IV&V, machine learning, microcode, minimum V&V tasks, ML, nth of a kind, objective evidence, operating procedure evaluation, qualification testing, quality assurance, regression analysis, regression testing, requirements allocation analysis, requirements evaluation, reuse software, risk analysis, security analysis, SaaS software life cycle, software as a service, software quality assurance, software V&V, software Verification and Validation, source code documentation evaluation, source code evaluation, SQA, stakeholder needs and requirements evaluation, system element interaction analysis, system life cycle, system maintenance strategy assessment, system of interest, system requirements evaluation, system V&V, system Verification and Validation, testing, traceability analysis, V&V, V&V measures, Validation, Verification, vignette

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## Introduction

This introduction is not part of IEEE Std 1012-2024, IEEE Standard for System, Software, and Hardware Verification and Validation.

The Verification and Validation (V&V) processes are technical processes of systems, software and hardware engineering. The Verification process and the Validation process are interrelated and complementary processes, and are referenced together as Verification and Validation. The purpose of V&V is to help the organization build quality into the system during the life cycle. V&V processes provide an objective assessment of products and processes throughout the life cycle. This assessment demonstrates whether the requirements are correct, complete, accurate, consistent, and testable. The V&V processes determine whether the development products of a given activity conform to the requirements of that activity and whether the product satisfies its intended use and user needs. The determination includes the assessment, analysis, evaluation, review, inspection, and testing of products and processes. V&V is performed in parallel with all life cycle stages, not at their conclusion.

V&V is an extension of program management and systems, software, and hardware engineering that employs a rigorous methodology to identify objective data and conclusions to provide feedback about quality, performance, and schedule to the supplier. This feedback consists of anomaly resolutions, performance improvements, and quality improvements not only for expected operating conditions but also across the full spectrum of the system and its interfaces. Early feedback results allow the organization to modify the products in a timely fashion and thereby reduce overall project and schedule impacts. Without a proactive approach, the anomalies and associated system changes are typically delayed to later in the program schedule, resulting in greater program costs and schedule delays.

IEEE Std 1012<sup>TM</sup> is a process standard that defines the V&V processes in terms of specific activities and related tasks. The standard also defines the contents of the V&V plan (VVP), including example formats.

V&V may be performed at the level of the system, software element, or hardware element, or on any combination of these. V&V may also be performed on an element of a system, including a subordinate system (i.e., subsystem). Throughout this standard, the term hardware means an electronic or mechanical hardware element. In each case, the V&V processes are invoked, either in parallel or recursively, across the full life cycle of the system or element.

This version of the standard is a revision to IEEE Std 1012<sup>TM</sup>-2016/Cor1-2017 [B19].<sup>6</sup> The earliest version of this standard (1986) described the content of a software V&V plan, with subsequent versions (1998 and 2004) changing the focus from the software V&V plan to software V&V processes. The 2012 revision expanded the scope of the V&V processes to include systems and hardware as well as software. The 2016 revision aligned the system V&V process more completely with the terminology and structure of ISO/IEC/IEEE 15288:2023(E) [B41]. This revision aligns the software and hardware V&V processes with the terminology and structure of ISO/IEC/IEEE 12207:2017(E) [B40] and adds several new annexes, including a set of vignettes to provide examples of applying this standard under various scenarios. The following is a summary of the changes made in this version:

- No new V&V activities or tasks have been added; however, some activities and tasks have been rearranged to align with the new or modified processes from ISO/IEC/IEEE 12207:2017(E) [B40]. Some V&V activities and tasks have been rearranged to facilitate understanding and ease of use.
- The terminology and structure were revised to be consistent with ISO/IEC/IEEE 12207:2017(E) [B40].
- Annexes were added on the topics of test concepts for V&V, V&V for artificial intelligence/machine learning systems, V&V of field programmable gate arrays, and guidance on V&V in an adaptive development environment such as Agile.

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<sup>6</sup> The numbers in brackets correspond to those of the bibliography in Annex P.

- Revisions were made to the annexes on independence and reuse.
- An annex was added to provide vignettes that illustrate the application of this standard in various scenarios, with special consideration of criticality analysis and its impact.

The following key concepts are emphasized in this standard:

- Integrity levels: Defines four integrity levels to describe the importance of the system, software, and hardware, varying from high integrity to low integrity, to the user.
- Minimum V&V tasks for each integrity level: Defines the minimum V&V tasks required for each of the four integrity levels.
- Optional V&V tasks: Includes a table of optional V&V tasks for tailoring the V&V effort to address the project needs and application-specific characteristics.
- Intensity and rigor applied to V&V tasks: Includes the concept that the intensity and rigor applied to the V&V tasks vary according to the integrity level. Higher integrity levels require the application of greater intensity and rigor to the V&V task. Intensity includes a greater scope of analysis across all normal and abnormal system operating conditions. Rigor includes more formal techniques and recording procedures.
- Detailed criteria for V&V tasks: Defines specific criteria for each V&V task, including minimum criteria for correctness, consistency, completeness, accuracy, readability, and testability. The V&V task descriptions include a list of the required task inputs and outputs.
- Systems viewpoints: Includes minimum software and hardware V&V tasks to address system issues. These tasks include hazard analysis, security analysis, risk analysis, migration assessment, and retirement assessment. Specific system issues are contained in individual V&V task criteria.
- Conformance to international and IEEE standards: Defines the V&V processes to conform to life cycle process standards such as ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#) and ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#), as well as the family of IEEE systems and software engineering standards. This standard addresses all system and software life cycle processes, including those in the Agreement, Organizational Project-Enabling, Technical Management, and Technical process groups. This standard is compatible with all life cycle models; however, not all life cycle models use all of the life cycle processes described in this standard.

## Acknowledgements

The IEEE P1012 Working Group would like to acknowledge the strong contribution by Dr. Michael J. Cox in the development of this revision of IEEE Std 1012. Dr. Cox passed away on March 2, 2020, and his commitment to improving professionalism in systems engineering and V&V is reflected in this standard.

## Contents

|                                                                     |    |
|---------------------------------------------------------------------|----|
| 1. Overview .....                                                   | 15 |
| 1.1 Scope .....                                                     | 15 |
| 1.2 Purpose .....                                                   | 16 |
| 1.3 Word usage.....                                                 | 16 |
| 1.4 Field of application.....                                       | 17 |
| 1.5 V&V objectives.....                                             | 18 |
| 1.6 Organization of the standard.....                               | 18 |
| 1.7 Audience.....                                                   | 21 |
| 1.8 Conformance .....                                               | 21 |
| 1.9 Disclaimer.....                                                 | 21 |
| 2. Normative references.....                                        | 21 |
| 3. Definitions and acronyms.....                                    | 22 |
| 3.1 Definitions .....                                               | 22 |
| 3.2 Acronyms .....                                                  | 27 |
| 4. Relationships between V&V and life cycle processes.....          | 28 |
| 5. Integrity levels.....                                            | 32 |
| 6. Overview of V&V processes.....                                   | 34 |
| 6.1 General .....                                                   | 34 |
| 6.2 V&V team characteristics.....                                   | 35 |
| 6.3 V&V testing.....                                                | 36 |
| 7. Common V&V processes.....                                        | 37 |
| 7.1 V&V Management Process .....                                    | 37 |
| 7.2 Acquisition Support V&V Process .....                           | 39 |
| 7.3 Supply Planning V&V Process.....                                | 39 |
| 7.4 Project Planning V&V Process.....                               | 40 |
| 7.5 Configuration Management V&V Process .....                      | 40 |
| 8. System V&V processes.....                                        | 51 |
| 8.1 Business or Mission Analysis V&V Process .....                  | 51 |
| 8.2 Stakeholder Needs and Requirements Definition V&V Process ..... | 51 |
| 8.3 System Requirements Definition V&V Process .....                | 52 |
| 8.4 Architecture Definition V&V Process.....                        | 53 |
| 8.5 Design Definition V&V Process.....                              | 54 |
| 8.6 System Analysis V&V Process.....                                | 55 |
| 8.7 Implementation V&V Process .....                                | 55 |
| 8.8 Integration V&V Process.....                                    | 56 |
| 8.9 Verification Process.....                                       | 57 |
| 8.10 Transition V&V Process.....                                    | 58 |
| 8.11 Validation Process .....                                       | 58 |
| 8.12 Operation V&V Process .....                                    | 59 |
| 8.13 Maintenance V&V Process.....                                   | 60 |
| 8.14 Disposal V&V Process .....                                     | 61 |
| 9. Software V&V processes.....                                      | 96 |
| 9.1 Business or Mission Analysis V&V Process .....                  | 96 |
| 9.2 Stakeholder Needs and Requirements Definition V&V Process ..... | 97 |



|                                                                               |     |
|-------------------------------------------------------------------------------|-----|
| 9.3 System/Software Requirements Definition V&V Process.....                  | 97  |
| 9.4 Architecture Definition V&V Process.....                                  | 98  |
| 9.5 Design Definition V&V Process.....                                        | 99  |
| 9.6 System Analysis V&V Process.....                                          | 100 |
| 9.7 Implementation V&V Process.....                                           | 101 |
| 9.8 Integration V&V Process.....                                              | 102 |
| 9.9 Verification Process.....                                                 | 103 |
| 9.10 Transition V&V Process.....                                              | 103 |
| 9.11 Validation Process.....                                                  | 104 |
| 9.12 Operation V&V Process.....                                               | 105 |
| 9.13 Maintenance V&V Process.....                                             | 105 |
| 9.14 Disposal V&V Process.....                                                | 107 |
| 10. Hardware V&V processes.....                                               | 148 |
| 10.1 Business or Mission Analysis V&V Process.....                            | 148 |
| 10.2 Stakeholder Needs and Requirements Definition V&V Process.....           | 149 |
| 10.3 System/Hardware Requirements Definition V&V Process.....                 | 149 |
| 10.4 Architecture Definition V&V Process.....                                 | 150 |
| 10.5 Design Definition V&V Process.....                                       | 151 |
| 10.6 System Analysis V&V Process.....                                         | 152 |
| 10.7 Implementation V&V Process.....                                          | 153 |
| 10.8 Integration V&V Process.....                                             | 154 |
| 10.9 Verification Process.....                                                | 155 |
| 10.10 Transition V&V Process.....                                             | 155 |
| 10.11 Validation Process.....                                                 | 156 |
| 10.12 Operation V&V Process.....                                              | 157 |
| 10.13 Maintenance V&V Process.....                                            | 157 |
| 10.14 Disposal V&V Process.....                                               | 158 |
| 11. V&V reporting, administrative, and documentation requirements.....        | 192 |
| 11.1 V&V reporting requirements.....                                          | 192 |
| 11.2 V&V administrative requirements.....                                     | 195 |
| 11.3 V&V documentation requirements.....                                      | 195 |
| 12. V&V plan.....                                                             | 196 |
| 12.1 Overview.....                                                            | 196 |
| 12.2 VVP section 1: Purpose.....                                              | 197 |
| 12.3 VVP section 2: Referenced documents.....                                 | 197 |
| 12.4 VVP section 3: Definitions.....                                          | 197 |
| 12.5 VVP section 4: V&V overview.....                                         | 197 |
| 12.6 VVP section 5: V&V processes.....                                        | 199 |
| 12.7 VVP section 6: V&V reporting requirements.....                           | 199 |
| 12.8 VVP section 7: V&V administrative requirements.....                      | 199 |
| 12.9 VVP section 8: V&V test documentation requirements.....                  | 200 |
| Annex A (informative) V&V test concepts.....                                  | 201 |
| Annex B (informative) A risk-based integrity level schema.....                | 210 |
| Annex C (informative) Definition of independent V&V (IV&V).....               | 212 |
| C.1 Independence parameters.....                                              | 212 |
| C.2 Forms of independence.....                                                | 212 |
| Annex D (informative) V&V of reuse software and hardware, including COTS..... | 215 |
| D.1 Purpose.....                                                              | 215 |
| D.2 V&V of software developed in a reuse process.....                         | 216 |

|                                                                                                                           |     |
|---------------------------------------------------------------------------------------------------------------------------|-----|
| D.3 V&V of software developed and reused outside of a reuse process .....                                                 | 216 |
| Annex E (informative) V&V measures .....                                                                                  | 221 |
| E.1 Introduction .....                                                                                                    | 221 |
| E.2 Measures for evaluating anomaly density .....                                                                         | 221 |
| E.3 Measures for evaluating V&V effectiveness.....                                                                        | 222 |
| E.4 Measures for evaluating V&V efficiency.....                                                                           | 222 |
| Annex F (informative) Example of V&V relationships to other project responsibilities .....                                | 224 |
| Annex G (informative) Optional V&V tasks.....                                                                             | 225 |
| Annex H (informative) Environmental factors consideration .....                                                           | 231 |
| H.1 Introduction .....                                                                                                    | 231 |
| H.2 In the agreement processes .....                                                                                      | 231 |
| H.3 In the organizational project-enabling processes .....                                                                | 231 |
| H.4 In the project processes.....                                                                                         | 232 |
| H.5 In the technical processes .....                                                                                      | 232 |
| Annex I (informative) V&V of system, software, and hardware integration .....                                             | 234 |
| I.1 Introduction .....                                                                                                    | 234 |
| I.2 Examples of system failures caused by integration issues .....                                                        | 234 |
| Annex J (informative) Hazard, security, and risk analyses .....                                                           | 239 |
| J.1 Introduction.....                                                                                                     | 239 |
| J.2 Hazard analysis .....                                                                                                 | 240 |
| J.3 Security analysis .....                                                                                               | 241 |
| J.4 Risk analysis .....                                                                                                   | 249 |
| Annex K (informative) Example of assigning and changing the system integrity level of “supporting system functions” ..... | 251 |
| Annex L (informative) Verification and Validation of n <sup>th</sup> of a kind systems.....                               | 253 |
| Annex M (informative) Guidance for applying IEEE 1012 in an adaptive environment.....                                     | 255 |
| M.1 Introduction.....                                                                                                     | 255 |
| M.2 Background.....                                                                                                       | 256 |
| M.3 V&V for adaptive (agile) activities.....                                                                              | 257 |
| Annex N (informative) Verification and Validation considerations for machine learning and artificial intelligence .....   | 260 |
| N.1 Introduction .....                                                                                                    | 260 |
| N.2 Requirements V&V .....                                                                                                | 260 |
| N.3 Hazard analysis.....                                                                                                  | 262 |
| N.4 Testing V&V .....                                                                                                     | 264 |
| N.5 Approaches to the V&V of AI/ML .....                                                                                  | 267 |
| N.6 Model evaluation .....                                                                                                | 267 |
| N.7 Operation V&V .....                                                                                                   | 268 |
| Annex O (informative) Vignettes to illustrate application of IEEE 1012.....                                               | 269 |
| O.1 Introduction .....                                                                                                    | 269 |
| O.2 Integrity level assignment of nuclear power system .....                                                              | 270 |
| O.3 Determining the integrity level for a CCTV monitoring system onboard a rail vehicle.....                              | 280 |
| O.4 Application of IEEE Std 1012 to a Non-defense government system .....                                                 | 287 |
| O.5 Determining integrity level for training simulators .....                                                             | 293 |

|                                                                    |     |
|--------------------------------------------------------------------|-----|
| O.6 V&V of field programmable gate array (FPGA) based systems..... | 298 |
| Annex P (informative) Bibliography .....                           | 304 |

# IEEE Standard for System, Software and Hardware Verification and Validation

## 1. Overview

### 1.1 Scope

This Verification and Validation (V&V) standard is a process standard that addresses all system, software, and hardware life cycle processes, including the Agreement, Organizational Project-Enabling, Technical Management, and Technical Process Groups. This standard is compatible with all life cycle models (e.g., system, software, and hardware); however, not all life cycle models use all of the processes listed in this standard.

V&V processes determine whether the development products of a given activity conform to the requirements of that activity and whether the product satisfies its intended use and user needs. This determination may include the analysis, evaluation, review, inspection, assessment, and testing of products and processes.

The user of this standard invokes the V&V processes associated with the life cycle processes used by the project. A description of system life cycle processes may be found in ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#), and a description of software life cycle processes may be found in ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#).

This standard defines the Verification and Validation Processes that are applied to the system, software, and hardware development throughout the life cycle, including acquisition, supply, development, operations, maintenance, and retirement. This standard applies to the system, software, and hardware being acquired, developed, maintained, or reused. The term software also includes firmware and microcode (e.g., field programmable gate arrays and programmable logic devices). Each of the terms system, software, and hardware includes its associated documentation.

The Verification Process provides objective evidence for whether the products:

- Conform to requirements (e.g., for correctness, completeness, consistency, and accuracy) for all activities during each life cycle process.
- Satisfy the standards, practices, and conventions during life cycle processes.
- Successfully complete each life cycle activity and satisfy all the criteria for initiating succeeding life cycle activities (i.e., builds the product correctly).

The Validation Process provides objective evidence for whether the products:

- Satisfy system requirements allocated to the products at the end of each life cycle activity.
- Solve the right problem (e.g., correctly model physical laws, implement business rules, and use the proper system assumptions).

— Satisfy intended use and user needs in the operational environment (i.e., builds the correct product). The Verification Process and the Validation Process are interrelated and complementary processes that use each other's process results to establish better completion criteria and analysis, evaluation, review, inspection, assessment, and test V&V tasks for each life cycle activity. The V&V task criteria described in [Table 1a](#) through [Table 1d](#) explicitly define the conformance requirements for V&V processes.

The development of a sufficient body of evidence requires a trade-off between the amount of time spent and a finite set of system conditions and assumptions against which to perform the V&V tasks. Each project should define criteria for a sufficient body of evidence (e.g., selecting an integrity level), the schedule, and the scope of the V&V analysis and test tasks.

This standard does not assign the responsibility for performing the V&V tasks to any specific organization. The analysis, evaluation, and test activities may be performed by multiple organizations; however, the methods and purpose will differ for each organization's functional objectives.

ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#) and ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#) include tasks for the supplier to execute the agreement according to established project plans and to deliver the product or service in accordance with the agreement criteria. The techniques described in this standard are useful in performing the supplier's tests and evaluations. Therefore, whenever this standard mentions the supplier's performance of a Verification or Validation activity, it is to be understood that the reference applies to the test and evaluation tasks of system development.

## 1.2 Purpose

The purpose of this standard is to achieve the following:

- Establish a common framework of the V&V processes, activities, and tasks in support of all system, software, and hardware life cycle processes.
- Define the V&V tasks, required inputs, and required outputs in each life cycle process.
- Identify the minimum V&V tasks corresponding to a four-level integrity schema.
- Define the content of the V&V plan.

## 1.3 Word usage

The word *shall* indicates mandatory requirements strictly to be followed in order to conform to the standard and from which no deviation is permitted (*shall equals is required to*).<sup>7,8</sup>

The word *should* indicates that among several possibilities one is recommended as particularly suitable, without mentioning or excluding others; or that a certain course of action is preferred but not necessarily required (*should equals is recommended that*).

The word *may* is used to indicate a course of action permissible within the limits of the standard (*may equals is permitted to*).

The word *can* is used for statements of possibility and capability, whether material, physical, or causal (*can equals is able to*).

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<sup>7</sup> The use of the word *must* is deprecated and cannot be used when stating mandatory requirements; *must* is used only to describe unavoidable situations.

<sup>8</sup> The use of *will* is deprecated and cannot be used when stating mandatory requirements; *will* is only used in statements of fact.

## 1.4 Field of application

This standard applies to all applications of systems. When conducting V&V of a system, software, or hardware element, it is important to examine the interactions with the system of which the element is a part. This standard identifies the important system considerations that V&V processes, activities and tasks address in determining correctness and other V&V attributes (e.g., completeness, accuracy, consistency, and testability).

The dynamics of complex systems and the multitude of different logic paths available within the system in response to varying stimuli and conditions demand that the V&V effort examines the correctness of the system for each possible variation in conditions. The ability to model complex, real-world conditions will be limited, and thus, the V&V effort examines whether the limits of the modeling are realistic and reasonable for the desired solution. The unlimited combination of system conditions presents the V&V effort with the challenge of using a finite set of analytical, test, simulation, and demonstration techniques to establish a reasonable body of evidence that the system is correct.

A system provides the capability to satisfy a stated need or objective by combining one or more of the following: processes, hardware, software, facilities, and people. These relationships require that V&V processes consider interactions among all system elements (software and hardware). The V&V processes address the following interactions with the system:

- Environment: Determines that the system correctly accounts for all conditions, natural phenomena, physical laws of nature, business rules, and physical properties and the full ranges of the system operating environment.
- Operators/users: Determines that the system communicates the proper status/condition of the system to the operator/user and correctly processes all operator/user inputs to produce the required results. For incorrect operator/user inputs, assures that the system is protected from entering into a dangerous or uncontrolled state. Validates that operator/user policies and procedures (e.g., security, interface protocols, data representations, and system assumptions) are consistently applied and used across each component interface.
- Other software, hardware, and systems: Determines that the software or hardware component interfaces correctly with other components in the system of interest in accordance with requirements and that errors are not propagated between components of the system of interest.

The scope of V&V processes includes the operational environment, operators and users, hardware, software, data processes (e.g., processes for providing service to users), procedures (e.g., operator instructions), and facilities. The user of this standard should consider V&V as part of the life cycle processes defined by industry standards, such as ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#) or ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#).

To address the systems perspective, the system, software, and hardware V&V efforts should provide an integrated analysis where the V&V tasks are interrelated, providing input and insight into other V&V tasks. The results from completed life cycle processes provide valuable and necessary inputs to V&V tasks in other life cycle processes. The results and findings from one V&V task may cause previously completed V&V tasks to be analyzed again with the new data. This relationship among V&V tasks (including feedback to the technical and software/hardware specific processes) employing rigorous systems engineering techniques is a key approach to an integrated systems, software, and hardware V&V. The V&V results provide the other ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#) and ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#) life cycle processes (agreement, organizational project-enabling, technical management, and technical) with early detection of anomalies and potential process trends that may be used for process improvement. The V&V processes and tasks described in this standard may be used in conjunction with systems engineering and process improvement models, such as the CMU/SEI-2010-TR-033 [\[B6\]](#).

## 1.5 V&V objectives

V&V processes provide an objective assessment of products and processes throughout the life cycle. This assessment demonstrates whether the requirements are correct, complete, accurate, consistent, and testable. The V&V processes determine whether the products of a given activity conform to the requirements of that activity and whether the product satisfies its intended use and user needs. The determination includes assessment, analysis, evaluation, review, inspection, and testing of products and processes. V&V tasks are performed in parallel with all life cycle stages, not at their conclusion.

The results of V&V provide the following benefits to the program:

- Facilitate early detection and correction of anomalies
- Enhance management insight into process and product risks
- Support the life cycle processes to assure conformance to program performance, schedule, and budget
- Provide an early assessment of performance
- Provide objective evidence of conformance to support a formal certification process
- Improve the products from the acquisition, supply, development, and maintenance processes
- Support process improvement activities

## 1.6 Organization of the standard

The V&V processes described in this standard are constructed to be consistent with ISO/IEC/IEEE 15288:2023(E) [B41] and ISO/IEC Std 12207:2008(E) [B40].

This standard is organized into clauses (Clause 1 through Clause 12), tables [Table 1 through Table 3, and their respective subparts a) through d)], figures [Figure 1 and Figure 2, and their respective subparts a) through d)], and annexes (Annex A through Annex P). Clause 2 through Clause 12, [Table 1a](#) through [Table 1d](#), and [Table 2a](#) through [Table 2d](#) provide the mandatory V&V requirements for this standard. [Table 1a](#) through [Table 1d](#) and [Table 2a](#) through [Table 2d](#) are the focal points of this standard, containing detailed V&V activity and task requirements. [Table 1a](#) through [Table 1d](#) and [Table 2a](#) through [Table 2d](#) address the following:

- Common V&V tasks in [Table 1a](#) and [Table 2a](#)
- System V&V tasks in [Table 1b](#) and [Table 2b](#)
- Software V&V tasks in [Table 1c](#) and [Table 2c](#)
- Hardware V&V tasks in [Table 1d](#) and [Table 2d](#)

Each clause listed below contains V&V activities and tasks and has the subset of [Table 1a](#) through [Table 1d](#) and [Table 2a](#) through [Table 2d](#) associated with the V&V requirements for that clause. [Figure 1a](#) through [Figure 1d](#), [Figure 2a](#) through [Figure 2d](#), and [Table 3a](#) through [Table 3d](#) contain informative material that provide examples of V&V processes and guidance for using this standard, and they are similarly organized.

- [Clause 1](#) provides guidance for using this standard.
- [Clause 2](#) is reserved for normative references; however, this standard does not prescribe any normative references.
- [Clause 3](#) provides a definition of terms, abbreviations, and conventions.
- [Clause 4](#) describes the relationships between the V&V processes and the life cycle processes from ISO/IEC/IEEE 15288:2023(E) and ISO/IEC/IEEE 12207:2017(E) [B40], and it describes how the V&V standard is applied recursively within the concept of a system of systems and from system to software or hardware components.
- [Clause 5](#) describes the use of integrity levels to determine the scope and rigor of V&V processes.

- [Clause 6](#) explains how V&V tasks are described within this standard.
- [Clause 7](#) describes common V&V tasks. The common V&V tasks are invoked for any V&V that is performed (system, software, or hardware).
  - 1) [Table 1a](#) contains the V&V tasks and activities and their criteria for common V&V tasks and activities.
  - 2) [Table 2a](#) contains the minimum tasks for each integrity level for common V&V tasks.
  - 3) [Table 3a](#) contains the optional common V&V tasks.
  - 4) [Figure 1a](#) depicts a summary of common V&V activities and tasks.
  - 5) [Figure 2a](#) depicts a summary of V&V test products and tasks.
- [Clause 8](#) describes system V&V tasks.
  - 1) [Table 1b](#) contains the V&V tasks and activities and their criteria for system V&V tasks and activities.
  - 2) [Table 2b](#) contains the minimum tasks for each integrity level for system V&V tasks.
  - 3) [Table 3b](#) contains the optional system V&V tasks.
  - 4) [Figure 1b](#) depicts a summary of system V&V activities and tasks.
  - 5) [Figure 2b](#) depicts a summary of system V&V test products and tasks.
- [Clause 9](#) describes software V&V tasks.
  - 1) [Table 1c](#) contains the V&V tasks and activities and their criteria for software V&V tasks and activities.
  - 2) [Table 2c](#) contains the minimum tasks for each integrity level for software V&V tasks.
  - 3) [Table 3c](#) contains the optional software V&V tasks.
  - 4) [Figure 1c](#) depicts a summary of software V&V activities and tasks.
  - 5) [Figure 2c](#) depicts a summary of software V&V test products and tasks.
- [Clause 10](#) describes hardware V&V tasks.
  - 1) [Table 1d](#) contains the V&V tasks and activities and their criteria for hardware V&V tasks and activities.
  - 2) [Table 2d](#) contains the minimum tasks for each integrity level for hardware V&V tasks.
  - 3) [Table 3d](#) contains the optional hardware V&V tasks.
  - 4) [Figure 1d](#) depicts a summary of hardware V&V activities and tasks.
  - 5) [Figure 2d](#) depicts a summary of hardware V&V test products and tasks.
- [Clause 11](#) describes V&V reporting, administrative, and documentation requirements.
- [Clause 12](#) describes the content of a V&V plan.

This standard is organized so that the V&V processes (system, software, or hardware) may be invoked separately or in any combination. V&V processes may be accomplished in any of the following combinations: system V&V ([Clause 8](#)), software V&V ([Clause 9](#)), hardware V&V ([Clause 10](#)), system/software V&V ([Clause 8](#) and [Clause 9](#)), system/hardware V&V ([Clause 8](#) and [Clause 10](#)), software/hardware V&V ([Clause 9](#) and [Clause 10](#)), and system/software/hardware V&V ([Clause 8](#) through [Clause 10](#)). Some V&V tasks are common to systems, software, and hardware V&V; in order not to repeat these V&V tasks in each part, the common V&V tasks are listed once in [Clause 7](#). The common V&V tasks are invoked for any V&V that is performed. These combinations are illustrated in [Figure 3](#).



| V&V Scope                      | Common<br><a href="#">Clause 7</a> | System<br><a href="#">Clause 8</a> | Software<br><a href="#">Clause 9</a> | Hardware<br><a href="#">Clause 10</a> |
|--------------------------------|------------------------------------|------------------------------------|--------------------------------------|---------------------------------------|
| System only                    | X                                  | X                                  |                                      |                                       |
| Software only                  | X                                  |                                    | X                                    |                                       |
| Hardware only                  | X                                  |                                    |                                      | X                                     |
| System and software            | X                                  | X                                  | X                                    |                                       |
| System and hardware            | X                                  | X                                  |                                      | X                                     |
| Software and hardware          | X                                  |                                    | X                                    | X                                     |
| System, software, and hardware | X                                  | X                                  | X                                    | X                                     |

**Figure 3—V&V effort combination**

NOTE—References to Table 1 and Table 2 without a suffix a, b, c, or d are understood to include those parts of the table that correspond to the parts of the V&V scope being invoked (i.e., Common, System, Software, or Hardware). References to Figure 1 and Figure 2 that do not have a suffix a, b, c, or d are similarly understood to include those parts of the figure that correspond to the parts of the V&V scope being invoked.<sup>9</sup>

[Clause 4](#) through [Clause 6](#) contain guidance that is applicable to all combinations of the V&V scope. [Clause 11](#) contains reporting requirements for all combinations of V&V scope. [Clause 12](#) describes the content of the V&V plan and provides a sample V&V plan outline.

[Table 1a](#) through [Table 1d](#) provide V&V task descriptions, inputs, and outputs for each life cycle process. [Table 2a](#) through [Table 2d](#) list minimum V&V tasks required for different integrity levels. [Table 3a](#) through [Table 3d](#) provide a list of optional V&V tasks and their suggested applications in the system life cycle. These optional V&V tasks may be added to the minimum V&V tasks to tailor the V&V effort to project needs.

[Figure 1a](#) through [Figure 1d](#) provide an example of an overview of the V&V inputs, outputs, and minimum V&V tasks for integrity level 4. [Figure 2a](#) through [Figure 2d](#) provide guidelines for scheduling V&V test planning, execution, and Verification activities. An example of a phased life cycle model was used in [Figure 1a](#) through [Figure 1d](#) and [Figure 2a](#) through [Figure 2d](#) to illustrate a mapping of the ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#) life cycle processes to the V&V activities and tasks described in this standard.

All annexes are informative. [Annex A](#) discusses concepts related to V&V testing. [Annex B](#) provides an example of a risk-based, four-level integrity schema. [Annex C](#) describes the degrees of independence in V&V. [Annex D](#) provides guidelines for conducting V&V of reuse software. [Annex E](#) describes V&V measures. [Annex F](#) illustrates an example of the V&V organizational relationship to other project responsibilities. [Annex G](#) describes optional V&V tasks. [Annex H](#) describes environmental factors that should be considered when conducting V&V. [Annex I](#) discusses potential issues with interactions among systems, software, and hardware. [Annex J](#) describes hazard analysis, security analysis, and risk analysis and their role in V&V. [Annex K](#) provides an example of assigning and changing the integrity level of supporting system functions. [Annex L](#) describes V&V for  $n^{\text{th}}$  of a kind systems. [Annex M](#) provides guidance for conducting V&V in an adaptive environment, such as Agile. [Annex N](#) discusses V&V considerations for machine learning and artificial intelligence (AI). [Annex O](#) contains vignettes (Integrity level assignment of nuclear power system, Determining the integrity level for a CCTV monitoring system onboard a rail vehicle, Application of IEEE Std 1012 to a non-defense government system, Determining integrity level for training simulators, and V&V of field programmable gate array (FPGA) based systems) to provide examples and considerations in determining criticality levels and applying V&V within varied scenarios. [Annex P](#) provides a bibliography of the informative standards referenced in this standard.

<sup>9</sup> Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement the standard.

## 1.7 Audience

The audience for this standard includes system, software, and hardware suppliers, acquirers, developers, maintainers, V&V practitioners, operators, users, and managers in both the supplier and acquirer organizations.

## 1.8 Conformance

The word *shall* identifies the mandatory requirements to be followed to conform to this standard. The words *should* and *may* indicate optional tasks that are not required to claim conformance to this standard.

Not all V&V efforts are initiated at the start of the life cycle process of Acquisition and continued through the Maintenance Process. If a project uses only selected life cycle processes, then conformance to this standard is achieved if the minimum V&V tasks are implemented for all of the associated life cycle processes selected for the project. A claim of conformance to this standard includes identification of the applicable life cycle processes. As in all cases, the minimum V&V tasks are defined by the integrity level assigned to the system, software, or hardware.

For life cycle processes that are not used by the project, the V&V requirements and tasks for those life cycle processes are optional V&V tasks invoked as needed at the discretion of the project. Specific development methods and technologies (such as automated code generation from detailed design) may eliminate development steps or combine several development steps into one; therefore, a corresponding adaptation of the minimum V&V tasks is permitted and is documented in any claim of conformance to this standard.

When this standard is invoked for existing systems, software, or hardware and the required V&V inputs are not available, then the V&V tasks may use other available project input sources or may reconstruct the needed inputs to achieve conformance to this standard.

[Clause 2](#) through [Clause 12](#), [Table 1a](#) through [Table 1d](#) and [Table 2a](#) through [Table 2d](#) provide the mandatory V&V requirements for this standard. System, software, or hardware conformance to this standard can be achieved by demonstrating that all the “shall” requirements in the specified clauses are met as defined by the following:

- System V&V conformance: [Clause 7](#) and [Clause 8](#)
- Software V&V conformance: [Clause 7](#) and [Clause 9](#)
- Hardware V&V conformance: [Clause 7](#) and [Clause 10](#)

All conformance waivers need to be approved by the acquiring organization and described in the Verification and Validation plan (VVP). The information required for deviations and waivers includes the task identification, rationale, and the effect on quality.

## 1.9 Disclaimer

This standard establishes minimum criteria for V&V processes, activities, and tasks. However, implementing these criteria does not automatically ensure conformance to system or mission objectives, or prevent adverse consequences (e.g., loss of life, mission failure, loss of system safety or security, or financial or social loss). Conformance to this standard does not absolve any party from any social, moral, financial, or legal obligations.

## 2. Normative references

This standard does not require the use of any normative references.

## 3. Definitions and acronyms

### 3.1 Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* [B16] or the IEEE Standards Dictionary [B15] should be consulted for terms not defined in this clause.<sup>10</sup>

**acceptance testing:** **(A)** Testing conducted to determine whether a system satisfies its acceptance criteria and to enable the customer to determine whether to accept the system. **(B)** Formal testing conducted to enable a user, customer, or other authorized entity to determine whether to accept a system or component.

NOTE—See ISO/IEC/IEEE 24765:2017 [B43].

**acquirer:** Stakeholder that acquires or procures a product or service from a supplier. Other terms commonly used for an acquirer are buyer, customer, owner, or purchaser.

NOTE—See ISO/IEC/IEEE 24765:2017 [B43].

**anomaly:** Anything observed in the documentation or operation of a system that deviates from expectations based on previously verified system, software, or hardware products or reference documents.

NOTE—See ISO/IEC/IEEE 24765:2017 [B43].

**artificial intelligence (AI):** <discipline> research and development of mechanisms and applications of AI systems.

NOTE 1—AI systems are designed to operate with varying levels of automation.

NOTE 2—Predictions can refer to various kinds of data analysis or production (including translating text, creating synthetic images, or diagnosing a previous power failure). It does not imply anteriority.

NOTE 3—See ISO 22989-2021 [B38].

**artificial intelligence system (AIS):** Engineered system that generates outputs such as content, forecasts, recommendations, or decisions for a given set of human-defined objectives.

NOTE—See ISO 22989-2021 [B38].

**asset:** An item (e.g., design, specifications, source code, documentation, test suites, or manual procedures) that has been designed for use in multiple contexts.

NOTE—See IEEE Std 1517™-2010 [B24].

**baseline:** Any agreement or result designated and fixed at a given time, from which changes require justification and approval.

NOTE—See ISO/IEC/IEEE 24765:2017 [B43].

**commercial off the shelf (COTS):** Software and/or hardware product available for purchase and use without the need to conduct development activities.

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<sup>10</sup>IEEE Standards Dictionary Online is available at: <http://dictionary.ieee.org>. An IEEE account is required for access to the dictionary, and one can be created at no charge on the *dictionary sign-in page*.

NOTE 1—COTS software product includes the product description (including all cover information, data sheet, web site information, etc.), the user documentation (necessary to install and use the software), the software contained on a computer sensible media (disk, CD-ROM, internet downloadable, etc.). Software is mainly composed of programs and data. This definition applies also to product descriptions, user documentation and software which are produced and supported as separate manufactured goods, but for which typical commercial fees and licensing considerations do not apply.

NOTE 2—See ISO/IEC/IEEE 24765:2017 [\[B43\]](#).

**component:** One part that makes up a system. A component may be hardware or software and may be subdivided into other components.

NOTE 1—The terms module, component, and unit are often used interchangeably or defined to be subelements of one another in different ways depending on the context. The relationship of these terms is not yet standardized.

NOTE 2—See ISO/IEC/IEEE 24765:2017 [\[B43\]](#).

**component testing:** Testing of individual hardware or software components.

NOTE—See ISO/IEC/IEEE 24765:2017 [\[B43\]](#).

**criticality:** The degree of impact that a requirement, module, error, fault, failure, or other item has on the development or operation of a system.

NOTE—See ISO/IEC/IEEE 24765:2017 [\[B43\]](#).

**documentation:** A record (e.g., report, calculation, specification) or other similar product that provides an evidentiary basis of a single or series of activities, tasks, or events and their associated plans, results, or outcomes that may support a larger product, process, or program. Documentation may be printed, electronic, or presented in another suitable media that allows the viewer to confirm that specific activities or tasks occurred and that any associated plans, outcomes, or results have been accurately recorded.

NOTE—This standard uses the term ‘documentation’ in an equivalent manner to the term document that may serve as a single information item or may be a part of a larger information item.

**domain:** A problem space.

NOTE—See IEEE Std 1517-2010 [\[B24\]](#).

**domain analysis:** (A) The analysis of systems within a domain to discover commonalities and differences among them. (B) The process by which information used in developing software systems is identified, captured, and organized so that it can be reused to create new systems, within a domain. (C) The result of the process in (A) and (B).

NOTE—See IEEE Std 1517-2010 [\[B24\]](#).

**domain engineering:** A reuse-based approach to defining the scope (i.e., domain definition), specifying the structure (i.e., domain architecture), and building the assets (e.g., requirements, designs, software code, and documentation) for a class of systems, subsystems, or applications. Domain engineering may include the following activities: domain definition, domain analysis, developing the domain architecture, and domain implementation.

NOTE—See IEEE Std 1517-2010 [\[B24\]](#).

**firmware:** The combination of a hardware device and computer instructions and data that reside as read-only software on that device.

NOTE 1—This term is sometimes used to refer only to the hardware device or only to the computer instructions or data, but these meanings are deprecated.

NOTE 2—The confusion surrounding this term has led some to suggest that it be avoided altogether.

NOTE 3—See ISO/IEC/IEEE 24765:2017 [B43].

**government off the shelf (GOTS):** software and/or hardware product supplied by the government for reuse in another project.

NOTE—See ISO/IEC/IEEE 24765:2017 [B43].

**hazard: (A)** An intrinsic property or condition that has the potential to cause harm or damage. **(B)** A source of potential harm or a situation with a potential for harm in terms of human injury, damage to health, property, or the environment, or some combination of these.

NOTE—See ISO/IEC/IEEE 24765:2017 [B43].

**hazard identification:** The process of recognizing that a hazard exists and defining its characteristics.

NOTE—See ISO/IEC/IEEE 24765:2017 [B43].

**independent Verification and Validation (IV&V):** Verification and Validation (V&V) performed by an organization that is technically, managerially, and financially independent of the development organization.

NOTE 1—See Annex C for a description of the degrees of independence for V&V.

NOTE 2—See ISO/IEC/IEEE 24765:2017 [B43].

**information item:** Separately identifiable body of information that is produced, stored, and delivered for human use.

NOTE 1—An information item can be produced in several versions during a project or system life cycle.

NOTE 2—See ISO/IEC/IEEE 15289:2019 [B42].

**integration testing:** Testing in which software components, hardware components, or both are combined and tested to evaluate the interaction between them.

NOTE—See ISO/IEC/IEEE 24765:2017 [B43].

**integrity level:** A value representing project-unique characteristics (e.g., complexity, criticality, risk, safety, security, desired performance, and reliability) that define the importance of the system, software, or hardware to the user.

**interface design document (IDD):** Documentation that describes the architecture and design interfaces between the system and components. These descriptions include control algorithms, protocols, data contents and formats, and performance.

NOTE—See ISO/IEC/IEEE 24765:2017 [B43].

**interface requirements specification (IRS):** Documentation that specifies the requirements for the interfaces between systems and components. These requirements include constraints on formats and timing.

NOTE—See ISO/IEC/IEEE 24765:2017 [B43].

**life cycle processes:** A set of interrelated or interacting activities that result in the development or assessment of system, software, or hardware products. Each activity consists of tasks. The life cycle processes may overlap one another. For Verification and Validation (V&V) purposes, no life cycle process is concluded until its development products are verified and validated according to the defined tasks in the Verification and Validation plan (VVP).

NOTE—See ISO/IEC/IEEE 24765:2017 [\[B43\]](#).

**machine learning (ML):** Process of optimizing model parameters through computational techniques, such that the model's behavior reflects the data or experience.

NOTE— See ISO 22989-2021 [\[B38\]](#)

**microcode:** A collection of microinstructions, comprising part of, all of, or a set of microprograms.

NOTE—See ISO/IEC/IEEE 24765:2017 [\[B43\]](#).

**microprogram:** A sequence of instructions, called microinstructions, specifying the basic operations needed to carry out a machine language instruction.

NOTE—See ISO/IEC/IEEE 24765:2017 [\[B43\]](#).

**minimum tasks:** Verification and Validation (V&V) tasks required by the integrity level assigned to the system, software, or hardware to be verified and validated.

**n<sup>th</sup> of a kind component/system:** A re-manufacturing or re-installation of a previously verified and validated hardware and/or software design. The n<sup>th</sup> of a kind component/system is equivalent to the first application in all relevant aspects, including functional and performance requirements, design documentation, environment, and regulatory constraints.

NOTE—See Annex L for a discussion of Verification and Validation (V&V) of n<sup>th</sup> of a kind systems.

**optional tasks:** Verification and Validation (V&V) tasks that may be added to the minimum required V&V tasks to address specific application requirements.

**qualification testing:** Testing conducted on a hardware element, software element or system to evaluate conformance with specified requirements.

**required inputs:** The set of items necessary to perform the minimum Verification and Validation (V&V) tasks mandated within any life cycle activity.

**required outputs:** The set of items produced as a result of performing the minimum Verification and Validation (V&V) tasks mandated within any life cycle activity.

**reusable product:** A system, software, or hardware product developed for one use but having other uses, or one developed specifically to be usable on multiple projects or in multiple roles on one project. Examples include, but are not limited to, commercial off-the-shelf (COTS) software products, acquirer-furnished software products, software products in reuse libraries, and preexisting developer software products. Each use may include all or part of the software product and may involve its modification. This term can be applied to any software product (for example, requirements, or architectures), not just to software itself.

NOTE—See ISO/IEC/IEEE 24765:2017 [\[B43\]](#).

**risk: (A)** The combination of the likelihood of occurrence and the consequences of a given future undesirable event. Risk can be associated with products and/or projects. **(B)** The combination of the likelihood of an abnormal event or failure and the consequence(s) of that event or failure to a system's components, operators, users, or environment.

NOTE—See ISO/IEC 16085-2006 [\[B37\]](#). ISO/IEC 16085 Systems and software engineering—Life cycle processes—Risk management provides examples of risk categories, but does not provide a complete taxonomy. By its nature, V&V will typically identify risks that are technical, safety, or engineering, but may also identify other types, including legal, organizational, cost, or schedule risks.

**security:** (A) The protection of computer hardware or software from accidental or malicious access, use, modification, destruction, or disclosure. Security also pertains to personnel, data, communications, and the physical protection of computer installations. (B) The protection of information and data so that unauthorized persons or systems cannot read or modify them, and authorized persons or systems are not denied access to them.

NOTE—See ISO/IEC/IEEE 24765:2017 [B43].

**software as a service (SaaS):** Access to resources from client devices through a thin client interface or program interface.

NOTE 1—The level of control over the resource provided to the consumer can vary with the service provider. In DevOps, SaaS can be automated as part of the DevOps pipeline.

NOTE 2—See IEEE Standards Dictionary Online [B16].

**software design description (SDD):** A representation of software created to facilitate analysis, planning, implementation, and decision-making. The software design description is used as a medium for communicating software design information and may be thought of as a blueprint or model of the system.

NOTE—See ISO/IEC/IEEE 24765:2017 [B43].

**software requirements specification (SRS):** Documentation of the essential requirements (functions, performance, design constraints, and attributes) of the software and its external interfaces.

NOTE—See ISO/IEC/IEEE 24765:2017 [B43].

**system of interest:** System whose life cycle is under consideration in the application of one instance of this Verification and Validation (V&V) standard.

NOTE—See ISO/IEC/IEEE 15288:2023(E) [B41].

**test case:** (A) A set of test inputs, execution conditions, and expected results developed for a particular objective, such as to exercise a particular program path or to verify compliance with a specific requirement. (B) Documentation specifying inputs, predicted results, and a set of execution conditions for a test item.

NOTE—See ISO/IEC/IEEE 24765:2017 [B43].

**test design:** Documentation specifying the details of the test approach for a system, software, or hardware feature or combination of features and identifying the associated tests.

NOTE—See ISO/IEC/IEEE 24765:2017 [B43].

**test plan:** (A) A document describing the scope, approach, resources, and schedule of intended test activities. It identifies test items, the features to be tested, the testing tasks, who will do each task, and any risks requiring contingency planning. (B) A document that describes the technical and management approach to be followed for testing a system or component. Typical contents identify the items to be tested, tasks to be performed, responsibilities, schedules, and required resources for the testing activity.

NOTE—See ISO/IEC/IEEE 24765:2017 [B43].

**test procedure:** (A) Detailed instructions for the setup, execution, and evaluation of results for a given test case. (B) A document containing a set of associated instructions as in (A). (C) Documentation that specifies a sequence of actions for the execution of a test.

NOTE—See IEEE Std 982.1™-2005 [B18].



**Validation: (A)** The process of providing evidence that the system, software, or hardware and its associated products satisfy requirements allocated to it at the end of each life cycle activity, solve the right problem (e.g., correctly model physical laws, implement business rules, and use the proper system assumptions), and satisfy intended use and user needs. **(B)** confirmation, through the provision of objective evidence, that the requirements for a specific intended use or application have been fulfilled

NOTE—For (B), see ISO/IEC/IEEE 24765:2017 [\[B43\]](#).

**Verification: (A)** The process of evaluating a system or component to determine whether the products of a given development phase satisfy the conditions imposed at the start of that phase. **(B)** The process of providing objective evidence that the system, software, or hardware and its associated products conform to requirements (e.g., for correctness, completeness, consistency, and accuracy) for all life cycle activities during each life cycle process (acquisition, supply, development, operation, and maintenance); satisfy standards, practices, and conventions during life cycle processes; and successfully complete each life cycle activity and satisfy all the criteria for initiating succeeding life cycle activities. Verification of interim work products is essential for proper understanding and assessment of the life cycle phase product(s).

NOTE—For (A), see ISO/IEC/IEEE 24765:2017 [\[B38\]](#).

**Verification and Validation (V&V) effort:** The work associated with performing the V&V processes, activities, and tasks.

### 3.2 Acronyms

The following acronyms appear in this standard:

|         |                                                           |
|---------|-----------------------------------------------------------|
| AI      | artificial intelligence                                   |
| AIS     | artificial intelligence system                            |
| ATWS    | anticipated transient without scram                       |
| AUC     | area under the curve                                      |
| BWR     | boiling water reactor                                     |
| CCTV    | closed-circuit television                                 |
| COOP    | continuity of operations plan                             |
| COTS    | commercial off-the-shelf                                  |
| CPU     | central processing unit                                   |
| CRC     | cyclic redundancy check                                   |
| CSAS    | containment spray actuation signal                        |
| DCS     | digital computer system                                   |
| DRP     | disaster recovery plan                                    |
| ECCS    | emergency core cooling systems                            |
| ECFS    | electronic contract filing system                         |
| EMC     | electromagnetic compatibility                             |
| EMI/RFI | electromagnetic interference/radio frequency interference |
| ESF     | engineered safety features                                |
| FEDRAMP | Federal Risk and Authorization Management Program         |
| FMEA    | failure modes and effects analysis                        |
| FNR     | false negative rate                                       |
| FPDS    | Federal Procurement Data System                           |
| FPGA    | field programmable gate array                             |
| FPR     | false positive rate                                       |
| GAN     | generative adversarial networks                           |
| GOTS    | government off-the-shelf                                  |
| GPS     | global positioning system                                 |
| HAZOPS  | hazards and operability studies                           |
| HILS    | hardware in the loop simulation                           |

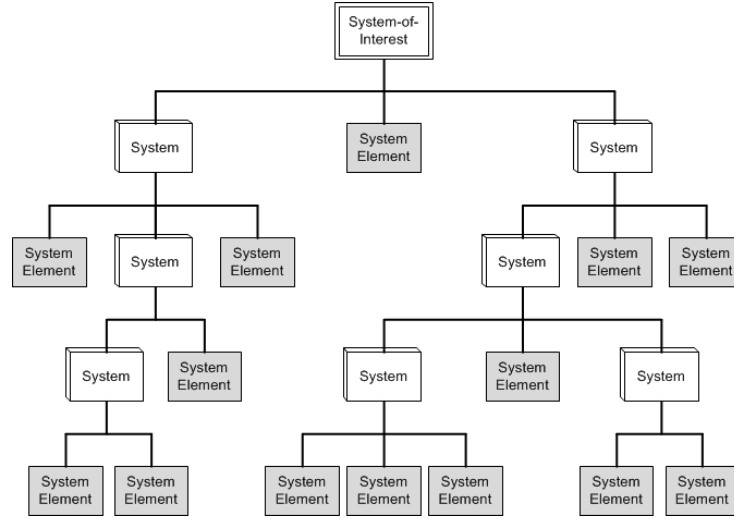


|      |                                         |
|------|-----------------------------------------|
| HRS  | hardware requirements specification     |
| HW   | hardware                                |
| I&C  | instrumentation and control             |
| IAEA | International Atomic Energy Agency      |
| IDD  | interface design document               |
| I/O  | input/output                            |
| IRS  | interface requirements specification    |
| IT   | information technology                  |
| IV&V | independent Verification and Validation |
| LCL  | local coincidence logic                 |
| LOCA | loss of coolant accident                |
| ML   | machine learning                        |
| MR   | metamorphic relation                    |
| MVP  | minimum viable product                  |
| NPP  | nuclear power plant                     |
| N/R  | not required                            |
| NVR  | network video recorder                  |
| OCD  | operational concept document            |
| PIA  | privacy impact assessment               |
| PII  | personally identifiable information     |
| PLC  | programmable logic controller           |
| PRNM | power range neutron monitor             |
| PWR  | pressurized water reactor               |
| QA   | quality assurance                       |
| RC   | radio control                           |
| RCS  | reactor coolant system                  |
| RFP  | request for proposal (tender)           |
| ROC  | receiver operating characteristic       |
| RPS  | reactor protection system               |
| SaaS | software as a service                   |
| SDD  | software design description             |
| SILS | software in the loop simulation         |
| SIS  | safety injection signal                 |
| SRS  | software requirements specification     |
| SSE  | subsystem element                       |
| SW   | software                                |
| TNR  | true negative rate                      |
| TPR  | true positive rate                      |
| TRA  | threat and risk assessment              |
| UAT  | user acceptance testing                 |
| VMS  | video management software               |
| VVP  | Verification and Validation plan        |
| V&V  | Verification and Validation             |
| WBS  | work breakdown structure                |
| XAI  | explainable artificial intelligence     |

## 4. Relationships between V&V and life cycle processes

This standard is structured to enable a V&V practitioner to use [Clause 7](#) (common V&V) and any of [Clause 8](#) (system V&V), [Clause 9](#) (software V&V), or [Clause 10](#) (hardware V&V) in performing V&V for a particular product. V&V on a system is performed using [Clause 7](#) and [Clause 8](#) (common and system), on software using [Clause 7](#) and [Clause 9](#) (common and software), and on hardware using [Clause 7](#) and [Clause 10](#) (common and hardware).

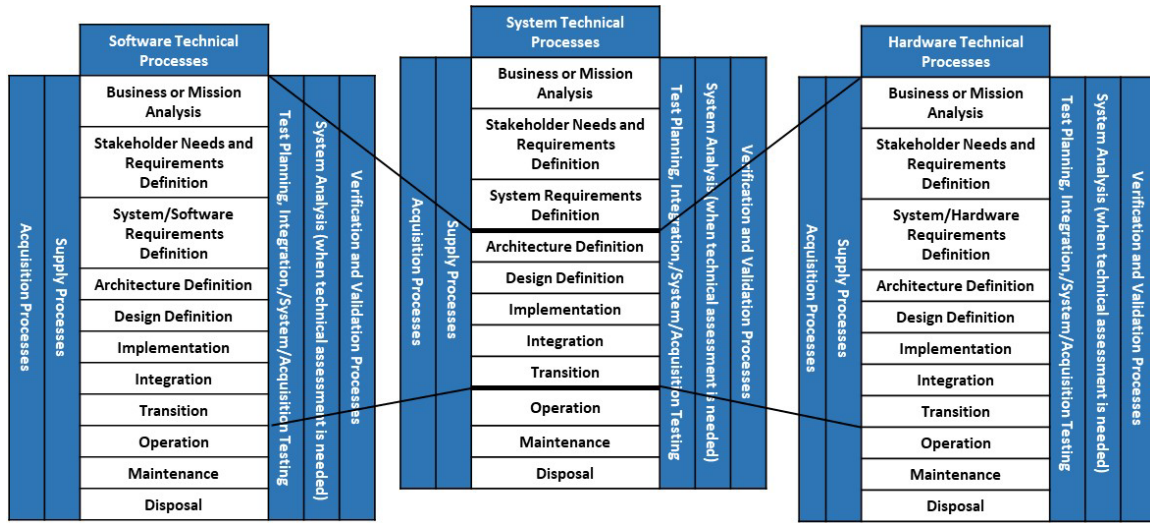
Most systems consist of more than a single hardware or a single software element. A depiction of a more complex system with subsystems and system elements (i.e., software or hardware, shown as shaded boxes) is shown in [Figure 4](#). The system life cycle processes are applied recursively from the system of interest to the systems at the next level, with the recursion continuing until a system element is reached. For software system elements, the software life cycle processes are applied, and for hardware system elements, the hardware life cycle processes are applied. The life cycle processes at the system, software, or hardware levels may be conducted in parallel for systems or system elements at the same level. The full set of life cycle processes (system, software, or hardware, as appropriate) are applied to each product. In the same way, system V&V is applied recursively for each system of the system of interest, and software or hardware V&V is applied to each system element. For V&V purposes, no life cycle process is concluded until its development products are verified and validated according to the defined tasks in the VVP (see [Clause 12](#)).



**Figure 4—Example system-of-interest structure**

The Verification Process and the Validation Process are two processes within the technical processes group of the system life cycle processes defined in ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#). System V&V, as described in this V&V standard, is a conforming instance of the V&V processes in ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#) (i.e., conducting system V&V as described in this V&V standard enables the V&V practitioner to claim full conformance with those two system life cycle processes, although other approaches may also be in conformance). In a similar fashion, software V&V as described in this V&V standard is a conforming instance of the V&V processes within the technical processes group of the life cycle processes defined in ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#).

The purpose of the system Implementation Process is to produce a specified system element implemented as a software and hardware product or service. Within the system Implementation Process, each software and hardware element is developed using the full software or hardware life cycle (requirements, design, implementation/construction, integration, and test). This concept is reflected in [Figure 5](#), which also illustrates that the Technical Processes for software and hardware overlap with the System Architecture Definition, Design Definition, System Analysis, Integration, and Transition processes. At the system element level, the implementation will be realized by the Implementation Process (described in [Clause 9](#)) or the Implementation process (described in [Clause 10](#)).



NOTE 1—The layout of the processes is not intended to specify an ordering of processes. V&V activities may be applied to any life cycle process.

NOTE 2—Management of V&V activity is concurrent with all V&V activities.

NOTE 3—The task description, inputs, and outputs of all V&V tasks are included in [Table 1a](#) through [Table 1d](#).

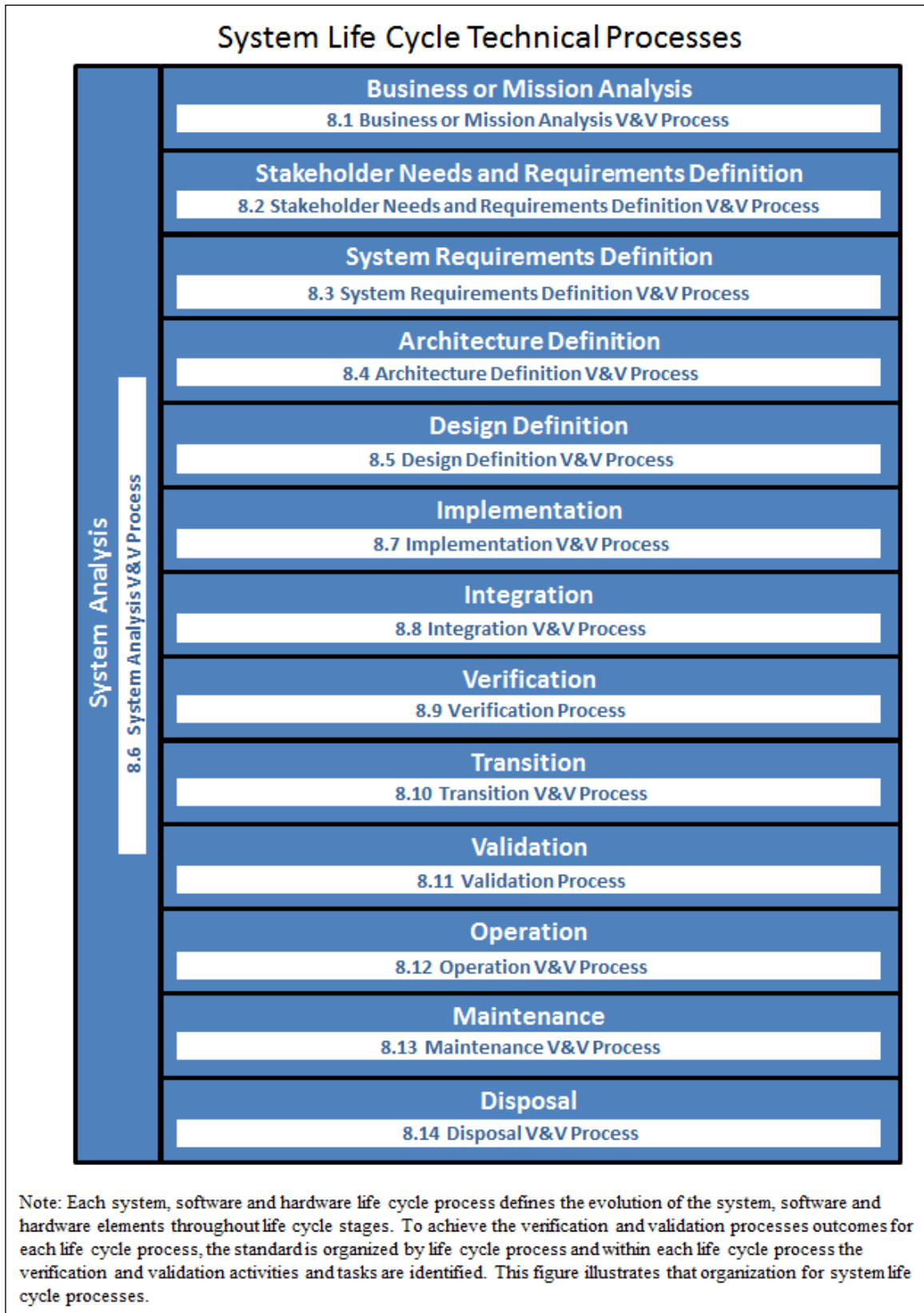
**Figure 5—Relationship of system, software, and hardware processes**

Not all life cycle processes from ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#) and ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#) have V&V tasks specified in this standard. Each Technical process has a corresponding V&V process, as do the Acquisition, Supply, Project Planning, and Configuration Management processes. The V&V processes also interact with the other system and software processes not included in this table, such as providing inputs and receiving outputs used in the V&V effort.

Each ISO/IEC/IEEE 12207 software-specific process is a specialization of its corresponding process in ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#). Users may consider claiming conformance to the ISO/IEC/IEEE 15288 process rather than the ISO/IEC/IEEE 12207 process.

Each hardware process corresponding to an ISO/IEC/IEEE 12207 software-specific process is a specialization of its related process in ISO/IEC/IEEE 15288.

The life cycle model, including selected processes, defines the evolution of the system, software, and hardware elements throughout the life cycle stages. To achieve the outcomes of the V&V processes for each of the other life cycle process, this standard is organized by life cycle process, and within each life cycle process, the V&V activities and tasks for that life cycle process are identified. [Figure 6](#) illustrates that organization for system life cycle processes. The listing of all V&V activities and tasks to specific system, software, or hardware life cycle processes are depicted in [Figure 1a](#), [Figure 1b](#), [Figure 1c](#), and [Figure 1d](#).



**Figure 6—V&V processes align with other technical life cycle processes**

## 5. Integrity levels

Integrity level determination establishes the importance of the system to the user and acquirer based on complexity, criticality, risk, safety, security, desired performance, reliability, or other system-unique characteristics. Integrity levels are used to determine the V&V tasks, activities, rigor, and the level of intensity of the V&V to be performed.

The following is a result of the integrity level determination:

- a) An integrity schema is defined.
- b) Integrity levels for all system elements are assigned.
- c) Integrity assignments are used to determine V&V activities and tasks using [Table 2a](#) through [Table 2d](#).
- d) A basis is established to review and revalidate the integrity levels throughout the V&V effort.

The degree of rigor and intensity in performing and documenting any V&V task associated with a system, software, or hardware work product shall be commensurate with the integrity level. As the integrity level decreases, so does the required scope, intensity, and degree of rigor associated with the V&V task. For example, system integration testing for integrity level 4 systems must be performed independently by the V&V organization, whereas for an integrity level 3 system, the V&V organization need only review the results from systems integration testing.

Use the integrity schema established for the system, software, or hardware, if one exists. An integrity level schema shall be specified if one is not already defined. Establish gradations of criticality or integrity to assure complete coverage of the risk classes from possible system behaviors from no risk to worst-case risk. The integrity levels established for a system element by the developer should result from agreements between the acquirer and the supplier and remain consistent with regulatory requirements.

This standard was developed with a four-level schema to explain the minimum normative V&V requirements. Other integrity schemas are acceptable. For any selected integrity schema, the selected integrity levels shall be mapped into this standard's four-level schema (and associated V&V criteria) to demonstrate that the minimum V&V requirements are satisfied. Because V&V is applied recursively from the system of interest down to each of the subsystems or elements at the next level, it is not necessary to use the same integrity level schema for all the subsystems or elements of the system of interest.

NOTE—A process for developing an integrity schema can be found in ISO/IEC 15026-3:2013 [\[B33\]](#).

The V&V criticality analysis task is used to classify the integrity level of every system (e.g., subsystem) or element in the system of interest. Integrity levels are assigned to requirements, functions, groups of functions, components, or subsystems. The integrity levels characterize and quantify the potential for undesirable effects and consequences resulting from integrity lapse, unintended effects, failure mode effects, and unverified performance effects. The characteristics that determine integrity level vary depending on the intended application and use of the system.

Integrity level assignments shall be applied recursively to the components of each system element. As the development stages progressively decompose the solution into more detail, the criticality analysis performed at each development stage assigns integrity levels to each function. By recursively assigning integrity levels to individual solution parts, greater rigor and intensity are applied to the higher integrity level elements. Progressive decomposition for criticality analysis and assignment of integrity levels shall be applied to segregate parts that warrant high-integrity V&V tasks from those parts that do not. This recursive assignment of integrity levels to solution parts keeps the V&V effort focused on the high-integrity elements, making the V&V effort cost-efficient and technically effective.

From the system perspective, during integrity level analysis, software and hardware are treated as functional elements. Subsystem components cannot have a higher criticality or integrity level than the parent subsystem. The system shall be assigned the same integrity level as the highest level assigned to any individual element.

The integrity level assigned to the parent shall be as high as the highest integrity level of the system elements. The system elements can have the integrity level of the system parent or lower, depending on the critical functions they support.

The integrity level assigned to reused, COTS, and government off-the-shelf (GOTS) components shall be in accordance with the integrity level schema adopted for the system element into which COTS or GOTS may be integrated for the project. The reused COTS or GOTS component shall be evaluated for use in the context of its application. The design, development, procedural, and technology features implemented in the system can raise or lower the assigned integrity levels.

The V&V criticality analysis task is used to classify the integrity level of enabling systems. Examples of enabling systems include tools that insert or translate code (e.g., optimizing compilers and auto-code generators). All enabling systems shall be assigned the same integrity level as the integrity level assigned to the software element that the tool affects. If the tool cannot be verified and validated, then the tool's output shall be subject to the same level of V&V as the software element.

The mapping of the integrity level schema and the associated minimum V&V tasks shall be documented in the VVP. The basis for assigning integrity levels to components shall be documented in a V&V task report and V&V final report. [Table 2a](#) through [Table 2d](#) of this standard identify the necessary and sufficient V&V activities and tasks to perform for each integrity level. Once the integrity levels have been determined, [Table 2a](#) through [Table 2d](#) are used to identify the activities and tasks to adapt and plan the V&V effort. High integrity requires a larger set of V&V processes and a more rigorous application of V&V tasks. The V&V processes should be tailored to specific system elements with a combination of the minimum V&V tasks and addition of optional V&V tasks.

The addition of optional V&V tasks allows the V&V effort to address application-specific characteristics. [Table 3a](#) through [Table 3d](#) provide a list of optional V&V tasks and their suggested applications in the life cycle of the system of interest. These optional V&V tasks may be added to the minimum V&V tasks, as necessary, to tailor the V&V effort to project needs to provide additional assurance and confidence in the operational use of the system.

The integrity level assignment shall be continually reviewed and updated by conducting the V&V criticality analysis task throughout the life cycle. Changes to assignments shall be used to reassess the V&V plan and performance for appropriate integrity assurance throughout the life cycle. If the integrity level of any component is revised, the effect of the revision on existing requirements shall be evaluated to identify additional activities and tasks to be performed on the system, software, and hardware. This includes adopting V&V tasks for the revised component integrity level and performing those V&V tasks for previous life cycle stages.

Critical functions are assigned a high system integrity level at the initiation of the project and shall not be lowered at any stage of the development process unless authorized by the system acquirer or governing regulatory organization. Other system functions may be assigned lower system integrity levels than the critical functions at the beginning stages. However, if any lower system integrity level functions can modify or alter critical data or can create an improper condition or system state to exist causing critical system functions to take incorrect actions, then those lower system functions shall be elevated to critical system integrity levels for V&V analysis and test. For discussion purposes, these system functions shall be designated "critical supporting system functions."

During the evolution of the system through requirements, design, implementation, and integration, the selection of technology or design/implementation techniques could mitigate or eliminate the impact that a "critical supporting system function" may have on critical system functions. In such cases and only with extreme care, the integrity level for a "critical supporting system function" may be lowered for V&V analysis and test purposes. In contrast, there may be means to obtain a higher integrity level by a combination of components having lower integrity levels (e.g., duplication or redundancy—see ISO/IEC 15026-3:2013 [\[B33\]](#)).



Annex K describes an example of how system integrity levels are assigned to “supporting system functions” and how system integrity levels may be changed during the system development stages for the “supporting system functions” as a result of the selection of design/implementation techniques.

## 6. Overview of V&V processes

### 6.1 General

V&V processes have activities and tasks defined for the processes in the Agreement processes and Technical processes group, the Project Planning Process, and the Configuration Management Process of ISO/IEC/IEEE 15288:2023(E) [B41]. V&V processes have activities and tasks defined for the processes in the Software Implementation Processes group and for the software Configuration Management Process of ISO/IEC/IEEE 12207:2017(E) [B40]. Although there is no corresponding IEEE standard that addresses the hardware life cycle, V&V activities and tasks are defined for processes analogous to those for the software life cycle. The minimum V&V activities and tasks supporting these processes are referenced in the following clauses and are defined in [Table 1a](#) through [Table 1d](#) of each part of this standard. The subclause titles in this clause are the same as the column headings in [Table 1a](#) through [Table 1d](#) to correlate the requirements of the following subclauses with [Table 1a](#) through [Table 1d](#) tasks. Not all projects include each of the life cycle processes listed. To conform to this standard, the V&V processes shall address all those life cycle processes used by the project.

V&V activities that are common across system, software, and hardware are in [Clause 7](#). These include the V&V Management activity, which is not specifically described by ISO/IEC/IEEE 15288:2023(E) [B41] or ISO/IEC/IEEE 12207:2017(E) [B40] but is essential to the implementation of V&V tasks. Other common V&V activities in [Clause 7](#) are associated with the Acquisition Process, the Supply Process, the Project Planning Process, and the Configuration Management Process.

V&V activities that are associated with system life cycle processes are in [Clause 8](#) (System V&V). The system processes are those in the technical processes group of ISO/IEC/IEEE 15288:2023(E) [B41].

V&V activities that are associated with software life cycle processes are in [Clause 9](#) (Software V&V). The software processes are those in the technical processes group of ISO/IEC/IEEE 12207:2017(E) [B40].

V&V activities that are associated with hardware life cycle processes are in [Clause 10](#) (Hardware V&V). The hardware processes are based on the technical processes group of ISO/IEC/IEEE 12207:2017(E) [B40].

The V&V effort shall conform to the task descriptions, inputs, and outputs as described in [Table 1a](#) through [Table 1d](#). The V&V effort shall include the minimum V&V tasks specified in [Table 2a](#) through [Table 2d](#) for the assigned integrity levels. If the user of this standard has selected a different integrity level schema, then this standard’s integrity level schema and associated minimum V&V tasks of [Table 2a](#) through [Table 2d](#) shall be mapped to their selected integrity level schema.

Optional V&V tasks may also be performed to augment the V&V effort to satisfy project needs. Optional V&V tasks are listed in [Table 3a](#) through [Table 3d](#) and described in [Annex G](#). The list in [Table 3a](#) through [Table 3d](#) is illustrative and not exhaustive.

Some V&V activities and tasks include analysis, evaluations, and tests that may be performed by multiple organizations (e.g., development, project management, quality assurance, and V&V). For example, risk analysis and hazard analysis may be performed by project management, the development organization, and the V&V effort. The V&V effort performs or uses the outputs from these tasks from other sources to develop the supporting basis of evidence showing whether the product satisfies its requirements and to mitigate risks to the project. These V&V analyses are complementary to other analyses and do not eliminate or replace the analyses performed by other organizations. The degree to which these analysis efforts will be coordinated with other organizations shall be documented in the organizational responsibility section of the VVP.

## 6.2 V&V team characteristics

This standard does not require that V&V be performed by any specific organization and even allows V&V activities to be performed by multiple organizations. However, V&V is often conducted by a well-defined organization, and even when not, the collection of individuals performing V&V activities may be considered to be a team. The term “V&V team” is used in this context.

The V&V team, as a combined group, should have the technical knowledge and expertise to enable the assessment of the system to the rigor and depth indicated by the integrity level of the system. In general, the team should have knowledge and expertise in the following:

- Domain of the system of interest
- Systems engineering
- V&V
- System perspective

The expertise of the team should include not only the engineering of the system itself, but also the overall environment in which the system is intended to operate: general business or mission area, functions and interactions with interfacing systems, support systems, physical environment, and regulatory constraints.

The V&V team needs systems engineering knowledge and expertise to enable assessment of the system's work products, system elements (e.g., software, hardware), interactions among the system elements, and interfaces with other systems and the environment. In this context, systems engineering includes requirements engineering, software engineering, hardware engineering, and systems, software and hardware architecture, design, implementation, and testing.

In addition, the V&V team needs to understand the V&V processes and corresponding life cycle processes (see ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#) and ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)) and the tasks and activities that will be conducted during the V&V effort. This includes V&V planning, V&V management, and interfaces with other stakeholders. The V&V team, and particularly V&V team management, needs to be able to communicate information about the V&V results and about discrepancies discovered during the V&V effort, understanding that different stakeholders need different levels and methods of communication.

In addition to the technical knowledge and expertise regarding the domain of the system of interest, systems engineering, and V&V, the V&V team needs a perspective that allows an understanding of the complete system and an ability to conceptualize how the proposed system might behave in varied circumstances. While performing V&V tasks, the team will need to consider alternatives (e.g., think outside the box, conceive of missing conditions or actions, alternate or abnormal scenarios), perceive relationships and correlations (especially those not explicitly defined), and have a holistic view of the system.

While not each member of the V&V team has to have this complete body of knowledge, the team as a whole should have expertise and knowledge in all these areas. The team may include a mixture of experience, with more senior members giving guidance to more junior members. In the case of cutting-edge technologies, the V&V team might not start with a strong understanding of some of these aspects, since the technologies are still being developed. In this case, a deliberate effort should be conducted to gain expertise in the new technologies to enable assessment of the work products.

Some members of the V&V team may have specific expertise that is utilized only sporadically during the life cycle, and they will engage in the V&V effort at specific times during the life cycle. However, it is best if some members of the V&V team are engaged throughout the life cycle in order to form an understanding of the full system, its elements and their interactions, and the interactions of the system with its environment. As the V&V effort evolves, the members of the team will develop detailed system knowledge and expertise.

Other skills, such as the ability to communicate within the V&V team and with developers, development management, and the customer, are also important. Communication should be tailored to each recipient, and



should be based on factual findings from V&V tasks and directed at successful development of the system. The V&V team should display high ethical standards in both the performance of V&V tasks and in communication with all parties.

### 6.3 V&V testing

Testing is an activity that may be performed by various organizations within the development effort. This standard requires a minimum level of V&V testing dependent on the integrity level, as reflected in [Table 1a](#) (common V&V activities and tasks), [Table 1b](#) (system V&V activities and tasks), [Table 1c](#) (software V&V activities and tasks), and [Table 1d](#) (hardware V&V activities and tasks). [Figure 7](#) summarizes the minimum level for V&V testing for the types of testing and integrity level. The term perform means that the V&V effort specifies and creates its testing products (i.e., test plan, test design, test cases, and test procedures) and either conducts that testing or analyzes the results of that testing if it is conducted by another organization. The term review means that the V&V effort reviews the testing plans and analyzes the results of tests.

| Software                           | V&V testing by integrity level |         |        |           |
|------------------------------------|--------------------------------|---------|--------|-----------|
|                                    | 4                              | 3       | 2      | 1         |
| Software component testing V&V     | Perform                        | Perform | Review | No action |
| Software integration testing V&V   | Perform                        | Perform | Review | No action |
| Software qualification testing V&V | Perform                        | Perform | Review | No action |
| Software acceptance testing V&V    | Perform                        | Perform | Review | No action |

| Hardware                           | V&V testing by integrity level |         |        |           |
|------------------------------------|--------------------------------|---------|--------|-----------|
|                                    | 4                              | 3       | 2      | 1         |
| Hardware component testing V&V     | Review                         | Review  | Review | No action |
| Hardware integration testing V&V   | Review                         | Review  | Review | No action |
| Hardware qualification testing V&V | Perform                        | Perform | Review | No action |
| Hardware acceptance testing V&V    | Perform                        | Perform | Review | No action |

| System                           | V&V testing by integrity level |        |        |           |
|----------------------------------|--------------------------------|--------|--------|-----------|
|                                  | 4                              | 3      | 2      | 1         |
| System integration testing V&V   | Perform                        | Review | Review | No action |
| System qualification testing V&V | Perform                        | Review | Review | No action |
| System acceptance testing V&V    | Perform                        | Review | Review | No action |

NOTE 1—The requirement to “perform” testing means that the V&V effort creates its testing criteria, confirms that the testing is conducted appropriately, and reviews the results. The term allows but does not require that the V&V effort conducts tests if the identical V&V criteria are already specified in other testing. Alternatives to conducting tests include methods such as witnessing a test conducted by the supplier or developer.

Having the V&V effort “perform” its own testing increases the detection of system/software/hardware errors, especially for high integrity levels, because V&V testing can provide the following benefits:

- Examines the stress, abnormal, and fault recovery scenarios that development testing often does not have sufficient time for such test case scenarios.
- Creates additional testing coverage of the system/software/hardware from the V&V perspective of how the system should perform with inputs and stimuli different from the development test scenarios such that those different inputs/stimuli may expose errors undetected in development test scenarios.

NOTE 2—For V&V testing, it is ideal for the V&V processes to have their own test drivers, test simulations, test models, and system test facility. It is not the intended purpose for V&V processes to repeat or duplicate the same or similar testing as accomplished by the development processes. Rather, V&V testing provides objective evidence that the high integrity level requirements perform as intended under all operational scenarios, including stress and abnormal conditions, error recovery situations, and high-risk hazardous situations. Having a separate V&V test capability allows for development testing to be uninterrupted (not shared); permits special test scenarios, error injection, and test monitoring capability to be integrated into the test facility; and provides for an independent test capability to investigate erroneous and unusual scenarios without interference with the development test schedule.

However, because of budgetary considerations, the development testing and V&V testing often share the same system test capabilities (e.g., resources or facilities). When sharing the same test capabilities with the development testing, the V&V effort shall establish the following procedures on common facility usage to maintain the integrity of both the V&V and development test processes:

- Maintain a controlled copy of system and hardware settings and software elements used in the test facility, such as test drivers, test simulations, interface drivers, and software items that are being tested and undergoing V&V; (These configured items and descriptions are required to allow V&V testing to set the test facility to a known state before beginning V&V testing since items could have been changed during the development testing period.)
- Initialize the test facility to the V&V test configuration at the start of V&V testing; (The test facility configuration may be restored to the prior test configuration at the conclusion of V&V testing.)
- Use V&V test procedures and V&V-provided system test database configurations and scenarios.
- Monitor and record the V&V test processes and results (although V&V testing may share the test facility operators used by development testing).

The V&V effort generates the input data and interfacing data inputs in accordance with the V&V test cases/procedures. The objectives of V&V testing are to identify the system/software/hardware element errors. Often, the small differences in test data and interface data inputs may cause the error source(s) to be evident. Testing system/software/hardware during key system transition states and at all boundary conditions is a method of exercising the key requirements of the system. Satisfactory V&V testing results and the absence of major errors constitute the objective evidence for verifying and validating the system.

NOTE 3—Hardware integration takes place at many levels. For electronics, this can range from the chip level and board level to a fully integrated computer system that can be integrated into a larger computer network. For a mechanical system, the levels of integration can range from a nut-and-bolt assembly to a complex multi-degree-of-freedom system. At the lowest level of integration, the issues are so elemental or the integration processes so standardized and well understood that V&V of the integration may not be required or applicable. In some cases, as in the assembly of a circuit board, the V&V activities may be carried out by the manufacturer during the integration of the assembled product.

### Figure 7—Minimum level for V&V testing by integrity level

[Annex A](#) describes V&V testing concepts and methodology to verify and validate the system/software component(s) for correctness, completeness, safety, security, and accuracy of the entire system/software performance requirements through all phases of the operational environment. The V&V testing is complementary to the development testing rather than duplicative or redundant. The annex describes the variety of different testing strategies that can be used by V&V testing to expose any errors or deficiencies as the system/software evolves through the development life cycle. The V&V testing methodology begins early in a system/software life cycle where the first test planning strategies (e.g., test plan, test scenarios, test design, and test cases) are formulated by the analytic V&V tasks performed in life cycle stages, as early as system requirements definition. Often system/software errors or incomplete solutions are discovered during analysis of test strategies across all operating conditions, which may reveal a deficiency or incompleteness pertaining to the requirements or design solution. By testing against stress and boundary conditions and conducting a comprehensive testing strategy of attempting to surface errors or deficiencies across the full performance spectrum of conditions, the V&V testing provides a comprehensive and supportive testing approach to development testing.

## 7. Common V&V processes

### 7.1 V&V Management Process

#### 7.1.1 Purpose

The purpose of the V&V Management Process is to develop and maintain the V&V plan, determine the status of the V&V effort, and direct the V&V effort to perform according to plans and satisfy technical objectives.

### 7.1.2 Outcomes

The following outcomes occur as a result of the successful implementation of the V&V Management Process:

- a) The VVP is developed.
- b) The V&V effort is continuously reviewed.
- c) Revisions of the VVP are made as necessary based on updated project schedules, development status, or changes in the V&V approach.
- d) V&V results are coordinated with other parties performing life cycle activities.
- e) V&V task results are reviewed for conformance to task requirements.
- f) Recommendations are made to program management regarding readiness to proceed to the next stage of the project life cycle, acceptance, and certification.
- g) Process improvement opportunities in the conduct of V&V are identified.

NOTE—V&V management is focused on management activities unique to V&V. Other project management activities that are common across all projects, such as cost management and human resource management, are not addressed by this standard.

The V&V Management Process monitors and evaluates all V&V outputs. Management of the V&V effort is performed for all life cycle V&V processes and activities. All the V&V reporting, administrative, and documentation requirements are defined in [Clause 11](#). The outline and content of the VVP are described in [Clause 12](#).

V&V management assesses each proposed change to the system, software, or hardware; identifies the requirements that are affected by the change; and plans V&V tasks to address the change. For each proposed change, management assesses whether any new hazards or risks are introduced in the development process and identifies the impact of the change on the assigned integrity levels. V&V task planning is revised by adding new V&V tasks or changing the scope or intensity of existing V&V tasks if integrity levels, hazards, or risks are changed. A baseline change results from changes allocated to releases in an incremental development process (e.g., planned baseline versions).

Through the use of V&V measures and other qualitative and quantitative measures, program trend data and possible risk issues are developed and provided to the developer and acquirer to effect timely notification and resolution. At key program milestones (e.g., requirements review, design review, and test readiness), V&V management consolidates the V&V results to establish supporting evidence of whether to proceed to the next stage of the project life cycle. V&V management determines whether a V&V task should be repeated as a result of changes in the program.

### 7.1.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2a](#) for the selected integrity level, the following V&V management activity and tasks described in [Table 1a, Activity 7.1](#):

- a) [V&V Management](#). This activity consists of the following tasks:
  - 1) [VVP generation](#)
  - 2) [Interface with other processes](#)
  - 3) [Proposed/baseline change assessment](#)
  - 4) [Management review of the V&V effort](#)
  - 5) [Management and technical review support](#)
  - 6) [Identify process improvement opportunities in the conduct of V&V](#)
  - 7) [V&V final report generation](#)

## 7.2 Acquisition Support V&V Process

### 7.2.1 Purpose

The purpose of the Acquisition Support V&V Process is to provide assurance that the outcomes of the Acquisition Process (ISO/IEC/IEEE 15288:2023(E) [B41] or ISO/IEC/IEEE 12207:2018(E) [B40]) have been achieved.

### 7.2.2 Outcomes

The following outcomes occur as a result of the successful implementation of the Acquisition Support V&V Process:

- a) Interfaces are planned with the supplier and acquirer.
- b) System requirements to be included in the request for proposal (RFP) are reviewed.
- c) V&V task results are provided to support the acquirer's acceptance of the system.

### 7.2.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2a](#) for the selected integrity level, the following acquisition support V&V activity and tasks described in [Table 1a, Activity 7.2](#):

- a) [Acquisition support V&V](#). This activity consists of the following tasks:
  - 1) [Scoping the V&V effort](#)
  - 2) [Planning the interface between the V&V effort and supplier](#)
  - 3) [System requirements review](#)
  - 4) [Acceptance support](#)

Acquirer acceptance of the system of interest culminates after acceptance testing and installation. The V&V acquisition acceptance support activities occur throughout the life cycle, in conjunction with other interrelated development and V&V tasks, inputs, and outputs.

The acquisition support V&V activity addresses project initiation, RFP, contract preparation, supplier monitoring, and acceptance and completion.

For a system of interest that is determined to be an identical copy of a system that has been previously verified and validated, and the environment is also identical, then no further V&V tasks are needed beyond those to provide the body of evidence that the system and environment are identical. This copy of a system is often referred to as an “n<sup>th</sup> of a kind system.” See [Annex L](#) for a more detailed discussion.

## 7.3 Supply Planning V&V Process

### 7.3.1 Purpose

The purpose of the Supply Planning V&V Process is to provide assurance that the outcomes of the Supply Planning Process (ISO/IEC/IEEE 15288:2023(E) [B41] or ISO/IEC/IEEE 12207:2018(E) [B40]) have been achieved.

### 7.3.2 Outcomes

The following outcomes occur as a result of the successful implementation of the Supply Planning V&V Process:

- a) Interfaces are planned with the supplier and acquirer.
- b) RFP requirements and contract requirements are determined to be consistent and satisfy user needs before the contract is finalized.

### 7.3.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2a](#) for the selected integrity level, the following supply planning V&V activity and tasks described in [Table 1a, Activity 7.3](#):

- a) [Supply planning V&V](#). This activity consists of the following tasks:
  - 1) [Planning the interface between the V&V effort and supplier](#)
  - 2) [Contract Verification](#)

The supply planning V&V activity addresses the initiation, preparation of response, contract, planning, execution and control, review, and evaluation, as well as delivery and completion activities.

## 7.4 Project Planning V&V Process

### 7.4.1 Purpose

The purpose of the Project Planning V&V Process is to provide assurance that the project scope is complete and that all activities are defined.

### 7.4.2 Outcomes

As a result of the successful implementation of the Project Planning V&V Process, the VVP is coordinated with the project plan.

### 7.4.3 Activity and task

The V&V effort shall perform, as specified in [Table 2a](#) for the selected integrity level, the following Project Planning Process V&V activity and task described in [Table 1a, Activity 7.4](#):

- a) [Project planning V&V](#). This activity consists of the following task:
  - 1) [Project planning strategy assessment](#)

## 7.5 Configuration Management V&V Process

### 7.5.1 Purpose

The purpose of the Configuration Management V&V Process is to provide assurance that the Configuration Management Process (ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#) or ISO/IEC/IEEE 12207:2018(E) [\[B40\]](#)) supports the V&V processes.

### 7.5.2 Outcomes

The following outcomes occur as a result of the successful implementation of the Configuration Management V&V Process:

- a) A configuration management strategy is defined for the program that will assure configuration controls are in place to maintain and document configuration item baselines with unique identifiers. The strategy should include notification to the V&V effort for all changes made to the configuration item baselines.
- b) Items requiring configuration management are defined and documented. The items controlled should include enabling the systems, tools, and processes that are integral to system development and life cycle support that will be subject to V&V to demonstrate conformance to this standard.
- c) The status of items under configuration management is made available throughout the life cycle. Provisions should be included to assure that the V&V effort receives the current status for all configuration items.

### 7.5.3 Activity and task

The V&V effort shall perform, as specified in [Table 2a](#) for the selected system integrity level, the following Configuration Management Process V&V activity and task described in [Table 1a, Activity 7.5](#):

- a) [Configuration management V&V](#): This activity consists of the following task:
  - 1) [Configuration management assessment](#)

If configuration management issues are discovered during the V&V effort that indicate configuration baselines are not established or controlled, or if the configuration of released items is not controlled, the issues shall be documented and provided to the system developer and to the V&V customer for resolution.

**Table 1a—V&V tasks, inputs, and outputs**

| <b>7.1 Activity: V&amp;V management (common)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                 |                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Required inputs</b>                                                                                                                                                                                                                                          | <b>Required outputs</b>                                                                                                                             |
| <p><b>(1) <u>VVP generation</u></b></p> <p>a) Generate a VVP (see <a href="#">Clause 12</a>) for all life cycle processes. The VVP may require updating throughout the life cycle. Outputs of other activities are inputs to the VVP.</p> <p>b) Establish a baseline VVP prior to the requirements V&amp;V activities.</p> <p>c) Identify project milestones in the VVP.</p> <p>d) Schedule V&amp;V tasks to support project management reviews and technical reviews.</p> <p>NOTE—The required input of concept documentation includes statement of need, advance planning report, project initiation memo, feasibility studies, system requirements, governing regulations, procedures, policies, customer acceptance criteria and requirements, acquisition documentation, business rules, and draft system architecture</p> | <p>VVP (previous update)</p> <p>Contract</p> <p>Concept documentation</p> <p>Supplier development plans and schedules</p>                                                                                                                                       | <p>VVP and updates</p>                                                                                                                              |
| <p><b>(2) <u>Interface with other processes</u></b></p> <p>a) Coordinate the V&amp;V effort with organizational project-enabling processes and project processes.</p> <p>b) Identify the V&amp;V data to be exchanged with these processes.</p> <p>c) Document the data exchange requirements in the VVP.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>VVP</p> <p>Data identified in the VVP from the organizational project-enabling processes and project processes</p>                                                                                                                                           | <p>Updated VVP</p>                                                                                                                                  |
| <p><b>(3) <u>Proposed/baseline change assessment</u></b></p> <p>a) Evaluate proposed changes (i.e., modifications, enhancements, and additions as a result of anomaly corrections or requirement changes) for effects on the system and previously completed V&amp;V tasks.</p> <p>b) Plan iteration of affected tasks or initiate new tasks to address proposed changes or baseline changes associated with an iterative development process.</p> <p>c) Verify and validate that changes are consistent with requirements and do not adversely affect requirements directly or indirectly. An adverse effect is a change that could create new system hazards and risks or impact previously resolved hazards and risks.</p>                                                                                                   | <p>VVP</p> <p>Proposed changes</p> <p>Hazard analysis results</p> <p>Risks identified by V&amp;V tasks</p> <p>Supplier development plans and schedules</p> <p>Developer products (produced to date)</p>                                                         | <p>Task report(s)—Proposed/baseline change assessment</p> <p>Updated VVP</p> <p>Anomaly report(s)</p>                                               |
| <p><b>(4) <u>Management review of the V&amp;V effort</u></b></p> <p>a) Review and summarize the V&amp;V effort to define changes to V&amp;V tasks or to redirect the V&amp;V effort.</p> <p>b) Evaluate each anomaly for its criticality (e.g., IEEE Std 1044™-2009 [B21]). Assess anomalies and risks for trends and tendencies. The scope and application of V&amp;V activities and tasks shall be revised to address the causes of these anomalies and risks.</p> <p>c) Recommend whether to proceed to the next stage of the project life cycle and provide task reports, anomaly reports, and V&amp;V activity summary reports to the organizations identified in the VVP.</p> <p>d) Verify that all V&amp;V tasks conform to task requirements defined in the VVP.</p>                                                    | <p>VVP and updates</p> <p>Supplier development plans and schedules</p> <p>Anomaly reports</p> <p>V&amp;V task results [e.g., technical accomplishments, V&amp;V reports, resource utilization, V&amp;V measures (see Annex E), plans, and identified risks]</p> | <p>Task report(s)—Recommendations</p> <p>Updated VVP</p> <p>V&amp;V activity summary reports</p> <p>Recommendations to the V&amp;V final report</p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                     |                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <p>e) Verify that V&amp;V task results have a basis of evidence supporting the results.</p> <p>f) Assess all V&amp;V results and provide recommendations for program acceptance and certification as input to the V&amp;V final report.</p> <p>g) Use the results of the review to identify process improvement opportunities in the conduct of V&amp;V.</p> <p>h) Review the quality of the products and services to assure they meet customer requirements.</p> <p>i) Review the program risks and initiate actions to mitigate risks that exceed thresholds.</p> <p>j) Review program measures (e.g., V&amp;V measures, test results, anomaly reports, timing, and capacity results) to assure the quality of products and processes.</p> <p>NOTE—For additional information and guidance on review methods, see IEEE Std 1028-2008 [B20].</p>                                                                                                                                                                                                                                                         |                                                                                                                                                                                                     |                                                               |
| <p>(5) <b><u>Management and technical review support</u></b></p> <p>a) Support project management reviews and provide V&amp;V results at project-defined technical reviews (e.g., preliminary design review and critical design review) by assessing the review materials, attending the reviews, and providing task reports and anomaly reports. Provide objective evidence on the status of development products (system, software, hardware) to project management authorities, including recommendations and assessment of positive trends and anomaly trends. This evidence can be used to determine whether the criteria of major program milestones have been achieved and whether to proceed to the next stage of the project life cycle. Provide an assessment of the positive trends and anomaly trends to allow greater focus toward the functional program requirements warranting more resources and additional analysis support or to require changes in V&amp;V resources or tasks.</p> <p>b) Verify timely delivery according to the approved schedule of all products and documents.</p> | <p>V&amp;V task results</p> <p>Materials for review (e.g., operational concept document [OCD], architectural and design documents, system requirements, SRS, IRS, SDD, IDD, and test documents)</p> | <p>Task report(s)—Review results</p> <p>Anomaly report(s)</p> |
| <p>(6) <b><u>Identify process improvement opportunities in the conduct of V&amp;V</u></b></p> <p>a) Gather and analyze the lessons learned.</p> <p>b) Gather and analyze the risks identified.</p> <p>c) Gather and analyze the V&amp;V measures.</p> <p>d) Identify and analyze deficiencies in the V&amp;V processes.</p> <p>e) Determine and implement corrective actions using best practices (e.g., repeat V&amp;V tasks, conduct a new V&amp;V task to address corrective actions, or use a different method/technique for executing a V&amp;V task).</p> <p>f) Monitor the efficacy of the corrective actions.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>V&amp;V task results</p> <p>materials for review (e.g., OCD, architectural and design documents, system requirements, SRS, IRS, SDD, IDD, and test documents)</p>                                | <p>Task report(s)—Review results</p> <p>Anomaly report(s)</p> |
| <p>(7) <b><u>V&amp;V final report generation</u></b></p> <p>a) Summarize in the V&amp;V final report the V&amp;V activities, tasks, and results, including status and disposition of anomalies.</p> <p>b) Provide an assessment of the overall quality and provide recommendations.</p> <p>c) Document V&amp;V process improvement opportunities.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>V&amp;V activity summary report(s)</p>                                                                                                                                                           | <p>Task report(s)—V&amp;V final report</p>                    |



| NOTE—The V&V final report is intended to be produced at the completion of V&V following development, operations, maintenance, or retirement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                       |                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>7.2 Activity: Acquisition support V&amp;V (common, 15288—Acquisition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                       |                                                                     |
| V&V tasks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Required inputs                                                                                       | Required outputs                                                    |
| <b>(1) <u>Scoping the V&amp;V effort</u></b><br>a) Determine the characteristics (e.g., complexity, criticality, risk, safety, security, desired performance, reliability, or other project-unique characteristics) that define the importance of the system of interest to the user.<br>b) Adopt the system integrity schema assigned to the project. If no integrity level schema exists, then one is developed.<br>c) Assign an integrity level to the system of interest.<br>d) Establish the degree of independence (see <a href="#">Annex C</a> ), if any, required for the V&V.<br>e) Determine the minimum V&V tasks for the integrity level using Table 2a through Table 2d and the selected integrity level schema.<br>f) Determine the extent of V&V on reuse items selected for the program (see <a href="#">Annex D</a> ).<br>g) Determine the extent of V&V on an n <sup>th</sup> of a kind system (see <a href="#">Annex M</a> ).<br>h) Determine the extent of V&V for tools that insert or translate code (e.g., optimizing compilers and auto-code generators).<br>i) Augment the minimum V&V tasks with optional V&V tasks, as necessary.<br>j) Provide an estimate of the V&V budget, including test facilities, and tools as required. | Preliminary system description<br>Statement of need<br>RFP or tender<br>System integrity level schema | VVP                                                                 |
| <b>(2) <u>Planning the interface between the V&amp;V effort and supplier</u></b><br>(Preparing the preliminary data and processes for the interface with a supplier to be selected in the Supply Process.)<br>a) Incorporate the project integrity level schema into the planning process.<br>b) Plan the V&V schedule for each V&V task.<br>c) Identify the preliminary list of development processes and products to be evaluated by the V&V processes.<br>d) Describe V&V access rights to proprietary and classified information.<br>e) Coordinate the plan with the acquirer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | VVP<br>Draft RFP or tender<br>Contract                                                                | Task report(s)—<br>Recommendations for RFP or tender<br>Updated VVP |
| <b>(3) <u>System requirements review</u></b><br>a) Review the system requirements (e.g., system requirements specification, feasibility study report, business rules description) in the RFP or tender to perform the following:<br>b) Verify the consistency of requirements to user needs.<br>c) Validate whether the requirements can be satisfied by the defined technologies, methods, and algorithms defined for the project (feasibility).<br>d) Verify whether objective information that can be demonstrated by testing is provided in the requirements (testability).<br>e) Review other requirements, such as deliverable definitions, listing of appropriate compliance standards and regulations,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Preliminary system description<br>Statement of need<br>User needs<br>Draft RFP or tender              | Task report(s)—<br>System requirements review<br>Anomaly report(s)  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                   |                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| performance, safety, security and quality attributes, and user needs for completeness, correctness, and accuracy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                   |                                                                                                                                                           |
| <p><b>(4) <u>Acceptance support</u></b><br/>(The following V&amp;V activities support acceptance in the Acquisition Process. The activities are described in various V&amp;V tasks across the life cycle where required inputs for the V&amp;V activities are generated to aid the understanding of the activity flow.)</p> <p>a) System acceptance test plan V&amp;V [8.3, Task 6].</p> <p>b) System acceptance test design V&amp;V [8.4, Task 8].</p> <p>c) System acceptance test case V&amp;V [8.5, Task 8].</p> <p>d) System acceptance test procedure V&amp;V [8.7, Task 7].</p> <p>e) System acceptance test execution V&amp;V [8.10, Task 3].</p> <p>f) Software acceptance test plan V&amp;V [9.3, Task 6].</p> <p>g) Software acceptance test design V&amp;V [9.4, Task 9].</p> <p>h) Software acceptance test case V&amp;V [9.5, Task 10].</p> <p>i) Software acceptance test procedure V&amp;V [9.7, Task 9].</p> <p>j) Software acceptance test execution V&amp;V [9.10, Task 3].</p> <p>k) Hardware acceptance test plan V&amp;V [10.3, Task 6].</p> <p>l) Hardware acceptance test design V&amp;V [10.4, Task 9].</p> <p>m) Hardware acceptance test case V&amp;V [10.5, Task 10].</p> <p>n) Hardware acceptance test procedure V&amp;V [10.7, Task 9].</p> <p>o) Hardware acceptance test execution V&amp;V [10.10, Task 3].</p> | <p>Concept documentation</p> <p>SDD</p> <p>IDD</p> <p>SRS</p> <p>IRS</p> <p>Source code</p> <p>Hardware items</p> <p>Executable code</p> <p>User documentation</p> <p>Test plans, designs, cases, procedures, results</p> <p>Acceptance test plan</p> <p>V&amp;V task results</p> | <p>Task report(s)—Acceptance Support V&amp;V</p> <p>Acceptance V&amp;V test plan, design(s), cases, procedures, test results</p> <p>Anomaly report(s)</p> |
| <b>7.3 Activity: Supply planning V&amp;V (common, 15288—Supply Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                   |                                                                                                                                                           |
| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Required inputs</b>                                                                                                                                                                                                                                                            | <b>Required outputs</b>                                                                                                                                   |
| <p><b>(1) <u>Planning the interface between the V&amp;V effort and supplier</u></b><br/>(Coordinating and documenting the interface data and processes with the selected supplier)</p> <p>a) Review the supplier development plans and schedules to coordinate the V&amp;V effort with development activities.</p> <p>b) Establish procedures to exchange V&amp;V data and results with the development effort.</p> <p>c) Coordinate the plan with the supplier.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>VVP</p> <p>Contract</p> <p>Supplier development plans and schedules</p>                                                                                                                                                                                                        | <p>Updated VVP</p>                                                                                                                                        |
| <p><b>(2) <u>Contract Verification</u></b><br/>Verify the following characteristics of the contract:</p> <p>a) System requirements (from RFP or tender, and contract) satisfy and are consistent with user needs.</p> <p>b) Procedures are documented for managing requirement changes and for identifying the management hierarchy to address problems.</p> <p>c) Procedures for interface and cooperation among the parties are documented, including ownership, warranty, copyright, and confidentiality.</p> <p>d) Acceptance criteria and procedures are documented in accordance with requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>VVP</p> <p>RFP or tender</p> <p>Contract</p> <p>User needs</p> <p>Supplier development plans and schedules</p>                                                                                                                                                                 | <p>Task report(s)—Contract Verification</p> <p>Updated VVP</p> <p>Anomaly report(s)</p>                                                                   |
| <b>7.4 Activity: Project planning V&amp;V (common, 15288—Project Planning Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                   |                                                                                                                                                           |
| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Required inputs</b>                                                                                                                                                                                                                                                            | <b>Required outputs</b>                                                                                                                                   |
| <p><b>(1) <u>Project planning strategy assessment</u></b></p> <p>a) Verify the project plan strategy, including objectives, and constraints.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Project plan strategy</p> <p>Schedules and tasks</p>                                                                                                                                                                                                                           | <p>Task report(s)—Project planning strategy assessment</p>                                                                                                |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                   |                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------|
| b) Verify project scope includes all relevant activities.<br>c) Verify life cycle model is defined.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Requirements<br>Life cycle model<br>Project scope | Anomaly report(s)                                                           |
| <b>7.5 Activity: Configuration management V&amp;V (common, 15288—Configuration Management Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                   |                                                                             |
| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Required inputs</b>                            | <b>Required outputs</b>                                                     |
| <b>(1) <u>Configuration management assessment</u></b><br>Verify that the Configuration Management Process is complete and adequate. The task criteria are as follows:<br>a) Completeness<br>Verify that there is a process for describing the system, software, and hardware product functionality, tracking program versions, generating baselines (including parameters and settings), and managing changes.<br>b) Adequacy<br>Verify that the Configuration Management Process is adequate for the development complexity, system size, integrity level, project plans, and user needs. | Configuration Management Process documentation    | Task report(s)—<br>Configuration management assessment<br>Anomaly report(s) |
| NOTE (for Table 1a)—Other inputs may be used. For any V&V activity and task, all of the required inputs and outputs from preceding activities and tasks may be used, but for conciseness, only the primary inputs are listed.                                                                                                                                                                                                                                                                                                                                                              |                                                   |                                                                             |

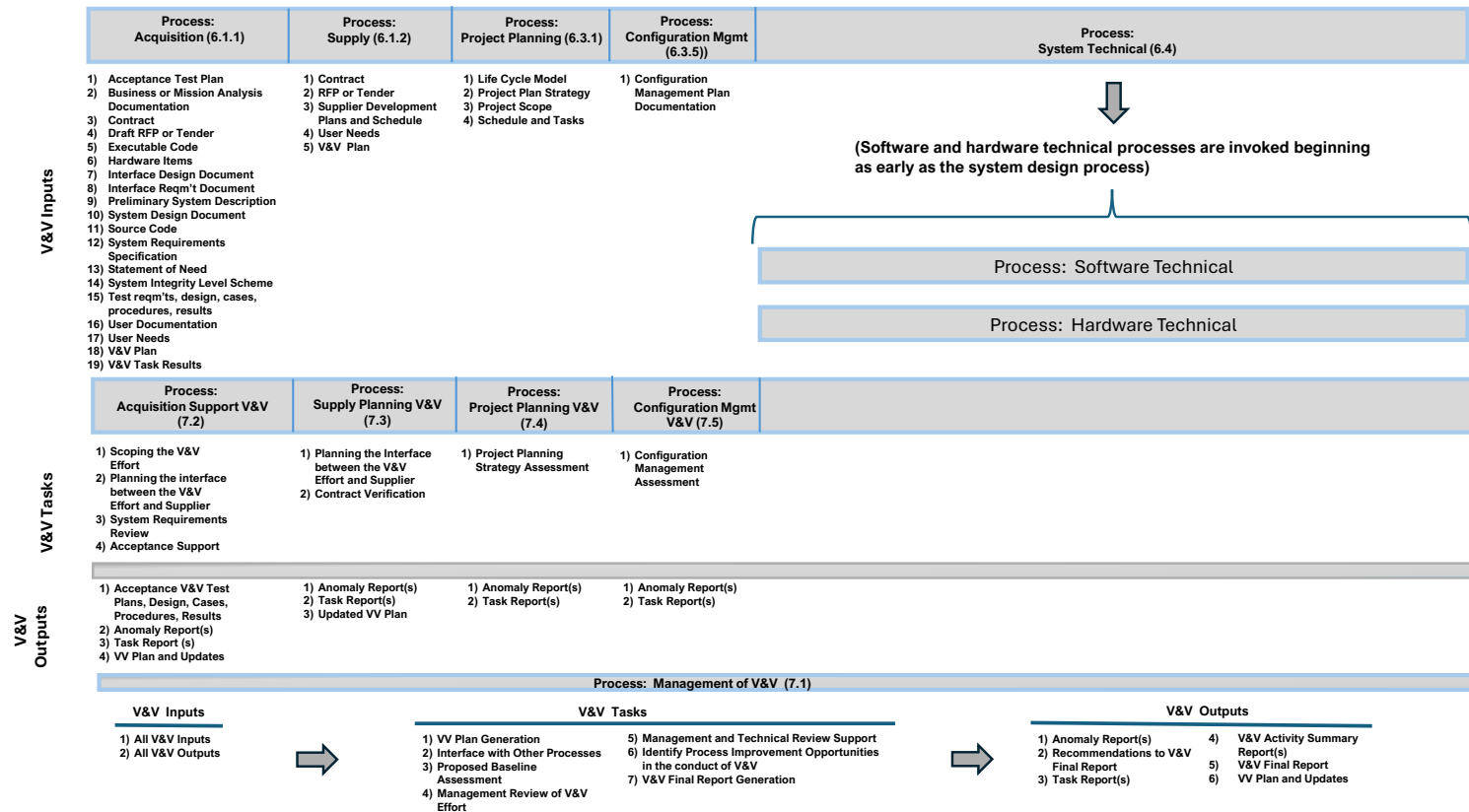
**Table 2a—Minimum V&V tasks assigned to each integrity level for common V&V**

| V&V processes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | V&V management<br>(see 7.1) |   |   |   | Acquisition support<br>V&V<br>(see 7.2) |   |   |   | Supply planning V&V<br>(see 7.3) |   |   |   | Project planning V&V<br>(see 7.4) |   |   |   | Configuration<br>management V&V<br>(see 7.5) |   |   |   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|---|---|---|-----------------------------------------|---|---|---|----------------------------------|---|---|---|-----------------------------------|---|---|---|----------------------------------------------|---|---|---|
| Integrity levels                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Levels                      |   |   |   | Levels                                  |   |   |   | Levels                           |   |   |   | Levels                            |   |   |   | Levels                                       |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4                           | 3 | 2 | 1 | 4                                       | 3 | 2 | 1 | 4                                | 3 | 2 | 1 | 4                                 | 3 | 2 | 1 | 4                                            | 3 | 2 | 1 |
| Acceptance support                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | —                           | — | — | — | X                                       | X | X | X | —                                | — | — | — | —                                 | — | — | — | —                                            | — | — | — |
| Configuration management assessment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | —                           | — | — | — | —                                       | — | — | — | —                                | — | — | — | —                                 | — | — | — | X                                            | X | X | X |
| Contract Verification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | —                           | — | — | — | —                                       | — | — | — | X                                | X | X | X | —                                 | — | — | — | —                                            | — | — | — |
| Identify process improvement opportunities in the conduct of V&V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | X                           | X | X | X | —                                       | — | — | — | —                                | — | — | — | —                                 | — | — | — | —                                            | — | — | — |
| Interface with other processes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | X                           | X | X | X | —                                       | — | — | — | —                                | — | — | — | —                                 | — | — | — | —                                            | — | — | — |
| Management and technical review support                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | X                           | X | X | X | —                                       | — | — | — | —                                | — | — | — | —                                 | — | — | — | —                                            | — | — | — |
| Management review of the V&V effort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | X                           | X | X | X | —                                       | — | — | — | —                                | — | — | — | —                                 | — | — | — | —                                            | — | — | — |
| Planning the interface between the V&V effort and the supplier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | —                           | — | — | — | X                                       | X | X | X | X                                | X | X | X | —                                 | — | — | — | —                                            | — | — | — |
| Project planning strategy assessment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | —                           | — | — | — | —                                       | — | — | — | —                                | — | — | — | X                                 | X | X | X | —                                            | — | — | — |
| Proposed/baseline change assessment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | X                           | X | X | X | —                                       | — | — | — | —                                | — | — | — | —                                 | — | — | — | —                                            | — | — | — |
| Scoping the V&V effort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                           | — | — | — | X                                       | X | X | X | —                                | — | — | — | —                                 | — | — | — | —                                            | — | — | — |
| System requirements review                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | —                           | — | — | — | X                                       | X | X | X | —                                | — | — | — | —                                 | — | — | — | —                                            | — | — | — |
| V&V final report generation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | X                           | X | X | X | —                                       | — | — | — | —                                | — | — | — | —                                 | — | — | — | —                                            | — | — | — |
| VVP generation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | X                           | X | X | X | —                                       | — | — | — | —                                | — | — | — | —                                 | — | — | — | —                                            | — | — | — |
| NOTE—Whenever a V&V task is selected as a mandatory requirement for multiple integrity levels, the V&V task implementation is dictated by the rigor, intensity, and depth of the analysis or test. A higher integrity level implementation requires greater rigor (e.g., formal methods or structured analysis methods), intensity (e.g., consideration of all system conditions and system environment states), and depth (e.g., abnormal cases, boundary conditions, or comprehensive fault and recovery scenarios) of the analysis or test than the lower integrity level implementation. |                             |   |   |   |                                         |   |   |   |                                  |   |   |   |                                   |   |   |   |                                              |   |   |   |

The recommended applicability of optional tasks to the common V&V processes described in [Clause 7](#) is shown in [Table 3a](#). [Annex G](#) provides a description of each of the optional V&V tasks.

**Table 3a—Optional V&V tasks and suggested applications in System Agreement and Project processes**

|                                                           | <a href="#">Acquisition<br/>(7.2)</a> | <a href="#">Supply<br/>(7.3)</a> | <a href="#">Project<br/>planning<br/>(7.4)</a> |
|-----------------------------------------------------------|---------------------------------------|----------------------------------|------------------------------------------------|
| <a href="#">Algorithm analysis</a>                        | —                                     | X                                | —                                              |
| <a href="#">Audit performance</a>                         | —                                     | —                                | —                                              |
| <a href="#">Audit support</a>                             | —                                     | —                                | X                                              |
| <a href="#">Control flow analysis</a>                     | —                                     | —                                | —                                              |
| <a href="#">Cost analysis</a>                             | X                                     | X                                | X                                              |
| <a href="#">Database analysis</a>                         | —                                     | —                                | —                                              |
| <a href="#">Data flow analysis</a>                        | —                                     | —                                | —                                              |
| <a href="#">Disaster recovery plan assessment</a>         | —                                     | —                                | X                                              |
| <a href="#">Distributed architecture assessment</a>       | —                                     | —                                | —                                              |
| <a href="#">Exploratory Testing</a>                       | —                                     | —                                | —                                              |
| <a href="#">Feasibility study evaluation</a>              | X                                     | X                                | X                                              |
| <a href="#">Independent risk assessment</a>               | —                                     | —                                | —                                              |
| <a href="#">Inspection</a>                                | —                                     | —                                | —                                              |
| <a href="#">Inspection—Concept</a>                        | —                                     | —                                | —                                              |
| <a href="#">Inspection—Requirements</a>                   | —                                     | —                                | —                                              |
| <a href="#">Inspection—Design</a>                         | —                                     | —                                | —                                              |
| <a href="#">Inspection—Source code</a>                    | —                                     | —                                | —                                              |
| <a href="#">Inspection—Test plan</a>                      | —                                     | —                                | —                                              |
| <a href="#">Inspection—Test design</a>                    | —                                     | —                                | —                                              |
| <a href="#">Inspection—Test case</a>                      | —                                     | —                                | —                                              |
| <a href="#">Operational evaluation</a>                    | —                                     | X                                | —                                              |
| <a href="#">Performance monitoring</a>                    | X                                     | —                                | —                                              |
| <a href="#">Post-installation Validation</a>              | —                                     | —                                | —                                              |
| <a href="#">Project management oversight support</a>      | X                                     | X                                | X                                              |
| <a href="#">Proposal evaluation support</a>               | X                                     | —                                | —                                              |
| <a href="#">Regression analysis and testing</a>           | —                                     | —                                | —                                              |
| <a href="#">Reusability analysis</a>                      | X                                     | X                                | X                                              |
| <a href="#">Reuse analysis</a>                            | X                                     | X                                | X                                              |
| <a href="#">Simulation analysis</a>                       | —                                     | X                                | —                                              |
| <a href="#">Sizing and timing analysis</a>                | —                                     | —                                | —                                              |
| <a href="#">System software assessment</a>                | —                                     | —                                | —                                              |
| <a href="#">Test certification</a>                        | —                                     | —                                | —                                              |
| <a href="#">Test evaluation</a>                           | —                                     | —                                | —                                              |
| <a href="#">Test witnessing</a>                           | —                                     | —                                | —                                              |
| <a href="#">Training documentation evaluation</a>         | —                                     | —                                | —                                              |
| <a href="#">Usability analysis</a>                        | —                                     | —                                | X                                              |
| <a href="#">User documentation evaluation</a>             | —                                     | —                                | X                                              |
| <a href="#">User training</a>                             | X                                     | —                                | X                                              |
| <a href="#">V&amp;V tool plan generation</a>              | X                                     | X                                | X                                              |
| <a href="#">V&amp;V tool qualification</a>                | —                                     | —                                | —                                              |
| <a href="#">Walkthrough</a>                               | —                                     | —                                | —                                              |
| <a href="#">Walkthrough—Design</a>                        | —                                     | —                                | —                                              |
| <a href="#">Walkthrough—Requirements</a>                  | —                                     | —                                | —                                              |
| <a href="#">Walkthrough—Source code</a>                   | —                                     | —                                | —                                              |
| <a href="#">Walkthrough—Test</a>                          | —                                     | —                                | X                                              |
| <a href="#">Work breakdown structure (WBS) evaluation</a> | —                                     | —                                | X                                              |



NOTE 1—Clause references in the process definitions (top graphic bar) are ISO/IEC/IEEE 15288:2023(E) [B41] clause numbers.

NOTE 2—The V&V tasks may be performed concurrently. The sequential (waterfall) model is shown as an example of a life cycle model.

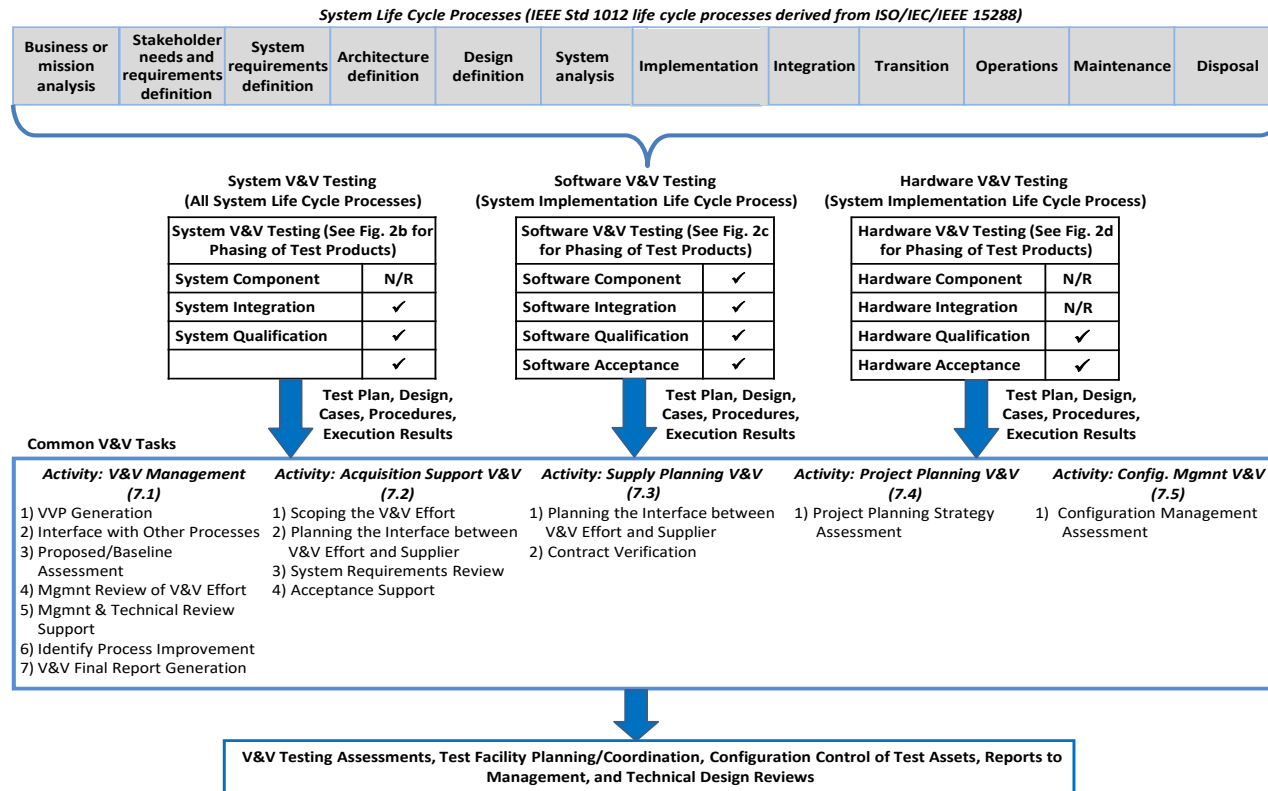
NOTE 3—V&V tasks listed in the figure are the minimum required for integrity level 4 (highest integrity level).

NOTE 4—Table 1a (common V&V tasks), Table 1b (system V&V tasks), Table 1c (software V&V tasks), and Table 1d (hardware V&V tasks) contain a complete list of minimum V&V tasks. Table 2a (common V&V tasks), Table 2b (system V&V tasks), Table 2c (software V&V tasks), and Table 2d (hardware V&V tasks) define the minimum V&V tasks for each integrity level.

NOTE 5—The clause numbers in the activity V&V graphic bar(s) correspond to Clause 7 of this standard, where the common V&V processes and activities can be found.

**Figure 1a—Summary of common V&V activities and tasks**

IEEE Std 1012-2024  
IEEE Standard for System, Software and Hardware Verification and Validation



NOTE 1—N/R = not required for integrity level 4; ✓ = required for integrity level 4. The diagram illustrates integrity level 4 V&V testing tasks.

NOTE 2—System component consists of software or hardware elements. Because those are tested as part of software and hardware V&V testing, no system component V&V testing is specified.

NOTE 3—The V&V activity clauses shown in the diagram are IEEE Std 1012 clauses.

NOTE 4—System V&V testing covers test planning to execution (spanning system life cycle stages of system requirements definition through transition). Software/hardware V&V testing covers test planning to execution (spanning software/hardware life cycle stages of requirements analysis through transition). Software/hardware life cycle stages all occur within the system implementation life cycle stage.

**Figure 2a—Summary of V&V test products and tasks**

## 8. System V&V processes

### 8.1 Business or Mission Analysis V&V Process

#### 8.1.1 Purpose

The purpose of the Business or Mission Analysis V&V Process is to provide assurance that the outcomes of the Business or Mission Analysis Process (ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#) have been achieved.

#### 8.1.2 Outcomes

As a result of the successful implementation of the Business or Mission Analysis V&V Process, objective evidence is developed to assess whether:

- a) The problem or opportunity space adheres to the organizational strategy.
- b) The solution space is consistent with the problem or opportunity space.
- c) Preliminary life-cycle concepts are consistent with the solution space.
- d) The preferred candidate solution alternative(s) are selected.
- e) Any enabling systems or services needed for business or mission analysis are available.
- f) The root of traceability is established, starting with the organization strategy and tracing to the problem or opportunity space, then to the solution space, and finally to the candidate solutions.

#### 8.1.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2b](#) for the selected integrity level, the following business or mission analysis V&V activity and tasks described in [Table 1b, Activity 8.1](#):

- a) [Business or mission analysis V&V](#): This activity consists of the following tasks:
  - 1) [Business or mission analysis documentation evaluation](#)
  - 2) [Traceability analysis](#)
  - 3) [Criticality analysis](#)
  - 4) [Hazard analysis](#)
  - 5) [Security analysis](#)
  - 6) [Risk analysis](#)

The business or mission analysis V&V tasks listed shall be used to verify and validate the Business or Mission Analysis Process outcomes described in ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#).

### 8.2 Stakeholder Needs and Requirements Definition V&V Process

#### 8.2.1 Purpose

The purpose of the Stakeholder Needs and Requirements Definition V&V Process is to provide assurance that the outcomes of the Stakeholder Needs and Requirements Definition Process (ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#)) have been achieved.

#### 8.2.2 Outcomes

As a result of the successful implementation of the Stakeholder Needs and Requirements Definition V&V Process, objective evidence is developed to assess whether the stakeholder requirements should be as follows:

- a) Specify the required characteristics and context of the use of services and operational concepts.



- b) Define all constraints on a system solution.
- c) Are traceable to the originating stakeholders.
- d) Are complete, unambiguous, correct, and accurate.
- e) Can be validated by measurable analyses or tests.

### 8.2.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2b](#) for the selected integrity level, the following stakeholder needs and requirements definition V&V activity and tasks described in [Table 1b, Activity 8.2](#):

- a) [Stakeholder needs and requirements definition V&V](#): This activity consists of the following tasks:
  - 1) [Stakeholder needs and requirements evaluation](#)
  - 2) [Traceability analysis](#)
  - 3) [Criticality analysis](#)
  - 4) [Hazard analysis](#)
  - 5) [Security analysis](#)
  - 6) [Risk analysis](#)

The stakeholder needs and requirements definition V&V tasks listed shall be used to verify and validate the Stakeholder Needs and Requirements Definition Process outcomes described in ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#).

## 8.3 System Requirements Definition V&V Process

### 8.3.1 Purpose

The purpose of the System Requirements Definition V&V Process is to provide assurance that the outcomes of the System Requirements Definition Process (ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#)) have been achieved.

### 8.3.2 Outcomes

As a result of the successful implementation of the System Requirements Definition V&V Process, objective evidence is developed to assess whether the system requirements are as follows:

- a) Specify all required characteristics, attributes, functional and performance requirements, interface requirements, and requirements for qualification, safety and security, human factors engineering, and user documentation for the system.
- b) Specify all constraints that will affect the architecture of the system and the means to realize the system.
- c) Are unique, complete, unambiguous, consistent with all other requirements, implementable, and verifiable.
- d) Are traceable to the stakeholder requirements.
- e) Provide a basis (through analysis or test planning) for verifying that each system requirement can be satisfied.

### 8.3.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2b](#) for the selected integrity level, the following system requirements definition V&V activity and tasks described in [Table 1b, Activity 8.3](#):

- a) [System requirements definition V&V](#): This activity consists of the following tasks:
  - 1) [Requirements evaluation](#)
  - 2) [Interface analysis](#)

- 3) [Traceability analysis](#)
- 4) [Criticality analysis](#)
- 5) [System qualification test plan V&V](#)
- 6) [System acceptance test plan V&V](#)
- 7) [Hazard analysis](#)
- 8) [Security analysis](#)
- 9) [Risk analysis](#)

The system requirements definition V&V tasks listed shall be used to verify and validate the System Requirements Definition Process outcomes described in ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#).

## 8.4 Architecture Definition V&V Process

### 8.4.1 Purpose

The purpose of the Architecture Definition V&V Process is to provide assurance that the outcomes of the Architecture Definition Process (ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#)) have been achieved.

### 8.4.2 Outcomes

As a result of the successful implementation of the Architecture Definition V&V Process, objective evidence is developed to assess whether:

- a) The system architecture (i.e., hardware, software, interfaces, and communication) satisfies the system requirements.
- b) The system architecture is realizable.
- c) The system architecture is based on specified selection criteria.
- d) The basis for verifying the system elements is defined.
- e) The basis for integration of the system elements is established.

### 8.4.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2b](#) for the selected integrity level, the following architecture definition V&V activity and tasks described in [Table 1b, Activity 8.4](#):

- a) [Architecture definition V&V](#). This activity consists of the following tasks:
  - 1) [Architecture evaluation](#)
  - 2) [Interface analysis](#)
  - 3) [Requirements allocation analysis](#)
  - 4) [Traceability analysis](#)
  - 5) [Criticality analysis](#)
  - 6) [System integration test plan V&V](#)
  - 7) [System qualification test design V&V](#)
  - 8) [System acceptance test design V&V](#)
  - 9) [Hazard analysis](#)
  - 10) [Security analysis](#)
  - 11) [Risk analysis](#)

The architecture definition V&V tasks listed above shall be used to verify and validate the Architecture Definition Process outcomes described in ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#).

## 8.5 Design Definition V&V Process

### 8.5.1 Purpose

The purpose of the Design Definition V&V Process is to provide assurance that the outcomes of the Design Definition Process (ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#)) have been achieved.

### 8.5.2 Outcomes

As a result of the successful implementation of the Design Definition V&V Process, objective evidence is developed to assess whether:

- a) The design characteristics of each system element are defined.
- b) The design enablers necessary for design definition are selected or defined.
- c) Each interface between system elements is described and includes purpose, parameters, data format, and performance criteria (e.g., timing, bandwidth, accuracy, safety, and security).
- d) The system design is established.
- e) Inputs for requirements of any enabling systems or system elements that serve the design definition activities are identified.
- f) Any enabling systems or services needed for design definition are available.
- g) Traceability of the design characteristics to the architectural elements of the system architecture is established.

### 8.5.3 Activity and Tasks

The V&V effort shall perform, as specified in [Table 2b](#) for the selected integrity level, the following design definition V&V activity and tasks described in [Table 1b, Activity 8.5](#):

- a) [Design definition V&V](#). This activity consists of the following tasks:
  - 1) [Design evaluation](#)
  - 2) [Interface analysis](#)
  - 3) [Traceability analysis](#)
  - 4) [Criticality analysis](#)
  - 5) [System integration test design V&V](#)
  - 6) [System integration test case V&V](#)
  - 7) [System qualification test case V&V](#)
  - 8) [System acceptance test case V&V](#)
  - 9) [Hazard analysis](#)
  - 10) [Security analysis](#)
  - 11) [Risk analysis](#)

Design definition V&V culminates in defining periodic assessment of the design for managing the design from the evolution of the system architecture to the obsolescence of components and technologies.

The Design Definition Process arrives at a solution that satisfies system requirements by defining system elements, design enablers, and interfaces. Requirements for enabling systems are defined, the design is baselined, and traceability of the design characteristics to the elements of the system architecture is established.

The design definition V&V activity analyzes the characteristics, elements, interfaces, baseline, and inputs, as well as traceability activities for the system under review and enabling systems.

The design definition V&V tasks listed shall be used to verify and validate the Design Definition Process outcomes described in ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#).

## 8.6 System Analysis V&V Process

### 8.6.1 Purpose

The purpose of the System Analysis V&V Process is to provide assurance that the outcomes of the System Analysis Process (ISO/IEC/IEEE 15288:2023(E) [B41]) have been achieved.

### 8.6.2 Outcomes

As a result of the successful implementation of the System Analysis V&V Process, objective evidence is developed to assess whether:

- a) The strategy for the system analysis is complete and is appropriate for the importance of the analysis.
- b) The results of the system analysis support its conclusions and recommendations.

### 8.6.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2b](#) for the selected integrity level, the following System Analysis V&V activity and tasks described in [Table 1b, Activity 8.6](#):

- a) [System analysis V&V](#). This activity consists of the following tasks:
  - 1) [System analysis strategy evaluation](#)
  - 2) [System analysis results evaluation](#)

The System Analysis V&V Process shall be performed for any specific implementation of the system analysis process where the conclusions and recommendations of the analysis could affect a system or system element of the integrity level indicated in [Table 2b](#).

The system analysis V&V tasks listed shall be used to verify and validate the System Analysis Process outcomes described in ISO/IEC/IEEE 15288:2023(E) [B41].

## 8.7 Implementation V&V Process

### 8.7.1 Purpose

The purpose of the Implementation V&V Process is to provide assurance that the outcomes of the Implementation Process (ISO/IEC/IEEE 15288:2023(E) [B41]) have been achieved.

### 8.7.2 Outcomes

As a result of the successful implementation of the Implementation V&V Process, objective evidence is developed to assess whether:

- a) The implementation activities performed produce a system element that conforms to the system requirements.
- b) The implementation correctly implements the design definition.
- c) The system element has been implemented within the defined constraints.
- d) The recorded evidence of implementation is complete and correct.
- e) The system element has been packaged and stored in accordance with defined requirements.

### 8.7.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2b](#) for the selected integrity level, the following implementation V&V activity and tasks described in [Table 1b, Activity 8.7](#):

- a) [Implementation V&V](#). This activity consists of the following tasks:
  - 1) [Implementation strategy assessment](#)
  - 2) [System element implementation analysis](#)
  - 3) [System element interaction analysis](#)
  - 4) [Criticality analysis](#)
  - 5) [System integration test procedure V&V](#)
  - 6) [System qualification test procedure V&V](#)
  - 7) [System acceptance test procedure V&V](#)
  - 8) [Hazard analysis](#)
  - 9) [Security analysis](#)
  - 10) [Risk analysis](#)

The implementation V&V tasks are to monitor, review, audit, and analyze hardware and software products. If a hardware or software V&V is being conducted, the system V&V activities will use those V&V results to perform, in part or in whole, the system V&V activities.

The behavior (e.g., performance, security, hazard, interfaces) of each element may impact the behavior (e.g., by realized performance or lack of performance, or through direct or indirect coupling) of other elements in the system. Therefore, as each system element undergoes its own V&V, the V&V results assessing the actual behavior of that system element should be used to assess the behavior of other coupled system elements.

The Implementation V&V Process should monitor the V&V results of each system element to assess if the elements are meeting their requirements. If a given system element is determined not to be able to meet all of its allocated requirements, requirements may need to be reallocated to other elements as a part of the systems engineering function. This reallocation scenario includes situations when COTS and GOTS products are selected as system elements and fall short of meeting all requirements. When this occurs, V&V tasks shall be reiterated (repeated, redone) or recursively applied to the affected elements.

The implementation V&V tasks listed shall be used to verify and validate the Implementation Process outcomes described in ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#).

## 8.8 Integration V&V Process

### 8.8.1 Purpose

The purpose of the Integration V&V Process is to provide assurance that the outcomes of the Integration Process (ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#)) have been achieved.

### 8.8.2 Outcomes

As a result of the successful implementation of the Integration V&V Process, objective evidence is developed to assess whether:

- a) The system elements, as described by the architecture definition, design definition, and implementation are integrated correctly.
- b) The integrated system meets the system requirements.
- c) The system integration strategy is consistent with the system architecture.
- d) The integration test plan, design, cases, and procedures are traceable to the system architecture.
- e) The unavoidable constraints of integration that influence requirements are addressed correctly.

- f) The integration of human performance into systems and their operation is correct.
- g) Nonconformances due to integration actions are recorded and addressed.

### 8.8.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2b](#) for the selected integrity level, the following integration V&V activity and tasks described in [Table 1b, Activity 8.8](#):

- a) [Integration V&V](#). This activity consists of the following tasks:
  - 1) [System integration strategy assessment](#)
  - 2) [System element interaction analysis](#)
  - 3) [System integration test execution V&V](#)
  - 4) [System qualification test execution V&V](#)
  - 5) [Traceability analysis](#)
  - 6) [Hazard analysis](#)
  - 7) [Security analysis](#)
  - 8) [Risk analysis](#)

The Integration V&V Process should monitor each system element through element V&V results to assess if the elements are meeting their requirements, in a manner similar to implementation V&V.

The integration V&V tasks listed shall be used to verify and validate the Integration Process outcomes described in ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#).

## 8.9 Verification Process

### 8.9.1 Purpose

The purpose of the Verification Process is to provide objective evidence for whether the outcomes achieve the following:

- a) Conform to requirements (e.g., for correctness, completeness, consistency, and accuracy) for all activities during each life cycle process.
- b) Satisfy the standards, practices, and conventions during life cycle processes.
- c) Successfully complete each life cycle activity and satisfy all the criteria for initiating succeeding life cycle activities (i.e., the product is built correctly).

### 8.9.2 Outcomes

The following outcomes occur as a result of the successful implementation of the Verification Process:

- a) A V&V plan is developed and implemented.
- b) The system of interest and all components of the system of interest are assigned integrity levels that are reevaluated throughout the life cycle of the system.
- c) The system and each of its components are evaluated for requirements satisfaction based on assigned integrity levels.
- d) Objective evidence is developed to determine whether the system and each of its components conform to requirements and satisfy all the criteria for each successive life cycle activity.

### 8.9.3 Activities and tasks

The activities and tasks for the Verification Process applied to the technical processes of the system life cycle processes from ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#) are described in [subclause 8.1](#) (Business or Mission Analysis V&V Process), [subclause 8.2](#) (Stakeholder Needs and Requirements Definition V&V Process),

[subclause 8.3](#) (System Requirements Definition V&V Process), [subclause 8.4](#) (Architecture Definition V&V Process), [subclause 8.5](#) (Design Definition V&V Process), [subclause 8.6](#) (System Analysis V&V Process), [subclause 8.7](#) (Implementation V&V Process), [subclause 8.8](#) (Integration V&V Process), [subclause 8.10](#) (Transition V&V Process), [subclause 8.12](#) (Operation V&V Process), [subclause 8.13](#) (Maintenance V&V Process), and [subclause 8.14](#) (Disposal V&V Process).

## 8.10 Transition V&V Process

### 8.10.1 Purpose

The purpose of the Transition V&V Process is to provide assurance that the outcomes of the Transition Process (ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#)) have been achieved.

### 8.10.2 Outcomes

As a result of the successful implementation of the Transition V&V Process, objective evidence is developed to assess whether:

- a) The system transition strategy is comprehensive and explicitly documented.
- b) The system is installed in its operational location in accordance with the transition plan.
- c) The system delivers all specified services per its governing requirements.
- d) The system is sustainable by enabling systems.
- e) The system satisfies its acceptance criteria.

### 8.10.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2b](#) for the selected integrity level, the following transition V&V activity and tasks described in [Table 1b, Activity 8.10](#):

- a) [Transition V&V](#). This activity consists of the following tasks:
  - 1) [Transition strategy evaluation](#)
  - 2) [Transition demonstration assessment](#)
  - 3) [System acceptance test execution V&V](#)
  - 4) [Installation configuration audit](#)
  - 5) [Installation checkout](#)
  - 6) [Hazard analysis](#)
  - 7) [Security analysis](#)
  - 8) [Risk analysis](#)

The Transition V&V tasks listed shall be used to verify and validate the Transition Process outcomes described in ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#).

## 8.11 Validation Process

### 8.11.1 Purpose

The purpose of the Validation Process is to provide objective evidence for whether the outcomes achieve the following:

- a) Satisfy system requirements allocated to the products at the end of each life cycle activity.
- b) Solve the right problem (e.g., correctly model physical laws, implement business rules, and use the proper system assumptions).
- c) Satisfy intended use and user needs in the operational environment (i.e., the correct product is built).

### 8.11.2 Outcomes

The following outcomes occur as a result of the successful implementation of the Validation Process:

- a) A V&V plan is developed and implemented.
- b) The system of interest and all components of the system of interest are assigned integrity levels that are reevaluated throughout the life cycle of the system.
- c) The system and each of its components are evaluated for satisfaction of allocated system requirements and of intended use and user needs based on assigned integrity levels.
- d) Objective evidence is developed to determine whether the system and each of its components satisfy all allocated system requirements and meet intended use and user needs.

### 8.11.3 Activities and tasks

The activities and tasks for the Validation Process applied to the technical processes of the system life cycle processes from ISO/IEC/IEEE 15288:2023(E) [B41] are described in [subclause 8.1](#) (Business or Mission Analysis V&V Process), [subclause 8.2](#) (Stakeholder Needs and Requirements Definition V&V Process), [subclause 8.3](#) (System Requirements Definition V&V Process), [subclause 8.4](#) (Architecture Definition V&V Process), [subclause 8.5](#) (Design Definition V&V Process), [subclause 8.6](#) (System Analysis V&V Process), [subclause 8.7](#) (Implementation V&V Process), [subclause 8.8](#) (Integration V&V Process), [subclause 8.10](#) (Transition V&V Process), [subclause 8.12](#) (Operation V&V Process), [subclause 8.13](#) (Maintenance V&V Process), and [subclause 8.14](#) (Disposal V&V Process).

## 8.12 Operation V&V Process

### 8.12.1 Purpose

The purpose of the Operation V&V Process is to provide assurance that the outcomes of the Operation Process (ISO/IEC/IEEE 15288:2023(E) [B41]) have been achieved.

### 8.12.2 Outcomes

As a result of the successful implementation of the Operation V&V Process, objective evidence is developed to assess whether:

- a) The operation strategy satisfies stakeholder requirements and needs.
- b) New constraints in the system are evaluated.
- c) Proposed system changes and their impacts are assessed.
- d) Operating procedures are evaluated for correctness and usability.

### 8.12.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2b](#) for the selected integrity level, the following operation V&V activity, and tasks described in [Table 1b, Activity 8.12](#):

- a) [Operation V&V](#). This activity consists of the following tasks:
  - 1) [Evaluation of new constraints](#)
  - 2) [Operating procedures evaluation](#)
  - 3) [Hazard analysis](#)
  - 4) [Security analysis](#)
  - 5) [Risk analysis](#)

The operation V&V tasks listed shall be used to verify and validate the Operation Process outcomes described in ISO/IEC/IEEE 15288:2023(E) [B41].



## 8.13 Maintenance V&V Process

### 8.13.1 Purpose

The purpose of the Maintenance V&V Process is to provide assurance that the outcomes of the Maintenance Process (ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#)) have been achieved. The Maintenance Process is to sustain the capability of the system to provide a service. This process monitors the system's capability to deliver services; records problems for analysis; takes corrective, adaptive, perfective, and preventive actions; and confirms restored capability (ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#)).

System modifications may be derived from requirements specified to correct errors (e.g., corrective), to adapt to a changed operating environment (e.g., adaptive), or to respond to additional user requests or enhancements (e.g., perfective). Modifications of the system shall be treated as development efforts and shall be verified and validated by performing V&V tasks corresponding to the modifications. Integrity level assignments shall be assessed as described in [Clause 5](#). The integrity-level assignments shall be revised as appropriate to reflect requirements derived from the Maintenance Process.

The integrity level could be revised on the basis of corrective, perfective, or adaptive changes to the system if these changes modify the original integrity level of the system. If the integrity level is revised, then the effect of the revision on existing system requirements shall be evaluated to identify additional activities and tasks to be performed regressively on the system, software, and hardware. Corresponding V&V activities and tasks shall be identified to confirm the system conforms to its required integrity level.

If system V&V were performed in accordance with this standard, then the Maintenance Process shall continue to conform to this standard. If the system was not verified and validated using this standard and appropriate documentation is not available or adequate, then the maintenance V&V effort shall determine whether the missing or incomplete documentation should be generated. In making this determination of whether to generate missing documentation, the minimum V&V requirements of the assigned integrity level shall be taken into consideration.

The objectives of maintenance V&V are to 1) assess that the maintenance strategy is comprehensive and explicitly documented; 2) evaluate the corrective actions to assure they resolve the issue or negative trend; 3) assess the effect of the corrective, adaptive, perfective, and preventive changes on stakeholder and system requirements, architectural design, and implementation; and 4) evaluate problem reports, corrective actions, and trends to determine the possible corrective, adaptive, perfective, and preventive actions.

### 8.13.2 Outcomes

As a result of the successful implementation of the Maintenance V&V Process, objective evidence is developed to assess whether:

- a) The maintenance strategy is comprehensive and explicitly documented.
- b) Corrective actions resolve the issue or negative trend.
- c) The effect is assessed for corrective, adaptive, perfective, and preventive changes on stakeholder and system requirements, architecture, design, and implementation.
- d) Evaluation is performed on problem reports, corrective actions, and trends to determine the possible corrective, adaptive, perfective, and preventive actions.

### 8.13.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2b](#) for the selected integrity level, the following maintenance V&V activity and tasks described in [Table 1b, Activity 8.13](#):

- a) [Maintenance V&V](#). This activity consists of the following tasks:

- 1) [System maintenance strategy assessment](#)
- 2) [System maintenance execution assessment](#)

The Maintenance V&V tasks listed shall be used to verify and validate the Maintenance Process outcomes described in ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#).

## 8.14 Disposal V&V Process

### 8.14.1 Purpose

The purpose of the Disposal V&V Process is to provide assurance that the outcomes of the Disposal Process (ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#)) have been achieved.

### 8.14.2 Outcomes

As a result of the successful implementation of the Disposal V&V Process, objective evidence is developed to assess whether the disposal plan:

- a) Defines system boundaries and identifies system elements.
- b) Is commensurate with the complexity and risk of the disposal and accounts for all system elements.
- c) Addresses environmental considerations, applicable laws, regulations, and organizational policies and procedures.

### 8.14.3 Activity and Task

The V&V effort shall perform, as specified in [Table 2b](#) for the selected integrity level, the following disposal V&V activity and task described in [Table 1b, Activity 8.14](#):

- a) [Disposal V&V](#). This activity consists of the following task:
  - 1) [Disposal strategy evaluation](#)

The Disposal V&V task listed shall be used to verify and validate the Disposal Process outcomes described in ISO/IEC/IEEE 15288:2023(E) [\[B41\]](#).

**Table 1b—V&V tasks, inputs and outputs**

| <b>8.1 Activity: Business or mission analysis V&amp;V (System, 15288—Business or Mission Analysis Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                  |                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Required inputs</b>                                                                                                                                                                                                                                                                                                                           | <b>Required outputs</b>                                                                   |
| <b>(1) <u>Business or mission analysis documentation evaluation</u></b><br>a) Evaluate the results of business or mission analysis for correctness, consistency, and completeness.<br>b) Verify and validate that the problem or opportunity space definition adheres to the organizational strategy.<br>c) Verify that the solution space characterization is consistent with the problem or opportunity space.<br>d) Verify that the preliminary life cycle concepts of the candidate solutions are consistent with the solution space.<br>e) Verify that the preferred candidate solution alternative(s) are selected.<br>f) Verify and validate that inputs for requirements of any enabling systems or system elements that serve the business or mission analysis activities are identified.<br>g) Verify the availability of enabling systems or services needed for business or mission analysis. | Business or mission analysis documentation<br>Organization strategy<br>Organizational concept of operations<br>Organizational strategic goals and plans<br>New market or mission elements<br>Identified problem or opportunity space<br>Identified solution space<br>Identified candidate solution(s)<br>Organizational processes and procedures | Task report(s)—Business or mission analysis documentation evaluation<br>Anomaly report(s) |
| <b>(2) <u>Traceability analysis</u></b><br>a) Establish the root of traceability, starting with the organization strategy.<br>b) Verify that the problem or opportunity space traces back to the organization strategy.<br>c) Verify that the solution space traces back to the problem or opportunity space.<br>d) Verify that the candidate solution(s) traces back to the solution space.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Business or mission analysis documentation<br>Organization strategy<br>Organizational concept of operations<br>Identified problem or opportunity space<br>Identified solution space<br>Preferred candidate solution(s)                                                                                                                           | Task report(s)—Root of traceability<br>Anomaly report(s)                                  |
| <b>(3) <u>Criticality analysis</u></b><br>a) Determine whether integrity levels are established for the identified solution space.<br>b) Verify that the assigned integrity levels are correct. If integrity levels are not assigned, then assign integrity levels to the solution space.<br>c) Determine whether integrity levels are established for the preferred candidate solutions.<br>d) Verify that the assigned integrity levels are correct. If integrity levels are not assigned, then assign integrity levels to the preferred candidate solution(s). Preferred candidate solutions.                                                                                                                                                                                                                                                                                                          | Identified problem or opportunity space<br>Identified solution space<br>Preferred candidate solution(s)                                                                                                                                                                                                                                          | Task report(s)—Criticality analysis<br>Anomaly report(s)                                  |
| <b>(4) <u>Hazard analysis</u></b><br>Analyze the potential hazards to and from the preferred candidate solutions. The analysis shall perform the following:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Preferred candidate solution(s)                                                                                                                                                                                                                                                                                                                  | Task report(s)—Hazard analysis<br>Anomaly report(s)                                       |

| <b>8.1 Activity: Business or mission analysis V&amp;V (System, 15288—Business or Mission Analysis Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                             |                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Required inputs</b>                                                                                      | <b>Required outputs</b>                               |
| a) Identify the potential solution hazards.<br>b) Assess the consequences of each hazard.<br>c) Assess the probability of each hazard.<br>d) Identify mitigation strategies for each hazard.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                             |                                                       |
| <b>(5) <u>Security analysis</u></b><br>a) Review the preferred candidate solution(s) to determine an acceptable level of security risk.<br>b) Analyze the preferred candidate solution(s) from a security perspective and assure that potential security risks with respect to confidentiality (disclosure of sensitive information/data), integrity (modification of information/data), availability (withholding of information or services), and accountability (attributing actions to an individual/process) have been identified. Include an assessment of the sensitivity of the information/data to be processed.<br>c) Analyze security risks introduced by the preferred candidate solution(s). | Preferred candidate solution(s)<br>Preliminary threat and risk assessment (TRA)                             | Task report(s)—Security analysis<br>Anomaly report(s) |
| <b>(6) <u>Risk analysis</u></b><br>a) Identify the technical and management risks.<br>b) Provide recommendations to eliminate, reduce, or mitigate the risks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Preferred candidate solution(s)<br>Hazard analysis results<br>Security analysis results<br>V&V task results | Task report(s)—Risk analysis<br>Anomaly report(s)     |

| <b>8.2 Activity: Stakeholder needs and requirements definition V&amp;V (System, 15288—Stakeholder Needs and Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                      |                                                                         |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Required inputs</b>                                                                                                                               | <b>Required outputs</b>                                                 |
| <b>(1) <u>Stakeholder needs and requirements evaluation</u></b><br>Evaluate the stakeholder requirements for correctness, consistency, completeness, readability, and testability. The task criteria are as follows:<br>a) Correctness<br>b) Verify and validate that the requirements satisfy the stakeholder needs for the system. <ol style="list-style-type: none"> <li>1) Verify that the requirements comply with standards, references, regulations, policies, physical laws, and business rules.</li> <li>2) Validate that the requirements define the intended interaction of the system with its operating environment and other interfacing systems.</li> <li>3) Validate the suitability of the preferred candidate solution(s) to meet the stakeholder requirements.</li> </ol> c) Consistency <ol style="list-style-type: none"> <li>1) Verify that all terms, concepts, and requirements are documented consistently in accordance with accepted syntax and structure (e.g., style guides and requirements modeling structure).</li> <li>2) Verify that there is consistency between assumptions, requirements, and between groups of requirements.</li> </ol> | Preferred candidate solution(s)<br>Stakeholder requirements<br>Business or mission analysis documentation<br>Organizational processes and procedures | Task report(s)—Stakeholder requirements evaluation<br>Anomaly report(s) |

| <b>8.2 Activity: Stakeholder needs and requirements definition V&amp;V (System, 15288—Stakeholder Needs and Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                    |                                                                       |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Required inputs</b>                                                                                             | <b>Required outputs</b>                                               |
| <p>d) Completeness</p> <ol style="list-style-type: none"> <li>1) Validate that performance criteria and functionality are described in the requirements, within the assumptions and constraints of the operating environment and system boundaries.</li> <li>2) Verify that all stakeholders, or stakeholder classes (used here as groupings of stakeholders), are identified.</li> <li>3) Verify that the stakeholder requirements satisfy specified configuration management procedures. Verify stakeholder requirements are in a form suitable for requirements management throughout the life cycle.</li> </ol> <p>e) Readability</p> <ol style="list-style-type: none"> <li>1) Verify that the documentation is legible, understandable, and unambiguous to the intended audience.</li> <li>2) Verify that the documentation defines all acronyms, mnemonics, abbreviations, terms, and symbols.</li> </ol> <p>f) Testability</p> <ol style="list-style-type: none"> <li>1) Verify that objective acceptance criteria can be developed to validate the requirements.</li> </ol> |                                                                                                                    |                                                                       |
| <p>(2) <a href="#">Traceability analysis</a></p> <ol style="list-style-type: none"> <li>a) Verify that all requirements are traceable to one or more preferred candidate solutions.</li> <li>b) Verify that all requirements are traceable to one or more stakeholders or stakeholder classes.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>Preferred candidate solution(s)<br/>Stakeholder requirements<br/>Business or mission analysis documentation</p> | <p>Task report(s)—<br/>Traceability report<br/>Anomaly report(s)</p>  |
| <p>(3) <a href="#">Criticality analysis</a></p> <ol style="list-style-type: none"> <li>a) Determine whether system integrity levels are established for stakeholder and system requirements, detailed functions, subsystems, or other partitions.</li> <li>b) Verify that the assigned system integrity levels are correct. If system integrity levels are not assigned, then assign system integrity levels to the stakeholder requirements.</li> <li>c) Document the system integrity level assigned to stakeholder requirements. For V&amp;V planning purposes, the system shall be assigned the same integrity level as the highest level assigned to any individual stakeholder requirement.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                         | <p>Stakeholder requirements<br/>Business or mission analysis documentation</p>                                     | <p>Task report(s)—<br/>Criticality analysis<br/>Anomaly report(s)</p> |
| <p>(4) <a href="#">Hazard analysis</a></p> <ol style="list-style-type: none"> <li>a) Analyze the potential hazards to and from the conceptual system. The analysis shall perform the following:</li> <li>b) Identify the potential system hazards.</li> <li>c) Assess the consequences of each hazard.</li> <li>d) Assess the probability of each hazard.</li> <li>e) Identify mitigation strategies for each hazard.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Stakeholder requirements<br/>Business or mission analysis documentation</p>                                     | <p>Task report(s)—<br/>Hazard analysis<br/>Anomaly report(s)</p>      |
| <p>(5) <a href="#">Security analysis</a></p> <ol style="list-style-type: none"> <li>a) Review the system owner’s definition of an acceptable level of security risk.</li> <li>b) Analyze the system concept from a security perspective and assure that potential security risks with respect to confidentiality (disclosure of sensitive information/data),</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Stakeholder requirements<br/>Business or mission analysis documentation</p>                                     | <p>Task report(s)—<br/>Security analysis<br/>Anomaly report(s)</p>    |

| <b>8.2 Activity: Stakeholder needs and requirements definition V&amp;V (System, 15288—Stakeholder Needs and Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                               |                                                              |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Required inputs</b>                                                                                                                                                                                                        | <b>Required outputs</b>                                      |
| <p>integrity (modification of information/data), availability (withholding of information or services), and accountability (attributing actions to an individual/process) have been identified. Include an assessment of the sensitivity of the information/data to be processed.</p> <p>c) Analyze security risks introduced by the system itself as well as those associated with the environment with which the system interfaces.</p> | Preliminary threat and risk assessment (TRA)                                                                                                                                                                                  |                                                              |
| <p>(6) <a href="#">Risk analysis</a></p> <p>a) Identify the technical and management risks.</p> <p>b) Provide recommendations to eliminate, reduce, or mitigate the risks.</p>                                                                                                                                                                                                                                                            | <p>Stakeholder requirements</p> <p>Business or mission analysis documentation</p> <p>Supplier development plans and schedules</p> <p>Hazard analysis results</p> <p>Security analysis results</p> <p>V&amp;V task results</p> | <p>Task report(s)—Risk analysis</p> <p>Anomaly report(s)</p> |

| <b>8.3 Activity: System requirements definition V&amp;V (System, 15288—System Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                             |                                                                        |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Required inputs</b>                                                                                                                                      | <b>Required outputs</b>                                                |
| <p>(1) <a href="#">Requirements evaluation</a></p> <p>Evaluate the system requirements for correctness, consistency, completeness, readability, and testability. The task criteria are as follows:</p> <p>a) Correctness</p> <ol style="list-style-type: none"> <li>1) Verify and validate that the required characteristics, attributes, constraints (e.g., mechanical, electrical, mass, thermal, data, procedural flows), and functional and performance requirements for a product solution are correct (e.g., security, ergonomics, human-machine interface, safety, reliability, maintainability, response time).</li> <li>2) Verify and validate that the system requirements satisfy the stakeholder requirements.</li> <li>3) Verify that the system requirements comply with standards, references, regulations, policies, physical laws, and business rules.</li> <li>4) Validate that the system requirements define the intended interaction of the system with its operating environment and other interfacing systems.</li> </ol> <p>b) Consistency</p> <ol style="list-style-type: none"> <li>1) Verify that all terms, concepts, and requirements are documented consistently in accordance with accepted syntax and structure (e.g., style guides and requirements modeling structure).</li> <li>2) Verify that there is consistency among the requirements, groups of requirements (functional interaction), and assumptions.</li> </ol> | <p>Stakeholder requirements</p> <p>System requirements</p> <p>Business or mission analysis documentation</p> <p>Organizational processes and procedures</p> | <p>Task report(s)—Requirements evaluation</p> <p>Anomaly report(s)</p> |

| <b>8.3 Activity: System requirements definition V&amp;V (System, 15288—System Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                       |                                                                                                                                                                                     |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Required inputs</b>                                                                                                                | <b>Required outputs</b>                                                                                                                                                             |
| <p>c) Completeness</p> <ol style="list-style-type: none"> <li>1) Validate that all stakeholder needs are satisfied by the set of system requirements.</li> <li>2) Validate that performance criteria and functionality are described in the requirements, within the assumptions and constraints of the operating environment and system boundaries.</li> <li>3) Verify that the System requirements satisfy specified configuration management procedures. Verify that system requirements are in a form suitable for requirements management throughout the life cycle.</li> </ol> <p>d) Readability</p> <ol style="list-style-type: none"> <li>1) Verify that the documentation is legible, understandable, and unambiguous to the intended audience.</li> <li>2) Verify that the documentation defines all acronyms, mnemonics, abbreviations, terms, and symbols.</li> </ol> <p>e) Testability</p> <ol style="list-style-type: none"> <li>1) Verify that objective acceptance criteria can be developed to validate the requirements.</li> </ol> |                                                                                                                                       |                                                                                                                                                                                     |
| <p><b>(2) <u>Interface analysis</u></b></p> <p>Verify and validate that the requirements for system interfaces with other systems are correct, complete, and testable. The task criteria are as follows:</p> <ol style="list-style-type: none"> <li>a) Correctness<br/>Validate the external interface requirements, including information to be exchanged.</li> <li>b) Completeness<br/>Verify that all external interface requirements are defined.</li> <li>c) Testability<br/>Verify that objective acceptance criteria can be developed to validate the requirements.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>System requirements<br/>Stakeholder requirements<br/>System interface requirements<br/>Organizational processes and procedures</p> | <p>Task report(s)—<br/>Interface analysis<br/>Anomaly report(s)</p>                                                                                                                 |
| <p><b>(3) <u>Traceability analysis</u></b></p> <p>Verify that the traceability analysis is complete. The task criteria are as follows:</p> <ol style="list-style-type: none"> <li>a) All system requirements are traceable to one or more stakeholder requirements.</li> <li>b) All stakeholder requirements are traceable to one or more system requirements.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>System requirements<br/>Stakeholder requirements</p>                                                                               | <p>Task report(s)—<br/>Traceability analysis<br/>Anomaly report(s)</p>                                                                                                              |
| <p><b>(4) <u>Criticality analysis</u></b></p> <p>Review and update the existing criticality analysis results from the prior criticality task report using the stakeholder requirements and system requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Criticality task report<br/>System requirements<br/>Stakeholder requirements</p>                                                   | <p>Task report(s)—<br/>Criticality analysis<br/>Anomaly report(s)</p>                                                                                                               |
| <p><b>(5) <u>System qualification test plan V&amp;V</u></b></p> <ol style="list-style-type: none"> <li>a) System integrity level 4 <ol style="list-style-type: none"> <li>1) Plan V&amp;V system qualification testing to validate the system against system requirements. The resulting V&amp;V system qualification test plan shall address the following: <ol style="list-style-type: none"> <li>i) Tracing of system requirements to test designs, cases, procedures, and results.</li> </ol> </li> </ol> </li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>System requirements<br/>Stakeholder requirements<br/>System qualification test plan (developer's)</p>                              | <p>V&amp;V system qualification test plan (integrity level 4)<br/>Task report(s)—<br/>Review of system qualification test plan (integrity levels 3 and 2)<br/>Anomaly report(s)</p> |

| <b>8.3 Activity: System requirements definition V&amp;V (System, 15288—System Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                            |                                                                                                                                                                                |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Required inputs</b>                                                                                     | <b>Required outputs</b>                                                                                                                                                        |
| <ul style="list-style-type: none"> <li>ii) Documentation of test designs, cases, procedures, and results.</li> <li>iii) Satisfaction of all system requirements (e.g., functional, performance, security, operation, and maintenance) in the system environment.</li> <li>iv) Adequacy of user documentation (e.g., training materials, procedural changes, and user guides).</li> <li>v) Performance at boundaries (e.g., data and interfaces) and under stress conditions.</li> <li>vi) Conformance to project-defined test document purpose, format, and content (e.g., IEEE Std 829™-2008 [B17]).</li> <li>vii) Test coverage of system requirements.</li> <li>viii) Expected results.</li> </ul> <p>b) System integrity levels 3 and 2</p> <ul style="list-style-type: none"> <li>1) Verify that the developer's system qualification test plan satisfies the following criteria: <ul style="list-style-type: none"> <li>i) Conformance to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>ii) Test coverage of system requirements.</li> </ul> </li> <li>2) Verify that the developer's system qualification test plan addresses the following: <ul style="list-style-type: none"> <li>i) Appropriateness of test methods and standards used.</li> <li>ii) Expected results.</li> <li>iii) Feasibility of system qualification testing.</li> <li>iv) Capability to be operated and maintained.</li> </ul> </li> </ul> <p>c) System integrity level 1</p> <p>There are no system qualification V&amp;V test plan requirements.</p> |                                                                                                            |                                                                                                                                                                                |
| <p><b>(6) <u>System acceptance test plan V&amp;V</u></b></p> <p>a) System integrity level 4</p> <ul style="list-style-type: none"> <li>1) Plan V&amp;V system acceptance testing to validate that the system correctly implements system requirements in the intended operational environment or as close to the intended operational environment as possible. The resulting V&amp;V system acceptance test plan shall address the following: <ul style="list-style-type: none"> <li>i) Tracing of acceptance test requirements to test design, cases, procedures, and execution results.</li> <li>ii) Documentation of test tasks and results.</li> <li>iii) Conformance to acceptance requirements in the operational environment (limitations of testing).</li> <li>iv) Adequacy of user documentation (e.g., training materials, procedural changes, user guides).</li> <li>v) Conformance to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>vi) Test coverage of acceptance requirements.</li> <li>vii) Expected results.</li> </ul> </li> </ul> <p>b) System integrity levels 3 and 2</p> <ul style="list-style-type: none"> <li>1) Verify that the acquirer's system acceptance test plan conforms to the project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> </ul>                                                                                                                                                                                                                | <p>System requirements</p> <p>Stakeholder requirements</p> <p>System acceptance test plan (acquirer's)</p> | <p>V&amp;V system acceptance test plan (integrity level 4)</p> <p>Task report(s)—Review of system acceptance test plan (integrity levels 3 and 2)</p> <p>Anomaly report(s)</p> |



| <b>8.3 Activity: System requirements definition V&amp;V (System, 15288—System Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                       |                                                                  |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Required inputs</b>                                                                                                                                                                                                                                                                | <b>Required outputs</b>                                          |
| <p>2) Verify that the acquirer's system acceptance test plan addresses the following:</p> <ul style="list-style-type: none"> <li>i) Test coverage of acceptance requirements.</li> <li>ii) Expected results.</li> <li>iii) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs).</li> </ul> <p>c) System integrity level 1</p> <p>There are no system acceptance V&amp;V test plan requirements.</p>                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                       |                                                                  |
| <p>(7) <a href="#">Hazard analysis</a></p> <p>Analyze the potential hazards to and from the stakeholder requirements. The analysis shall perform the following:</p> <ul style="list-style-type: none"> <li>a) Identify the potential system hazards.</li> <li>b) Assess the consequences of each hazard.</li> <li>c) Assess the likelihood of each hazard.</li> <li>d) Identify mitigation strategies for each hazard.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                      | <p>System requirements</p> <p>Stakeholder requirements</p> <p>Business or mission analysis documentation</p> <p>Hazard analysis results</p>                                                                                                                                           | <p>Task report(s)—Hazard analysis</p> <p>Anomaly report(s)</p>   |
| <p>(8) <a href="#">Security analysis</a></p> <ul style="list-style-type: none"> <li>a) Review the system owner's definition of an acceptable level of security risk.</li> <li>b) Analyze the stakeholder requirements from a security perspective and assure that potential security risks with respect to confidentiality (disclosure of sensitive information/data), integrity (modification of information/data), availability (withholding of information or services), and accountability (attributing actions to an individual/process) have been identified. Include an assessment of the sensitivity of the information/data to be processed.</li> <li>c) Analyze security risks introduced by the system itself as well as those associated with the environment with which the system interfaces.</li> </ul> | <p>System requirements</p> <p>Stakeholder requirements</p> <p>Business or mission analysis documentation</p> <p>Preliminary threat and risk assessment (TRA)</p> <p>Security analysis results</p>                                                                                     | <p>Task report(s)—Security analysis</p> <p>Anomaly report(s)</p> |
| <p>(9) <a href="#">Risk analysis</a></p> <ul style="list-style-type: none"> <li>a) Identify the technical and management risks.</li> <li>b) Provide recommendations to eliminate, reduce, or mitigate the risks.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>System requirements</p> <p>Stakeholder requirements</p> <p>Business or mission analysis documentation</p> <p>Supplier development plans and schedules</p> <p>Hazard analysis results</p> <p>Security analysis results</p> <p>Risk analysis results</p> <p>V&amp;V task results</p> | <p>Task report(s)—Risk analysis</p> <p>Anomaly report(s)</p>     |

| <b>8.4 Activity: Architecture definition V&amp;V (System, 15288—Architecture Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                         |                                                                        |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Required inputs</b>                                                                                                                                                                                  | <b>Required outputs</b>                                                |
| <p><b>(1) <a href="#">Architecture evaluation</a></b><br/>Evaluate the system architecture(s) for correctness, consistency, completeness, and testability. The task criteria are as follows:</p> <p>a) Correctness</p> <ol style="list-style-type: none"> <li>1) Verify the characteristics, attributes, constraints, and functional and performance requirements of the selected architecture(s) correctly implement the system requirements.</li> <li>2) Verify the selected architecture(s) complies with standards, regulations, policies, physical laws, and business rules.</li> <li>3) Validate the product solution(s) defined by the system architecture(s) satisfies the stakeholder needs.</li> <li>4) Validate the selected architecture(s) and its element interactions do not result in unnecessary, unintended, or deleterious consequences.</li> <li>5) Validate the selected architecture(s) defines the intended interaction of the system with its operating environment.</li> </ol> <p>b) Consistency</p> <ol style="list-style-type: none"> <li>1) Verify the selected architecture(s) conform to the architectural guidance, principles, and tenets of the organization processes and procedures (e.g., service-oriented architecture, modular open-systems architecture).</li> <li>2) Verify the architectural principles, characteristics, and rules established for the system are being applied across the selected architecture(s).</li> </ol> <p>c) Completeness</p> <ol style="list-style-type: none"> <li>1) Verify the system functions are allocated to the elements of the selected architecture(s).</li> <li>2) Verify all system requirements are included in the selected architecture(s).</li> </ol> <p>NOTE—Typically, the stakeholder needs and requirements definition, system requirements definition, architecture definition, and design definition processes are recursively applied to successive levels of detail in the system architecture and design until elements are completely defined.</p> | <p>System requirements</p> <p>Business or mission analysis documentation</p> <p>Architecture models and views</p> <p>Architecture trade-off analyses</p> <p>Organizational processes and procedures</p> | <p>Task report(s)—Architecture evaluation</p> <p>Anomaly report(s)</p> |
| <p><b>(2) <a href="#">Interface analysis</a></b><br/>Verify and validate that the architectural interfaces between system elements and with other systems are correct and complete. The task criteria are:</p> <p>a) Correctness</p> <ol style="list-style-type: none"> <li>1) Verify the system architecture satisfies the system interface requirements between system elements.</li> <li>2) Validate that the system architecture satisfies the system interface requirements with external systems (external system boundaries).</li> </ol> <p>b) Completeness</p> <p>Verify the architecture describes all internal and external interfaces.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>System requirements</p> <p>Architecture models and views</p>                                                                                                                                         | <p>Task report(s)—Interface analysis</p> <p>Anomaly report(s)</p>      |
| <p><b>(3) <a href="#">Requirements allocation analysis</a></b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>User needs</p>                                                                                                                                                                                       | <p>Task report(s)—Requirements allocation analysis</p>                 |

| <b>8.4 Activity: Architecture definition V&amp;V (System, 15288—Architecture Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                   |                                                                                                                                                                                  |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Required inputs</b>                                                                                            | <b>Required outputs</b>                                                                                                                                                          |
| <p>Verify the correctness and completeness of the system requirements allocation to the hardware, software, and user interfaces against user needs.</p> <p>a) <b>Correctness</b></p> <p>Verify that the system requirements allocated to software and hardware elements can be accomplished by the designated element. Review the allocations to verify that tradeoff considerations of significance (e.g., reliability, maintainability, safety, robustness, and cost-effectiveness) were factored into architecture decisions.</p> <p>b) <b>Completeness</b></p> <p>Verify that each system requirement is allocated to at least one system element or user.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>Business or mission analysis documentation</p> <p>System requirements</p> <p>Architecture models and views</p> | <p>Anomaly report(s)</p>                                                                                                                                                         |
| <p><b>(4) <u>Traceability analysis</u></b></p> <p>Verify that the traceability analysis is complete. The task criteria are as follows:</p> <p>a) Verify that all architectural elements are traceable to one or more system requirements.</p> <p>b) Verify that all system requirements are traceable to one or more architectural elements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>System requirements</p> <p>Architecture models and views</p>                                                   | <p>Task report(s)—Traceability analysis</p> <p>Anomaly report(s)</p>                                                                                                             |
| <p><b>(5) <u>Criticality analysis</u></b></p> <p>a) Review and update the existing criticality analysis results from the prior criticality task report using the system architectural definition to assign a system integrity level to each architectural element.</p> <p>b) Implementation methods and interfacing technologies may cause previously assigned system integrity levels to be raised or lowered for a given stakeholder or system requirement. Verify that no inconsistent or undesired system integrity consequences are introduced by reviewing the revised system integrity levels.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Criticality task report</p> <p>System requirements</p> <p>Architecture models and views</p>                    | <p>Task report(s)—Criticality analysis</p> <p>Anomaly report(s)</p>                                                                                                              |
| <p><b>(6) <u>System integration test plan V&amp;V</u></b></p> <p>a) System elements of integrity level 4</p> <ol style="list-style-type: none"> <li>1) Plan V&amp;V system integration testing to validate that the system element correctly implements the system requirements and design as each system element is incrementally integrated with each other.</li> <li>2) Plan tracing of system requirements to test designs, cases, procedures, and results.</li> <li>3) Plan documentation of test designs, cases, procedures, and results.</li> <li>4) The V&amp;V system integration test plan shall address the following: <ol style="list-style-type: none"> <li>i) Conformance to an increasingly larger set of functional requirements at each stage of integration.</li> <li>ii) Assessment of timing, sizing, and accuracy.</li> <li>iii) Performance at boundaries and under stress conditions.</li> <li>iv) Measures of requirements test coverage and system reliability.</li> </ol> </li> <li>5) Verify that the V&amp;V system integration test plan conforms to the project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> </ol> | <p>System requirements</p> <p>Stakeholder requirements</p> <p>System integration test plan</p>                    | <p>V&amp;V system integration test plan (integrity level 4)</p> <p>Task report(s)—Review of system integration test plan (integrity levels 3 and 2)</p> <p>Anomaly report(s)</p> |

| <b>8.4 Activity: Architecture definition V&amp;V (System, 15288—Architecture Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                             |                                                                                                                                                                                             |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Required inputs</b>                                                                                                                                                                      | <b>Required outputs</b>                                                                                                                                                                     |
| <p>6) Validate that the V&amp;V system integration test plan satisfies the following criteria:</p> <ul style="list-style-type: none"> <li>i) Traceable to the system requirements.</li> <li>ii) External consistency with the system requirements.</li> <li>iii) Internal consistency.</li> <li>iv) Test coverage of the system requirements.</li> <li>v) Appropriateness of test standards and methods used.</li> <li>vi) Conformance to expected test results.</li> <li>vii) Feasibility of system integration testing.</li> <li>viii) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs).</li> </ul> <p>b) System elements of integrity levels 3 and 2</p> <ul style="list-style-type: none"> <li>1) Verify that the developer's system integration test plan conforms to the project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>2) Validate that the developer's system integration test plan satisfies the following criteria: <ul style="list-style-type: none"> <li>i) Traceable to the system requirements.</li> <li>ii) External consistency with the system requirements.</li> <li>iii) Internal consistency.</li> <li>iv) Test coverage of the system requirements.</li> <li>v) Appropriateness of test standards and methods.</li> <li>vi) Conformance to expected test results.</li> </ul> </li> </ul> <p>c) System elements of integrity level 1<br/>There are no system integration V&amp;V test plan requirements.</p> |                                                                                                                                                                                             |                                                                                                                                                                                             |
| <p>(7) <b><u>System qualification test design V&amp;V</u></b></p> <p>a) System integrity level 4</p> <ul style="list-style-type: none"> <li>1) Design tests for V&amp;V system qualification testing.</li> <li>2) Continue tracing required by the V&amp;V system qualification test plan. Verify that the V&amp;V system acceptance test designs conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>3) Validate that the V&amp;V system qualification test designs satisfy the criteria in V&amp;V activity <a href="#">8.3, Task 5</a>.</li> </ul> <p>b) System integrity levels 3 and 2</p> <ul style="list-style-type: none"> <li>1) Verify that the developer's test designs for system qualification testing conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>2) Verify that the developer's system qualification test designs satisfy the criteria in V&amp;V activity <a href="#">8.3, Task 5</a>.</li> </ul> <p>c) System integrity level 1<br/>There are no system qualification V&amp;V test design requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                  | <p>System requirements</p> <p>Architecture models and views</p> <p>System qualification test plan</p> <p>System qualification test design from the developer (integrity levels 3 and 2)</p> | <p>V&amp;V system qualification test design(s) (integrity level 4)</p> <p>Task report(s)—Review of system qualification test design (integrity levels 3 and 2)</p> <p>Anomaly report(s)</p> |
| <p>(8) <b><u>System acceptance test design V&amp;V</u></b></p> <p>a) System integrity level 4</p> <ul style="list-style-type: none"> <li>1) Design tests for V&amp;V system acceptance testing.</li> <li>2) Continue tracing required by the V&amp;V system acceptance test plan. Verify that the V&amp;V system acceptance test</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>System requirements</p> <p>Architecture models and views</p>                                                                                                                             | <p>V&amp;V system acceptance test design(s) (integrity level 4)</p>                                                                                                                         |

| <b>8.4 Activity: Architecture definition V&amp;V (System, 15288—Architecture Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                               |                                                                                                                        |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Required inputs</b>                                                                                                                                                                                        | <b>Required outputs</b>                                                                                                |
| <p>designs conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</p> <p>3) Validate that the V&amp;V system acceptance test designs satisfy the criteria in V&amp;V activity <a href="#">8.3, Task 7</a>.</p> <p>b) System integrity levels 3 and 2</p> <p>1) Verify that the acquirer's test designs for system acceptance testing conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</p> <p>2) Validate that the acquirer's system acceptance test designs satisfy the criteria in V&amp;V activity <a href="#">8.3, Task 7</a>.</p> <p>c) System integrity level 1</p> <p>There are no system acceptance V&amp;V test plan requirements.</p>                                                                                                                                                                                                                                             | <p>System acceptance test plan</p> <p>System acceptance test design from the developer (integrity levels 3 and 2)</p>                                                                                         | <p>Task report(s)—<br/>Review of system acceptance test design (integrity levels 3 and 2)</p> <p>Anomaly report(s)</p> |
| <p><b>(9) <a href="#">Hazard analysis</a></b></p> <p>Analyze the potential hazards to and from the conceptual system. The analysis shall perform the following:</p> <p>a) Identify the potential system hazards.</p> <p>b) Assess the consequences of each hazard.</p> <p>c) Assess the probability of each hazard.</p> <p>d) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled. (Any unmitigated hazards are documented and addressed as part of system operations.)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Architecture models and views</p> <p>Hazard analysis results</p>                                                                                                                                           | <p>Task report(s)—<br/>Hazard analysis</p> <p>Anomaly report(s)</p>                                                    |
| <p><b>(10) <a href="#">Security analysis</a></b></p> <p>a) Review the system owner's definition of an acceptable level of security risk.</p> <p>b) Analyze the system requirements from a security perspective and assure that potential security risks with respect to confidentiality (disclosure of sensitive information/data), integrity (modification of information/data), availability (withholding of information or services), and accountability (attributing actions to an individual/process) have been identified. Include an assessment of the sensitivity of the information/data to be processed.</p> <p>c) Analyze security risks introduced by the system itself as well as those associated with the environment with which the system interfaces.</p> <p>d) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system operations).</p> | <p>Architecture models and views</p> <p>Preliminary threat and risk assessment (TRA)</p> <p>Security analysis results</p>                                                                                     | <p>Task report(s)—<br/>Security analysis</p> <p>Anomaly report(s)</p>                                                  |
| <p><b>(11) <a href="#">Risk analysis</a></b></p> <p>a) Identify the technical and management risks.</p> <p>b) Provide recommendations to eliminate, reduce, or mitigate the risks.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>Architecture models and views</p> <p>Supplier development plans and schedules</p> <p>Hazard analysis results</p> <p>Security analysis results</p> <p>Risk analysis results</p> <p>V&amp;V task results</p> | <p>Task report(s)—<br/>Risk analysis</p> <p>Anomaly report(s)</p>                                                      |

| <b>8.5 Activity: Design definition V&amp;V (System, 15288—Design Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                 |                                                                      |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Required inputs</b>                                                                                                                                                                                                          | <b>Required outputs</b>                                              |
| <p><b>(1) <u>Design evaluation</u></b></p> <p>Evaluate the system design for correctness, consistency, completeness, and testability. The task criteria are as follows:</p> <ul style="list-style-type: none"> <li>a) Correctness <ul style="list-style-type: none"> <li>1) Verify the characteristics, attributes, constraints, and functional and performance requirements of the system design correctly implement the architecture definition.</li> <li>2) Verify the system design complies with standards, regulations, policies, physical laws, and business rules.</li> <li>3) Validate the product solution(s) defined by the system design satisfies the stakeholder needs.</li> <li>4) Validate the system design and its element interactions do not result in unnecessary, unintended, or deleterious consequences.</li> <li>5) Validate the system design defines the intended interaction of the system with its operating environment.</li> </ul> </li> <li>b) Consistency <ul style="list-style-type: none"> <li>1) Verify the system design conforms to the design guidance, principles, and tenets of the organization processes and procedures (e.g., service-oriented architecture, modular open-systems architecture).</li> <li>2) Verify the design principles, characteristics, and rules established for the system are being applied across the system design.</li> </ul> </li> <li>c) Completeness <ul style="list-style-type: none"> <li>1) Verify the system requirements are allocated to the elements of the system design.</li> <li>2) Verify all system requirements are included in the system design.</li> </ul> </li> </ul> <p>NOTE—Typically, the stakeholder needs and requirements definition, system requirements definition, architecture definition, and design definition processes are recursively applied to successive levels of detail in the system architecture and design until elements are completely defined.</p> | <p>System requirements</p> <p>Business or mission analysis documentation</p> <p>Architecture models and views</p> <p>Architecture trade-off analyses</p> <p>Design documents</p> <p>Organizational processes and procedures</p> | <p>Task report(s)—Design evaluation</p> <p>Anomaly report(s)</p>     |
| <p><b>(2) <u>Interface analysis</u></b></p> <p>Verify the design interfaces between system elements and with other systems are correct and complete. The task criteria are as follows:</p> <ul style="list-style-type: none"> <li>a) Correctness <ul style="list-style-type: none"> <li>1) Verify the system design satisfies the system interface requirements between system elements.</li> <li>2) Validate the system design satisfies the system interface requirements with external systems (external system boundaries).</li> </ul> </li> <li>b) Completeness <ul style="list-style-type: none"> <li>1) Verify the system design describes all internal and external interfaces.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>System requirements</p> <p>Design documents</p>                                                                                                                                                                              | <p>Task report(s)—Interface analysis</p> <p>Anomaly report(s)</p>    |
| <p><b>(3) <u>Traceability analysis</u></b></p> <p>Verify that the traceability analysis is complete. The task criteria are as follows:</p> <ul style="list-style-type: none"> <li>a) Verify that all design elements are traceable to one or more architectural elements.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Architecture models and views</p> <p>Design documents</p>                                                                                                                                                                    | <p>Task report(s)—Traceability analysis</p> <p>Anomaly report(s)</p> |

| <b>8.5 Activity: Design definition V&amp;V (System, 15288—Design Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                      |                                                                                                                                                                      |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Required inputs</b>                                                                                                                                               | <b>Required outputs</b>                                                                                                                                              |
| b) Verify that all architectural elements are traceable to one or more design elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                      |                                                                                                                                                                      |
| <b>(4) <u>Criticality analysis</u></b><br>a) Review and update the existing criticality analysis results from the prior criticality task report using the system design to assign a system integrity level to each design element.<br>b) Implementation methods and interfacing technologies may cause previously assigned system integrity levels to be raised or lowered for a given stakeholder or system requirement. Verify that no inconsistent or undesired system integrity consequences are introduced by reviewing the revised system integrity levels.                                                                                                                                                                                                                                                                                                                                                                                                                   | Criticality task report<br>System requirements<br>Design documents                                                                                                   | Task report(s)—Criticality analysis<br>Anomaly report(s)                                                                                                             |
| <b>(5) <u>System integration test design V&amp;V</u></b><br>a) System integrity level 4<br>1) Design tests for V&V system integration testing.<br>2) Continue tracing required by the V&V system integration test plan. Verify that the V&V system integration test designs conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).<br>3) Validate that the V&V system integration test designs satisfy the criteria in V&V activity <a href="#">8.4, Task 6</a> .<br>b) System integrity levels 3 and 2<br>1) Verify that the developer's test designs for system integration testing conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).<br>2) Validate that the developer's system integration test designs satisfy the criteria in V&V activity <a href="#">8.4, Task 6</a> .<br>c) System integrity level 1<br>There are no system integration V&V test design requirements. | System requirements<br>Architecture models and views<br>System integration test plan<br>System integration test design from the developer (integrity levels 3 and 2) | V&V system integration test design(s) (integrity level 4)<br>Task report(s)—Review of system integration test design (integrity levels 3 and 2)<br>Anomaly report(s) |
| <b>(6) <u>System integration test case V&amp;V</u></b><br>a) System integrity level 4<br>1) Develop test cases for V&V system integration testing.<br>2) Continue tracing required by the V&V system integration test plan.<br>3) Verify that the system integration V&V test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).<br>4) Validate that the V&V system integration test cases satisfy the criteria in V&V activity <a href="#">8.5, Task 5</a> .<br>b) System integrity levels 3 and 2<br>1) Verify that the developer's system integration test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).<br>2) Validate that the developer's system integration test cases satisfy the criteria in V&V activity <a href="#">8.5, Task 5</a> .<br>c) System integrity level 1<br>There are no system integration test case V&V requirements.             | System requirements<br>Design documents<br>System integration test design<br>System integration test cases from the developer (integrity levels 3 and 2)             | V&V system integration test cases (integrity level 4)<br>Task report(s)—Review of system integration test cases (integrity levels 3 and 2)<br>Anomaly report(s)      |
| <b>(7) <u>System qualification test case V&amp;V</u></b><br>a) System integrity level 4<br>1) Develop test cases for V&V system qualification testing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | System requirements<br>Design documents                                                                                                                              | V&V system qualification test cases (integrity level 4)                                                                                                              |



| <b>8.5 Activity: Design definition V&amp;V (System, 15288—Design Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                        |                                                                                                                                                                   |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Required inputs</b>                                                                                                                                 | <b>Required outputs</b>                                                                                                                                           |
| 2) Continue tracing required by the V&V system qualification test plan.<br>3) Verify that the V&V system qualification test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).<br>4) Validate that the V&V system qualification test cases satisfy the criteria in V&V activity <a href="#">8.4, Task 7</a> .<br>b) System integrity levels 3 and 2<br>1) Verify that the developer's system qualification test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).<br>2) Validate that the developer's system qualification test cases satisfy the criteria in V&V activity <a href="#">8.4, Task 7</a> .<br>c) System integrity level 1<br>There are no system qualification test case V&V requirements.                                                                                                                                        | System qualification test design<br>System qualification test cases from the developer (integrity levels 3 and 2)                                      | Task report(s)—<br>Review of system qualification test cases (integrity levels 3 and 2)<br>Anomaly report(s)                                                      |
| <b>(8) <a href="#">System acceptance test case V&amp;V</a></b><br>a) System integrity level 4<br>1) Develop test cases for V&V system acceptance testing.<br>2) Continue tracing required by the V&V system acceptance test plan.<br>3) Verify that the V&V system acceptance test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).<br>4) Validate that the V&V system acceptance test cases satisfy the criteria in V&V activity <a href="#">8.4, Task 8</a> .<br>b) System integrity levels 3 and 2<br>1) Verify that the acquirer's system acceptance test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).<br>2) Verify that the acquirer's system acceptance test cases satisfy the criteria in V&V activity <a href="#">8.4, Task 8</a> .<br>c) System integrity level 1<br>There are no system acceptance test case V&V requirements. | System requirements<br>Design documents<br>System acceptance test design<br>System acceptance test cases from the developer (integrity levels 3 and 2) | V&V system acceptance test cases (integrity level 4)<br>Task report(s)—<br>Review of system acceptance test cases (integrity levels 3 and 2)<br>Anomaly report(s) |
| <b>(9) <a href="#">Hazard analysis</a></b><br>a) Analyze the potential hazards to and from the conceptual system. The analysis shall perform the following:<br>b) Identify the potential system hazards.<br>c) Assess the consequences of each hazard.<br>d) Assess the probability of each hazard.<br>e) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system operations).                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Design documents<br>Hazard analysis results                                                                                                            | Task report(s)—<br>Hazard analysis<br>Anomaly report(s)                                                                                                           |
| <b>(10) <a href="#">Security analysis</a></b><br>a) Review the system owner's definition of an acceptable level of security risk.<br>b) Analyze the system concept from a security perspective and assure that potential security risks with respect to confidentiality (disclosure of sensitive information/data), integrity (modification of information/data), availability (withholding of information or services), and accountability (attributing actions to an individual/process) have been                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Design documents<br>Preliminary threat and risk assessment (TRA)<br>Subsystems security analysis<br>Security analysis results                          | Task report(s)—<br>Security analysis<br>Anomaly report(s)                                                                                                         |



| <b>8.5 Activity: Design definition V&amp;V (System, 15288—Design Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                  |                                                                      |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Required inputs</b>                                                                                                                                                                           | <b>Required outputs</b>                                              |
| <p>identified. Include an assessment of the sensitivity of the information/data to be processed.</p> <p>c) Analyze security risks introduced by the system itself as well as those associated with the environment with which the system interfaces.</p> <p>d) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system operations).</p> |                                                                                                                                                                                                  |                                                                      |
| <p><b>(11) <a href="#">Risk analysis</a></b></p> <p>a) Identify the technical and management risks.</p> <p>b) Provide recommendations to eliminate, reduce, or mitigate the risks.</p>                                                                                                                                                                                                                                                                                         | <p>Design documents</p> <p>Supplier development plans and schedules</p> <p>Hazard analysis results</p> <p>Security analysis results</p> <p>Risk analysis results</p> <p>V&amp;V task results</p> | <p>Task report(s)—</p> <p>Risk analysis</p> <p>Anomaly report(s)</p> |

| <b>8.6 Activity: System analysis V&amp;V (System, 15288—System Analysis Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                |                                                                   |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Required inputs</b>                                         | <b>Required outputs</b>                                           |
| <p><b>(1) <a href="#">System analysis strategy evaluation</a></b></p> <p>Evaluate the system analysis strategy for completeness and consistency with the criticality of the system or system element. The task criteria are as follows:</p> <p>a) Completeness</p> <ol style="list-style-type: none"> <li>1) Verify that the scope and objectives of the system analysis address the correct needs.</li> <li>2) Verify that the scope of the system analysis addresses all aspects related to the decision.</li> <li>3) Verify that all assumptions have been identified and are correct.</li> <li>4) Verify that any inputs for enabling systems or services have been identified and are available.</li> <li>5) Verify that any inputs for enabling the system have been identified and are available in the system analysis strategy plan.</li> </ol> <p>b) Consistency with Criticality</p> <p>Verify that the rigor and approach specified in the system analysis strategy are consistent with that needed for the system or system elements that could be affected by the decision.</p> | <p>System analysis strategy</p>                                | <p>Task report(s)—</p> <p>System analysis strategy evaluation</p> |
| <p><b>(2) <a href="#">System analysis results evaluation</a></b></p> <p>Evaluate the results of the system analysis to determine that the analysis was conducted as specified in the strategy, and the results support its conclusions and recommendations.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>System analysis strategy</p> <p>System analysis results</p> | <p>Task report(s)—</p> <p>System analysis results evaluation</p>  |

| <b>8.7 Activity: Implementation V&amp;V (System, 15288—Implementation Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Required inputs</b>                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Required outputs</b>                                                               |
| <p><b>(1) <a href="#">Implementation strategy assessment</a></b></p> <p>a) Verify that the implementation strategy accounts for all implementation procedures (e.g., technical manuals, organizational policies, or standards), fabrication processes, tools and equipment, implementation tolerances, and Verification uncertainties needed to construct the system element.</p> <p>b) Verify that the implementation strategy defines the limitations and constraints imposed on the system element design solution.</p> <p>c) Verify that execution of the implementation strategy will produce a system element that conforms to the system element description in the system design.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Implementation strategy</p> <p>Implementation procedures</p> <p>Fabrication processes, tools, and equipment</p> <p>System element description from the system design</p>                                                                                                                                                                                                                                             | <p>Task report(s)—Implementation strategy assessment</p> <p>Anomaly report(s)</p>     |
| <p><b>(2) <a href="#">System element implementation analysis</a></b></p> <p>a) V&amp;V of system element requirements</p> <p>1) System elements of integrity levels 4 and 3</p> <p>i) Verify (e.g., through analysis, reviews, and audits) that integrity level 4 and 3 element requirements trace to system requirements.</p> <p>ii) Verify (e.g., through analysis, reviews, and audits) that the element requirements satisfy the integrity level 4 and 3 system requirements.</p> <p>2) System elements of integrity level 2</p> <p>Verify (e.g., through reviews and audits) that the element requirements satisfy the level 2 system requirements.</p> <p>3) System elements of integrity level 1</p> <p>There are no system element V&amp;V requirements.</p> <p>b) V&amp;V of system element design</p> <p>1) System elements of integrity level 4</p> <p>i) Verify that the design of the integrity level 4 element is consistent with the system design.</p> <p>ii) Verify that the design of the integrity level 4 element satisfies the requirements of the system element.</p> <p>2) System elements of integrity levels 3 and 2</p> <p>Verify that the design of each element is consistent with the system design.</p> <p>3) System elements of integrity level 1</p> <p>There are no system element design V&amp;V requirements.</p> <p>c) V&amp;V of system element implementation</p> <p>1) System elements of integrity level 4</p> <p>Begin pre-integration testing using early partial deliveries of system elements (software and hardware) to derive early visibility into potential system integration testing issues and to initiate corrective changes to the system element's development to correct or mitigate the potential system integration issue(s).</p> <p>2) System elements of integrity levels 4 and 3</p> <p>Evaluate the system element artifacts (requirements, design, anomaly reports/trends, element performance metrics, schedule compliance, resource usage actuals, and projected resource needs) produced from the</p> | <p>Business or mission analysis documentation</p> <p>System requirements</p> <p>Software requirements specification(s)</p> <p>Software interface requirements specification(s)</p> <p>Hardware requirements specification(s)</p> <p>System design</p> <p>Software design documents(s)</p> <p>Hardware drawings</p> <p>System element(s)</p> <p>Software traceability analysis</p> <p>Hardware traceability analysis</p> | <p>Task report(s)—System element implementation analysis</p> <p>Anomaly report(s)</p> |

| <b>8.7 Activity: Implementation V&amp;V (System, 15288—Implementation Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                        |                         |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Required inputs</b> | <b>Required outputs</b> |
| <p>software/hardware development life cycle process to perform the following:</p> <ul style="list-style-type: none"> <li>i) Derive an evolving assessment of each system element's performance (i.e., a preview of how well the system element is meeting its requirements) and recommend corrective actions (i.e., feedback to the element development process or change to system architecture) to mitigate any projected performance shortfalls.</li> <li>ii) Assess each system element's project schedule performance (i.e., estimated system element schedule completion dates for system element milestone reviews and delivery dates) and recommend system schedule changes if the system element project completion dates are slipping.</li> <li>iii) Estimate remaining schedule resource requirements (i.e., labor or equipment budget) if performance or schedule shortfalls are projected.</li> </ul> <p>3) System elements of integrity levels 2 and 1<br/>There are no system element implementation V&amp;V requirements.</p> <p>d) V&amp;V of system element integration testing</p> <ul style="list-style-type: none"> <li>1) System elements of integrity level 4<br/>Verify through independently produced objective evidence (e.g., independent test) that integrity level 4 requirements in that element are satisfied by the system element hardware/software integration testing.</li> <li>2) System elements of integrity levels 3 and 2<br/>Verify through the data produced by the development organization (e.g., reviews, audits) that the integrity levels 3 and 2 requirements in that element are satisfied by the system element hardware/software integration testing.</li> <li>3) System elements of integrity levels 1<br/>There are no system element integration testing V&amp;V requirements.</li> </ul> <p>e) V&amp;V of system element qualification testing</p> <ul style="list-style-type: none"> <li>1) System elements of integrity level 4 <ul style="list-style-type: none"> <li>i) Develop V&amp;V test cases for system element qualification testing.</li> <li>ii) Continue tracing required by the system element V&amp;V test plan.</li> <li>iii) Verify that the system element V&amp;V test cases conform to the project-defined test document purpose, format, and content.</li> <li>iv) Validate that the system element V&amp;V test cases satisfy the criteria in the system element V&amp;V qualification test plan generation activity.</li> </ul> </li> <li>2) System elements of integrity levels 3 and 2<br/>Review the results of the developer's qualification testing.</li> <li>3) System elements of integrity level 1<br/>There are no system element qualification testing V&amp;V requirements.</li> </ul> <p>f) V&amp;V of system element acceptance testing</p> |                        |                         |

| <b>8.7 Activity: Implementation V&amp;V (System, 15288—Implementation Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                       |                                                                                                                                                                                              |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Required inputs</b>                                                                                                                                                                | <b>Required outputs</b>                                                                                                                                                                      |
| <ol style="list-style-type: none"> <li>1) System elements of integrity level 4 <ol style="list-style-type: none"> <li>i) Plan acceptance V&amp;V testing to validate that the system element correctly implements system requirements in an operational environment.</li> <li>ii) Design tests for acceptance V&amp;V testing.</li> <li>iii) Develop test cases for acceptance V&amp;V testing.</li> <li>iv) Perform acceptance V&amp;V testing.</li> <li>v) Analyze test results to validate that the system element satisfies the V&amp;V test acceptance criteria.</li> </ol> </li> <li>2) System elements of integrity levels 3 and 2<br/>Review the acquirer's acceptance test results to validate that the system element satisfies the test acceptance criteria.</li> <li>3) System elements of integrity level 1<br/>There are no system element acceptance testing V&amp;V requirements.</li> </ol>                                                                                                                                                                                                               |                                                                                                                                                                                       |                                                                                                                                                                                              |
| <p><b>(3) <a href="#">System element interaction analysis</a></b><br/>The System Element Interaction Analysis shall analyze the results of each element V&amp;V to determine whether directly or indirectly coupled system elements require their V&amp;V tasks to be reiterated (repeated, redone) or new V&amp;V tasks performed. As each element V&amp;V results are determined, V&amp;V may have to be recursively applied to all directly or indirectly coupled system elements. The System Analysis process may be used to establish the criteria of how to recursively apply V&amp;V to system elements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>System element implementation analysis results</p> <p>System architecture</p> <p>System design</p>                                                                                 | <p>Task report(s)—System element interaction analysis</p> <p>Anomaly report(s)</p> <p>V&amp;V Plan update (if tasks are added or changed)</p>                                                |
| <p><b>(4) <a href="#">Criticality analysis</a></b></p> <ol style="list-style-type: none"> <li>a) Review and update the existing criticality levels of the system elements during the implementation process based on the implementation of the software and hardware components.</li> <li>b) Verify that no inconsistent or undesired system element integrity consequences have been introduced due to implementation methods and interfacing technologies that may cause previously assigned system element integrity levels to be raised or lowered.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>Criticality task report</p> <p>System element requirements, designs, and implementations</p>                                                                                       | <p>Task report(s)—Criticality analysis</p> <p>Anomaly report(s)</p>                                                                                                                          |
| <p><b>(5) <a href="#">System integration test procedure V&amp;V</a></b></p> <ol style="list-style-type: none"> <li>a) System integrity level 4 <ol style="list-style-type: none"> <li>1) Develop test procedures for V&amp;V system integration testing.</li> <li>2) Continue tracing required by the V&amp;V system integration test plan.</li> <li>3) Verify that the V&amp;V system integration test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>4) Validate that the V&amp;V system integration test procedures satisfy the criteria in V&amp;V activity <a href="#">8.5, Task 6</a>.</li> </ol> </li> <li>b) System integrity levels 3 and 2 <ol style="list-style-type: none"> <li>1) Verify that the developer's system integration test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>2) Validate that the developer's system integration test procedures satisfy the criteria in V&amp;V activity <a href="#">8.5, Task 6</a>.</li> </ol> </li> </ol> | <p>System requirements</p> <p>System design document</p> <p>System integration test cases</p> <p>System integration test procedures from the developer (integrity levels 3 and 2)</p> | <p>V&amp;V system integration test procedures (integrity level 4)</p> <p>Task report(s)—Review of system integration test procedures (integrity levels 3 and 2)</p> <p>Anomaly report(s)</p> |

| <b>8.7 Activity: Implementation V&amp;V (System, 15288—Implementation Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                  |                                                                                                                                                                                       |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Required inputs</b>                                                                                                                                           | <b>Required outputs</b>                                                                                                                                                               |
| c) System integrity level 1<br>There are no system integration test procedure V&V requirements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                  |                                                                                                                                                                                       |
| <b>(6) <u>System qualification test procedure V&amp;V</u></b><br>a) System integrity level 4<br>1) Develop test procedures for V&V system qualification testing.<br>2) Continue tracing required by the V&V system qualification test plan.<br>3) Verify that the V&V system qualification test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).<br>4) Validate that the V&V system qualification test procedures satisfy the criteria in V&V activity <a href="#">8.5, Task 7</a> .<br>b) System integrity levels 3 and 2<br>1) Verify that the developer's system qualification test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).<br>2) Validate that the developer's system qualification test procedures satisfy the criteria in V&V activity <a href="#">8.5, Task 7</a> .<br>c) System integrity level 1<br>There are no system qualification test case V&V requirements. | System requirements<br>Design documents<br>System qualification test cases<br>System qualification test procedures from the developer (integrity levels 3 and 2) | V&V system qualification test procedures (integrity level 4)<br>Task report(s)—<br>Review of system qualification test procedures (integrity levels 3 and 2)<br><br>Anomaly report(s) |
| <b>(7) <u>System acceptance test procedure V&amp;V</u></b><br>a) System integrity level 4<br>1) Develop test procedures for V&V system acceptance testing.<br>2) Continue the tracing required by the V&V system acceptance test plan.<br>3) Verify that the V&V system acceptance test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).<br>4) Validate that the V&V system acceptance test procedures satisfy the criteria in V&V activity <a href="#">8.5, Task 8</a> .<br>b) System integrity levels 3 and 2<br>1) Verify that the acquirer's system acceptance test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).<br>2) Verify that the acquirer's system acceptance test procedures satisfy the criteria in V&V activity <a href="#">8.5, Task 8</a> .<br>c) System integrity level 1<br>There are no system acceptance test case V&V requirements.                         | System requirements<br>Design documents<br>System acceptance test cases<br>System acceptance test procedures from the developer (integrity levels 3 and 2)       | V&V system acceptance test procedures (integrity level 4)<br>Task report(s)—<br>Review of system acceptance test procedures (integrity levels 3 and 2)<br><br>Anomaly report(s)       |
| <b>(8) <u>Hazard analysis</u></b><br>a) Verify the system elements that implement critical requirements introduce no new hazards.<br>b) Update the hazard analysis.<br>c) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system operations).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Subsystems<br>Hazard analysis results                                                                                                                            | Task report(s)—<br>Hazard analysis<br>Anomaly report(s)                                                                                                                               |

| <b>8.7 Activity: Implementation V&amp;V (System, 15288—Implementation Process)</b>                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                             |                                                           |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Required inputs</b>                                                                      | <b>Required outputs</b>                                   |
| <b>(9) <u>Security analysis</u></b><br>a) Verify that the system element is implemented in accordance with the system architecture in that it addresses the identified security risks and that the implementation does not introduce new security risks.<br>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system operations). | System architecture<br>Subsystems<br>Security analysis results<br>V&V task results          | Task report(s)—<br>Security analysis<br>Anomaly report(s) |
| <b>(10) <u>Risk analysis</u></b><br>Review and update risk analysis using prior task reports. Provide recommendations to eliminate, reduce, or mitigate the risks.                                                                                                                                                                                                                                                                                                         | Subsystems<br>Hazard analysis results<br>Security analysis results<br>Risk analysis results | Task report(s)—<br>Risk analysis<br>Anomaly report(s)     |

| <b>8.8 Activity: Integration V&amp;V (System, 15288—Integration Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                           |                                                                                                                                                                                     |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Required inputs</b>                                                                                                                                                    | <b>Required outputs</b>                                                                                                                                                             |
| <b>(1) <u>System integration strategy assessment</u></b><br>a) Verify the system integration strategy.<br>b) Verify that any constraints specified or implied by the system integration strategy are included in the system requirements and design.<br>c) Integration test specification and procedure generation.                                                                                                                                                                                                                                                                                                                                                                                                                           | System integration strategy<br>System architecture<br>System requirements                                                                                                 | Task report(s)—<br>System integration strategy assessment<br>Anomaly report(s)                                                                                                      |
| <b>(2) <u>System element interaction analysis</u></b><br>The system element interaction analysis shall analyze the results of system integration V&V to determine whether directly or indirectly coupled system elements require their V&V tasks to be reiterated (repeated, redone) or new V&V tasks performed. As integration results are determined, V&V may have to be recursively applied to all directly or indirectly coupled system elements. The System Analysis Process may be used to establish the criteria of how to recursively apply V&V to system elements.                                                                                                                                                                   | System integration results<br>System architecture<br>System design                                                                                                        | Task report(s)—<br>System element interaction analysis<br>Anomaly report(s)<br>V&V Plan update (if tasks are added or changed)                                                      |
| <b>(3) <u>System integration test execution V&amp;V</u></b><br>a) System integrity level 4<br>1) Perform V&V system integration testing.<br>2) Use the V&V system integration test results to validate that the system satisfies the test acceptance criteria.<br>3) Document the results as required by the V&V system integration test plan.<br>4) Use the V&V system integration test results to validate that the system satisfies the test acceptance criteria.<br>5) Document discrepancies between the actual and expected test results.<br>b) System integrity levels 3 and 2<br>Use the developer's system integration test results to verify that the system satisfies the test acceptance criteria.<br>c) System integrity level 1 | Integration sequence and strategy (e.g., test scenario inputs, test procedure sequence)<br>Hardware element(s)<br>Software element(s)<br>Other external system element(s) | V&V system integration test execution results (integrity level 4)<br>Task report(s)—<br>Review of system integration test execution (integrity levels 3 and 2)<br>Anomaly report(s) |

| <b>8.8 Activity: Integration V&amp;V (System, 15288—Integration Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                           |                                                                                                                                                                                                        |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Required inputs</b>                                                                                                                                                                                                                                    | <b>Required outputs</b>                                                                                                                                                                                |
| There are no system integration test execution V&V requirements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                           |                                                                                                                                                                                                        |
| <p><b>(4) <a href="#">System qualification test execution V&amp;V</a></b></p> <p>a) System integrity level 4</p> <ol style="list-style-type: none"> <li>1) Perform V&amp;V system qualification testing.</li> <li>2) Use the V&amp;V system qualification test results to validate that the system satisfies the test acceptance criteria.</li> <li>3) Document discrepancies between the V&amp;V system qualification test results and the test acceptance criteria.</li> <li>4) Complete the traceability by documenting the location of the V&amp;V system qualification test results.</li> <li>5) Analyze test results to validate that the system delivers required services when installed in its operational location and staffed by operators. Additionally, the often less formally expressed but sometimes overriding attitudes, experience, and subjective tests that comprise customer satisfaction also need consideration.</li> <li>6) Document the results as required by the V&amp;V system qualification test plan.</li> </ol> <p>b) System integrity levels 3 and 2</p> <p>Use the developer's system qualification test results to verify that the acceptance criteria are satisfied.</p> <p>c) System integrity level 1</p> <p>There are no system qualification test execution V&amp;V requirements.</p> | <p>Stakeholder requirements</p> <p>System requirements</p> <p>Business or mission analysis documentation</p> <p>System qualification test procedures</p> <p>System qualification test execution results from the developer (integrity levels 3 and 2)</p> | <p>V&amp;V system qualification test execution results (integrity level 4)</p> <p>Task report(s)—Review of system qualification test execution (integrity levels 3 and 2)</p> <p>Anomaly report(s)</p> |
| <p><b>(5) <a href="#">Traceability analysis</a></b></p> <p>Analyze relationships in the V&amp;V test plans, designs, cases, and procedures for correctness and completeness. The task criteria are as follows:</p> <p>a) Correctness</p> <p>Verify that there is a valid relationship between the V&amp;V test plans, designs, cases, and procedures.</p> <p>b) Completeness</p> <p>Verify that all V&amp;V test procedures are traceable to the V&amp;V test plans.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>V&amp;V test plans</p> <p>V&amp;V test designs</p> <p>V&amp;V test cases</p> <p>V&amp;V test procedures</p>                                                                                                                                            | <p>Task report(s)—Traceability analysis</p> <p>Anomaly report(s)</p>                                                                                                                                   |
| <p><b>(6) <a href="#">Hazard analysis</a></b></p> <p>a) Verify that the test instrumentation does not introduce new hazards.</p> <p>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of the system and software operations).</p> <p>c) Update the hazard analysis.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Hardware element(s)</p> <p>Software element(s)</p> <p>Other external system element(s)</p> <p>Test results</p> <p>Hazard analysis results</p>                                                                                                          | <p>Task report(s)—Hazard analysis</p> <p>Anomaly report(s)</p>                                                                                                                                         |
| <p><b>(7) <a href="#">Security analysis</a></b></p> <p>a) Verify that the implemented system does not increase the security risk.</p> <p>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of the system and software operations).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>Hardware element(s)</p> <p>Software element(s)</p> <p>Other external system element(s)</p> <p>Security analysis results</p>                                                                                                                            | <p>Task report(s)—Security analysis</p> <p>Anomaly report(s)</p>                                                                                                                                       |



| <b>8.8 Activity: Integration V&amp;V (System, 15288—Integration Process)</b>                                                                                                        |                                                                                                                                               |                                                   |
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| <b>V&amp;V tasks</b>                                                                                                                                                                | <b>Required inputs</b>                                                                                                                        | <b>Required outputs</b>                           |
| <b>(8) <a href="#">Risk analysis</a></b><br>a) Review and update risk analysis using prior task reports.<br>b) Provide recommendations to eliminate, reduce, or mitigate the risks. | Supplier development plans and schedules<br>Hazard analysis results<br>Security analysis results<br>Risk analysis results<br>V&V task results | Task report(s)—Risk analysis<br>Anomaly report(s) |

| <b>8.9 Verification (System, 15288—Verification Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                |                                                                                                                  |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Required inputs</b>                                                                                         | <b>Required outputs</b>                                                                                          |
| The activities and tasks for the system Verification Process are conducted in system technical life cycle processes and are contained in Table 1b, <a href="#">Activity 8.1</a> (business or mission analysis V&V), <a href="#">Activity 8.2</a> (stakeholder needs and requirements definition V&V), <a href="#">Activity 8.3</a> (system requirements definition V&V), <a href="#">Activity 8.4</a> (architecture definition V&V), <a href="#">Activity 8.5</a> (design definition V&V), <a href="#">Activity 8.6</a> (system analysis V&V), <a href="#">Activity 8.7</a> (implementation V&V), <a href="#">Activity 8.8</a> (integration V&V), <a href="#">Activity 8.10</a> (transition V&V), <a href="#">Activity 8.12</a> (operation V&V), <a href="#">Activity 8.13</a> (maintenance V&V) and <a href="#">Activity 8.14</a> (disposal V&V). The system Verification activities and tasks in <a href="#">Table 1b</a> and the common Verification activities and tasks in <a href="#">Table 1a</a> represent all Verification activities and tasks needed to perform system Verification. | The required inputs for the system Verification Process are found in <a href="#">Table 1b</a> required inputs. | The required outputs for the system Verification Process are found in <a href="#">Table 1b</a> required outputs. |

| <b>8.10 Activity: Transition V&amp;V (System, 15288—Transition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                       |                                                                |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Required inputs</b>                                                                                                                                                                                                                                                | <b>Required outputs</b>                                        |
| NOTE—If system transition is taking place by parts, then the Transition Process V&V tasks should be executed for each of the parts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                       |                                                                |
| <b>(1) <a href="#">Transition strategy evaluation</a></b><br>a) Verify that the transition strategy is comprehensive. Verify that the transition strategy includes the following: <ol style="list-style-type: none"> <li>1) All the system parts and the system as a whole (e.g., hardware [HW]/software [SW] functionality and databases) are considered in the transition execution.</li> <li>2) Transition schedule and sequence.</li> <li>3) Identification of transition hardware and software tools, equipment, and instructions.</li> <li>4) Archiving system artifacts, such as documentation and code.</li> <li>5) Impact of interacting systems in terms of transition timing and transition impacts.</li> <li>6) Continuity of capabilities when replacing or upgrading a legacy system.</li> <li>7) Site preparation for installation and legacy system retirement, storage, and/or incorporation.</li> </ol> | Transition strategy<br>Installation package (e.g., user documentation, system requirements, SDD, IDD, SRS, IRS, business or mission analysis documentation, installation procedures, site-specific parameters, installation tests, and configuration management data) | Task report(s)—Transition plan evaluation<br>Anomaly report(s) |



| <b>8.10 Activity: Transition V&amp;V (System, 15288—Transition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                               |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Required inputs</b>                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Required outputs</b>                                                                                                                                                                                                                       |
| <ul style="list-style-type: none"> <li>8) Provisions for documentation of the process results.</li> <li>9) A fallback position, in case of transition failure (e.g., going back to the old system) and other risk mitigation considerations.</li> <li>b) Verify that the transition strategy has a defined approach to establishing the system in the operational environment that is consistent with stakeholder requirements (e.g., by running the old and new systems in parallel or by well-defined system characteristics that can be compared).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                               |
| <p><b>(2) <u>Transition demonstration assessment</u></b></p> <ul style="list-style-type: none"> <li>a) Validate that acceptance tests defined in the acquirer agreement (as documented in the acceptance test plan) demonstrate satisfactory installation at the specified location(s). Where the exact location or environment of operation is not available, a representative example is selected.</li> <li>b) Verify and validate that the installed system (software and hardware elements) and related site-specific parameters are identical to the final as-verified and as-validated system (software and hardware elements).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Transitioning system (operational system under test)</li> <li>User documentation</li> <li>Installation package (e.g., user documentation, system requirements, SDD, IDD, SRS, IRS, concept documentation, installation procedures, site-specific parameters, installation tests, and configuration management data)</li> <li>Acquirer agreement</li> <li>Acceptance test plan</li> </ul> | <ul style="list-style-type: none"> <li>Task report(s)—Transition demonstration assessment</li> <li>Anomaly report(s)</li> </ul>                                                                                                               |
| <p><b>(3) <u>System acceptance test execution V&amp;V</u></b></p> <ul style="list-style-type: none"> <li>a) System integrity level 4 <ul style="list-style-type: none"> <li>1) Perform V&amp;V system acceptance testing.</li> <li>2) Use the system V&amp;V acceptance test results to validate that the system satisfies the test acceptance criteria.</li> <li>3) Document discrepancies between the V&amp;V system acceptance test results and the test acceptance criteria.</li> <li>4) Complete the traceability by documenting the location of the V&amp;V system acceptance test results.</li> <li>5) Analyze test results to validate that the system delivers required services when installed in its operational location and staffed by operators. Additionally, the often less formally expressed but sometimes overriding attitudes, experience, and subjective tests that comprise customer satisfaction also need consideration.</li> <li>6) Document the results as required by the V&amp;V system acceptance test plan.</li> </ul> </li> <li>b) System integrity levels 3 and 2 <ul style="list-style-type: none"> <li>Use the acquirer's system acceptance test results to verify that the system satisfies the test acceptance criteria.</li> </ul> </li> <li>c) System integrity level 1 <ul style="list-style-type: none"> <li>There are no system acceptance test execution V&amp;V requirements.</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>Stakeholder requirements</li> <li>System requirements</li> <li>Business or mission analysis documentation</li> <li>System acceptance test procedures</li> <li>System acceptance test results from the developer (integrity levels 3 and 2)</li> </ul>                                                                                                                                    | <ul style="list-style-type: none"> <li>V&amp;V system acceptance test execution results (integrity level 4)</li> <li>Task report(s)—System acceptance test execution V&amp;V (integrity levels 3 and 2)</li> <li>Anomaly report(s)</li> </ul> |

| <b>8.10 Activity: Transition V&amp;V (System, 15288—Transition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                         |                                                                          |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Required inputs</b>                                                                                                                                                                                                                                                                                                  | <b>Required outputs</b>                                                  |
| <b>(4) <a href="#">Installation configuration audit</a></b><br>a) Verify that all products required to correctly install and operate the system are present in the installation package.<br>b) Validate that all site-dependent settings, parameters, or conditions to verify supplied values are correct.                                                                                                                                                                                                                                                                                                                 | Installation package<br>(e.g., user information system design, interface design, system requirements, interface requirements, system drawings, business needs or mission analysis documentation, installation procedures, site-specific settings and parameters, installation tests, and configuration management data) | Task report(s)—<br>Installation configuration audit<br>Anomaly report(s) |
| <b>(5) <a href="#">Installation checkout</a></b><br>a) Conduct analyses or tests to verify that the installed system corresponds to the system subjected to V&V.<br>b) Verify that the system initializes, executes, and terminates as specified.<br>c) In the transition from one version of the system to the next, validate that the system can be replaced with the new version without adversely affecting or degrading the functionality of the remaining system components.<br>d) Verify the requirements for continuous operation and service during the transition, including requirements for user notification. | Installation package                                                                                                                                                                                                                                                                                                    | Task report(s)—<br>Installation checkout<br>Anomaly report(s)            |
| <b>(6) <a href="#">Hazard analysis</a></b><br>a) Verify that the installation procedures and installation environment do not introduce new hazards.<br>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system operations).<br>c) Update the hazard analysis.                                                                                                                                                                                                                                  | Installation package<br>Hazard analysis results                                                                                                                                                                                                                                                                         | Task report(s)—<br>Hazard analysis<br>Anomaly report(s)                  |
| <b>(7) <a href="#">Security analysis</a></b><br>a) Verify that the installed system does not introduce new or increased vulnerabilities or security risks.<br>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system operations).                                                                                                                                                                                                                                               | Installation package<br>Security analysis results                                                                                                                                                                                                                                                                       | Task report(s)—<br>Security analysis<br>Anomaly report(s)                |
| <b>(8) <a href="#">Risk analysis</a></b><br>a) Review and update risk analysis using prior task reports.<br>b) Provide recommendations to eliminate, reduce, or mitigate the risks.                                                                                                                                                                                                                                                                                                                                                                                                                                        | Installation package<br>Supplier development plans and schedules<br>Security analysis results<br>Risk analysis results<br>V&V task results                                                                                                                                                                              | Task report(s)—<br>Risk analysis<br>Anomaly report(s)                    |

| <b>8.11 Validation (System, 15288—Validation Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                              |                                                                                                                |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Required inputs</b>                                                                                       | <b>Required outputs</b>                                                                                        |
| The activities and tasks for the system Validation Process are conducted in system technical life cycle processes and are contained in <a href="#">Table 1b</a> , <a href="#">Activity 8.1</a> (business or mission analysis V&V), <a href="#">Activity 8.2</a> (stakeholder needs and requirements definition V&V), <a href="#">Activity 8.3</a> (system requirements definition V&V), <a href="#">Activity 8.4</a> (architecture definition V&V), <a href="#">Activity 8.5</a> (design definition V&V), <a href="#">Activity 8.6</a> (system analysis V&V), <a href="#">Activity 8.7</a> (implementation V&V), <a href="#">Activity 8.8</a> (integration V&V), <a href="#">Activity 8.10</a> (transition V&V), <a href="#">Activity 8.12</a> (operation V&V), <a href="#">Activity 8.13</a> (maintenance V&V) and <a href="#">Activity 8.14</a> (disposal V&V). The system Validation activities and tasks in <a href="#">Table 1b</a> and the common Validation activities and tasks in <a href="#">Table 1a</a> represent all Validation activities and tasks needed to perform system Validation. | The required inputs for the system Validation Process are found in <a href="#">Table 1b</a> required inputs. | The required outputs for the system Validation Process are found in <a href="#">Table 1b</a> required outputs. |

| <b>8.12 Activity: Operation V&amp;V (System, 15288—Operation Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                |                                                                                                      |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Required inputs</b>                                                                                                                         | <b>Required outputs</b>                                                                              |
| (1) <a href="#">Evaluation of new constraints</a><br>Evaluate new constraints (e.g., operational requirements, platform characteristics, and operating environment) on the system requirements to verify the applicability of the VVP.                                                                                                                                                                                                                                                                                                           | System VVP<br>New constraints                                                                                                                  | Task report(s)—<br>Evaluation of new constraints                                                     |
| (2) <a href="#">Operating procedures evaluation</a><br>a) Verify that the operating procedures are consistent with the user documentation and conform to the system requirements.<br>b) Evaluate new or revised constraints or changes in the environment (e.g., operational requirements, platform characteristics, and operating environment) on the system requirements to verify the applicability of the system VVP (see Annex H).                                                                                                          | Business or mission analysis documentation<br>Operating procedures<br>User documentation<br>System VVP<br>Constraints<br>Environmental changes | Task report(s)—<br>Operating procedures evaluation<br>Anomaly report(s)<br>Updated security analysis |
| (3) <a href="#">Hazard analysis</a><br>a) Verify that the operating procedures (e.g., instructions to the operator or user, standard operating procedures, abnormal operating procedures, safety procedures, and emergency operating procedures) and operational environment do not introduce new hazards.<br>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system operations).<br>c) Update the hazard analysis. | Operating procedures<br>Hazard analysis results                                                                                                | Task report(s)—<br>Hazard analysis<br>Anomaly report(s)                                              |
| (4) <a href="#">Security analysis</a><br>a) Verify that no new security risks are introduced due to changes in the operational environment.<br>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system operations).                                                                                                                                                                                    | New constraints<br>Environmental changes<br>Operating procedures<br>Security analysis results                                                  | Task Reports—<br>Security analysis                                                                   |

| <b>8.12 Activity: Operation V&amp;V (System, 15288—Operation Process)</b>                                                                                                           |                                                                                                                                                                                                                        |                                                       |
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| <b>V&amp;V tasks</b>                                                                                                                                                                | <b>Required inputs</b>                                                                                                                                                                                                 | <b>Required outputs</b>                               |
| c) Over time, changes in external interfaces, threats, or technology in general require that an updated security analysis be performed to determine an updated residual risk.       |                                                                                                                                                                                                                        |                                                       |
| <b>(5) <a href="#">Risk analysis</a></b><br>a) Review and update risk analysis using prior task reports.<br>b) Provide recommendations to eliminate, reduce, or mitigate the risks. | Installation package<br>Proposed changes<br>Hazard analysis results<br>Security analysis results<br>Risk analysis results<br>Supplier development plans and schedules<br>Operation problem reports<br>V&V task results | Task report(s)—<br>Risk Analysis<br>Anomaly report(s) |

| <b>8.13 Activity: Maintenance V&amp;V (System, 15288—Maintenance Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                   |                                                                                                               |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Required inputs</b>                                                                                            | <b>Required outputs</b>                                                                                       |
| <b>(1) <a href="#">System maintenance strategy assessment</a></b><br>a) Verify the system maintenance strategy.<br>b) Verify that any constraints specified or implied by the system maintenance strategy are included in the system requirements and design.                                                                                                                                                                                                                                                                                                                                                                                                                                                         | System maintenance strategy<br>System performance data<br>System requirements                                     | Task report(s)—<br>System maintenance strategy analysis<br>Corrective actions assessment<br>Anomaly report(s) |
| <b>(2) <a href="#">System maintenance execution assessment</a></b><br>a) Verify the maintenance operational strategy results by monitoring the system maintenance execution.<br>b) Evaluate the maintenance operational strategy results.<br>c) Stakeholder requirements evaluation.<br>d) Requirements.<br>e) Traceability analysis.<br>f) Criticality analysis.<br>g) Hazard analysis.<br>h) Security analysis.<br>i) Risk analysis.<br>j) Interface analysis.<br>k) Design evaluation.<br>l) System element implementation analysis.<br>m) Integration analysis.<br>n) System test analysis (integration, qualification, and acceptance).<br>o) System installation analysis.<br>p) Operation transition analysis. | System maintenance strategy (e.g., operational availability requirements, system failure, suspension of services) | Task report(s)—<br>System maintenance execution assessment and quality metrics<br>Anomaly report(s)           |
| NOTE—System changes are maintenance activities (see <a href="#">subclause 8.13</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                   |                                                                                                               |

| <b>8.14 Activity: Disposal V&amp;V (System, 15288 – Disposal Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                              |                                                                                                                                                                  |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Required inputs</b>                                                                                                                                                                                                                                       | <b>Required outputs</b>                                                                                                                                          |
| <p><b>(1) <u>Disposal strategy evaluation</u></b></p> <p>a) Verify that system boundaries are defined and all relevant system elements (information, hardware, media, etc.) are identified.</p> <p>b) Verify that disposal has been planned to a degree commensurate with the complexity and risk of the disposal and all that identified elements are accounted for in the plan.</p> <p>c) Verify that environmental considerations, all applicable laws, regulations, organizational policies, and procedures as well as other constraints, are identified and communicated to relevant stakeholders.</p> <p>d) Verify that system elements to be stored have available billets and established retention criteria.</p> | <p>Business or mission analysis<br/>documentation<br/>Configuration repository<br/>Inventory records<br/>Disposal strategy<br/>Applicable laws<br/>Organizational policies and procedures<br/>List of constraints<br/>Management and technical decisions</p> | <p>Task report(s)—<br/>System description<br/>Task report(s)—<br/>Constraints<br/>Task report(s)—<br/>Disposal Strategy<br/>Evaluation<br/>Anomaly report(s)</p> |
| NOTE (for Table 1b)—Other inputs may be used. For any V&V activity and task, all of the required inputs and outputs from preceding activities and tasks may be used, but for conciseness, only the primary inputs are listed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                              |                                                                                                                                                                  |

**Table 2b—Minimum V&V tasks assigned to each integrity level for system V&V**

| V&V Processes                                         | Business or Mission Analysis V&V (see 8.1) |   |   |   | Stakeholder Needs and Requirements Definition V&V (see 8.2) |   |   |   | System Requirements Definition (see 8.3) |   |   |   | Architecture Definition V&V (see 8.4) |   |   |   | Design Definition V&V (see 8.5) |   |   |   | System Analysis V&V (see 8.6) |   |   |   | Implementation V&V (see 8.7) |   |   |   | Integration V&V (see 8.8) |   |   |   | Transition V&V (see 8.10) |   |   |   | Operation V&V (see 8.12) |   |   |   | Maintenance V&V (see 8.13) |   |   |   | Disposal V&V (see 8.14) |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|-------------------------------------------------------|--------------------------------------------|---|---|---|-------------------------------------------------------------|---|---|---|------------------------------------------|---|---|---|---------------------------------------|---|---|---|---------------------------------|---|---|---|-------------------------------|---|---|---|------------------------------|---|---|---|---------------------------|---|---|---|---------------------------|---|---|---|--------------------------|---|---|---|----------------------------|---|---|---|-------------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| Integrity Levels                                      | Levels                                     |   |   |   | Levels                                                      |   |   |   | Levels                                   |   |   |   | Levels                                |   |   |   | Levels                          |   |   |   | Levels                        |   |   |   | Levels                       |   |   |   | Levels                    |   |   |   | Levels                    |   |   |   | Levels                   |   |   |   | Levels                     |   |   |   | Levels                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                       | 4                                          | 3 | 2 | 1 | 4                                                           | 3 | 2 | 1 | 4                                        | 3 | 2 | 1 | 4                                     | 3 | 2 | 1 | 4                               | 3 | 2 | 1 | 4                             | 3 | 2 | 1 | 4                            | 3 | 2 | 1 | 4                         | 3 | 2 | 1 | 4                         | 3 | 2 | 1 | 4                        | 3 | 2 | 1 | 4                          | 3 | 2 | 1 |                         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Architecture Evaluation                               |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   | X                                     | X | X | X |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Business or Mission Analysis Documentation Evaluation | X                                          | X | X | X |                                                             |   |   |   |                                          |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Criticality Analysis                                  | X                                          | X | X | X | X                                                           | X | X | X | X                                        | X | X | X | X                                     | X | X |   | X                               | X | X |   |                               |   |   | X | X                            | X |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Design Evaluation                                     |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   | X | X                               | X | X |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Disposal Strategy Evaluation                          |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

| V&V Processes                          | Business or Mission Analysis V&V (see 8.1) |   |   |   | Stakeholder Needs and Requirements Definition V&V (see 8.2) |   |   |   | System Requirements Definition (see 8.3) |   |   |   | Architecture Definition V&V (see 8.4) |   |   |   | Design Definition V&V (see 8.5) |   |   |   | System Analysis V&V (see 8.6) |   |   |   | Implementation V&V (see 8.7) |   |   |   | Integration V&V (see 8.8) |   |   |   | Transition V&V (see 8.10) |   |   |   | Operation V&V (see 8.12) |   |   |   | Maintenance V&V (see 8.13) |   |   |   | Disposal V&V (see 8.14) |  |  |  |
|----------------------------------------|--------------------------------------------|---|---|---|-------------------------------------------------------------|---|---|---|------------------------------------------|---|---|---|---------------------------------------|---|---|---|---------------------------------|---|---|---|-------------------------------|---|---|---|------------------------------|---|---|---|---------------------------|---|---|---|---------------------------|---|---|---|--------------------------|---|---|---|----------------------------|---|---|---|-------------------------|--|--|--|
| Integrity Levels                       | Levels                                     |   |   |   | Levels                                                      |   |   |   | Levels                                   |   |   |   | Levels                                |   |   |   | Levels                          |   |   |   | Levels                        |   |   |   | Levels                       |   |   |   | Levels                    |   |   |   | Levels                    |   |   |   | Levels                   |   |   |   | Levels                     |   |   |   | Levels                  |  |  |  |
|                                        | 4                                          | 3 | 2 | 1 | 4                                                           | 3 | 2 | 1 | 4                                        | 3 | 2 | 1 | 4                                     | 3 | 2 | 1 | 4                               | 3 | 2 | 1 | 4                             | 3 | 2 | 1 | 4                            | 3 | 2 | 1 | 4                         | 3 | 2 | 1 | 4                         | 3 | 2 | 1 | 4                        | 3 | 2 | 1 | 4                          | 3 | 2 | 1 |                         |  |  |  |
| Risk Analysis                          | X                                          | X |   |   | X                                                           | X |   |   | X                                        | X |   |   | X                                     | X |   |   |                                 |   |   | X | X                             |   |   | X | X                            |   |   |   |                           | X | X |   |                           | X | X |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |
| Security Analysis                      | X                                          | X |   |   | X                                                           | X |   |   | X                                        | X |   |   | X                                     | X |   |   |                                 |   |   | X | X                             |   |   | X | X                            |   |   |   |                           | X | X |   |                           | X | X |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |
| Stakeholder Requirements Evaluation    |                                            |   |   |   | X                                                           | X | X | X |                                          |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |
| System Acceptance Test Case V&V        |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   | X | X                               | X |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |
| System Acceptance Test Design V&V      |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   | X                                     | X | X |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |
| System Acceptance Test Execution V&V   |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   |                              |   | X | X | X                         |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |
| System Acceptance Test Plan V&V        |                                            |   |   |   |                                                             |   |   |   | X                                        | X | X |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |
| System Acceptance Test Procedure V&V   |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   |   |                                 |   | X | X | X                             |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |
| System Analysis Strategy Evaluation    |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   |   |                                 |   | X | X |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |
| System Analysis Results Evaluation     |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   |   |                                 |   | X | X |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |
| System Element Implementation Analysis |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   |   |                                 |   |   |   |                               | X | X | X |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |
| System Element Interaction Analysis    |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   |   |                                 |   |   |   |                               | X | X | X |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |
| System Integration Strategy Assessment |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   |                              | X | X | X |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |
| System Integration Test Case V&V       |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   | X | X                               | X |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |
| System Integration Test Design V&V     |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   | X | X | X                               |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |
| System Integration Test Execution V&V  |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   | X                            | X | X |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |

| V&V Processes                           | Business or Mission Analysis V&V (see 8.1) |   |   |   | Stakeholder Needs and Requirements Definition V&V (see 8.2) |   |   |   | System Requirements Definition (see 8.3) |   |   |   | Architecture Definition V&V (see 8.4) |   |   |   | Design Definition V&V (see 8.5) |   |   |   | System Analysis V&V (see 8.6) |   |   |   | Implementation V&V (see 8.7) |   |   |   | Integration V&V (see 8.8) |   |   |   | Transition V&V (see 8.10) |   |   |   | Operation V&V (see 8.12) |   |   |   | Maintenance V&V (see 8.13) |   |   |   | Disposal V&V (see 8.14) |   |  |  |  |  |
|-----------------------------------------|--------------------------------------------|---|---|---|-------------------------------------------------------------|---|---|---|------------------------------------------|---|---|---|---------------------------------------|---|---|---|---------------------------------|---|---|---|-------------------------------|---|---|---|------------------------------|---|---|---|---------------------------|---|---|---|---------------------------|---|---|---|--------------------------|---|---|---|----------------------------|---|---|---|-------------------------|---|--|--|--|--|
| Integrity Levels                        | Levels                                     |   |   |   | Levels                                                      |   |   |   | Levels                                   |   |   |   | Levels                                |   |   |   | Levels                          |   |   |   | Levels                        |   |   |   | Levels                       |   |   |   | Levels                    |   |   |   | Levels                    |   |   |   | Levels                   |   |   |   | Levels                     |   |   |   | Levels                  |   |  |  |  |  |
|                                         | 4                                          | 3 | 2 | 1 | 4                                                           | 3 | 2 | 1 | 4                                        | 3 | 2 | 1 | 4                                     | 3 | 2 | 1 | 4                               | 3 | 2 | 1 | 4                             | 3 | 2 | 1 | 4                            | 3 | 2 | 1 | 4                         | 3 | 2 | 1 | 4                         | 3 | 2 | 1 | 4                        | 3 | 2 | 1 | 4                          | 3 | 2 | 1 |                         |   |  |  |  |  |
| System Integration Test Plan V&V        |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   | X                                     | X | X |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |   |  |  |  |  |
| System Integration Test Procedure V&V   |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   |   |                                 |   |   |   | X                             | X | X |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |   |  |  |  |  |
| System Maintenance Execution Assessment |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   | X | X                       | X |  |  |  |  |
| System Maintenance Strategy Assessment  |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   | X | X                       | X |  |  |  |  |
| System Qualification Test Case V&V      |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   |   | X                               | X | X |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |   |  |  |  |  |
| System Qualification Test Design V&V    |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   | X                                     | X | X |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |   |  |  |  |  |
| System Qualification Test Execution V&V |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   | X | X                            | X |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |   |  |  |  |  |
| System Qualification Test Plan V&V      |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   | X                                     | X | X |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |   |  |  |  |  |
| System Qualification Test Procedure V&V |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |   |  |  |  |  |
| Traceability Analysis                   | X                                          | X | X |   | X                                                           | X | X |   | X                                        | X | X |   | X                                     | X | X |   |                                 |   |   |   |                               |   |   |   |                              |   |   | X | X                         | X |   | X | X                         | X |   |   |                          |   |   |   |                            |   |   |   |                         |   |  |  |  |  |
| Transition Demonstration Assessment     |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |   |  |  |  |  |
| Transition Strategy Evaluation          |                                            |   |   |   |                                                             |   |   |   |                                          |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |   |  |  |  |  |

NOTE (for Table 2b)—Whenever a V&V task is selected as a mandatory requirement for multiple integrity levels, the V&V task implementation is dictated by the rigor, intensity, and depth of analysis or test. Higher integrity level implementation requires greater rigor (e.g., formal methods and structured analysis methods), intensity (e.g., consideration of all system conditions and system environment states), and depth (e.g., abnormal cases, boundary conditions, and comprehensive fault and recovery scenarios) of analysis or test than the lower integrity level implementation.



The recommended applicability of optional tasks to the system V&V processes described in Clause 8 is shown in Table 3b. [Annex G](#) provides a description of each of the optional V&V tasks.

**Table 3b—Optional V&V tasks and suggested applications in system technical processes**

|                                                      | <a href="#">Business or mission analysis (8.1)</a> | <a href="#">Stakeholder needs and requirements definition (8.2)</a> | <a href="#">System requirements definition (8.3)</a> | <a href="#">architecture definition (8.4)</a> | <a href="#">Design definition (8.5)</a> | <a href="#">System analysis (8.6)</a> | <a href="#">Implementation (8.7)</a> | <a href="#">Integration (8.8)</a> | <a href="#">Transition (8.10)</a> | <a href="#">Operation (8.12)</a> | <a href="#">Maintenance (8.13)</a> | <a href="#">Disposal (8.14)</a> |
|------------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------|-----------------------------------------|---------------------------------------|--------------------------------------|-----------------------------------|-----------------------------------|----------------------------------|------------------------------------|---------------------------------|
| <a href="#">Algorithm analysis</a>                   | —                                                  | —                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Audit performance</a>                    | —                                                  | —                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | —                                | X                                  | —                               |
| <a href="#">Audit support</a>                        | —                                                  | —                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | —                                | X                                  | —                               |
| <a href="#">Control flow analysis</a>                | —                                                  | —                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Cost analysis</a>                        | —                                                  | X                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | —                                | X                                  | —                               |
| <a href="#">Database analysis</a>                    | —                                                  | —                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | —                                | X                                  | —                               |
| <a href="#">Data flow analysis</a>                   | —                                                  | —                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Disaster recovery plan assessment</a>    | —                                                  | X                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | —                                 | —                                 | X                                | X                                  | X                               |
| <a href="#">Distributed architecture assessment</a>  | —                                                  | X                                                                   | X                                                    | X                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Exploratory testing</a>                  | X                                                  | X                                                                   | X                                                    | X                                             | X                                       | X                                     | X                                    | X                                 | X                                 | X                                | X                                  | —                               |
| <a href="#">Feasibility study evaluation</a>         | —                                                  | X                                                                   | X                                                    | X                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Independent risk assessment</a>          | —                                                  | —                                                                   | —                                                    | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Inspection</a>                           | —                                                  | —                                                                   | —                                                    | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | —                                  | —                               |
| <a href="#">Inspection—Concept</a>                   | —                                                  | X                                                                   | —                                                    | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Inspection—Requirements</a>              | —                                                  | —                                                                   | X                                                    | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Inspection—Design</a>                    | —                                                  | —                                                                   | —                                                    | X                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Inspection—Source code</a>               | —                                                  | —                                                                   | —                                                    | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | —                                  | —                               |
| <a href="#">Inspection—Test plan</a>                 | —                                                  | —                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | —                                | X                                  | —                               |
| <a href="#">Inspection—Test design</a>               | —                                                  | —                                                                   | —                                                    | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | —                                | X                                  | —                               |
| <a href="#">Inspection—Test case</a>                 | —                                                  | —                                                                   | —                                                    | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | —                                | X                                  | —                               |
| <a href="#">Operational evaluation</a>               | —                                                  | —                                                                   | —                                                    | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | X                                | —                                  | —                               |
| <a href="#">Performance monitoring</a>               | —                                                  | X                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | X                               |
| <a href="#">Post-installation Validation</a>         | —                                                  | —                                                                   | —                                                    | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | X                                | X                                  | —                               |
| <a href="#">Project management oversight support</a> | —                                                  | X                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | X                               |
| <a href="#">Proposal evaluation support</a>          | —                                                  | —                                                                   | —                                                    | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | —                                  | —                               |
| <a href="#">Regression analysis and testing</a>      | —                                                  | —                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | —                                | X                                  | —                               |
| <a href="#">Reusability analysis</a>                 | —                                                  | X                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Reuse analysis</a>                       | —                                                  | X                                                                   | X                                                    | X                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Simulation analysis</a>                  | —                                                  | X                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | X                               |
| <a href="#">Sizing and timing analysis</a>           | —                                                  | —                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | —                                | X                                  | —                               |
| <a href="#">System software assessment</a>           | —                                                  | —                                                                   | —                                                    | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | —                               |
| <a href="#">Test certification</a>                   | —                                                  | —                                                                   | —                                                    | —                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | X                               |
| <a href="#">Test evaluation</a>                      | —                                                  | —                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | X                               |
| <a href="#">Test witnessing</a>                      | —                                                  | —                                                                   | —                                                    | —                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | X                               |
| <a href="#">Training documentation evaluation</a>    | —                                                  | —                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | X                               |
| <a href="#">Usability analysis</a>                   | —                                                  | X                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | —                               |
| <a href="#">User documentation evaluation</a>        | —                                                  | X                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | —                               |
| <a href="#">User training</a>                        | —                                                  | —                                                                   | —                                                    | —                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | —                               |
| <a href="#">V&amp;V tool plan generation</a>         | —                                                  | —                                                                   | —                                                    | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | —                                  | —                               |
| <a href="#">V&amp;V tool qualification</a>           | —                                                  | X                                                                   | X                                                    | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | —                               |
| <a href="#">Walkthrough</a>                          | —                                                  | —                                                                   | —                                                    | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | —                                  | —                               |

|                                                           | <a href="#">Business or mission analysis (8.1)</a> | <a href="#">Stakeholder needs and requirements definition (8.2)</a> | <a href="#">System requirements definition (8.3)</a> | <a href="#">architecture definition (8.4)</a> | <a href="#">Design definition (8.5)</a> | <a href="#">System analysis (8.6)</a> | <a href="#">Implementation (8.7)</a> | <a href="#">Integration (8.8)</a> | <a href="#">Transition (8.10)</a> | <a href="#">Operation (8.12)</a> | <a href="#">Maintenance (8.13)</a> | <a href="#">Disposal (8.14)</a> |
|-----------------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------|-----------------------------------------|---------------------------------------|--------------------------------------|-----------------------------------|-----------------------------------|----------------------------------|------------------------------------|---------------------------------|
| <a href="#">Walkthrough—Design</a>                        | —                                                  | —                                                                   | —                                                    | X                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Walkthrough—Requirements</a>                  | —                                                  | —                                                                   | X                                                    | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Walkthrough—Source code</a>                   | —                                                  | —                                                                   | —                                                    | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | —                                  | —                               |
| <a href="#">Walkthrough—Test</a>                          | —                                                  | —                                                                   | —                                                    | —                                             | —                                       | —                                     | X                                    | X                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Work breakdown structure (WBS) evaluation</a> | —                                                  | —                                                                   | —                                                    | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | —                                  | —                               |

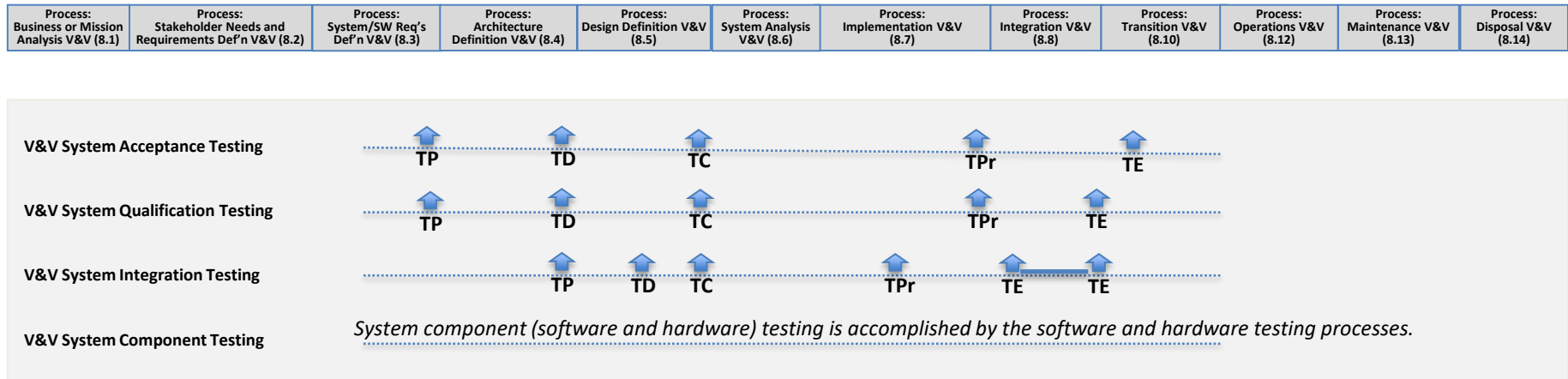
IEEE Std 1012-2024  
IEEE Standard for System, Software and Hardware Verification and Validation

| ISO/IEC/IEEE 15288 System Life Cycle Processes                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                      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| Process:<br>Business or Mission<br>Analysis                                                                                                                      | Process:<br>Stakeholder Needs and<br>Requirements Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Process:<br>System Requirements<br>Def'n                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Process:<br>Architecture<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Process:<br>Design Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Process:<br>System<br>Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Process:<br>Implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Process:<br>Integration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Process:<br>Transition                                                                                                                                                                                                                                                                                                                                                                                                                                              | Process:<br>Operations                                                                                                                                                                                                                                                                                                                                                                                                                             | Process:<br>Maintenance                                                                                                                                                                                                                                                                             | Process:<br>Disposal                                                                                                                                                                                                                                               |
| V&V Inputs                                                                                                                                                       | 1) Business or mission analysis documentation<br>2) Hazard analysis results<br>3) Identified candidate solutions<br>4) Identified problem or opportunity space<br>5) Identified solution space<br>6) New market or mission elements<br>7) Organization strategy<br>8) Organizational concept of operations<br>9) Organizational processes and procedures<br>10) Organizational strategic goals and plans<br>11) Preferred candidate solution(s)<br>12) Preliminary threat and risk assessment (TRA)<br>13) Security analysis results<br>14) V&V task results | 1) Business or mission analysis documentation<br>2) Criticality task report<br>3) Hazard analysis results<br>4) Organizational processes and procedures<br>5) Preliminary threat and risk assessment (TRA)<br>6) Security analysis results<br>7) Stakeholders requirements<br>8) Supplier development plans and schedules<br>9) System acceptance test plan (acquirer's)<br>10) System interface requirements<br>11) System qualification test plan V&V (developer's)<br>12) System requirements<br>13) V&V task results | 1) Architecture models and views<br>2) Architecture tradeoff analyses<br>3) Business or mission analysis documentation<br>4) Criticality task report<br>5) Hazard analysis results<br>6) Organizational processes and procedures<br>7) Preliminary threat and risk assessment (TRA)<br>8) Risk analysis results<br>9) Security analysis results<br>10) Stakeholders requirements<br>11) Supplier development plans and schedules<br>12) System acceptance test plan V&V<br>13) System qualification test plan V&V<br>14) System requirements<br>15) User needs<br>16) V&V task results | 1) Architecture models and views<br>2) Architecture tradeoff analyses<br>3) Business or mission analysis documentation<br>4) Criticality task report<br>5) Design documents<br>6) Hazard analysis results<br>7) Organizational processes and procedures<br>8) Preliminary threat and risk assessment (TRA)<br>9) Risk analysis results<br>10) Security analysis results<br>11) Subsystems security analysis<br>12) Supplier development plans and schedules<br>13) System acceptance test design V&V<br>14) System integration test design V&V<br>15) System integration test design V&V<br>16) System qualification test design V&V<br>17) System requirements<br>18) V&V task results | 1) System analysis strategy<br>2) System analysis results<br>3) Business or mission analysis results<br>4) Hardware drawings<br>5) Hardware requirements specification(s)<br>6) HW traceability analysis<br>7) Hazard analysis results<br>8) Implementation procedures<br>9) Implementation strategy<br>10) Risk analysis results<br>11) Security analysis results<br>12) Software design docs<br>13) Software requirements specification(s)<br>14) SW traceability analysis<br>15) Subsystems<br>16) System accept test cases V&V<br>17) System architecture<br>18) System design documents<br>19) System element description<br>20) System implementation analysis results<br>21) System element requirements, designs, and implementation<br>22) System element(s)<br>23) System integration test cases V&V<br>24) System qual test cases V&V<br>25) System requirements<br>26) V&V task results | 1) Business or mission analysis documentation<br>2) Criticality task report<br>3) Fabrication processes, tools, and equipment<br>4) Hardware drawings<br>5) Hardware requirements specification(s)<br>6) HW traceability analysis<br>7) Hazard analysis results<br>8) Implementation procedures<br>9) Implementation strategy<br>10) Risk analysis results<br>11) Security analysis results<br>12) Software design docs<br>13) Software requirements specification(s)<br>14) SW traceability analysis<br>15) Subsystems<br>16) System accept test cases V&V<br>17) System architecture<br>18) System design documents<br>19) System element description<br>20) System implementation analysis results<br>21) System element requirements, designs, and implementation<br>22) System element(s)<br>23) System integration test cases V&V<br>24) System qual test cases V&V<br>25) System requirements<br>26) V&V task results | 1) All V&V test documentation<br>2) Business or mission analysis documentation<br>3) External documentation<br>4) HW elements<br>5) Hazard analysis results<br>6) Integration sequence and strategy<br>7) Security analysis results<br>8) SW elements<br>9) Stakeholders requirements<br>10) System architecture<br>11) System design<br>12) System integration strategy<br>13) System integration test procedures<br>14) System interaction results<br>15) System qual test procedure V&V<br>16) System requirements | 1) Acceptance requirements<br>2) Acquirer agreement<br>3) Business or mission analysis<br>4) Hazard analysis results<br>5) Installation package<br>6) New constraints<br>7) Operating procedures<br>8) Operation problem reports<br>9) Proposed changes<br>10) Security analysis results<br>11) Security analysis results<br>12) Supplier development plans and schedule<br>13) System acceptance results<br>14) Transitioning system items<br>15) V&V task results | 1) Business or mission analysis documentation<br>2) Constraints<br>3) Environmental changes<br>4) Hazard analysis results<br>5) Installation package<br>6) New constraints<br>7) Operating procedures<br>8) Operation problem reports<br>9) Proposed changes<br>10) Security analysis results<br>11) Security analysis results<br>12) Supplier development plans and schedules<br>13) System VVP<br>14) User documentation<br>15) V&V task results | 1) System maintenance strategy<br>2) System performance data<br>3) System requirements<br>4) Hazard analysis results<br>5) Inventory records<br>6) List of constraints<br>7) Management and technical decisions<br>8) Organizational policies and procedures                                        | 1) Applicable laws<br>2) Business or mission analysis documentation<br>3) Configuration repository<br>4) Disposal strategy<br>5) Inventory records<br>6) List of constraints<br>7) Management and technical decisions<br>8) Organizational policies and procedures |
| IEEE Std 1012 System V&V Tasks (All tasks are minimum required for integrity level 4)                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                           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                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                    |
| Process:<br>Business or Mission<br>Analysis V&V (8.1)                                                                                                            | Process:<br>Stakeholder Needs and<br>Requirements Def'n V&V (8.2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Process:<br>System/SW Req's<br>Def'n V&V (8.3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Process:<br>Architecture<br>Definition V&V (8.4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Process:<br>Design Definition V&V<br>(8.5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Process:<br>System Analysis<br>V&V (8.6)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Process:<br>Implementation V&V<br>(8.7)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Process:<br>Integration V&V<br>(8.8)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Process:<br>Transition V&V<br>(8.10)                                                                                                                                                                                                                                                                                                                                                                                                                                | Process:<br>Operations V&V<br>(8.12)                                                                                                                                                                                                                                                                                                                                                                                                               | Process:<br>Maintenance V&V<br>(8.13)                                                                                                                                                                                                                                                               | Process:<br>Disposal V&V<br>(8.14)                                                                                                                                                                                                                                 |
| 1) Business or Mission Analysis Results<br>2) Traceability Analysis<br>3) Criticality Analysis<br>4) Hazard Analysis<br>5) Security Analysis<br>6) Risk Analysis | 1) Stakeholder Needs and Requirements Evaluation<br>2) Traceability Analysis<br>3) Criticality Analysis<br>4) Hazard Analysis<br>5) Security Analysis<br>6) Risk Analysis                                                                                                                                                                                                                                                                                                                                                                                    | 1) Requirements Evaluation<br>2) Interface Analysis<br>3) Traceability Analysis<br>4) Criticality Analysis<br>5) System Qualification Test Plan V&V<br>6) System Acceptance Test Plan V&V<br>7) Hazard Analysis<br>8) Security Analysis<br>9) Risk Analysis                                                                                                                                                                                                                                                              | 1) Architecture Evaluation<br>2) Interface Analysis<br>3) Req Allocation Analysis<br>4) Traceability Analysis<br>5) Criticality Analysis<br>6) System Int Test Plan V&V<br>7) System Qual Test Design V&V<br>8) System Accept Test Design V&V<br>9) Hazard Analysis<br>10) Security Analysis<br>11) Risk Analysis                                                                                                                                                                                                                                                                      | 1) Design Evaluation<br>2) Interface Analysis<br>3) Traceability Analysis<br>4) Criticality Analysis<br>5) System Integration Test Design V&V<br>6) System Integration Test Case V&V<br>7) System Qual Test Case V&V<br>8) Hazard Analysis<br>9) Security Analysis<br>10) Risk Analysis                                                                                                                                                                                                                                                                                                                                                                                                 | 1) System Analysis Strategy Evaluation<br>2) System Analysis Results Evaluation<br>3) System Element Interaction Analysis<br>4) Criticality Analysis<br>5) System Int Test Procedure V&V<br>6) System Qual Test Procedure V&V<br>7) System Accept Test Procedure V&V<br>8) Hazard Analysis<br>9) Security Analysis<br>10) Risk Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1) Implementation strategy assessment<br>2) System Element Implementation Analysis<br>3) System Element Interaction Analysis<br>4) Criticality Analysis<br>5) System Int Test Procedure V&V<br>6) System Qual Test Procedure V&V<br>7) System Accept Test Procedure V&V<br>8) Hazard Analysis<br>9) Security Analysis<br>10) Risk Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1) System Integration Strategy Assessment<br>2) System Element Interaction Analysis<br>3) System Int Test Execution V&V<br>4) System Qual Test Execution V&V<br>5) Traceability Analysis<br>6) Hazard Analysis<br>7) Security Analysis<br>8) Risk Analysis                                                                                                                                                                                                                                                            | 1) Transition Strategy Evaluation<br>2) Transition Demonstration Assessment<br>3) System Accept Test Execution V&V<br>4) Installation Configuration Audit<br>5) Installation Checkout<br>6) Hazard Analysis<br>7) Security Analysis<br>8) Risk Analysis                                                                                                                                                                                                             | 1) Evaluation of New Constraints<br>2) Operating Procedures Evaluation<br>3) Hazard Analysis<br>4) Security Analysis<br>5) Risk Analysis                                                                                                                                                                                                                                                                                                           | 1) System Maintenance Strategy Assessment<br>2) System Maintenance Execution Assessment<br>• VVP Revision<br>• Anomaly Evaluation<br>• Criticality Analysis<br>• Migration Assessment<br>• Retirement Assessment<br>• Hazard Analysis<br>• Security Analysis<br>• Risk Analysis<br>• Task Iteration | 1) Disposal Strategy Evaluation                                                                                                                                                                                                                                    |
| V&V Outputs                                                                                                                                                      | 1) Anomaly report(s)<br>2) Task report(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1) Anomaly report(s)<br>2) Task report(s)<br>3) V&V System test plan(s)<br>- Acceptance<br>- Qualification                                                                                                                                                                                                                                                                                                                                                                                                               | 1) Anomaly report(s)<br>2) Task report(s)<br>3) V&V System test plan (P)/design (D)<br>- Acceptance (D)<br>- Integration (P)<br>- Qualification (D)                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1) Anomaly report(s)<br>2) Task report(s)<br>3) V&V System test design (D)/cases (C)<br>- Acceptance (D)<br>- Integration (C)<br>- Qualification (C)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1) Task report(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1) Anomaly report(s)<br>2) Task report(s)<br>3) V&V System test procedures (Pr)<br>- Acceptance (Pr)<br>- Qualification (Pr)<br>- Integration (Pr)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1) Anomaly report(s)<br>2) Task report(s)<br>3) V&V System test execution results<br>- Qualification<br>- Integration                                                                                                                                                                                                                                                                                                                                                                                                 | 1) Anomaly report(s)<br>2) Task report(s)<br>3) V&V System test execution results<br>- Acceptance                                                                                                                                                                                                                                                                                                                                                                   | 1) Anomaly report(s)<br>2) Task report(s)                                                                                                                                                                                                                                                                                                                                                                                                          | 1) Anomaly report(s)<br>2) Task report(s)<br>3) Updated VVP                                                                                                                                                                                                                                         | 1) Anomaly report(s)<br>2) Task report(s)                                                                                                                                                                                                                          |

Figure 1b—Summary of system V&V activities and tasks

IEEE Std 1012-2024  
IEEE Standard for System, Software and Hardware Verification and Validation

IEEE Std 1012 System V&V Activities



**Note:**

TP = Test Plan  
TD = Test Design  
TC = Test Cases  
TPr = Test Procedures  
TE = Test Execution

NOTE 1—All V&V system test products and tasks represent the activities and products required as a minimum for integrity level 4.

NOTE 2—This is an example of the phasing of system V&V test products and tasks across the system life cycle. The system V&V test products (upward arrows) are shown in the system life cycle processes when the products are generated (per IEEE Std 1012). System test execution tasks are shown to occur during one or more system life cycle processes indicated by “activity bars” in the diagram. The life cycle process (in which each test product is generated) and phasing of each test product and task can vary from this diagram in accordance with project-specific needs.

NOTE 3—The V&V activity clauses referenced in the system V&V life cycle stages are IEEE Std 1012 clauses.

**Figure 2b—Summary of system V&V test products and tasks**

## 9. Software V&V processes

### 9.1 Business or Mission Analysis V&V Process

#### 9.1.1 Purpose

The purpose of the Business or Mission Analysis V&V Process is to provide assurance that the outcomes of the Business or Mission Analysis Process (ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)) have been achieved.

#### 9.1.2 Outcomes

As a result of the successful implementation of the Business or Mission Analysis V&V Process, objective evidence is developed to assess whether:

- a) The problem or opportunity space adheres to the organizational strategy.
- b) The solution space is consistent with the problem or opportunity space.
- c) Preliminary life-cycle concepts are consistent with the solution space.
- d) The preferred candidate solution alternative(s) are selected.
- e) Any enabling systems or services needed for business or mission analysis are available.
- f) The root of traceability is established, starting with the organization strategy and tracing the problem or opportunity space to the solution space to the candidate solutions.

#### 9.1.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2c](#) for the selected integrity level, the following Business or Mission Analysis V&V activity and tasks described in [Table 1c, Activity 9.1](#):

- a) [Business or mission analysis V&V](#). This activity consists of the following tasks:
  - 1) [Business or mission analysis documentation evaluation](#)
  - 2) [Requirements allocation analysis](#)
  - 3) [Traceability analysis](#)
  - 4) [Criticality analysis](#)
  - 5) [Hazard analysis](#)
  - 6) [Security analysis](#)
  - 7) [Risk analysis](#)

The Business or Mission Analysis V&V tasks listed shall be used to verify and validate the Business or Mission Analysis Process outcomes described in ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#).

The primary focus is on software with consideration of the interactions with hardware and user allocations to verify the allocation of system requirements, validate the selected solution, and assure that no false assumptions have been incorporated in the solution.

During the Business or Mission Analysis Process, different software concepts are investigated, and trade studies are conducted on each concept before a final concept is selected. These trade studies may involve assessing the performance features of each concept, estimating the cost of development, determining the performance efficiency, identifying technology risks, and estimating the schedule to develop the software. Software models and prototypes may be constructed to conduct these trade studies, in conjunction with simulations and analytic analyses. The business or mission analysis stage may provide preliminary software models and prototypes to the system business or mission analysis stage to support trade studies for system concept definition. In such cases, the software business or mission analysis stage may start before system/software requirements are allocated.

## 9.2 Stakeholder Needs and Requirements Definition V&V Process

### 9.2.1 Purpose

The purpose of the Stakeholder Needs and Requirements Definition V&V Process is to provide assurance that outcomes of the Stakeholder Needs and Requirements Definition Process (ISO/IEC/IEEE 12207:2017(E) [B40]) have been achieved.

### 9.2.2 Outcomes

As a result of the successful implementation of the Stakeholder Needs and Requirements Definition V&V Process, objective evidence is developed to assess whether the stakeholder requirements:

- a) Specify the required characteristics and context of the use of services and operational concepts.
- b) Define all constraints on a software solution.
- c) Are traceable to the originating stakeholders.
- d) Are complete, unambiguous, correct, and accurate.
- e) Can be validated by measurable analyses or tests.

### 9.2.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2c](#) for the selected integrity level, the following Stakeholder Needs and Requirements Definition V&V activity and tasks described in [Table 1c, Activity 9.2](#):

- a) [Stakeholder needs and requirements definition V&V](#). This activity consists of the following tasks:
  - 1) [Stakeholder needs and requirements evaluation](#)
  - 2) [Traceability analysis](#)
  - 3) [Criticality analysis](#)
  - 4) [Hazard analysis](#)
  - 5) [Security analysis](#)
  - 6) [Risk analysis](#)

The Stakeholder Needs and Requirements Definition V&V tasks listed shall be used to verify and validate the Stakeholder Needs and Requirements Definition Process outcomes described in ISO/IEC/IEEE 12207:2017(E) [B40]. These tasks focus on needs and requirements related to software, with consideration of the interactions with hardware and user allocations, to verify that the identified needs and requirements are consistent and sufficient for the purpose of the system.

## 9.3 System/Software Requirements Definition V&V Process

### 9.3.1 Purpose

The purpose of the System/Software Requirements Definition V&V Process is to provide assurance that outcomes of the System/Software Requirements Definition Process (ISO/IEC/IEEE 12207:2017(E) [B40]) have been achieved.

### 9.3.2 Outcomes

As a result of the successful implementation of the System/Software Requirements Definition V&V Process, objective evidence is developed to assess whether the system/software requirements:

- a) Specify all required characteristics, attributes, functional and performance requirements, interface requirements, and requirements for qualification, safety and security, human factors engineering, and user documentation for the software.

- b) Specify all constraints that will affect the architecture of the software and the means to realize the software.
- c) Are unique, complete, unambiguous, consistent with all other requirements, implementable, and verifiable.
- d) Are traceable to the stakeholder requirements.
- e) Provide a basis (through analysis or test planning) for verifying that each software requirement can be satisfied.

### 9.3.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2c](#) for the selected integrity level, the following System/Software Requirements Definition V&V activity and tasks described in [Table 1c, Activity 9.3](#):

- a) [System/software requirements definition V&V](#): This activity consists of the following tasks:
  - 1) [Requirements evaluation](#)
  - 2) [Interface analysis](#)
  - 3) [Traceability analysis](#)
  - 4) [Criticality analysis](#)
  - 5) [Software qualification test plan V&V](#)
  - 6) [Software acceptance test plan V&V](#)
  - 7) [Hazard analysis](#)
  - 8) [Security analysis](#)
  - 9) [Risk analysis](#)

The System/Software Requirements Definition V&V Process begins after the system requirements have been defined and allocated to each software and hardware element of the system. Given the system requirements and software concept architecture, this stage of the software life cycle further defines the allocation of system requirements into specific requirements of the software element. Any constraints imposed by the system/software requirements for the software element. Any constraints or limitations imposed by the system/software requirements on system performance are identified and verified with the system acquirer that these limitations or constraints are acceptable. This life cycle process addresses system/software requirements analysis of the functional and performance requirements; interfaces external to the software; and requirements for qualification, safety and security, human factors engineering, data definitions, user documentation for the software, installation and acceptance, user operation and execution, and user maintenance. V&V test planning begins in the System/Software Requirements Definition V&V process, and test activities span several software V&V processes.

The System/Software Requirements Definition V&V tasks listed shall be used to verify and validate the System/Software Requirements Definition Process outcomes described in ISO/IEC/IEEE 12207:2017(E) [\[B31\]](#).

## 9.4 Architecture Definition V&V Process

### 9.4.1 Purpose

The purpose of the Architecture Definition V&V Process is to provide assurance that the outcomes of the Architecture Definition Process (ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)) have been achieved.

### 9.4.2 Outcomes

As a result of the successful implementation of the Architecture Definition V&V Process, objective evidence is developed to assess whether:

- a) The software architecture (i.e., software, interfaces, and communication) satisfies the system/software requirements.

- b) The software architecture is realizable.
- c) The software architecture is based on specified selection criteria.
- d) The basis for verifying the software elements is defined.
- e) The basis for integration of the system elements is established.

### 9.4.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2c](#) for the selected integrity level, the following Architecture Definition V&V activity and tasks described in [Table 1c, Activity 9.4](#):

- a) [Architecture definition V&V](#). This activity consists of the following tasks:
  - 1) [Architecture evaluation](#)
  - 2) [Interface analysis](#)
  - 3) [Requirements allocation analysis](#)
  - 4) [Traceability analysis](#)
  - 5) [Criticality analysis](#)
  - 6) [Software component test plan V&V](#)
  - 7) [Software integration test plan V&V](#)
  - 8) [Software qualification test design V&V](#)
  - 9) [Software acceptance test design V&V](#)
  - 10) [Hazard analysis](#)
  - 11) [Security analysis](#)
  - 12) [Risk analysis](#)

After system/software requirements have been developed, the software architecture process is initiated. Architectural solutions are developed that will satisfy the specific system/software requirements. Any constraints or limitations imposed by the architecture on the system performance are identified and verified with the system acquirer that these limitations or constraints are acceptable. V&V test design begins during the Architecture Definition V&V Process and spans several software V&V processes.

The Architecture Definition V&V tasks listed shall be used to verify and validate the Architecture Definition Process outcomes described in ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#).

## 9.5 Design Definition V&V Process

### 9.5.1 Purpose

The purpose of the Design Definition V&V Process is to provide assurance that the outcomes of the Design Definition Process (ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)) have been achieved.

### 9.5.2 Outcomes

As a result of the successful implementation of the Design Definition V&V Process, objective evidence is developed to assess whether:

- a) The design characteristics of each software element are defined.
- b) The design enablers necessary for design definition are selected or defined.
- c) Each interface between software elements is described and includes purpose, parameters, data format, and performance criteria (e.g., timing, bandwidth, accuracy, safety, and security).
- d) The software design is established.
- e) Inputs for requirements of any enabling systems or system elements that serve the design definition activities are identified.
- f) Any enabling systems or services needed for design definition are available.
- g) Traceability of the design characteristics to the architectural elements of the software architecture is established.



### 9.5.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2c](#) for the selected integrity level, the following software Design Definition V&V activity and tasks described in [Table 1c, Activity 9.5](#):

- a) [Design definition V&V](#). This activity consists of the following tasks:
  - 1) [Design evaluation](#)
  - 2) [Interface analysis](#)
  - 3) [Traceability analysis](#)
  - 4) [Criticality analysis](#)
  - 5) [Software component test design V&V](#)
  - 6) [Software component test case V&V](#)
  - 7) [Software integration test design V&V](#)
  - 8) [Software integration test case V&V](#)
  - 9) [Software qualification test case V&V](#)
  - 10) [Software acceptance test case V&V](#)
  - 11) [Hazard analysis](#)
  - 12) [Security analysis](#)
  - 13) [Risk analysis](#)

Using the defined software architecture, the software Design Definition Process is initiated. Detailed software design solutions are developed that will satisfy the specific system/software requirements. Also, during this process, software V&V test outcomes for component test design and case V&V, integration test design and case V&V, qualification test case V&V, and acceptance test case V&V are developed.

The Design Definition V&V tasks listed shall be used to verify and validate the Design Definition Process outcomes described in ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#).

## 9.6 System Analysis V&V Process

### 9.6.1 Purpose

The purpose of the System Analysis V&V Process is to provide assurance that outcomes of the System Analysis Process (ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)), have been achieved.

### 9.6.2 Outcomes

As a result of the successful implementation of the System Analysis V&V Process, objective evidence is developed to assess whether:

- a) The strategy for the system analysis is complete and is appropriate for the importance of the analysis.
- b) The results of the system analysis support its conclusions and recommendations.

### 9.6.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2c](#) for the selected integrity level, the following System Analysis V&V activity and tasks described in [Table 1c, Activity 9.6](#):

- a) [System analysis V&V](#). This activity consists of the following tasks:
  - 1) [System analysis strategy evaluation](#)
  - 2) [System analysis results evaluation](#)

The System Analysis V&V Process shall be performed for any specific implementation of the System Analysis Process where the associated conclusions and recommendations of the analysis could affect a software element of the integrity level indicated in [Table 2c](#).

The System Analysis V&V tasks listed shall be used to verify and validate the System Analysis Process outcomes described in ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#).

## 9.7 Implementation V&V Process

### 9.7.1 Purpose

The purpose of the Implementation V&V Process is to provide assurance that outcomes of the Implementation Process (ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)) have been achieved.

### 9.7.2 Outcomes

As a result of the successful implementation of the Implementation V&V Process, objective evidence is developed to assess whether:

- a) The implementation activities performed produce a software element that conforms to the system/software requirements.
- b) The implementation correctly implements the design definition.
- c) The software element has been implemented within the defined constraints.
- d) The recorded evidence of implementation is complete and correct.
- e) The software element has been packaged and stored in accordance with defined requirements.

### 9.7.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2c](#) for the selected integrity level, the following Implementation V&V activity and tasks described in [Table 1c, Activity 9.7](#):

- a) [Implementation V&V](#). This activity consists of the following tasks:
  - 1) [Source code and source code information evaluation](#)
  - 2) [Interface analysis](#)
  - 3) [Traceability analysis](#)
  - 4) [Criticality analysis](#)
  - 5) [Software component test procedure V&V](#)
  - 6) [Software component test execution V&V](#)
  - 7) [Software integration test procedure V&V](#)
  - 8) [Software qualification test procedure V&V](#)
  - 9) [Software acceptance test procedure V&V](#)
  - 10) [Hazard analysis](#)
  - 11) [Security analysis](#)
  - 12) [Risk analysis](#)

The Implementation V&V tasks are to monitor, review, audit, and analyze software products. The system V&V activities will use the software V&V results to perform, in part or in whole, the system V&V activities.

The Implementation V&V activity evaluates software coding to assess compliance with system/software requirements and design, and implements V&V test procedures for software component test V&V, software integration test V&V, software qualification test V&V, and software acceptance test V&V. software component test V&V execution is initiated in this life cycle stage.

The Implementation V&V tasks listed shall be used to verify and validate the Implementation Process outcomes described in ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#).

## 9.8 Integration V&V Process

### 9.8.1 Purpose

The purpose of the Integration V&V Process is to provide assurance that the outcomes of the Integration Process (ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)) have been achieved.

### 9.8.2 Outcomes

As a result of the successful implementation of the Integration V&V Process, objective evidence is developed to assess whether:

- a) The software elements, as described by the architecture definition, design definition, and implementation, are integrated correctly.
- b) The integrated software meets the system/software requirements.
- c) The software integration strategy is consistent with the software architecture.
- d) The integration test plan and procedures are traceable to the software architecture.
- e) The unavoidable constraints of integration that influence requirements are addressed correctly.
- f) The integration of human performance into systems and their operation is correct.
- g) Nonconformances due to integration actions are recorded and addressed.
- h) The software element does not cause unintended or undesired consequences.

### 9.8.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2c](#) for the selected integrity level, the following software Integration V&V activity and tasks described in [Table 1c, Activity 9.8](#):

- a) [Integration V&V](#). This activity consists of the following tasks:
  - 1) [Software integration strategy assessment](#)
  - 2) [Software element interaction analysis](#)
  - 3) [Software integration test execution V&V](#)
  - 4) [Software qualification test execution V&V](#)
  - 5) [Traceability analysis](#)
  - 6) [Hazard analysis](#)
  - 7) [Security analysis](#)
  - 8) [Risk analysis](#)

The Integration V&V Process occurs in parallel with software implementation as soon as sufficient related software components are available. When sufficient software components are available to demonstrate compliance with system and system/software requirements, software integration test execution V&V is performed so that those collective software components are subjected to software integration testing. With the most recently developed software component and continuing with increasingly larger number of components as they are developed. When all software components have been implemented and tested (component, integration), software qualification test V&V is conducted to demonstrate compliance of the entire software element to both individual system/software requirements and total system requirements. The software qualification test V&V occurs in a qualified test environment replicating the exact operational environment as much as possible.

The Integration V&V tasks listed shall be used to verify and validate the Integration Process outcomes described in ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#).

## 9.9 Verification Process

### 9.9.1 Purpose

The purpose of the Verification Process is to provide objective evidence for whether the outcomes achieve the following:

- a) Conform to requirements (e.g., for correctness, completeness, consistency, and accuracy) for all activities during each life cycle process.
- b) Satisfy the standards, practices, and conventions during life cycle processes.
- c) Successfully complete each life cycle activity and satisfy all the criteria for initiating succeeding life cycle activities (i.e., the product is built correctly).

### 9.9.2 Outcomes

The following outcomes occur as a result of the successful implementation of the Verification Process:

- a) A V&V plan is developed and implemented.
- b) The system of interest (software) and all components of the system of interest are assigned integrity levels that are reevaluated throughout the life cycle of the system.
- c) The software and each of its components are evaluated for requirements satisfaction based on assigned integrity levels.
- d) Objective evidence is developed to determine whether the software and each of its components conform to requirements and satisfy all the criteria for each successive life cycle activity.

### 9.9.3 Activities and tasks

The activities and tasks for the Verification Process applied to the technical processes of the software life cycle processes from ISO/IEC/IEEE 12207:2017(E) [B40] are described in subclause 9.1 (Business or Mission Analysis V&V Process) subclause, 9.2 (Stakeholder Needs and Requirements Analysis V&V process), subclause 9.3 (System/Software Requirements Definition V&V process), subclause 9.4 (Architecture Definition V&V Process), subclause 9.5 (Design Definition V&V Process), subclause 9.6 (System Analysis V&V Process) subclause, 9.7 (Implementation V&V Process), subclause 9.8 (Integration V&V Process), subclause 9.10 (Transition V&V Process), subclause 9.12 (Operation V&V Process), subclause 9.13 (Maintenance V&V Process), and subclause 9.14 (Disposal V&V Process).

## 9.10 Transition V&V Process

### 9.10.1 Purpose

The purpose of the Transition V&V Process is to provide assurance that the outcomes of the Transition Process (ISO/IEC/IEEE 12207:2017(E) [B40]) have been achieved.

### 9.10.2 Outcomes

As a result of the successful implementation of the Transition V&V Process, objective evidence is developed to assess whether:

- a) The software transition strategy is comprehensive and explicitly documented.
- b) The software is installed in its operational location in accordance with the transition plan.
- c) The software delivers all specified services per its governing requirements.
- d) The software is sustainable by enabling systems.
- e) The software satisfies its acceptance criteria.

### 9.10.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2c](#) for the selected integrity level, the following Transition V&V activity and tasks described in [Table 1c, Activity 9.10](#):

- a) [Transition V&V](#). This activity consists of the following tasks:
  - 1) [Transition strategy evaluation](#)
  - 2) [Transition demonstration assessment](#)
  - 3) [Software acceptance test execution](#)
  - 4) [Installation configuration audit](#)
  - 5) [Installation checkout](#)
  - 6) [Hazard analysis](#)
  - 7) [Security analysis](#)
  - 8) [Risk analysis](#)

In transition, the software product is installed and tested in the target environment (acceptance test). The Transition V&V activity supports the software system installation activities.

The Transition V&V tasks listed shall be used to verify and validate the Transition Process outcomes described in ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#).

## 9.11 Validation Process

### 9.11.1 Purpose

The following outcomes occur as a result of the successful implementation of the Validation Process:

- a) Satisfy requirements allocated to the software at the end of each life cycle activity.
- b) Solve the right problem (e.g., correctly model physical laws, implement business rules, and use the proper system assumptions).
- c) Satisfy intended use and user needs in the operational environment (i.e., the correct product is built).

### 9.11.2 Outcomes

The following outcomes occur as a result of the successful implementation of the Validation Process:

- a) A V&V plan is developed and implemented.
- b) The system of interest (software) and all components of the system of interest are assigned integrity levels that are maintained throughout the life cycle of the system.
- c) The software and each of its components are evaluated for satisfaction of allocated system requirements and of intended use and user needs based on assigned integrity levels.
- d) Objective evidence is developed to determine whether the software and each of its components satisfy all system requirements allocated to software and meet intended use and user needs.

### 9.11.3 Activities and tasks

The activities and tasks for the Validation Process applied to the technical processes of the software life cycle processes from ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#) are described in subclause [9.1](#) (Business or Mission Analysis V&V Process) subclause, [9.2](#) (Stakeholder Needs and Requirements Analysis V&V process), subclause [9.3](#) (System/Software Requirements Definition V&V process), subclause [9.4](#) (Architecture Definition V&V Process), subclause [9.5](#) (Design Definition V&V Process), subclause [9.6](#) (System Analysis V&V Process) subclause, [9.7](#) (Implementation V&V Process), subclause [9.8](#) (Integration V&V Process), subclause [9.10](#) (Transition V&V Process), subclause [9.12](#) (Operation V&V Process), subclause [9.13](#) (Maintenance V&V Process), and subclause [9.14](#) (Disposal V&V Process).

## 9.12 Operation V&V Process

### 9.12.1 Purpose

The purpose of the Operation V&V Process is to provide assurance that the outcomes of the Operation Process (ISO/IEC/IEEE 12207:2017(E) [B40]) have been achieved.

### 9.12.2 Outcomes

As a result of the successful implementation of the Operation V&V Process, objective evidence is developed to assess whether:

- a) The operation strategy satisfies stakeholder requirements and needs.
- b) New constraints in the system are evaluated.
- c) Proposed system changes and their impacts on the software are assessed.
- d) Operating procedures are evaluated for correctness and usability.

### 9.12.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2c](#) for the selected integrity level, the following Operation V&V activity and tasks described in [Table 1c, Activity 9.12](#):

- a) [Operation V&V](#)—This activity consists of the following tasks:
  - 1) [Evaluation of new constraints](#)
  - 2) [Operating procedures evaluation](#)
  - 3) [Hazard analysis](#)
  - 4) [Security analysis](#)
  - 5) [Risk analysis](#)

The Operation V&V activity evaluates the impact of changes in the operating environment, assesses the effect on the system of any proposed changes, evaluates operating procedures for adherence with the intended use, and analyzes risks affecting the user and the system.

The Operation V&V tasks listed shall be used to verify and validate the Operation Process outcomes described in ISO/IEC/IEEE 12207:2017(E) [B40].

## 9.13 Maintenance V&V Process

### 9.13.1 Purpose

The purpose of the Maintenance V&V Process is to provide assurance that the outcomes of the Maintenance Process (ISO/IEC/IEEE 12207:2017(E) [B40]) have been achieved. The Maintenance Process sustains the capability of the software to provide a service. This process monitors the software's capability to deliver services; records problems for analysis; takes corrective, adaptive, perfective, and preventive actions; and confirms restored capability (ISO/IEC/IEEE 15288:2023(E) [B41]).

System and software modifications may be derived from requirements specified to correct errors (e.g., corrective), to adapt to a changed operating environment (e.g., adaptive), or to respond to additional user requests or enhancements (e.g., perfective). Modifications of the software shall be treated as development efforts and shall be verified and validated by performing V&V tasks corresponding to the modifications. Integrity level assignments shall be assessed as described in [Clause 5](#). The integrity level assignments shall be revised as appropriate to reflect requirements derived from the Maintenance Process.

The integrity level could be revised on the basis of corrective, perfective, or adaptive changes to the software if these changes modify the original integrity level of the system. If the integrity level is revised, then the

effect of the revision on existing system/software requirements shall be evaluated to identify additional activities and tasks to be performed regressively on the system, software, and hardware. Corresponding V&V activities and tasks shall be identified to confirm the software conforms to its required integrity level.

If software V&V were performed in accordance with this standard, then the Maintenance Process shall continue to conform to this standard. If the software was not verified and validated using this standard and appropriate documentation is not available or adequate, then the maintenance V&V effort shall determine whether the missing or incomplete documentation should be generated. In making this determination of whether to generate missing documentation, the minimum V&V requirements of the assigned integrity level shall be taken into consideration.

The objectives of maintenance V&V are to 1) assess that the maintenance strategy is comprehensive and explicitly documented; 2) evaluate the corrective actions to assure they resolve the issue or negative trend; 3) assess the effect of the corrective, adaptive, perfective, and preventive changes on stakeholder and system/software requirements, architectural design, and implementation; and 4) evaluate problem reports, corrective actions, and trends to determine the possible corrective, adaptive, perfective, and preventive actions.

### 9.13.2 Outcomes

As a result of the successful implementation of the Maintenance V&V Process, objective evidence is developed to assess whether:

- a) The maintenance strategy is comprehensive and explicitly documented.
- b) Corrective actions resolve the issue or negative trend.
- c) The effect is assessed for corrective, adaptive, perfective, and preventive changes on stakeholder and system/software requirements, architecture, design, and implementation.
- d) Evaluation is performed on problem reports, corrective actions, and trends to determine the possible corrective, adaptive, perfective, and preventive actions.
- e) Proposed software changes and their impact on the system are assessed.
- f) Anomalies that are discovered during operation are evaluated.
- g) Migration requirements are assessed.
- h) Retirement requirements are assessed.
- i) V&V tasks are re-performed.

### 9.13.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2c](#) for the selected integrity level, the following Maintenance V&V activity and tasks described in [Table 1c, Activity 9.13](#):

- a) [Maintenance V&V](#)—This activity consists of the following tasks:
  - 1) [Software maintenance strategy assessment](#)
  - 2) [Software maintenance execution assessment](#)
  - 3) [VVP revision](#)
  - 4) [Anomaly evaluation](#)
  - 5) [Criticality analysis](#)
  - 6) [Migration assessment](#)
  - 7) [Retirement assessment](#)
  - 8) [Hazard analysis](#)
  - 9) [Security analysis](#)
  - 10) [Risk analysis](#)
  - 11) [Task iteration](#)

Proposed changes are assessed by the proposed/baseline change assessment task of the V&V management activity.

The Maintenance Process is activated when the software or associated information is changed in response to a need for system maintenance. The Maintenance V&V activity addresses the following:

- Software modifications (i.e., corrective, adaptive, or perfective changes).
- Software migration (i.e., the movement of software to a new operational environment).
- Software retirement (i.e., the withdrawal of active support by the operation and maintenance organization, partial or total replacement by a new system, or installation of an upgraded system).

The Maintenance V&V tasks listed shall be used to verify and validate the Maintenance Process outcomes described in ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#).

## 9.14 Disposal V&V Process

### 9.14.1 Purpose

The purpose of the Disposal V&V Process is to provide assurance that the outcomes of the Disposal Process (ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)) have been achieved.

### 9.14.2 Outcomes

As a result of the successful implementation of the Disposal V&V Process, objective evidence is developed to assess whether:

- a) The constraints in the software disposal strategy are included in system/software requirements.
- b) Disposal leaves the system in an agreed-on state.

### 9.14.3 Activity and task

The V&V effort shall perform, as specified in [Table 2c](#) for the selected integrity level, the following Disposal V&V activity and task described in [Table 1c, Activity 9.14](#):

- a) [Disposal V&V](#)—This activity consists of the following task:

- 1) [Software disposal strategy evaluation](#)

This process ends active support by the operation and maintenance organization or deactivates, disassembles, and removes the affected software products, consigning them to a final condition and leaving the environment in an acceptable condition. This process destroys or stores system software elements and related products in a sound manner, in accordance with legislation, agreements, organizational constraints, and stakeholder requirements. Where required, it maintains records that may be monitored.

The Disposal V&V task listed shall be used to verify and validate the Disposal Process outcomes described in ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#).



**Table 1c—V&V tasks, inputs, and outputs**

| <b>9.1 Activity: Business or mission analysis V&amp;V (Software, 12207—Business or Mission Analysis Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                 |                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Required inputs</b>                                                                                                                                                          | <b>Required outputs</b>                                                                              |
| <p><b>(1) <u>Business or mission analysis documentation evaluation</u></b></p> <p>a) Validate that the concept documentation satisfies user needs and is consistent with acquisition needs.</p> <p>b) Validate constraints of interfacing systems and constraints or limitations of the proposed approach.</p> <p>c) Analyze system requirements and validate that the following satisfy user needs:</p> <ol style="list-style-type: none"> <li>1) System functions.</li> <li>2) End-to-end system performance.</li> <li>3) Feasibility and testability of the functional requirements.</li> <li>4) System architecture design.</li> <li>5) Operation and maintenance requirements and environments.</li> <li>6) Migration requirements from an existing system where applicable.</li> </ol>                                                                                                                                                                                                                                                                                                                                      | <p>Business or mission Analysis documentation</p> <p>System architectural design</p> <p>Supplier development plans and schedules</p> <p>User needs</p> <p>Acquisition needs</p> | <p>Task report(s)—Business or Mission Analysis documentation evaluation</p> <p>Anomaly report(s)</p> |
| <p><b>(2) <u>Requirements allocation analysis</u></b></p> <p>Verify the correctness, accuracy, and completeness of the system requirements allocation to the software against user needs.</p> <p>a) Correctness</p> <p>Verify that performance requirements (e.g., timing, response time, and throughput) allocated to the hardware, software, and user interfaces satisfy user needs.</p> <p>b) Accuracy</p> <p>Verify that the internal and external interfaces specify the data formats, interface protocols, frequency of data exchange at each interface, and other performance requirements to demonstrate satisfaction with user requirements.</p> <p>c) Completeness</p> <ol style="list-style-type: none"> <li>1) Verify that application-specific requirements such as functional diversity, fault detection, fault isolation, and diagnostic and error recovery satisfy user needs.</li> <li>2) Verify that the user's maintenance requirements for the system are completely specified.</li> <li>3) Verify that the migration from the existing system and replacement of the system satisfies user needs.</li> </ol> | <p>User needs</p> <p>Business or mission Analysis documentation</p> <p>System requirements</p> <p>System architecture</p> <p>System design</p>                                  | <p>Task report(s)—Requirements allocation analysis</p> <p>Anomaly report(s)</p>                      |
| <p><b>(3) <u>Traceability analysis</u></b></p> <p>a) Identify all system requirements.</p> <p>b) Verify that these system requirements are traceable to acquisition needs.</p> <p>c) Start the requirements traceability analysis with system requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Business or mission analysis documentation</p>                                                                                                                               | <p>Task report(s)—Traceability analysis</p> <p>Anomaly report(s)</p>                                 |
| <p><b>(4) <u>Criticality analysis</u></b></p> <p>a) Determine whether integrity levels are established for requirements, detailed functions, software modules, hardware elements, subsystems, or other partitions.</p> <p>b) Verify that the assigned integrity levels are correct. If integrity levels are not assigned, then assign integrity levels to the system requirements.</p> <p>c) Document the integrity level assigned to individual components (e.g., requirements, detailed functions, software modules, hardware elements, subsystems, or other partitions).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Business or mission analysis documentation</p> <p>System requirements</p> <p>Developer integrity level assignments</p>                                                       | <p>Task report(s)—Criticality analysis</p> <p>Anomaly report(s)</p>                                  |

| <b>9.1 Activity: Business or mission analysis V&amp;V (Software, 12207—Business or Mission Analysis Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                    |                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Required inputs</b>                                                                                                                                             | <b>Required outputs</b>                                   |
| <p>For V&amp;V planning purposes, the system shall be assigned the same integrity level as the highest level assigned to any individual element.</p> <p>d) Verify whether any component can influence the individual components assigned a higher software integrity level, and if such conditions exist, then assign that component the same higher integrity level.</p>                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                    |                                                           |
| <p><b>(5) <u>Hazard analysis</u></b></p> <p>Analyze the potential hazards to and from the conceptual system. The analysis shall perform the following:</p> <p>a) Identify the potential system hazards.</p> <p>b) Assess the consequences of each hazard.</p> <p>c) Assess the probability of each hazard.</p> <p>d) Identify mitigation strategies for each hazard.</p>                                                                                                                                                                                                                                                                                                                                                                                        | Business or mission analysis documentation                                                                                                                         | Task report(s)—<br>Hazard analysis<br>Anomaly report(s)   |
| <p><b>(6) <u>Security analysis</u></b></p> <p>a) Review the system owner's definition of an acceptable level of security risk.</p> <p>b) Analyze the system concept from a security perspective and assure that potential security risks with respect to confidentiality (disclosure of sensitive information/data), integrity (modification of information/data), availability (withholding of information or services), and accountability (attributing actions to an individual/process) have been identified. Include an assessment of the sensitivity of the information/data to be processed.</p> <p>c) Analyze the security risks introduced by the system itself as well as those associated with the environment with which the system interfaces.</p> | Business or mission analysis documentation<br>Preliminary TRA                                                                                                      | Task report(s)—<br>Security analysis<br>Anomaly report(s) |
| <p><b>(7) <u>Risk analysis</u></b></p> <p>a) Identify the technical and management risks.</p> <p>b) Provide recommendations to eliminate, reduce, or mitigate the risks.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Business or mission analysis documentation<br>Supplier development plans and schedules<br>Hazard analysis results<br>Security analysis results<br>V&V task results | Task report(s)—<br>Risk analysis<br>Anomaly report(s)     |

| <b>9.2 Activity: Stakeholder needs and requirements definition V&amp;V (Software, 12207—Stakeholder Needs and Requirements Definition Process)</b>                                                                                                                                                                                                      |                                                                                                                     |                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                    | <b>Required inputs</b>                                                                                              | <b>Required outputs</b>                                                               |
| <p><b>(1) <u>Stakeholder needs and requirements evaluation</u></b></p> <p>Evaluate the stakeholder requirements for correctness, consistency, completeness, readability, and testability. The task criteria are as follows:</p> <p>a) Correctness</p> <p>1) Verify and validate that the requirements satisfy the stakeholder needs for the system.</p> | Preferred candidate solution(s)<br>Stakeholder needs and requirements<br>Business or mission analysis documentation | Task report(s)—<br>Stakeholder needs and requirements evaluation<br>Anomaly report(s) |

| <b>9.2 Activity: Stakeholder needs and requirements definition V&amp;V (Software, 12207—Stakeholder Needs and Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                |                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Required inputs</b>                                                                                                                                         | <b>Required outputs</b>                                   |
| <ul style="list-style-type: none"> <li>2) Verify that the requirements comply with standards, references, regulations, policies, physical laws, and business rules.</li> <li>3) Validate that the requirements define the intended interaction of the system with its operating environment and other interfacing systems.</li> <li>b) Validate the suitability of the preferred candidate solution(s) to meet the stakeholder requirements.</li> <li>c) Consistency <ul style="list-style-type: none"> <li>1) Verify that all terms, concepts, and requirements are documented consistently in accordance with accepted syntax and structure (e.g., style guides and requirements modeling structure).</li> <li>2) Verify that there is consistency between assumptions, requirements, and between groups of requirements.</li> </ul> </li> <li>d) Completeness <ul style="list-style-type: none"> <li>1) Validate that performance criteria and functionality are described in the requirements, within the assumptions and constraints of the operating environment and system boundaries.</li> <li>2) Verify that all stakeholders, or stakeholder classes (used here as groupings of stakeholders), are identified.</li> <li>3) Verify that the stakeholder requirements satisfy specified configuration management procedures. Verify stakeholder requirements are in a form suitable for requirements management throughout the life cycle.</li> </ul> </li> <li>e) Readability <ul style="list-style-type: none"> <li>1) Verify that the documentation is legible, understandable, and unambiguous to the intended audience.</li> <li>2) Verify that the documentation defines all acronyms, mnemonics, abbreviations, terms, and symbols.</li> </ul> </li> <li>f) Testability <ul style="list-style-type: none"> <li>1) Verify that objective acceptance criteria can be developed to validate the requirements.</li> </ul> </li> </ul> |                                                                                                                                                                |                                                           |
| <p><b>(2) <u>Traceability analysis</u></b></p> <ul style="list-style-type: none"> <li>a) Establish the root of traceability for software, starting with the organization strategy.</li> <li>b) Verify that the candidate hardware solution(s) traces back to the solution space.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Business or mission analysis documentation<br>Identified problem or opportunity space<br>Identified solution space<br>Preferred candidate software solution(s) | Task report(s)—Traceability analysis<br>Anomaly report(s) |
| <p><b>(3) <u>Criticality analysis</u></b></p> <ul style="list-style-type: none"> <li>a) Verify that the assigned integrity levels are correct. If integrity levels are not assigned, then assign integrity levels to the system requirements related to software, detailed functions, software components, subsystems, or other partitions.</li> <li>b) Document the integrity level assigned to individual components (e.g., requirements, detailed functions, software components, subsystems, or other partitions). For V&amp;V planning purposes, the software component shall be assigned</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Identified problem or opportunity space<br>Identified solution space<br>Preferred candidate software solution(s)<br>Developer integrity level assignments      | Task report(s)—Criticality analysis<br>Anomaly report(s)  |

| <b>9.2 Activity: Stakeholder needs and requirements definition V&amp;V (Software, 12207—Stakeholder Needs and Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                  |                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Required inputs</b>                                                                                                                                           | <b>Required outputs</b>                                   |
| <p>the same integrity level as the highest level assigned to any individual element.</p> <p>c) Verify whether any component can influence individual components assigned a higher integrity level, and if such conditions exist, then assign that component the same higher integrity level.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                  |                                                           |
| <p><b>(4) <u>Hazard analysis</u></b></p> <p>Analyze the potential hazards to and from the preferred candidate software solution(s). The analysis shall perform the following:</p> <p>a) Identify the potential system hazards [e.g., environmental conditions such as pressure, radiation, temperature, and humidity, electrostatic sensitivity, electromagnetic interference/radio frequency interference (EMI/RFI), seismic, electro-mechanical properties, aging].</p> <p>b) Assess the consequences of each hazard.</p> <p>c) Assess the probability of each hazard.</p> <p>d) Identify mitigation strategies for each hazard.</p>                                                                                                                                                                                                                                                                                                    | Preferred candidate software solution(s)                                                                                                                         | Task report(s)—<br>Hazard analysis<br>Anomaly report(s)   |
| <p><b>(5) <u>Security analysis</u></b></p> <p>a) Review the system owner's definition of an acceptable level of security risk.</p> <p>b) Analyze the preferred candidate hardware solution(s) from a security perspective and assure that potential security risks with respect to confidentiality (disclosure of sensitive information/data), integrity (modification of information/data), availability (withholding of information or services), and accountability (attributing actions to an individual/process) have been identified. Include an assessment of the sensitivity of the information/data to be processed. Security attributes include access control, electrical/optical isolation, location tracking, and network medium control.</p> <p>c) Analyze security risks introduced by the preferred candidate hardware solution(s) as well as those associated with the environment with which the system interfaces.</p> | Preferred candidate software solution(s)<br>Preliminary TRA                                                                                                      | Task report(s)—<br>Security analysis<br>Anomaly report(s) |
| <p><b>(6) <u>Risk analysis</u></b></p> <p>a) Identify software technical and management risks.</p> <p>b) Provide recommendations to eliminate, reduce, or mitigate the risks.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Preferred candidate software solution(s)<br>Supplier development plans and schedules<br>Hazard analysis results<br>Security analysis results<br>V&V task results | Task report(s)—<br>Risk analysis<br>Anomaly report(s)     |

| <b>9.3 Activity: System/software requirements definition V&amp;V (Software, 12207—System/Software Requirements Definition Process)</b>                                                                                                                                                                                                |                                                                                     |                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                  | <b>Required inputs</b>                                                              | <b>Required outputs</b>                                                         |
| <p><b>(1) <u>Requirements evaluation</u></b></p> <p>Evaluate the requirements (e.g., functional, capability, interface, qualification, safety, security, human factors, data definitions, user documentation, installation and acceptance, user operation, and user maintenance) of the SRS and IRS for correctness, consistency,</p> | Business or mission analysis<br>documentation<br>Stakeholder needs and requirements | Task report(s)—<br>System/software requirements evaluation<br>Anomaly report(s) |

| <b>9.3 Activity: System/software requirements definition V&amp;V (Software, 12207—System/Software Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                               |                         |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Required inputs</b>                                                        | <b>Required outputs</b> |
| <p>completeness, accuracy, readability, and testability. The task criteria are as follows:</p> <p>a) Correctness</p> <ol style="list-style-type: none"> <li>1) Verify and validate that the system/software requirements satisfy the system requirements allocated to software within the assumptions, constraints, and operating environment for the system.</li> <li>2) Verify that the system/software requirements comply with standards, references, regulations, policies, physical laws, and business rules.</li> <li>3) Validate the sequences of states and state changes using logic and data flows coupled with domain expertise, prototyping results, engineering principles, or other basis.</li> <li>4) Validate that the flow of data and control satisfy functionality and performance requirements.</li> <li>5) Validate data usage and format.</li> </ol> <p>b) Consistency</p> <ol style="list-style-type: none"> <li>1) Verify that all terms and concepts are documented consistently.</li> <li>2) Verify that the function interactions and assumptions are consistent and satisfy system requirements and acquisition needs.</li> <li>3) Verify that there is internal consistency between the system/software requirements and external consistency with the system requirements.</li> </ol> <p>c) Completeness</p> <ol style="list-style-type: none"> <li>1) Verify that the following elements are in the SRS or IRS, within the assumptions and constraints of the system: <ol style="list-style-type: none"> <li>i) Functionality (e.g., algorithms, state/mode definitions, input/output Validation, exception handling, reporting and logging).</li> <li>ii) Process definition and scheduling.</li> <li>iii) Hardware, software, and user-interface descriptions.</li> <li>iv) Performance criteria (e.g., timing, sizing, speed, capacity, accuracy, precision, safety, and security).</li> <li>v) Critical configuration data.</li> <li>vi) System, device, and software control (e.g., initialization, transaction and state monitoring, and self-testing).</li> </ol> </li> <li>2) Verify that the SRS and IRS satisfy specified configuration management procedures.</li> </ol> <p>d) Accuracy</p> <ol style="list-style-type: none"> <li>1) Validate that the logic, computational, and interface precision (e.g., truncation and rounding) satisfy the requirements in the system environment.</li> <li>2) Validate that the modeled physical phenomena conform to system accuracy requirements and physical laws.</li> </ol> <p>e) Readability</p> <ol style="list-style-type: none"> <li>1) Verify that the documentation is legible, understandable, and unambiguous to the intended audience.</li> <li>2) Verify that the documentation defines all acronyms, mnemonics, abbreviations, terms, and symbols.</li> </ol> <p>f) Testability</p> | <p>System/software requirements (SRS)</p> <p>Interface requirements (IRS)</p> |                         |

| <b>9.3 Activity: System/software requirements definition V&amp;V (Software, 12207—System/Software Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                      |                                                                      |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Required inputs</b>                                                                                                                                                                               | <b>Required outputs</b>                                              |
| Verify that there are objective acceptance criteria for validating the requirements of the SRS and IRS.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                      |                                                                      |
| <p><b>(2) <a href="#">Interface analysis</a></b></p> <p>Verify and validate that the requirements for software interfaces with hardware, user, operator, and other systems are correct, consistent, complete, accurate, and testable. The task criteria are as follows:</p> <ul style="list-style-type: none"> <li>a) <b>Correctness</b><br/>Validate the external and internal system and software interface requirements.</li> <li>b) <b>Consistency</b><br/>Verify that the interface descriptions are consistent between the SRS and IRS.</li> <li>c) <b>Completeness</b><br/>Verify that each interface is described and includes data format and performance criteria (e.g., timing, bandwidth, accuracy, safety, and security).</li> <li>d) <b>Accuracy</b><br/>Verify that each interface provides information with the required accuracy.</li> <li>e) <b>Testability</b><br/>Verify that there are objective acceptance criteria for validating the interface requirements.</li> </ul>                                                                                                                                                                                                                                                                                                                                      | <p>Business or mission analysis documentation</p> <p>Stakeholder needs and requirements</p> <p>System/software requirements (SRS)</p> <p>Interface requirements (IRS)</p>                            | <p>Task report(s)—Interface analysis</p> <p>Anomaly report(s)</p>    |
| <p><b>(3) <a href="#">Traceability analysis</a></b></p> <p>Trace the system/software requirements (SRS and IRS) to the system requirements (stakeholder needs and requirements documentation and business or mission analysis documentation) and the system requirements to the system/software requirements. Analyze identified relationships for correctness, consistency, completeness, and accuracy. The task criteria are as follows:</p> <ul style="list-style-type: none"> <li>a) <b>Correctness</b><br/>Validate that the relationships between each software requirement and its system requirement are correct.</li> <li>b) <b>Consistency</b><br/>Verify that the relationships between the software and system requirements are specified to a consistent level of detail.</li> <li>c) <b>Completeness</b> <ul style="list-style-type: none"> <li>1) Verify that every software requirement is traceable to a system requirement with sufficient detail to show conformance to the system requirement.</li> <li>2) Verify that all system requirements related to software are traceable to system/software requirements.</li> </ul> </li> <li>d) <b>Accuracy</b><br/>Validate that the system performance and operating characteristics are accurately specified by the traced system/software requirements.</li> </ul> | <p>Business or mission analysis documentation</p> <p>Stakeholder needs and requirements</p> <p>System requirements</p> <p>System/software requirements (SRS)</p> <p>Interface requirements (IRS)</p> | <p>Task report(s)—Traceability analysis</p> <p>Anomaly report(s)</p> |
| <p><b>(4) <a href="#">Criticality analysis</a></b></p> <ul style="list-style-type: none"> <li>a) Review and update the existing criticality analysis results from the prior criticality task report using the SRS and IRS.</li> <li>b) Implementation methods and interfacing technologies may cause previously assigned integrity levels to be raised or lowered for a given software element (i.e., requirement,</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>Criticality task report</p> <p>System/software requirements (SRS)</p> <p>Interface requirements (IRS)</p>                                                                                         | <p>Task report(s)—Criticality analysis</p> <p>Anomaly report(s)</p>  |

| <b>9.3 Activity: System/software requirements definition V&amp;V (Software, 12207—System/Software Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                               |                                                                                                                                                                               |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Required inputs</b>                                                                                                                                                                                                                        | <b>Required outputs</b>                                                                                                                                                       |
| module, function, subsystem, and other software partition).<br>Verify that no inconsistent or undesired integrity consequences are introduced by reviewing the revised integrity levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                               |                                                                                                                                                                               |
| <p><b>(5) <u>Software qualification test plan V&amp;V</u></b></p> <p>a) Integrity levels 4 and 3</p> <ol style="list-style-type: none"> <li>1) Plan V&amp;V software qualification testing to validate system/software requirements.</li> <li>2) Plan tracing of system requirements to software qualification test designs, cases, procedures, and results.</li> <li>3) Plan documentation of V&amp;V software qualification test designs, cases, procedures, and results.</li> <li>4) The V&amp;V software qualification test plan shall address the following: <ol style="list-style-type: none"> <li>i) Conformance to all system requirements (e.g., functional, performance, security, operation, and maintenance) as complete software end items in the system environment.</li> <li>ii) Adequacy of user documentation (e.g., training materials and procedural changes).</li> <li>iii) Performance at boundaries (e.g., data and interfaces) and under stress conditions.</li> </ol> </li> <li>5) Verify that the V&amp;V software qualification test plan satisfies the following criteria: <ol style="list-style-type: none"> <li>i) Conformance to project-defined test document purpose, format, and content (e.g., IEEE Std 829™-2008 [B17]).</li> <li>ii) Test coverage of system requirements.</li> </ol> </li> <li>6) Validate that the V&amp;V software qualification test plan satisfies the following criteria: <ol style="list-style-type: none"> <li>i) Appropriateness of test methods and standards used.</li> <li>ii) Conformance to expected results.</li> <li>iii) Feasibility of system qualification testing.</li> <li>iv) Feasibility and testability of operation and maintenance requirements.</li> </ol> </li> </ol> <p>b) Integrity level 2</p> <ol style="list-style-type: none"> <li>1) Verify that the developer's software qualification test plan satisfies the following criteria: <ol style="list-style-type: none"> <li>i) Conformance to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>ii) Test coverage of system requirements.</li> </ol> </li> <li>2) Validate that the developer's software qualification test plan satisfies the following criteria: <ol style="list-style-type: none"> <li>i) Appropriateness of test methods and standards used.</li> <li>ii) Conformance to expected results.</li> <li>iii) Feasibility of system qualification testing.</li> <li>iv) Capability to be operated and maintained.</li> </ol> </li> </ol> <p>c) Integrity level 1</p> <p>There are no software qualification test plan V&amp;V requirements.</p> | Business or mission analysis<br>documentation<br>Stakeholder needs and requirements<br>System requirements<br>requirements<br>System/software (SRS)<br>Interface requirements (IRS)<br>User documentation<br>Software qualification test plan | V&V software qualification test plan (integrity levels 4 and 3)<br><br>Task report(s)—<br>Review of software qualification test plan (integrity level 2)<br>Anomaly report(s) |



| <b>9.3 Activity: System/software requirements definition V&amp;V (Software, 12207—System/Software Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                          |                                                                                                                                                                                    |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Required inputs</b>                                                                                                                                                                                                                   | <b>Required outputs</b>                                                                                                                                                            |
| <p><b>(6) <u>Software acceptance test plan V&amp;V</u></b></p> <p>a) Integrity levels 4 and 3</p> <ol style="list-style-type: none"> <li>1) Plan V&amp;V software acceptance testing to validate that the software correctly implements system and system/software requirements in an operational environment.</li> <li>2) Plan tracing of test requirements to test software acceptance design, cases, procedures, and execution results.</li> <li>3) Plan documentation of test tasks and results.</li> <li>4) The V&amp;V software acceptance test plan shall address the following: <ol style="list-style-type: none"> <li>i) Conformance to acceptance requirements in the operational environment.</li> <li>ii) Adequacy of user documentation.</li> </ol> </li> <li>5) Verify that the V&amp;V software acceptance test plan satisfies the following criteria: <ol style="list-style-type: none"> <li>i) Conformance to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>ii) Test coverage of acceptance requirements.</li> </ol> </li> <li>6) Validate that the V&amp;V software acceptance test plan satisfies the following criteria: <ol style="list-style-type: none"> <li>i) Conformance to expected results.</li> <li>ii) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs).</li> </ol> </li> </ol> <p>b) Integrity level 2</p> <ol style="list-style-type: none"> <li>1) Verify that the acquirer's software acceptance test plan conforms to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>2) Validate that the acquirer's software acceptance test plan satisfies the following criteria: <ol style="list-style-type: none"> <li>i) Test coverage of acceptance requirements.</li> <li>ii) Conformance to expected results.</li> <li>iii) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs).</li> </ol> </li> </ol> <p>c) Integrity level 1</p> <p>There are no software acceptance V&amp;V test requirements.</p> | <p>Business or mission analysis documentation</p> <p>Stakeholder needs and requirements</p> <p>System/software requirements (SRS)</p> <p>Interface requirements (IRS)</p> <p>User documentation</p> <p>Software acceptance test plan</p> | <p>V&amp;V software acceptance test plan (integrity levels 4 and 3)</p> <p>Task report(s)—Review of software acceptance test plan (integrity level 2)</p> <p>Anomaly report(s)</p> |
| <p><b>(7) <u>Hazard analysis</u></b></p> <p>a) Determine software contributions to system hazards. The hazard analysis shall perform the following:</p> <p>b) Identify the system/software requirements that contribute to each system hazard.</p> <p>c) Validate that the software addresses, controls, or mitigates each hazard.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>System/software requirements (SRS)</p> <p>Interface requirements (IRS)</p> <p>Hazard analysis results</p>                                                                                                                             | <p>Task report(s)—Hazard analysis</p> <p>Anomaly report(s)</p>                                                                                                                     |
| <p><b>(8) <u>Security analysis</u></b></p> <p>a) Determine that the security requirements identified in the SRS and IRS address the security risks introduced by the system concept.</p> <p>b) Verify that the system security requirements will mitigate the identified security risks to an acceptable level.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>System/software requirements (SRS)</p> <p>Interface requirements (IRS)</p> <p>Preliminary TRA</p> <p>Security analysis results</p>                                                                                                    | <p>Task report(s)—Security analysis</p> <p>Anomaly report(s)</p>                                                                                                                   |



| <b>9.3 Activity: System/software requirements definition V&amp;V (Software, 12207—System/Software Requirements Definition Process)</b>                                                             |                                                                                                                                                                                                                                                                                                          |                                                              |
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| <b>V&amp;V tasks</b>                                                                                                                                                                               | <b>Required inputs</b>                                                                                                                                                                                                                                                                                   | <b>Required outputs</b>                                      |
| <p><b>(9) <a href="#">Risk analysis</a></b></p> <p>a) Review and update risk analysis using prior task reports.</p> <p>b) Provide recommendations to eliminate, reduce, or mitigate the risks.</p> | <p>Business or mission analysis documentation</p> <p>System/software requirements (SRS)</p> <p>Interface requirements (IRS)</p> <p>Supplier development plans and schedules</p> <p>Hazard analysis results</p> <p>Security analysis results</p> <p>Risk analysis results</p> <p>V&amp;V task results</p> | <p>Task report(s)—Risk analysis</p> <p>Anomaly report(s)</p> |

| <b>9.4 Activity: Architecture definition V&amp;V (Software, 12207—Architecture Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                    |                                                                                                       |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Required inputs</b>                                                                                                                                                                                             | <b>Required outputs</b>                                                                               |
| <p><b>(1) <a href="#">Architecture evaluation</a></b></p> <p>Evaluate the software architecture for correctness, consistency, completeness, accuracy, readability, testability, and design margins. Verify and validate the allocation of system/hardware requirements to the hardware elements.</p>                                                                                                                                                                                                                                                                                                                                                                                      | <p>Stakeholder needs and requirements</p> <p>requirements</p> <p>System/software (SRS)</p> <p>Interface requirements (IRS)</p> <p>Software architecture</p>                                                        | <p>Task report(s)—Architecture evaluation</p> <p>Anomaly report(s)</p>                                |
| <p><b>(2) <a href="#">Interface analysis</a></b></p> <p>Verify and validate the software interfaces with software, other hardware elements, user, operator, and conditioning and enabling software/systems [e.g., radio control (RC) filter network, gear reduction, communication protocol], for correctness, consistency, completeness, accuracy, and testability.</p>                                                                                                                                                                                                                                                                                                                  | <p>Business or mission analysis documentation</p> <p>System requirements</p> <p>System/software requirements (SRS)</p> <p>Interface requirements (IRS)</p> <p>System architecture</p> <p>Software architecture</p> | <p>Task report(s)—Interface analysis</p> <p>Anomaly report(s)</p>                                     |
| <p><b>(3) <a href="#">Requirements allocation analysis</a></b></p> <p>Verify the correctness, accuracy, and completeness of the system requirements allocation to the software against user needs.</p> <p>a) Correctness</p> <p>Verify that performance requirements (e.g., timing, response time, and throughput) allocated to the hardware, software, and user interfaces satisfy user needs.</p> <p>b) Accuracy</p> <p>Verify that the internal and external interfaces specify the data formats, interface protocols, frequency of data exchange at each interface, and other performance requirements to demonstrate satisfaction with user requirements.</p> <p>c) Completeness</p> | <p>Stakeholder needs and requirements</p> <p>System/software requirements (SRS)</p> <p>Interface requirements (IRS)</p>                                                                                            | <p>Software architecture task report(s)—Requirements allocation analysis</p> <p>Anomaly report(s)</p> |

| <b>9.4 Activity: Architecture definition V&amp;V (Software, 12207—Architecture Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                       |                                                                                                                                                                   |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Required inputs</b>                                                                                                                | <b>Required outputs</b>                                                                                                                                           |
| <ol style="list-style-type: none"> <li>1) Verify that application-specific requirements such as functional diversity, fault detection, fault isolation, and diagnostic and error recovery satisfy user needs.</li> <li>2) Verify that the user's maintenance requirements for the system are completely specified.</li> <li>3) Verify that the migration from the existing system and replacement of the system satisfies user needs.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                       |                                                                                                                                                                   |
| <p><b>(4) <u>Traceability analysis</u></b></p> <p>Trace design elements (SDD and IDD) to requirements (SRS and IRS) and the requirements to design elements. Analyze relationships for correctness, consistency, and completeness. The task criteria are as follows:</p> <ol style="list-style-type: none"> <li>a) <b>Correctness</b><br/>Validate the relationship between each design element and the software requirement(s).</li> <li>b) <b>Consistency</b><br/>Verify that the relationships between the design elements and the system/software requirements are specified to a consistent level of detail.</li> <li>c) <b>Completeness</b> <ol style="list-style-type: none"> <li>1) Verify that all design elements are traceable from the system/software requirements.</li> <li>2) Verify that all system/software requirements are traceable to the design elements.</li> </ol> </li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>System/software requirements (SRS)</p> <p>Interface requirements (IRS)</p> <p>System architecture</p> <p>Software architecture</p> | <p>Task report(s)—Traceability analysis</p> <p>Anomaly report(s)</p>                                                                                              |
| <p><b>(5) <u>Criticality analysis</u></b></p> <ol style="list-style-type: none"> <li>a) Review and update the existing criticality analysis results from the prior Criticality Task Report using the SDD and IDD.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Criticality task report</p> <p>Software architecture</p>                                                                           | <p>Task report(s)—Criticality analysis</p> <p>Anomaly report(s)</p>                                                                                               |
| <p><b>(6) <u>Software component test plan V&amp;V</u></b></p> <ol style="list-style-type: none"> <li>a) Integrity levels 4 and 3 <ol style="list-style-type: none"> <li>1) Plan V&amp;V software component testing to validate that the software components (e.g., units and source code modules) correctly implement component requirements.</li> <li>2) Plan tracing of design requirements to test design, cases, procedures, and results.</li> <li>3) Plan documentation of test tasks and results.</li> <li>4) The V&amp;V software component test plan shall address the following: <ol style="list-style-type: none"> <li>i) Conformance to design requirements.</li> <li>ii) Assessment of timing, sizing, and accuracy.</li> <li>iii) Performance at boundaries and interfaces and under stress and error conditions.</li> <li>iv) Measures of requirements test coverage and software reliability and maintainability.</li> </ol> </li> <li>5) Verify that the V&amp;V software component test plan conforms to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>6) Validate that the V&amp;V software component test plan satisfies the following criteria: <ol style="list-style-type: none"> <li>i) Traceable to the system/software requirements and design.</li> </ol> </li> </ol> </li> </ol> | <p>System/software requirements (SRS)</p> <p>Interface requirements (IRS)</p> <p>Software architecture</p>                            | <p>V&amp;V software component test plan (integrity levels 4 and 3)</p> <p>Review of software component test plan (integrity level 2)</p> <p>Anomaly report(s)</p> |

| <b>9.4 Activity: Architecture definition V&amp;V (Software, 12207—Architecture Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                  |                                                                                                                                                                       |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Required inputs</b>                                                                                                                                           | <b>Required outputs</b>                                                                                                                                               |
| <ul style="list-style-type: none"> <li>ii) External consistency with the system/software requirements and design.</li> <li>iii) Internal consistency between unit requirements.</li> <li>iv) Test coverage of requirements in each unit.</li> <li>v) Feasibility of software integration and testing.</li> <li>vi) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs).</li> </ul> <p>b) Integrity level 2</p> <ul style="list-style-type: none"> <li>1) Verify that the developer's software component test plan conforms to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>2) Validate that the developer's software component test plan satisfies the following criteria: <ul style="list-style-type: none"> <li>i) Traceable to the system/software requirements and design.</li> <li>ii) External consistency with the system/software requirements and design.</li> <li>iii) Internal consistency between unit requirements.</li> <li>iv) Test coverage of units.</li> <li>v) Feasibility of software integration and testing.</li> <li>vi) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs).</li> </ul> </li> </ul> <p>c) Integrity level 1</p> <p>There are no software component V&amp;V test requirements.</p> |                                                                                                                                                                  |                                                                                                                                                                       |
| <p>(7) <a href="#">Software integration test plan V&amp;V</a></p> <p>a) For Integrity levels 4 and 3</p> <ul style="list-style-type: none"> <li>1) Plan V&amp;V software integration testing to validate that the software correctly implements the system/software requirements and design as each software component (e.g., units or modules) is incrementally integrated with each other.</li> <li>2) Plan tracing of requirements to test design, cases, procedures, and results.</li> <li>3) Plan documentation of test tasks and results.</li> <li>4) The V&amp;V software integration test plan shall address the following: <ul style="list-style-type: none"> <li>i) Conformance to an increasingly larger set of functional requirements at each stage of integration.</li> <li>ii) Assessment of timing, sizing, and accuracy.</li> <li>iii) Performance at boundaries and under stress conditions.</li> <li>iv) Measures of requirements test coverage and software reliability.</li> </ul> </li> <li>5) Verify that the V&amp;V software integration test plan satisfies the following criteria: <ul style="list-style-type: none"> <li>Conformance to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> </ul> </li> </ul>                                                                                                                                    | System/software requirements (SRS)<br>Interface requirements (IRS)<br>Software architecture<br>Software integration test plan from developer (integrity level 2) | V&V software integration test plan (integrity levels 4 and 3)<br>Task report(s)—<br>Review of software integration test plan (integrity level 2)<br>Anomaly report(s) |

| <b>9.4 Activity: Architecture definition V&amp;V (Software, 12207—Architecture Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                    |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Required inputs</b>                                                                                                                                                                                                                                                                                                    | <b>Required outputs</b>                                                                                                                                                                            |
| <p>6) Validate that the V&amp;V software integration test plan satisfies the following criteria:</p> <ul style="list-style-type: none"> <li>i) Traceable to the system requirements.</li> <li>ii) External consistency with the system requirements.</li> <li>iii) Internal consistency.</li> <li>iv) Test coverage of the system/software requirements.</li> <li>v) Appropriateness of test standards and methods used.</li> <li>vi) Conformance to expected results.</li> <li>vii) Feasibility of software qualification testing.</li> <li>viii) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs).</li> </ul> <p>b) Integrity level 2</p> <ul style="list-style-type: none"> <li>1) Verify that the developer's software integration test plan conforms to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>2) Validate that the developer's software integration test plan satisfies the following criteria: <ul style="list-style-type: none"> <li>i) Traceable to the system requirements.</li> <li>ii) External consistency with the system requirements.</li> <li>iii) Internal consistency.</li> <li>iv) Test coverage of the system/software requirements.</li> <li>v) Appropriateness of test standards and methods.</li> <li>vi) Conformance to expected results.</li> <li>vii) Feasibility of software qualification testing.</li> <li>viii) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs).</li> </ul> </li> </ul> <p>c) Integrity level 1</p> <p>There are no software integration test plan V&amp;V requirements.</p> |                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                    |
| <p><b>(8) <u>Software qualification test design V&amp;V</u></b></p> <p>a) Integrity levels 4 and 3</p> <ul style="list-style-type: none"> <li>1) Design tests for V&amp;V software qualification testing.</li> <li>2) Continue tracing required by the V&amp;V software qualification test plan. Verify that the V&amp;V software qualification test designs conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>3) Validate that the V&amp;V software qualification test designs satisfy the criteria in V&amp;V activity <a href="#">9.3, Task 5</a>.</li> </ul> <p>b) Integrity level 2</p> <ul style="list-style-type: none"> <li>1) Verify that the developer's designs for software qualification testing conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>2) Validate that the developer's software qualification test designs satisfy the criteria in V&amp;V activity <a href="#">9.3, Task 5</a>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>System/software requirements (SRS)</p> <p>Interface requirements (IRS)</p> <p>System architecture</p> <p>Software architecture</p> <p>User information</p> <p>V&amp;V software qualification test plan (integrity levels 4 and 3)</p> <p>Software qualification test design from the developer (integrity level 2)</p> | <p>V&amp;V software qualification test design(s) (integrity levels 4 and 3)</p> <p>Task report(s)—Review of software qualification test design(s) (integrity level 2)</p> <p>Anomaly report(s)</p> |

| <b>9.4 Activity: Architecture definition V&amp;V (Software, 12207—Architecture Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                              |                                                                                                                                                                           |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Required inputs</b>                                                                                                                                                                                                                                                       | <b>Required outputs</b>                                                                                                                                                   |
| c) Integrity level 1<br>There are no software integration test plan V&V requirements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                              |                                                                                                                                                                           |
| <b>(9) <a href="#">Software acceptance test design V&amp;V</a></b><br>a) Integrity levels 4 and 3 <ol style="list-style-type: none"> <li>Design tests for V&amp;V software acceptance testing.</li> <li>Continue tracing required by the V&amp;V software acceptance test plan. Verify that the V&amp;V software acceptance test designs conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>Validate that the V&amp;V software acceptance test designs satisfy the criteria in V&amp;V activity <a href="#">9.3, Task 6</a>.</li> </ol> b) Integrity level 2 <ol style="list-style-type: none"> <li>Verify that the acquirer's test designs for software acceptance testing conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>Validate that the acquirer's software acceptance test designs satisfy the criteria in V&amp;V activity <a href="#">9.3, Task 6</a>.</li> </ol> c) Integrity level 1<br>There are no software acceptance V&V test requirements. | System/software requirements (SRS)<br>Interface requirements (IRS)<br>System architecture<br>Software architecture<br>User information<br>V&V software acceptance test plan (integrity levels 4 and 3)<br>Software acceptance test design from developer (integrity level 2) | V&V software acceptance test design(s) (integrity levels 4 and 3)<br>Task report(s)—Review of software acceptance test design(s) (integrity level 2)<br>Anomaly report(s) |
| <b>(10) <a href="#">Hazard Analysis</a></b><br>a) Verify the logic design and associated data elements that implement critical requirements introduce no new hazards.<br>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of the system and software operations).<br>c) Update the hazard analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | System architecture<br>Software architecture<br>Hazard analysis results                                                                                                                                                                                                      | Task report(s)—Hazard analysis<br>Anomaly report(s)                                                                                                                       |
| <b>(11) <a href="#">Security analysis</a></b><br>a) Verify that the architecture and detailed design outputs adequately address the identified security requirements. This Verification includes both the system itself and security risks introduced as a result of interfacing with external components.<br>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of the system and software operations).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | System architecture<br>Software architecture<br>Security analysis results<br>V&V task results                                                                                                                                                                                | Task report(s)—Security analysis<br>Anomaly report(s)                                                                                                                     |
| <b>(12) <a href="#">Risk analysis</a></b><br>a) Review and update risk analysis using prior task reports.<br>b) Provide recommendations to eliminate, reduce, or mitigate the risks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | System architecture<br>Software architecture<br>Supplier development plans and schedules<br>Hazard analysis results<br>Security analysis results<br>V&V task results                                                                                                         | Task report(s)—Risk analysis<br>Anomaly report(s)                                                                                                                         |

| <b>9.5 Activity: Design definition V&amp;V (Software, 12207—Design Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                  |                                                                       |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Required inputs</b>                                                                                                                                                           | <b>Required outputs</b>                                               |
| <p><b>(1) <u>Design evaluation</u></b><br/>Evaluate the design elements (SDD and IDD) for correctness, consistency, completeness, accuracy, readability, and testability. The task criteria are as follows:</p> <p>a) Correctness</p> <ol style="list-style-type: none"> <li>1) Verify and validate the allocation of system/software requirements to the software design elements.</li> <li>2) Verify and validate that the software design satisfies the system/software requirements.</li> <li>3) Verify that the software design complies with standards, references, regulations, policies, physical laws, and business rules.</li> <li>4) Validate the design sequences of states and state changes using logic and data flows coupled with domain expertise, prototyping results, engineering principles, or other bases.</li> <li>5) Validate that the detailed design and its element interactions do not result in unnecessary, unintended, or deleterious consequences.</li> <li>6) Validate that the flow of data and control satisfy functionality and performance requirements.</li> <li>7) Validate data usage and format.</li> <li>8) Assess the appropriateness of design methods and standards used.</li> </ol> <p>b) Consistency</p> <ol style="list-style-type: none"> <li>1) Verify that all terms and design concepts are documented consistently.</li> <li>2) Verify that there is internal consistency between the design elements and external consistency with architectural design.</li> </ol> <p>c) Completeness</p> <ol style="list-style-type: none"> <li>1) Verify that the following elements are in the SDD, within the assumptions and constraints of the system: <ol style="list-style-type: none"> <li>i) Functionality (e.g., algorithms, state/mode definitions, input/output Validation, exception handling, reporting, and logging).</li> <li>ii) Process definition and scheduling.</li> <li>iii) Hardware, software, and user interface descriptions.</li> <li>iv) Performance criteria (e.g., timing, sizing, speed, capacity, accuracy, precision, safety, and security).</li> <li>v) Critical configuration data.</li> <li>vi) System, device, and software control (e.g., initialization, transaction and state monitoring, and self-testing).</li> </ol> </li> <li>2) Verify that the SDD and IDD satisfy specified configuration management procedures.</li> </ol> <p>d) Accuracy</p> <ol style="list-style-type: none"> <li>1) Validate that the logic, computational, and interface precision (e.g., truncation and rounding) satisfy the requirements in the system environment.</li> </ol> | <p>System/software requirements (SRS)</p> <p>Interface requirements (IRS)</p> <p>Software design (SDD)</p> <p>Design standards (e.g., standards, practices, and conventions)</p> | <p>Task report(s)—<br/>Design evaluation</p> <p>Anomaly report(s)</p> |

| <b>9.5 Activity: Design definition V&amp;V (Software, 12207—Design Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                         |                                                                              |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Required inputs</b>                                                                                                                                                                  | <b>Required outputs</b>                                                      |
| <p>2) Validate that the modeled physical phenomena conform to system accuracy requirements and physical laws.</p> <p>e) Readability</p> <p>1) Verify that the documentation is legible, understandable, and unambiguous to the intended audience.</p> <p>2) Verify that the documentation defines all acronyms, mnemonics, abbreviations, terms, symbols, and design language, if any.</p> <p>f) Testability</p> <p>1) Verify that there are objective acceptance criteria for validating each software design element and the system design.</p> <p>2) Verify that each software design element is testable to objective acceptance criteria.</p>                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                         |                                                                              |
| <p><b>(2) <u>Interface analysis</u></b></p> <p>Verify and validate that the software design interfaces with hardware, user, operator, software, and other systems for correctness, consistency, completeness, accuracy, and testability. The task criteria are as follows:</p> <p>a) Correctness</p> <p>Validate the external and internal software interface design in the context of system requirements.</p> <p>b) Consistency</p> <p>Verify that the interface design is consistent between the SDD and IDD.</p> <p>c) Completeness</p> <p>Verify that each interface is described and includes purpose, parameters, data format, and performance criteria (e.g., timing, bandwidth, accuracy, safety, and security).</p> <p>d) Accuracy</p> <p>Verify that each interface provides information with the required accuracy.</p> <p>e) Testability</p> <p>Verify that there are objective acceptance criteria for validating the interface design.</p> | <p>Business or mission analysis documentation</p> <p>System requirements</p> <p>System/software requirements (SRS)</p> <p>Interface requirements (IRS)</p> <p>Software design (SDD)</p> | <p>Task report(s)—</p> <p>Interface analysis</p> <p>Anomaly report(s)</p>    |
| <p><b>(3) <u>Traceability analysis</u></b></p> <p>Trace design elements (SDD and IDD) to requirements (SRS and IRS) and the requirements to design elements. Analyze relationships for correctness, consistency, and completeness. The task criteria are as follows:</p> <p>a) Correctness</p> <p>Validate the relationship between each design element and the software requirement(s).</p> <p>b) Consistency</p> <p>Verify that the relationships between the design elements and the system/software requirements are specified to a consistent level of detail.</p> <p>c) Completeness</p> <p>1) Verify that all design elements are traceable from the system/software requirements.</p>                                                                                                                                                                                                                                                             | <p>System/software requirements (SRS) Software design (SDD)</p> <p>Interface requirements (IRS)</p> <p>Interface design (IDD)</p>                                                       | <p>Task report(s)—</p> <p>Traceability analysis</p> <p>Anomaly report(s)</p> |



| <b>9.5 Activity: Design definition V&amp;V (Software, 12207—Design Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                  |                                                                                                                                                                          |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Required inputs</b>                                                                                                                                                                                                                                                           | <b>Required outputs</b>                                                                                                                                                  |
| 2) Verify that all system/software requirements are traceable to the design elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                  |                                                                                                                                                                          |
| <b>(4) <u>Criticality analysis</u></b><br>a) Review and update the existing criticality analysis results from the prior criticality task report using the SDD and IDD.<br>b) Implementation methods and interfacing technologies may cause previously assigned integrity levels to be raised or lowered for a given software element (i.e., requirement, module, function, subsystem, or other software partition). Verify that no inconsistent or undesired integrity consequences are introduced by reviewing the revised integrity levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Criticality task report<br>Software design (SDD)<br>Interface design (IDD)                                                                                                                                                                                                       | Task report(s)—<br>Criticality analysis<br>Anomaly report(s)                                                                                                             |
| <b>(5) <u>Software component test design V&amp;V</u></b><br>a) Integrity levels 4 and 3 <ol style="list-style-type: none"> <li>Design tests for V&amp;V software component testing.</li> <li>Continue tracing required by the V&amp;V software component test plan.</li> <li>Verify that the V&amp;V software component test designs conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>Validate that the V&amp;V software component test designs satisfy the criteria in V&amp;V activity <a href="#">9.4, Task 6</a>.</li> </ol> b) Integrity level 2 <ol style="list-style-type: none"> <li>Verify that the developer's test designs for software component testing conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>Validate that the developer's software component test designs satisfy the criteria in V&amp;V activity <a href="#">9.4, Task 6</a>.</li> </ol> c) Integrity level 1<br>There are no software component V&V test requirements. | Software design (SDD)<br>Interface design (IDD)<br>User documentation<br>V&V software component test plans (integrity levels 4 and 3)<br>Software component test design from the developer (integrity level 2)                                                                   | V&V software component test design(s) (integrity levels 4 and 3)<br>Task report(s)—<br>Review of software component test design (integrity level 2)<br>Anomaly report(s) |
| <b>(6) <u>Software component test case V&amp;V</u></b><br>a) Integrity levels 4 and 3 <ol style="list-style-type: none"> <li>Develop test cases for V&amp;V software component testing (e.g., statistical sampling, boundary conditions, and code coverage).</li> <li>Continue tracing required by the V&amp;V software component test plan.</li> <li>Verify that the software component test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>Validate that the software component test cases satisfy the criteria in V&amp;V activity <a href="#">9.5, Task 5</a>.</li> </ol> b) Integrity level 2 <ol style="list-style-type: none"> <li>Verify that the developer's software component test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> </ol>                                                                                                                                                                                      | System/software requirements (SRS)<br>Interface requirements (IRS)<br>Software design (SDD)<br>Interface design (IDD)<br>User documentation<br>V&V software component test design (integrity levels 4 and 3)<br>Software component test cases from developer (integrity level 2) | V&V software component test case(s) (integrity levels 4 and 3)<br>Task report(s)—<br>Review of software component test cases (integrity level 2)<br>Anomaly report(s)    |



| <b>9.5 Activity: Design definition V&amp;V (Software, 12207—Design Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Required inputs</b>                                                                                                                                                                                                                                                                                                      | <b>Required outputs</b>                                                                                                                                                                        |
| <p>2) Validate that the developer's software component test cases satisfy the criteria in V&amp;V activity <a href="#">9.5, Task 5</a>.</p> <p>c) Integrity level 1</p> <p>There are no software component V&amp;V test requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                |
| <p><b>(7) <a href="#">Software integration test design V&amp;V</a></b></p> <p>a) Integrity levels 4 and 3</p> <ol style="list-style-type: none"> <li>Design tests for V&amp;V software integration testing.</li> <li>Continue tracing required by the V&amp;V software integration test plan. Verify that the V&amp;V software integration test designs conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 <a href="#">[B17]</a>).</li> <li>Validate that the V&amp;V software integration test designs satisfy the criteria in V&amp;V activity <a href="#">9.4, Task 7</a>.</li> </ol> <p>b) Integrity level 2</p> <ol style="list-style-type: none"> <li>Verify that the developer's designs for software integration testing conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 <a href="#">[B17]</a>).</li> <li>Validate that the developer's software integration test designs satisfy the criteria in V&amp;V activity <a href="#">9.4, Task 7</a>.</li> </ol> <p>c) Integrity level 1</p> <p>There are no software integration test design V&amp;V requirements.</p> | <p>Software design (SDD)</p> <p>Interface design (IDD)</p> <p>User documentation</p> <p>V&amp;V software integration test plans (integrity levels 4 and 3)</p> <p>Software integration test design from the developer (integrity level 2)</p>                                                                               | <p>V&amp;V software integration test design(s) (integrity levels 4 and 3)</p> <p>Task report(s)—Review of software integration test design(s) (integrity level 2)</p> <p>Anomaly report(s)</p> |
| <p><b>(8) <a href="#">Software integration test case V&amp;V</a></b></p> <p>a) Integrity levels 4 and 3</p> <ol style="list-style-type: none"> <li>Develop test cases for V&amp;V software integration testing.</li> <li>Continue tracing required by the V&amp;V software integration test plan.</li> <li>Verify that the V&amp;V software integration test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 <a href="#">[B17]</a>).</li> <li>Validate that the V&amp;V software integration test cases satisfy the criteria in V&amp;V activity <a href="#">9.5, Task 7</a>.</li> </ol> <p>b) Integrity level 2</p> <ol style="list-style-type: none"> <li>Verify that the developer's software integration test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 <a href="#">[B17]</a>).</li> <li>Validate that the developer's software integration test cases satisfy the criteria in V&amp;V activity <a href="#">9.5, Task 7</a>.</li> </ol> <p>c) Integrity level 1</p> <p>There are no software integration test case V&amp;V requirements.</p>     | <p>System/software Requirements (SRS)</p> <p>Interface requirements (IRS)</p> <p>Software design (SDD)</p> <p>Interface design (IDD)</p> <p>User documentation</p> <p>V&amp;V software integration test design (integrity levels 4 and 3)</p> <p>Software integration test cases from the developer (integrity level 2)</p> | <p>V&amp;V software integration test cases (integrity levels 4 and 3)</p> <p>Task report(s)—Review of software integration test cases (integrity level 2)</p> <p>Anomaly report(s)</p>         |
| <p><b>(9) <a href="#">Software qualification test case V&amp;V</a></b></p> <p>a) Integrity levels 4 and 3</p> <ol style="list-style-type: none"> <li>Develop test cases for V&amp;V software qualification testing.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>System/software Requirements (SRS)</p> <p>Interface requirements (IRS)</p>                                                                                                                                                                                                                                               | <p>V&amp;V software qualification test cases (integrity levels 4 and 3)</p>                                                                                                                    |

| <b>9.5 Activity: Design definition V&amp;V (Software, 12207—Design Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                      |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Required inputs</b>                                                                                                                                                                                                                                                                                                    | <b>Required outputs</b>                                                                                                                                                              |
| <p>2) Continue tracing required by the V&amp;V software qualification test plan.</p> <p>3) Verify that the V&amp;V software qualification test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</p> <p>4) Validate that the V&amp;V software qualification test cases satisfy the criteria in V&amp;V activity <a href="#">9.4, Task 8</a></p> <p>b) Integrity level 2</p> <p>1) Verify that the developer's software qualification test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</p> <p>2) Validate that the developer's software qualification test cases satisfy the criteria in V&amp;V activity <a href="#">9.4, Task 8</a>.</p> <p>c) Integrity level 1</p> <p>There are no software qualification test case V&amp;V requirements.</p>                                                                                                                                                           | <p>Software design (SDD)</p> <p>Interface design (IDD)</p> <p>User documentation</p> <p>V&amp;V software qualification test design (integrity levels 4 and 3)</p> <p>Software qualification test cases from the developer (integrity level 2)</p>                                                                         | <p>Task report(s)—Review of software qualification test cases (integrity level 2)</p> <p>Anomaly report(s)</p>                                                                       |
| <p><b>(10) <a href="#">Software acceptance test case V&amp;V</a></b></p> <p>a) Integrity levels 4 and 3</p> <p>1) Develop test cases for V&amp;V software acceptance testing.</p> <p>2) Continue tracing required by the V&amp;V software acceptance test plan.</p> <p>3) Verify that the V&amp;V software acceptance test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</p> <p>4) Validate that the V&amp;V software acceptance test cases satisfy the criteria in V&amp;V activity <a href="#">9.4, Task 9</a>.</p> <p>b) Integrity level 2</p> <p>1) Verify that the acquirer's software acceptance test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</p> <p>2) Validate that the acquirer's software acceptance test cases satisfy the criteria in V&amp;V activity <a href="#">9.4, Task 9</a>.</p> <p>c) Integrity level 1</p> <p>There are no software acceptance V&amp;V test requirements.</p> | <p>System/software Requirements (SRS)</p> <p>Interface requirements (IRS)</p> <p>Software design (SDD)</p> <p>Interface design (IDD)</p> <p>User documentation</p> <p>V&amp;V software acceptance test design (integrity levels 4 and 3)</p> <p>Software acceptance test cases from the developer (integrity level 2)</p> | <p>V&amp;V software acceptance test cases (integrity levels 4 and 3)</p> <p>Task report(s)—Review of software acceptance test cases (integrity level 2)</p> <p>Anomaly report(s)</p> |
| <p><b>(11) <a href="#">Hazard analysis</a></b></p> <p>a) Verify the logic design and associated data elements that implement critical requirements introduce no new hazards.</p> <p>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of the system and software operations).</p> <p>c) Update the hazard analysis.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>Software design (SDD)</p> <p>Interface design (IDD)</p> <p>Hazard analysis results</p>                                                                                                                                                                                                                                 | <p>Task report(s)—Hazard analysis</p> <p>Anomaly report(s)</p>                                                                                                                       |

| <b>9.5 Activity: Design definition V&amp;V (Software, 12207—Design Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                  |                                                           |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Required inputs</b>                                                                                                                                                                           | <b>Required outputs</b>                                   |
| <b>(12) <a href="#">Security analysis</a></b><br>a) Verify that the architecture and detailed design outputs adequately address the identified security requirements. This Verification includes both the system itself and security risks introduced as a result of interfacing with external components.<br>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of the system and software operations). | Software design (SDD)<br>Interface design (IDD)<br>Security analysis results<br>V&V task results                                                                                                 | Task report(s)—<br>Security analysis<br>Anomaly report(s) |
| <b>(13) <a href="#">Risk analysis</a></b><br>a) Review and update risk analysis using prior task reports.<br>b) Provide recommendations to eliminate, reduce, or mitigate the risks.                                                                                                                                                                                                                                                                                                                                                          | Software design (SDD)<br>Interface design (IDD)<br>Supplier development plans and schedules<br>Hazard analysis results<br>Security analysis results<br>Risk analysis results<br>V&V task results | Task report(s)—<br>Risk analysis<br>Anomaly report(s)     |

| <b>9.6 Activity: System analysis V&amp;V (Software, 12207—System Analysis Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                     |                                                        |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Required inputs</b>                              | <b>Required outputs</b>                                |
| <b>(1) <a href="#">System analysis strategy evaluation</a></b><br>Evaluate the system analysis strategy for completeness and for consistency with the criticality of the software. The task criteria are as follows:<br>a) Completeness<br>1) Verify that the scope and objectives of the system analysis address the correct needs.<br>2) Verify that the scope of the system analysis addresses all aspects related to the decision.<br>3) Verify that all assumptions have been identified and are correct.<br>4) Verify that any inputs for enabling systems or services have been identified and are available.<br>5) Verify that any inputs for enabling the system have been identified and are available in the system analysis strategy plan.<br>b) Consistency with Criticality<br>Verify that the rigor and approach specified in the system analysis strategy are consistent with that needed for the software elements that could be affected by the decision. | System analysis strategy                            | Task report(s)—<br>System analysis strategy evaluation |
| <b>(2) <a href="#">System analysis results evaluation</a></b><br>Evaluate the results of the system analysis to determine that the analysis was conducted as specified in the strategy, and the results support its conclusions and recommendations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | System analysis strategy<br>System analysis results | Task report(s)—<br>System analysis results evaluation  |

| <b>9.7 Activity: Implementation V&amp;V (Software, 12207—Implementation Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                     |                                                                                                     |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Required inputs</b>                                                                                                                                                                              | <b>Required outputs</b>                                                                             |
| <p><b>(1) <u>Source code and source code documentation evaluation</u></b><br/>Evaluate the source code components (source code and source code documentation) for correctness, consistency, completeness, accuracy, readability, and testability. The task criteria are as follows:</p> <p>a) Correctness</p> <ol style="list-style-type: none"> <li>1) Verify and validate that the source code component satisfies the software design.</li> <li>2) Verify that the source code components comply with standards, references, regulations, policies, physical laws, and business rules.</li> <li>3) Validate the source code component sequences of states and state changes using logic and data flows coupled with domain expertise, prototyping results, engineering principles, or other basis.</li> <li>4) Validate the software code and its interactions with other elements do not result in unnecessary, unintended, or deleterious consequences.</li> <li>5) Validate that the flow of data and control satisfy functionality and performance requirements.</li> <li>6) Validate data usage and format.</li> <li>7) Assess the appropriateness of coding methods and standards.</li> </ol> <p>b) Consistency</p> <ol style="list-style-type: none"> <li>1) Verify that all terms and code concepts are documented consistently.</li> <li>2) Verify that there is internal consistency between the source code components.</li> <li>3) Validate external consistency with the software design and requirements.</li> </ol> <p>c) Completeness</p> <ol style="list-style-type: none"> <li>1) Verify that the following elements are in the source code, within the assumptions and constraints of the system: <ol style="list-style-type: none"> <li>i) Functionality (e.g., algorithms, state/mode definitions, input/output Validation, exception handling, reporting, and logging).</li> <li>ii) Process definition and scheduling.</li> <li>iii) Hardware, software, and user interface descriptions.</li> <li>iv) Performance criteria (e.g., timing, sizing, speed, capacity, accuracy, precision, safety, and security).</li> <li>v) Critical configuration data.</li> <li>vi) System, device, and software control (e.g., initialization, transaction, and state monitoring, defensive programming practices, and self-testing).</li> </ol> </li> <li>2) Verify that the source code documentation satisfies specified coding standards.</li> </ol> <p>d) Accuracy</p> <ol style="list-style-type: none"> <li>1) Validate the logic, computational, and interface precision (e.g., truncation and rounding) in the system environment.</li> <li>2) Validate that the modeled physical phenomena conform to system accuracy requirements and physical laws.</li> </ol> <p>e) Readability</p> | <p>Source code</p> <p>Software design (SDD)</p> <p>Interface design (IDD)</p> <p>Coding standards (e.g., standards, practices, project restrictions, and conventions)</p> <p>User documentation</p> | <p>Task report(s)—Source code and source code documentation evaluation</p> <p>Anomaly report(s)</p> |

| <b>9.7 Activity: Implementation V&amp;V (Software, 12207—Implementation Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                           |                                                               |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Required inputs</b>                                                                                                                                    | <b>Required outputs</b>                                       |
| 1) Verify that the documentation is legible, understandable, and unambiguous to the intended audience.<br>2) Verify that the documentation defines all acronyms, mnemonics, abbreviations, terms, and symbols.<br>f) Testability<br>1) Verify that there are objective acceptance criteria for validating each source code component.<br>2) Verify that each source code component is testable against objective acceptance criteria.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                           |                                                               |
| <b>(2) <u>Interface analysis</u></b><br>Verify and validate that the software source code interfaces with hardware, user, operator, software, and other systems for correctness, consistency, completeness, accuracy, and testability. The task criteria are as follows:<br>a) Correctness<br>Validate the external and internal software interface code in the context of system requirements.<br>b) Consistency<br>Verify that the interface code is consistent between source code components and to external interfaces (i.e., hardware, user, operator, and other software).<br>c) Completeness<br>Verify that each interface is described and includes data format and performance criteria (e.g., timing, bandwidth, accuracy, safety, and security).<br>d) Accuracy<br>Verify that each interface provides information with the required accuracy.<br>e) Testability<br>Verify that there are objective acceptance criteria for validating the interface code. | Business or mission analysis documentation<br>System requirements<br>Software design (SDD)<br>Interface design (IDD)<br>Source code<br>User documentation | Task report(s)—<br>Interface analysis<br>Anomaly report(s)    |
| <b>(3) <u>Traceability analysis</u></b><br>a) Trace the source code components to corresponding design specification(s), and design specification(s) to source code components.<br>b) Analyze identified relationships for correctness, consistency, and completeness. The task criteria are as follows:<br>1) Correctness<br>Validate the relationship between the source code components and design element(s).<br>2) Consistency<br>Verify that the relationships between the source code components and design elements are specified to a consistent level of detail.<br>3) Completeness<br>i) Verify that all source code components are traceable from the design elements.<br>ii) Verify that all design elements are traceable to the source code components.                                                                                                                                                                                                 | Software design (SDD)<br>Interface design (IDD)<br>Source code                                                                                            | Task report(s)—<br>Traceability analysis<br>Anomaly report(s) |
| <b>(4) <u>Criticality analysis</u></b><br>a) Review and update the existing criticality analysis results from the prior criticality task report using the source code.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Criticality report<br>Source code                                                                                                                         | Task report(s)—<br>Criticality analysis<br>Anomaly report(s)  |

| <b>9.7 Activity: Implementation V&amp;V (Software, 12207—Implementation Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                    |                                                                                                                                                                                              |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Required inputs</b>                                                                                                                                                                                                                             | <b>Required outputs</b>                                                                                                                                                                      |
| b) Implementation methods and interfacing technologies may cause previously assigned integrity levels to be raised or lowered for a given software element (i.e., requirement, module, function, subsystem, or other software partition). Verify that no inconsistent or undesired integrity consequences are introduced by reviewing the revised integrity levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                    |                                                                                                                                                                                              |
| <b>(5) <u>Software component test procedure V&amp;V</u></b><br>a) Integrity levels 4 and 3 <ol style="list-style-type: none"> <li>Develop test procedures for V&amp;V software component testing.</li> <li>Continue tracing required by the V&amp;V software component test plan.</li> <li>Verify that the V&amp;V software component test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>Validate that the V&amp;V software component test procedures satisfy the criteria in V&amp;V activity <a href="#">9.5, Task 6</a>.</li> </ol> b) Integrity level 2 <ol style="list-style-type: none"> <li>Verify that the developer's software component test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</li> <li>Validate that the developer's software component test procedures satisfy the criteria in V&amp;V activity <a href="#">9.5, Task 6</a>.</li> </ol> c) Integrity level 1<br>There are no software component V&V test requirements. | Software requirements (SRS)<br>Interface requirements (IRS)<br>Software design (SDD)<br>Interface design (IDD)<br>User documentation<br>Software component test cases<br>Software component test procedures from the developer (integrity level 2) | V&V software component test procedures (integrity levels 4 and 3)<br>Task report(s)—Review of software component test procedures (integrity level 2)<br>Anomaly report(s)                    |
| <b>(6) <u>Software component test execution V&amp;V</u></b><br>a) Integrity levels 4 and 3 <ol style="list-style-type: none"> <li>Perform V&amp;V software component testing.</li> <li>Analyze test results to validate that the software correctly implements the design.</li> <li>Validate that the test results trace to test criteria established by the test traceability in the test planning documents.</li> <li>Document the results as required by the V&amp;V software component test plan.</li> <li>Use the V&amp;V software component test results to validate that the software satisfies the test acceptance criteria.</li> <li>Document discrepancies between the actual and expected test results.</li> </ol> b) Integrity level 2<br>Use the developer's software component test results to validate that the software satisfies the test acceptance criteria.<br>c) Integrity level 1<br>There are no software component V&V test requirements.                                                                                                                               | Source code<br>Executable code<br>Software design (SDD)<br>Interface design (IDD)<br>Software component test procedures<br>Software component test results<br>Software component test results from the developer (integrity level 2)               | Task report(s)—V&V software component test results (integrity levels 4 and 3)<br>Task report(s)—Review of software component test execution results (integrity level 2)<br>Anomaly report(s) |
| <b>(7) <u>Software integration test procedure V&amp;V</u></b><br>a) Integrity levels 4 and 3 <ol style="list-style-type: none"> <li>Develop test procedures for V&amp;V software integration testing.</li> <li>Continue tracing required by the V&amp;V software integration test plan.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Software requirements (SRS)<br>Interface requirements (IRS)<br>Software design (SDD)<br>Interface design (IDD)                                                                                                                                     | V&V software integration test procedures (integrity levels 4 and 3)<br>Task report(s)—Review of software integration test                                                                    |

| <b>9.7 Activity: Implementation V&amp;V (Software, 12207—Implementation Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                      |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Required inputs</b>                                                                                                                                                                                                                                                                                                                  | <b>Required outputs</b>                                                                                                                                                                              |
| <p>3) Verify that the V&amp;V software integration test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</p> <p>4) Validate that the V&amp;V software integration test procedures satisfy the criteria in V&amp;V activity <a href="#">9.5, Task 8</a>.</p> <p>b) Integrity level 2</p> <p>1) Verify that the developer's software integration test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</p> <p>2) Validate that the developer's software integration test procedures satisfy the criteria in V&amp;V activity <a href="#">9.5, Task 8</a>.</p> <p>c) Integrity level 1</p> <p>There are no software integration test procedure V&amp;V requirements.</p>                                                                                                                                                                                                                                                                                                 | <p>User documentation</p> <p>Software integration test cases</p> <p>Software integration test procedures</p> <p>Software integration test procedures from the developer (integrity level 2)</p>                                                                                                                                         | <p>procedures (integrity level 2)</p> <p>Anomaly report(s)</p>                                                                                                                                       |
| <p><b>(8) <a href="#">Software qualification test procedure V&amp;V</a></b></p> <p>a) Integrity levels 4 and 3</p> <p>1) Develop test procedures for V&amp;V software qualification testing.</p> <p>2) Continue tracing required by the V&amp;V software qualification test plan.</p> <p>3) Verify that the V&amp;V software qualification test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</p> <p>4) Validate that the V&amp;V software qualification test procedures satisfy the criteria in V&amp;V activity <a href="#">9.5, Task 9</a>.</p> <p>b) Integrity level 2</p> <p>1) Verify that the developer's software qualification test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</p> <p>2) Validate that the developer's software qualification test procedures satisfy the criteria in V&amp;V activity <a href="#">9.5, Task 9</a>.</p> <p>c) Integrity level 1</p> <p>There are no software qualification test procedure V&amp;V requirements.</p> | <p>Software requirements (SRS)</p> <p>Interface requirements (IRS)</p> <p>Software design (SDD)</p> <p>Interface design (IDD)</p> <p>User documentation</p> <p>Software qualification test cases</p> <p>Software qualification test procedures</p> <p>Software qualification test procedures from the developer (integrity level 2)</p> | <p>V&amp;V software qualification test procedures (integrity levels 4 and 3)</p> <p>Task report(s)—Review of software qualification test procedures (integrity level 2)</p> <p>Anomaly report(s)</p> |
| <p><b>(9) <a href="#">Software acceptance test procedure V&amp;V</a></b></p> <p>a) Integrity levels 4 and 3</p> <p>1) Develop test procedures for V&amp;V software acceptance testing.</p> <p>2) Continue the tracing required by the V&amp;V software acceptance test plan.</p> <p>3) Verify that the V&amp;V software acceptance test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</p> <p>4) Validate that the V&amp;V software acceptance test procedures satisfy the criteria in the V&amp;V software acceptance test cases (V&amp;V activity <a href="#">9.5, Task 10</a>).</p> <p>b) Integrity level 2</p> <p>1) Verify that the acquirer's software acceptance test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]).</p>                                                                                                                                                                                                                                   | <p>System/software Requirements (SRS)</p> <p>Interface requirements (IRS)</p> <p>Software design (SDD)</p> <p>Interface design (IDD)</p> <p>User documentation</p> <p>V&amp;V software acceptance test procedures (integrity levels 4 and 3)</p> <p>Software acceptance test procedures from developer (integrity level 2)</p>          | <p>V&amp;V hardware acceptance test procedures (integrity levels 4 and 3)</p> <p>Task report(s)—Software acceptance test procedure V&amp;V (integrity level 2)</p> <p>Anomaly report(s)</p>          |



| <b>9.7 Activity: Implementation V&amp;V (Software, 12207—Implementation Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                              |                                                       |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Required inputs</b>                                                                                                                                       | <b>Required outputs</b>                               |
| 2) Validate that the acquirer's software acceptance test procedures satisfy the criteria in the V&V software acceptance test cases (V&V activity <a href="#">9.5, Task 10</a> ).<br>c) Integrity level 1<br>There are no software acceptance V&V test requirements.                                                                                                                                                                                                                                                                     |                                                                                                                                                              |                                                       |
| <b>(10) <a href="#">Hazard analysis</a></b><br>a) Verify that the implementation and associated data elements correctly implement the critical requirements and introduce no new hazards.<br>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of the system and software operations).<br>c) Update the hazard analysis.                                                                                        | Source code<br>Software design (SDD)<br>Interface design (IDD)<br>Hazard analysis results                                                                    | Task report(s)—Hazard analysis<br>Anomaly report(s)   |
| <b>(11) <a href="#">Security analysis</a></b><br>a) Verify that the implementation is completed in accordance with the system design in that it addresses the identified security risks, and that the implementation does not introduce new security risks through coding flaws, or compiler errors.<br>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of the system and software operations). | Source code<br>Software design (SDD)<br>Interface design (IDD)<br>Security analysis results                                                                  | Task report(s)—Security analysis<br>Anomaly report(s) |
| <b>(12) <a href="#">Risk analysis</a></b><br>Review and update risk analysis using prior task reports. Provide recommendations to eliminate, reduce, or mitigate the risks.                                                                                                                                                                                                                                                                                                                                                             | Source code<br>Supplier development plans and schedules<br>Hazard analysis results<br>Security analysis results<br>Risk analysis results<br>V&V task results | Task report(s)—Risk analysis<br>Anomaly report(s)     |

| <b>9.8 Activity: Integration V&amp;V (Software, 12207—Integration Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                              |                                                                                                                              |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Required inputs</b>                                                                       | <b>Required outputs</b>                                                                                                      |
| <b>(1) <a href="#">Software integration strategy assessment</a></b><br>a) Verify the software integration strategy.<br>b) Verify that any constraints specified or implied by the software integration strategy are included in the system/software requirements and design.                                                                                                                                                                                                                                                                   | Software integration strategy<br>Software architecture<br>System/software requirements (SRS) | Task report(s)—Software integration strategy assessment<br>Anomaly report(s)                                                 |
| <b>(2) <a href="#">Software element interaction analysis</a></b><br>Analyze the results of system integration V&V to determine whether directly or indirectly coupled software elements require their V&V tasks to be reiterated (repeated, redone) or new V&V tasks performed. As integration results are determined, V&V may have to be recursively applied to all directly or indirectly coupled system elements. The Software Analysis Process may be used to establish the criteria of how to recursively apply V&V to software elements. | Software integration results<br>Software architecture<br>Software design (SDD)               | Task report(s)—Software element interaction analysis<br>Anomaly report(s)<br>V&V plan update (if tasks are added or changed) |



| <b>9.8 Activity: Integration V&amp;V (Software, 12207—Integration Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                     |                                                                                                                                                                                                                         |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Required inputs</b>                                                                                                                                                              | <b>Required outputs</b>                                                                                                                                                                                                 |
| <p><b>(3) <u>Software integration test execution V&amp;V</u></b></p> <p>a) Integrity levels 4 and 3</p> <ol style="list-style-type: none"> <li>1) Perform V&amp;V software integration testing.</li> <li>2) Analyze test results to verify that the software components are integrated correctly.</li> <li>3) Validate that the test results trace to test criteria established by the test traceability in the test planning documents.</li> <li>4) Document the results as required by the V&amp;V software integration test plan.</li> <li>5) Use the V&amp;V software integration test results to validate that the software satisfies the test acceptance criteria.</li> <li>6) Document discrepancies between the actual and expected test results.</li> </ol> <p>b) Integrity level 2</p> <p>Use the developer's software integration test results to verify that the software satisfies the test acceptance criteria.</p> <p>c) Integrity level 1</p> <p>There are no software integration test execution V&amp;V requirements.</p>             | <p>Source code</p> <p>Executable code</p> <p>Software integration test procedures</p> <p>Software integration test results from the developer (integrity level 2)</p>               | <p>Task report(s)—V&amp;V software integration test results (integrity levels 4 and 3)</p> <p>Task report(s)—Review of software integration test execution results (integrity level 2)</p> <p>Anomaly report(s)</p>     |
| <p><b>(4) <u>Software qualification test execution V&amp;V</u></b></p> <p>a) Integrity levels 4 and 3</p> <ol style="list-style-type: none"> <li>1) Perform V&amp;V software qualification testing.</li> <li>2) Analyze test results to validate that the software satisfies the system requirements.</li> <li>3) Validate that the test results trace to test criteria established by the test traceability in the test planning documents.</li> <li>4) Document the results as required by the V&amp;V software qualification test plan.</li> <li>5) Use the V&amp;V software qualification test results to validate that the software satisfies the test acceptance criteria.</li> <li>6) Document discrepancies between the actual and expected test results.</li> </ol> <p>b) Integrity level 2</p> <p>Use the developer's software qualification test results to verify that the software satisfies the test acceptance criteria.</p> <p>c) Integrity level 1</p> <p>There are no software qualification test execution V&amp;V requirements.</p> | <p>Source code</p> <p>Executable code</p> <p>Software qualification test procedures</p> <p>Software qualification test execution results from the developer (integrity level 2)</p> | <p>Task report(s)—V&amp;V software qualification test results (integrity levels 4 and 3)</p> <p>Task report(s)—Review of software qualification test execution results (integrity level 2)</p> <p>Anomaly report(s)</p> |
| <p><b>(5) <u>Traceability analysis</u></b></p> <p>Analyze relationships in the V&amp;V test plans, designs, cases, and procedures for correctness and completeness. The task criteria are as follows:</p> <p>a) Correctness</p> <p>Verify that there is a valid relationship between the V&amp;V test plans, designs, cases, and procedures.</p> <p>b) Completeness</p> <p>Verify that all V&amp;V test procedures are traceable to the V&amp;V test plans.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>V&amp;V test plans</p> <p>V&amp;V test designs</p> <p>V&amp;V test cases</p> <p>V&amp;V test procedures</p>                                                                      | <p>Task report(s)—Traceability analysis</p> <p>Anomaly report(s)</p>                                                                                                                                                    |
| <p><b>(6) <u>Hazard analysis</u></b></p> <p>a) Verify that the test instrumentation does not introduce new hazards.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Source code</p> <p>Executable code</p> <p>Test results</p>                                                                                                                       | <p>Task report(s)—Hazard analysis</p> <p>Anomaly report(s)</p>                                                                                                                                                          |

| <b>9.8 Activity: Integration V&amp;V (Software, 12207—Integration Process)</b>                                                                                                                                                                                                                                                                                |                                                                                                                                               |                                                           |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                          | <b>Required inputs</b>                                                                                                                        | <b>Required outputs</b>                                   |
| b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of the system and software operations).<br>c) Update the hazard analysis.                                                                                                           | Hazard analysis results                                                                                                                       |                                                           |
| <b>(7) <a href="#">Security analysis</a></b><br>a) Verify that the implemented system does not increase the security risk.<br>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of the system and software operations). | Source code<br>Executable code<br>Security analysis results                                                                                   | Task report(s)—<br>Security analysis<br>Anomaly report(s) |
| <b>(8) <a href="#">Risk analysis</a></b><br>a) Review and update risk analysis using prior task reports.<br>b) Provide recommendations to eliminate, reduce, or mitigate the risks.                                                                                                                                                                           | Supplier development plans and schedules<br>Hazard analysis results<br>Security analysis results<br>Risk analysis results<br>V&V task results | Task report(s)—<br>Risk analysis<br>Anomaly report(s)     |

| <b>9.9 Activity: Verification (Software, 12207—Verification Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                  |                                                                                                                    |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Required inputs</b>                                                                                           | <b>Required outputs</b>                                                                                            |
| The activities and tasks for the Verification Process are conducted in software technical life cycle processes and are contained in <a href="#">Table 1c</a> , <a href="#">Activity 9.1</a> (business or mission analysis V&V), <a href="#">Activity 9.2</a> (stakeholder needs and requirements definition V&V), <a href="#">Activity 9.3</a> (system/software requirements V&V), <a href="#">Activity 9.4</a> (architecture definition V&V), <a href="#">Activity 9.5</a> (design definition V&V), <a href="#">Activity 9.6</a> (system analysis V&V), <a href="#">Activity 9.7</a> (implementation V&V), <a href="#">Activity 9.8</a> (integration V&V), <a href="#">Activity 9.10</a> (transition V&V), <a href="#">Activity 9.12</a> (operation V&V), <a href="#">Activity 9.13</a> (maintenance V&V), and <a href="#">Activity 9.14</a> (disposal V&V). The software Verification activities and tasks in <a href="#">Table 1c</a> and the common Verification activities and tasks in <a href="#">Table 1a</a> represent all activities and tasks needed to perform software Verification. | The required inputs for the Software Verification Process are found in <a href="#">Table 1c</a> required inputs. | The required outputs for the software Verification Process are found in <a href="#">Table 1c</a> required outputs. |

| <b>9.10 Activity: Transition V&amp;V (Software, 12207—Transition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                              |                                                                        |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Required inputs</b>                                                                                                                                                                                       | <b>Required outputs</b>                                                |
| <b>(1) <a href="#">Transition strategy evaluation</a></b><br>a) Verify that the transition strategy is comprehensive. Verify that the transition strategy includes the following: <ol style="list-style-type: none"> <li>1) All the software parts and the system (e.g., hardware [HW]/software [SW] functionality and databases) are considered in the transition execution.</li> <li>2) Transition schedule and sequence.</li> <li>3) Identification of transition hardware and software tools, equipment, and instructions.</li> <li>4) Archiving software artifacts, such as documentation and code.</li> </ol> | Transition strategy<br>Installation package (e.g., user documentation, system requirements, software design (SDD), interface design (IDD), system/software requirements (SRS), interface requirements (IRS), | Task report(s)—<br>Transition strategy evaluation<br>Anomaly report(s) |

| <b>9.10 Activity: Transition V&amp;V (Software, 12207—Transition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                   |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Required inputs</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Required outputs</b>                                                                                                                                                                                                                           |
| <ul style="list-style-type: none"> <li>5) Impact of interacting systems in terms of transition timing and transition impacts.</li> <li>6) Continuity of capabilities when replacing or upgrading a legacy system.</li> <li>7) Site preparation for installation and legacy system retirement, storage, and/or incorporation.</li> <li>8) Provisions for documentation of the process results.</li> <li>9) A fallback position, in case of transition failure (e.g., going back to the old system) and other risk mitigation considerations.</li> <li>b) Verify that the transition strategy has a defined approach to establishing the software in the operational environment that is consistent with stakeholder requirements (e.g., by running the old and new systems in parallel or by well-defined software characteristics that can be compared).</li> </ul> | <ul style="list-style-type: none"> <li>mission or business analysis</li> <li>documentation, installation procedures, site-specific parameters, installation tests, and configuration management data)</li> </ul>                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                   |
| <p><b>(2) <u>Transition demonstration assessment</u></b></p> <ul style="list-style-type: none"> <li>a) Validate that acceptance tests defined in the acquirer agreement (as documented in the acceptance test plan) demonstrate satisfactory installation at the specified location(s). Where the exact location or environment of operation is not available, a representative example is selected.</li> <li>b) Verify and validate that the installed software elements and related site-specific parameters are identical to the final as-verified and as-validated software elements.</li> </ul>                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Transitioning software (operational software under test)</li> <li>User documentation</li> <li>Installation package (e.g., user documentation, system requirements, software design (SDD), interface design (IDD), system/software requirements (SRS), interface requirements (IRS), business or mission analysis documentation, installation procedures, site-specific parameters, installation tests, and configuration management data)</li> <li>Acquirer agreement</li> <li>Acceptance test plan</li> </ul> | <ul style="list-style-type: none"> <li>Task report(s)—Transition demonstration assessment</li> <li>Anomaly report(s)</li> </ul>                                                                                                                   |
| <p><b>(3) <u>Software acceptance test execution V&amp;V</u></b></p> <ul style="list-style-type: none"> <li>a) Integrity levels 4 and 3 <ul style="list-style-type: none"> <li>1) Perform V&amp;V software acceptance testing.</li> <li>2) Analyze test results to validate that the software satisfies the system requirements.</li> <li>3) Validate that the test results trace to test criteria established by the test traceability in the V&amp;V software acceptance test planning documents.</li> <li>4) Document the results as required by the V&amp;V software acceptance test plan.</li> <li>5) Use the V&amp;V software acceptance test results to validate that the software satisfies the V&amp;V test acceptance criteria.</li> </ul> </li> </ul>                                                                                                     | <ul style="list-style-type: none"> <li>Source code</li> <li>Executable code</li> <li>User documentation</li> <li>Software acceptance test procedures</li> <li>Software acceptance test results from the developer (integrity level 2)</li> <li>V&amp;V task results</li> </ul>                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>V&amp;V software acceptance test results (integrity levels 4 and 3)</li> <li>Task report(s)—Review of software acceptance test execution results (integrity level 2)</li> <li>Anomaly report(s)</li> </ul> |

| <b>9.10 Activity: Transition V&amp;V (Software, 12207—Transition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                    |                                                                          |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Required inputs</b>                                                                                                                                                                                                             | <b>Required outputs</b>                                                  |
| <p>6) Document discrepancies between the actual and expected test results.</p> <p>b) Integrity level 2<br/>Use the acquirer's software acceptance test results to verify that the software satisfies the test acceptance criteria.</p> <p>c) Integrity level 1<br/>There are no software acceptance V&amp;V test requirements.</p>                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                    |                                                                          |
| <p><b>(4) <u>Installation configuration audit</u></b></p> <p>a) Verify that all software products required to correctly install and operate the software are present in the installation package.</p> <p>b) Validate that all site-dependent parameters or conditions to verify supplied values are correct.</p>                                                                                                                                                                                                                                                                                                                                                   | Installation package (e.g., source code, executable code, user documentation, SDD, IDD, SRS, IRS, concept documentation, installation procedures, site-specific parameters, installation tests, and configuration management data) | Task report(s)—<br>Installation configuration audit<br>Anomaly report(s) |
| <p><b>(5) <u>Installation checkout</u></b></p> <p>a) Conduct analyses or tests to verify that the installed software corresponds to the software subjected to V&amp;V.</p> <p>b) Verify that the software code and databases initialize, execute, and terminate as specified.</p> <p>c) In the transition from one version of software to the next, validate that the software can be replaced with the new version without adversely affecting or degrading the functionality of the remaining system components.</p> <p>d) Verify the requirements for continuous operation and service during the transition, including requirements for user notification.</p> | Installation package                                                                                                                                                                                                               | Task report(s)—<br>Installation checkout<br>Anomaly report(s)            |
| <p><b>(6) <u>Hazard analysis</u></b></p> <p>a) Verify that the installation procedures and installation environment do not introduce new hazards.</p> <p>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of the system and software operations).</p> <p>c) Update the hazard analysis.</p>                                                                                                                                                                                                                                               | Installation package<br>Hazard analysis results                                                                                                                                                                                    | Task report(s)—<br>Hazard analysis<br>Anomaly report(s)                  |
| <p><b>(7) <u>Security analysis</u></b></p> <p>a) Verify that the installed software does not introduce new or increased vulnerabilities or security risks to the overall system.</p> <p>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of the system and software operations).</p>                                                                                                                                                                                                                                        | Installation package<br>Security analysis results                                                                                                                                                                                  | Task report(s)—<br>Security analysis<br>Anomaly report(s)                |
| <p><b>(8) <u>Risk analysis</u></b></p> <p>a) Review and update risk analysis using prior task reports.</p> <p>b) Provide recommendations to eliminate, reduce, or mitigate the risks.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Installation package<br>Supplier development plans and schedules<br>Hazard analysis results                                                                                                                                        | Task report(s)—<br>Risk analysis<br>Anomaly report(s)                    |

| <b>9.10 Activity: Transition V&amp;V (Software, 12207—Transition Process)</b> |                                                                        |                         |
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| <b>V&amp;V tasks</b>                                                          | <b>Required inputs</b>                                                 | <b>Required outputs</b> |
|                                                                               | Security analysis results<br>Risk analysis results<br>V&V task results |                         |

| <b>9.11 Activity: Software Validation (Software, 12207—Software Validation Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                |                                                                                                                  |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Required inputs</b>                                                                                         | <b>Required outputs</b>                                                                                          |
| The activities and tasks for the software Validation Process are conducted in software technical life cycle processes and are contained in <a href="#">Table 1c</a> , <a href="#">Activity 9.1</a> (business or mission analysis V&V), <a href="#">Activity 9.2</a> (stakeholder needs and requirements definition V&V), <a href="#">Activity 9.3</a> (system/software requirements V&V), <a href="#">Activity 9.4</a> (architecture definition V&V), <a href="#">Activity 9.5</a> (design definition V&V), <a href="#">Activity 9.6</a> (system analysis V&V), <a href="#">Activity 9.7</a> (implementation V&V), <a href="#">Activity 9.8</a> (integration V&V), <a href="#">Activity 9.10</a> (transition V&V), <a href="#">Activity 9.12</a> (operation V&V), <a href="#">Activity 9.13</a> (maintenance V&V), and <a href="#">Activity 9.14</a> (disposal V&V). The software Validation activities and tasks in <a href="#">Table 1c</a> and the common Validation activities and tasks in <a href="#">Table 1a</a> represent all activities and tasks needed to perform software Validation. | The required inputs for the software Validation Process are found in <a href="#">Table 1c</a> required inputs. | The required outputs for the software Validation Process are found in <a href="#">Table 1c</a> required outputs. |

| <b>9.12 Activity: Operation V&amp;V (Software, 12207—Operation Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                               |                                                                         |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Required inputs</b>                                                                        | <b>Required outputs</b>                                                 |
| <b>(1) <a href="#">Evaluation of new constraints</a></b><br>Evaluate new constraints (e.g., operational requirements, platform characteristics, and operating environment) on the system or system/software requirements to verify the applicability of the VVP.                                                                                                                                                                                                                                                                                                       | VVP<br>New constraints                                                                        | Task report(s)—<br>Evaluation of new constraints                        |
| <b>(2) <a href="#">Operating procedures evaluation</a></b><br>Verify that the operating procedures are consistent with the user documentation and conform to the system requirements.                                                                                                                                                                                                                                                                                                                                                                                  | Operating procedures<br>User documentation<br>Business or mission analysis documentation      | Task report(s)—<br>Operating procedures evaluation<br>Anomaly report(s) |
| <b>(3) <a href="#">Hazard analysis</a></b><br>a) Verify that the operating procedures and operational environment do not introduce new hazards.<br>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of the system and software operations).<br>c) Update the hazard analysis.                                                                                                                                                                 | Operating procedures<br>Hazard analysis results                                               | Task report(s)—<br>Hazard analysis<br>Anomaly report(s)                 |
| <b>(4) <a href="#">Security analysis</a></b><br>a) Verify that no new security risks are introduced due to changes in the operational environment.<br>b) Over time, changes in external interfaces, threats, or technology in general require that an updated security analysis be performed to determine an updated residual risk.<br>c) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of the system and software operations). | New constraints<br>Environmental changes<br>Operating procedures<br>Security analysis results | Task report(s)—<br>Security analysis                                    |

| <b>9.12 Activity: Operation V&amp;V (Software, 12207—Operation Process)</b>                                                                                                         |                                                                                                                                                                                                                        |                                                       |
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| <b>V&amp;V tasks</b>                                                                                                                                                                | <b>Required inputs</b>                                                                                                                                                                                                 | <b>Required outputs</b>                               |
| <b>(5) <a href="#">Risk analysis</a></b><br>a) Review and update risk analysis using prior task reports.<br>b) Provide recommendations to eliminate, reduce, or mitigate the risks. | Installation package<br>Proposed changes<br>Hazard analysis results<br>Security analysis results<br>Risk analysis results<br>Supplier development plans and schedules<br>Operation problem reports<br>V&V task results | Task report(s)—<br>Risk analysis<br>Anomaly report(s) |

| <b>9.13 Activity: Maintenance V&amp;V (Software, 12207—Maintenance Process)</b>                                                                                                                                                                                                                                  |                                                                                                                     |                                                                                                                 |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                             | <b>Required inputs</b>                                                                                              | <b>Required outputs</b>                                                                                         |
| <b>(1) <a href="#">Software maintenance strategy assessment</a></b><br>a) Verify the software maintenance strategy.<br>b) Verify that any constraints specified or implied by the software maintenance strategy are included in the system/software requirements and design.                                     | Software maintenance strategy<br>Software performance data<br>System/software requirements                          | Task report(s)—<br>Software maintenance strategy analysis<br>Corrective actions assessment<br>Anomaly report(s) |
| <b>(2) <a href="#">Software maintenance execution assessment</a></b><br>a) Verify the maintenance operational strategy results by monitoring the software maintenance execution.<br>b) Evaluate the maintenance operational strategy results.                                                                    | System maintenance strategy (e.g., operational availability requirements, software failure, suspension of services) | Task report(s)—<br>Software maintenance execution assessment and quality metrics<br>Anomaly report(s)           |
| <b>(3) <a href="#">VVP revision</a></b><br>a) Revise the VVP to conform to the approved changes.<br>b) When the development documentation required by this standard is not available, generate a new VVP and consider the methods in Annex D for deriving the required development documentation.                | VVP<br>Approved changes<br>Installation package<br>Supplier development plans and schedules                         | Updated VVP                                                                                                     |
| <b>(4) <a href="#">Anomaly evaluation</a></b><br>Evaluate the effect of software operation anomalies.                                                                                                                                                                                                            | Anomaly report(s)                                                                                                   | Task report(s)—<br>Anomaly evaluation                                                                           |
| <b>(5) <a href="#">Criticality analysis</a></b><br>a) Determine the integrity levels for the proposed modifications.<br>b) Validate the integrity levels provided by the maintainer. For V&V planning purposes, the highest integrity level assigned to the software shall be the integrity level of the system. | Proposed changes<br>Installation package<br>Assigned integrity levels                                               | Task report(s)—<br>Criticality analysis<br>Anomaly report(s)                                                    |
| <b>(6) <a href="#">Migration assessment</a></b><br>Assess whether the system/software requirements and implementation address the following:<br>a) Specific migration requirements<br>b) Migration tools<br>c) Conversion of software products and data                                                          | Installation package<br>Approved changes                                                                            | Task report(s)—<br>Migration assessment<br>Anomaly report(s)                                                    |

| <b>9.13 Activity: Maintenance V&amp;V (Software, 12207—Maintenance Process)</b>                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                        |                                                               |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Required inputs</b>                                                                                                                                                                                                 | <b>Required outputs</b>                                       |
| d) Software archiving<br>e) Support for the prior environment<br>f) User notification                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                        |                                                               |
| <b>(7) <a href="#">Retirement assessment</a></b><br>Assess whether the installation package addresses the following:<br>a) Software support<br>b) Impact on existing systems and databases<br>c) Software archiving<br>d) Transition to a new software product<br>e) User notification                                                                                                                                 | Installation package<br>Approved changes                                                                                                                                                                               | Task report(s)—<br>Retirement assessment<br>Anomaly report(s) |
| <b>(8) <a href="#">Hazard analysis</a></b><br>a) Verify that software modifications correctly implement the critical requirements and introduce no new hazards.<br>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of the system and software operations).<br>c) Update the hazard analysis. | Proposed changes<br>Installation package<br>Hazard analysis results                                                                                                                                                    | Task report(s)—<br>Hazard analysis<br>Anomaly report(s)       |
| <b>(9) <a href="#">Security analysis</a></b><br>a) Verify that the proposed changes/updates to the software do not introduce new or increased security risks to the overall system.<br>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of the system and software operations). | Proposed changes<br>Installation package<br>Security analysis results                                                                                                                                                  | Task reports—<br>Security analysis                            |
| <b>(10) <a href="#">Risk analysis</a></b><br>a) Review and update risk analysis using prior task reports.<br>b) Provide recommendations to eliminate, reduce, or mitigate the risks.                                                                                                                                                                                                                                   | Installation package<br>Proposed changes<br>Hazard analysis results<br>Security analysis results<br>Risk analysis results<br>Supplier development plans and schedules<br>Operation problem reports<br>V&V task results | Task report(s)—<br>Risk analysis<br>Anomaly report(s)         |
| <b>(11) <a href="#">Task iteration</a></b><br>Perform V&V tasks, as needed, to assure the following are performed:<br>a) Planned changes are implemented correctly.<br>b) Documentation is complete and current.<br>c) Changes do not cause unacceptable or unintended system behaviors.                                                                                                                               | Approved changes<br>Installation package                                                                                                                                                                               | Task report(s)<br>Anomaly report(s)                           |
| NOTE—Software changes are maintenance activities (see <a href="#">subclause 9.13</a> ).                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                        |                                                               |

| <b>9.14 Activity: Disposal V&amp;V (Software, 12207—Disposal Process)</b>                                                                                                                                                                                                                                                                                                                                                    |                            |                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------|
| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Required inputs</b>     | <b>Required outputs</b>                                                       |
| <b>(1) <u>Software disposal strategy evaluation</u></b><br>a) Verify that any constraints specified or implied by the software disposal strategy are included in the system/software requirements, including software element destruction/storage and recording disposal actions and analysis of disposal impacts on the system.<br>b) Validate that the software disposal strategy leaves the system in an agreed-on state. | Software disposal strategy | Task report(s)—<br>Software disposal strategy evaluation<br>Anomaly report(s) |
| NOTE (for Table 1c)—Other inputs may be used. For any V&V activity and task, all of the required inputs and outputs from preceding activities and tasks may be used, but for conciseness, only the primary inputs are listed.                                                                                                                                                                                                |                            |                                                                               |



**Table 2c—Minimum V&V tasks assigned to each integrity level for software V&V**

| V&V Processes      | Business or Mission Analysis V&V (see 9.1) |   |   |   | Stakeholder Needs and Requirements Definition V&V (see 9.2) |   |   |   | System/ Software Requirements Definition V&V (see 9.3) |   |   |   | Architecture Definition V&V (see 9.4) |   |   |   | Design Definition V&V (see 9.5) |   |   |   | System Analysis V&V (see 9.6) |   |   |   | Implementation V&V (see 9.7) |   |   |   | Integration V&V (see 9.8) |   |   |   | Transition V&V (see 9.10) |   |   |   | Operation V&V (see 9.12) |   |   |   | Maintenance V&V (see 9.13) |   |   |   | Disposal V&V (see 9.14) |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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| Integrity Levels   | Levels                                     |   |   |   | Levels                                                      |   |   |   | Levels                                                 |   |   |   | Levels                                |   |   |   | Levels                          |   |   |   | Levels                        |   |   |   | Levels                       |   |   |   | Levels                    |   |   |   | Levels                    |   |   |   | Levels                   |   |   |   | Levels                     |   |   |   | Levels                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                    | 4                                          | 3 | 2 | 1 | 4                                                           | 3 | 2 | 1 | 4                                                      | 3 | 2 | 1 | 4                                     | 3 | 2 | 1 | 4                               | 3 | 2 | 1 | 4                             | 3 | 2 | 1 | 4                            | 3 | 2 | 1 | 4                         | 3 | 2 | 1 | 4                         | 3 | 2 | 1 | 4                        | 3 | 2 | 1 | 4                          | 3 | 2 | 1 |                         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Anomaly Evaluation |                                            |   |   |   |                                                             |   |   |   |                                                        |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

[illegible]

| V&V Processes                                 | Business or Mission Analysis V&V (see 9.1) |   |   |   | Stakeholder Needs and Requirements Definition V&V (see 9.2) |   |   |   | System/ Software Requirements Definition V&V (see 9.3) |   |   |   | Architecture Definition V&V (see 9.4) |   |   |   | Design Definition V&V (see 9.5) |   |   |   | System Analysis V&V (see 9.6) |   |   |   | Implementation V&V (see 9.7) |   |   |   | Integration V&V (see 9.8) |   |   |   | Transition V&V (see 9.10) |   |   |   | Operation V&V (see 9.12) |   |   |   | Maintenance V&V (see 9.13) |   |   |   | Disposal V&V (see 9.14) |  |  |  |  |  |
|-----------------------------------------------|--------------------------------------------|---|---|---|-------------------------------------------------------------|---|---|---|--------------------------------------------------------|---|---|---|---------------------------------------|---|---|---|---------------------------------|---|---|---|-------------------------------|---|---|---|------------------------------|---|---|---|---------------------------|---|---|---|---------------------------|---|---|---|--------------------------|---|---|---|----------------------------|---|---|---|-------------------------|--|--|--|--|--|
| Integrity Levels                              | Levels                                     |   |   |   | Levels                                                      |   |   |   | Levels                                                 |   |   |   | Levels                                |   |   |   | Levels                          |   |   |   | Levels                        |   |   |   | Levels                       |   |   |   | Levels                    |   |   |   | Levels                    |   |   |   | Levels                   |   |   |   | Levels                     |   |   |   | Levels                  |  |  |  |  |  |
|                                               | 4                                          | 3 | 2 | 1 | 4                                                           | 3 | 2 | 1 | 4                                                      | 3 | 2 | 1 | 4                                     | 3 | 2 | 1 | 4                               | 3 | 2 | 1 | 4                             | 3 | 2 | 1 | 4                            | 3 | 2 | 1 | 4                         | 3 | 2 | 1 | 4                         | 3 | 2 | 1 | 4                        | 3 | 2 | 1 | 4                          | 3 | 2 | 1 |                         |  |  |  |  |  |
| Software Integration Test Procedure V&V       |                                            |   |   |   |                                                             |   |   |   |                                                        |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |  |  |
| Software Qualification Test Case V&V          |                                            |   |   |   |                                                             |   |   |   |                                                        |   |   |   |                                       |   |   |   | X                               | X | X |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |  |  |
| Software Qualification Test Design V&V        |                                            |   |   |   |                                                             |   |   |   |                                                        |   |   |   | X                                     | X | X |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |  |  |
| Software Qualification Test Execution V&V     |                                            |   |   |   |                                                             |   |   |   |                                                        |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   | X                            | X | X |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |  |  |
| Software Qualification Test Plan V&V          |                                            |   |   |   |                                                             |   |   |   |                                                        | X | X | X |                                       |   |   |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |  |  |
| Software Qualification Test Procedure V&V     |                                            |   |   |   |                                                             |   |   |   |                                                        |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   | X                            | X | X |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |  |  |
| Stakeholder Needs and Requirements Evaluation |                                            |   |   |   |                                                             | X | X |   |                                                        |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |  |  |
| System Analysis Strategy Evaluation           |                                            |   |   |   |                                                             |   |   |   |                                                        |   |   |   |                                       |   |   |   |                                 |   |   |   | X                             | X |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |  |  |
| System Analysis Results Evaluation            |                                            |   |   |   |                                                             |   |   |   |                                                        |   |   |   |                                       |   |   |   |                                 |   |   |   | X                             | X |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |  |  |
| Task Iteration                                |                                            |   |   |   |                                                             |   |   |   |                                                        |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |  |  |
| Traceability Analysis                         | X                                          | X | X |   | X                                                           | X | X |   | X                                                      | X | X |   | X                                     | X | X |   | X                               | X | X |   |                               |   |   | X | X                            | X |   | X | X                         | X |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |  |  |
| VVP Revision                                  |                                            |   |   |   |                                                             |   |   |   |                                                        |   |   |   |                                       |   |   |   |                                 |   |   |   |                               |   |   |   |                              |   |   |   |                           |   |   |   |                           |   |   |   |                          |   |   |   |                            |   |   |   |                         |  |  |  |  |  |

NOTE—Whenever a V&V task is selected as a mandatory requirement for multiple integrity levels, the V&V task implementation is dictated by the rigor, intensity, and depth of the analysis or test. Higher integrity level implementation requires greater rigor (e.g., formal methods and structured analysis methods), intensity (e.g., consideration of all system conditions and system environment states), and depth (e.g., abnormal cases, boundary conditions, and comprehensive fault and recovery scenarios) of the analysis or test than the lower integrity level implementation.

The recommended applicability of optional tasks to the software V&V processes described in Clause 9 is shown in Table 3c. [Annex G](#) provides a description of each of the optional V&V tasks.

**Table 3c—Optional V&V tasks and suggested applications in software technical and implementation processes**

|                                                      | <a href="#">Business or Mission Analysis (9.1)</a> | <a href="#">Stakeholder Needs and Requirements Definition (9.2)</a> | <a href="#">System/Software Requirements Definition (9.3)</a> | <a href="#">Architecture Definition (9.4)</a> | <a href="#">Design Definition (9.5)</a> | <a href="#">System Analysis (9.6)</a> | <a href="#">Implementation (9.7)</a> | <a href="#">Integration (9.8)</a> | <a href="#">Transition (9.10)</a> | <a href="#">Operation (9.12)</a> | <a href="#">Maintenance (9.13)</a> | <a href="#">Disposal (9.14)</a> |
|------------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------|-----------------------------------------|---------------------------------------|--------------------------------------|-----------------------------------|-----------------------------------|----------------------------------|------------------------------------|---------------------------------|
| <a href="#">Algorithm analysis</a>                   | —                                                  | —                                                                   | X                                                             | X                                             | —                                       | —                                     | X                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Audit performance</a>                    | —                                                  | —                                                                   | X                                                             | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | —                                | X                                  | —                               |
| <a href="#">Audit support</a>                        | —                                                  | —                                                                   | X                                                             | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | —                                | X                                  | —                               |
| <a href="#">Control flow analysis</a>                | —                                                  | —                                                                   | X                                                             | X                                             | —                                       | —                                     | X                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Cost analysis</a>                        | —                                                  | X                                                                   | X                                                             | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | —                                | X                                  | —                               |
| <a href="#">Data flow analysis</a>                   | —                                                  | —                                                                   | X                                                             | X                                             | —                                       | —                                     | X                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Disaster recovery plan assessment</a>    | —                                                  | X                                                                   | X                                                             | X                                             | —                                       | —                                     | X                                    | —                                 | —                                 | X                                | X                                  | X                               |
| <a href="#">Distributed architecture assessment</a>  | —                                                  | X                                                                   | X                                                             | X                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Exploratory testing</a>                  | X                                                  | X                                                                   | X                                                             | X                                             | X                                       | X                                     | X                                    | X                                 | X                                 | X                                | X                                  | —                               |
| <a href="#">Feasibility study evaluation</a>         | —                                                  | X                                                                   | X                                                             | X                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Independent risk assessment</a>          | —                                                  | —                                                                   | —                                                             | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Inspection</a>                           | —                                                  | —                                                                   | —                                                             | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | —                                  | —                               |
| <a href="#">Inspection—Concept</a>                   | —                                                  | X                                                                   | —                                                             | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Inspection—Requirements</a>              | —                                                  | —                                                                   | X                                                             | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Inspection—Design</a>                    | —                                                  | —                                                                   | —                                                             | X                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Inspection—Source Code</a>               | —                                                  | —                                                                   | —                                                             | —                                             | —                                       | —                                     | X                                    | —                                 | —                                 | —                                | —                                  | —                               |
| <a href="#">Inspection—Test plan</a>                 | —                                                  | —                                                                   | X                                                             | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | —                                | X                                  | —                               |
| <a href="#">Inspection—Test design</a>               | —                                                  | —                                                                   | —                                                             | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | —                                | X                                  | —                               |
| <a href="#">Inspection—Test case</a>                 | —                                                  | —                                                                   | —                                                             | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | —                                | X                                  | —                               |
| <a href="#">Operational evaluation</a>               | —                                                  | —                                                                   | —                                                             | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | X                                | —                                  | —                               |
| <a href="#">Performance monitoring</a>               | —                                                  | X                                                                   | X                                                             | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | X                               |
| <a href="#">Post-installation Validation</a>         | —                                                  | —                                                                   | —                                                             | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | X                                | X                                  | —                               |
| <a href="#">Project management oversight support</a> | —                                                  | X                                                                   | X                                                             | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | X                               |
| <a href="#">Proposal evaluation support</a>          | —                                                  | —                                                                   | —                                                             | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | —                                  | —                               |
| <a href="#">Regression analysis and testing</a>      | —                                                  | —                                                                   | X                                                             | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | —                                | X                                  | —                               |
| <a href="#">Reusability analysis</a>                 | —                                                  | X                                                                   | X                                                             | X                                             | —                                       | —                                     | X                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Reuse analysis</a>                       | —                                                  | X                                                                   | X                                                             | X                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Simulation analysis</a>                  | —                                                  | X                                                                   | X                                                             | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | X                               |
| <a href="#">Test certification</a>                   | —                                                  | —                                                                   | —                                                             | —                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | X                               |
| <a href="#">Test evaluation</a>                      | —                                                  | —                                                                   | X                                                             | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | X                               |
| <a href="#">Test witnessing</a>                      | —                                                  | —                                                                   | —                                                             | —                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | X                               |
| <a href="#">Training documentation evaluation</a>    | —                                                  | —                                                                   | X                                                             | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | X                               |
| <a href="#">Usability analysis</a>                   | —                                                  | X                                                                   | X                                                             | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | —                               |
| <a href="#">User documentation evaluation</a>        | —                                                  | X                                                                   | X                                                             | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | —                               |
| <a href="#">User training</a>                        | —                                                  | —                                                                   | —                                                             | —                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | —                               |
| <a href="#">V&amp;V tool plan generation</a>         | —                                                  | —                                                                   | —                                                             | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | —                                  | —                               |
| <a href="#">V&amp;V tool qualification</a>           | —                                                  | X                                                                   | X                                                             | X                                             | —                                       | —                                     | X                                    | X                                 | X                                 | X                                | X                                  | —                               |

|                                                           | <a href="#">Business or Mission Analysis (9.1)</a> | <a href="#">Stakeholder Needs and Requirements Definition (9.2)</a> | <a href="#">System/Software Requirements Definition (9.3)</a> | <a href="#">Architecture Definition (9.4)</a> | <a href="#">Design Definition (9.5)</a> | <a href="#">System Analysis (9.6)</a> | <a href="#">Implementation (9.7)</a> | <a href="#">Integration (9.8)</a> | <a href="#">Transition (9.10)</a> | <a href="#">Operation (9.12)</a> | <a href="#">Maintenance (9.13)</a> | <a href="#">Disposal (9.14)</a> |
|-----------------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------|-----------------------------------------|---------------------------------------|--------------------------------------|-----------------------------------|-----------------------------------|----------------------------------|------------------------------------|---------------------------------|
| <a href="#">Walkthrough</a>                               | —                                                  | —                                                                   | —                                                             | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | —                                  | —                               |
| <a href="#">Walkthrough—Design</a>                        | —                                                  | —                                                                   | —                                                             | X                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Walkthrough—Requirements</a>                  | —                                                  | —                                                                   | X                                                             | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Walkthrough—Test</a>                          | —                                                  | —                                                                   | —                                                             | —                                             | —                                       | —                                     | X                                    | X                                 | —                                 | —                                | X                                  | —                               |
| <a href="#">Work breakdown structure (WBS) Evaluation</a> | —                                                  | —                                                                   | —                                                             | —                                             | —                                       | —                                     | —                                    | —                                 | —                                 | —                                | —                                  | —                               |

IEEE Std 1012-2024  
IEEE Standard for System, Software and Hardware Verification and Validation

ISO/IEC/IEEE 12207 System/Software Life Cycle Processes

| Process:<br>Business or Mission<br>Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Process:<br>Stakeholder Needs and<br>Requirements Definition                                                                                                                                                                                                                                                                                                                                                              | Process:<br>Sys/SW Requirements<br>Def'n                                                                                                                                                                                                                                                                                                                                                                                                   | Process:<br>Architecture<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                             | Process:<br>Design Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Process:<br>System<br>Analysis                                                          | Process:<br>Implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Process:<br>Integration                                                                                                                                                                                                                                                                                                                                                         | Process:<br>Transition                                                                                                                                                                                                                                                                                                                                                                                            | Process:<br>Operations                                                                                                                                                                                                                                                                                                                                                                     | Process:<br>Maintenance                                                                                                                                                                                                                                                                                                                 | Process:<br>Disposal          |
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| <b>V&amp;V Inputs</b><br>1) Acquisition needs<br>2) Business or mission documentation<br>3) Developer integrity level assignment<br>4) Hazard analysis report<br>5) Identified problem or opportunity space<br>6) Hazard analysis report<br>7) Identified problem or opportunity space<br>8) Identified solution space<br>9) Preferred candidate solution(s)<br>10) Preliminary threat and risk assessment<br>11) Security analysis report<br>12) Supplier development plans and schedules<br>13) System architecture design<br>14) System requirements<br>15) User needs<br>16) V&V task results | 1) Business or mission analysis documentation<br>2) Developer integrity level assignments<br>3) Hazard Analysis Report<br>4) Identified problem or opportunity space<br>5) Identified solution space<br>6) Preferred candidate software solution(s)<br>7) Preliminary TRA<br>8) Security analysis report<br>9) Stakeholder needs and requirements<br>10) Supplier development plans and schedules<br>11) V&V task results | 1) Business or mission analysis results<br>2) Criticality task report<br>3) Hazard analysis report<br>4) Interface requirements<br>5) Preliminary TRA<br>6) Risk analysis report<br>7) Security analysis results<br>8) Stakeholder needs and requirements<br>9) Supplier development plans and schedules<br>10) System architecture design<br>11) System/SW requirements (SRS)<br>12) System/SW requirements (SRS)<br>13) V&V task results | 1) Criticality task report<br>2) SW architecture/ diagrams<br>3) SW acceptance test plan V&V<br>4) SW qualification test plan V&V<br>5) Hazard analysis results<br>6) Interface requirements (IRS)<br>7) Security analysis results<br>8) Stakeholder needs and requirements<br>9) Supplier development plans and schedules<br>10) System architecture design<br>11) System/SW requirements (SRS)<br>12) User documentation<br>13) V&V task results | 1) Criticality task report<br>2) Design standards<br>3) SW design (SDD)<br>4) SW Component Test Plan V&V<br>5) SW Component Test Design V&V<br>6) SW Integration Test Plan V&V<br>7) SW Integration Test Design V&V<br>8) SW qualification test design V&V<br>9) SW acceptance test design V&V<br>10) Hazard analysis Report<br>11) Interface requirements specification (IRS)<br>12) Interface design (IDD)<br>13) Risk analysis report<br>14) Security analysis report<br>15) Supplier development plans and schedules<br>16) System architecture design<br>17) System requirements<br>18) System/SW requirements (SRS)<br>19) User documentation<br>20) V&V task results | 1) System analysis strategy<br>2) System analysis results<br>3) System analysis results | 1) Software Source Code and Source Component information<br>2) Business or Mission Analysis Documentation<br>3) Criticality Task Report<br>4) Coding Standards<br>5) SW acceptance test cases V&V<br>6) SW component test cases V&V<br>7) SW Component test procedures V&V<br>8) SW design (SDD)<br>9) SW integration test cases V&V<br>10) SW qualification test cases V&V<br>11) Hazard analysis report<br>12) Interface design (IDD)<br>13) Risk analysis report<br>14) Security analysis report<br>15) Supplier development plans and schedules<br>16) System/SW requirements<br>17) User Documentation<br>18) V&V task results | 1) SW integration strategy<br>2) SW architecture, requirements/ design (SRS/SDD)<br>3) Source Code/ Executable<br>4) SW integration test procedures V&V<br>5) SW qualification test procedures V&V<br>6) Hazard analysis report (test results)<br>7) Risk analysis report<br>8) Security analysis report<br>9) Supplier development plans and schedules<br>10) V&V task results | 1) Acquirer agreement acceptance test plan<br>2) SW acceptance test procedures V&V<br>3) Hazard analysis report<br>4) Installation package<br>5) New constraints<br>6) Operating procedures<br>7) Security analysis report<br>8) Supplier development plans and schedules<br>9) Transitioning software source code and executable<br>10) Transitioning strategy<br>11) User documentation<br>12) V&V task results | 1) Business or mission analysis documentation<br>2) Environmental changes<br>3) Hazard analysis report<br>4) Installation package<br>5) Assigned integrity levels<br>6) Operation problem reports<br>7) Proposed changes<br>8) Risk analysis report<br>9) Security analysis report<br>10) Security analysis report<br>11) Supplier development plans and schedules<br>12) V&V task results | 1) Anomaly report(s)<br>2) Approved changes<br>3) Hazard analysis report<br>4) Installation package<br>5) Assigned integrity levels<br>6) Operation problem reports<br>7) Proposed changes<br>8) Risk analysis report<br>9) Security analysis report<br>10) Supplier development plans and schedules<br>11) V&V task results<br>12) VVP | 1) Software disposal strategy |

IEEE Std 1012 Software V&V Activities and Tasks (All tasks are minimum required for integrity level 4)

| Process:<br>Business or Mission<br>Analysis V&V (9.1)                                                                                                                                               | Process:<br>Stakeholder Needs and<br>Requirements Def'n V&V (9.2)                                                                                                         | Process:<br>System/SW Req's<br>Def'n V&V (9.3)                                                                                                                                                                                                      | Process:<br>Architecture<br>Definition V&V (9.4)                                                                                                                                                                                                                                                                                                           | Process:<br>Design Definition V&V<br>(9.5)                                                                                                                                                                                                                                                                                                                                         | Process:<br>System Analysis<br>V&V (9.6)                                                                                 | Process:<br>Implementation V&V<br>(9.7)                                                                                                                                                                                                                                                                                                                                                    | Process:<br>Integration V&V<br>(9.8)                                                                                                                                                                                                                   | Process:<br>Transition V&V<br>(9.10)                                                                                                                                                                                                                 | Process:<br>Operations V&V<br>(9.12)                                                                                                     | Process:<br>Maintenance V&V<br>(9.13)                                                                                                                                                                                                                                                                              | Process:<br>Disposal V&V<br>(9.14)        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>V&amp;V Tasks</b><br>1) Business or Mission Analysis Results Evaluation<br>2) Traceability Analysis<br>3) Criticality Analysis<br>4) Hazard Analysis<br>5) Security Analysis<br>6) Risk Analysis | 1) Stakeholder Needs and Requirements Evaluation<br>2) Traceability Analysis<br>3) Criticality Analysis<br>4) Hazard Analysis<br>5) Security Analysis<br>6) Risk Analysis | 1) Requirements Evaluation<br>2) Interface Analysis<br>3) Traceability Analysis<br>4) Criticality Analysis<br>5) SW Qualification Test Plan V&V<br>6) SW Acceptance Test Plan V&V<br>7) Hazard Analysis<br>8) Security Analysis<br>9) Risk Analysis | 1) Architecture Evaluation<br>2) Interface Analysis<br>3) Req Allocation Analysis<br>4) Traceability Analysis<br>5) Criticality Analysis<br>6) SW Component Test Plan V&V<br>7) SW Integration Test Design V&V<br>8) SW Qual Test Design V&V<br>9) SW Accept Test Design V&V<br>10) Hazard Analysis Case V&V<br>11) Security Analysis<br>12) Risk Analysis | 1) Design Evaluation<br>2) Interface Analysis<br>3) Traceability Analysis<br>4) Criticality Analysis<br>5) SW Component Test Design V&V<br>6) SW Component Test Case V&V<br>7) SW Integration Test Design V&V<br>8) SW Integration Test Case V&V<br>9) SW Qual Test Case V&V<br>10) SW Accept Test Case V&V<br>11) Security Analysis<br>12) Security Analysis<br>13) Risk Analysis | 1) System Analysis Strategy Evaluation<br>2) System Analysis Results Evaluation<br>3) System Analysis Results Evaluation | 1) Source Code and Source Code Information Evaluation<br>2) Interface Analysis<br>3) Traceability Analysis<br>4) Criticality Analysis<br>5) SW Component Test Procedure V&V<br>6) SW Component Test Execution V&V<br>7) SW Int Test Procedure V&V<br>8) SW Qual Test Procedure V&V<br>9) SW Accept Test Procedure V&V<br>10) Hazard Analysis<br>11) Security Analysis<br>12) Risk Analysis | 1) Software Integration Strategy Assessment<br>2) Software Element Interaction Analysis<br>3) SW Int Test Execution V&V<br>4) SW Qual Test Execution V&V<br>5) Traceability Analysis<br>6) Hazard Analysis<br>7) Security Analysis<br>8) Risk Analysis | 1) Transition Strategy Evaluation<br>2) Transition Demonstration Assessment<br>3) SW Accept Test Execution V&V<br>4) Installation Configuration Audit<br>5) Installation Check-out<br>6) Hazard Analysis<br>7) Security Analysis<br>8) Risk Analysis | 1) Evaluation of New Constraints<br>2) Operating Procedures Evaluation<br>3) Hazard Analysis<br>4) Security Analysis<br>5) Risk Analysis | 1) Software Maintenance Strategy Assessment<br>2) Software Maintenance Execution Assessment<br>3) VVP Revision<br>4) Anomaly Evaluation<br>5) Criticality Analysis<br>6) Migration Assessment<br>7) Retirement Assessment<br>8) Hazard Analysis<br>9) Security Analysis<br>10) Risk Analysis<br>11) Task Iteration | 1) SW Disposal Strategy Evaluation        |
| <b>V&amp;V Outputs</b><br>1) Anomaly report(s)<br>2) Task report(s)                                                                                                                                 | 1) Anomaly report(s)<br>2) Task report(s)                                                                                                                                 | 1) Anomaly report(s)<br>2) Task report(s)<br>3) V&V SW test plan(s)<br>- Acceptance<br>- Qualification                                                                                                                                              | 1) Anomaly report(s)<br>2) Task report(s)<br>3) V&V SW test plan (P)/design (D)<br>- Acceptance (D)<br>- Component (D)<br>- Integration (P)<br>- Qualification (D)                                                                                                                                                                                         | 1) Anomaly report(s)<br>2) Task report(s)<br>3) V&V SW test design (D)/cases (C)<br>- Acceptance (D)<br>- Component (D)<br>- Integration (P)<br>- Qualification (C)                                                                                                                                                                                                                | 1) Task report(s)                                                                                                        | 1) Anomaly report(s)<br>2) Task report(s)<br>3) V&V SW test procedures(Pr)/ Execution Report (E)<br>- Acceptance (Pr)<br>- Qualification (Pr)<br>- Integration (Pr)<br>- Component (Pr,E)                                                                                                                                                                                                  | 1) Anomaly report(s)<br>2) Task report(s)<br>3) V&V SW test execution results<br>- Qualification<br>- Integration                                                                                                                                      | 1) Anomaly report(s)<br>2) Task report(s)<br>3) V&V SW test execution results<br>- Acceptance                                                                                                                                                        | 1) Anomaly report(s)<br>2) Task report(s)                                                                                                | 1) Anomaly report(s)<br>2) Task report(s)<br>3) Updated VVP                                                                                                                                                                                                                                                        | 1) Anomaly report(s)<br>2) Task report(s) |

NOTE 1—Clause references in the process definitions (top graphic bar) are ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#) clause numbers.

NOTE 2—Clause references in the activity V&V definitions (middle graphic bar) are IEEE Std 1012 clause numbers.

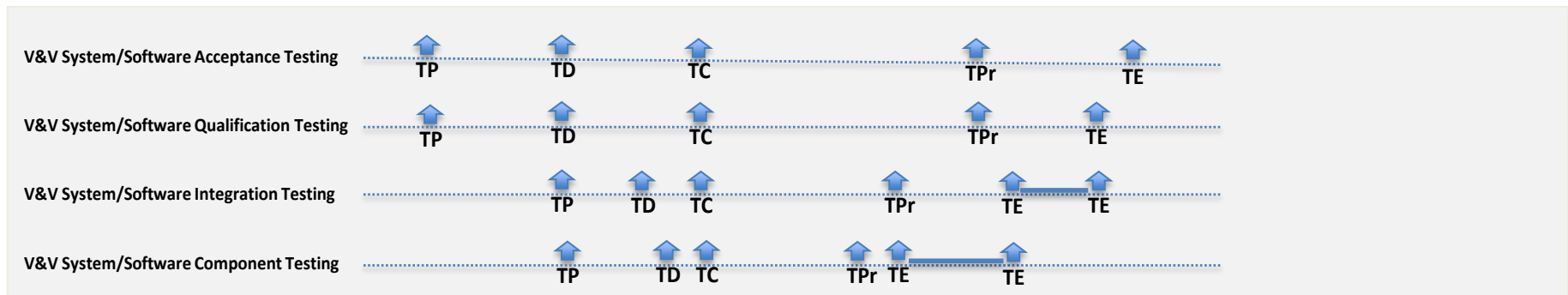
NOTE 3—V&V tasks listed in the figure are the minimum required for integrity level 4 (highest integrity level).

NOTE 4—Software acceptance testing process supports the systems integration testing process.

NOTE 5—Software installation and checkout process supports the system transition process.

**Figure 1c—Summary of software V&V activities and tasks**

| IEEE Std 1012 System/Software V&V Activities          |                                                                   |                                                |                                                  |                                            |                                          |                                         |                                      |                                      |                                      |                                       |                                    |
|-------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------|--------------------------------------------------|--------------------------------------------|------------------------------------------|-----------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|---------------------------------------|------------------------------------|
| Process:<br>Business or Mission<br>Analysis V&V (9.1) | Process:<br>Stakeholder Needs and<br>Requirements Def'n V&V (9.2) | Process:<br>System/SW Req's<br>Def'n V&V (9.3) | Process:<br>Architecture<br>Definition V&V (9.4) | Process:<br>Design Definition V&V<br>(9.5) | Process:<br>System Analysis<br>V&V (9.6) | Process:<br>Implementation V&V<br>(9.7) | Process:<br>Integration V&V<br>(9.8) | Process:<br>Transition V&V<br>(9.10) | Process:<br>Operations V&V<br>(9.12) | Process:<br>Maintenance V&V<br>(9.13) | Process:<br>Disposal V&V<br>(9.14) |



**Note:**

TP = Test Plan  
TD = Test Design  
TC = Test Cases  
TPr = Test Procedures  
TE = Test Execution

NOTE 1—All V&V software test products and tasks represent the activities and products required as a minimum for integrity level 4.

NOTE 2—This is an example of the phasing of software V&V test products and tasks across the software life cycle. The software V&V test products (upward arrows) are shown in the software life cycle stages when the products are generated. Software test execution tasks are shown to occur during one or more software life cycle stages, as indicated by "activity bars" in the diagram. The life cycle stage (in which each test product is generated) and phasing of each test product and task can vary from this diagram in accordance with project-specific needs.

NOTE 3—The V&V activity clauses referenced in the software V&V life cycle stages are IEEE Std 1012 clauses.

**Figure 2c—Summary of software V&V test products and tasks**



## 10. Hardware V&V processes

### 10.1 Business or Mission Analysis V&V Process

#### 10.1.1 Purpose

The purpose of the Business or Mission Analysis V&V Process is to provide assurance that the outcomes of the Business or Mission Analysis Process (ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)) related to hardware, modeled on the Business or Mission Analysis Process (ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)) have been achieved.

#### 10.1.2 Outcomes

As a result of the successful implementation of the Business or Mission Analysis V&V Process, objective evidence is developed to assess whether:

- a) The problem or opportunity space adheres to the organizational strategy.
- b) The solution space is consistent with the problem or opportunity space.
- c) Preliminary life-cycle concepts are consistent with the solution space.
- d) The preferred candidate solution alternative(s) are selected.
- e) Any enabling systems or services needed for business or mission analysis are available.
- f) The root of traceability is established, starting with the organization strategy and tracing the problem or opportunity space to the solution space to the candidate solutions.

#### 10.1.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2d](#) for the selected integrity level, the following Business or Mission Analysis V&V activity and tasks described in [Table 1d, Activity 10.1](#):

- a) [Business or mission analysis V&V](#). This activity consists of the following tasks:
  - 1) [Business or mission analysis documentation evaluation](#)
  - 2) [Traceability analysis](#)
  - 3) [Criticality analysis](#)
  - 4) [Hazard analysis](#)
  - 5) [Security analysis](#)
  - 6) [Risk analysis](#)

The primary focus is on hardware with consideration of the interactions with software and user allocations to verify the allocation of system requirements, validate the selected solution, and assure that no false assumptions have been incorporated in the solution.

During the Mission or Business Analysis Process, different hardware concepts are investigated, and trade studies are conducted on each concept before a final concept is selected. These trade studies may involve assessing the performance features of each concept, estimating the cost of parts, determining the manufacturing efficiency, identifying the technology risks, and estimating the schedule to develop the hardware. Hardware models and prototypes may be constructed to conduct these trade studies in conjunction with simulations and analytic analyses. The hardware-level Mission or Business Analysis Process may provide preliminary hardware models and prototypes to support trade studies for the system concept.

## 10.2 Stakeholder Needs and Requirements Definition V&V Process

### 10.2.1 Purpose

The purpose of the Stakeholder Needs and Requirements Definition V&V Process is to provide assurance that outcomes of the Stakeholder Needs and Requirements Definition Process related to hardware, modeled on the Stakeholder Needs and Requirements Definition Process (ISO/IEC/IEEE 12207:2017(E) [B40]), have been achieved.

### 10.2.2 Outcomes

As a result of the successful implementation of the Stakeholder Needs and Requirements Definition V&V Process, objective evidence is developed to assess whether the stakeholder requirements:

- a) Specify the required characteristics and context of the use of services and operational concepts.
- b) Define all constraints on a hardware solution.
- c) Are traceable to the originating stakeholders.
- d) Are complete, unambiguous, correct, and accurate.
- e) Can be validated by measurable analyses or tests.

### 10.2.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2d](#) for the selected integrity level, the following Stakeholder Needs and Requirements Definition V&V activity and tasks described in [Table 1d, Activity 10.2](#):

- a) [Stakeholder needs and requirements definition V&V](#). This activity consists of the following tasks:
  - 1) [Stakeholder needs and requirements evaluation](#)
  - 2) [Traceability analysis](#)
  - 3) [Criticality analysis](#)
  - 4) [Hazard analysis](#)
  - 5) [Security analysis](#)
  - 6) [Risk analysis](#)

The Stakeholder Needs and Requirements Definition V&V Process focuses on needs and requirements related to hardware, with consideration of the interactions with software and user allocations, to verify that the identified needs and requirements are consistent and sufficient for the purpose of the system.

## 10.3 System/Hardware Requirements Definition V&V Process

### 10.3.1 Purpose

The purpose of the System/Hardware Requirements Definition V&V Process is to provide assurance that outcomes of the System/Hardware Requirements Definition Process related to hardware, modeled on the System/Software Requirements Definition Process (ISO/IEC/IEEE 12207:2017(E) [B40]), have been achieved.

### 10.3.2 Outcomes

As a result of the successful implementation of the System/Hardware Requirements Definition V&V Process, objective evidence is developed to assess whether the system/hardware requirements:

- a) Specify all required characteristics, attributes, functional and performance requirements, interface requirements, and requirements for qualification, safety and security, human factors engineering, and user documentation for the hardware.

- b) Specify all constraints that will affect the architecture of the hardware and the means to realize the hardware.
- c) Are unique, complete, unambiguous, consistent with all other requirements, implementable, and verifiable.
- d) Are traceable to the stakeholder requirements.
- e) Provide a basis (through analysis or test planning) for verifying that each hardware requirement can be satisfied.

### 10.3.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2d](#) for the selected integrity level, the following System/Hardware Requirements Definition V&V activity and tasks described in [Table 1d, Activity 10.3](#):

- a) [System/hardware requirements definition V&V](#). This activity consists of the following tasks:
  - 1) [Requirements evaluation](#)
  - 2) [Interface analysis](#)
  - 3) [Traceability analysis](#)
  - 4) [Criticality analysis](#)
  - 5) [Hardware qualification test plan V&V](#)
  - 6) [Hardware acceptance test plan V&V](#)
  - 7) [Hazard analysis](#)
  - 8) [Security analysis](#)
  - 9) [Risk analysis](#)

The System/Hardware Requirements Definition V&V Process begins after the system/hardware requirements have been defined and allocated to each hardware element of the system. Given the system and system/hardware requirements and hardware business or mission analysis, the requirements are evaluated to determine if the allocation of the requirements to the hardware elements satisfies the entire set of system/hardware requirements. Any constraints or limitations imposed by the system/hardware requirements on the system performance are identified and verified with the system acquirer that these limitations or constraints are acceptable. V&V test planning begins during the System/Hardware Requirements Definition V&V Process, and test activities span several hardware V&V processes.

## 10.4 Architecture Definition V&V Process

### 10.4.1 Purpose

The purpose of the Architecture Definition V&V Process is to provide assurance that the outcomes of the Architecture Definition Process related to hardware, modeled on the Architecture Definition Process (ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)), have been achieved.

### 10.4.2 Outcomes

As a result of the successful implementation of the Architecture Definition V&V Process, objective evidence is developed to assess whether:

- a) The hardware architecture (i.e., hardware, interfaces, and communication) satisfies the system/hardware requirements.
- b) The hardware architecture is realizable.
- c) The hardware architecture is based on specified selection criteria.
- d) The basis for verifying the hardware elements is defined.
- e) The basis for integration of the system elements is established.

### 10.4.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2c](#) for the selected integrity level, the following Architecture Definition V&V activity and tasks described in [Table 1d, Activity 10.4](#):

- a) [Architecture definition V&V](#). This activity consists of the following tasks:
  - 1) [Architecture evaluation](#)
  - 2) [Interface analysis](#)
  - 3) [Requirements allocation analysis](#)
  - 4) [Traceability analysis](#)
  - 5) [Criticality analysis](#)
  - 6) [Hardware component test plan V&V](#)
  - 7) [Hardware integration test plan V&V](#)
  - 8) [Hardware qualification test design V&V](#)
  - 9) [Hardware acceptance test design V&V](#)
  - 10) [Hazard analysis](#)
  - 11) [Security analysis](#)
  - 12) [Risk analysis](#)

As system/hardware requirements are being developed, the hardware architecture process can be initiated. Architectural solutions are developed that will satisfy the specific system/hardware requirements. Any constraints or limitations imposed by the hardware architecture on the system performance are identified and verified with the system acquirer that these limitations or constraints are acceptable. V&V test design is performed during the Architecture Definition V&V Process.

## 10.5 Design Definition V&V Process

### 10.5.1 Purpose

The purpose of the Design Definition V&V Process is to provide assurance that outcomes of the Design Definition Process related to hardware, modeled on the Design Definition Process (ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)), have been achieved.

### 10.5.2 Outcomes

As a result of the successful implementation of the Design Definition V&V Process, objective evidence is developed to assess whether:

- a) The design characteristics of each hardware element are defined.
- b) The design enablers necessary for design definition are selected or defined.
- c) Each interface between hardware elements is described and includes purpose, parameters, data format, and performance criteria (e.g., timing, bandwidth, accuracy, safety, and security).
- d) The hardware design is established.
- e) Inputs for requirements of any enabling systems or system elements that serve the design definition activities are identified.
- f) Any enabling systems or services needed for design definition are available.
- g) Traceability of the design characteristics to the architectural elements of the hardware architecture is established.

### 10.5.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2d](#) for the selected integrity level, the following Design Definition V&V activity and tasks described in [Table 1d, Activity 10.5](#):

- a) [Design definition V&V](#). This activity consists of the following tasks:

- 1) [Design evaluation](#)
- 2) [Interface analysis](#)
- 3) [Traceability analysis](#)
- 4) [Criticality analysis](#)
- 5) [Hardware component test design V&V](#)
- 6) [Hardware component test case V&V](#)
- 7) [Hardware integration test design V&V](#)
- 8) [Hardware integration test case V&V](#)
- 9) [Hardware qualification test case V&V](#)
- 10) [Hardware acceptance test case V&V](#)
- 11) [Hazard analysis](#)
- 12) [Security analysis](#)
- 13) [Risk analysis](#)

As the system/hardware architecture is being developed, the hardware Design Definition Process can be initiated. Detailed design solutions are developed that will satisfy the specific system/hardware requirements. Depending on the hardware, the design can take the form of a selection of materials or composites, drawings showing dimensions, circuit diagrams, and other electromechanical descriptions. Often, the hardware element involves subtiered design components from a variety of suppliers. These subtiered designs are documented and evaluated as part of the entire hardware element.

## 10.6 System Analysis V&V Process

### 10.6.1 Purpose

The purpose of the System Analysis V&V Process is to provide assurance that outcomes of the System Analysis Process related to hardware, modeled on the System Analysis Process (ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)), have been achieved.

### 10.6.2 Outcomes

As a result of the successful implementation of the System Analysis V&V Process, objective evidence is developed to assess whether:

- a) The strategy for the system analysis is complete and is appropriate for the importance of the analysis.
- b) The results of the system analysis support its conclusions and recommendations.

### 10.6.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2d](#) for the selected integrity level, the following System Analysis V&V activity and tasks described in [Table 1d, Activity 10.6](#):

- a) [System analysis V&V](#). This activity consists of the following tasks:
  - 1) [System analysis strategy evaluation](#)
  - 2) [System analysis results evaluation](#)

The System Analysis V&V Process shall be performed for any specific implementation of the System Analysis Process where the conclusions and recommendations of the analysis could affect a hardware element of the integrity level indicated in [Table 2d](#).

## 10.7 Implementation V&V Process

### 10.7.1 Purpose

The purpose of the Implementation V&V Process is to provide assurance that outcomes of the Implementation Process related to hardware, modeled on the Implementation Process (ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)), have been achieved.

### 10.7.2 Outcomes

As a result of the successful implementation of the Implementation V&V Process, objective evidence is developed to assess whether:

- a) The implementation activities performed produce a hardware element that conforms to the system/hardware requirements.
- b) The implementation correctly implements the design definition.
- c) The hardware element has been implemented within the defined constraints.
- d) The recorded evidence of implementation is complete and correct.
- e) The hardware element has been packaged and stored in accordance with defined requirements.

### 10.7.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2d](#) for the selected integrity level, the following Implementation V&V activity and tasks described in [Table 1d, Activity 10.7](#):

- a) [Implementation V&V](#). This activity consists of the following tasks:
  - 1) [Fabricated component information evaluation](#)
  - 2) [Interface analysis](#)
  - 3) [Traceability analysis](#)
  - 4) [Criticality analysis](#)
  - 5) [Hardware component test procedure V&V](#)
  - 6) [Hardware component test execution V&V](#)
  - 7) [Hardware integration test procedure V&V](#)
  - 8) [Hardware qualification test procedure V&V](#)
  - 9) [Hardware acceptance test procedure V&V](#)
  - 10) [Hazard analysis](#)
  - 11) [Security analysis](#)
  - 12) [Risk analysis](#)

The Implementation V&V tasks are to monitor, review, audit, and analyze hardware products. The system V&V activities will use the hardware V&V results to perform, in part or in whole, the system V&V activities.

The methods used in this activity include visual analysis/inspection, quality sampling testing, physical measurements, form/fit checks, and complete component analysis/test. As the hardware design is being defined, the hardware implementation (fabrication) process can be initiated. During fabrication, the hardware is built from the raw materials, electronic components are selected and integrated onto circuit boards, mechanical elements are shaped and interconnected, and manufactured parts are created from molds and milling processes and other hardware fabrication methods. The Implementation V&V Process determines whether or not deviations have been introduced into the solution as the hardware progressed from the hardware design to actual fabrication. For example, an electronic component selected from a catalog may be similar to the specified part but not identical. The Implementation V&V Process investigates whether the small difference in the final fabrication creates any limitations or constraints that affect the overall system performance. The hardware components received from subtiered suppliers are validated by either the manufacturer or the supplier before being included in the overall hardware element fabrication.

## 10.8 Integration V&V Process

### 10.8.1 Purpose

The purpose of the Integration V&V Process is to provide assurance that outcomes of the Integration Process related to hardware, modeled on the Integration Process (ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)) have been achieved.

### 10.8.2 Outcomes

As a result of the successful implementation of the Integration V&V Process, objective evidence is developed to assess whether:

- a) The hardware elements, as described by the architecture definition, design definition, and implementation, are integrated correctly.
- b) The integrated hardware meets the system/hardware requirements.
- c) The hardware integration strategy is consistent with the hardware architecture.
- d) The integration test plan, design, cases, and procedures are traceable to the hardware architecture.
- e) The unavoidable constraints of integration that influence requirements are addressed correctly.
- f) The integration of human performance into hardware and its operation is correct.
- g) Nonconformances due to integration actions are identified and addressed.
- h) The hardware element does not cause unintended or undesired consequences.

### 10.8.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2d](#) for the selected integrity level, the following Integration V&V activity and tasks described in [Table 1d, Activity 10.8](#):

- a) [Integration V&V](#). This activity consists of the following tasks:
  - 1) [Hardware integration strategy assessment](#)
  - 2) [Hardware element interaction analysis](#)
  - 3) [Hardware integration test execution V&V](#)
  - 4) [Hardware qualification test execution V&V](#)
  - 5) [Traceability analysis](#)
  - 6) [Hazard analysis](#)
  - 7) [Security analysis](#)
  - 8) [Risk analysis](#)

The Integration V&V Process occurs in parallel with hardware fabrication. As fabrication progresses, hardware parts are integrated into components, components are integrated into the hardware element, and a series of tests are conducted to verify and validate conformance with requirements.

During the initial Integration life cycle stage, remaining activities of hardware component test V&V are finishing execution. Hardware integration test V&V is initiated to demonstrate that individual components integrate properly and satisfy system/hardware requirements. After individual components are demonstrated to be correctly integrated, hardware qualification test V&V execution is conducted to demonstrate that system/hardware requirements satisfy total system requirements in a qualified system test environment.

NOTE—In some programs, the integration testing determines the efficiency of the fabrication and integration manufacturing steps, especially when multiple hardware copies (hardware assembly line) are produced.

## 10.9 Verification Process

### 10.9.1 Purpose

The purpose of the Verification Process is to provide objective evidence for whether the outcomes achieve the following:

- a) Conform to requirements (e.g., for correctness, completeness, consistency, and accuracy) for all activities during each life cycle process.
- b) Satisfy the standards, practices, and conventions during life cycle processes.
- c) Successfully complete each life cycle activity and satisfy all the criteria for initiating succeeding life cycle activities (i.e., the product is built correctly).

### 10.9.2 Outcomes

The following outcomes occur as a result of successful implementation of the Verification Process:

- a) A V&V plan is developed and implemented.
- b) The system of interest (i.e., hardware) and all its components are assigned integrity levels that are reevaluated throughout the life cycle.
- c) The hardware and each of its components are evaluated for requirements satisfaction based on assigned integrity levels.
- d) Objective evidence is developed to determine whether the hardware and each of its components conform to requirements and satisfy all the criteria for each successive life cycle activity.

### 10.9.3 Activities and tasks

The activities and tasks for the Verification Process as applied to the technical processes of the hardware life cycle processes are described in subclause [10.1](#) (Business or Mission Analysis V&V Process), subclause [10.2](#) (Stakeholder Needs and Requirements Definition V&V Process), subclause [10.3](#) (System/Hardware Requirements Definition V&V process), subclause [10.4](#) (Architecture Definition V&V Process), subclause [10.5](#) (Design Definition V&V process), subclause [10.6](#) (System Analysis V&V Process), subclause [10.7](#) (Implementation V&V Process), subclause [10.8](#) (Integration V&V Process), subclause [10.10](#) (Transition V&V Process), subclause [10.12](#) (Operation V&V Process), subclause [10.13](#) (Maintenance V&V Process) and subclause [10.14](#) (Disposal V&V Process).

## 10.10 Transition V&V Process

### 10.10.1 Purpose

The purpose of the Transition V&V Process is to provide assurance that outcomes of the Transition Process related to hardware, modeled on the Transition Process (ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)), have been achieved.

### 10.10.2 Outcomes

As a result of the successful implementation of the Transition V&V Process, objective evidence is developed to assess whether:

- a) The hardware transition strategy is comprehensive and explicitly documented.
- b) The hardware is installed in its operational location in accordance with the transition plan.
- c) The hardware delivers all specified services per its governing requirements.
- d) The hardware element meets its requirements and associated objective acceptance criteria.
- e) The hardware is sustainable by enabling systems.



### 10.10.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2d](#) for the selected integrity level, the following hardware Transition V&V activity and tasks described in [Table 1d, Activity 10.10](#):

- a) [Transition V&V](#). This activity consists of the following tasks:
  - 1) [Transition strategy evaluation](#)
  - 2) [Transition demonstration assessment](#)
  - 3) [Hardware acceptance test execution V&V](#)
  - 4) [Installation configuration audit](#)
  - 5) [Installation checkout](#)
  - 6) [Hazard analysis](#)
  - 7) [Security analysis](#)
  - 8) [Risk analysis](#)

Before the hardware element transitions to operational use, hardware acceptance testing V&V and installation and checkout are executed to demonstrate that the hardware is ready for operational use.

## 10.11 Validation Process

### 10.11.1 Purpose

The purpose of the Validation Process is to provide objective evidence for whether the outcomes achieve the following:

- a) Satisfy requirements allocated to the hardware at the end of each life cycle activity.
- b) Solve the right problem (e.g., correctly model physical laws, implement business rules, and use the proper system assumptions).
- c) Satisfy intended use and user needs in the operational environment (i.e., the correct product is built).

### 10.11.2 Outcomes

The following outcomes occur as a result of the successful implementation of the Validation Process:

- a) A V&V plan is developed and implemented.
- b) The system of interest (i.e., hardware) and all its components are assigned integrity levels that are reevaluated throughout the life cycle.
- c) The hardware and each of its components are evaluated for satisfaction of allocated system requirements and of intended use and user needs based on assigned integrity levels.
- d) Objective evidence is developed to determine whether the hardware and each of its components satisfy all system requirements allocated to hardware and meet intended use and user needs.

### 10.11.3 Activities and tasks

The activities and tasks for the Validation Process as applied to the technical processes of the hardware life cycle processes are described in subclause [10.1](#) (Business or Mission Analysis V&V Process), subclause [10.2](#) (Stakeholder Needs and Requirements Definition V&V Process), subclause [10.3](#) (System/Hardware Requirements Definition V&V process), subclause [10.4](#) (Architecture Definition V&V Process), subclause [10.5](#) (Design Definition V&V process), subclause [10.6](#) (System Analysis V&V Process), subclause [10.7](#) (Implementation V&V Process), subclause [10.8](#) (Integration V&V Process), subclause [10.10](#) (Transition V&V Process), subclause [10.12](#) (Operation V&V Process), subclause [10.13](#) (Maintenance V&V Process) and subclause [10.14](#) (Disposal V&V Process).

## 10.12 Operation V&V Process

### 10.12.1 Purpose

The purpose of the Operation V&V Process is to provide assurance that outcomes of the Operation Process related to hardware, modeled on the Operation Process (ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)), have been achieved.

### 10.12.2 Outcomes

As a result of the successful implementation of the Operation V&V Process, objective evidence is developed to assess whether:

- a) New constraints in the system are evaluated.
- b) Proposed system changes and their impacts on the hardware are assessed.
- c) Operating procedures are evaluated for correctness and usability.

### 10.12.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2d](#) for the selected integrity level, the following Operation V&V activity and tasks described in [Table 1d, Activity 10.12](#):

- a) [Operation V&V](#). This activity consists of the following tasks:
  - 1) [Evaluation of new constraints](#)
  - 2) [Operating procedures evaluation](#)
  - 3) [Hazard analysis](#)
  - 4) [Security analysis](#)
  - 5) [Risk analysis](#)

The Operation V&V activity evaluates the impact of changes in the operating environment, assesses the effect on the system of any proposed changes, evaluates operating procedures for adherence with the intended use, and analyzes the risks affecting the user and the system.

## 10.13 Maintenance V&V Process

### 10.13.1 Purpose

The purpose of the Maintenance V&V Process is to provide assurance that outcomes of the Maintenance Process related to hardware, modeled on the Maintenance Process (ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)), have been achieved.

### 10.13.2 Outcomes

As a result of the successful implementation of the Maintenance V&V Process, objective evidence is developed to assess whether:

- a) The maintenance strategy is comprehensive and explicitly documented.
- b) Corrective actions resolve the issue or negative trend.
- c) The effect is assessed for corrective, adaptive, perfective, and preventive changes on stakeholder and system/hardware requirements, architecture, design, and implementation.
- d) Evaluation is performed on problem reports, corrective actions, and trends to determine the possible corrective, adaptive, perfective, and preventive actions.
- e) Proposed hardware changes and their impact on the system are assessed.
- f) Anomalies that are discovered during operation are evaluated.
- g) Migration requirements are assessed.

- h) Retirement requirements are assessed.
- i) V&V tasks are re-performed.

### 10.13.3 Activity and tasks

The V&V effort shall perform, as specified in [Table 2d](#) for the selected integrity level, the following Maintenance V&V activity and tasks described in [Table 1d, Activity 10.13](#):

- a) [Maintenance V&V](#). This activity consists of the following tasks:
  - 1) [Hardware maintenance strategy assessment](#)
  - 2) [Hardware maintenance execution assessment](#)
  - 3) [VVP revision](#)
  - 4) [Anomaly evaluation](#)
  - 5) [Criticality analysis](#)
  - 6) [Migration assessment](#)
  - 7) [Retirement assessment](#)
  - 8) [Hazard analysis](#)
  - 9) [Security analysis](#)
  - 10) [Risk analysis](#)
  - 11) [Task iteration](#)

Proposed changes are assessed by the proposed/baseline change assessment task of the V&V management activity.

The Maintenance Process is activated when the hardware or associated information is changed in response to a need for system maintenance. The maintenance V&V activity addresses the following:

- Hardware modifications (i.e., corrective, adaptive, or perfective changes).
- Hardware migration (i.e., the movement of hardware to a new operational environment).
- Hardware retirement (i.e., the withdrawal of active support by the operation and maintenance organization, partial or total replacement by a new system, or installation of an upgraded system).

System modifications may be derived from the requirements specified to correct hardware errors (e.g., corrective), to adapt to a changed operating environment (e.g., adaptive), or to respond to additional user requests or enhancements (e.g., perfective). Modifications of the hardware shall be treated as development processes and shall be verified and validated by performing V&V tasks corresponding to the modifications. Integrity level assignments shall be assessed as described in [Clause 5](#). The integrity level assignments shall be revised as appropriate to reflect the requirements derived from the Maintenance Process.

If V&V for the hardware was performed in accordance with this standard, then the Maintenance Process shall continue to conform to this standard. If the hardware was not verified and validated using this standard and appropriate information is not available or adequate, then the maintenance V&V effort shall determine whether the missing or incomplete information should be generated. In making this determination of whether to generate missing information, the minimum V&V requirements of the assigned integrity level shall be taken into consideration.

## 10.14 Disposal V&V Process

### 10.14.1 Purpose

The purpose of the Disposal V&V Process is to provide assurance that outcomes of the Disposal Process related to hardware, modeled on the Disposal Process (ISO/IEC/IEEE 12207:2017(E) [\[B40\]](#)), have been achieved.

### 10.14.2 Outcomes

As a result of the successful implementation of the Disposal V&V Process, objective evidence is developed to assess whether:

- a) The constraints in the hardware disposal strategy are included in system/hardware requirements.
- b) Disposal leaves the system in an agreed-on state.

### 10.14.3 Activity and task

The V&V effort shall perform, as specified in [Table 2d](#) for the selected integrity level, the following disposal V&V activity and task described in [Table 1d, Activity 10.14](#):

- a) [Disposal V&V](#). This activity consists of the following task:
  - 1) [Hardware disposal strategy evaluation](#)

This process ends active support by the operation and maintenance organization or deactivates, disassembles, and removes the affected hardware products, consigning them to a final condition and leaving the environment in an acceptable condition. This process destroys or stores system hardware elements and related products in a sound manner, in accordance with legislation, agreements, organizational constraints, and stakeholder requirements. Where required, it maintains records that may be monitored.

**Table 1d—V&V tasks, inputs, and outputs**

| <a href="#">10.1</a> Activity: Business or mission analysis V&V (Hardware, based on 12207—Business or Mission Analysis Process)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                           |
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| V&V tasks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Required inputs                                                                                                                                                                                                                                                                                                                                                                                             | Required outputs                                                                          |
| <b>(1) <a href="#">Business or mission analysis documentation evaluation</a></b><br>a) Evaluate the results of business or mission analysis for correctness, consistency, and completeness.<br>b) Verify and validate that the problem or opportunity space definition adheres to the organizational strategy.<br>c) Verify that the solution space characterization is consistent with the problem or opportunity space.<br>d) Verify that the preliminary life cycle concepts of the candidate solutions are consistent with the solution space.<br>e) Verify that the preferred candidate solution alternative(s) are selected.<br>f) Verify and validate that inputs for requirements of any enabling systems or system elements that serve the business or mission analysis activities are identified.<br>g) Verify the availability of enabling systems or services needed for business or mission analysis. | System-level business or mission analysis documentation<br>Business or mission analysis documentation<br>Organization strategy<br>Organizational concept of operations<br>Organizational strategic goals and plans<br>New market or mission elements<br>Identified problem or opportunity space<br>Identified solution space<br>Identified candidate solution(s)<br>Organizational processes and procedures | Task report(s)—Business or mission analysis documentation evaluation<br>Anomaly report(s) |
| <b>(2) <a href="#">Traceability analysis</a></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Business or mission analysis documentation                                                                                                                                                                                                                                                                                                                                                                  | Task report(s)—Root of traceability                                                       |

| <b>10.1 Activity: Business or mission analysis V&amp;V (Hardware, based on 12207—Business or Mission Analysis Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                          |                                                              |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Required inputs</b>                                                                                                                                                   | <b>Required outputs</b>                                      |
| a) Establish the root of traceability, starting with the organization strategy.<br>b) Verify that the problem or opportunity space traces back to the organization strategy.<br>c) Verify that the solution space traces back to the problem or opportunity space.<br>d) Verify that the candidate solution(s) traces back to the solution space.                                                                                                                                                                                                                                                                                                                                                                  | Organization strategy<br>Organizational concept of operations<br>Identified problem or opportunity space<br>Identified solution space<br>Preferred candidate solution(s) | Anomaly report(s)                                            |
| <b>(3) <a href="#">Criticality analysis</a></b><br>a) Determine whether integrity levels are established for the identified solution space.<br>b) Verify that the assigned integrity levels are correct. If integrity levels are not assigned, then assign integrity levels to the solution space.<br>c) Determine whether integrity levels are established for the preferred candidate solutions.<br>d) Verify that the assigned integrity levels are correct. If integrity levels are not assigned, then assign integrity levels to the preferred candidate solution(s). Preferred candidate solutions.                                                                                                          | Identified problem or opportunity space<br>Identified solution space<br>Preferred candidate solution(s)                                                                  | Task report(s)—<br>Criticality analysis<br>Anomaly report(s) |
| <b>(4) <a href="#">Hazard analysis</a></b><br>Analyze the potential hazards to and from the preferred candidate solutions. The analysis shall perform the following:<br>a) Identify the potential solution hazards.<br>b) Assess the consequences of each hazard.<br>c) Assess the probability of each hazard.<br>d) Identify mitigation strategies for each hazard.                                                                                                                                                                                                                                                                                                                                               | Preferred candidate solution(s)                                                                                                                                          | Task report(s)—<br>Hazard analysis<br>Anomaly report(s)      |
| <b>(5) <a href="#">Security analysis</a></b><br>a) Review the preferred candidate solution(s) to determine an acceptable level of security risk.<br>b) Analyze the preferred candidate solution(s) from a security perspective and assure that potential security risks with respect to confidentiality (disclosure of sensitive information/data), integrity (modification of information/data), availability (withholding of information or services), and accountability (attributing actions to an individual/process) have been identified. Include an assessment of the sensitivity of the information/data to be processed.<br>c) Analyze security risks introduced by the preferred candidate solution(s). | Preferred candidate solution(s)<br>Preliminary threat and risk assessment (TRA)                                                                                          | Task report(s)—<br>Security analysis<br>Anomaly report(s)    |
| <b>(6) <a href="#">Risk analysis</a></b><br>a) Identify the technical and management risks.<br>b) Provide recommendations to eliminate, reduce, or mitigate the risks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Preferred candidate solution(s)<br>Hazard analysis results<br>Security analysis results<br>V&V task results                                                              | Task report(s)—<br>Risk analysis<br>Anomaly report(s)        |

| <b>10.2 Activity: Stakeholder needs and requirements definition V&amp;V (Hardware, based on 12207—Stakeholder Needs and Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                   |                                                                                              |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Required inputs</b>                                                                                                                                                            | <b>Required outputs</b>                                                                      |
| <p><b>(1) <u>Stakeholder needs and requirements evaluation</u></b></p> <p>Evaluate the stakeholder requirements for correctness, consistency, completeness, readability, and testability. The task criteria are as follows:</p> <p>a) Correctness</p> <ol style="list-style-type: none"> <li>1) Verify and validate that the requirements satisfy the stakeholder needs for the system.</li> <li>2) Verify that the requirements comply with standards, references, regulations, policies, physical laws, and business rules.</li> <li>3) Validate that the requirements define the intended interaction of the system with its operating environment and other interfacing systems.</li> <li>4) Validate the suitability of the preferred candidate solution(s) to meet the stakeholder requirements.</li> </ol> <p>b) Consistency</p> <ol style="list-style-type: none"> <li>1) Verify that all terms, concepts, and requirements are documented consistently in accordance with accepted syntax and structure (e.g., style guides and requirements modeling structure).</li> <li>2) Verify that there is consistency between assumptions, requirements, and between groups of requirements.</li> </ol> <p>c) Completeness</p> <ol style="list-style-type: none"> <li>1) Validate that performance criteria and functionality are described in the requirements, within the assumptions and constraints of the operating environment and system boundaries.</li> <li>2) Verify that all stakeholders, or stakeholder classes (used here as groupings of stakeholders), are identified.</li> <li>3) Verify that the stakeholder requirements satisfy specified configuration management procedures. Verify stakeholder requirements are in a form suitable for requirements management throughout the life cycle.</li> </ol> <p>d) Readability</p> <ol style="list-style-type: none"> <li>1) Verify that the documentation is legible, understandable, and unambiguous to the intended audience.</li> <li>2) Verify that the documentation defines all acronyms, mnemonics, abbreviations, terms, and symbols.</li> </ol> <p>e) Testability</p> <ol style="list-style-type: none"> <li>1) Verify that objective acceptance criteria can be developed to validate the requirements.</li> </ol> | <p>Preferred candidate solution(s)</p> <p>Stakeholder needs and requirements</p> <p>Business or mission analysis documentation</p>                                                | <p>Task report(s)—Stakeholder needs and requirements evaluation</p> <p>Anomaly report(s)</p> |
| <p><b>(2) <u>Traceability analysis</u></b></p> <p>a) Establish the root of traceability for hardware, starting with the organization strategy.</p> <p>b) Verify that the candidate hardware solution(s) traces back to the solution space.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Business or mission analysis documentation</p> <p>Identified problem or opportunity space</p> <p>Identified solution space</p> <p>Preferred candidate hardware solution(s)</p> | <p>Task report(s)—Traceability analysis</p> <p>Anomaly report(s)</p>                         |
| <p><b>(3) <u>Criticality analysis</u></b></p> <p>a) Verify that the assigned integrity levels are correct. If integrity levels are not assigned, then assign integrity levels</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Identified problem or opportunity space</p>                                                                                                                                    | <p>Task report(s)—Criticality analysis</p> <p>Anomaly report(s)</p>                          |

| <b>10.2 Activity: Stakeholder needs and requirements definition V&amp;V (Hardware, based on 12207—Stakeholder Needs and Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                             |                                                                  |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Required inputs</b>                                                                                                                                                                      | <b>Required outputs</b>                                          |
| <p>to the system requirements related to hardware, detailed functions, hardware components, subsystems, or other partitions.</p> <p>b) Document the integrity level assigned to individual components (e.g., requirements, detailed functions, hardware components, subsystems, or other partitions). For V&amp;V planning purposes, the hardware component shall be assigned the same integrity level as the highest level assigned to any individual element.</p> <p>c) Verify whether any component can influence individual components assigned a higher integrity level, and if such conditions exist, then assign that component the same higher integrity level.</p>                                                                                                                                                                                                                                                               | <p>Identified solution space</p> <p>Preferred candidate hardware solution(s)</p> <p>Developer integrity level assignments</p>                                                               |                                                                  |
| <p><b>(4) <u>Hazard analysis</u></b></p> <p>Analyze the potential hazards to and from the preferred candidate hardware solution(s). The analysis shall perform the following:</p> <p>a) Identify the potential system hazards (e.g., environmental conditions such as pressure, radiation, temperature, humidity, electrostatic sensitivity, EMI/RFI, seismic, electro-mechanical properties, aging).</p> <p>b) Assess the consequences of each hazard.</p> <p>c) Assess the probability of each hazard.</p> <p>d) Identify mitigation strategies for each hazard.</p>                                                                                                                                                                                                                                                                                                                                                                    | <p>Preferred candidate hardware solution(s)</p>                                                                                                                                             | <p>Task report(s)—Hazard analysis</p> <p>Anomaly report(s)</p>   |
| <p><b>(5) <u>Security analysis</u></b></p> <p>a) Review the system owner's definition of an acceptable level of security risk.</p> <p>b) Analyze the preferred candidate hardware solution(s) from a security perspective and assure that potential security risks with respect to confidentiality (disclosure of sensitive information/data), integrity (modification of information/data), availability (withholding of information or services), and accountability (attributing actions to an individual/process) have been identified. Include an assessment of the sensitivity of the information/data to be processed. Security attributes include access control, electrical/optical isolation, location tracking, and network medium control.</p> <p>c) Analyze security risks introduced by the preferred candidate hardware solution(s) as well as those associated with the environment with which the system interfaces.</p> | <p>Preferred candidate hardware solution(s)</p> <p>Preliminary TRA</p>                                                                                                                      | <p>Task report(s)—Security analysis</p> <p>Anomaly report(s)</p> |
| <p><b>(6) <u>Risk analysis</u></b></p> <p>a) Identify hardware technical and management risks.</p> <p>b) Provide recommendations to eliminate, reduce, or mitigate the risks.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>Preferred candidate hardware solution(s)</p> <p>Supplier development plans and schedules</p> <p>Hazard analysis results</p> <p>Security analysis results</p> <p>V&amp;V task results</p> | <p>Task report(s)—Risk analysis</p> <p>Anomaly report(s)</p>     |

| <b>10.3 Activity: System/hardware requirements definition V&amp;V (Hardware, based on 12207—<br/>System/Hardware Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                               |                                                                                                        |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Required inputs</b>                                                                                                                                        | <b>Required outputs</b>                                                                                |
| <p><b>(1) Requirements evaluation</b></p> <p>Evaluate the requirements (e.g., functional, capability, interface, qualification, safety, security, human factors, data definitions, user documentation, installation and acceptance, user operation, and user maintenance) of the system/hardware requirements and interface requirements for correctness, consistency, completeness, accuracy, readability, and testability. The task criteria are as follows:</p> <p>a) Correctness</p> <ol style="list-style-type: none"> <li>1) Verify and validate that the system/hardware requirements satisfy the system requirements allocated to hardware within the assumptions, constraints, and operating environment for the system.</li> <li>2) Verify that the system/hardware requirements comply with standards, references, regulations, policies, physical laws, and business rules.</li> <li>3) Validate the sequences of states and state changes using logic flows coupled with domain expertise, prototyping results, engineering principles, or other basis.</li> <li>4) Validate that the flow of control satisfies functionality and performance requirements.</li> </ol> <p>b) Consistency</p> <ol style="list-style-type: none"> <li>1) Verify that all terms and concepts are documented consistently.</li> <li>2) Verify that the hardware interactions and assumptions are consistent and satisfy system requirements and acquisition needs.</li> <li>3) Verify that there is internal consistency between the system/hardware requirements and external consistency with the system requirements.</li> </ol> <p>c) Completeness</p> <ol style="list-style-type: none"> <li>1) Verify that the following elements are in the system/hardware requirements or interface requirements, within the assumptions and constraints of the system: <ol style="list-style-type: none"> <li>i) Hardware, software, and user-interface descriptions.</li> <li>ii) Performance criteria (e.g., timing, sizing, speed, capacity, accuracy, precision, safety, and security).</li> <li>iii) System, device, and hardware control (e.g., initialization, transaction and state monitoring, and self-testing).</li> </ol> </li> <li>2) Verify that the system/hardware requirements and interface requirements satisfy specified configuration management procedures.</li> </ol> <p>d) Accuracy</p> <ol style="list-style-type: none"> <li>1) Validate that the modeled physical phenomena conform to system accuracy requirements and physical laws.</li> </ol> <p>e) Readability</p> <ol style="list-style-type: none"> <li>1) Verify that the documentation is legible, understandable, and unambiguous to the intended audience.</li> </ol> | <p>Business or mission analysis documentation</p> <p>Stakeholder needs and requirements</p> <p>System/hardware requirements</p> <p>Interface requirements</p> | <p>Task report(s)—<br/>System/hardware requirements definition evaluation</p> <p>Anomaly report(s)</p> |



| <b>10.3 Activity: System/hardware requirements definition V&amp;V (Hardware, based on 12207—<br/>System/Hardware Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                           |                                                                                                                                                                                               |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Required inputs</b>                                                                                                                                                                                                                    | <b>Required outputs</b>                                                                                                                                                                       |
| <p>2) Verify that the documentation defines all acronyms, mnemonics, abbreviations, terms, and symbols.</p> <p>f) Testability</p> <p>Verify that there are objective acceptance criteria for validating the requirements of the system/hardware requirements and interface requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                           |                                                                                                                                                                                               |
| <p><b>(2) <u>Interface analysis</u></b></p> <p>Verify and validate that the requirements for hardware interfaces with software, user, operator, other hardware, and other systems are correct, consistent, complete, accurate, and testable.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Business or mission analysis documentation</p> <p>Stakeholder needs and requirements</p> <p>System/hardware requirements</p> <p>Interface requirements</p>                                                                             | <p>Task report(s)—<br/>Interface analysis</p> <p>Anomaly report(s)</p>                                                                                                                        |
| <p><b>(3) <u>Traceability analysis</u></b></p> <p>a) Trace the system/hardware requirements to system requirements and stakeholder needs and requirements. Trace system requirements and stakeholder needs and requirements to the system/hardware requirements.</p> <p>b) Analyze identified relationships for correctness, consistency, completeness, and accuracy.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>System requirements</p> <p>System/hardware requirements</p> <p>Interface requirements</p>                                                                                                                                              | <p>Task report(s)—<br/>Traceability analysis</p> <p>Anomaly report(s)</p>                                                                                                                     |
| <p><b>(4) <u>Criticality analysis</u></b></p> <p>Review and update the existing criticality analysis results from the prior criticality task report using the hardware requirements specification (HRS) and IRS.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Criticality task report</p> <p>System/hardware requirements</p> <p>Interface requirements</p>                                                                                                                                          | <p>Task report(s)—<br/>Criticality analysis</p> <p>Anomaly report(s)</p>                                                                                                                      |
| <p><b>(5) <u>Hardware qualification test plan V&amp;V</u></b></p> <p>a) Integrity levels 4 and 3</p> <p>1) Plan V&amp;V hardware qualification testing to validate system/hardware requirements (e.g., static, transient, or dynamic loads; interface compatibility; electromagnetic compatibility; electromagnetic interference; thermal, mechanical, electrical, acoustic, environmental [humidity, water, saltwater, etc.], acceleration, vibration, shock, and pressure).</p> <p>2) Plan tracing of system/hardware requirements to test designs, cases, procedures, and results.</p> <p>3) Plan documentation of test designs, cases, procedures, and results.</p> <p>4) The V&amp;V hardware qualification test plan shall address the following:</p> <p>i) Conformance to all system/hardware requirements (e.g., functional, performance, security, operation, and maintenance) as complete hardware end items in the system environment.</p> <p>ii) Adequacy of user information (e.g., training materials and procedural changes).</p> <p>iii) Performance at boundaries (e.g., data and interfaces) and under stress conditions.</p> <p>5) Verify that the V&amp;V hardware qualification test plan satisfies the following criteria:</p> | <p>Stakeholder needs and requirements</p> <p>System requirements</p> <p>System/hardware requirements</p> <p>Interface requirements</p> <p>User information</p> <p>Hardware qualification test plan from developer (integrity level 2)</p> | <p>V&amp;V hardware qualification test plan (integrity levels 4 and 3)</p> <p>Task report(s)—<br/>Review of hardware qualification test plan (integrity level 2)</p> <p>Anomaly report(s)</p> |

| <b>10.3 Activity: System/hardware requirements definition V&amp;V (Hardware, based on 12207—<br/>System/Hardware Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                   |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Required inputs</b>                                                                                                                                                                                                                                                                          | <b>Required outputs</b>                                                                                                                                                                                                           |
| <ul style="list-style-type: none"> <li>i) Conformance to project-defined test document purpose, format, and content.</li> <li>ii) Test coverage of system/hardware requirements.</li> <li>6) Validate that the V&amp;V hardware qualification test plan satisfies the following criteria: <ul style="list-style-type: none"> <li>i) Appropriateness of test methods and standards used.</li> <li>ii) Conformance to expected results.</li> <li>iii) Feasibility of hardware qualification testing.</li> <li>iv) Feasibility and testability of operation and maintenance requirements.</li> </ul> </li> <li>b) Integrity level 2 <ul style="list-style-type: none"> <li>1) Verify that the developer's hardware qualification test plan satisfies the following criteria: <ul style="list-style-type: none"> <li>i) Conformance to project-defined test document purpose, format, and content.</li> <li>ii) Test coverage of system/hardware requirements.</li> </ul> </li> <li>2) Validate that the developer's hardware qualification test plan satisfies the following criteria: <ul style="list-style-type: none"> <li>i) Appropriateness of test methods and standards used.</li> <li>ii) Conformance to expected results.</li> <li>iii) Feasibility of hardware qualification testing.</li> <li>iv) Capability to be operated and maintained.</li> </ul> </li> </ul> </li> <li>c) Integrity level 1<br/>There are no hardware qualification V&amp;V test requirements.</li> </ul>      |                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                   |
| <p><b>(6) <u>Hardware acceptance test plan V&amp;V</u></b></p> <ul style="list-style-type: none"> <li>a) Integrity levels 4 and 3 <ul style="list-style-type: none"> <li>1) Plan V&amp;V hardware acceptance testing to validate that the hardware correctly implements system and system/hardware requirements (e.g., static, transient, or dynamic loads; interface compatibility; electromagnetic compatibility; electromagnetic interference; thermal; mechanical; electrical; acoustic; environmental (humidity, water, saltwater, etc.); acceleration; vibration; and shock, pressure) in an operational environment.</li> <li>2) Plan tracing of acceptance test requirements to test design, cases, procedures, and execution results.</li> <li>3) Plan documentation of test tasks and results.</li> <li>4) The V&amp;V hardware acceptance test plan shall address the following: <ul style="list-style-type: none"> <li>i) Conformance to acceptance requirements in the operational environment.</li> <li>ii) Adequacy of user information.</li> </ul> </li> <li>5) Verify that the V&amp;V hardware acceptance test plan satisfies the following criteria: <ul style="list-style-type: none"> <li>i) Conformance to project-defined test document purpose, format, and content.</li> <li>ii) Test coverage of acceptance requirements.</li> </ul> </li> <li>6) Validate that the V&amp;V hardware acceptance test plan satisfies the following criteria:</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>Stakeholder needs and requirements</li> <li>System requirements</li> <li>System/hardware requirements</li> <li>Interface requirements</li> <li>User information</li> <li>Hardware acceptance test plan from the developer (integrity level 2)</li> </ul> | <ul style="list-style-type: none"> <li>V&amp;V hardware acceptance test plan (integrity levels 4 and 3)</li> <li>Task report(s)—Review of hardware acceptance test plan (integrity level 2)</li> <li>Anomaly report(s)</li> </ul> |

| <b>10.3 Activity: System/hardware requirements definition V&amp;V (Hardware, based on 12207—System/Hardware Requirements Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                      |                                                       |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Required inputs</b>                                                                                                                                                                                                                                               | <b>Required outputs</b>                               |
| <ul style="list-style-type: none"> <li>i) Conformance to expected results.</li> <li>ii) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs).</li> </ul> <p>b) Integrity level 2</p> <ul style="list-style-type: none"> <li>1) Verify that the acquirer's hardware acceptance test plan conforms to the project-defined test document purpose, format, and content.</li> <li>2) Validate that the acquirer's hardware acceptance test plan satisfies the following criteria: <ul style="list-style-type: none"> <li>i) Test coverage of acceptance requirements.</li> <li>ii) Conformance to expected results.</li> <li>iii) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs).</li> </ul> </li> </ul> <p>c) Integrity level 1</p> <p>There are no hardware acceptance V&amp;V test requirements.</p> |                                                                                                                                                                                                                                                                      |                                                       |
| <p>(7) <a href="#">Hazard analysis</a></p> <p>Determine hardware contributions to system hazards. The hazard analysis shall perform the following:</p> <ul style="list-style-type: none"> <li>a) Identify the system/hardware requirements that contribute to each system hazard.</li> <li>b) Validate that the hardware addresses, controls, or mitigates each hazard.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | System/hardware requirements<br>Interface requirements<br>Hazard analysis results                                                                                                                                                                                    | Task report(s)—Hazard analysis<br>Anomaly report(s)   |
| <p>(8) <a href="#">Security analysis</a></p> <ul style="list-style-type: none"> <li>a) Determine that the security requirements identified in the system/hardware requirements and interface requirements address the security risks introduced by the system concept.</li> <li>b) Verify that the system security requirements will mitigate the identified security risks to an acceptable level.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | System/hardware requirements<br>Interface requirements<br>Preliminary TRA<br>Security analysis results                                                                                                                                                               | Task report(s)—Security analysis<br>Anomaly report(s) |
| <p>(9) <a href="#">Risk analysis</a></p> <ul style="list-style-type: none"> <li>a) Review and update risk analysis using prior task reports.</li> <li>b) Provide recommendations to eliminate, reduce, or mitigate the risks.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Stakeholder needs and requirements<br>System requirements<br>System/hardware requirements<br>Interface requirements<br>Supplier development plans and schedules<br>Hazard analysis results<br>Security analysis results<br>Risk analysis results<br>V&V task results | Task report(s)—Risk analysis<br>Anomaly report(s)     |

| <b>10.4 Activity: Architecture definition V&amp;V (Hardware, based on 12207—Architecture Definition Process)</b>                                                                                 |                                    |                                        |
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| <b>V&amp;V tasks</b>                                                                                                                                                                             | <b>Required inputs</b>             | <b>Required outputs</b>                |
| <p>(1) <a href="#">Architecture evaluation</a></p> <p>Evaluate the hardware architecture for correctness, consistency, completeness, accuracy, readability, testability, and design margins.</p> | Stakeholder needs and requirements | Task report(s)—Architecture evaluation |

| <b>10.4 Activity: Architecture definition V&amp;V (Hardware, based on 12207—Architecture Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                            |                                                                          |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Required inputs</b>                                                                                                                     | <b>Required outputs</b>                                                  |
| Verify and validate the allocation of system/hardware requirements to the hardware elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | System/hardware requirements<br>Interface requirements<br>Hardware architecture<br>Hardware drawings                                       | Anomaly report(s)                                                        |
| (2) <a href="#"><u>Interface analysis</u></a><br>Verify and validate the hardware interfaces with software, other hardware elements, user, operator, and conditioning and enabling hardware/systems (e.g., RC filter network, gear reduction, communication protocol), for correctness, consistency, completeness, accuracy, and testability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | System/hardware requirements<br>Interface requirements<br>System architecture<br>Hardware architecture<br>Hardware drawings                | Task report(s)—<br>Interface analysis<br>Anomaly report(s)               |
| (3) <a href="#"><u>Requirements allocation analysis</u></a><br>Verify the correctness, accuracy, and completeness of the system requirements allocation to the hardware against user needs.<br>a) Correctness<br>Verify that performance requirements (e.g., timing, response time, and throughput) allocated to the hardware, software, and user interfaces satisfy user needs.<br>b) Accuracy<br>Verify that the internal and external interfaces specify the data formats, interface protocols, frequency of data exchange at each interface, and other performance requirements to demonstrate satisfaction of user requirements.<br>c) Completeness<br>1) Verify that application-specific requirements such as functional diversity, fault detection, fault isolation, and diagnostic and error recovery satisfy user needs.<br>2) Verify that the user's maintenance requirements for the system are completely specified.<br>3) Verify that the migration from the existing system and replacement of the system satisfies user needs. | Stakeholder needs and requirements<br>System/hardware requirements<br>Interface requirements<br>Hardware architecture<br>Hardware drawings | Task report(s)—<br>Requirements allocation analysis<br>Anomaly report(s) |
| (4) <a href="#"><u>Traceability analysis</u></a><br>Trace the hardware architectural elements to the system/hardware requirements and the system architecture, and the system/hardware requirements and the system architecture to the hardware architectural elements. Analyze the relationships for correctness, consistency, and completeness.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | System/hardware requirements<br>Interface requirements<br>System architecture<br>Hardware architecture<br>Hardware drawings                | Task report(s)—<br>Traceability analysis<br>Anomaly report(s)            |
| (5) <a href="#"><u>Criticality analysis</u></a><br>Review and update the existing criticality analysis results from the prior criticality task report using the hardware architecture.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Criticality task report<br>Hardware architecture<br>Hardware drawings                                                                      | Task report(s)—<br>Criticality analysis<br>Anomaly report(s)             |
| (6) <a href="#"><u>Hardware component test plan V&amp;V</u></a><br>a) Integrity levels 4, 3 and 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | System/hardware requirements                                                                                                               | V&V hardware component test plan                                         |

| <b>10.4 Activity: Architecture definition V&amp;V (Hardware, based on 12207—Architecture Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                       |                                                                                                                                                                                            |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Required inputs</b>                                                                                                                                                                                                                                                | <b>Required outputs</b>                                                                                                                                                                    |
| <ol style="list-style-type: none"> <li>1) Verify that the developer's hardware component test plan conforms to the project-defined test document purpose, format, and content.</li> <li>2) Validate that the developer's hardware component test plan satisfies the following criteria: <ol style="list-style-type: none"> <li>i) Traceable to the system/hardware requirements and design.</li> <li>ii) External consistency with the system/hardware requirements and design.</li> <li>iii) Internal consistency between unit requirements.</li> <li>iv) Test coverage of units.</li> <li>v) Feasibility of hardware integration and testing.</li> <li>vi) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs).</li> </ol> </li> <li>b) Integrity level 1 <ol style="list-style-type: none"> <li>1) There are no hardware component V&amp;V test requirements.</li> </ol> </li> </ol>                                                                                                                                                                                | <p>Interface requirements</p> <p>Hardware drawings</p> <p>Hardware component test plan from the developer (integrity levels 4, 3, and 2)</p>                                                                                                                          | <p>(integrity levels 4, 3, and 2)</p> <p>Anomaly report(s)</p>                                                                                                                             |
| <p><b>(7) <u>Hardware integration test plan V&amp;V</u></b></p> <ol style="list-style-type: none"> <li>a) Integrity levels 4, 3 and 2 <ol style="list-style-type: none"> <li>1) Verify that the developer's hardware integration test plan conforms to the project-defined test document purpose, format, and content.</li> <li>2) Validate that the developer's hardware integration test plan satisfies the following criteria: <ol style="list-style-type: none"> <li>i) Traceable to the system/hardware requirements and design.</li> <li>ii) External consistency with the system/hardware requirements and design.</li> <li>iii) Internal consistency between unit requirements.</li> <li>iv) Test coverage of system/hardware requirements.</li> <li>v) Feasibility of hardware integration and testing.</li> <li>vi) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs).</li> </ol> </li> </ol> </li> <li>b) Integrity level 1 <ol style="list-style-type: none"> <li>1) There are no hardware Integration V&amp;V test requirements.</li> </ol> </li> </ol> | <p>System/hardware requirements</p> <p>Interface requirements</p> <p>Hardware drawings</p> <p>Hardware integration test plan from the developer (integrity levels 4, 3, and 2)</p>                                                                                    | <p>V&amp;V hardware integration test plan (integrity levels 4, 3, and 2)</p> <p>Anomaly report(s)</p>                                                                                      |
| <p><b>(8) <u>Hardware qualification test design V&amp;V</u></b></p> <ol style="list-style-type: none"> <li>a) Integrity levels 4 and 3 <ol style="list-style-type: none"> <li>1) Design tests for hardware qualification testing.</li> <li>2) Continue tracing required by the V&amp;V hardware qualification test plan. Verify that the V&amp;V hardware qualification test designs conform to the project-defined test document purpose, format, and content.</li> <li>3) Validate that the V&amp;V hardware qualification test designs satisfy the criteria in the V&amp;V hardware qualification test plan (V&amp;V activity <a href="#">10.3, Task 5</a>).</li> </ol> </li> <li>b) Integrity level 2 <ol style="list-style-type: none"> <li>1) Verify that the developer's test designs for hardware qualification testing conform to the project-defined test document purpose, format, and content.</li> </ol> </li> </ol>                                                                                                                                                                                                          | <p>System/hardware requirements</p> <p>Interface requirements</p> <p>System architecture</p> <p>Hardware architecture</p> <p>Hardware drawings</p> <p>User information</p> <p>Hardware qualification test plan</p> <p>Hardware qualification test design from the</p> | <p>V&amp;V hardware qualification test design (integrity levels 4 and 3)</p> <p>Task report(s)—Hardware qualification test design V&amp;V (integrity level 2)</p> <p>Anomaly report(s)</p> |

| <b>10.4 Activity: Architecture definition V&amp;V (Hardware, based on 12207—Architecture Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                      |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Required inputs</b>                                                                                                                                                                                                                                                                        | <b>Required outputs</b>                                                                                                                                                              |
| <p>2) Validate that the developer's hardware qualification test designs satisfy the criteria in the qualification test plan (V&amp;V activity <a href="#">10.3, Task 5.</a>)</p> <p>c) Integrity level 1</p> <p>There are no hardware V&amp;V qualification test requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | developer (integrity level 2)                                                                                                                                                                                                                                                                 |                                                                                                                                                                                      |
| <p><b>(9) <a href="#">Hardware acceptance test design V&amp;V</a></b></p> <p>a) Integrity levels 4 and 3</p> <p>1) Design tests for V&amp;V hardware acceptance testing.</p> <p>2) Continue tracing required by the V&amp;V hardware acceptance test plan. Verify that the V&amp;V hardware acceptance test designs conform to the project-defined test document purpose, format, and content.</p> <p>3) Validate that the V&amp;V hardware acceptance test designs satisfy the criteria in the V&amp;V hardware acceptance test plan (V&amp;V activity <a href="#">10.3, Task 6.</a>)</p> <p>b) Integrity level 2</p> <p>1) Verify that the acquirer's hardware test designs for acceptance testing conform to the project-defined test document purpose, format, and content.</p> <p>2) Validate that the acquirer's hardware acceptance test designs satisfy the criteria in the hardware acceptance test plan (V&amp;V activity <a href="#">10.3, Task 6.</a>)</p> <p>c) Integrity level 1</p> <p>There are no hardware acceptance V&amp;V test requirements.</p> | <p>System/hardware requirements</p> <p>Interface requirements</p> <p>System architecture</p> <p>Hardware architecture</p> <p>Hardware drawings</p> <p>User information</p> <p>Hardware acceptance test plan</p> <p>Hardware acceptance test design from the developer (integrity level 2)</p> | <p>V&amp;V hardware acceptance test design (integrity levels 4 and 3)</p> <p>Task report(s)—Hardware acceptance test design V&amp;V (integrity level 2)</p> <p>Anomaly report(s)</p> |
| <p><b>(10) <a href="#">Hazard analysis</a></b></p> <p>a) Verify the hardware architectural elements that implement critical requirements and introduce no new hazards.</p> <p>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and hardware operations).</p> <p>c) Update the hazard analysis.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>System architecture</p> <p>Hardware architecture</p> <p>Hardware drawings</p> <p>Hazard analysis results</p>                                                                                                                                                                               | <p>Task report(s)—Hazard analysis</p> <p>Anomaly report(s)</p>                                                                                                                       |
| <p><b>(11) <a href="#">Security analysis</a></b></p> <p>a) Verify that the hardware element complies with the security features of the hardware architecture and that the hardware element does not introduce new security risks.</p> <p>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and hardware operations).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>System architecture</p> <p>Hardware architecture</p> <p>Hardware drawings</p> <p>Security analysis results</p> <p>V&amp;V task results</p>                                                                                                                                                 | <p>Task report(s)—Security analysis</p> <p>Anomaly report(s)</p>                                                                                                                     |
| <p><b>(12) <a href="#">Risk analysis</a></b></p> <p>a) Review and update risk analysis using prior task reports.</p> <p>b) Provide recommendations to eliminate, reduce, or mitigate the risks.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>System architecture</p> <p>Hardware architecture</p> <p>Hardware drawings</p> <p>Supplier development plans and schedules</p> <p>Hazard analysis results</p> <p>Security analysis results</p> <p>V&amp;V task results</p>                                                                  | <p>Task report(s)—Risk analysis</p> <p>Anomaly report(s)</p>                                                                                                                         |

| <b>10.5 Activity: Design definition V&amp;V (Hardware, based on 12207—Design Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                   |                                                                                                        |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Required inputs</b>                                                                                                                                                                                                                                                                            | <b>Required outputs</b>                                                                                |
| <b>(1) <a href="#">Design evaluation</a></b><br>Evaluate the hardware design elements for correctness, consistency, completeness, accuracy, readability, testability, and design margins. Verify and validate the allocation of system/hardware requirements to the hardware architecture.                                                                                                                                                                                                                                | System/hardware requirements<br>Interface requirements<br>Hardware architecture<br>Hardware design<br>Hardware drawings<br>Design standards (e.g., standards, practices, and conventions)                                                                                                         | Task report(s)—<br>Design evaluation<br>Anomaly report(s)                                              |
| <b>(2) <a href="#">Interface analysis</a></b><br>Verify and validate the hardware design interfaces with software, user, operator, and conditioning and enabling hardware/systems (e.g., RC filter network, gear reduction, communication protocol), for correctness, consistency, completeness, accuracy, and testability.                                                                                                                                                                                               | Stakeholder needs and requirements<br>System/hardware requirements<br>Interface requirements<br>Hardware architecture<br>Hardware design<br>Hardware drawings                                                                                                                                     | Task report(s)—<br>Interface analysis<br>Anomaly report(s)                                             |
| <b>(3) <a href="#">Traceability analysis</a></b><br>Trace the hardware design elements to the hardware architectural elements, and the hardware architectural elements to the hardware design elements. Analyze the relationships for correctness, consistency, and completeness.                                                                                                                                                                                                                                         | Hardware architecture<br>Hardware design<br>Hardware drawings                                                                                                                                                                                                                                     | Task report(s)—<br>Traceability analysis<br>Anomaly report(s)                                          |
| <b>(4) <a href="#">Criticality analysis</a></b><br>Review and update the existing criticality analysis results from the prior Criticality Task Report using the hardware design.                                                                                                                                                                                                                                                                                                                                          | Criticality task report<br>Hardware design                                                                                                                                                                                                                                                        | Task report(s)—<br>Criticality analysis<br>Anomaly report(s)                                           |
| <b>(5) <a href="#">Hardware component test design V&amp;V</a></b><br>a) Integrity levels 4, 3, and 2<br>1) Verify that the developer's test designs for hardware component testing conform to the project-defined test document purpose, format, and content.<br>2) Validate that the developer's hardware component test designs satisfy the criteria in the hardware component test plan (V&V activity <a href="#">10.4, Task 6</a> ).<br>b) Integrity level 1<br>There are no hardware component V&V test requirements | System/hardware requirements<br>Interface requirements<br>System architecture<br>Hardware architecture<br>Hardware drawings<br>Hardware component test plan from the developer (integrity levels 4, 3, and 2)<br>Hardware component test design from the developer (integrity levels 4, 3, and 2) | V&V hardware component test design (integrity levels 4, 3, and 2)<br>Anomaly report(s)                 |
| <b>(6) <a href="#">Hardware component test case V&amp;V</a></b><br>a) Integrity levels 4, 3 and 2<br>1) Verify that the developer's test designs for hardware component testing conform to the project-defined test document purpose, format, and content.<br>2) Validate that the developer's hardware component test cases satisfy the criteria in the hardware component test design (V&V activity <a href="#">10.5, Task 5</a> ).<br>b) Integrity level 1<br>There are no hardware component V&V test requirements.   | Hardware design<br>Hardware drawings<br>User information<br>Hardware component test design from the developer (integrity levels 4, 3, and 2)<br>Hardware component test cases from the developer (integrity levels 4, 3, and 2)                                                                   | Task report(s)—<br>Hardware component test case V&V (integrity levels 4, 3 and 2)<br>Anomaly report(s) |



| <b>10.5 Activity: Design definition V&amp;V (Hardware, based on 12207—Design Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                   |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Required inputs</b>                                                                                                                                                                                                                                                                                                               | <b>Required outputs</b>                                                                                                                                                                           |
| <p><b>(7) <a href="#">Hardware integration test design V&amp;V</a></b></p> <p>a) Integrity levels 4, 3, and 2</p> <ol style="list-style-type: none"> <li>1) Verify that the developer's test designs for hardware integration testing conform to the project-defined test document purpose, format, and content.</li> <li>2) Validate that the developer's hardware integration test designs satisfy the criteria in the hardware integration test plan (V&amp;V activity <a href="#">10.4, Task 7</a>).</li> </ol> <p>b) Integrity level 1</p> <p>There are no hardware integration V&amp;V test requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>System/hardware requirements</p> <p>Interface requirements</p> <p>System architecture</p> <p>Hardware architecture</p> <p>Hardware drawings</p> <p>Hardware integration test plan from the developer (integrity levels 4, 3, and 2)</p> <p>Hardware integration test design from the developer (integrity levels 4, 3, and 2)</p> | <p>V&amp;V hardware integration test design (integrity levels 4, 3, and 2)</p> <p>Anomaly report(s)</p>                                                                                           |
| <p><b>(8) <a href="#">Hardware integration test case V&amp;V</a></b></p> <p>a) Integrity levels 4, 3, and 2</p> <ol style="list-style-type: none"> <li>1) Verify that the developer's test designs for hardware integration testing conform to the project-defined test document purpose, format, and content.</li> <li>2) Validate that the developer's hardware integration test designs satisfy the criteria in the hardware integration test design (V&amp;V activity <a href="#">10.5, Task 7</a>).</li> </ol> <p>b) Integrity level 1</p> <p>There are no hardware integration V&amp;V test requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>Hardware design</p> <p>Hardware drawings</p> <p>User information</p> <p>Hardware integration test design from the developer (integrity levels 4, 3, and 2)</p> <p>Hardware integration test cases from the developer (integrity levels 4, 3, and 2)</p>                                                                           | <p>Task report(s)—</p> <p>Hardware integration test case V&amp;V (integrity levels 4, 3 and 2)</p> <p>Anomaly report(s)</p>                                                                       |
| <p><b>(9) <a href="#">Hardware qualification test case V&amp;V</a></b></p> <p>a) Integrity levels 4 and 3</p> <ol style="list-style-type: none"> <li>1) Design tests for qualification testing.</li> <li>2) Continue tracing required by the V&amp;V qualification test plan. Verify that the V&amp;V qualification test designs conform to the project-defined test document purpose, format, and content.</li> <li>3) Validate that the V&amp;V qualification test designs satisfy the criteria in the hardware qualification test design (V&amp;V activity <a href="#">10.4, Task 8</a>).</li> </ol> <p>b) Integrity level 2</p> <ol style="list-style-type: none"> <li>1) Verify that the developer's test designs for hardware qualification testing conform to the project-defined test document purpose, format, and content.</li> <li>2) Validate that the developer's hardware qualification test designs satisfy the criteria in the hardware qualification test design (V&amp;V activity <a href="#">10.4, Task 8</a>).</li> </ol> <p>c) Integrity level 1</p> <p>There are no hardware V&amp;V qualification test requirements.</p> | <p>Hardware design</p> <p>Hardware drawings</p> <p>User information</p> <p>Hardware qualification test design</p> <p>Hardware qualification test cases from the developer (integrity level 2)</p>                                                                                                                                    | <p>V&amp;V hardware qualification test case(s) (integrity levels 4 and 3)</p> <p>Task report(s)—</p> <p>Hardware qualification test case V&amp;V (integrity level 2)</p> <p>Anomaly report(s)</p> |
| <p><b>(10) <a href="#">Hardware acceptance test case V&amp;V</a></b></p> <p>a) Integrity levels 4 and 3</p> <ol style="list-style-type: none"> <li>1) Design tests for V&amp;V hardware acceptance testing.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Hardware design</p> <p>Hardware drawings</p> <p>User documentation</p>                                                                                                                                                                                                                                                            | <p>V&amp;V hardware acceptance test case(s) (integrity levels 4 and 3)</p>                                                                                                                        |



| <b>10.5 Activity: Design definition V&amp;V (Hardware, based on 12207—Design Definition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                           |                                                                                                          |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Required inputs</b>                                                                                                                                                                                                                                    | <b>Required outputs</b>                                                                                  |
| <p>2) Continue tracing required by the V&amp;V hardware acceptance test plan. Verify that the V&amp;V hardware acceptance test designs conform to the project-defined test document purpose, format, and content.</p> <p>3) Validate that the V&amp;V hardware acceptance test designs satisfy the criteria in the hardware acceptance test design (V&amp;V activity <a href="#">10.4, Task 9</a>).</p> <p>b) Integrity level 2</p> <p>1) Verify that the acquirer's hardware test designs for acceptance testing conform to the project-defined test document purpose, format, and content.</p> <p>2) Validate that the acquirer's hardware acceptance test designs satisfy the criteria in the hardware acceptance test design (V&amp;V activity <a href="#">10.4, Task 9</a>).</p> <p>c) Integrity level 1</p> <p>There are no hardware acceptance V&amp;V test requirements.</p> | <p>Hardware acceptance test design</p> <p>Hardware qualification test cases from developer (integrity level 2)</p>                                                                                                                                        | <p>Task report(s)—Hardware acceptance test case V&amp;V (integrity level 2)</p> <p>Anomaly report(s)</p> |
| <p><b>(11) <a href="#">Hazard analysis</a></b></p> <p>a) Verify the design elements that implement critical requirements introduce no new hazards.</p> <p>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and hardware operations).</p> <p>c) Update the hazard analysis.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>System architecture</p> <p>Hardware architecture</p> <p>Hardware drawings</p> <p>Hazard analysis results</p>                                                                                                                                           | <p>Task report(s)—Hazard analysis</p> <p>Anomaly report(s)</p>                                           |
| <p><b>(12) <a href="#">Security analysis</a></b></p> <p>a) Verify that the design element complies with the security features of the architectural and that the design element does not introduce new security risks.</p> <p>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and hardware operations).</p>                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>System architecture</p> <p>Hardware architecture</p> <p>Hardware drawings</p> <p>Security analysis results</p> <p>V&amp;V task results</p>                                                                                                             | <p>Task report(s)—Security analysis</p> <p>Anomaly report(s)</p>                                         |
| <p><b>(13) <a href="#">Risk analysis</a></b></p> <p>a) Review and update risk analysis using prior task reports.</p> <p>b) Provide recommendations to eliminate, reduce, or mitigate the risks.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>System architecture</p> <p>Hardware architecture</p> <p>Hardware drawings</p> <p>Supplier development plans and schedules</p> <p>Hazard analysis results</p> <p>Security analysis results</p> <p>Risk analysis results</p> <p>V&amp;V task results</p> | <p>Task report(s)—Risk analysis</p> <p>Anomaly report(s)</p>                                             |

| <b>10.6 Activity: System analysis V&amp;V (Hardware, based on 12207—System Analysis Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                     |                                                        |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Required inputs</b>                              | <b>Required outputs</b>                                |
| <p><b>(1) <u>System analysis strategy evaluation</u></b></p> <p>Evaluate the system analysis strategy for completeness and for consistency with the criticality of the hardware. The task criteria are as follows:</p> <p>a) Completeness</p> <ol style="list-style-type: none"> <li>1) Verify that the scope and objectives of the system analysis address the correct needs.</li> <li>2) Verify that the scope of the system analysis addresses all aspects related to the decision.</li> <li>3) Verify that all assumptions have been identified and are correct.</li> <li>4) Verify that any inputs for enabling systems or services have been identified and are available.</li> <li>5) Verify that any inputs for enabling the system have been identified and are available in the system analysis strategy plan.</li> </ol> <p>b) Consistency with criticality</p> <p>Verify that the rigor and approach specified in the system analysis strategy is consistent with that needed for the hardware elements that could be affected by the decision.</p> | System analysis strategy                            | Task report(s)—<br>System analysis strategy evaluation |
| <p><b>(2) <u>System analysis results evaluation</u></b></p> <p>Evaluate the results of the system analysis to determine that the analysis was conducted as specified in the strategy, and the results support its conclusions and recommendations.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | System analysis strategy<br>System analysis results | Task report(s)—<br>System analysis results evaluation  |

| <b>10.7 Activity: Implementation V&amp;V (Hardware, based on 12207—Implementation Process)</b>                                                                                                                                                      |                                                                                                                                                                                |                                                                                     |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                | <b>Required inputs</b>                                                                                                                                                         | <b>Required outputs</b>                                                             |
| <p><b>(1) <u>Fabricated component information evaluation</u></b></p> <p>Evaluate the fabricated component information for correctness, consistency, completeness, accuracy, and readability.</p>                                                    | Component information (e.g., documentation)<br>Hardware design<br>Interface design<br>Hardware drawings (e.g., shop drawings, bill of materials)<br>User information           | Task report(s)—<br>Fabricated component information evaluation<br>Anomaly report(s) |
| <p><b>(2) <u>Interface analysis</u></b></p> <p>Verify and validate the fabricated component interfaces with software, user, operator, other hardware, and other systems for correctness, consistency, completeness, accuracy, and testability.</p>  | Concept documentation<br>System/hardware requirements<br>Hardware design<br>Interface design<br>Hardware drawings (e.g., shop drawings, bill of materials)<br>User information | Task report(s)—<br>Interface analysis<br>Anomaly report(s)                          |
| <p><b>(3) <u>Traceability analysis</u></b></p> <p>a) Trace the fabricated component(s) to the design, and the design to the fabricated component(s).</p> <p>b) Analyze identified relationships for correctness, consistency, and completeness.</p> | Hardware design<br>Interface design<br>Hardware drawings (e.g., shop drawings, bill of materials)<br>Component documentation                                                   | Task report(s)—<br>Traceability analysis<br>Anomaly report(s)                       |

| <b>10.7 Activity: Implementation V&amp;V (Hardware, based on 12207—Implementation Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                     |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Required inputs</b>                                                                                                                                                                                                                                                                                                                                         | <b>Required outputs</b>                                                                                             |
| <b>(4) <u>Criticality analysis</u></b><br>Review and update the existing criticality analysis results from the prior criticality task report using the fabricated component and its associated documentation.                                                                                                                                                                                                                                                                                                            | Criticality task report<br>Component documentation                                                                                                                                                                                                                                                                                                             | Task report(s)—<br>Criticality analysis<br>Anomaly report(s)                                                        |
| <b>(5) <u>Hardware component test procedure V&amp;V</u></b><br>a) Integrity levels 4, 3, and 2<br>1) Verify that the developer's hardware component test procedures conform to the project-defined test document purpose, format, and content.<br>2) Validate that the developer's hardware component test procedures satisfy the criteria in the hardware component test cases (V&V activity <a href="#">10.5, Task 6</a> ).<br>b) Integrity level 1<br>There are no hardware component V&V test requirements.          | System/hardware requirements<br>Interface requirements<br>Hardware design<br>Interface design<br>Hardware drawings (e.g., shop drawings, bill of materials)<br>User information<br>Hardware component test cases from developer (integrity levels 4, 3, and 2)<br>Hardware component test procedures from the developer (integrity levels 4, 3, and 2)         | Task report(s)—<br>Hardware component test procedure V&V (integrity levels 4, 3, and 2)<br>Anomaly report(s)        |
| <b>(6) <u>Hardware component test execution V&amp;V</u></b><br>a) Integrity levels 4, 3, and 2<br>Use the developer's hardware component test results to validate that the hardware component satisfies the test acceptance criteria.<br>b) Integrity level 1<br>There are no hardware component V&V test requirements.                                                                                                                                                                                                  | Component documentation<br>Hardware design<br>Interface design<br>Hardware drawings (e.g., shop drawings, bill of materials)<br>Component test plans<br>Hardware component test procedures from the developer (integrity levels 4, 3, and 2)<br>Hardware component test results from the developer (integrity levels 4, 3, and 2)                              | Task report(s)—<br>Hardware component test execution assessment (integrity levels 4, 3, and 2)<br>Anomaly report(s) |
| <b>(7) <u>Hardware integration test procedure V&amp;V</u></b><br>a) Integrity levels 4, 3 and 2<br>1) Verify that the developer's hardware integration test procedures conform to the project-defined test document purpose, format, and content.<br>2) Validate that the developer's hardware integration test procedures satisfy the criteria in the hardware integration test cases (V&V activity <a href="#">10.5, Task 8</a> ).<br>b) Integrity level 1<br>There are no hardware integration V&V test requirements. | System/hardware requirements<br>Interface requirements<br>Hardware design<br>Interface design<br>Hardware drawings (e.g., shop drawings, bill of materials)<br>User information<br>Hardware integration test cases from the developer (integrity levels 4, 3, and 2)<br>Hardware integration test procedures from the developer (integrity levels 4, 3, and 2) | Task report(s)—<br>Hardware integration test procedure V&V (integrity levels 4, 3, and 2)<br>Anomaly report(s)      |

| <b>10.7 Activity: Implementation V&amp;V (Hardware, based on 12207—Implementation Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                   |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Required inputs</b>                                                                                                                                                                                                                                                                                                               | <b>Required outputs</b>                                                                                                                                                                           |
| <p><b>(8) <a href="#">Hardware qualification test procedure V&amp;V</a></b></p> <p>a) Integrity levels 4 and 3</p> <ol style="list-style-type: none"> <li>1) Develop test procedures for V&amp;V hardware qualification testing.</li> <li>2) Continue tracing required by the V&amp;V hardware qualification test plan.</li> <li>3) Verify that the V&amp;V hardware qualification test procedures conform to the project-defined test document purpose, format, and content.</li> <li>4) Validate that the V&amp;V hardware qualification test procedures satisfy the criteria in the V&amp;V hardware qualification test cases (V&amp;V activity <a href="#">10.5, Task 9</a>).</li> </ol> <p>b) Integrity level 2</p> <ol style="list-style-type: none"> <li>1) Verify that the developer's hardware qualification test procedures conform to the project-defined test document purpose, format, and content.</li> <li>2) Validate that the developer's hardware qualification test procedures satisfy the criteria in the V&amp;V hardware qualification test cases (V&amp;V activity <a href="#">10.5, Task 9</a>).</li> </ol> <p>c) Integrity level 1</p> <p>There are no hardware qualification V&amp;V test requirements.</p>                                                               | <p>System/hardware requirements</p> <p>Interface requirements</p> <p>Hardware design</p> <p>Interface design</p> <p>Hardware drawings (e.g., shop drawings, bill of materials)</p> <p>User information</p> <p>Hardware qualification test cases</p> <p>Hardware qualification test procedures from developer (integrity level 2)</p> | <p>V&amp;V hardware qualification test procedures (integrity levels 4 and 3)</p> <p>Task report(s)—Hardware qualification test procedure V&amp;V (integrity level 2)</p> <p>Anomaly report(s)</p> |
| <p><b>(9) <a href="#">Hardware acceptance test procedure V&amp;V</a></b></p> <p>a) Integrity levels 4 and 3</p> <ol style="list-style-type: none"> <li>1) Develop test procedures for V&amp;V hardware acceptance testing.</li> <li>2) Continue the tracing required by the V&amp;V hardware acceptance test plan.</li> <li>3) Verify that the V&amp;V hardware acceptance test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 <a href="#">[B17]</a>).</li> <li>4) Validate that the V&amp;V hardware acceptance test procedures satisfy the criteria in the V&amp;V hardware acceptance test cases (V&amp;V activity <a href="#">10.5, Task 10</a>).</li> </ol> <p>b) Integrity level 2</p> <ol style="list-style-type: none"> <li>1) Verify that the acquirer's hardware acceptance test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 <a href="#">[B17]</a>).</li> <li>2) Validate that the acquirer's hardware acceptance test procedures satisfy the criteria in the V&amp;V hardware acceptance test cases (V&amp;V activity <a href="#">10.5, Task 10</a>).</li> </ol> <p>c) Integrity level 1</p> <p>There are no hardware acceptance V&amp;V test requirements.</p> | <p>Hardware design</p> <p>Interface design</p> <p>Hardware drawings</p> <p>Hardware component specifications</p> <p>User information</p> <p>Hardware acceptance test plan</p> <p>Hardware acceptance test procedures from the developer (integrity level 2)</p>                                                                      | <p>V&amp;V hardware acceptance test procedures (integrity levels 4 and 3)</p> <p>Task report(s)—Hardware acceptance test procedure V&amp;V (integrity level 2)</p> <p>Anomaly report(s)</p>       |
| <p><b>(10) <a href="#">Hazard analysis</a></b></p> <p>a) Verify that the fabricated component correctly implements the critical requirements and introduces no new hazards.</p> <p>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and hardware operations).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>Component documentation</p> <p>Hardware design</p> <p>Interface design</p> <p>Hardware drawings (e.g., shop drawings, bill of materials)</p> <p>Hazard analysis results</p>                                                                                                                                                       | <p>Task report(s)—Hazard analysis</p> <p>Anomaly report(s)</p>                                                                                                                                    |

| <b>10.7 Activity: Implementation V&amp;V (Hardware, based on 12207—Implementation Process)</b>                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                        |                                                           |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Required inputs</b>                                                                                                                                                                                                                 | <b>Required outputs</b>                                   |
| c) Update the hazard analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                        |                                                           |
| <b>(11) <a href="#">Security analysis</a></b><br>a) Verify that the fabricated component is completed in accordance with the security features of the system design and that the fabricated component does not introduce new security risks.<br>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and hardware operations). | Component documentation<br>Hardware design<br>Interface design<br>Hardware drawings (e.g., shop drawings, bill of materials)<br>Security analysis results                                                                              | Task report(s)—<br>Security analysis<br>Anomaly report(s) |
| <b>(12) <a href="#">Risk analysis</a></b><br>Review and update risk analysis using prior task reports. Provide recommendations to eliminate, reduce, or mitigate the risks.                                                                                                                                                                                                                                                                                                 | Component documentation<br>Hardware drawings (e.g., shop drawings, bill of materials)<br>Supplier development plans and schedules<br>Hazard analysis results<br>Security analysis results<br>Risk analysis results<br>V&V task results | Task report(s)—<br>Risk analysis<br>Anomaly report(s)     |

| <b>10.8 Activity: Integration V&amp;V (Hardware, based on 12207—Integration Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                    |                                                                                                                                  |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Required inputs</b>                                                                                                                                                                             | <b>Required outputs</b>                                                                                                          |
| <b>(1) <a href="#">Hardware integration strategy assessment</a></b><br>a) Verify the hardware integration strategy.<br>b) Verify that any constraints specified or implied by the hardware integration strategy are included in the system/hardware requirements and design.                                                                                                                                                                                                                                                                   | Hardware integration strategy<br>Hardware architecture<br>System/hardware requirements                                                                                                             | Task report(s)—<br>Hardware integration strategy assessment<br>Anomaly report(s)                                                 |
| <b>(2) <a href="#">Hardware element interaction analysis</a></b><br>Analyze the results of system integration V&V to determine whether directly or indirectly coupled hardware elements require their V&V tasks to be reiterated (repeated, redone) or new V&V tasks performed. As integration results are determined, V&V may have to be recursively applied to all directly or indirectly coupled system elements. The hardware analysis process may be used to establish the criteria of how to recursively apply V&V to hardware elements. | Hardware integration results<br>Hardware architecture<br>Hardware design                                                                                                                           | Task report(s)—<br>Hardware element interaction analysis<br>Anomaly report(s)<br>V&V Plan update (if tasks are added or changed) |
| <b>(3) <a href="#">Hardware integration test execution V&amp;V</a></b><br>a) Integrity levels 4, 3, and 2<br>Use the developer's hardware integration test results to verify that the hardware satisfies the test acceptance criteria.<br>b) Integrity level 1<br>There are no hardware acceptance V&V test requirements.                                                                                                                                                                                                                      | Hardware component specifications<br>Hardware integration test plan<br>Hardware integration test procedures<br>Hardware integration test results from the developer (integrity levels 4, 3, and 2) | Task report(s)—<br>Review of hardware integration test execution results (integrity levels 4, 3, and 2)<br>Anomaly report(s)     |

| <b>10.8 Activity: Integration V&amp;V (Hardware, based on 12207—Integration Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                     |                                                                                                                                                                                                                 |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Required inputs</b>                                                                                                                                                                                              | <b>Required outputs</b>                                                                                                                                                                                         |
| <p><b>(4) <u>Hardware qualification test execution V&amp;V</u></b></p> <p>a) Integrity levels 4 and 3</p> <ol style="list-style-type: none"> <li>1) Perform V&amp;V hardware qualification testing.</li> <li>2) Analyze test results to validate that the hardware satisfies the system requirements.</li> <li>3) Validate that the test results trace to test criteria established by the test traceability in the test planning documents.</li> <li>4) Document the results as required by the V&amp;V hardware qualification test plan.</li> <li>5) Use the V&amp;V hardware qualification test results to validate that the hardware satisfies the test acceptance criteria.</li> <li>6) Document discrepancies between the actual and expected test results.</li> </ol> <p>b) Integrity levels 2 and 1</p> <p>Use the developer's hardware qualification test results to verify that the hardware satisfies the test acceptance criteria</p> | <p>Hardware component specifications</p> <p>Hardware qualification test plan</p> <p>Hardware qualification test procedures</p> <p>Hardware qualification test results from developer (integrity levels 2 and 1)</p> | <p>V&amp;V hardware qualification test execution results (integrity levels 4 and 3)</p> <p>Task report(s)—Hardware qualification test execution V&amp;V (integrity levels 2 and 1)</p> <p>Anomaly report(s)</p> |
| <p><b>(5) <u>Traceability analysis</u></b></p> <p>Analyze relationships in the V&amp;V test plans, designs, cases, and procedures for correctness and completeness. The task criteria are as follows:</p> <p>a) Correctness</p> <p>Verify that there is a valid relationship between the V&amp;V test plans, designs, cases, and procedures.</p> <p>b) Completeness</p> <p>Verify that all V&amp;V test procedures are traceable to the V&amp;V test plans.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>V&amp;V test plans</p> <p>V&amp;V test designs</p> <p>V&amp;V test procedures</p>                                                                                                                                | <p>Task report(s)—Traceability analysis</p> <p>Anomaly report(s)</p>                                                                                                                                            |
| <p><b>(6) <u>Hazard analysis</u></b></p> <p>a) Verify that the test instrumentation does not introduce new hazards.</p> <p>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and hardware operations).</p> <p>c) Update the hazard analysis.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>Hardware component specifications</p> <p>Test results</p> <p>Hazard analysis results</p>                                                                                                                         | <p>Task report(s)—Hazard analysis</p> <p>Anomaly report(s)</p>                                                                                                                                                  |
| <p><b>(7) <u>Security analysis</u></b></p> <p>a) Verify that the implemented system does not increase the security risk.</p> <p>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and hardware operations).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Hardware component specifications</p> <p>Security analysis results</p>                                                                                                                                           | <p>Task report(s)—Security analysis</p> <p>Anomaly report(s)</p>                                                                                                                                                |
| <p><b>(8) <u>Risk analysis</u></b></p> <p>a) Review and update risk analysis using prior task reports.</p> <p>b) Provide recommendations to eliminate, reduce, or mitigate the risks.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>Supplier development plans and schedules</p> <p>Hazard analysis results</p> <p>Security analysis results</p> <p>Risk analysis results</p> <p>V&amp;V task results</p>                                            | <p>Task report(s)—Risk analysis</p> <p>Anomaly report(s)</p>                                                                                                                                                    |

| <b>10.9 Activity: Verification (Hardware, based on 12207—Verification Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                      |                                                                                                                        |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Required inputs</b>                                                                                               | <b>Required outputs</b>                                                                                                |
| The activities and tasks for the hardware Verification Process are conducted in hardware technical life cycle processes, and are contained in <a href="#">Table 1d</a> , <a href="#">Activity 10.1</a> (business or mission analysis V&V), <a href="#">Activity 10.2</a> (stakeholder needs and requirements definition V&V), <a href="#">Activity 10.3</a> (system/hardware requirements definition V&V), <a href="#">Activity 10.4</a> (architecture definition V&V), <a href="#">Activity 10.5</a> (design definition V&V), <a href="#">Activity 10.6</a> (system analysis V&V), <a href="#">Activity 10.7</a> (implementation V&V), <a href="#">Activity 10.8</a> (integration V&V), <a href="#">Activity 10.10</a> (transition V&V), <a href="#">Activity 10.12</a> (operation V&V), <a href="#">Activity 10.13</a> (maintenance V&V), and <a href="#">Activity 10.14</a> (disposal V&V). The hardware Verification activities and tasks in <a href="#">Table 1d</a> and the common Verification activities and tasks in <a href="#">Table 1a</a> represent all activities and tasks needed to perform hardware Verification. | The required inputs for the Verification Process for hardware are found in <a href="#">Table 1d</a> required inputs. | The required outputs for the Verification Process for hardware are found in <a href="#">Table 1d</a> required outputs. |

| <b>10.10 Activity: Transition V&amp;V (Hardware, based on 12207—Transition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                        |                                                                             |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Required inputs</b>                                                                                                                                                                                                                                                                                                                                 | <b>Required outputs</b>                                                     |
| <b>(1) <a href="#">Transition strategy evaluation</a></b><br>a) Verify that the transition strategy is comprehensive. Verify that the transition strategy includes the following: <ol style="list-style-type: none"> <li>1) All the hardware parts and the system (e.g., hardware [HW]/software [SW] functionality and databases) are considered in the transition execution.</li> <li>2) Transition schedule and sequence.</li> <li>3) Identification of transition hardware and software tools, equipment, and instructions.</li> <li>4) Archiving hardware artifacts, such as documentation and code.</li> <li>5) Impact of interacting systems in terms of transition timing and transition impacts.</li> <li>6) Continuity of capabilities when replacing or upgrading a legacy system.</li> <li>7) Site preparation for installation and legacy system retirement, storage, and/or incorporation.</li> <li>8) Provisions for documentation of the process results.</li> <li>9) A fallback position, in case of transition failure (e.g., going back to the old system) and other risk mitigation considerations.</li> </ol> b) Verify that the transition strategy has a defined approach to establishing the hardware in the operational environment that is consistent with stakeholder requirements (e.g., by running the old and new systems in parallel or by well-defined hardware characteristics that can be compared). | Transition strategy<br>Installation package (e.g., user documentation, system requirements, hardware design, interface design (IDD), system/hardware requirements, interface requirements (IRS), mission or business analysis documentation, installation procedures, site-specific parameters, installation tests, and configuration management data) | Task report(s)—<br>Transition strategy evaluation<br>Anomaly report(s)      |
| <b>(2) <a href="#">Transition demonstration assessment</a></b><br>a) Validate that acceptance tests defined in the acquirer agreement (as documented in the acceptance test plan) demonstrate satisfactory installation at the specified location(s). Where the exact location or environment of operation is not available, a representative example is selected.<br>b) Verify and validate that the installed hardware elements and related site-specific parameters are identical to the final as-verified and as-validated hardware elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Transitioning hardware (operational hardware under test)<br>User documentation<br>Installation package (e.g., user documentation, system requirements, hardware design, interface design (IDD), system/hardware                                                                                                                                        | Task report(s)—<br>Transition demonstration assessment<br>Anomaly report(s) |



| <b>10.10 Activity: Transition V&amp;V (Hardware, based on 12207—Transition Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                             |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Required inputs</b>                                                                                                                                                                                                                                                                                                | <b>Required outputs</b>                                                                                                                                                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | requirements, interface requirements (IRS), business or mission analysis documentation, installation procedures, site-specific parameters, installation tests, and configuration management data)<br>Acquirer agreement<br>Acceptance test plan                                                                       |                                                                                                                                                                             |
| <b>(3) <u>Hardware acceptance test execution V&amp;V</u></b><br>a) Integrity levels 4 and 3<br>1) Perform V&V hardware acceptance testing.<br>2) Analyze test results to validate that the hardware satisfies the system requirements.<br>3) Validate that the test results trace to test criteria established by the test traceability in the test planning documents.<br>4) Document the results as required by the V&V hardware acceptance test plan.<br>5) Use the V&V hardware acceptance test results to validate that the hardware satisfies the test acceptance criteria.<br>6) Document discrepancies between the actual and expected test results.<br>b) Integrity level 2<br>Use the acquirer's hardware acceptance test results to verify that the hardware satisfies the test acceptance criteria.<br>c) Integrity level 1<br>There are no hardware acceptance V&V test requirements. | Hardware component specifications<br>User information<br>Hardware acceptance test plan<br>Hardware acceptance test procedures<br>Hardware acceptance test results from the developer (integrity level 2)                                                                                                              | V&V hardware acceptance test execution results (integrity levels 4 and 3)<br>Task report(s)—Hardware acceptance test execution V&V (integrity level 2)<br>Anomaly report(s) |
| <b>(4) <u>Installation configuration audit</u></b><br>a) Verify that all hardware products required to correctly install and operate the hardware are present in the installation package.<br>b) Validate that all site-dependent settings, parameters, or conditions to verify supplied values are correct.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Installation package (e.g., user information, hardware design, interface design, system/hardware requirements, interface requirements, hardware drawings, business needs or mission analysis documentation, installation procedures, site-specific parameters, installation tests, and configuration management data) | Task report(s)—Installation configuration audit<br>Anomaly report(s)                                                                                                        |
| <b>(5) <u>Installation Checkout</u></b><br>a) Conduct analyses or tests to verify that the installed hardware corresponds to the hardware subjected to V&V.<br>b) Verify that the hardware initializes, executes, and terminates as specified.<br>c) In the transition from one version of hardware to the next, validate that the hardware can be replaced with the new version without adversely affecting or degrading the functionality of the remaining system components.                                                                                                                                                                                                                                                                                                                                                                                                                    | Installation package                                                                                                                                                                                                                                                                                                  | Task report(s)—Installation checkout<br>Anomaly report(s)                                                                                                                   |



| <b>10.10 Activity: Transition V&amp;V (Hardware, based on 12207—Transition Process)</b>                                                                                                                                                                                                                                                                                                                    |                                                                                                                                            |                                                           |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                       | <b>Required inputs</b>                                                                                                                     | <b>Required outputs</b>                                   |
| d) Verify the requirements for continuous operation and service during the transition, including requirements for user notification.                                                                                                                                                                                                                                                                       |                                                                                                                                            |                                                           |
| (6) <a href="#">Hazard analysis</a><br>a) Verify that the installation procedures and installation environment do not introduce new hazards.<br>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and hardware operations).<br>c) Update the hazard analysis.            | Installation package<br>Hazard analysis results                                                                                            | Task report(s)—<br>Hazard analysis<br>Anomaly report(s)   |
| (7) <a href="#">Security analysis</a><br>a) Verify that the installed hardware does not introduce new or increased vulnerabilities or security risks to the overall system.<br>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and hardware operations). | Installation package<br>Security analysis results                                                                                          | Task report(s)—<br>Security analysis<br>Anomaly report(s) |
| (8) <a href="#">Risk analysis</a><br>a) Review and update risk analysis using prior task reports.<br>b) Provide recommendations to eliminate, reduce, or mitigate the risks.                                                                                                                                                                                                                               | Installation package<br>Supplier development plans and schedules<br>Security analysis results<br>Risk analysis results<br>V&V task results | Task report(s)—<br>Risk analysis<br>Anomaly report(s)     |

| <b>10.11 Activity: Hardware Validation (Hardware, based on 12207—Validation Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                    |                                                                                                                      |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Required inputs</b>                                                                                             | <b>Required outputs</b>                                                                                              |
| The activities and tasks for the hardware Validation Process are conducted in hardware technical life cycle processes, and are contained in <a href="#">Table 1d</a> , <a href="#">Activity 10.1</a> (business or mission analysis V&V), <a href="#">Activity 10.2</a> (stakeholder needs and requirements definition V&V), <a href="#">Activity 10.3</a> (system/hardware requirements definition V&V), <a href="#">Activity 10.4</a> (architecture definition V&V), <a href="#">Activity 10.5</a> (design definition V&V), <a href="#">Activity 10.6</a> (system analysis V&V), <a href="#">Activity 10.7</a> (implementation V&V), <a href="#">Activity 10.8</a> (integration V&V), <a href="#">Activity 10.10</a> (transition V&V), <a href="#">Activity 10.12</a> (operation V&V), <a href="#">Activity 10.13</a> (maintenance V&V), and <a href="#">Activity 10.14</a> (disposal V&V). The hardware Validation activities and tasks in <a href="#">Table 1d</a> and the common Validation activities and tasks in <a href="#">Table 1a</a> represent all activities and tasks needed to perform hardware Validation. | The required inputs for the Validation Process for hardware are found in <a href="#">Table 1d</a> required inputs. | The required outputs for the Validation Process for hardware are found in <a href="#">Table 1d</a> required outputs. |

| <b>10.12 Activity: Operation V&amp;V (Hardware, based on 12207—Operation Process)</b>                                                                                                                                                                     |                        |                                                  |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                      | <b>Required inputs</b> | <b>Required outputs</b>                          |
| (1) <a href="#">Evaluation of new constraints</a><br>Evaluate new constraints (e.g., operational requirements, platform characteristics, and operating environment) on the system or system/hardware requirements to verify the applicability of the VVP. | VVP<br>New constraints | Task report(s)—<br>Evaluation of new constraints |

| <b>10.12 Activity: Operation V&amp;V (Hardware, based on 12207—Operation Process)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                        |                                                                         |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Required inputs</b>                                                                                                                                                                                                 | <b>Required outputs</b>                                                 |
| <b>(2) <u>Operating procedures evaluation</u></b><br>Verify that the operating procedures are consistent with the user documentation and conform to the system requirements.                                                                                                                                                                                                                                                                                                                                                                              | Operating procedures<br>User information<br>Business or mission analysis documentation                                                                                                                                 | Task report(s)—<br>Operating procedures evaluation<br>Anomaly report(s) |
| <b>(3) <u>Hazard analysis</u></b><br>a) Verify that the operating procedures and operational environment do not introduce new hazards.<br>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and hardware operations).<br>c) Update the hazard analysis.                                                                                                                                                                 | Operating procedures<br>Hazard analysis results                                                                                                                                                                        | Task report(s)—<br>Hazard analysis<br>Anomaly report(s)                 |
| <b>(4) <u>Security analysis</u></b><br>a) Verify that no new security risks are introduced due to changes in the operational environment.<br>b) Over time, changes in external interfaces, threats, or technology in general require that an updated security analysis be performed to determine an updated residual risk.<br>c) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and hardware operations). | New constraints<br>Environmental changes<br>Operating procedures<br>Security analysis results                                                                                                                          | Task Reports—<br>Security analysis                                      |
| <b>(5) <u>Risk analysis</u></b><br>a) Review and update risk analysis using prior task reports.<br>b) Provide recommendations to eliminate, reduce, or mitigate the risks.                                                                                                                                                                                                                                                                                                                                                                                | Installation package<br>Proposed changes<br>Hazard analysis results<br>Security analysis results<br>Risk analysis results<br>Supplier development plans and schedules<br>Operation problem reports<br>V&V task results | Task report(s)—<br>Risk analysis<br>Anomaly report(s)                   |

| <b>10.13 Activity: Maintenance V&amp;V (Hardware, based on 12207—Maintenance Process)</b>                                                                                                                                                                           |                                                                                            |                                                                                                                 |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                | <b>Required inputs</b>                                                                     | <b>Required outputs</b>                                                                                         |
| <b>(1) <u>Hardware maintenance strategy assessment</u></b><br>a) Verify the hardware maintenance strategy.<br>b) Verify that any constraints specified or implied by the hardware maintenance strategy are included in the system/hardware requirements and design. | Hardware maintenance strategy<br>Hardware performance data<br>System/hardware requirements | Task report(s)—<br>Hardware maintenance strategy analysis<br>Corrective actions assessment<br>Anomaly report(s) |
| <b>(2) <u>Hardware maintenance execution assessment</u></b><br>a) Verify the maintenance operational strategy results by monitoring the hardware maintenance execution.                                                                                             | System maintenance strategy (e.g., operational availability)                               | Task report(s)—<br>Hardware maintenance execution                                                               |

| <b>10.13 Activity: Maintenance V&amp;V (Hardware, based on 12207—Maintenance Process)</b>                                                                                                                                                                                                                                                                                                                 |                                                                                                   |                                                                  |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                      | <b>Required inputs</b>                                                                            | <b>Required outputs</b>                                          |
| b) Evaluate the maintenance operational strategy results.                                                                                                                                                                                                                                                                                                                                                 | requirements,<br>hardware failure,<br>suspension of<br>services)                                  | assessment and<br>quality metrics<br>Anomaly report(s)           |
| (3) <b><u>VVP revision</u></b><br>a) Revise the VVP to conform to approved changes.<br>b) When the development documentation required by this standard is not available, generate a new VVP and consider the methods in Annex D for deriving the required development documentation.                                                                                                                      | VVP<br>Approved changes<br>Installation package<br>Supplier<br>development plans<br>and schedules | Updated VVP                                                      |
| (4) <b><u>Anomaly evaluation</u></b><br>Evaluate the effect of hardware operation anomalies.                                                                                                                                                                                                                                                                                                              | Anomaly report(s)                                                                                 | Task report(s)—<br>Anomaly evaluation                            |
| (5) <b><u>Criticality analysis</u></b><br>a) Determine the integrity levels for proposed modifications.<br>b) Validate the integrity levels provided by the maintainer. For V&V planning purposes, the highest integrity level assigned to the hardware shall be the integrity level of the system.                                                                                                       | Proposed changes<br>Installation package<br>Maintainer integrity<br>levels                        | Task report(s)—<br>Criticality analysis<br>Anomaly report(s)     |
| (6) <b><u>Migration assessment</u></b><br>Assess whether the system/hardware requirements and fabrication address the following:<br>a) Specific migration requirements.<br>b) Migration tools.<br>c) Conversion of hardware products and data.<br>d) Hardware archiving.<br>e) Support for the prior environment.<br>f) User notification.                                                                | Installation package<br>Approved changes                                                          | Task report(s)—<br>Migration<br>assessment<br>Anomaly report(s)  |
| (7) <b><u>Retirement assessment</u></b><br>Assess whether the installation package addresses the following:<br>a) Hardware support.<br>b) Impact on existing systems and databases.<br>c) Hardware archiving.<br>d) Transition to a new hardware product.<br>e) User notification.                                                                                                                        | Installation package<br>Approved changes                                                          | Task report(s)—<br>Retirement<br>assessment<br>Anomaly report(s) |
| (8) <b><u>Hazard analysis</u></b><br>a) Verify that hardware modifications correctly implement the critical requirements and introduce no new hazards.<br>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and hardware operations).<br>c) Update the hazard analysis. | Proposed changes<br>Installation package<br>Hazard analysis<br>results                            | Task report(s)—<br>Hazard analysis<br>Anomaly report(s)          |
| (9) <b><u>Security analysis</u></b><br>a) Verify that proposed changes/updates to the hardware do not introduce new or increased security risks to the overall system.<br>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and hardware operations).     | Proposed changes<br>Installation package<br>Security analysis<br>results                          | Task reports—<br>Security analysis                               |

| <b>10.13 Activity: Maintenance V&amp;V (Hardware, based on 12207—Maintenance Process)</b>                                                                                                                                                                                           |                                                                                                                                                                                                                        |                                                       |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                | <b>Required inputs</b>                                                                                                                                                                                                 | <b>Required outputs</b>                               |
| <b>(10) <u>Risk analysis</u></b><br>a) Review and update risk analysis using prior task reports.<br>b) Provide recommendations to eliminate, reduce, or mitigate the risks.                                                                                                         | Installation package<br>Proposed changes<br>Hazard analysis results<br>Security analysis results<br>Risk analysis results<br>Supplier development plans and schedules<br>Operation problem reports<br>V&V task results | Task report(s)—<br>Risk analysis<br>Anomaly report(s) |
| <b>(11) <u>Task iteration</u></b><br>Perform V&V tasks, as needed, to assure that the following is performed:<br>a) Planned changes are implemented correctly.<br>b) Documentation is complete and current.<br>c) Changes do not cause unacceptable or unintended system behaviors. | Approved changes<br>Installation package                                                                                                                                                                               | Task report(s)<br>Anomaly report(s)                   |
| NOTE—Hardware changes are maintenance activities (see subclause 10.13).                                                                                                                                                                                                             |                                                                                                                                                                                                                        |                                                       |

| <b>10.14 Activity: Hardware Disposal V&amp;V (Hardware, Based on 12207—Disposal Process)</b>                                                                                                                                                                                                                                                                                                                                  |                            |                                                                               |
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| <b>V&amp;V tasks</b>                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Required inputs</b>     | <b>Required outputs</b>                                                       |
| <b>(1) <u>Hardware disposal strategy evaluation</u></b><br>a) Verify that any constraints specified or implied by the hardware disposal strategy are included in the system/hardware requirements, including hardware element destruction/storage and recording disposal actions, and analysis of disposal impacts on the system.<br>b) Validate that the hardware disposal strategy leaves the system in an agreed-on state. | Hardware disposal strategy | Task report(s)—<br>Hardware disposal strategy evaluation<br>Anomaly report(s) |
| NOTE (for Table 1d)—Other inputs may be used. For any V&V activity and task, all of the required inputs and outputs from preceding activities and tasks may be used, but for conciseness, only the primary inputs are listed.                                                                                                                                                                                                 |                            |                                                                               |

**Table 2d—Minimum V&V tasks assigned to each integrity level for hardware V&V**

[illegible]

| V&V Processes                             | Business or Mission Analysis V&V (see 10.1) |   |   |   | Stakeholder Needs and Requirements Definition V&V (see 10.2) |   |   |   | System/ Hardware Requirements Definition V&V (see 10.3) |   |   |   | Architecture Definition V&V (see 10.4) |   |   |   | Design Definition V&V (see 10.5) |   |   |   | System Analysis V&V (see 10.6) |   |   |   | Implementation V&V (see 10.7) |   |   |   | Integration V&V (see 10.8) |   |   |   | Transition V&V (see 10.10) |   |   |   | Operation V&V (see 10.12) |   |   |   | Maintenance V&V (see 10.13) |   |   |   | Disposal V&V (see 10.14) |   |   |  |  |  |  |
|-------------------------------------------|---------------------------------------------|---|---|---|--------------------------------------------------------------|---|---|---|---------------------------------------------------------|---|---|---|----------------------------------------|---|---|---|----------------------------------|---|---|---|--------------------------------|---|---|---|-------------------------------|---|---|---|----------------------------|---|---|---|----------------------------|---|---|---|---------------------------|---|---|---|-----------------------------|---|---|---|--------------------------|---|---|--|--|--|--|
| Integrity Levels                          | Levels                                      |   |   |   | Levels                                                       |   |   |   | Levels                                                  |   |   |   | Levels                                 |   |   |   | Levels                           |   |   |   | Levels                         |   |   |   | Levels                        |   |   |   | Levels                     |   |   |   | Levels                     |   |   |   | Levels                    |   |   |   | Levels                      |   |   |   | Levels                   |   |   |  |  |  |  |
|                                           | 4                                           | 3 | 2 | 1 | 4                                                            | 3 | 2 | 1 | 4                                                       | 3 | 2 | 1 | 4                                      | 3 | 2 | 1 | 4                                | 3 | 2 | 1 | 4                              | 3 | 2 | 1 | 4                             | 3 | 2 | 1 | 4                          | 3 | 2 | 1 | 4                          | 3 | 2 | 1 | 4                         | 3 | 2 | 1 | 4                           | 3 | 2 | 1 |                          |   |   |  |  |  |  |
| Hardware Component Test Plan V&V          |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   | X                                      | X | X |   |                                  |   |   |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |   |   |  |  |  |  |
| Hardware Component Test Procedure V&V     |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   |                                  |   |   |   | X                              | X | X |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |   |   |  |  |  |  |
| Hardware Disposal Strategy Evaluation     |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   |                                  |   |   |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          | X | X |  |  |  |  |
| Hardware Integration Test Case V&V        |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   | X                                | X | X |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |   |   |  |  |  |  |
| Hardware Integration Test Design V&V      |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   | X                                | X | X |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |   |   |  |  |  |  |
| Hardware Integration Test Execution V&V   |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   |                                  |   |   |   |                                |   |   |   | X                             | X | X |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |   |   |  |  |  |  |
| Hardware Integration Test Plan V&V        |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   | X                                      | X | X |   |                                  |   |   |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |   |   |  |  |  |  |
| Hardware Integration Test Procedure V&V   |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   |                                  |   |   |   | X                              | X | X |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |   |   |  |  |  |  |
| Hardware Qualification Test Case V&V      |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   | X                                | X | X |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |   |   |  |  |  |  |
| Hardware Qualification Test Design V&V    |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   | X                                      | X | X |   |                                  |   |   |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |   |   |  |  |  |  |
| Hardware Qualification Test Execution V&V |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   |                                  |   |   |   |                                |   |   | X | X                             | X |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |   |   |  |  |  |  |
| Hardware Qualification Test Plan V&V      |                                             |   |   |   |                                                              |   |   |   | X                                                       | X | X |   |                                        |   |   |   |                                  |   |   |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |   |   |  |  |  |  |
| Hardware Qualification Test Procedure V&V |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   |                                  |   |   |   | X                              | X | X |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |   |   |  |  |  |  |
| Hazard Analysis                           | X                                           | X |   |   | X                                                            | X |   |   | X                                                       | X |   |   | X                                      | X | X |   | X                                | X |   |   |                                |   |   | X | X                             |   |   | X | X                          |   |   | X | X                          |   |   | X | X                         |   |   |   |                             |   |   |   |                          |   |   |  |  |  |  |
| Installation Checkout                     |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   |                                  |   |   |   |                                |   |   |   |                               |   | X | X |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |   |   |  |  |  |  |
| Installation Configuration Audit          |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   |                                  |   |   |   |                                |   |   |   |                               |   | X | X |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |   |   |  |  |  |  |
| Interface Analysis                        |                                             |   |   |   |                                                              |   |   |   | X                                                       | X | X |   | X                                      | X | X |   | X                                | X | X |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |   |   |  |  |  |  |
| Migration Assessment                      |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   |                                  |   |   |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |   |   |  |  |  |  |

| V&V Processes                                 | Business or Mission Analysis V&V (see 10.1) |   |   |   | Stakeholder Needs and Requirements Definition V&V (see 10.2) |   |   |   | System/ Hardware Requirements Definition V&V (see 10.3) |   |   |   | Architecture Definition V&V (see 10.4) |   |   |   | Design Definition V&V (see 10.5) |   |   |   | System Analysis V&V (see 10.6) |   |   |   | Implementation V&V (see 10.7) |   |   |   | Integration V&V (see 10.8) |   |   |   | Transition V&V (see 10.10) |   |   |   | Operation V&V (see 10.12) |   |   |   | Maintenance V&V (see 10.13) |   |   |   | Disposal V&V (see 10.14) |  |  |  |  |  |
|-----------------------------------------------|---------------------------------------------|---|---|---|--------------------------------------------------------------|---|---|---|---------------------------------------------------------|---|---|---|----------------------------------------|---|---|---|----------------------------------|---|---|---|--------------------------------|---|---|---|-------------------------------|---|---|---|----------------------------|---|---|---|----------------------------|---|---|---|---------------------------|---|---|---|-----------------------------|---|---|---|--------------------------|--|--|--|--|--|
| Integrity Levels                              | Levels                                      |   |   |   | Levels                                                       |   |   |   | Levels                                                  |   |   |   | Levels                                 |   |   |   | Levels                           |   |   |   | Levels                         |   |   |   | Levels                        |   |   |   | Levels                     |   |   |   | Levels                     |   |   |   | Levels                    |   |   |   | Levels                      |   |   |   | Levels                   |  |  |  |  |  |
|                                               | 4                                           | 3 | 2 | 1 | 4                                                            | 3 | 2 | 1 | 4                                                       | 3 | 2 | 1 | 4                                      | 3 | 2 | 1 | 4                                | 3 | 2 | 1 | 4                              | 3 | 2 | 1 | 4                             | 3 | 2 | 1 | 4                          | 3 | 2 | 1 | 4                          | 3 | 2 | 1 | 4                         | 3 | 2 | 1 | 4                           | 3 | 2 | 1 |                          |  |  |  |  |  |
| Interface Analysis                            |                                             |   |   |   |                                                              |   |   |   | X                                                       | X | X |   | X                                      | X | X |   | X                                | X | X |   |                                |   |   | X | X                             | X |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |  |  |  |  |  |
| Migration Assessment                          |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   |                                  |   |   |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |  |  |  |  |  |
| Operation Procedures Evaluation               |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   |                                  |   |   |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |  |  |  |  |  |
| Requirements Evaluation                       |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   | X                                      | X | X | X |                                  |   |   |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |  |  |  |  |  |
| Requirements Allocation Analysis              |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   | X                                | X |   |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |  |  |  |  |  |
| Retirement Assessment                         |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   |                                  |   |   |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |  |  |  |  |  |
| Risk Analysis                                 | X                                           | X |   |   | X                                                            | X |   |   | X                                                       | X |   |   | X                                      | X | X |   | X                                | X |   |   |                                |   |   | X | X                             |   |   | X | X                          |   | X | X |                            | X | X |   | X                         | X |   | X | X                           |   |   |   |                          |  |  |  |  |  |
| Security Analysis                             | X                                           | X |   |   | X                                                            | X |   |   | X                                                       | X |   |   | X                                      | X | X |   | X                                | X |   |   |                                |   |   | X | X                             |   |   | X | X                          |   | X | X |                            | X | X |   | X                         | X |   |   |                             |   |   |   |                          |  |  |  |  |  |
| Stakeholder Needs and Requirements Evaluation |                                             |   |   |   | X                                                            | X |   |   |                                                         |   |   |   |                                        |   |   |   |                                  |   |   |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |  |  |  |  |  |
| System Analysis Strategy Evaluation           |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   |                                  |   |   |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |  |  |  |  |  |
| System Analysis Results Evaluation            |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   |                                  |   |   |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |  |  |  |  |  |
| Task Iteration                                |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   |                                  |   |   |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |  |  |  |  |  |
| Traceability Analysis                         | X                                           | X | X |   | X                                                            | X | X |   | X                                                       | X | X |   | X                                      | X | X |   | X                                | X | X |   |                                |   |   | X | X                             | X |   | X | X                          | X |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |  |  |  |  |  |
| VVP Revision                                  |                                             |   |   |   |                                                              |   |   |   |                                                         |   |   |   |                                        |   |   |   |                                  |   |   |   |                                |   |   |   |                               |   |   |   |                            |   |   |   |                            |   |   |   |                           |   |   |   |                             |   |   |   |                          |  |  |  |  |  |

NOTE—Whenever a V&V task is selected as a mandatory requirement for multiple integrity levels, the V&V task implementation is dictated by the rigor, intensity, and depth of analysis or test. A higher integrity level implementation requires greater rigor (e.g., formal methods, structured analysis methods), intensity (e.g., consideration of all system conditions and system environment states), and depth (e.g., abnormal cases, boundary conditions, comprehensive fault, and recovery scenarios) of analysis or test than the lower integrity level implementation.

The recommended applicability of optional tasks to the hardware V&V processes described in Clause 10 is shown in [Table 3d](#). [Annex G](#) provides a description of each of the optional V&V tasks.

**Table 3d—Optional V&V tasks and suggested applications in hardware technical processes**

|                                                      | <a href="#">Business or Mission Analysis (10.1)</a> | <a href="#">Stakeholder needs and Requirements Definition (10.2)</a> | <a href="#">System/Hardware Requirements Definition (10.3)</a> | <a href="#">Architecture Definition (10.4)</a> | <a href="#">Design Definition (10.5)</a> | <a href="#">System Analysis (10.6)</a> | <a href="#">Implementation (10.7)</a> | <a href="#">Integration (10.8)</a> | <a href="#">Transition (10.10)</a> | <a href="#">Operation (10.12)</a> | <a href="#">Maintenance (10.13)</a> | <a href="#">Disposal (10.14)</a> |
|------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------|------------------------------------------|----------------------------------------|---------------------------------------|------------------------------------|------------------------------------|-----------------------------------|-------------------------------------|----------------------------------|
| <a href="#">Algorithm analysis</a>                   | —                                                   | —                                                                    | X                                                              | X                                              | —                                        | —                                      | X                                     | —                                  | —                                  | —                                 | X                                   | —                                |
| <a href="#">Audit performance</a>                    | —                                                   | —                                                                    | X                                                              | X                                              | —                                        | —                                      | X                                     | X                                  | X                                  | —                                 | X                                   | —                                |
| <a href="#">Audit support</a>                        | —                                                   | —                                                                    | X                                                              | X                                              | —                                        | —                                      | X                                     | X                                  | X                                  | —                                 | X                                   | —                                |
| <a href="#">Control flow analysis</a>                | —                                                   | —                                                                    | X                                                              | X                                              | —                                        | —                                      | X                                     | —                                  | —                                  | —                                 | X                                   | —                                |
| <a href="#">Cost analysis</a>                        | —                                                   | X                                                                    | X                                                              | X                                              | —                                        | —                                      | X                                     | X                                  | X                                  | —                                 | X                                   | —                                |
| <a href="#">Data flow analysis</a>                   | —                                                   | —                                                                    | X                                                              | X                                              | —                                        | —                                      | X                                     | —                                  | —                                  | —                                 | X                                   | —                                |
| <a href="#">Disaster recovery plan assessment</a>    | —                                                   | X                                                                    | X                                                              | X                                              | —                                        | —                                      | X                                     | —                                  | —                                  | X                                 | X                                   | X                                |
| <a href="#">Distributed architecture assessment</a>  | —                                                   | X                                                                    | X                                                              | X                                              | —                                        | —                                      | —                                     | —                                  | —                                  | —                                 | X                                   | —                                |
| <a href="#">Exploratory testing</a>                  | X                                                   | X                                                                    | X                                                              | X                                              | X                                        | X                                      | X                                     | X                                  | X                                  | X                                 | X                                   | —                                |
| <a href="#">Feasibility study evaluation</a>         | —                                                   | X                                                                    | X                                                              | X                                              | —                                        | —                                      | —                                     | —                                  | —                                  | —                                 | X                                   | —                                |
| <a href="#">Independent risk assessment</a>          | —                                                   | —                                                                    | —                                                              | —                                              | —                                        | —                                      | —                                     | —                                  | —                                  | —                                 | X                                   | —                                |
| <a href="#">Inspection</a>                           | —                                                   | —                                                                    | —                                                              | —                                              | —                                        | —                                      | —                                     | —                                  | —                                  | —                                 | —                                   | —                                |
| <a href="#">Inspection—Concept</a>                   | —                                                   | X                                                                    | —                                                              | —                                              | —                                        | —                                      | —                                     | —                                  | —                                  | —                                 | X                                   | —                                |
| <a href="#">Inspection—Requirements</a>              | —                                                   | —                                                                    | X                                                              | —                                              | —                                        | —                                      | —                                     | —                                  | —                                  | —                                 | X                                   | —                                |
| <a href="#">Inspection—Design</a>                    | —                                                   | —                                                                    | —                                                              | X                                              | —                                        | —                                      | —                                     | —                                  | —                                  | —                                 | X                                   | —                                |
| <a href="#">Inspection—Source Code</a>               | —                                                   | —                                                                    | —                                                              | —                                              | —                                        | —                                      | —                                     | —                                  | —                                  | —                                 | —                                   | —                                |
| <a href="#">Inspection—Test plan</a>                 | —                                                   | —                                                                    | X                                                              | X                                              | —                                        | —                                      | X                                     | X                                  | X                                  | —                                 | X                                   | —                                |
| <a href="#">Inspection—Test design</a>               | —                                                   | —                                                                    | —                                                              | X                                              | —                                        | —                                      | X                                     | X                                  | X                                  | —                                 | X                                   | —                                |
| <a href="#">Inspection—Test case</a>                 | —                                                   | —                                                                    | —                                                              | X                                              | —                                        | —                                      | X                                     | X                                  | X                                  | —                                 | X                                   | —                                |
| <a href="#">Operational evaluation</a>               | —                                                   | —                                                                    | —                                                              | —                                              | —                                        | —                                      | —                                     | —                                  | —                                  | X                                 | —                                   | —                                |
| <a href="#">Performance monitoring</a>               | —                                                   | X                                                                    | X                                                              | X                                              | —                                        | —                                      | X                                     | X                                  | X                                  | X                                 | X                                   | X                                |
| <a href="#">Post-installation Validation</a>         | —                                                   | —                                                                    | —                                                              | —                                              | —                                        | —                                      | —                                     | —                                  | —                                  | X                                 | X                                   | —                                |
| <a href="#">Project management oversight support</a> | —                                                   | X                                                                    | X                                                              | X                                              | —                                        | —                                      | X                                     | X                                  | X                                  | X                                 | X                                   | X                                |
| <a href="#">Proposal evaluation support</a>          | —                                                   | —                                                                    | —                                                              | —                                              | —                                        | —                                      | —                                     | —                                  | —                                  | —                                 | —                                   | —                                |
| <a href="#">Regression analysis and testing</a>      | —                                                   | —                                                                    | X                                                              | X                                              | —                                        | —                                      | X                                     | X                                  | X                                  | —                                 | X                                   | —                                |
| <a href="#">Reusability analysis</a>                 | —                                                   | X                                                                    | X                                                              | X                                              | —                                        | —                                      | X                                     | —                                  | —                                  | —                                 | X                                   | —                                |
| <a href="#">Reuse analysis</a>                       | —                                                   | X                                                                    | X                                                              | X                                              | —                                        | —                                      | —                                     | —                                  | —                                  | —                                 | X                                   | —                                |
| <a href="#">Simulation analysis</a>                  | —                                                   | X                                                                    | X                                                              | X                                              | —                                        | —                                      | X                                     | X                                  | X                                  | X                                 | X                                   | X                                |
| <a href="#">Test certification</a>                   | —                                                   | —                                                                    | —                                                              | —                                              | —                                        | —                                      | X                                     | X                                  | X                                  | X                                 | X                                   | X                                |
| <a href="#">Test evaluation</a>                      | —                                                   | —                                                                    | X                                                              | X                                              | —                                        | —                                      | X                                     | X                                  | X                                  | X                                 | X                                   | X                                |
| <a href="#">Test witnessing</a>                      | —                                                   | —                                                                    | —                                                              | —                                              | —                                        | —                                      | X                                     | X                                  | X                                  | X                                 | X                                   | X                                |
| <a href="#">Training documentation evaluation</a>    | —                                                   | —                                                                    | X                                                              | X                                              | —                                        | —                                      | X                                     | X                                  | X                                  | X                                 | X                                   | X                                |
| <a href="#">Usability analysis</a>                   | —                                                   | X                                                                    | X                                                              | X                                              | —                                        | —                                      | X                                     | X                                  | X                                  | X                                 | X                                   | —                                |
| <a href="#">User documentation evaluation</a>        | —                                                   | X                                                                    | X                                                              | X                                              | —                                        | —                                      | X                                     | X                                  | X                                  | X                                 | X                                   | —                                |
| <a href="#">User training</a>                        | —                                                   | —                                                                    | —                                                              | —                                              | —                                        | —                                      | X                                     | X                                  | X                                  | X                                 | X                                   | —                                |
| <a href="#">V&amp;V tool plan generation</a>         | —                                                   | —                                                                    | —                                                              | —                                              | —                                        | —                                      | —                                     | —                                  | —                                  | —                                 | —                                   | —                                |
| <a href="#">V&amp;V tool qualification</a>           | —                                                   | X                                                                    | X                                                              | X                                              | —                                        | —                                      | X                                     | X                                  | X                                  | X                                 | X                                   | —                                |



|                                                           | <a href="#">Business or Mission Analysis (10.1)</a> | <a href="#">Stakeholder needs and Requirements Definition (10.2)</a> | <a href="#">System/Hardware Requirements Definition (10.3)</a> | <a href="#">Architecture Definition (10.4)</a> | <a href="#">Design Definition (10.5)</a> | <a href="#">System Analysis (10.6)</a> | <a href="#">Implementation (10.7)</a> | <a href="#">Integration (10.8)</a> | <a href="#">Transition (10.10)</a> | <a href="#">Operation (10.12)</a> | <a href="#">Maintenance (10.13)</a> | <a href="#">Disposal (10.14)</a> |
|-----------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------|------------------------------------------|----------------------------------------|---------------------------------------|------------------------------------|------------------------------------|-----------------------------------|-------------------------------------|----------------------------------|
| <a href="#">Walkthrough</a>                               | —                                                   | —                                                                    | —                                                              | —                                              | —                                        | —                                      | —                                     | —                                  | —                                  | —                                 | —                                   | —                                |
| <a href="#">Walkthrough—Design</a>                        | —                                                   | —                                                                    | —                                                              | X                                              | —                                        | —                                      | —                                     | —                                  | —                                  | —                                 | X                                   | —                                |
| <a href="#">Walkthrough—Requirements</a>                  | —                                                   | —                                                                    | X                                                              | —                                              | —                                        | —                                      | —                                     | —                                  | —                                  | —                                 | X                                   | —                                |
| <a href="#">Walkthrough—Test</a>                          | —                                                   | —                                                                    | —                                                              | —                                              | —                                        | —                                      | X                                     | X                                  | —                                  | —                                 | X                                   | —                                |
| <a href="#">Work breakdown structure (WBS) evaluation</a> | —                                                   | —                                                                    | —                                                              | —                                              | —                                        | —                                      | —                                     | —                                  | —                                  | —                                 | —                                   | —                                |

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Hardware Technical Life Cycle Processes

|            | Process:<br>Business or Mission<br>Analysis  | Process:<br>Stakeholder Needs and<br>Requirements Definition | Process:<br>Sys/HW<br>Requirements Def'n  | Process:<br>Architecture<br>Definition | Process:<br>Design Definition    | Process:<br>System<br>Analysis | Process:<br>Implementation                               | Process:<br>Integration                      | Process:<br>Transition                       | Process:<br>Operations                       | Process:<br>Maintenance                      | Process:<br>Disposal          |
|------------|----------------------------------------------|--------------------------------------------------------------|-------------------------------------------|----------------------------------------|----------------------------------|--------------------------------|----------------------------------------------------------|----------------------------------------------|----------------------------------------------|----------------------------------------------|----------------------------------------------|-------------------------------|
| V&V Inputs | 1) Business or mission analysis results      | 1) Business or mission analysis results                      | 1) Acquirer's HW acceptance test plan     | 1) Criticality task report             | 1) Criticality task report       | 1) System analysis strategy    | 1) Component information (e.g., documentation)           | 1) HW component specifications               | 1) HW component specifications               | 1) Business or mission analysis results      | 1) Anomaly report(s)                         | 1) Hardware disposal strategy |
|            | 2) Hazard analysis report                    | 2) Developer integrity level assignments                     | 2) Business or mission analysis results   | 2) HW acceptance test design           | 2) Design standards              | 2) System analysis results     | 2) Component documentation                               | 2) HW component specifications               | 2) HW acceptance test plan                   | 2) Environmental changes                     | 2) Approved changes                          |                               |
|            | 3) Identified candidate solution(s)          | 3) Hazard Analysis Report                                    | 3) Criticality task                       | 3) HW acceptance test plan             | 3) HW architecture               |                                | 3) Concept Documentation                                 | 3) HW component specifications               | 3) HW acceptance test procedures             | 3) Hazard analysis report                    | 3) Hazard analysis report                    |                               |
|            | 4) Identified problem or opportunity space   | 4) Identified problem or opportunity space                   | 4) Developer's HW qualification test plan | 4) HW architecture                     | 4) HW architecture               |                                | 5) Criticality Task Report                               | 4) HW integration test plan                  | 4) HW acceptance test results                | 4) Installation package                      | 4) Installation package                      |                               |
|            | 5) Identified solution space                 | 5) Identified solution space                                 | 5) HW component test plan                 | 5) HW component test design            | 5) HW component test cases       |                                | 6) HW acceptance test plan                               | 5) HW integration test results               | 5) Hazard analysis report                    | 5) New constraints                           | 5) Maintainer integrity levels               |                               |
|            | 6) New market or mission elements            | 6) Preferred candidate hardware solution(s)                  | 6) HW drawings                            | 6) HW drawings                         | 6) HW component test plans       |                                | 7) HW acceptance test procedures                         | 6) HW qualification test plan                | 6) Installation package                      | 6) Operating procedures                      | 6) Operation problem reports                 |                               |
|            | 7) Organization strategy                     | 7) Preferred candidate solution(s)                           | 7) HW integration test plan               | 7) HW integration test plan            | 7) HW component test design      |                                | 8) HW component specifications                           | 7) HW qualification test procedures          | 7) Installation package                      | 7) Operation problem reports                 | 7) Proposed changes                          |                               |
|            | 8) Organizational concept of operations      | 8) Preliminary TRA                                           | 8) Hazard analysis report                 | 8) HW qualification test plan          | 8) HW design                     |                                | 9) HW component test cases                               | 8) HW qualification test results             | 8) Risk analysis report                      | 8) Proposed changes                          | 8) Risk analysis report                      |                               |
|            | 9) Organizational processes and procedures   | 9) Security analysis report                                  | 9) Interface requirements                 | 9) HW integration test design          | 9) HW drawings                   |                                | 10) HW component test procedures                         | 9) Hazard analysis report                    | 9) Security report                           | 9) Risk analysis report                      | 9) Security analysis report                  |                               |
|            | 10) Organizational strategic goals and plans | 10) Stakeholder needs and requirements                       | 10) Preliminary TRA                       | 10) HW qualification test design       | 10) HW integration test cases    |                                | 11) HW component test results                            | 10) Risk analysis report                     | 10) Supplier development plans and schedules | 10) Supplier development plans and schedules | 10) Supplier development plans and schedules |                               |
|            | 11) Preferred candidate solution(s)          | 11) Supplier development plans and schedules                 | 11) Risk analysis report                  | 11) HW qualification test results      | 11) HW qualification test design |                                | 12) HW design                                            | 11) Security analysis report                 | 11) User information                         | 11) Supplier development plans and schedules | 11) V&V task results                         |                               |
|            | 12) Preliminary threat and risk assessment   | 12) Security analysis report                                 | 12) Security analysis report              | 12) Hazard analysis results            | 12) HW integration test design   |                                | 13) HW drawings                                          | 12) Hazard analysis report                   | 12) V&V task results                         | 12) User information                         | 12) V&V task results                         |                               |
|            | 13) Security analysis report                 | 13) Stakeholder needs and requirements                       | 13) Stakeholder needs and requirements    | 13) Interface requirements             | 13) Hazard analysis results      |                                | 14) HW drawings (e.g., shop drawings, bill of materials) | 13) Supplier development plans and schedules | 13) Test results                             | 13) V&V task results                         | 13) V&V task results                         |                               |
|            | 14) V&V task results                         | 14) Supplier development plans and schedules                 | 14) Security analysis report              | 14) Security analysis results          | 14) HW qualification test design |                                | 15) HW integration test cases                            | 14) V&V task results                         | 14) V&V test designs                         | 14) V&V test plans                           | 14) V&V test procedures                      |                               |

IEEE Std 1012 Hardware V&V Activities and Clause Numbers (All tasks are minimum required for integrity level 4)

|                        | Process:<br>Business or Mission<br>Analysis V&V (10.1)                                                                                                                      | Process:<br>Stakeholder Needs and<br>Requirements Def'n V&V (10.2)                                                                                                        | Process:<br>System/ HW Req's<br>Def'n V&V (10.3)                                                                                                                                                                                                                                                     | Process:<br>Architecture<br>Definition V&V (10.4)                                                                                                                                                                                                                                                                                   | Process:<br>Design Definition V&V<br>(10.5)                                                                                                                                                                                                                                                    | Process:<br>System Analysis<br>V&V (10.6)                                                                                                                                                                                                                                                                                                | Process:<br>Implementation V&V<br>(10.7)                                                                                                                                                                                                                                                                                       | Process:<br>Integration V&V<br>(10.8)                                                                                                                       | Process:<br>Transition V&V<br>(10.10)                                                                                                                                | Process:<br>Operations V&V<br>(10.12)                                                                                                    | Process:<br>Maintenance V&V<br>(10.13)                                                                                                                                                                            | Process:<br>Disposal V&V<br>(10.14)       |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>V&amp;V Tasks</b>   | 1) Business or Mission Analysis Results<br>2) Evaluation Traceability Analysis<br>3) Criticality Analysis<br>4) Hazard Analysis<br>5) Security Analysis<br>6) Risk Analysis | 1) Stakeholder Needs and Requirements Evaluation<br>2) Traceability Analysis<br>3) Criticality Analysis<br>4) Hazard Analysis<br>5) Security Analysis<br>6) Risk Analysis | 1) Requirements Evaluation<br>2) Interface Analysis<br>3) Traceability Analysis<br>4) Criticality Analysis<br>5) HW Component Test Plan V&V<br>6) HW Int Test Plan V&V<br>7) HW Qual Test Plan V&V<br>8) HW Accept Test Plan V&V<br>9) Hazard Analysis<br>10) Security Analysis<br>11) Risk Analysis | 1) Architecture Evaluation<br>2) Interface Analysis<br>3) Req Allocation Analysis<br>4) Traceability Analysis<br>5) Criticality Analysis<br>6) HW Component Test Case V&V<br>7) HW Int Test Case V&V<br>8) HW Qual Test Case V&V<br>9) HW Accept Test Case V&V<br>10) Security Analysis<br>11) Hazard Analysis<br>12) Risk Analysis | 1) Design Evaluation<br>2) Interface Analysis<br>3) Traceability Analysis<br>4) Criticality Analysis<br>5) HW Component Test Case V&V<br>6) HW Int Test Case V&V<br>7) HW Qual Test Case V&V<br>8) HW Accept Test Case V&V<br>9) Hazard Analysis<br>10) Security Analysis<br>11) Risk Analysis | 1) System Analysis Strategy Evaluation<br>2) System Analysis Results Evaluation<br>3) Component Test Case V&V<br>4) HW Component Test Case V&V<br>5) HW Component Test Case V&V<br>6) HW Int Test Case V&V<br>7) HW Qual Test Case V&V<br>8) HW Accept Test Case V&V<br>9) Hazard Analysis<br>10) Security Analysis<br>11) Risk Analysis | 1) Fabricated Component Information Evaluation<br>2) Interface Analysis<br>3) Traceability Analysis<br>4) Criticality Analysis<br>5) Hardware Component Test Case V&V<br>6) HW Int Test Case V&V<br>7) HW Qual Test Case V&V<br>8) HW Accept Test Case V&V<br>9) Hazard Analysis<br>10) Security Analysis<br>11) Risk Analysis | 1) HW Int Test Execution V&V<br>2) HW Qual Test Execution V&V<br>3) Traceability Analysis<br>4) Hazard Analysis<br>5) Security Analysis<br>6) Risk Analysis | 1) HW Accept Test Execution V&V<br>2) Installation Configuration Audit<br>3) Installation Checkout<br>4) Hazard Analysis<br>5) Security Analysis<br>6) Risk Analysis | 1) Evaluation of New Constraints<br>2) Operating Procedures Evaluation<br>3) Hazard Analysis<br>4) Security Analysis<br>5) Risk Analysis | 1) VVP Revision<br>2) Anomaly Evaluation<br>3) Criticality Analysis<br>4) Migration Assessment<br>5) Retirement Assessment<br>6) Hazard Analysis<br>7) Security Analysis<br>8) Risk Analysis<br>9) Task Iteration | 1) HW Disposal Strategy Evaluation        |
| <b>V&amp;V Outputs</b> | 1) Anomaly report(s)<br>2) Task report(s)                                                                                                                                   | 1) Anomaly report(s)<br>2) Task report(s)                                                                                                                                 | 1) Anomaly report(s)<br>2) Task report(s)<br>3) V&V HW test plan(s)<br>- Acceptance<br>- Component<br>- Integration<br>- Qualification                                                                                                                                                               | 1) Anomaly report(s)<br>2) Task report(s)<br>3) V&V HW test design(s)<br>- Acceptance<br>- Component<br>- Qualification                                                                                                                                                                                                             | 1) Anomaly report(s)<br>2) Task report(s)<br>3) V&V HW test cases<br>- Acceptance<br>- Qualification                                                                                                                                                                                           | 1) Task report(s)                                                                                                                                                                                                                                                                                                                        | 1) Anomaly report(s)<br>2) Task report(s)<br>3) V&V HW test procedures<br>- Acceptance<br>- Qualification                                                                                                                                                                                                                      | 1) Anomaly report(s)<br>2) Task report(s)<br>3) V&V HW test execution results<br>- Qualification                                                            | 1) Anomaly report(s)<br>2) Task report(s)<br>3) V&V HW test execution results<br>- Qualification                                                                     | 1) Anomaly report(s)<br>2) Task report(s)                                                                                                | 1) Anomaly report(s)<br>2) Task report(s)<br>3) Updated VVP                                                                                                                                                       | 1) Anomaly report(s)<br>2) Task report(s) |

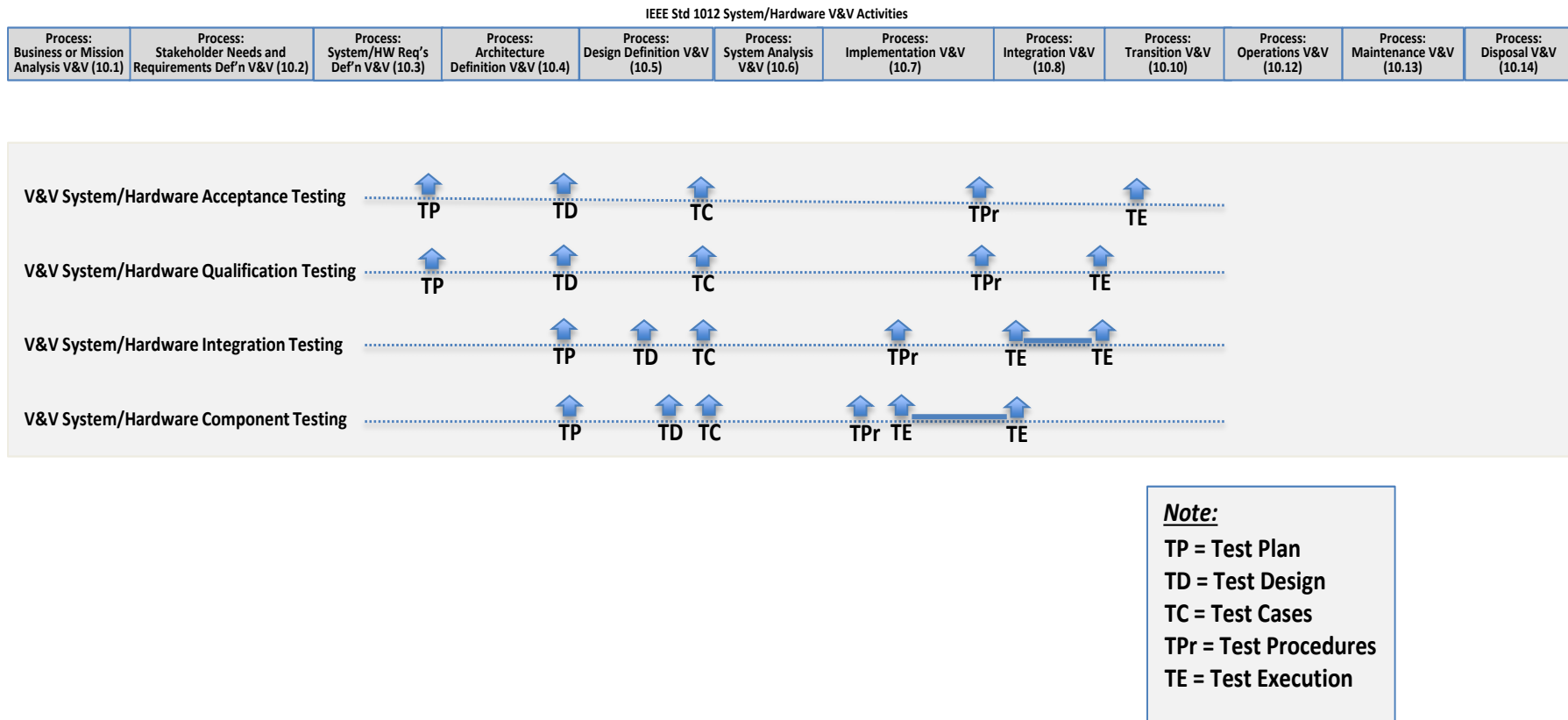
NOTE 1—No clause references are listed in the process definitions (top graphic bar) as there are no ISO/IEC or IEEE hardware life cycle definition standards used.

NOTE 2—Clause references in the activity V&V definitions (middle graphic bar) are IEEE Std 1012 clause numbers.

NOTE 3—V&V tasks listed in the figure are the minimum required for integrity level 4 (highest integrity level).

**Figure 1d—Summary of hardware V&V activities and tasks**

IEEE Std 1012-2024  
IEEE Standard for System, Software and Hardware Verification and Validation



NOTE 1—All V&V hardware test products and tasks represent the activities and products required as a minimum for integrity level 4.

NOTE 2—This is an example of the phasing of hardware V&V test products and tasks across the hardware life cycle. The hardware V&V test products (upward arrows) are shown in the hardware life cycle processes when the products are generated. Hardware test execution tasks are shown to occur during one or more hardware life cycle processes. The life cycle process (in which each test product is generated) and phasing of each test product and task can vary from this diagram in accordance with project-specific needs.

NOTE 3—The V&V activity clauses referenced in the hardware V&V life cycle stages are IEEE Std 1012 clauses.

**Figure 2d—Summary of hardware V&V test products and tasks**

## 11. V&V reporting, administrative, and documentation requirements

### 11.1 V&V reporting requirements

V&V reporting occurs throughout the system, software, or hardware life cycle. The V&V effort shall produce the required outputs listed in [Table 1a](#) through Table 1d for each V&V task performed. The format and grouping of the V&V reports may be user-defined. The V&V activity and task reports shall constitute the V&V report.

The V&V report shall consist of the following:

- a) V&V task reports. The V&V effort shall document V&V task results and status. A task report may be combined with its associated activity report or with other task reports. Task reports are generated for the following tasks:
  - 1) Acceptance support
  - 2) Anomaly evaluation
  - 3) Architecture evaluation
  - 4) Business or mission analysis documentation evaluation
  - 5) Configuration management assessment
  - 6) Contract Verification
  - 7) Criticality analysis
  - 8) Design evaluation
  - 9) Disposal strategy evaluation
  - 10) Evaluation of new constraints
  - 11) Fabricated component documentation evaluation
  - 12) Hardware acceptance test case V&V
  - 13) Hardware acceptance test design V&V
  - 14) Hardware acceptance test execution V&V
  - 15) Hardware acceptance test plan V&V
  - 16) Hardware acceptance test procedure V&V
  - 17) Hardware component test case assessment
  - 18) Hardware component test design assessment
  - 19) Hardware component test execution assessment
  - 20) Hardware component test plan assessment
  - 21) Hardware component test procedure assessment
  - 22) Hardware disposal strategy evaluation
  - 23) Hardware element interaction
  - 24) Hardware integration strategy assessment
  - 25) Hardware integration test case assessment
  - 26) Hardware integration test design assessment
  - 27) Hardware integration test execution assessment
  - 28) Hardware integration test plan assessment
  - 29) Hardware integration test procedure assessment
  - 30) Hardware maintenance execution assessment
  - 31) Hardware maintenance strategy assessment
  - 32) Hardware qualification test case V&V
  - 33) Hardware qualification test design V&V
  - 34) Hardware qualification test execution V&V
  - 35) Hardware qualification test plan V&V
  - 36) Hardware qualification test procedure V&V
  - 37) Hazard analysis
  - 38) Identify process improvement opportunities in the conduct of V&V
  - 39) Implementation strategy assessment
  - 40) Installation checkout

- 41) Installation configuration audit
- 42) Interface analysis
- 43) Management and technical review support
- 44) Management review of the V&V effort
- 45) Migration assessment
- 46) Operating procedures evaluation
- 47) Planning the interface between the V&V effort and supplier
- 48) Project planning strategy assessment
- 49) Proposed/baseline change assessment
- 50) Requirements allocation analysis
- 51) Requirements evaluation
- 52) Retirement assessment
- 53) Risk analysis
- 54) Security analysis
- 55) Software acceptance test case V&V
- 56) Software acceptance test design V&V
- 57) Software acceptance test execution V&V
- 58) Software acceptance test plan V&V
- 59) Software acceptance test procedure V&V
- 60) Software component test case V&V
- 61) Software component test design V&V
- 62) Software component test execution V&V
- 63) Software component test plan V&V
- 64) Software component test procedure V&V
- 65) Software disposal strategy evaluation
- 66) Software element interaction analysis
- 67) Software element interaction analysis
- 68) Software integration test case V&V
- 69) Software integration test design V&V
- 70) Software integration test execution V&V
- 71) Software integration test plan V&V
- 72) Software integration test procedure V&V
- 73) Software maintenance execution assessment
- 74) Software maintenance strategy assessment
- 75) Software qualification test case V&V
- 76) Software qualification test design V&V
- 77) Software qualification test execution V&V
- 78) Software qualification test plan V&V
- 79) Software qualification test procedure V&V
- 80) Source code and source code documentation evaluation
- 81) Stakeholder needs and requirements evaluation
- 82) System acceptance test case V&V
- 83) System acceptance test design V&V
- 84) System acceptance test execution V&V
- 85) System acceptance test plan V&V
- 86) System acceptance test procedure V&V
- 87) System analysis results evaluation
- 88) System analysis strategy evaluation
- 89) System element implementation analysis
- 90) System element interaction analysis
- 91) System integration strategy assessment
- 92) System integration test case V&V
- 93) System integration test design V&V
- 94) System integration test execution V&V
- 95) System integration test plan V&V
- 96) System integration test procedure V&V
- 97) System maintenance execution assessment

- 98) System maintenance strategy assessment
- 99) System qualification test case V&V
- 100) System qualification test design V&V
- 101) System qualification test execution V&V
- 102) System qualification test plan V&V
- 103) System qualification test procedure V&V
- 104) System requirements review
- 105) Traceability analysis
- 106) Transition demonstration assessment
- 107) Transition strategy evaluation
- 108) V&V final report generation

- b) V&V activity summary reports. An activity summary report shall summarize the results of V&V tasks performed for the following V&V life cycle activities:

#### **System V&V**

- 1) [Acquisition support](#)
- 2) [Supply planning](#)
- 3) [Project planning](#)
- 4) [Configuration management](#)
- 5) [Business or mission analysis](#)
- 6) [Stakeholder needs and requirements definition](#)
- 7) [System requirements definition](#)
- 8) [Architecture definition](#)
- 9) [Design definition](#)
- 10) [System analysis](#)
- 11) [Implementation](#)
- 12) [Integration](#)
- 13) [Transition](#)
- 14) [Operation](#)
- 15) [Maintenance](#)
- 16) [Disposal](#)

#### **Software V&V**

- 1) [Acquisition support](#)
- 2) [Supply planning](#)
- 3) [Project planning](#)
- 4) [Configuration management](#)
- 5) [Business or mission analysis](#)
- 6) [Stakeholder needs and requirements definition](#)
- 7) [System/software requirements definition](#)
- 8) [Architecture definition](#)
- 9) [Design definition](#)
- 10) [System analysis](#)
- 11) [Implementation](#)
- 12) [Integration](#)
- 13) [Transition](#)
- 14) [Operation](#)
- 15) [Maintenance](#)
- 16) [Disposal](#)

#### **Hardware V&V**

- 1) [Acquisition support](#)
- 2) [Supply planning](#)

- 3) [Project planning](#)
- 4) [Configuration management](#)
- 5) [Business or mission analysis](#)
- 6) [Stakeholder needs and requirements definition](#)
- 7) [System/hardware requirements definition](#)
- 8) [Architecture definition](#)
- 9) [Design definition](#)
- 10) [System analysis](#)
- 11) [Implementation](#)
- 12) [Integration](#)
- 13) [Transition](#)
- 14) [Operation](#)
- 15) [Maintenance](#)
- 16) [Disposal](#)

For the operation and maintenance life cycle activities, V&V activity summary reports may be either updates to previous V&V activity summary reports or separate documents.

- a) V&V anomaly reports. The V&V effort shall document in an anomaly report each anomaly it detects.
- b) V&V final report. The V&V final report shall be issued at the conclusion of the V&V effort.
- c) Optional V&V reports. The V&V reports may also include optional reports (i.e., special assessment reports and other reports). The V&V effort shall document in a special assessment report any special V&V assessments conducted during the life cycle. The V&V effort shall document in a report the results of tasks conducted but not defined in the VVP. These other task reports may include, for example, quality assurance results, end-user testing results, safety assessment reports, or configuration and data management status results. The title of the report may vary according to the subject matter.

Task report(s) and anomaly report(s) should be provided as feedback to the development process regarding the technical quality of each product and process.

## 11.2 V&V administrative requirements

The V&V administrative requirements shall consist of the following:

- a) Anomaly resolution and reporting policy
- b) Task iteration policy
- c) Deviation policy
- d) Control procedures
- e) Standards, practices, and conventions

These administrative requirements shall be documented in the VVP.

## 11.3 V&V documentation requirements

### 11.3.1 V&V test documentation

V&V test documentation requirements shall include the test plans, designs, cases, procedures, and results for component, integration, qualification, and acceptance testing developed by the V&V effort. The V&V test documentation shall conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B17]). If the V&V effort uses test documentation or test types different from those in this standard (i.e., component, integration, qualification, and acceptance), the V&V effort shall show a mapping of the proposed test documentation and execution to the test items defined in this standard. Test planning tasks defined in [Table 1a](#) through [Table 1d](#) shall be documented in the test plan, test design(s), test case(s), and test procedure(s).



### 11.3.2 VVP documentation

The V&V effort shall generate a VVP that addresses the topics described in [Clause 12](#) of this standard. If there is no information pertinent to a topic, then the VVP shall contain the phrase “This topic is not applicable to this plan” and shall state an appropriate reason for the exclusion. Additional topics may be added to the plan. If some VVP material appears in other documents, the VVP may repeat the material or make reference to the material. The VVP shall be maintained throughout the life cycle of the system, software, or hardware.

## 12. V&V plan

### 12.1 Overview

The VVP shall contain the content described in this subclause. The user of this standard may adopt any format and section numbering system for the VVP. The VVP section numbers listed in this clause are provided to assist readability. An example VVP outline is shown in the following boxed text:

| VVP outline (example)                                   |  |
|---------------------------------------------------------|--|
| 1. Purpose                                              |  |
| 2. Referenced documents                                 |  |
| 3. Definitions                                          |  |
| 4. V&V overview                                         |  |
| 4.1 Organization                                        |  |
| 4.2 Master schedule                                     |  |
| 4.3 Integrity level schema                              |  |
| 4.4 Resources summary                                   |  |
| 4.5 Responsibilities                                    |  |
| 4.6 Tools, techniques, and methods                      |  |
| 5. V&V processes                                        |  |
| 5.1 Common V&V processes, activities and tasks          |  |
| 5.1.1 Acquisition support V&V                           |  |
| 5.1.2 Supply planning V&V                               |  |
| 5.1.3 Project planning V&V                              |  |
| 5.1.4 Configuration management V&V                      |  |
| 5.2 System V&V processes, activities and tasks          |  |
| 5.2.1 Business or mission analysis V&V                  |  |
| 5.2.2 Stakeholder needs and requirements definition V&V |  |
| 5.2.3 System requirements definition V&V                |  |
| 5.2.4 Architecture definition V&V                       |  |
| 5.2.5 Design definition V&V                             |  |
| 5.2.6 System analysis V&V                               |  |
| 5.2.7 Implementation V&V                                |  |
| 5.2.8 Integration V&V                                   |  |
| 5.2.9 Transition V&V                                    |  |
| 5.2.10 Operation V&V                                    |  |
| 5.2.11 Maintenance V&V                                  |  |
| 5.2.12 Disposal V&V                                     |  |
| 5.3 Software V&V processes, activities and tasks        |  |
| 5.2.1 Business or mission analysis V&V                  |  |
| 5.2.2 Stakeholder needs and requirements definition V&V |  |
| 5.2.3 System/software requirements definition V&V       |  |
| 5.2.4 Architecture definition V&V                       |  |
| 5.2.5 Design definition V&V                             |  |
| 5.2.6 System analysis V&V                               |  |
| 5.2.7 Implementation V&V                                |  |
| 5.2.8 Integration V&V                                   |  |
| 5.2.9 Transition V&V                                    |  |
| 5.3.9 Operation V&V                                     |  |
| 5.3.10 Maintenance V&V                                  |  |
| 5.3.11 Disposal V&V                                     |  |
| 5.4 Hardware V&V processes, activities and tasks        |  |

|     |                                                         |
|-----|---------------------------------------------------------|
|     | 5.2.1 Business or mission analysis V&V                  |
|     | 5.2.2 Stakeholder needs and requirements definition V&V |
|     | 5.2.3 System/hardware requirements definition V&V       |
|     | 5.2.4 Architecture definition V&V                       |
|     | 5.2.5 Design definition V&V                             |
|     | 5.2.6 System analysis V&V                               |
|     | 5.2.7 Implementation V&V                                |
|     | 5.2.8 Integration V&V                                   |
|     | 5.2.9 Transition V&V                                    |
|     | 5.4.9 Operation V&V                                     |
|     | 5.4.10 Maintenance V&V                                  |
|     | 5.4.11 Disposal V&V                                     |
| 6.  | V&V reporting requirements                              |
| 6.1 | Task reports                                            |
| 6.2 | Anomaly reports                                         |
| 6.3 | V&V final report                                        |
| 6.4 | Special studies reports (optional)                      |
| 6.5 | Other reports (optional)                                |
| 7.  | V&V administrative requirements                         |
| 7.1 | Anomaly resolution and reporting                        |
| 7.2 | Task iteration policy                                   |
| 7.3 | Deviation policy                                        |
| 7.4 | Control procedures                                      |
| 7.5 | Standards, practices, and conventions                   |
| 8.  | V&V test documentation requirements                     |

## 12.2 VVP section 1: Purpose

The VVP shall describe the purpose, goals, and scope of the V&V effort, including waivers from this standard. The date of issue and status, identification of the issuing organization, and identification of approval authority shall be provided.

## 12.3 VVP section 2: Referenced documents

The VVP shall identify the compliance documents, documents referenced by the VVP, and any supporting documents supplementing or implementing the VVP.

## 12.4 VVP section 3: Definitions

The VVP shall define or reference all terms used in the VVP, including the criteria for classifying an anomaly as a critical anomaly. All abbreviations and notations used in the VVP also shall be described.

## 12.5 VVP section 4: V&V overview

### 12.5.1 VVP section 4.1: Organization

The VVP shall describe the organization of the V&V effort, including the degree of independence required (see [Annex C](#)). The VVP shall describe the relationship of the V&V processes to other processes, such as development, project management, quality assurance, and configuration management. The VVP shall describe the lines of communication within the V&V effort, the authority for resolving issues raised by V&V tasks, and the authority for approving V&V products. [Annex F](#) illustrates a sample organizational interrelationship chart.

### 12.5.2 VVP section 4.2: Master schedule

The VVP shall describe the project life cycle and milestones and shall summarize the schedule of V&V tasks and task results as feedback to the development, organizational project-enabling, and supporting processes

(e.g., quality assurance and configuration management). V&V tasks should be scheduled to be re-performed according to the task iteration policy. The detailed V&V schedule (e.g., task start and end dates, dependencies, assigned resources) may be maintained outside the VVP in order to accommodate the inevitable changes that occur in a project schedule.

If the life cycle used in the VVP differs from the life cycle model in this standard, this section shall describe how all requirements of the standard are satisfied (e.g., by cross-referencing to this standard).

### 12.5.3 VVP section 4.3: Integrity level schema

The VVP shall describe the agreed-on integrity level schema established for the system, software, or hardware, and the mapping of the selected schema to the model used in this standard. Where different integrity levels are assigned within the system of interest, the VVP shall document (by inclusion or by reference to the criticality analysis) the assignment of integrity levels to individual components (e.g., requirements, detailed functions, subsystems, software modules, hardware component, or other components).

### 12.5.4 VVP section 4.4: Resources summary

The VVP shall summarize the V&V resources, including staffing, facilities, tools, finances, and special procedural requirements (e.g., security, access rights, and documentation control).

### 12.5.5 VVP section 4.5: Responsibilities

The VVP shall identify an overview of the organizational element(s) and responsibilities for V&V tasks.

### 12.5.6 VVP section 4.6: Tools, techniques, and methods<sup>11</sup>

The VVP shall describe documents, hardware and software V&V tools, techniques, methods, and operating and test environment to be used in the V&V processes. Acquisition, training, support, and qualification information for each tool, technology, and method shall be included.

The VVP should document the measures to be used by V&V (see [Annex E](#)) and should describe how these measures support the V&V objectives.

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<sup>11</sup> The information in this footnote is based on excerpts from ISO/IEC/IEEE 29148-2011 [B45].

Verification and validation methods define how (including success criteria and closure approach), where, and when each requirement's conformance will be proven. A verification or validation method is associated with each requirement to define activities that yield objective information to prove satisfaction of the requirement. A verification or validation method definition addresses the following content considerations:

- How: Identify which method(s) will be applied (e.g., inspection, analysis, demonstration, and test).
- Who: Identify the organization/person with the lead responsibility for performing the method, such as a contractor, subcontractor, vendor, product team, or supplier.
- When: Designate a time in the program plan when the method is to be done. This should be an event-based, and not a calendar date or accomplishment.
- Where: Specify any unique venue and environment needed for the method.
- Examples of verification and validation methods to use to obtain the objective evidence that the requirements have been fulfilled are inspection, analysis or simulation, demonstration, and test. The definitions are as follows:
  - Inspection: An examination of the item against applicable documentation to confirm compliance with requirements.
  - Analysis (including modeling and simulation): Use of analytical data or simulations under defined conditions to show theoretical compliance.
  - Demonstration: A qualitative exhibition of functional performance, usually accomplished with no or minimal instrumentation or test equipment.
  - Test: An action by which the operability, supportability, or performance capability of an item is quantitatively verified when subjected to controlled conditions that are real or simulated.

## 12.6 VVP section 5: V&V processes

The VVP also shall identify the specific processes and products covered by the V&V effort and the activities and tasks performed as derived from the Table 2 minimum V&V requirements and the optional tasks selected for the project.

### 12.6.1 VVP section 5.1: Common V&V processes, activities, and tasks

The VVP shall identify and describe the common V&V processes, activities and tasks as identified in [Table 1a](#) to be implemented and performed for this V&V project. The minimum common V&V tasks for the selected system integrity level are defined in [Table 2a](#).

### 12.6.2 VVP section 5.2: System V&V processes, activities, and tasks

The VVP shall identify and describe the system V&V processes, activities, and tasks as identified in Table 1b to be implemented and performed for this V&V project. The minimum system V&V tasks for the selected system integrity level are defined in [Table 2b](#).

### 12.6.3 VVP section 5.3: Software V&V processes, activities, and tasks

The VVP shall identify and describe the software V&V processes, activities, and tasks as identified in Table 1c to be implemented and performed for this V&V project. The minimum software V&V tasks for the selected system integrity level are defined in [Table 2c](#).

### 12.6.4 VVP section 5.4: Hardware V&V processes, activities, and tasks

The VVP shall identify and describe the hardware V&V processes, activities, and tasks as identified in Table 1d to be implemented and performed for this V&V project. The minimum hardware V&V tasks for the selected system integrity level are defined in [Table 2d](#).

## 12.7 VVP section 6: V&V reporting requirements

The VVP shall specify the purpose, content, format, recipients, and timing of all V&V reports. The V&V reporting requirements are specified in [Clause 11](#).

## 12.8 VVP section 7: V&V administrative requirements

### 12.8.1 General

The VVP shall describe the anomaly resolution and reporting, task iteration policy, deviation policy, control procedures, and standards, practices, and conventions.

### 12.8.2 VVP section 7.1: Anomaly resolution and reporting

The VVP shall describe the method of reporting and resolving anomalies, including the criteria for reporting an anomaly, the anomaly report distribution list, the authority and timelines for resolving anomalies, and the anomaly criticality levels. Classification for software anomalies may be found in IEEE Std 1044-2009 [\[B21\]](#).

### 12.8.3 VVP section 7.2: Task iteration policy

The VVP shall describe the criteria used to determine the extent to which a V&V task should be repeated when its input is changed, or task procedure is changed. These criteria may include assessments of change, integrity level, and effects on budget, schedule, and quality.

#### **12.8.4 VVP section 7.3: Deviation policy**

The VVP shall describe the procedures and criteria used to deviate from the plan. The information required for deviations shall include task identification, rationale, and effect on quality. The VVP shall identify the authorities responsible for approving deviations.

#### **12.8.5 VVP section 7.4: Control procedures**

The VVP shall identify control procedures applied to the V&V effort. These procedures shall describe how products and V&V results should be configured, protected, and stored.

These procedures may describe quality assurance, configuration management, data management, or other activities if they are not addressed by other efforts. The VVP shall describe how the V&V effort shall conform to existing security provisions and how the validity of V&V results shall be protected from unauthorized alterations.

#### **12.8.6 VVP section 7.5: Standards, practices, and conventions**

The VVP shall identify the standards, practices, and conventions that govern the performance of V&V tasks, including internal organizational standards, practices, and policies.

#### **12.9 VVP section 8: V&V test documentation requirements**

The VVP shall describe the purpose, format, and content for the following V&V test documents:

- a) Test plan
- b) Test design
- c) Test cases
- d) Test procedures
- e) Test results

The V&V effort may define the format for these documents. IEEE Std 829-2008 [\[B17\]](#) contains sample formats for these test documents.

## Annex A

(informative)

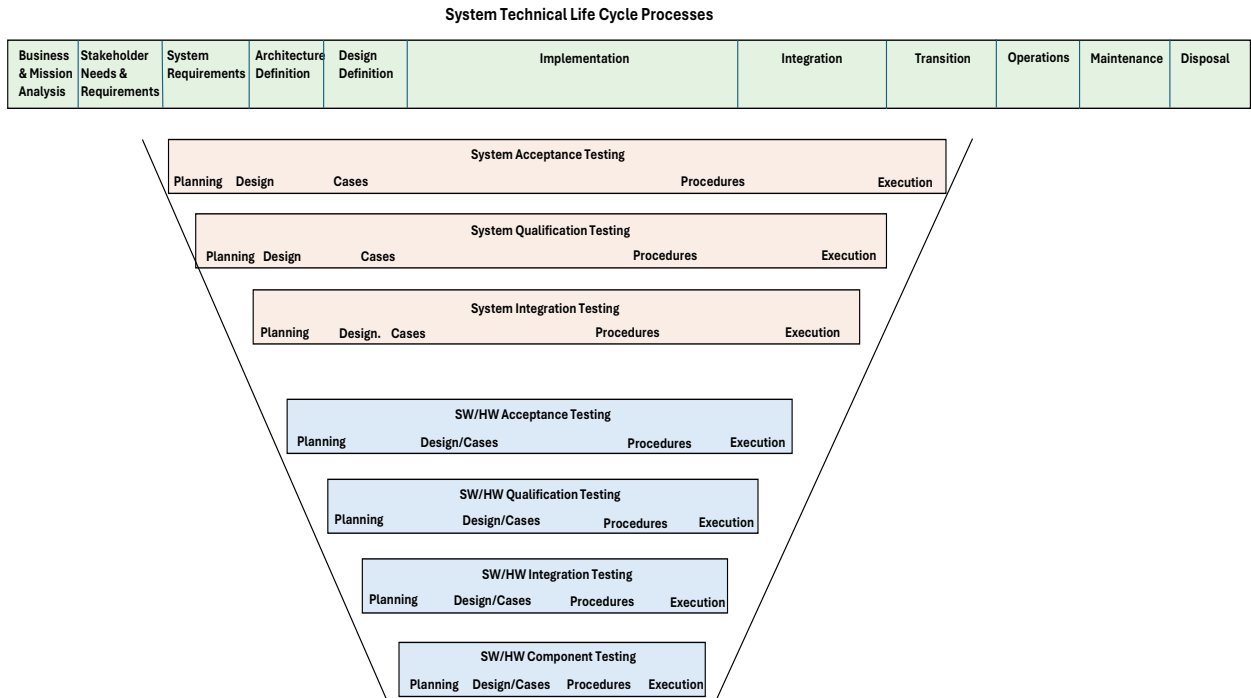
### V&V test concepts

Test activities span the entire system and the system's software/hardware elements life cycles starting as early as the mission and business/system requirements definition stages. Each stage's test activities should use the V&V analyses results from that stage's V&V tasks to formulate the optimal test approach, best scenarios to test the system and/or element characteristics/requirements, needed test methods and techniques (e.g., test facility requirements, simulation and test drivers, test analytical tools), and identification of potential system or element structural weaknesses and risk areas to focus testing. The strong interlinking of V&V analysis results with V&V test planning and knowledge of the system/element structure and requirements developed during each stage's V&V analysis have been shown to enhance a more comprehensive and successful V&V testing approach. When an organizational structure decouples the V&V testing members from the V&V analysis team, the inside knowledge of the system or element structure and its characteristics is bypassed, and the testing is less than efficient and effective. The transfer of knowledge of the program and V&V analysis between the V&V analyst and the test activities is accomplished by the V&V analysis team developing the test strategies, test scenarios, and test procedures. The actual conduct of the V&V test cases does not have to be performed by the V&V analysts, but rather often, those tests are conducted by test specialists knowledgeable about the test laboratory setup and simulation tools.

Effective testing requires a comprehensive understanding of the system. The foundation of system knowledge is developed by a systematic process of analyzing the software from concept, requirements, design, and implementation. Since program testing only demonstrates the presence of an error but can never show its absence (Dijkstra), the selection of the right test case is crucial to the success of testing. If each test case can excite errors in a given region of the software, then the secret to successful testing is to define test cases whose region of influence overlap each other such that all error classes are excited and detected. The difficulty is obviously in determining how the software functional interactions and execution paths are influenced by uniquely different test cases. However, by taking a point of view of looking specifically for errors, as well as compliance with requirements, V&V testing is likely to find errors.

Research literature and V&V experience suggest that the V&V test planning activities reveal a significant number of system/elements errors and deficiencies early in the life cycle stages, even before actual test execution occurs. The process of V&V test planning, in thinking about how to uncover software faults, often leads to the detection of errors (incomplete compliance with requirements, failure to address hazard or fault conditions, faulty or inaccurate assumptions leading to an incorrect solution).

[Figure A.1](#) illustrates how V&V test planning activities begin early in the system or element life cycles and conclude with the V&V test execution at the later stages of the life cycle. V&V testing uses the building-block approach to pyramid larger efforts from smaller efforts. Component testing focuses on driving out errors at the component level. This information is used by integration testing to prove the correctness of a combination of components. Special testing focus is applied to those components that revealed weaknesses during component testing. In a similar pyramid fashion, qualification and acceptance testing build from knowledge gained from prior V&V test levels.



**Figure A.1—Testing activities across the life cycle**

In IEEE standards testing, there are four types of test categories:

- **Component Test:** Tests the components of software or hardware. For software, this would be software subroutines or modules. For hardware, this would be the mechanical piece or electronic chip/board. For a system, the component is the entire software or hardware element. System component testing is the same as software or hardware acceptance test. Therefore, there is no system component test.
- **Integration Test:** Tests groupings of components generally starting with small groupings of components and building toward a larger integrated set of components. Tests the requirements within each component or grouping of components to demonstrate that the integrated set of components satisfies the component requirements.
- **Qualification Test:** Tests all system, software, or hardware requirements in its entirety, generally in a controlled test environment using simulators or special drivers to represent the real environment and other interfacing systems inputs. Qualification testing generally occurs within the development team's test environment and may have diagnostic tools and recorders attached to the test environment to capture data showing that the system or element requirements have been satisfied.
- **Acceptance Test:** Test conducted or directed by the authority accepting the system or software and hardware element. The tests are generally conducted in as close to the operational environment as possible.

Each test category has the following test documents/artifacts for each test category:

- **Test Plan:** Identifies the requirements to test and the criteria or results that demonstrate that the requirement was satisfied. The product is a test plan document.
- **Test Design:** Identifies the scenarios to be tested and how each test would be accomplished (e.g., simulation, test drivers, test facility). The product is a test design document.
- **Test Cases:** Identify the details of the test scenarios (e.g., range of values, specific conditions of the environment, or interfacing system or hardware). The product is a test case document.
- **Test Procedures:** Identify the specific inputs of each scenario and the specific state of the system and its environment. The product is a test procedures document.

- *Test Execution*: Conducts the execution of the test procedures and determines whether the expected test results were achieved.
- *Test Results*: Documents the test execution results for each test procedure and documents whether the results were achieved. If the results were not achieved, then the discrepancy is documented in an anomaly report. Products are test results and anomaly reports (if required).

A summary of the test activities (test plan, test design, test cases, test procedures, test execution, and test results document) for each element (system, software, and hardware) is shown in [Table A.1](#), [Table A.2](#), and [Table A.3](#).

**Table A.1—Alignment of system test activities with life cycle processes**

| Element | Test category | Test activities                                     | System life cycle processes                                            |
|---------|---------------|-----------------------------------------------------|------------------------------------------------------------------------|
| System  | Component     | See software or hardware acceptance test activities |                                                                        |
|         | Integration   | Test plan                                           | Architecture Definition Process (1 <sup>st</sup> draft)                |
|         |               | Test design                                         | Design Definition Process (1 <sup>st</sup> draft)                      |
|         |               | Test cases                                          | Design Definition Process (1 <sup>st</sup> draft)                      |
|         |               | Test procedures                                     | Implementation Process (1 <sup>st</sup> process)                       |
|         |               | Test execution                                      | Integration Process                                                    |
|         |               | Test results                                        | Completion of integration testing (including all regression testing)   |
|         | Qualification | Test plan                                           | System Requirements Definition Process (1 <sup>st</sup> draft)         |
|         |               | Test design                                         | Architecture Definition Process (1 <sup>st</sup> draft)                |
|         |               | Test cases                                          | Design Definition Process (1 <sup>st</sup> draft)                      |
|         |               | Test procedures                                     | Implementation Process (1 <sup>st</sup> process)                       |
|         |               | Test execution                                      | Integration Process                                                    |
|         |               | Test results                                        | Completion of qualification testing (including all regression testing) |
|         | Acceptance    | Test plan                                           | System Requirements Definition Process (1 <sup>st</sup> draft)         |
|         |               | Test design                                         | Architecture Definition Process (1 <sup>st</sup> draft)                |
|         |               | Test cases                                          | Design Definition Process (1 <sup>st</sup> draft)                      |
|         |               | Test procedures                                     | Implementation Process (1 <sup>st</sup> process)                       |
|         |               | Test execution                                      | Transition Process                                                     |
|         |               | Test results                                        | Completion of acceptance testing (including all regression testing)    |

**Table A.2—Alignment of software test activities with life cycle processes**

| Element  | Test category | Test activities | Software life cycle processes                                           |
|----------|---------------|-----------------|-------------------------------------------------------------------------|
| Software | Component     | Test plan       | Architecture Definition Process (1 <sup>st</sup> draft)                 |
|          |               | Test design     | Design Definition Process (1 <sup>st</sup> draft)                       |
|          |               | Test cases      | Design Definition Process                                               |
|          |               | Test procedures | Implementation Process                                                  |
|          |               | Test execution  | Implementation Process                                                  |
|          |               | Test results    | Completion of component testing (including all regression testing)      |
|          | Integration   | Test plan       | Architecture Definition Process (1 <sup>st</sup> draft)                 |
|          |               | Test design     | Design Definition Process (1 <sup>st</sup> draft)                       |
|          |               | Test cases      | Design Definition Process                                               |
|          |               | Test procedures | Implementation Process                                                  |
|          |               | Test execution  | Integration Process                                                     |
|          |               | Test results    | Completion of integration testing (including all regression testing)    |
|          | Qualification | Test plan       | System/Software Requirements Definition Process (1 <sup>st</sup> draft) |
|          |               | Test design     | Architecture Definition Process (1 <sup>st</sup> draft)                 |
|          |               | Test cases      | Design Definition Process                                               |
|          |               | Test procedures | Implementation Process                                                  |



|  |            |                 |                                                                         |
|--|------------|-----------------|-------------------------------------------------------------------------|
|  | Acceptance | Test execution  | Integration Process                                                     |
|  |            | Test results    | Completion of qualification testing (including all regression testing)  |
|  |            | Test plan       | System/Software Requirements Definition Process (1 <sup>st</sup> draft) |
|  |            | Test design     | Architecture Definition Process (1 <sup>st</sup> draft)                 |
|  |            | Test cases      | Design Definition Process                                               |
|  |            | Test procedures | Implementation Process                                                  |
|  |            | Test execution  | Transition Process                                                      |
|  |            | Test results    | Completion of acceptance testing (including all regression testing)     |

**Table A.3—Alignment of hardware test activities with life cycle processes**

| Element  | Test category | Test activities | Hardware life cycle processes                                           |
|----------|---------------|-----------------|-------------------------------------------------------------------------|
| Hardware | Component     | Test plan       | Architecture Definition Process (1 <sup>st</sup> draft)                 |
|          |               | Test design     | Design Definition Process (1 <sup>st</sup> draft)                       |
|          |               | Test cases      | Design Definition Process                                               |
|          |               | Test procedures | Implementation Process                                                  |
|          |               | Test execution  | Implementation Process                                                  |
|          |               | Test results    | Completion of component testing (including all regression testing)      |
|          | Integration   | Test plan       | Architecture Definition Process (1 <sup>st</sup> draft)                 |
|          |               | Test design     | Design Definition Process (1 <sup>st</sup> draft)                       |
|          |               | Test cases      | Design Definition Process                                               |
|          |               | Test procedures | Implementation Process                                                  |
|          |               | Test execution  | Integration Process                                                     |
|          |               | Test results    | Completion of integration testing (including all regression testing)    |
|          | Qualification | Test plan       | System/Hardware Requirements Definition Process (1 <sup>st</sup> draft) |
|          |               | Test design     | Architecture Definition Process (1 <sup>st</sup> draft)                 |
|          |               | Test cases      | Design Definition Process                                               |
|          |               | Test procedures | Implementation Process                                                  |
|          |               | Test execution  | Integration Process                                                     |
|          |               | Test results    | Completion of qualification testing (including all regression testing)  |
|          | Acceptance    | Test plan       | System/Hardware Requirements Definition Process (1 <sup>st</sup> draft) |
|          |               | Test design     | Architecture Definition Process (1 <sup>st</sup> draft)                 |
|          |               | Test cases      | Design Definition Process                                               |
|          |               | Test procedures | Implementation Process                                                  |
|          |               | Test execution  | Transition Process                                                      |
|          |               | Test results    | Completion of acceptance testing (including all regression testing)     |

V&V testing is a disciplined approach to verifying and validating the system, software, and hardware elements, starting with a comprehensive planning activity and progressively evolving the details of the V&V test strategy, comprehensive test design and cases, and detailed execution procedures. [Figure A.1](#) and [Table A.1](#) through [Table A.3](#) illustrate the detailed test activities starting early in the system, software and hardware element life cycle, and identify the products of each V&V test activity.

In addition to the disciplined V&V test approach, V&V testing tailors its testing techniques and test methods to provide objective evidence that the system, software, and hardware elements comply with all of its intended requirements and do not perform any unintended requirements across the full spectrum of test conditions. These test conditions include normal full spectrum conditions (e.g., computational accuracy, security protection scenarios), stress conditions (e.g., data volume, timing sequences, process sequences), anomalous requirements, boundary conditions, all hazardous scenarios, and mission-specific constraints. The use of simulations during the evolution of the software, beginning as early as the concept stage, is one of the most effective methods of using test methods to discover early errors. Use of simulation to reflect the

proposed approach (using the information available at each life cycle—conceptual ideas during the concept stage, overall system requirements during the system requirements/architecture stage, high level and then detailed component design details during the design stage, and finally actual implementation approach during the implementation stage) is a way to look into the future of how the final software/system performance would look like given the information available at each life cycle stage. This look into the future is a powerful perspective of what might be if the implementation solution used the proposed approach at each life cycle stage. Errors or oversights are more easily detected, and early correction of the concept, requirements, design, or implementation can be made.

[Figure A.2](#) illustrates how the lowest level software and hardware elements are integrated in increasingly larger combinations following the planned testing and integration strategies to prove that the functional set of requirements are correct. The Integration Process continues to build larger combinations of requirements until the full system is created. Different testing methods, as shown at the bottom row of [Figure A.2](#) illustrates how the V&V testing may select different test methods as the Integration Process continually builds a larger integrated set of requirements.

[Table A.4](#) lists the different test strategies that V&V testing may employ. A comprehensive V&V test strategy does not rely solely on one test strategy but uses a number of different strategies that are best to provide the objective evidence that requirements are correct. Key inputs to selecting the best test strategy are the V&V analysis results that provide detailed knowledge of the characteristics of the system/software/hardware solution. Sharing the knowledge of the solution between V&V analysis tasks and V&V testing creates the most effective V&V since this integrated approach focuses the V&V effort on those areas of the program element that are the most important and on program areas of complexity (i.e., greater likelihood of being an incomplete solution) as determined by the V&V analysis tasks accomplished through each stage of the life cycle. V&V might also consider using strategies that differ from the testing strategy of the developer, in order to provide a different perspective on the system.

The V&V testing strategies adopt a fundamental approach of assuming that there are errors in the program. Therefore, the test design/cases/procedures selected for V&V testing are to explore all aspects of program performance against all system scenarios, including stress conditions, abnormal events, security/hazard threats, performance limits, accidental/deliberate erroneous operator actions as well as determination of how well the program performs or exceeds its performance requirements. The basic philosophy of assuming that the program has errors creates different testing approaches from the nominal development test approach of demonstrating that the program is correct. The consequence of this type of examination and rigorous testing strategy complements the development testing, and if no errors are found through this vigorous approach, then one can draw a conclusion that the program is correct and safe. Naturally, this type of testing should test for compliance with performance and be comprehensive in all domains of program requirements assessment.

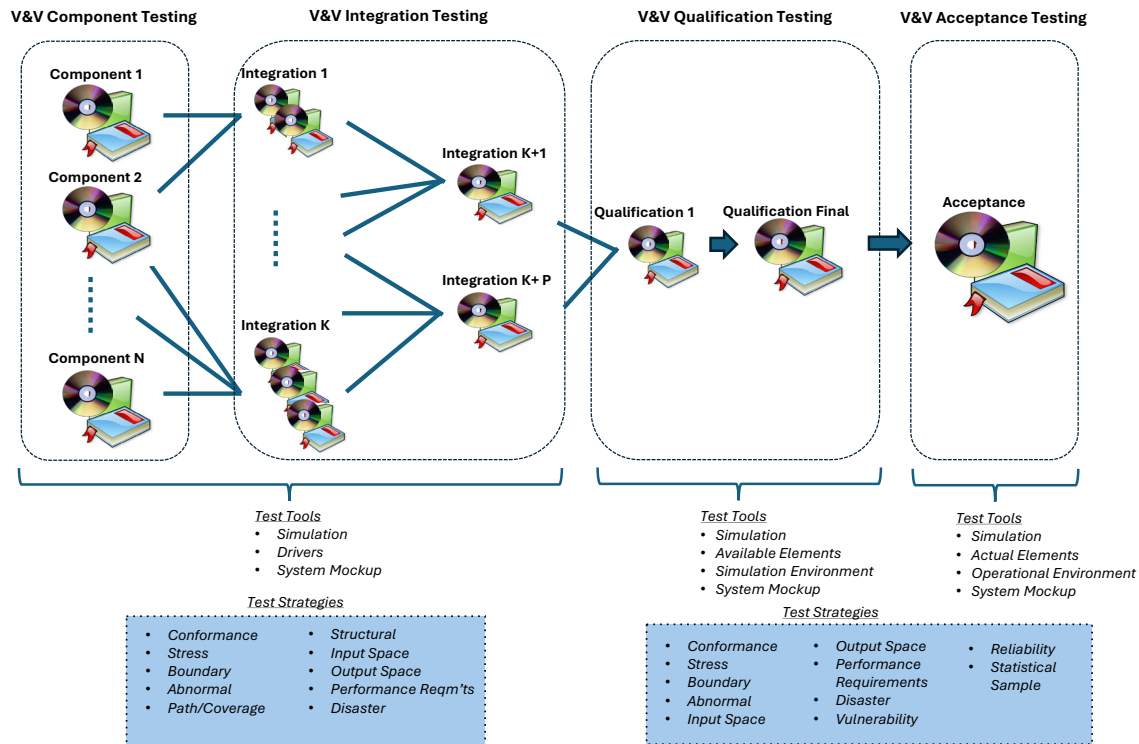


Figure A.2—Disciplined test activities leading to an integrated system element

Table A.4—V&V test strategies

| Test strategies                                                                                                                                                                                                                                | Advantages                                                                                                                                                                                                         | Disadvantages                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scenario-based</b> <ul style="list-style-type: none"> <li>Normal conditions</li> <li>Abnormal conditions</li> <li>Stress</li> <li>Boundary conditions (performance, inputs, operator idiosyncrasies)</li> <li>Anomaly conditions</li> </ul> | <ul style="list-style-type: none"> <li>Comprehensive</li> <li>Representative of operational scenarios (design and cases)</li> <li>Strong system views</li> <li>Detection of information coupling errors</li> </ul> | <ul style="list-style-type: none"> <li>Difficult to identify all data inputs for meaningful scenarios</li> <li>Requires extensive test capability and simulations</li> </ul>                                 |
| <b>Functional requirements</b>                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>Covers all requirements</li> </ul>                                                                                                                                          | <ul style="list-style-type: none"> <li>Difficult to specify all variations to inputs</li> <li>Test of singular functional requirement does not find information coupling errors between functions</li> </ul> |
| <b>Coverage</b>                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Executes each logic path</li> <li>Easier to set up a singular test case</li> </ul>                                                                                          | <ul style="list-style-type: none"> <li>Spotty coverage of system functions and scenarios</li> <li>Weak in comprehensive testing—single path only</li> </ul>                                                  |
| <b>Statistical campling</b>                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>Economical</li> </ul>                                                                                                                                                       | <ul style="list-style-type: none"> <li>Not comprehensive</li> </ul>                                                                                                                                          |

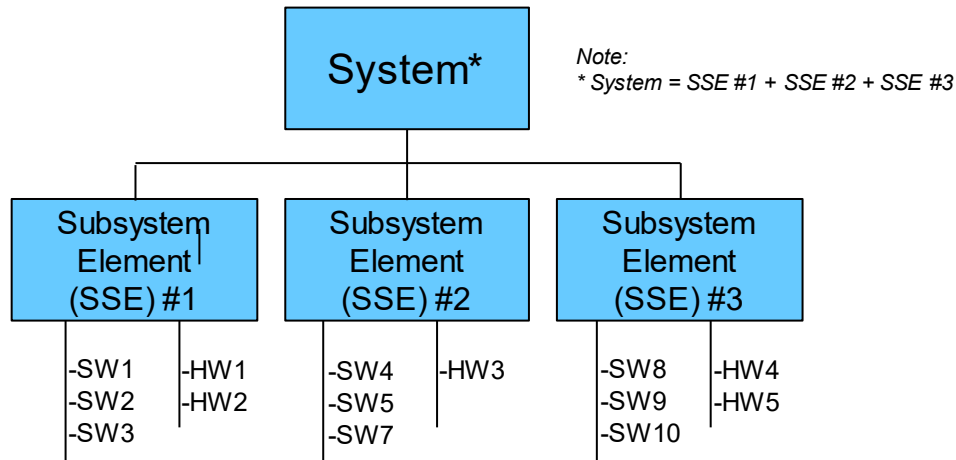
The important role of V&V integration testing: The system/software/hardware V&V integration testing starts when individual system/software/hardware components and elements start coming together to connect (integrate) with each individual software and hardware element to form increasingly larger components to implement increasing system functionalities. The V&V integration testing strategies utilize the V&V knowledge of all prior V&V life cycle processes (requirements, system/component architecture, detailed

design, and implementation characteristics) to formulate the V&V integration test cases/procedures which will show whether or not the system functionality/intent (end to end system requirements) are satisfied. The function of V&V integration testing is to provide objective test evidence that end-to-end system performance requirements are being correctly satisfied and, if the system performance requirements are not satisfied, to identify the error source and conditions causing the failure. The integration testing starts with small groupings of system elements demonstrating that the core architectural elements are correctly working and then increasingly, stepwise adding additional system elements to form larger integration elements to test system performance requirements that involve a greater number of system elements. This stepwise increasing integration of system elements continues until the entire system composed of all elements is combined to perform total end-to-end system integration testing. Whenever errors are discovered in system integration testing, the corrections must be re-tested, and any related prior system integration tests must be repeated with regression testing to provide assurance that the added correction has not adversely influenced the performance of other elements of the system.

In large systems consisting of multiple subsystems, as each subsystem completes the subsystem V&V integration testing, the subsystem must be integrated with other subsystems starting with a small grouping of subsystems and stepwise increasing, adding other subsystems to test those system performance requirements involving greater complexity. Any errors discovered and corrected during system integration testing will require a system integration regression testing of the affected subsystem to verify and validate the correction.

To illustrate how the different V&V tests analyze the implementation of the system and software/hardware requirements following a disciplined process of examining each solution as the system and software/hardware evolve from fundamental components and systematically being integrated into larger subsystem elements and finally integrating all subsystem elements (SSEs) into a total system solution. To illustrate the steps of the V&V testing,

[Figure A.3](#) illustrates a system consisting of three subsystems. Each subsystem has software and hardware components as indicated by software elements labeled SW, and hardware elements labelled HW.

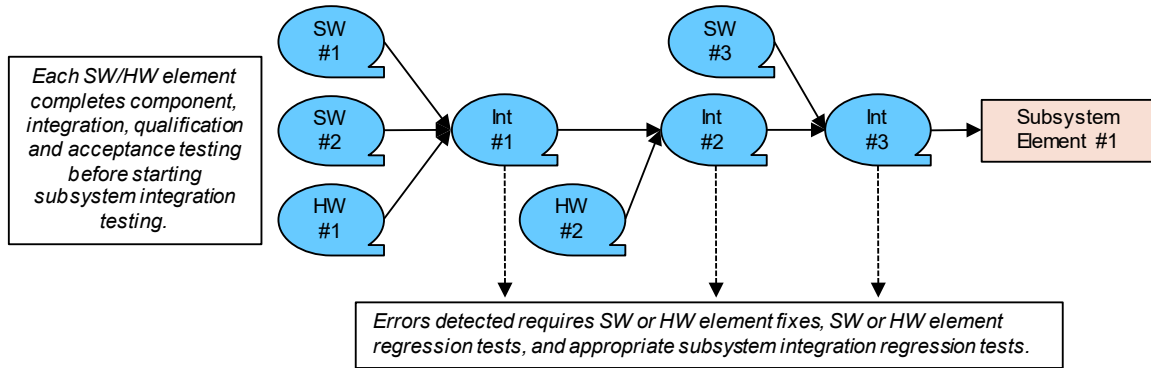


**Figure A.3—System composition of subsystem elements**

The V&V testing will first test each software and hardware (if included in the scope of V&V) element. Each element is verified and validated by V&V analysis through each of the element life cycle (concept, requirements analysis, design, construction). For the test life cycle phases, the V&V will test each component in the software/hardware during the construction phase. After V&V component testing is completed, the V&V testing will conduct software/hardware integration, acceptance, and qualification testing of the software element.

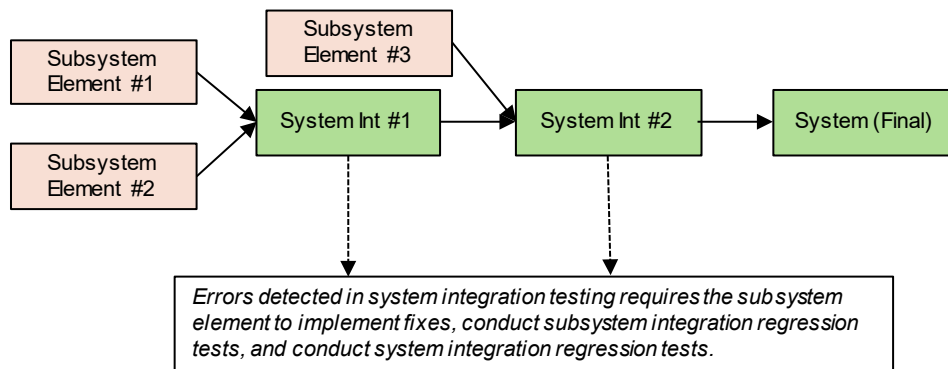
To verify and validate the subsystem element, the individual SW and HW elements are tested via an integration test strategy by bringing together the SW and HW elements by testing the combination of the SW and HW elements into larger and larger building blocks. An example subsystem integration sequence is shown in [Figure A.4](#). After all the integration testing is completed and all errors are corrected and regression

tested, the subsystem element is ready for testing with other subsystems to verify and validate the overall system requirements.



**Figure A.4—Subsystem element #1 integration test example**

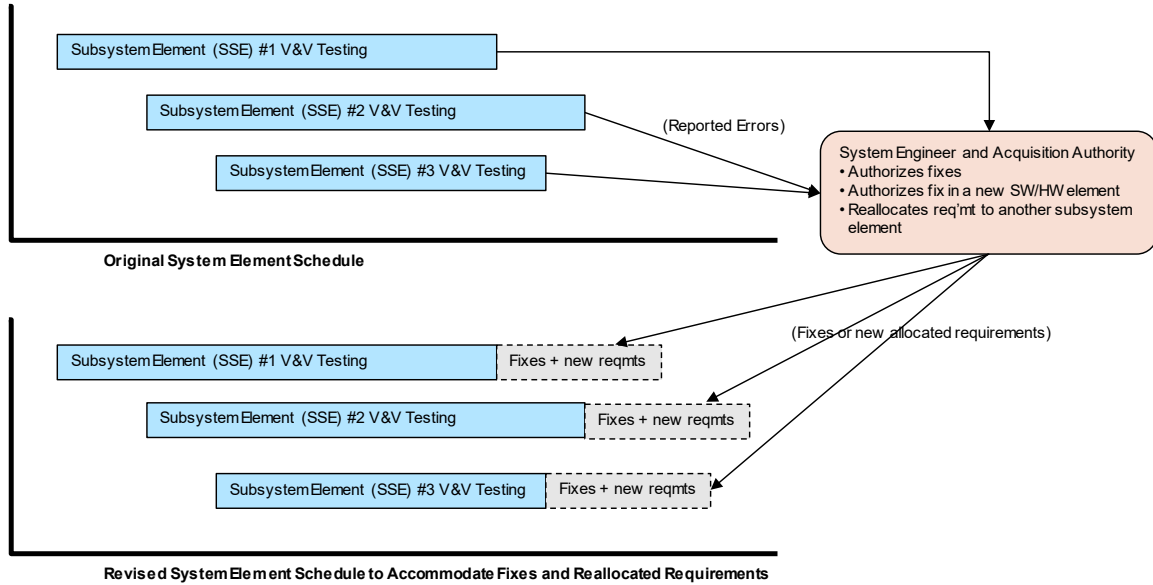
To test the entire system, the subsystem elements are integrated together in a building block approach, starting with a small set of subsystem elements being tested as an integrated element and progressively adding other subsystem elements until the entire system has been integrated and tested. [Figure A.5](#) illustrates this testing building block approach starting by testing an integrated group of a few subsystem elements and progressively adding other subsystem elements until the fully integrated system with all subsystem elements has been tested.



**Figure A.5—System integration test example**

If any subsystem has not fully addressed (satisfied) the system requirements assigned to the subsystem's individual elements (software or hardware), the development systems engineering team/acquisition authority may create new software or hardware elements within the subsystem to address the incomplete/missing system requirements or assign those incomplete system requirements to another subsystem or newly created subsystem. The consequence of any of these actions typically adds to the system development schedule, requires additional subsystem and system integration testing and regression testing of all subsystems to provide assurance that the new addition has not adversely affected other system and subsystem elements. The system integration testing life cycle stage becomes highly complex and requires careful replanning of regression testing to provide assurance that no adverse effects are inadvertently created by the revisions and corrections made to system and subsystem elements. The V&V subsystem/system integration testing becomes important to provide a common and cohesive independent system/subsystem integration source since the development subsystem integration testing may be performed by different organizational units.

[Figure A.6](#) illustrates how the complexity and dynamics of system/subsystem integration testing combined with regression testing can easily impact the total program schedule and testing completeness and the V&V of all system performance requirements.



**Figure A.6—Impacts of element changes due to fixes and newly allocated requirements**

Program fixes and reallocated requirements can occur at any point in the system development. The V&V analysis and testing strategies and approaches must constantly be able to adjust to the dynamics of systems development. Large, complex systems are continuously in a state of dynamic adjustment, and the V&V planning and strategies must be prepared to quickly adapt to these changes and to carefully determine the amount of regression analysis and testing necessary to validate the new fixes and newly allocated requirements to any of the system elements. By being involved in the system development early in the life cycle and using strong analytic and testing techniques, V&V can assist in minimizing the impact of required program fixes and newly allocated requirements by detecting errors early in each life cycle stage.

In summary, V&V testing is to demonstrate that not only the system requirements are correctly implemented but to demonstrate through exhaustive stress and boundary testing that no unintended behavior or abnormal performance exists in the requirements' implementation. The V&V testing for abnormal behavior provides objective evidence that no unintended functions exist within the system design and implementation, which can easily be overlooked by the development organization due to the shortness of time or the strong belief in the design approach (i.e., loss of objectivity).

## Annex B

(informative)

### A risk-based integrity level schema

[Table B.1](#) defines four integrity levels used for reference purposes by this standard. [Table B.2](#) describes the consequences of errors for each of the four integrity levels. There are overlaps between the integrity levels to allow for individual interpretations of acceptable risk depending on the application.

**Table B.1—Assignment of integrity levels**

| Integrity level | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4               | Behavior of the system, in combination with its environment, causes the following: <ul style="list-style-type: none"> <li>— Catastrophic consequences for which the likelihood of the behavior occurring is at most occasional</li> <li>— Critical consequences for which the likelihood of the behavior occurring is at most probable</li> </ul>                                                                                                           |
| 3               | Behavior of the system, in combination with its environment, causes the following: <ul style="list-style-type: none"> <li>— Catastrophic consequences for which the likelihood of the behavior occurring is at most infrequent</li> <li>— Critical consequences for which the likelihood of the behavior occurring is at most occasional</li> <li>— Marginal consequences for which the likelihood of the behavior occurring is at most probable</li> </ul> |
| 2               | Behavior of the system, in combination with its environment, causes the following: <ul style="list-style-type: none"> <li>— Critical consequences for which the likelihood of the behavior occurring is at most infrequent</li> <li>— Marginal consequences for which the likelihood of the behavior occurring is at most probably</li> <li>— Negligible consequences for which the likelihood of the behavior occurring is at most reasonable</li> </ul>   |
| 1               | Behavior of the system, in combination with its environment, causes the following: <ul style="list-style-type: none"> <li>— Critical consequences for which the likelihood of the behavior occurring is at most infrequent</li> <li>— Marginal consequences for which the likelihood of the behavior occurring is at most occasional</li> <li>— Negligible consequences for which the likelihood of the behavior occurring is at most probable</li> </ul>   |

**Table B.2—Definition of consequences**

| Consequence  | Definition                                                                                                               |
|--------------|--------------------------------------------------------------------------------------------------------------------------|
| Catastrophic | Loss of human life, complete mission failure, loss of system security and safety, or extensive financial or social loss. |
| Critical     | Major and permanent injury, partial loss of mission, major system damage, or major financial or social loss.             |
| Marginal     | Severe injury or illness, degradation of secondary mission, or some financial or social loss.                            |
| Negligible   | Minor injury or illness, minor impact on system performance, or operator inconvenience.                                  |

[Table B.3](#) illustrates the risk-based schema shown in [Table B.1](#) and [Table B.2](#). Each cell in the table assigns an integrity level based on the combination of an error consequence and the likelihood of the occurrence of

an operating state that contributes to the error. Some table cells reflect more than one integrity level, indicating that the final assignment of the integrity level can be selected to address the system application and risk mitigation recommendations. For some industry applications, the definition of likelihood of occurrence categories may be expressed as probability figures derived by analysis or from system requirements.

**Table B.3—Graphic illustration of the assignment of integrity levels**

| <b>Error</b>       | <b>Likelihood of occurrence of an operating state that contributes to the error<br/>(decreasing order of likelihood)</b> |                 |                   |                   |
|--------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------|-------------------|
| <b>Consequence</b> | <b>Reasonable</b>                                                                                                        | <b>Probable</b> | <b>Occasional</b> | <b>Infrequent</b> |
| Catastrophic       | 4                                                                                                                        | 4               | 4 or 3            | 3                 |
| Critical           | 4                                                                                                                        | 4 or 3          | 3                 | 2 or 1            |
| Marginal           | 3                                                                                                                        | 3 or 2          | 2 or 1            | 1                 |
| Negligible         | 2                                                                                                                        | 2 or 1          | 1                 | 1                 |



## **Annex C**

(informative)

### **Definition of independent V&V (IV&V)**

#### **C.1 Independence parameters**

##### **C.1.1 Introduction**

IV&V is defined by three parameters: technical independence, managerial independence, and financial independence.

##### **C.1.2 Technical independence**

Technical independence requires the V&V effort to use personnel who are not involved in the development of the system or its elements. The IV&V effort should formulate its own understanding of the problem and how the proposed system is solving the problem. Technical independence (“fresh viewpoint”) is an important method to detect subtle errors overlooked by those too close to the solution.

For system tools, technical independence means that the IV&V effort uses or develops its own set of test and analysis tools separate from the developer’s tools. Sharing of tools is allowable for computer support environments (e.g., compilers, assemblers, and utilities) or for system simulations where an independent version would be too costly. For shared tools, IV&V conducts qualification tests on tools to assure that the common tools do not contain errors that may mask errors in the system being analyzed and tested. Off-the-shelf tools that have an extensive history of use do not require qualification testing. The most important aspect for the use of these tools is to verify the input data used.

##### **C.1.3 Managerial independence**

This requires that the responsibility for the IV&V effort be vested in an organization separate from the development and program management organizations. Managerial independence also means that the IV&V effort independently selects the segments of the software, hardware, and system to analyze and test, chooses the IV&V techniques, defines the schedule of IV&V activities, and selects the specific technical issues and problems to act on. The IV&V effort provides its findings in a timely fashion simultaneously to both the development and program management organizations. The IV&V effort is allowed to submit to program management the IV&V results, anomalies, and findings without any restrictions (e.g., without requiring prior approval from the development group) or adverse pressures, direct or indirect, from the development group.

##### **C.1.4 Financial independence**

This requires that control of the IV&V budget be vested in an organization independent of the development organization. This independence prevents situations where the IV&V effort cannot complete its analysis test or deliver timely results because funds have been diverted or adverse financial pressures or influences have been exerted.

#### **C.2 Forms of independence**

##### **C.2.1 Introduction**

The extent to which each of the three independence parameters (technical, managerial, and financial) is vested in a V&V effort determines the degree of independence achieved.

Many forms of independence can be adopted for a V&V effort. The five most prevalent are as follows: 1) classical, 2) modified, 3) integrated, 4) internal, and 5) embedded.

### **C.2.2 Classical IV&V**

Classical IV&V embodies all three independence parameters. The IV&V responsibility is vested in an organization that is separate from the development organization. The IV&V effort establishes a close working relationship with the development organization to assure that IV&V findings and recommendations are integrated rapidly back into the development process. Typically, classical IV&V is performed by one organization (e.g., supplier), and the development is performed by a separate organization (i.e., separate supplier). Classical IV&V is generally required for integrity level 4 when the system performance has the potential for loss of life, loss of mission, significant social loss, or financial loss or when required by regulations and standards imposed on the system development.

### **C.2.3 Modified IV&V**

Modified IV&V is used in many large programs where the system prime integrator is selected to manage the entire system development including the IV&V. The prime integrator selects organizations to assist in the development of the system and to perform the IV&V. In the modified IV&V form, the acquirer reduces its own acquisition time by passing this responsibility to the prime integrator. Technical independence is preserved because the IV&V effort formulates an unbiased opinion of the system solution and uses an independent staff to perform the IV&V. Financial independence is preserved because a separate budget is set aside for the IV&V effort. Because the prime integrator performs all or some of the development, the managerial independence is compromised by having the IV&V effort report to the prime integrator who also may be performing some aspects of the system development. Managerial independence is compromised because the IV&V engineer may encounter coercion, undue pressure, retaliation, future promotion opportunities, etc., if IV&V reports system errors that negatively impact the development team schedule and costs. Modified IV&V effort would be appropriate for systems with integrity level 3 if there are adequate protection procedures and management structures in place to allow the IV&V team to develop an unbiased assessment of the system development and to permit unimpeded reporting of any errors or findings.

### **C.2.4 Integrated IV&V**

This form is focused on providing rapid feedback of V&V results into the development process and is performed by an organization that is financially and managerially independent of the development organization to minimize compromises with respect to independence. The rapid feedback of V&V results into the development process is facilitated by the integrated IV&V effort: working side by side with the development organization, reviewing interim work products, and providing V&V feedback during inspections, walkthroughs, and reviews conducted by the development staff (potential impact on technical independence). Impacts on technical independence are counterbalanced by benefits associated with a focus on interdependence between the integrated IV&V effort and the development organization. Interdependence means that the successes of the organizations are closely coupled, ensuring that they work together in a cooperative fashion.

### **C.2.5 Internal IV&V**

Internal IV&V exists when the developer conducts the IV&V with personnel from within its own organization, although preferably not the same personnel involved directly in the development effort. Technical, managerial, and financial independence are compromised. Technical independence is compromised because the IV&V analysis and test are vulnerable to overlooking errors by using the same assumptions or development environment that masked the error from the developers. Managerial independence is compromised because the internal IV&V effort uses the same common tools and corporate analysis procedures as the development group. Peer pressure from the development group may adversely influence how aggressively the system is analyzed and tested by the IV&V effort. Financial independence is compromised because the development group controls the IV&V budget. IV&V funds, resources, and schedules may be reduced as development pressures and needs redirect the IV&V funds into solving

development problems. The benefit of an internal IV&V effort is access to staff who know the system and its software. This form of IV&V is used when the degree of independence is not explicitly stated, and the benefits of preexisting staff knowledge outweigh the benefits of objectivity.

### C.2.6 Embedded V&V

This form is similar to internal IV&V in that it uses personnel from the development organization who should not be involved directly in the development effort. Embedded V&V is focused on ensuring conformance to the development procedures and processes. The embedded V&V effort works side by side with the development organization and attends the same inspections, walkthroughs, and reviews as the development staff (i.e., compromise of technical independence). Embedded V&V is not tasked specifically to assess independently the original solution or conduct independent tests (i.e., compromise of managerial independence). Financial independence is compromised because the V&V staff resource assignments are controlled by the development group. Embedded V&V allows rapid feedback of V&V results into the development process, but compromises the technical, managerial, and financial independence of the V&V effort.

### C.2.7 Summary table of different forms of V&V Independence

[Table C.1](#) illustrates the degree of independence achieved by the five prevalent forms of V&V independence.

- “I” requires that the independence parameter (i.e., technical, management, or financial) must be rigorously and fully satisfied.
- “i” indicates that the independence parameter is not fully satisfied, but has some conditional compromised compliance. The conditional compliance for any of the three independence criteria can range from rigorous compliance to minimal compliance.
- “e” indicates that the independence parameter is not fully satisfied and has an alternative minimal approach of compliance to the independence parameter requirements.

**Table C.1—Forms of IV&V**

| IV&V form                                                                               | Technical | Management | Financial |
|-----------------------------------------------------------------------------------------|-----------|------------|-----------|
| Classical                                                                               | I         | I          | I         |
| Modified                                                                                | I         | i          | I         |
| Integrated                                                                              | i         | I          | I         |
| Internal                                                                                | i         | i          | i         |
| Embedded                                                                                | e         | e          | e         |
| NOTE—I = rigorous independence; i = conditional independence; e = minimal independence. |           |            |           |

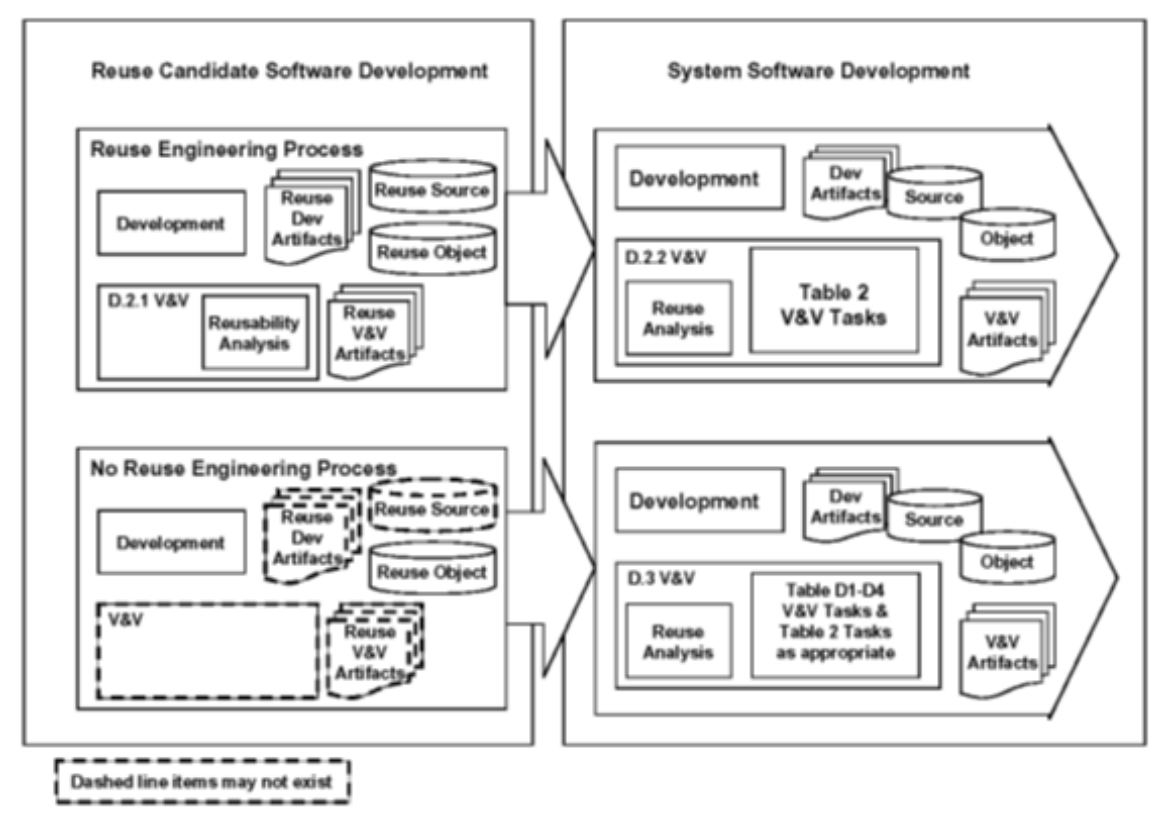
## Annex D

(informative)

### V&V of reuse software and hardware, including COTS

#### D.1 Purpose

The purpose of this annex is to provide options and suggestions to aid V&V of reuse software. Reuse software can take many forms and could include software from software libraries, custom software developed for other applications, COTS software, system/software requirements, software designs, or other artifacts from existing software. This annex addresses both reuse software developed and used as part of a reuse process, and reuse software developed and used outside of a reuse process. [Figure D.1](#) illustrates V&V activities and tasks for reuse software; whether it was developed under a reuse process, or outside of a reuse process.



**Figure D.1—V&V of reuse software**

[Figure D.1](#) identifies the following based upon whether the reuse asset was developed within a reuse engineering process or external to a reuse engineering process:

- Expected/required artifacts for each scenario
- V&V Tasks and Tables within the standard that should be referenced

Regardless of whether a critical application asset is developed within a reuse engineering process or external to a reuse engineering process, it is recommended that environmental updates (e.g., automatic or manual) of COTS assets be evaluated for potentially adverse impacts prior to allowing the updates to occur.

## D.2 V&V of software developed in a reuse process

### D.2.1 Introduction

A structured software reuse process develops assets (e.g., design, code, and documentation) intended for use in multiple contexts. The software reuse processes of IEEE Std 1517-2010 [B24] provide a framework for extending the software life cycle processes of ISO/IEC/IEEE 12207:2017(E) [B40] to include a systematic domain engineering process for software reuse. The domain scope and the domain analysis of an asset provide the requirements, the intended use, the interface parameters, and other information necessary for the V&V of the asset, or an understanding of previously performed V&V of the asset.

### D.2.2 V&V of assets in development

The V&V effort should analyze the artifacts (e.g., plans, models, and architecture) of the domain engineering as part of the required V&V tasks. Significant analysis of the domain engineering products should occur during system requirements review, system/software requirements evaluation, interface analysis, software design evaluation, source code and source code documentation evaluation, and all test planning. The V&V effort includes the assignment of an integrity level to the asset, in the context of the domain for its intended use, to determine the minimum V&V tasks to be performed (see Table 2). When planning the V&V effort, the optional task reusability analysis (see [Annex G](#)) should be included.

### D.2.3 V&V of reused assets

A domain engineering process assures that the information used in developing software systems is identified, captured, and organized so that it can be reused to create new systems within a domain. The V&V effort includes the assignment of an integrity level to the asset, in the context of its actual use, to determine the minimum V&V tasks to be performed (see Table 2). When planning the V&V effort, the optional task reuse analysis (see [Annex G](#)) should be included.

## D.3 V&V of software developed and reused outside of a reuse process

Some software systems are developed, operated, and maintained using software items that were not designed for use in multiple contexts or were not developed as part of a structured software reuse process (e.g., domain engineering products are not available). In these cases, the V&V effort should perform the optional task reuse analysis (see [Annex G](#)) to produce inputs for determining the suitability of the reuse candidate software. The V&V effort assigns an integrity level to the reuse candidate software to determine the minimum V&V tasks to be performed.

Reused software requires special consideration during the V&V effort when any one of the following is applicable:

- The inputs for a required V&V task are not available for the reused software.
- The reused software was developed as part of a system that is different in function or application from the system where it will be reused.
- The reused software was developed to meet different user needs from the current system.
- The original user needs are unknown.

In some cases, inputs for the V&V tasks may not be available, reducing visibility into the software products and processes. Options and techniques are available to compensate for the lack of inputs. Each technique has varying strengths and weaknesses—consideration should be given to performing multiple techniques to counter the weaknesses of one technique with strengths of another technique when high confidence is demanded. These options are addressed in decreasing order of desirability.

First, substitute tasks. Substitution for [Table 1a](#) through Table 1d V&V tasks is permitted if equivalent substitute V&V tasks can be shown to satisfy the same criteria as in Table 1. Two substitution task techniques are suggested in [Table D.1](#).

**Table D.1—Substitution tasks to establish V&V task inputs**

| Substitution tasks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description:</b> Substitute alternative analysis and test methods in lieu of the IEEE 1012 requirement V&V tasks to generate objective conclusions about the correctness, completeness, accuracy, and usability of the reused software.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Technique 1: Closed box testing</b><br><br><i>Closed box testing and Validation:</i> Execute the reused software against a spectrum of test case inputs and validate the correctness of the output.<br><br><i>User's manual analysis:</i> Derive system and system/software requirements from the user's manual and validate that the closed box testing results satisfy the requirements.<br><br><i>Limit checks in interfacing software:</i> Add limit checks within all interface software on all data and logical information received from the reused software to assure that no erroneous information is accepted.                                                                                                                                                          | <div>Pros</div> <ul style="list-style-type: none"> <li>— Test results reflect actual target software</li> <li>— Limits catastrophic errors from propagating into interfacing systems</li> <li>— Ability to check the major system and user requirements derived from the user's manual</li> <li>— Independent analysis</li> </ul> <div>Cons</div> <ul style="list-style-type: none"> <li>— Inability to detect all test errors if the presence of error is not observable in closed box outputs (e.g., latent errors)</li> <li>— Limit checks difficult to cover all execution scenarios</li> <li>— Limited by the thoroughness of the user's manual</li> </ul>                                                     |
| <b>Technique 2: Review developer's quality assurance (QA)</b><br><br><i>Developer's QA results:</i> Review the developer's QA results and confirm the evidence of data similar to those that would be generated from V&V tasks.<br><br><i>Developer's test results:</i> Review the developer's test results and confirm the evidence of data similar to those that would be generated from V&V tasks.<br><br><i>Review of developer's notebook:</i> Review the developer's notebook to derive additional insights and problems with the software during the early stages of development.<br><br><i>User's manual analysis:</i> Derive system and system/software requirements from the user's manual and validate that the developer's QA and test results satisfy the requirements. | <div>Pros</div> <ul style="list-style-type: none"> <li>— Ability to derive insight into the details of the software design and internal performance characteristics</li> <li>— Identification of possible program error characteristics warranting further analysis and testing by other methods</li> <li>— Observations of program performance using test results reflecting actual software execution characteristics</li> </ul> <div>Cons</div> <ul style="list-style-type: none"> <li>— Limited by the scope and extent of the QA analysis and the specific focus of the testing performed</li> <li>— Limited by the thoroughness of the user's manual</li> <li>— Not a totally independent analysis</li> </ul> |

Second, use alternative sources of information to perform the V&V tasks in [Table 1a](#) through [Table 1d](#) and [Table 2a](#) through [Table 2d](#) of this standard. Three alternative source techniques are suggested in [Table D.2](#).

**Table D.2—Alternative sources to establish V&V task inputs**

| Alternative sources                                                                                                                                                    |      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Description:</b> Use alternative sources of program data to derive conclusions about the correctness, completeness, accuracy, and usability of the reused software. |      |
| <b>Technique 3: Operational history</b>                                                                                                                                | Pros |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><i>Historical data analysis:</i> Examine and analyze the operational history of the reused software with particular attention to how the software performed in a system with similar characteristics to the new system being proposed.</p> <p><i>User interviews:</i> Conduct interviews with operational users. Focus data gathering on how the system performed in scenarios and conditions similar to those expected in the new system being proposed.</p>                                                                                                                                                                | <ul style="list-style-type: none"> <li>— Real data of the reused software in an operational environment.</li> <li>— User observations about the performance of the software and its related system.</li> <li>— Software burn-in established and track record of discrepancies recorded.</li> <li>— Independent analysis.</li> </ul> <p>Cons</p> <ul style="list-style-type: none"> <li>— Different characteristics, technologies, and user interfaces with the new proposed system could cause errors not observed in historical systems.</li> <li>— Not all interactions are recordable or observable in historical systems, so data completeness and accuracy are limited.</li> <li>— User observations can be subjective, biased, and error-prone, so correctness, completeness, and accuracy may be limited.</li> </ul> |
| <p><b>Technique 4: Audit results</b></p> <p><i>Developer's interview:</i> Conduct interviews with the development team to extract pertinent information about the design and performance characteristics of the reused software.</p> <p><i>Design walkthrough review analysis:</i> Analyze the design and code walkthrough data to determine how the reused software would interact in the new proposed system.</p> <p><i>Standard compliance analysis:</i> Review the results of any standards compliance audits to determine that the proper software standards were followed in the construction of the reused software.</p> | <p>Pros</p> <ul style="list-style-type: none"> <li>— Depending on the thoroughness of the development team's records, good insight into the design and code approaches can be obtained from the interviews.</li> <li>— Historical artifacts of design and code walkthroughs may be of sufficient quality to act as a substitute for the actual source code and design details.</li> </ul> <p>Cons</p> <ul style="list-style-type: none"> <li>— Limited by the thoroughness of the development team's documentation and recollection during interviews.</li> <li>— Not a totally independent analysis.</li> </ul>                                                                                                                                                                                                            |
| <p><b>Technique 5: Artifacts</b></p> <p><i>Product documentation analysis:</i> Review any product documentation to derive artifacts similar to requirements, design, and code (if possible—for example, if a pseudo-design language is used).</p> <p><i>Prior V&amp;V results analysis:</i> Analyze any prior V&amp;V results and develop inferences and extrapolation of the data to the new proposed system.</p>                                                                                                                                                                                                              | <p>Pros</p> <ul style="list-style-type: none"> <li>— Use actual artifacts representing some form of the reused software.</li> <li>— Prior V&amp;V results for an initial basis for formulating additional analysis and testing to conduct to fill in the analysis and testing voids caused by lack of program documentation.</li> <li>— Independent analysis.</li> </ul> <p>Cons</p> <ul style="list-style-type: none"> <li>— Overstating a conclusion without having a solid basis for knowing the system conditions could lead to faulty conclusions about the suitability in the new proposed system and its different system conditions.</li> <li>— Limited by the thoroughness of the product documentation.</li> </ul>                                                                                                |

Third, use reverse engineering to generate inputs to perform the V&V tasks in [Table 1a](#) through [Table 1d](#) and [Table 2a](#) through [Table 2d](#) of this standard. One reverse engineering technique is suggested in [Table D.3](#).

**Table D.3—Reverse engineering to establish V&V task inputs**

| Reverse engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description: Reverse engineer requirements, design, and code data to generate objective conclusions about the correctness, completeness, accuracy, and usability of the reused software.                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Technique 6: Closed-box testing</b><br><br><i>Reverse engineer source code:</i> Reverse compile “pseudo source” code from the program object file. Analyze the reverse compiled pseudo code using normal V&V procedures, including all the V&V test strategies and methods.<br><br><i>Reverse engineer requirements:</i> Derive the system and system/software requirements from the user’s manual. Analyze the requirements using the IEEE 1012 V&V tasks and test the reused software against these reverse-engineered requirements. | Pros <ul style="list-style-type: none"> <li>— Uses the actual code-no hidden or implied data.</li> <li>— Perform all of the IEEE 1012 V&amp;V tasks.</li> <li>— Test data reflects the actual performance of the reused software under the system conditions of the new proposed system.</li> <li>— Independent analysis.</li> </ul> Cons <ul style="list-style-type: none"> <li>— Time-consuming to reverse engineer data.</li> <li>— Pseudocode is hard to read.</li> </ul> |

Fourth, use independent prototyping and comparison of performance results with those of the reuse asset to perform the V&V tasks in [Table 1a](#) through [Table 1d](#) and [Table 2a](#) through [Table 2d](#) of this standard. Two independent prototyping and comparison techniques are suggested in [Table D.4](#).

**Table D.4—Independent prototyping and comparison to establish V&V task inputs**

| Independent prototyping and comparison                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description: Develop a model (prototype) of the proposed software or use portions of the prior system. Execute test scenarios on the prototype or prior system and compare the test results against the reused software. Analyze the results to generate objective conclusions about the correctness, completeness, accuracy, and usability of the reused software.                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Technique 7: Prototyping</b><br><br><i>Comparison of prototype code:</i> Develop a replication model (prototype) of the function or requirements in a user-friendly language. Execute test cases representing the range of system scenarios on both the model and reuse software. Compare the results and analyze the differences to determine whether the reused software is performing as intended. | Pros <ul style="list-style-type: none"> <li>— Useful for small functions and sets of requirements.</li> <li>— Easy to diagnose problems in the reused software.</li> <li>— Ability to run a wide range of system scenarios and compare against a benchmark program.</li> <li>— Independent analysis.</li> </ul> Cons <ul style="list-style-type: none"> <li>— Cost and time of building the model.</li> <li>— Errors in the model can mask similar errors in the reused software (the likelihood of two similar errors generated by two independent sources should be small or unlikely).</li> </ul> |
| <b>Technique 8: Prior system results</b><br><br><i>Comparison with prior system/function:</i> Execute test cases representing the range of system scenarios on the prior system/function and the reused software. Compare the results and analyze the differences to determine whether the reused software is performing as intended.                                                                    | Pros <ul style="list-style-type: none"> <li>— Inexpensive to execute test cases on prior systems and compare.</li> <li>— Proven track record of performance of the prior system/function establishes a performance baseline.</li> <li>— Differences in execution results leads to other analyses and testing to be conducted.</li> <li>— Independent analysis.</li> </ul>                                                                                                                                                                                                                            |



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>Cons</p> <ul style="list-style-type: none"> <li>— Limited in scope to smaller functions.</li> <li>— The ability to instrument prior system/function to extract diagnosis data about interim program steps may be difficult and make it harder to diagnose the exact location of a problem.</li> <li>— Any problems hidden in the prior system/function may go undetected or untested in the new proposed system/function (inheritance of errors).</li> </ul> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Last, a combination of the following circumstantial evidence provides visibility and insight into the reused software to perform the V&V tasks in [Table 1a](#) through [Table 1d](#) and Table 2 of this standard:

- a) Operational history
- b) Test history
- c) Audit results
- d) User interviews
- e) Engineering judgment
- f) Product documentation
- g) Prior hazard analysis results
- h) Prior V&V results
- i) Software developer's notebook
- j) Design process documentation
- k) Original developers' interviews
- l) Static code analysis results
- m) Standards compliance assessments

If V&V of reused software cannot be accomplished at the appropriate level, then the items may be used, as long as the risk associated with this use is recognized and accounted for in the risk mitigation strategy. The V&V effort should assure that the risks are thoroughly understood, properly documented, and properly tracked under the risk analysis tasks.

## Annex E

(informative)

### V&V measures

#### E.1 Introduction

The management of V&V activity uses measures to provide feedback for the continuous improvement of the V&V processes and to evaluate the system development processes and products. Trends can be identified and addressed by computing evaluation measures over a period of time. Threshold values of measures should be established, and trends should be evaluated to serve as indicators as to whether a process, product, or V&V task has been satisfactorily accomplished. No standard set of measures is applicable for all projects, so the use of measures may vary according to the application domain and software development environment.

No consensus exists on measures for evaluating the quality and coverage of the V&V tasks. IEEE Std 1061<sup>TM</sup>-1998 [B22] provides a standard definition of the available software quality measures. Another measure-related standard is IEEE Std 982.1-2005 [B18].

This standard considers three categories of measures associated with the V&V effort: 1) measures for evaluating anomaly density, 2) measures for assessing V&V effectiveness, and 3) measures for evaluating V&V efficiency.

#### E.2 Measures for evaluating anomaly density

Anomaly density measures can provide insightful information on the product quality, the quality of the system development processes, and the quality of the V&V effort to discover anomalies in the system/software/hardware and to facilitate the correction of the anomalies. Anomaly density measures are influenced by numerous variables (e.g., software complexity, hardware complexity, type of domain, and time-phase application of the V&V processes); consequently, the measures are analyzed to gain insight into the interdependencies between the development efforts and the V&V efforts.

If the V&V anomaly density measure value is low, then this suggests that the program development quality is high, that the V&V processes need to be improved, or a combination of both. If the measured value is high, then this suggests that the program development quality is low, that the V&V processes are effective, or a combination of both. Regardless of the measured value, the next step is to evaluate related program development measures to clarify and discern the measure trends to determine the need for process improvements.

Anomaly measures and trends can be used to improve the quality of the current project and can be used to improve the planning and execution of V&V processes for future projects with similar characteristics. The measures defined by Equation (E.1) through Equation (E.4) are applicable for four system life cycle phases:

$$\text{Requirements anomaly density} = \frac{\# \text{ Requirements anomalies found by V\&V effort}}{\# \text{ Requirements reviewed by V\&V effort}} \quad (\text{E.1})$$

$$\text{Design anomaly density} = \frac{\# \text{ Design statement anomalies found by V\&V effort}}{\# \text{ Design statements reviewed by V\&V effort}} \quad (\text{E.2})$$

$$\text{Implementation anomaly density} = \frac{\# \text{ Implementation anomalies found by V\&V effort}}{\# \text{ Implementation volume reviewed by V\&V effort}} \quad (\text{E.3})$$

$$\text{Test anomaly density} = \frac{\# \text{ Test anomalies found by V\&V effort}}{\# \text{ Tests reviewed by V\&V effort}} \quad (\text{E.4})$$

### E.3 Measures for evaluating V&V effectiveness

Measures associated with V&V effort effectiveness provide quantitative indications that characterize the added benefits of V&V to discover anomalies in system products and processes. These measures delineate the percentage of the total anomalies found by the V&V effort. The measures are influenced by numerous variables (e.g., complexity), and the measures are analyzed to gain insight into the interdependencies between the development efforts and the V&V efforts.

The V&V effectiveness measure values are highly influenced by the degree of parallelism between the software development effort and the V&V effort. Assuming that the efforts are parallel, a low V&V effectiveness measure value suggests that the development effort is effective, that the V&V effort may require improvement, or a combination of both. If the V&V effectiveness measure value is high, then this suggests that the development processes may require improvement, that the V&V processes are effective, or that only incremental changes to the V&V processes may be required. Regardless of the measured value, the next step is to evaluate related system development measures to further clarify and discern the measure trends to determine the need for process improvements. The measures defined by Equation (E.5) through Equation (E.8) are applicable for four system life cycle phases.

$$\text{Requirements V\&V effectiveness} = \frac{\# \text{ Requirements anomalies found by V\&V effort}}{\# \text{ Requirements anomalies found by all sources}} \quad (\text{E.5})$$

$$\text{Design V\&V effectiveness} = \frac{\# \text{ Design statement anomalies found by V\&V effort}}{\# \text{ Design statements anomalies found by all sources}} \quad (\text{E.6})$$

$$\text{Implementation V\&V effectiveness} = \frac{\# \text{ Implementation anomalies found by V\&V effort}}{\# \text{ Implementation anomalies found by all sources}} \quad (\text{E.7})$$

$$\text{Test V\&V effectiveness} = \frac{\# \text{ Test anomalies found by V\&V effort}}{\# \text{ Test anomalies found by all sources}} \quad (\text{E.8})$$

### E.4 Measures for evaluating V&V efficiency

The measures associated with V&V effort efficiency provide data that characterize the capability of the V&V effort to discover anomalies in software products and processes in the development activity in which they are injected. Maximum benefits are realized when anomalies are discovered as early as possible in the development life cycle, thereby minimizing rework and development costs. Analysis of these measures, the anomalies, and the causal factors that prevented the discovery of the anomaly in the phase in which it was injected can reveal needed improvements in methods, processes, tools, and skills to improve the overall V&V effort.

A low V&V efficiency measure value suggests that the V&V effort is not discovering anomalies in the earliest possible activity, that the development products are immature, or a combination of both. If the V&V efficiency measure value is high, then this suggests that the V&V effort is discovering anomalies in the earliest possible activity, that the development products are mature, or a combination of both. Regardless of the measured value, the next step is to evaluate related system program development measures to further clarify and discern the measure trends to determine the need for process improvements. The measures defined by Equation (E.9) through Equation (E.12) are applicable for four system life cycle phases.

$$\text{Requirements V\&V effectiveness} = \frac{\# \text{ Requirements anomalies found by V\&V in requirements activities}}{\# \text{ Requirements anomalies found by V\&V in all activities}} \times 100\% \quad (\text{E.9})$$

$$\text{Design V\&V effectiveness} = \frac{\# \text{ Design statement anomalies found by V\&V in design activities}}{\# \text{ Design statements anomalies found by V\&V in all activities}} \times 100\% \quad (\text{E.10})$$

$$\text{Implementation V\&V effectiveness} = \frac{\# \text{ Implementation anomalies found by V\&V in implementation activities}}{\# \text{ Implementation anomalies found by V\&V in all activities}} \times 100\% \quad (\text{E.11})$$

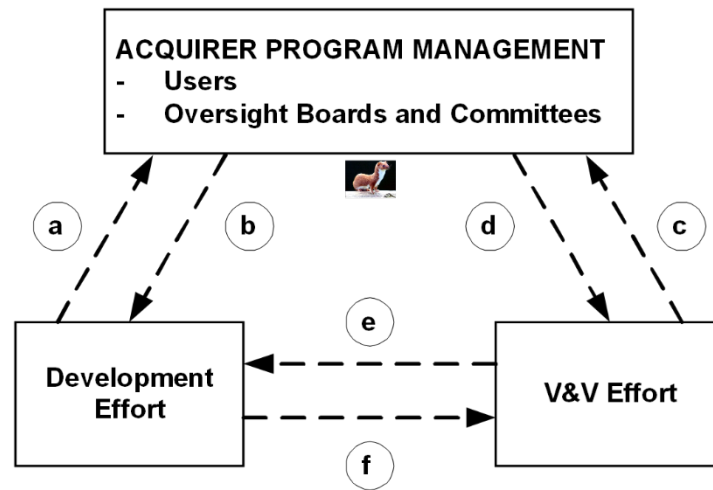
$$\text{Test V\&V effectiveness} = \frac{\# \text{ Test anomalies found by V\&V in test activities}}{\# \text{ Test anomalies found by V\&V in all activities}} \times 100\% \quad (\text{E.12})$$

## Annex F

(informative)

### Example of V&V relationships to other project responsibilities

[Figure F.1](#) provides an example of relationships among the V&V team, the acquirer, and the supplier, and it identifies the information and data flows throughout the V&V effort. Many other relationships will work well as long as the project responsibilities, data flows, and reporting flows are defined and documented.



NOTE—The lines in [Figure F.1](#) represent the flow of control and data as follows:

- Submittal of program documentation (e.g., concept, requirements, design, and user manuals), source code, program status, program budgets, and developmental plans and schedules.
- Approval, denial, and recommendations on development issues and deliverables transmitted in flow a.
- Submittal of VVP, V&V task results, anomaly reports, activity reports, and other special reports.
- Approval, denial, and recommendations on V&V issues and deliverables transmitted in flow c.
- Submittal of V&V task results, anomaly reports, activity reports, and special reports as directed by the acquirer program management.
- Submittal of program documentation (e.g., concept, requirements, design, user manuals, special reports, source code, and program schedules).

**Figure F.1—Relationship of V&V to other project responsibilities**

## Annex G

(informative)

### Optional V&V tasks

*Algorithm analysis.* Verify the correct implementation of algorithms, equations, mathematical formulations, or expressions. Re-derive any significant algorithms and equations from basic principles and theories. Compare against established references or proven historical data. Validate the algorithms, equations, mathematical formulations, or expressions with respect to the system and system/software requirements. Assure that the algorithms and equations are appropriate for the problem solution. Validate the correctness of any constraints or limitations, such as rounding, truncation, expression simplifications, best-fit estimations, and nonlinear solutions imposed by the algorithms and equations.

*Audit performance.* Provide an independent assessment of whether a software process and its products conform to applicable regulations, standards, plans, procedures, specifications, and guidelines. Audits may be applied to any software process or product at any development stage. Audits may be initiated by the supplier, the acquirer, the developer, or another involved party, such as a regulatory agency. The initiator of the audit selects the audit team and determines the degree of independence required. The initiator of the audit and the audit team leader establish the purpose, scope, plan, and reporting requirements for the audit.

The auditors collect sufficient evidence to decide whether the software processes and products meet the evaluation criteria. They identify major deviations; assess risk to quality, schedule, and cost; and report their findings. Examples of processes that could be audited include configuration management practices, use of software tools, degree of integration of the various software engineering disciplines, particularly in developing an architecture, security issues, training, and project management.

*Audit support.* Provide technical expertise to the auditors on request. They may represent the acquirer at audit proceedings and may assist in the V&V of remedial activities identified by the audit.

*Control flow analysis.* Assess the correctness of the software by diagramming the logical control. Examine the flow of the logic to identify missing, incomplete, or inaccurate requirements. Validate whether the flow of control among the functions represents a correct solution to the problem.

*Cost analysis.* Evaluate the cost status of the development processes. Compare budgeted costs against actual costs. Correlate cost expenditures with technical status and schedule progress. Identify program risks if actual costs indicate behind schedule and over cost estimates.

*Database analysis.* Evaluation of database design as part of a design review process could include the following:

- *Physical limitations analysis.* Identify the physical limitations of the database, such as the maximum number of records, maximum record length, largest numeric value, smallest numeric value, and maximum array length in a data structure, and compare them to designed values.
- *Index versus storage analysis.* Analyze the use of multiple indexes compared to the volume of stored data to determine whether the proposed approach meets the requirements for data retrieval performance and size constraints.
- *Data structures analysis.* Some database management systems have specific data structures within a record, such as arrays, tables, and date formats. Review the use of these structures for potential impact on requirements for data storage and retrieval.
- *Backup and disaster recovery analysis.* Review the methods employed for backup against the requirements for data recovery and system disaster recovery, and identify deficiencies.

*Data flow analysis.* Evaluation of data flow diagrams as part of a design review process could include the following:

- *Symbology consistency check.* The various methods used to depict data flow diagrams employ very specific symbology to represent the actions performed. Verify that each symbol is used consistently.
- *Flow balancing.* Compare the output data from each process to the data inputs and the data derived within the process to assure the data are available when required. This process does not specifically examine timing or sequence considerations.
- *Confirmation of derived data.* Examine the data derived within a process for correctness and format. Data designed to be entered into a process by operator action should be confirmed to assure availability.
- *Keys to index comparison.* Compare the data keys used to retrieve data from data stores within a process to the database index design to confirm that no invalid keys have been used and the uniqueness properties are consistent.

*Disaster recovery plan assessment.* Verify that the disaster recovery plan is adequate to restore critical operation of the system in the case of an extended system outage. The disaster recovery plan should include the following:

- Identification of the disaster recovery team and a contact list.
- Recovery operation procedures.
- Procedure for establishing an alternative site, including voice and data communications, mail, and support equipment.
- Plans for replacement of computer equipment.
- Establishment of a system backup schedule.
- Procedures for storage and retrieval of software, data, documentation, and vital records off-site.
- Logistics of moving staff, data, documentation, etc.

*Distributed architecture assessment.* Assess the distribution of data and processes in the proposed architecture for feasibility, timing conflicts, availability of telecommunications, cost, backup and restore features, downtime, system degradation, and provisions for installation of software updates.

*Exploratory testing.* Perform exploratory testing (i.e., simultaneous learning, test design, and execution rather than scripted testing) on the system, software, or hardware.

*Feasibility study evaluation.* Verify that the feasibility study is correct, accurate, and complete. Validate that all logical and physical assumptions (e.g., physical models, business rules, and logical processes), constraints, and user requirements are satisfied.

*Independent risk assessment.* Conduct an independent risk assessment on any aspect of the software project and report on the findings. Such risk assessments will be primarily from a system perspective. Examples of risk assessment include the appropriateness of the selected development methodology or tools for the project and quality risks associated with proposed development schedule alternatives.

*Inspection.* Inspect the software products to detect defects in the product at each selected development stage to assure the quality of the emerging software. The inspection process may consist of multiple steps for the segregation of the inspection functions of the following:

- Inspection planning
- Product overview
- Inspection preparation

- Examination meeting
- Defect rework
- Resolution follow-up

An inspection is performed by a small team of peer developers and includes but is not led by the author. The inspection team usually consists of three to six persons, and in some cases, includes personnel from the test group, quality assurance, or V&V. The participants assume specific roles to find, classify, report, and analyze defects in the product. Each type of inspection is specifically defined by its intended purpose, required entry criteria, defect classification, checklists, exit criteria, designated participants, and its preparation and examination procedures. Inspections do not debate engineering judgments, suggest corrections, or educate project members; they detect anomalies and problems and verify their resolution by the author.

*Inspection (concept).* Validate that the business or mission needs are completely stated and that they are internally consistent.

*Inspection (design).* Verify that the design can be implemented, is traceable to the requirements, all interface and procedural logic is complete and correct, and omissions, defects, and ambiguities in the design are detected.

*Inspection (requirements).* Validate that the requirements meet customer needs and can be implemented. Verify that they are complete, traceable, testable, and consistent so that omissions, defects, and ambiguities in the requirements are detected.

*Inspection (source code).* Verify that the source code implementation is traceable to the design; all interfaces and procedural logic are complete and correct; and omissions, defects, and ambiguities in the source code are detected.

*Inspection—Test case (component, integration, system, acceptance).* Verify that the (component, integration, system, acceptance) test plan has been followed accurately, that the set of component test cases is complete, and that all component test cases are correct.

*Inspection—Test design (component, integration, system, acceptance).* Verify that the (component, integration, system, acceptance) test design is consistent with the test plan and that the test design is correct, complete, and readable.

*Inspection—Test plan (component, integration, system, acceptance).* Verify that the scope, strategy, resources, and schedule of the (component, integration, system, acceptance) testing process have been completely and accurately specified, that all items to be tested and all required tasks to be performed have been defined, and to assure that all personnel and resources necessary to perform the testing have been identified.

*Operational evaluation.* Assess the deployment readiness and operational readiness of the software. Operational evaluation may include examining the results of operational tests, audit reviews, and anomaly reports. This evaluation verifies that the software is as follows:

- At a suitable point of correctness for mass production of that software.
- Valid and correct for site-specific configurations.

*Performance monitoring.* Collect information on the performance of software under operational conditions. Determine whether system and software performance requirements are satisfied. Performance monitoring is a continual process and may include evaluation of the following items:

- Database transaction rates to determine the need to reorganize or re-index the database.
- Central processing unit (CPU) performance monitoring for load balancing.



- Direct access storage utilization.
- Network traffic to assure adequate bandwidth.
- Critical outputs of a system (e.g., scheduled frequency, expected range of values, scheduled system reports, and reports of events).

*Post-installation Validation.* Execute a reference benchmark or periodic test for critical software when reliability is crucial, or there is a possibility of software corruption. By automatically or manually comparing results with the established benchmark results, the system can be validated prior to each execution of the software. When pre-use benchmark testing is impractical, such as for real-time, process control, and emergency-use software, a periodic test, conducted at a predetermined interval, can be used to assure continued reliability.

*Project management oversight support.* Assess project development status for technical and management issues, risks, and problems. Coordinate oversight assessment with the acquirer and development organization. Evaluate project plans, schedules, development processes, and status. Collect, analyze, and report on key project measures.

*Proposal evaluation support.* Participate in the development organization source selection process. Develop proposal evaluation factors and assessment criteria. Independently evaluate development organization proposals to assess conformance to the statement of work and performance requirements.

*Regression analysis and testing.* Determine the extent of V&V analyses and tests to be repeated when changes are made to any previously examined software products. Assess the nature of the change to determine the potential ripple or side effects and the impacts on other aspects of the system. Rerun test cases based on changes, error corrections, and impact assessment to detect errors spawned by software modifications.

*Reusability analysis.* Verify that the artifacts (products) of the domain engineering process conform to project-defined purpose, format, and content (e.g., IEEE Std 1517-2010 [B24]). Verify that the domain models and domain architecture are correct, consistent, complete, and accurate and that they conform to the domain engineering plan. Analyze the asset (software item intended for reuse) to verify that the asset is consistent with the domain model and domain architecture.

*Reuse analysis.* Analyze the developer's documentation to verify that the original domain of the candidate reuse software will satisfy the domain of the new system (e.g., integrity level, user needs, operating environment, safety, security, and interfaces). If the developer has performed no domain analysis, perform domain analysis (IEEE Std 1517-2010 [B24]) to compare the original domain and the new domain of the candidate reuse software. Verify that the developer's reuse plan dispositions and documents all domain differences.

*Simulation analysis.* Use a simulation to exercise the software or portions of the software to measure the performance of the software against predefined conditions and events. The simulation can take the form of a manual walkthrough of the software against specific program values and inputs. The simulation can also be another software program that provides the inputs and simulation of the environment to the software under examination. Simulation analysis is used to examine critical performance and response time requirements or the software's response to abnormal events and conditions.

*Sizing and timing analysis.* Collect and analyze data about the software functions and resource utilization to determine whether system and system/software requirements for speed and capacity are satisfied. The types of software functions and resource utilization issues include, but are not limited to, the following:

- CPU load
- Random access memory and secondary storage (e.g. disk, tape) utilization
- Network speed and capacity

— Input and output speed

Sizing and timing analysis is started at software design and iterated through acceptance testing.

*System software assessment.* Assess system software (e.g., operating system, computer-aided software engineering tools, database management system, repository, telecommunications software, and graphical user interface) for feasibility, impact on performance and functional requirements, maturity, supportability, adherence to standards, developer's knowledge of and experience with the system software and hardware, and software interface requirements.

*Test certification.* Certify the test results by verifying that the tests were conducted using baseline requirements, a configuration control process, and repeatable tests, and by witnessing the tests. Certification may be accomplished at a software configuration item level or at a system level.

*Test evaluation.* Evaluate the developer's tests for requirements coverage and test completeness. Assess coverage by assessing the extent of the software exercised. Assess test completeness by determining whether the set of inputs used during testing are a fair representative sample from the set of all possible inputs to the software. Assess whether test inputs include boundary condition inputs, rarely encountered inputs and invalid inputs. For some software, it may be necessary to have a set of sequential or simultaneous inputs on one or several processors to test the software adequately.

*Test witnessing.* Monitor the fidelity of test execution to the specified test procedures and witness the recording of test results. When a test failure occurs, the testing process can be continued by: (1) implementing a "workaround" to the failure, (2) inserting a temporary code patch, or (3) halting the testing process and implementing a software repair. In all cases, assess the test continuation process for test process breakage (e.g., some software is not tested or a patch is left in place permanently), adverse impact on other tests, and loss of configuration control. Regression analysis and testing should be done for all the software affected by the test failure.

*Training documentation evaluation.* Evaluate the training materials and procedures for completeness, correctness, readability, and effectiveness.

*Usability analysis.* Verify that stakeholder needs and interests are considered during development, operation, and maintenance process activities. The analysis will assure that human-centered design activities are performed; human factors and ergonomics considerations are incorporated into the design, potential adverse effects on human health and safety are addressed in the design; and user needs are satisfied in a manner that supports user effectiveness and efficiency.

*User documentation evaluation.* Evaluate the user documentation for its completeness, correctness, and consistency with respect to requirements for the user interface and for any functionality that can be invoked by the user. The review of the user documentation for its readability and effectiveness should include representative end users who are unfamiliar with the software. Employ the user documentation in planning an acceptance test that is representative of the operational environment.

*User training.* Assure that the user training includes rules that are specific to the administrative, operational, and application aspects, as well as the industry standards for that system. This training should be based on the technical user documentation and procedures provided by the manufacturer of the system. The organization responsible for the use of the system should be responsible for providing appropriate user training.

*V&V tool plan generation.* Prepare a plan that describes the tools needed to support the V&V effort. The plan includes a description of each tool's performance, required inputs and associated tools, outputs generated, need date, and cost of tool purchase or development. The tool plan should also describe test facilities and integration and system test laboratories supporting the V&V effort. The scope and rigor of the V&V effort, as defined by the selected integrity level should be considered in defining the performance required of each tool.

*V&V tool qualification.* Verify the features and usability of a V&V tool to assure that the tool is functioning properly and does not mask errors that it was designed to find. Validate correct operation, limitations, and workarounds for known problems and deficiencies.

*Walkthrough.* Participate in the evaluation processes in which development personnel lead others through a structured examination of a product. Assure that the participants are qualified to examine the products and are not subject to undue influence. See specific descriptions of the requirement walkthrough, design walkthrough, source code walkthrough, and test walkthrough.

*Walkthrough (design).* Participate in a walkthrough of the design and updates of the design to assure completeness, correctness, technical integrity, and quality.

*Walkthrough (requirements).* Participate in a walkthrough of the requirements specification to assure that the system/software requirements are correct, unambiguous, complete, verifiable, consistent, modifiable, traceable, testable, and usable throughout the life cycle.

*Walkthrough (source code).* Participate in a walkthrough of the source code to assure that the code is complete, correct, maintainable, and free from logic errors; also assure the code conforms to coding standards and conventions and will operate efficiently.

*Walkthrough (test).* Participate in a walkthrough of the test documentation to assure that the planned testing is correct and complete, and that the test results will be correctly analyzed.

*Work Breakdown Structure (WBS) Evaluation.* Verify that the WBS represents all of the project scope and captures all deliverables, including internal, external, and interim deliverables. Verify that the WBS decomposes the project scope into a set of deliverables that comprehensively defines the work to be performed. If a WBS dictionary exists, then verify that the WBS dictionary contains a brief definition of the project scope, each WBS component is included, each WBS component has a list of associated activities, and each major milestone is included.

## Annex H

(informative)

### Environmental factors consideration

#### H.1 Introduction

The performance of system, software, and hardware V&V should be planned and executed to account for environmental factors that will influence the life cycle stages of the system—from concept development through disposal. These environmental factors may drive and constrain activities and tasks in each of the V&V life cycle processes; therefore, all environmental factors should be assessed to determine relevance and influence on the V&V activities and tasks to be performed.

#### H.2 In the agreement processes

Environmental factor requirements and constraints should be defined during the Acquisition Process and specified for the Supply Process. These processes define the activities necessary to establish an agreement between two organizations. When preparing for the acquisition, the strategy development should consider the system, software, and hardware attributes, requirements, and/or constraints to verify that appropriate environmental factor requirements are included in the request for proposal/quotation and the resulting contract. The V&V for this stage should perform the following:

- Assess the need definition to verify that all environmental factors affecting system design, development, production, operation, and disposal throughout the life cycle are clearly and comprehensively stated.
- Review the solution approach selection process (such as analysis of alternatives) to verify that environmental factors are adequately incorporated into the decision process and criteria.
- Review the acquisition documentation for adequacy and sufficiency of definition/specification for the system operating environment(s), system non-operating environment(s), development environment(s), and maintenance environment(s) to assure that the resultant program or contract delivers a system that satisfies the defined need. The request for proposal/quotation should be reviewed to verify that all pertinent environmental requirements are incorporated.
- Review the acceptance documentation to assess planned and specified testing environment(s) to assure that the testing will provide full spectrum, operationally representative tests to determine whether the system is properly designed and produced for operation per the need definition.
- Assess the supplier qualifications to verify that the selected supplier has the industrial capability to design, develop, produce, and support the program in accordance with acquisition documentation requirements. This assessment should include development, testing, production, and repair environments, processes, and tools.
- Assess the supplier's proposal to verify that system design and development, testing and manufacturing approaches, and processes adequately and sufficiently address all environmental requirements of the acquisition documentation's request for proposal/quotation.
- Assess the contract documentation to verify that environmental requirements reflect defined needs and the supplier's proposed solution with clear and binding terms.

#### H.3 In the organizational project-enabling processes

Environmental considerations should be assessed to verify that the processes for infrastructure management, human resource management, and quality management adequately address the ability of the acquirer and

supplier to provision, execute, monitor, assess, maintain, and improve processes and products that satisfy the system, software, and hardware needs over the full range of environmental requirements and constraints. The V&V for this stage should perform the following:

- Assess the supplier’s infrastructure to be employed in the development and production of the system, software and hardware to verify the environmental requirements and constraints will be maintained throughout the acquisition contract. This assessment should include development, testing, production and repair environments, processes, and tools.
- Assess the human resources assigned to the contract to verify that they have the necessary education, experience, training, and certifications for the activities and tasks to which they are assigned. Assess the supplier’s strategy and plans to verify that qualified human resources will be assigned to the project throughout the contract life.
- Assess the quality management strategy for the project to verify that environmental requirements and constraints for the infrastructure and the product (system, software, and hardware) will be monitored and maintained compliant throughout the contract life.

#### **H.4 In the project processes**

The project processes should address environmental requirements and constraints in the project plan development, execution, and monitoring. The V&V for this stage should perform the following:

- Review the project technical management plan to verify that environmental factors will be included in technical reviews throughout the project.
- Assess the quality management plan to verify that environmental requirements and constraints for the infrastructure and the product (system, software, and hardware) will be monitored and maintained compliant throughout the contract life.
- Review the project performance measures to verify that environmental factors will be collected throughout the contract life.
- Verify that project assessments will include environmental factors on a continuing basis.
- Verify that the Configuration Management Process will include adequate controls for environmental requirements and constraints affecting infrastructure, facilities, manufacturing processes, and tools.

#### **H.5 In the technical processes**

The V&V should assess environmental factors throughout the technical processes as well as perform the following:

- Assess that all environmental requirements and constraints included in the acquirer’s request for proposal and in the resulting contract with the supplier are completely specified in the project requirements documents and are appropriately allocated to the system elements, software, and hardware.
- Assess the architectural documents to verify that the environmental requirements and constraints from the project requirements documents have been incorporated into the architectural design.
- Monitor the Implementation Process, the Integration Process, and the development qualification testing to verify that the system elements, software, and hardware conform to and perform throughout the complete range of specified environmental conditions. Review the test environments, tools, test plans/procedures, and results to assure the design is adequately tested to verify that the system will meet performance in all operating and non-operating environment requirements of the contract. Review anomaly-clearing procedures and retest procedures for adequacy and sufficiency. Specific test environments to be monitored and assessed include the following:

- 1) Component
  - 2) Integration
  - 3) Qualification
  - 4) Acceptance
  - 5) Manufacturing (parts, subassembly, process sampling, etc.)
  - 6) Certification and accreditation testing (interoperability, architecture and standards, security, etc.)
  - 7) Operational
- Review the transition plans to verify the environmental requirements and constraints are adequately addressed.
  - Assess the effectiveness and suitability of design, procedures, training, and support by reviewing operational use data, operational failure data, and user feedback to determine whether environmental factors are adversely impacting system performance, safety, or security or whether requested changes could improve system performance, safety, or security.
  - Perform evaluations of new constraints to determine whether changing environments are impacting or have the potential to impact system performance, safety, or security in an adverse way.
  - Assess operating procedures to determine whether changes in operational procedures or operational use profiles are impacting or have the potential to impact system performance, safety, or security adversely.
  - Review metrics from the maintenance processes, tools, staffing, and controls to verify predictable and consistent results are achieved, adverse trends are identified, and corrective action/process improvements are incorporated when warranted. Review parts/component/COTS substitution and technical refresh processes to verify that environmental considerations are incorporated in the decision process and criteria. Review repair/upgrade kits and procedures intended for operational site location incorporation to verify environmental factors are adequately and sufficiently addressed.
  - Assess plans for disposal of the system elements to verify that adequate provisions are delineated for the retirement of the elements and subsequent storage, destruction, or disposal to preclude or contain hazardous and environmentally destructive materials and to prevent hazardous conditions.

## Annex I

(informative)

### V&V of system, software, and hardware integration

#### I.1 Introduction

The disciplines required to develop software elements in digital systems are distinctly different from those required to develop the hardware components of digital systems in many important aspects. Developing software for a digital system requires knowledge of software engineering principles, the software language(s) to be used, the limitations of the language(s), and the processes by which the system/software requirements can be translated into a set of functions that meet a user's requirements. Optimally, some members of the software development team also should have system domain expertise to understand the system requirements allocated to software. Developing the hardware on which system software is to operate requires knowledge of electrical engineering, computer engineering, hardware performance characteristics, printed circuit board layout, equipment qualification, and similar considerations. Managing these separate disciplines and integrating the disciplines into a unified development process requires its own set of project management disciplines.

Given the diverse disciplines involved in the development of digital systems, misconceptions about the relationships between the system, software, and hardware should not be unexpected. Resolving these differences in specializations prevents incompatibilities arising between the three categories comprising a digital system (i.e., system, software, and hardware). Consequently, in addition to verifying and validating system, software, and hardware requirements, V&V efforts should verify and validate system, software, and hardware integration.

This annex provides examples of system, software, and hardware integration issues that should be addressed in the phases of the system development life cycle. The integration considerations discussed in this annex are not the full set of integration considerations a system developer should explore. Instead, the integration considerations are intended to encourage system developers to view the system development effort holistically instead of thinking of the software and hardware development efforts as tasks to be performed in isolation.

The annex provides summaries of digital system faults that were introduced during system development as a result of failing to identify potential adverse integration effects between the systems, software and hardware. The annex then provides in [Table I.1](#) examples of integration considerations that should be addressed in the following life cycle processes:

- Acquisition
- Requirements (stakeholder and system, software or hardware)
- Architecture and design definition
- Implementation
- Integration

#### I.2 Examples of system failures caused by integration issues

##### I.2.1 Introduction

Despite the individual skills of the specialists developing a system, there have been events in which integration of the software and hardware into the system did not result in the desired system performance

reliability goals. The following events histories highlight the necessity for tightly coupling the system development effort with the software and hardware development efforts.

### **I.2.2 Year 2000 system integration issue**

Hardware memory limitation constraints in conjunction with the historical period in which early date/time-usage systems were developed (e.g., 1970) encouraged system developers to represent calendar year dates with two digits instead of four (e.g., 70 instead of 1970) when calculating the elapsed time between two events. A contributing premise (when even considered) was that the system being developed would likely be updated or replaced before the year 2000 (i.e., 00) could affect elapsed time/date calculations over the transition period between December 31, 1999 and January 1, 2000.

In validating the software during development, developers used ranges of dates that did not include dates beyond December 31, 1999. The dates used to validate the software were, at the time reasonable, since the system/software requirements were developed with the hardware limitations in mind.

System performance considerations of the effect of dates transitioning from “99” to “00” arose circa 1998 to 1999 as industries realized the degree to which digital systems had been integrated into critical digital assets. The nuclear industry, as well as other industries, began an intensive effort to identify date-related dependencies in critical systems. This industry-wide effort required significant expenditures of resources and funds to remediate critical systems, implement software revisions, and install new equipment.

The issue in this case is not attributed to software or hardware deficiencies; the software and hardware, when integrated, performed as required during the initially planned performance period. The issue in this case was in not identifying the full set of parameters that constrained the system operating space. The system's temporal limits were defined for a date/time period that did not include dates in the following century.

The lesson from this example is that the full extent of a system boundary needs to be verified when developing software and hardware requirements. In the case of the year 2000 example, the system boundary should have included both physical and temporal constraints. For those cases in which hardware limitations force system/software requirements limits, it is not enough that software integrated with hardware performs in accordance with software and hardware requirements; a Verification Process has to include the system boundaries that completely constrain the system performance parameters.

### **I.2.3 System architecture integration issues**

In the late 1990s, a consortium developed system requirements for a digital safety system function to be installed at several industrial sites. The system developer was selected for this development effort on the basis of its extensive experience developing high-integrity digital systems for the defense industry.

The consortium developed requirements for a primary–replica system architecture to enhance system self-testing capabilities and thereby improve digital safety system availability. This type of architecture, which is common in high-integrity systems in many industries, uses a duplicate (i.e., replica) system to check the operability of the primary system functions. The assumption was that identical outputs from the primary and replica system functions implied the system was operating as required. If the two output signals were different, then the system was not operating as required, and a system reset would be needed to alert the operators that either the primary channel was inoperable, or the replica channel was inoperable (or perhaps both were inoperable).

To verify that the primary-replica calculations in this system were identical when the system was operating, the primary system processor synchronized the replica processor to verify that the input signal values were the same in both microprocessor channels. This synchronization required the replica processor to prioritize safety function checking calculations over internal self-testing functions such that self-testing functions were placed on hold until the checking calculations were completed. The self-testing functions then resumed as the highest priority function in the replica processor.



Both the primary processor channel and the replica processor channel performed channel self-testing to verify the two channels were operable. These self-testing functions operated cyclically such that the system was continuously tested when it was not performing safety function processing. Both channels used watchdog timers to verify their respective processor did not have an error that would prevent the channel from performing every self-testing function. If the self-testing was not completed within a set period, then the watchdog timer would reset the offending channel and thereby alert the operators that a channel was inoperable.

After the system was installed in several facilities, sporadic watchdog timer-initiated resets occurred in the replica processor channel. In response to these channel resets, the channel was required to be declared inoperable until the system could be placed back in service. The system developer was tasked with determining the cause of the sporadic watchdog timer resets.

After extensive root cause analyses, the system developer determined the fault to be in the priority baton passing function built into the microprocessor. This microprocessor fault only arose when a high-priority task acquired the priority baton from a lower-priority task before the lower-priority task was completed. Upon completion of the high-priority task in the replica processor (i.e., the safety function processing in the replica processor that was initiated by the primary processor), the replica microprocessor sometimes would not release the priority baton back to the interrupted self-testing function. This caused the self-testing function to fail to meet the watchdog timer restriction, and thereby caused the replica channel watchdog timer to reset the channel microprocessor. Approximately 10 months were required for the system developer to identify and correct this fault.

While the system developer had extensive experience developing systems using the microprocessor, that experience did not translate into the primary-replica architecture. Interestingly, if the developer had reviewed publicly available failure history reports from the microprocessor vendor's website at the start of the project, the priority baton passing error would have been discovered during the hardware acquisition life cycle phase instead of in the operations life cycle phase.

This event illustrates the necessity for verifying that the hardware acquisition requirements and the system/software requirements (both internal to the hardware and imposed by software on the hardware) are compatible with the system architecture.

#### **1.2.4 System, software, and hardware interaction issues**

The following example interaction issues provide a starting point for identifying issues that should be addressed during the system development process:

- Software size relative to hardware capability
- Hardware compatibility with software type (e.g., C, hardware design language, etc.)
- System timing constraints on software design and hardware capabilities
- System architecture and software architecture relative to hardware capabilities
- Capability to validate software in an environment representative of the hardware architecture
- Hardware restrictions relative to system environment (e.g., equipment qualification issues, etc.)
- Software integrity requirements relative to hardware architecture
- System maintainability relative to hardware acquisition constraints
- Software architecture relative to system maintainability (e.g., use of object-oriented structure or an arcane structure that becomes outdated over time, thereby limiting the number of personnel qualified to perform corrective, adaptive, or perfective changes on the system)
- Hardware availability over the projected lifetime of the system

This list of potential interaction issues is not exhaustive. The purpose of the list is to instill in system developers the realization that digital system performance issues can arise from interactions among software, hardware, and the system.

[Table I.1](#) graphically summarizes the potential interactions among the system, software, and hardware for the example issues described for each system development life cycle phase in which the software and hardware are developed relatively individually. The cells in [Table I.1](#) marked with an “X” corresponding to the system, software, and hardware potential contributors to interactions corresponding to the life cycle phases in which the potential interactions should be addressed during the development effort.

The purpose of [Table I.1](#) is to provide a sample framework system developers can use to summarize potential interaction issues identified or anticipated during life cycle development activities. By identifying these interactions in a graphical format, the developer can refine the scope of V&V activities to address system, software, and hardware interaction issues specifically. Other interaction concerns may need to be added to the list of examples depending on specific system architecture requirements.

**Table I.1—Examples of integration V&V issues**

| Interaction<br>issue                                                                                            | System development life cycle phases |    |    |              |    |    |                        |    |    |                |    |    |             |    |    |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------------|----|----|--------------|----|----|------------------------|----|----|----------------|----|----|-------------|----|----|
|                                                                                                                 | Acquisition                          |    |    | Requirements |    |    | Arch and design<br>def |    |    | Implementation |    |    | Integration |    |    |
|                                                                                                                 | Interactions to be addressed         |    |    |              |    |    |                        |    |    |                |    |    |             |    |    |
|                                                                                                                 | Sys                                  | SW | HW | Sys          | SW | HW | Sys                    | SW | HW | Sys            | SW | HW | Sys         | SW | HW |
| Software size<br>relative to<br>hardware<br>capability                                                          | X                                    | X  | X  | X            | X  | X  |                        | X  | X  |                | X  | X  |             | X  | X  |
| Hardware<br>compatibility<br>with software<br>language and<br>structure                                         |                                      | X  | X  |              | X  | X  |                        | X  | X  |                | X  | X  |             | X  | X  |
| System timing<br>relative to<br>software<br>design and<br>hardware<br>capabilities                              | X                                    | X  | X  | X            | X  | X  | X                      | X  | X  | X              | X  | X  | X           | X  | X  |
| System and<br>software<br>architecture<br>relative to<br>hardware<br>capabilities                               | X                                    | X  | X  | X            | X  | X  | X                      | X  | X  | X              | X  | X  | X           | X  | X  |
| Capability to<br>validate<br>software in an<br>environment<br>representative<br>of the hardware<br>architecture |                                      | X  | X  | X            | X  | X  |                        | X  | X  |                | X  | X  |             | X  | X  |
| Hardware<br>restrictions<br>relative to the<br>system<br>environment                                            | X                                    |    | X  | X            |    | X  | X                      |    | X  | X              |    | X  | X           |    | X  |
| Software<br>integrity<br>relative to<br>hardware<br>architecture                                                |                                      | X  | X  | X            | X  | X  |                        | X  | X  |                | X  | X  |             | X  | X  |
| System<br>maintainability<br>relative to<br>hardware<br>acquisition                                             | X                                    |    | X  | X            |    | X  | X                      |    | X  | X              |    | X  | X           |    | X  |
| System<br>maintainability<br>relative to<br>software<br>architecture                                            | X                                    | X  |    | X            | X  |    | X                      | X  |    | X              | X  |    | X           | X  |    |
| Hardware<br>availability<br>relative to<br>projected<br>system<br>lifetime                                      | X                                    |    | X  | X            |    | X  |                        |    |    |                |    |    |             |    |    |

## Annex J

(informative)

### Hazard, security, and risk analyses

#### J.1 Introduction

Hazard, security, and risk analysis are systems engineering disciplines that enable a systematic understanding of technical, operational, and environmental factors that can cause harm. [Table J.1](#) provides an overview of each type of analysis and discusses the relationships among the three disciplines.

**Table J.1—Examples of integration V&V issues**

| Discipline        | Overview                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Relationships                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hazard analysis   | Identifies hazardous conditions that could lead to adverse consequences. The analysis identifies hazard contributors or combinations of contributors. The hazard analysis allows the development project to control or mitigate identified hazards.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>• Hazard analysis enables system responses to realized events.</li> <li>• Security analysis focuses on protecting the system in the life cycle.</li> <li>• Preventing a threat can occur as part of either hazard or security analysis.</li> <li>• Both hazard analysis and security analysis are based on system risk assessment, so they support the risk analysis process. An output of both includes an understanding of residual risks in the respective areas after the technical, operational, or management steps are taken during system development, operation, and maintenance processes.</li> <li>• Risk analysis addresses all areas of the system life cycle.</li> </ul> |
| Security analysis | <p>Develops processes and mechanisms to protect the assets (system of interest). Analysis activities include the identification of assets and threats, assessing vulnerabilities, and developing countermeasures to reduce the security risk. Security analysis also includes an assessment of processes to determine that the development, V&amp;V, and operational environments are safe.</p> <p>The protection of the system includes sub disciplines including Physical Security, Computer Security, and Information Assurance. A quickly emerging field within Security Analysis is Cyber Security, which is the process of applying security measures for confidentiality, integrity, and availability of assets. The rapidity of cyber threats necessitates system 1) development and V&amp;V techniques, 2) processes (system life cycle) management to respond accordingly.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Risk analysis     | Identifies and defines actions for risks that have a measurable possibility of negative consequences to either the successful development or operation of the system. The risk process is also viewed as a management tool to understand potential consequences to the development or operation of the system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

Details including V&V objectives for each of the three areas are detailed in subsequent sections.

## J.2 Hazard analysis

V&V often uses the results of hazard analyses to target V&V activities in a manner that provides greater assurance that the controls for hazards work as needed and that the system does not contribute to a hazardous condition by behaving in an undesirable way. Generally, one or more additional conditions need to exist, or additional events need to occur in conjunction with the existence of the hazard in order for an accident or mishap with consequences adverse to safety to result. These additional events enable the hazard to proceed to the adverse consequence. Hazards analysis involves the application of systematic and replicable methods to identify and understand hazards, as well as to characterize the risk of mishaps that involve hazards.

The hazard analysis is a systems engineering activity that will account for the system design, operational conditions, system physical constraints, and regulations to identify hazardous conditions that could lead to adverse consequences. Once the end hazardous conditions are identified, the hazard analysis typically uses a fault tree analysis approach to identify the contributors or combination of contributors (within the bounds of required system fault tolerance) to reach the hazardous condition. The fault tree is a symbolic logic diagram showing the cause–effect relationship between a top undesired event (failure) and one or more contributing causes. It is a type of logic tree that is developed by deductive logic from a top undesired event to all sub-events that must occur to cause it. The contributing causes may be the result of hardware or software faults, human actions (e.g., procedures), or hostile environmental conditions. The hazard analysis is repeated in each life cycle phase and accounts for further elaboration of designs, changes to intended system use and operations, and the emergence of new hazardous conditions. The hazard analysis may be performed by any organization within the project, such as systems engineering, reliability, safety, or V&V. In any case, V&V reviews the hazard analysis for completeness and usability, and it assures that stated hazards and contributors are clearly identified to sufficient detail to affect engineering and mitigation activities properly and to develop V&V plans and evaluation criteria.

Not all hazards represent the same risk. Systems engineers, operations experts, and reliability engineers may all contribute to evaluating the magnitude of risk that a hazard may present. The risk-based integrity level schema presented in [Annex B](#) is suitable as well for the assessment of risk associated with a hazard.

As a result of the hazard analysis, the project will define requirements for the control or mitigation of hazards (as well as re-design the system to reduce risk exposure). These requirements may be in the form of system capabilities, redundancy, and fault tolerance requirements, use of engineering standards, or operational procedures. The traceability of these requirements and associated design elements to the specific hazard and the hazard's risk magnitude will serve to determine the V&V integrity level for the requirement or system element and the level of V&V rigor needed.

Specific V&V activities that may be appropriate for “critical” requirements necessary to control, mitigate, or prevent hazards may include the following activities:

- a) Traceability of critical requirements through the life cycle to verify implementation
- b) Evaluation of potential hazard contributors to validate that critical requirements are complete and are appropriate for the system's operational need
- c) Evaluation of architectures and designs to determine whether hazard mitigation functions meet required capabilities and whether additional mitigation strategies are needed
- d) Application of Verification methods (analyses, inspections, demonstrations, or tests) that are intended to determine whether the contributing conditions to a hazardous state are possible. These Verification methods may include the following:
  - 1) Statistical analyses to determine whether the probability of reaching a contributing condition is within acceptable levels. This may include random testing, Monte Carlo analysis, stress testing, duration testing, or mathematical models.

- 2) Inspections to verify that hazard causes are not present and that hazard controls are implemented as specified. This may also include inspections to confirm regulatory or engineering standards have been followed.
- 3) Demonstrations in an operational setting to show that operational and human factor-based hazard controls are reasonable and effective.
- 4) Tests to verify that specific hazard controls (physical, procedural, and automated controls) are capable of identifying the faults or hazard contributors and isolating them from propagating into a hazardous condition.

The objective for the V&V of hazards is to perform a sufficient set of V&V tasks for all contributors such that the likelihood of reaching a hazardous condition is known to a desired level of confidence. The V&V integrity levels and their use in tailoring V&V activities as described in this standard provide for more rigor as the integrity level increases. Additionally, the use of assurance cases provides the evidence to satisfy claims that certain desired properties or behaviors exist or undesired properties or behaviors do not exist as necessary to prevent a specific hazardous condition.

## J.3 Security analysis

### J.3.1 Summary

One of the objectives of security analysis performed by the V&V effort is to verify that the system-required security controls and safeguards are correctly implemented and to validate that they provide the desired levels of protection against system vulnerabilities. The other objective is to verify that there is a process for describing the system, software, and hardware process security. [Figure J.1](#) (reference ISO/IEC 15408-1:2009 [B35]) shows the basic concept for the security context of the system. Except for general threat-risk assessment, information security analysis becomes more important to assure confidence in countermeasures to reduce the security risk for the final product. A system should consider different security issues in each phase of the life cycle because the system owner may change as the product evolves. The V&V security analysis should consider:

- The context of the system (e.g., the development process and environment, the final operational environment, organization structures and management policy, operational and maintenance personnel roles, interfaces with other external systems or support systems).
- The system-of-interest and its elements, threats, vulnerabilities, and countermeasures.
- Tradeoffs between techniques, operations, and management to address security requirements.
- Identification of threats. These threats may be natural (e.g., inclement weather, earthquakes), human (e.g., unintended or malicious), or environmental (e.g., chemical leak, power loss).

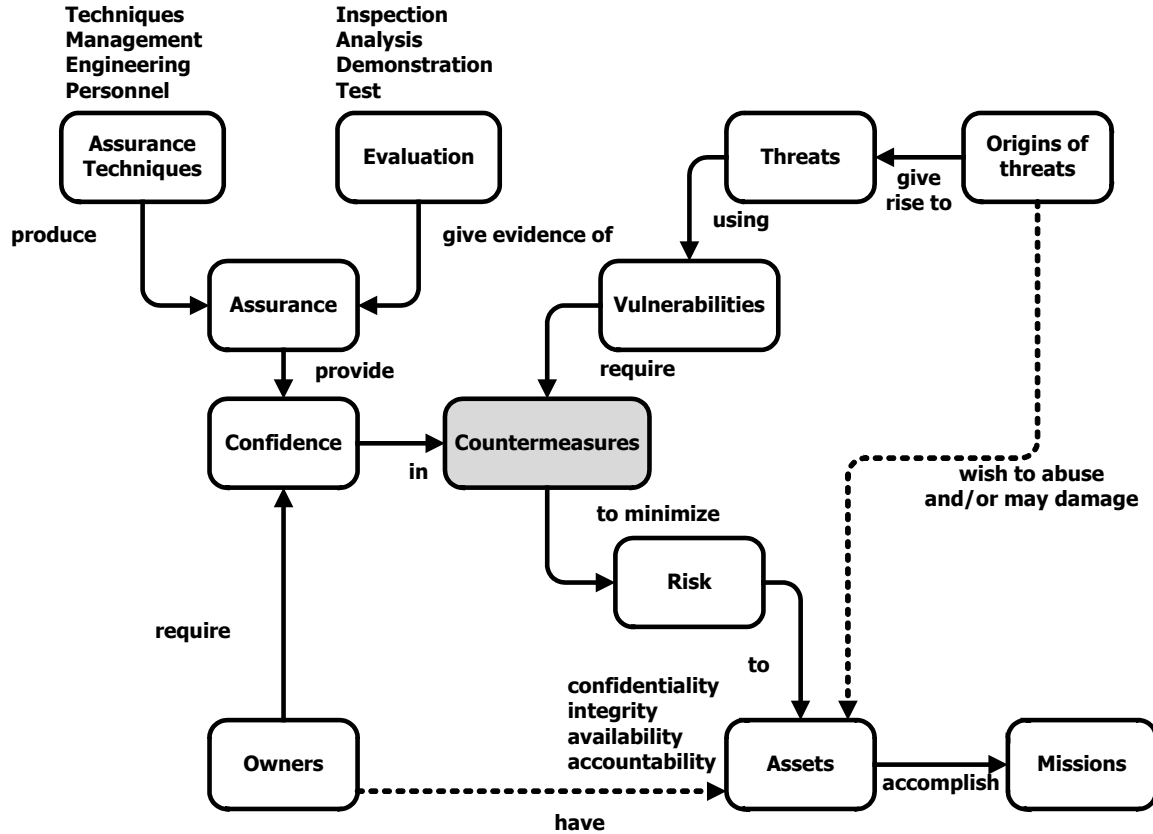


Figure J.1—The security context of the system

### J.3.2 Threat-based security analysis

The V&V security analysis task may rely on threat-based analyses. A threat has the potential to harm assets such as information, processes, and systems and, therefore, organizations. A vulnerability is a flaw or weakness in system security procedures, design, implementation, or internal controls that could be exercised (accidentally triggered or intentionally exploited) and result in a security breach or a violation of security policy. The desired levels of protection and security policies are unique to the stakeholder organization.

The V&V security analysis is performed in parallel with project system security risk assessments that generally include the following activities:

- Identification of assets. Assets are anything of value that will be managed by the system or associated with the system, such as sensitive data, critical function, or intellectual property.
- Identification of threats.
- An identification of system vulnerabilities. Vulnerabilities may include items such as system physical exposure, inability to recognize an intrusion, security perimeters that can be compromised or accessibility of assets.
- Evaluation of controls needed to prevent threats from exercising a potential vulnerability.
- Evaluation of the likelihood of a threat exercising a potential vulnerability according to current or planned controls.
- Evaluation of the impact of a security breach or security policy violation. Impacts may include items such as human safety, financial loss, social loss, environmental mission loss, or business continuity.

The system security risk assessment creates security controls to mitigate security risk to levels acceptable to the stakeholders. The security control requirements may be in the form of system functional capabilities, physical perimeter safeguards, redundancy, backup and recovery, or operational procedures. Much like the hazard analysis, the traceability of these requirements and associated design elements to the specific security impact severity will serve to determine the V&V integrity level for the requirement or system element and the level of V&V rigor needed.

Specific V&V activities that may be appropriate for “critical” security requirements necessary to control threats and exposure to vulnerabilities may include the following activities:

- a) Traceability of critical requirements through the life cycle to verify implementation.
- b) Evaluation of potential threat sources and vulnerabilities to validate that critical security requirements are complete and appropriate for the system’s operational need.
- c) Evaluation of architectures and designs to determine whether security functions meet required capabilities, whether additional security controls are needed, and whether design changes are needed to remove vulnerabilities.
- d) Application of verification methods (analyses, inspections, demonstrations, or tests) that are intended to determine whether plausible threats can exploit vulnerabilities. These verification methods may include the following:
  - 1) Statistical analyses to determine whether the probability of breaching a security control is within acceptable levels. This may include simulations or mathematical models (e.g., for encryption methods).
  - 2) Inspections to verify that security controls are implemented as specified. This may also include inspections that regulatory or policy standards have been followed.
  - 3) Demonstrations in an operational setting to show that security controls are reasonable and effective.
  - 4) Tests to verify that specific security controls (physical, procedural, and automated controls) cannot be breached. For information technology (IT) systems, this may also include vulnerability scanning and penetration testing.
- e) Review of the residual security risks (residual security risk can contain unidentified risk, or be known as “retained risk”) to validate whether the risks are acceptable and meet the organization’s risk acceptance criteria.

### J.3.2.1 Typical threats

Based on the V&V object, a system defense-in-depth protection strategy and its threat approach may be more systematic. The system elements may require different technical or management measures to support the system security design, so the risks that threats successfully use vulnerabilities through some approaches may be more specialized.

For any system or its element, [Table J.2](#) provides examples of types of typical threats, examples of specific threats within each type, and the origin of the types of threats. The list and following explanation, referenced from [Annex C](#), “Examples of typical threats” in ISO/IEC 27005:2011 [\[B39\]](#) could be helpful during V&V security analysis activities.

**Table J.2—Types of typical threats**

| Type            | Threats                           | Origin        |
|-----------------|-----------------------------------|---------------|
| Physical damage | Fire                              | Accidental    |
|                 | Water damage                      | Deliberate    |
|                 | Pollution                         | Environmental |
|                 | Major accident                    |               |
|                 | Destruction of equipment or media |               |
|                 | Climatic phenomenon               | Environmental |



| Type                         | Threats                                            | Origin                                    |
|------------------------------|----------------------------------------------------|-------------------------------------------|
| Natural events               | Seismic phenomenon                                 | Environmental                             |
|                              | Volcanic phenomenon                                | Environmental                             |
|                              | Meteorological phenomenon                          | Environmental                             |
|                              | Flood                                              | Environmental                             |
| Loss of essential services   | Failure of air-conditioning or water supply system | Accidental<br>Deliberate                  |
|                              | Loss of power supply                               | Accidental<br>Deliberate<br>Environmental |
|                              | Failure of telecommunication equipment             | Accidental<br>Deliberate                  |
| Disturbance due to radiation | Electromagnetic radiation                          | Accidental<br>Deliberate<br>Environmental |
|                              | Thermal radiation                                  |                                           |
|                              | Reactive radiation                                 |                                           |
|                              | Electromagnetic pulses                             |                                           |
| Compromise of information    | Interception of compromising interference signals  | Deliberate                                |
|                              | Remote spying                                      | Deliberate                                |
|                              | Eavesdropping                                      | Deliberate                                |
|                              | Theft of media or documents                        | Deliberate                                |
|                              | Theft of equipment                                 | Deliberate                                |
|                              | Retrieval of recycled or discarded media           | Deliberate                                |
|                              | Disclosure                                         | Accidental<br>Deliberate                  |
|                              | Data from untrustworthy sources                    | Accidental<br>Deliberate                  |
|                              | Tampering with hardware                            | Deliberate                                |
|                              | Tampering with software                            | Accidental<br>Deliberate                  |
|                              | Position detection                                 | Deliberate                                |
|                              |                                                    |                                           |
| Technical failures           | Equipment failure                                  | Accidental                                |
|                              | Equipment malfunction                              | Accidental                                |
|                              | Saturation of the information system               | Accidental<br>Deliberate                  |
|                              | Software malfunction                               | Accidental                                |
|                              | Breach of information system maintainability       | Accidental<br>Deliberate                  |
|                              |                                                    |                                           |
| Unauthorized actions         | Unauthorized use of equipment                      | Deliberate                                |
|                              | Fraudulent copying of software                     | Deliberate                                |
|                              | Use of counterfeit or copied software              | Accidental<br>Deliberate                  |
|                              | Corruption of data                                 | Deliberate                                |
|                              | Illegal processing of data                         | Deliberate                                |
|                              |                                                    |                                           |
| Compromise of function       | Error in use                                       | Accidental                                |
|                              | Abuse of rights                                    | Accidental<br>Deliberate                  |
|                              | Forging of rights                                  | Deliberate                                |
|                              | Denial of actions                                  | Deliberate                                |
|                              | Breach of personnel availability                   | Accidental<br>Deliberate<br>Environmental |
|                              |                                                    |                                           |

Threats may be deliberate, accidental, or environmental (natural), and may result in damage or loss of essential services. The origin of threats may be categorized as follows:

Deliberate threats (also called active threats) arise from deliberate actions aimed at assets. Updates, corrections, and other changes to operating systems, application programs, configurations, connectivity, and

equipment can pose an unexpected security threat to the systems or the respective production processes from deliberate threats. These may come in various forms:

- Communication attack
- Database injection
- Signals replay
- Spoofing and impersonation
- Social engineering
- Phishing
- Malicious code (virus, worm, Trojan horse, etc.)
- Denial of service
- Escalation of privileges
- Physical destruction attack

Accidental threats (also called passive threats) arise from human actions that can accidentally damage assets. For example, a person unfamiliar with proper procedures and policies or, through an honest oversight causes an accidental risk. Also, a user may not be aware of potential risks in a system and may trigger a threat by accident as the user operates a complex system. Environmental threats arise from conditions or external events that are not based on human actions.

The origins of threats come in many different forms. Examples from IEC/TS 62443-1-1: 2009—Security for industrial automation and control systems: Terminology, concepts, and models [B14] include:

- a) Insider—An insider is a “trusted” person, employee, contractor, or supplier who has information that is not generally known to the public. An insider can present a threat even if there is no intent to do harm. For example, the threat may arise as a result of an insider bypassing security controls “to get the job done.”
- b) Outsider—An outsider is a person or group not “trusted” with inside access, which may or may not be known to the targeted organization. Outsiders may or may not have been insiders at one time.
- c) Natural—Natural events include storms, earthquakes, floods, and tornadoes, and are generally considered a physical threat.

Particular attention (based on different businesses) should be paid to human threat sources. These are specifically itemized in [Table J.3](#).

**Table J.3—Example origins of threats**

| Origin of threat  | Motivation                                                                                                    | Possible consequences                                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hacker, cracker   | Challenge<br>Ego<br>Rebellion<br>Status<br>Money                                                              | Hacking<br>Social engineering<br>System intrusion break-ins<br>Unauthorized system access                                                                 |
| Computer criminal | Destruction of information<br>Illegal information disclosure<br>Monetary gain<br>Unauthorized data alteration | Computer crime (e.g., cyberstalking)<br>Fraudulent act (e.g., replay, impersonation, interception)<br>Information bribery<br>Spoofing<br>System intrusion |
| Terrorist         | Blackmail<br>Destruction of information<br>Exploitation<br>Revenge<br>Political gain                          | Bomb/Terrorism<br>Information warfare<br>System attack (e.g., distributed denial of service)<br>System penetration<br>System tampering                    |

| Origin of threat                                                                                       | Motivation                                                                                         | Possible consequences                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                        | Media coverage                                                                                     |                                                                                                                                                                                                                                                                                                                                                                            |
| Industrial espionage<br>(intelligence, companies, foreign governments, and other government interests) | Competitive advantage<br>Economic espionage                                                        | Defense advantage<br>Political advantage<br>Economic exploitation<br>Information theft<br>Intrusion on personal privacy<br>Social engineering<br>System penetration<br>Unauthorized system access (access to classified, proprietary, and/or technology-related information)                                                                                               |
| Insiders                                                                                               | Curiosity<br>Ego<br>Intelligence<br>Monetary gain<br>Revenge<br>Unintentional errors and omissions | Assault on an employee<br>Blackmail<br>Browsing of proprietary information<br>Computer abuse<br>Fraud and theft<br>Information bribery<br>Input of falsified, corrupted data<br>Interception<br>Malicious code (e.g., virus, logic bomb, Trojan horse)<br>Sale of personal information<br>System bugs<br>System intrusion<br>System sabotage<br>Unauthorized system access |

### J.3.2.2 Typical vulnerabilities

Vulnerabilities may be present objectively in any system. The following table (referenced from [Annex D](#), “Vulnerabilities and methods for vulnerability assessment” in ISO/IEC 27005:2011 [B39]) gives examples of typical vulnerabilities in various security areas, including examples of threats that might exploit these vulnerabilities. The table could provide help during the assessment of threats and vulnerabilities to determine relevant incident scenarios. It is emphasized that, in some cases, other threats may exploit these vulnerabilities as well. [Table J.4](#) provides examples of typical vulnerabilities.

**Table J.4—Example origins of threats**

| Origin   | Examples of vulnerabilities                                   | Examples of threats                          |
|----------|---------------------------------------------------------------|----------------------------------------------|
| Hardware | Insufficient maintenance/faulty installation of storage media | Breach of information system maintainability |
|          | Lack of periodic replacement schemas                          | Destruction of equipment or media            |
|          | Susceptibility to humidity, dust, soiling                     | Dust, corrosion, freezing                    |
|          | Sensitivity to electromagnetic radiation                      | Electromagnetic radiation                    |
|          | Lack of efficient configuration change control                | Error in use                                 |
|          | Susceptibility to voltage variations                          | Loss of power supply                         |
|          | Susceptibility to temperature variations                      | Meteorological phenomenon                    |
|          | Unprotected storage                                           | Theft of media or documents                  |
|          | Lack of care at disposal                                      | Theft of media or documents                  |
| Software | Uncontrolled copying                                          | Theft of media or documents                  |
|          | No or insufficient software testing                           | Abuse of rights                              |
|          | Well-known flaws in the software                              | Abuse of rights                              |
|          | No 'Logout' when leaving the workstation                      | Abuse of rights                              |
|          | Disposal or reuse of storage media without proper erasure     | Abuse of rights                              |
|          | Lack of audit trail                                           | Abuse of rights                              |
|          | Wrong allocation of access rights                             | Abuse of rights                              |
|          | Widely-distributed software                                   | Corruption of data                           |

| Origin       | Examples of vulnerabilities                                                                    | Examples of threats                    |
|--------------|------------------------------------------------------------------------------------------------|----------------------------------------|
|              | Applying application programs to the wrong data in terms of time                               | Corruption of data                     |
|              | Complicated user interface                                                                     | Error in use                           |
|              | Lack of documentation                                                                          | Error in use                           |
|              | Incorrect parameter setup                                                                      | Error in use                           |
|              | Incorrect dates                                                                                | Error in use                           |
| Network      | Lack of identification and authentication mechanisms like user authentication                  | Forging of rights                      |
|              | Unprotected password tables                                                                    | Forging of rights                      |
|              | Poor password management                                                                       | Forging of rights                      |
|              | Unnecessary services enabled                                                                   | Illegal processing of data             |
|              | Immature or new software                                                                       | Software malfunction                   |
|              | Unclear or incomplete specifications for developers                                            | Software malfunction                   |
|              | Lack of effective change control                                                               | Software malfunction                   |
|              | Uncontrolled downloading and use of software                                                   | Tampering with software                |
|              | Lack of back-up copies                                                                         | Tampering with software                |
|              | Lack of physical protection of the building, doors, and windows                                | Theft of media or documents            |
|              | Failure to produce management reports                                                          | Unauthorized use of equipment          |
|              | Lack of proof of sending or receiving a message                                                | Denial of actions                      |
|              | Unprotected communication lines                                                                | Eavesdropping                          |
|              | Unprotected sensitive traffic                                                                  | Eavesdropping                          |
|              | Poor joint cabling                                                                             | Failure of telecommunication equipment |
|              | Single point of failure                                                                        | Failure of telecommunication equipment |
|              | Lack of identification and authentication of sender and receiver                               | Forging of rights                      |
|              | Insecure network architecture                                                                  | Remote spying                          |
|              | Transfer of passwords in clear communication                                                   | Remote spying                          |
|              | Inadequate network management (resilience of routing)                                          | Saturation of the information system   |
| Personnel    | Unprotected public network connections                                                         | Unauthorized use of equipment          |
|              | Absence of personnel                                                                           | Breach of personnel availability       |
|              | Inadequate recruitment procedures                                                              | Destruction of equipment or media      |
|              | Insufficient security training                                                                 | Error in use                           |
|              | Incorrect use of software and hardware                                                         | Error in use                           |
|              | Lack of security awareness                                                                     | Error in use                           |
|              | Lack of monitoring mechanisms                                                                  | Illegal processing of data             |
|              | Unsupervised work by outside or cleaning staff                                                 | Theft of media or documents            |
| Site         | Lack of policies for the correct use of telecommunications media and messaging                 | Unauthorized use of equipment          |
|              | Inadequate or careless use of physical access control to buildings and rooms                   | Destruction of equipment or media      |
|              | Location in an area susceptible to flood                                                       | Flood                                  |
|              | Unstable power grid                                                                            | Loss of power supply                   |
| Organization | Lack of physical protection of the building, doors, and windows                                | Theft of equipment                     |
|              | Lack of formal procedure for user registration and de-registration                             | Abuse of rights                        |
|              | Lack of formal process for access right review (supervision)                                   | Abuse of rights                        |
|              | Insufficient provisions (concerning security) in contracts with customers and/or third parties | Abuse of rights                        |
|              | Lack of procedure for monitoring of information processing facilities                          | Abuse of rights                        |
|              | Lack of regular audits (supervision)                                                           | Abuse of rights                        |
|              | Lack of procedures for risk identification and assessment                                      | Abuse of rights                        |
|              | Lack of fault reports recorded in administrator and operator logs                              | Abuse of rights                        |
|              | Inadequate service maintenance response                                                        | Breach of maintainability              |
|              | Insufficient service level agreement                                                           | Breach of maintainability              |
|              | Lack of change control procedure                                                               | Breach of maintainability              |

| Origin | Examples of vulnerabilities                                                                                | Examples of threats                   |
|--------|------------------------------------------------------------------------------------------------------------|---------------------------------------|
|        | Lack of formal procedure for security documentation control                                                | Corruption of data                    |
|        | Lack of formal procedure for security record supervision                                                   | Corruption of data                    |
|        | Lack of formal process for authorization of publicly available information                                 | Data from untrustworthy sources       |
|        | Lack of proper allocation of information security responsibilities                                         | Denial of actions                     |
|        | Lack of continuity plans                                                                                   | Equipment failure                     |
|        | Lack of e-mail usage policy                                                                                | Error in use                          |
|        | Lack of procedures for introducing software into operational systems                                       | Error in use                          |
|        | Lack of records in administrator and operator logs                                                         | Error in use                          |
|        | Lack of procedures for classified information handling                                                     | Error in use                          |
|        | Lack of information security responsibilities in job descriptions                                          | Error in use                          |
|        | Insufficient provisions (concerning information security) in contracts with customers and/or third parties | Illegal processing of employee data   |
|        | Lack of defined disciplinary process in case of information security incident                              | Theft of equipment                    |
|        | Lack of formal policy on mobile computer usage                                                             | Theft of equipment                    |
|        | Lack of control of off-premise assets                                                                      | Theft of equipment                    |
|        | Insufficient 'clear desk and clear screen policy'                                                          | Theft of media or documents           |
|        | Lack of information processing facilities authorization                                                    | Theft of media or documents           |
|        | Lack of established monitoring mechanisms for security breaches                                            | Theft of media or documents           |
|        | Lack of regular management reviews                                                                         | Unauthorized use of equipment         |
|        | Lack of procedures for reporting security weaknesses                                                       | Unauthorized use of equipment         |
|        | Lack of procedures of provisions compliance with intellectual rights                                       | Use of counterfeit or copied software |

### J.3.3 Process assurance consideration in system life cycle

As shown in [Figure J.1](#), the system owner should consider the assurance of system security. Generally, the final users or operational organization undertakes the responsibility. But, when a certain system is under development or integration, the supplier or its vendor is the temporary owner who designs the system or system elements for operations. Unless the supplier or its vendor has considered the security problems, the development or integration process also has certain risks that could be introduced into the system. Furthermore, the V&V team evaluates the security implementations, so any result or issue of demonstration should be controlled under the Configuration Management Process.

Assurance requires that stakeholders of the system-of-interest should have the following:

- A robust development, agreement (acquisition or supply), and V&V process (see note) to achieve a product. For some critical systems, the supplier should verify no security issue (unwanted or unnecessary code and function) has been introduced.
- A secure development and operational environment against undocumented, unneeded, and unwanted modifications.
- A secure V&V environment to protect V&V result.
- Secure tools for development and V&V.
- Long-standing resource(human and budget) for security management, training, design, implementation, evaluation, operation, maintenance, and emergency preparedness.

NOTE—Subclause [12.8.5](#) requires that the VVP describe how the V&V effort conforms to existing security provisions and how the validity of V&V results is to be protected from unauthorized alterations. The security analysis V&V task should not only verify and validate the secure development and operational environment feature criteria and functions, but also assure protection from inadvertent manipulation of the test environment and test results.

## J.4 Risk analysis

### J.4.1 Risk analysis objectives

The objective of risk analyses performed by the V&V effort is to identify technical and engineering risks that have a measurable possibility of negative consequences to either the operation of the system or the successful development of the system. The risk analysis task includes the following:

- Identifying the initiating events, hazards, threats, or situations that create risks.
- Estimating the probability of occurrence, the consequences for each risk, and the expected timing of the risk.
- Evaluating each risk or defined combination of risks against its applicable threshold, generating alternatives to treat risks above their risk thresholds, and making recommendations for treatment based on a priority order.

Risk analysis is performed continuously throughout the life cycle.

### J.4.2 Risk analysis context

The V&V effort risk analysis task falls within a broader risk management approach. For example, the risk analysis steps of identify, estimate, and evaluate described in [J.3.3](#) follow the broader risk process described in ISO/IEC 16085-2006 [\[B37\]](#), which also addresses the risk management activities of managing the project risk portfolio, risk treatment, or risk monitoring.

Risks identified by the V&V effort often require management and mitigation by other project organizations. Accordingly, the V&V Plan should have clear provisions for communicating and escalating V&V identified risks to the organization with the responsibility to perform risk treatment.

Some risks of concern to the V&V effort are tightly coupled with hazard analyses and integrity levels. The severity of the consequence of a system operational risk will likely not exceed the worst-case severity of a system hazard or integrity level. Risks are often directly coupled to hazards. The likelihood that a hazard is reachable may be too high and cause re-engineering of the system to add hazard controls (or security controls for security hazards). There may be risk that a particular hazard control is unreliable or not verifiable. Both cases are of interest to the V&V effort. The hazard analysis is driven by the system design and operating environment. Probabilistic risk assessments evaluate the likelihood of reaching the hazardous condition, and risk management determines whether the probabilities exceed acceptable thresholds and require treatment. The V&V effort may identify risks when the probability of reaching a hazardous condition is perceived by the V&V effort to be beyond levels of acceptability for the system, and there is no plan to treat the risk. The V&V effort may also identify a risk when there is inadequate evidence that hazard controls provide the desired behavior/function or when there is uncertainty with regard to the conditions that may lead to a hazard. The same hazard-based risk analysis concepts apply to security-based risk analyses where undesired conditions may result from uncontrolled vulnerabilities or inadequate protection from threats.

### J.4.3 V&V risk analysis

#### J.4.3.1 Risk identification

Various approaches to identifying risks should be used. These approaches may include the use of risk questionnaires, taxonomies, brainstorming, scenario analysis, lessons learned, and prototyping or other knowledge acquisition approaches. Repeatable identification processes may be used to aid in the capture of lessons learned. Where possible, events, hazards, threats, or situations that can create risks should be identified to aid future risk treatment. Risks not identified are implicitly accepted. Risk categories should be used consistently for effective communication with stakeholders. Risks that are related may be combined for ease of analysis, monitoring, and treatment. System or software anomalies, reports on measures, and other indicators should be continuously reviewed as sources for risks.

NOTE—IEEE Std 1044-2009 [\[B21\]](#) provides useful information regarding anomaly classification. IEEE Std 982.1-2005 [\[B18\]](#) provides useful information regarding software measures related to reliability. ISO/IEC 15939-2008 [\[B36\]](#) provides a measurement process that can be used to help identify and characterize risks.

The V&V risk analysis is complementary to the risk analyses performed by the project management and developer organizations. The focus for V&V effort risk analysis is driven by the visibility the V&V effort has into life cycle products and is often concerned with the following types of risks:

- a) Risk that the system, as specified, will not meet stakeholder needs and requirements.
- b) Risk that the system may exhibit undesired behavior.
- c) Risk that engineering and integration plans will not enable accomplishment of V&V objectives through system integration/buildup.
- d) Risk that the acceptability of system or system elements cannot be determined.
- e) Risk that artifact quality or defect trends will impair the accomplishment of project milestones or system operational objectives.
- f) Risk that the likelihood of reaching a defined hazardous condition is too high.
- g) Risk that security controls are not adequate.
- h) Risk that security vulnerabilities are not adequately mitigated.
- i) Risk that the system is not maintainable.

Less frequently, but in some cases, the V&V effort may focus on programmatic risks such as cost and schedule performance.

#### **J.4.3.2 Risk estimation**

The likelihood of occurrence and consequences of each risk identified is estimated. Estimates may be either quantitative or qualitative. The stakeholders should define which risks will be evaluated using a qualitative scale and which will be evaluated using a quantitative scale. The scale(s) used for estimating risk likelihood and consequences should be used consistently. The descriptive and measurement uncertainty inherent in the scale used should be described in the risk management plan. The level of confidence in a risk estimate should be captured in its risk state.

#### **J.4.3.3 Risk evaluation**

Each risk is evaluated against its risk thresholds. Risks should be evaluated independently, in combination, and along with their interactions with system and enterprise risks. Risks should be evaluated against the project risk threshold to assure that a combination of risks, while below their individual thresholds, does not unacceptably place the project as a whole at risk. Different techniques may be used to evaluate the risks, such as decision trees, scenario planning, game theory, probabilistic analysis, and linear programming.

Risks are placed in a priority order where the ordering criteria are determined by the stakeholders. Priority may be based on when the risk is anticipated to become a problem, the risk exposure, risk-related measures, or some other consistent criteria. Various treatment alternatives to addressing risk should be considered to reduce or eliminate risks. For each risk that is above its risk threshold, recommended treatment strategies such as eliminating the risk, reducing its probability of occurrence or severity of consequence, or accepting the risk are defined and documented in a risk action request such as that found in [Annex B](#). Contingency plans should be developed for all risks above their thresholds. Measures indicating the effectiveness of the treatment alternatives are also defined. The risks, their recommended treatments, and measures of risk treatment effectiveness are communicated to the stakeholders for approval, rejection, or modification.

## Annex K

(informative)

### Example of assigning and changing the system integrity level of

#### “supporting system functions”

The integrity level of system functions that are designated as critical functions does not change during the system life cycle unless authorized by the authorized system acquirer or regulatory organization. Noncritical system functions may become critical if the system is configured such that the originally noncritical functions can modify or alter critical data or can create an improper condition or system state to exist causing critical system functions to take incorrect actions. We will refer to these system functions as “critical supporting system functions.” Any of the “critical supporting system functions” can lower their assigned system integrity level during development stages by the selection of technology or design/implementation techniques.

The V&V criticality analysis task performed at each development stage verifies the criticality of each system function and verifies that the correct system integrity level has been assigned. It is in this V&V task that a determination can occur where a “critical supporting system function” from a previous stage can be lowered or raised due to a technology selection or design/implementation technique being used by the subsequent development stage. The following two figures illustrate both the raising and the later lowering of system integrity levels for a supporting system function.

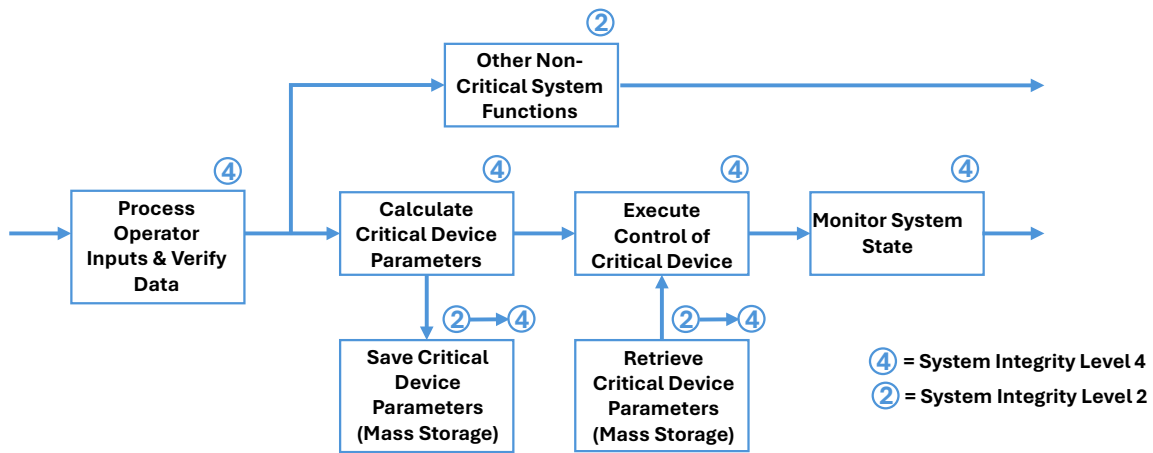
The example is a functional control flow diagram of the function of controlling a critical device. During the design stage of development, the control flow shows the receipt of device control inputs from an operator and critical device control parameters being calculated from the operator inputs. Those critical parameters are stored in a mass storage device and then retrieved by the critical device controller for execution of control of the critical device. Initial system integrity level assignment from the criticality analysis task designates the following system functions at the highest system criticality level 4 (in a four-level system integrity schema):

- Process operator inputs and verify data
- Calculate critical device parameters
- Execute control of critical device
- Monitor system

Supporting system functions in [Figure K.1](#) are initially designated as low criticality and thus are assigned a system integrity level 2. However, the criticality analysis of the control flow logic diagram in the design stage clearly indicates that the supporting system functions of “Save” and “Retrieve” critical device parameters from the mass storage device have the possibility of modifying or altering the critical parameters. Thus, these supporting system functions are raised to the high system integrity level 4 so that the level 4 V&V tasks are conducted.

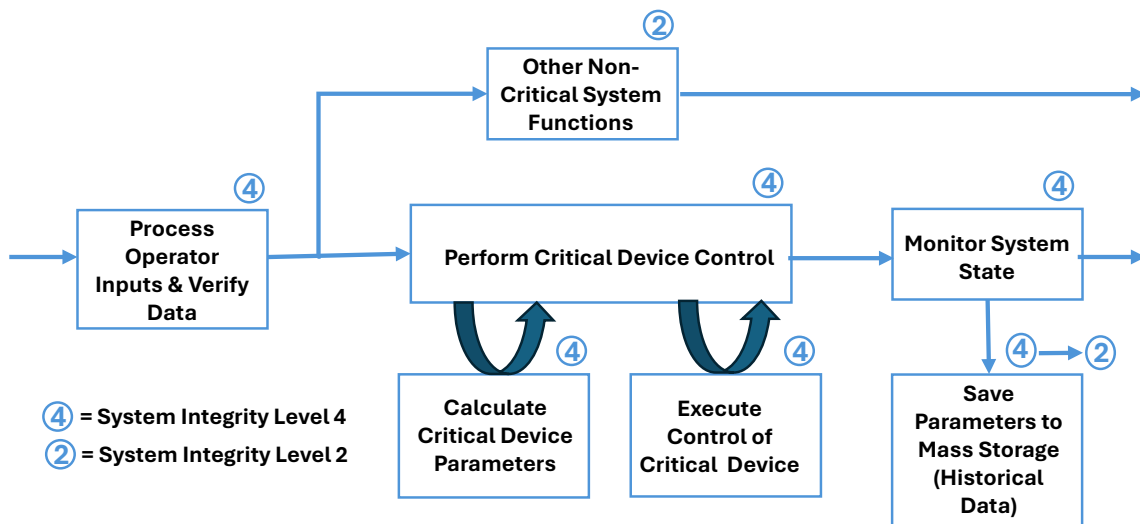
However, during the subsequent implementation stage (shown in [Figure K.2](#)), the developer has adopted an implementation technique whereby a “critical device controller” module is used to invoke the “calculate critical device parameters” and “execute control of critical devices” as subroutine calls. The critical parameters are passed parameters in the subroutine calls and thus eliminate the need to store and retrieve the critical data from a mass storage device. The storage of data on the mass storage device is later performed only for historical record keeping, and thus, the system integrity level is lowered from level 4 to level 2, being that it is no longer a critical execution function.





*Design Process Functional Diagram Illustrating the Elevating of System Integrity Level for Supporting System Design Functions* ②→④

**Figure K.1—Example of raising system criticality level**



*Design Process Functional Diagram Illustrating the Lowering of System Integrity Level for Supporting System Design Functions* ④→②

**Figure K.2—Example of lowering system integrity level**

## Annex L

### (informative)

#### Verification and Validation of $n^{\text{th}}$ of a kind systems

There may come a time when the same system is needed in multiple instances and therefore needs to be copied and installed numerous times to satisfy stakeholder needs. Examples include installing a redundant system for training, development, or continuous operation purposes; a second or third unit of a power plant with the same specifications; or manufacturing the same system for different customers under different projects. This copy of a system is often referred to as an “ $n^{\text{th}}$  of a kind system.”

The initial system design represents an identifiable set of functional and performance requirements, and its implementation is defined as the first application. An  $n^{\text{th}}$  of a kind system is defined as a re-manufacturing or re-installation of this first application. Consideration of what constitutes a first application versus an  $n^{\text{th}}$  application needs to take into consideration the functional and performance requirements for the project as well as the regulatory environment. Any significant differences should cause the project to be considered a first application. For a system of interest to be determined as an “ $n^{\text{th}}$  of a kind system,” it needs to be an identical copy of a system that has been previously verified and validated, with an identical operating and regulatory environment. In this case, there would be no further V&V tasks needed beyond those to provide the body of evidence that the system and environment are identical.

The first application performs the entire V&V activities, including testing for the hardware, software, and integration per this standard. All applications require a demonstration of required V&V activities performed. However, to the extent that the design does not change from application to application (i.e.,  $n^{\text{th}}$  of a kind system), these activities do not necessarily need to be repeated. Therefore, required V&V activities on an  $n^{\text{th}}$  of a kind system may be greatly reduced since the V&V plan could take credit for the previous life cycle activities performed on the first application system. A regression analysis of the requirements and the system/hardware/software design and implementation should be performed to demonstrate that the  $n^{\text{th}}$  of a kind system application is identical to the first application in all relevant aspects.

Even when all design requirements and specifications were identical and previously verified, and the first application was implemented and tested to validate functional and performance requirements, the target hardware components for the  $n^{\text{th}}$  of a kind system are likely to be procured and integrated separately from the first application. Hardware is subject to errors being introduced in the manufacturing process; hence, hardware testing is performed on each system instance to prove correctness. This necessitates a certain level of manufacturing and factory testing on each of the subsequent copies to validate that the system was integrated correctly and continues to meet the performance requirements with the new set of hardware used in that particular copy. Additional testing and/or analyses may need to be performed on each copy of the  $n^{\text{th}}$  of a kind system. For identical systems, these tests are expected to be focusing on correct integration and installation rather than a repeat of functional tests previously performed on the first application. Analysis may be performed to conclude equivalency of hardware components, for instance, when a hardware component needs to be replaced with another from a different manufacturer due to obsolescence or procurement issues. Software, on the other hand, can be installed from configuration-managed files and remain identical to the reference system. The compilation and final load assembly tools used need to be identical, as different or updated compiler and load file generators may create minor alterations to the final object code. Regression analysis is performed to confirm that the software and software tools are identical in all relevant aspects.

A regression analysis is defined as a series of activities performed under configuration management that demonstrates the two implementations are identical or identifies the differences between the implementations that need to be justified or re-qualified through further V&V activities. To confirm the system is equivalent to that of the first application, the V&V activities for the  $n^{\text{th}}$  of a kind system should consider the following:

- Equivalency of hardware (e.g., batch and version controlled, performance specifications)
- Equivalency of hardware qualifications (e.g., seismic evaluations, electromagnetic compatibility, temperature, humidity evaluations)
- Equivalency of software [i.e., cyclic redundancy checks (CRC), configuration, version control]
- Location and orientation of the follow-on systems' installation (i.e., operating environment)
- Regulatory environment (e.g., demands from licensing authorities and the customer)

The  $n^{\text{th}}$  of a kind system developer should also implement processes to support change notices and updates to correct identified deficiencies in any of the installations and have the means to keep track of system configurations after their delivery.

The regression analysis may also show that the two systems are not identical per the strict definition of  $n^{\text{th}}$  of a kind system. However, depending on the differences between the first application (reference system) and its next implementation with a certain level of differences/customization (repeat system), the V&V activities previously performed on the reference system may still be applicable and can be mapped to the life cycle activities of the repeat system. If these changes become large in the sense that user needs or the environment are different, the system should be considered a new “first application.” The key to credit V&V activities of a first application in another first application lies in the configuration management and body of evidence that shows direct and undisputable mapping of life cycle activities from one project to another. [Annex D](#) of this standard provides guidance on the V&V of reuse software, which could be seen to apply to the case of  $n^{\text{th}}$  of a kind systems as well as a new first application system.

## Annex M

(informative)

### Guidance for applying IEEE 1012 in an adaptive environment

#### M.1 Introduction

The software industry has evolved in the last two decades to more adaptive approaches to the development life cycle. This evolution is driven by customer expectations of higher quality software products at lower cost and the need for speed to market with product lines. Advancements in technologies and software tools have replaced prior manual processes that were labor intensive with automated life cycle capabilities, improving the overall quality and speed of releases. The V&V practitioner has likewise needed to adapt. This informative annex is intended to provide direction on how this IEEE standard may be better understood and applied within the many modern adaptive methodologies in use today.

There are many adaptive frameworks that seek to develop and deploy capabilities using techniques that do not align with traditional methodologies (e.g., waterfall, iterative). Of all the adaptive methodologies Agile is the most prominent. Therefore, we focus on Agile V&V while keeping the discussion broad enough to include other adaptive strategies.

The Agile methodology develops incremental software solutions to a set of desired outcomes (capabilities) and user stories (requirements) in a very short and rapidly repeating timeframe. Through the repeated development of these small incremental solutions, the expectation is that total software emerges in a shorter timeframe than the traditional software life cycle methodologies. The challenge for the V&V effort is to provide a meaningful and comprehensive evaluation of the system concept, selected requirements, design features, and implementation correctness using only the desired outcomes description, user stories, and software code of the Agile incremental software release. The V&V results/findings to the Agile development team need to be timely to support the next Agile software cycle (e.g., several weeks between Agile development cycles) or be placed on the Agile backlog list for incorporation in later Agile software releases. The V&V effort may need to reconstruct the implied design and requirements from the code, then extrapolate from this reverse engineering method and determine whether the Agile software code satisfies the system objectives and requirements as specified in the desired outcomes and user stories. The V&V task criteria in IEEE Std 1012 provides the roadmap for the V&V analysis effort to provide meaningful results for early identification of incomplete, missing, or erroneous requirements and implementation. The opportunity for early discovery of errors or incomplete solutions provides invaluable feedback to the Agile development team to develop a comprehensive and correct solution in its future Agile release cycles.

This annex focuses on the IEEE Std 1012, [Clause 9](#), V&V Software processes. [Table M.1](#) provides a mapping of the software technical processes described in ISO/IEC/IEEE 12207(E) [\[B40\]](#) (omitting the V&V processes) to the corresponding adaptive activities, and, more specifically, agile activities. The V&V processes are performed, as described in [Clause 9](#), to the associated adaptive activities.

**Table M.1—Software technical processes and adaptive activities**

| Software technical processes                  | Adaptive activities                                    |
|-----------------------------------------------|--------------------------------------------------------|
| Business or Mission Analysis                  | Capabilities                                           |
| Stakeholder Needs and Requirements Definition | User stories                                           |
| System/Software Requirements Definition       | Incremental implementation (e.g., iterations, sprints) |
| Architecture Definition                       |                                                        |
| Design Definition                             |                                                        |
| Implementation                                |                                                        |
| Integration                                   |                                                        |
| Transition                                    |                                                        |
| System Analysis                               | No change                                              |
| Operation                                     |                                                        |
| Maintenance                                   |                                                        |
| Disposal                                      |                                                        |

## M.2 Background

### M.2.1 Defining characteristics of adaptive approaches

The software industries have realized a driving force changing the face of the software engineering industry known as Agile and are taking steps to adjust standards, guidelines, policies, and processes to align with this new paradigm. This document seeks to inform the V&V practitioner on best practices for utilizing this standard within adaptive development practices.

### M.2.2 Defining the challenges

The V&V practitioner has to adjust to a smaller-iterations adaptive software development life cycle framework. As a result, the V&V practitioner may experience the following challenges with applying IEEE 1012 to adaptive approaches:

- Increase in effort required to adjust to ever-changing content of product iteration or release.
- Possibility of change in capabilities or user stories from interim user evaluation and acceptance.
- Less, insufficient, or outdated documentation.
- Less time to perform key assessments restricting the depth at which analysis can be performed.
- Creates significant execution challenges for Integrity Level 3 and Level 4 V&V practitioners.
- Greater reliance on developer’s tools/developer’s data to track dynamic backlog information.
- Communicating V&V strategies, benefits, and status to customers more familiar with traditional development methodologies.

### M.2.3 Requirements and user stories

Many of the agile frameworks involve user stories, instead of the traditional requirements, which include functional and non-functional (technical) requirements.

A user story contains the following three elements:

- Who wants the functionality?
- What is it they want?
- Why they want it.

In addition, a user story involves acceptance criteria, which refer to a set of predefined requirements that must be met to mark a user story complete. Acceptance criteria are also sometimes called the “definition of done” because they determine the scope and requirements that must be executed by developers to consider the user story complete.

## **M.2.4 Release and shippable product**

In traditional development, the V&V practitioner validates the final product against the functional and non-functional requirements. For each iteration in an adaptive environment, there will be a shippable product or executable, which the V&V practitioner may validate. The challenge is that sometimes not all user stories are successfully implemented in the shippable product for that specific iteration; they are usually moved to a future iteration. Therefore, the V&V practitioner continually evaluates the “product backlog” to assure that delayed requirements are correctly captured in user stories to be implemented in the future.

## **M.3 V&V for adaptive (agile) activities**

### **M.3.1 V&V for capabilities**

Capabilities activities in an adaptive effort generally align with the Business or Mission Analysis Process, and are centered around the description of large solution functionality. A capability should include a benefit hypothesis and acceptance criteria. The capability statement can be assessed using V&V activities and tasks in the business or Mission Analysis V&V Process (9.2), with criticality analysis being used to determine the scope, rigor, and intensity of the V&V effort:

- Business or mission analysis documentation evaluation
- Traceability analysis
- Hazard analysis
- Security analysis
- Risk analysis

The minimal viable product (MVP) contains the basic features of a new product to satisfy early adopters. The complete set of basic features is only designed and developed after considering feedback from the product's initial users. To understand the development and implementation of the basic features, the project management team should document a timeline of the essential elements on a product roadmap and a plan of action for how a product or solution will evolve. To provide assurance core capabilities are in the delivered product, the V&V practitioner assesses the utilization of the product roadmap to represent the MVP. In addition, the V&V practitioner assessed the traceability of the objectives and user stories to the MVP.

### **M.3.2 V&V for user stories**

User stories in an adaptive effort generally align with the Stakeholder Needs and Requirements Definition Process, where user stories provide an informal description of a feature and its corresponding characteristics from the customer's perspective. Each user story can be assessed using V&V activities and tasks in the Stakeholder Needs and Requirements Definition V&V Process (9.2), with criticality analysis being used to determine the scope, rigor, and intensity of the V&V effort:

- Stakeholder needs and requirements evaluation
- Traceability analysis
- Hazard analysis
- Security analysis
- Risk analysis

Specific considerations for assessing each user story include the following:

- a) Verify the user story is independent (of all others).
- b) Verify the user story is negotiable (not a specific contract for features).
- c) Verify the user story is valuable (or vertical).
- d) Verify the user story is estimable (to a good approximation).
- e) Verify the user story is small (fits within an iteration).

- f) Verify the user story is testable (in principle, even if there is not a test for it yet).
- g) Verify the user story includes a definition of done.
- h) Verify the acceptance criteria are written from a product owner's perspective.
- i) Verify the acceptance criteria are complete, clear, and concise.
- j) Verify that when successfully evaluated, the acceptance criteria achieve stated requirements.
- k) Verify the acceptance criteria consist of a set of test scenarios that are to be met to confirm the software is working as expected.

In addition to evaluating each user story, the V&V effort should assess if the set of user stories expresses all stakeholder needs and requirements. While not all user stories may be developed prior to the beginning of software implementation, eventually, all user needs should be described in the set of user stories.

The iterative development team commonly uses story points as the units of measure to estimate a user story. Consider understanding how the iterative development team calculates story points.

### M.3.3 V&V for incremental implementation

During iterative development cycles, incremental system features are built, integrated into the broader repository, and, for those incompletely implemented user stories, sent back to the user story process for any product backlog refinement and subsequent re-planning into future iterations. Incremental implementation incorporates tasks and activities from several software technical processes, including the following:

- System/software requirements definition
- Architecture definition
- Design definition
- Implementation
- Transition

The purpose of V&V during incremental implementation is to assure the planned iteration event is complete and reflects the correct product backlog and user stories allocation. V&V for incremental implementation also assures the iteration results have been achieved with any iterative metrics incorporated into future planning and assures the implementation increment meets the defined features added with each iteration. Each iteration can be assessed using V&V activities and tasks in the V&V processes associated with the corresponding software technical processes:

- System/software requirements definition V&V ([subclause 9.3](#))
- Architecture definition V&V ([subclause 9.4](#))
- Design definition V&V ([subclause 9.5](#))
- Implementation V&V ([subclause 9.7](#))
- Integration V&V ([subclause 9.8](#))
- Transition V&V ([subclause 9.10](#))

Not all traditional artifacts may be developed during an adaptive effort, and in particular, system/software requirements may be implied from user stories, and software architecture and software design may be found only in the structure of the software. Depending on the criticality of the software and the needs of the customer, the V&V effort could reverse engineer these artifacts or simply evaluate them based on the working software.

Most testing in an adaptive approach is automated and not necessarily separated into the traditional divisions of component testing, integration testing, qualification testing, and acceptance testing. However, all aspects of these types of testing should be included prior to software deployment, and the V&V effort should include activities to verify testing suitable to the criticality of the software.

Traditionally V&V evaluates work products from the development organization as they become available, with some time needed between the availability of the work product and the completion of the V&V on that product. The speed of adaptive incremental development presents challenges to the completion of the V&V activities and tasks in a timely fashion for the deployment of the software. Depending on the criticality and complexity of the software, different approaches for the execution of V&V might be used. Several of these approaches are described as follows, with others certainly possible:

- a) The V&V team could work alongside the development effort, evaluating the work products as soon as they are available (typically daily during the iteration). This still might require some period of time at the end of the iteration to complete the V&V effort.
- b) The V&V team could work one iteration after the development team, evaluating each full iteration after its completion. The V&V team would have a full iteration package to assess, but deployment would be delayed by up to a full iteration cycle.
- c) While each adaptive iteration produces software that could be deployed, many adaptive projects combine the results of multiple iterations into a single deployment. In this case, the V&V effort could be directed at the software to be deployed, starting with the product of each iteration as it becomes available.



## Annex N

(informative)

# Verification and Validation considerations for machine learning and artificial intelligence

## N.1 Introduction

Machine learning (ML) and artificial intelligence (AI) have recently gained popularity in numerous applications in machine vision, virtual assistants, self-driving vehicles, network analysis, and translation, in domains such as transportation, finance, energy, security, marketing, and manufacturing. Unfortunately, best practices and standards for the (V&V) of ML and AI have lagged behind their proliferation. Though some aspects of V&V for conventional systems/software can be used without modification, there remain important challenges in AI and ML (AI/ML) V&V. AI/ML-based systems or components bring new risks and challenges for Verification from, for example, (1) erosion of determinism, and (2) unanticipated, or ‘emergent,’ behaviors (evolution of the IEEE P7009 [B7]). The purpose of this annex is to raise awareness of the potential issues and differences from traditional systems and software V&V accompanying AI/ML. The primary focus is on V&V that occurs prior to system operation (as in traditional V&V) (subclause [N.2](#) through subclause [N.6](#)); however, operation V&V is required for mission-, safety-, or business-critical systems and is covered in subclause N.7.

This annex assumes that for critical systems using AI/ML, the AI/ML component is learning (and thus potentially changing) only during the development life cycle phases, prior to release. That is, currently the AI/ML software does not change during operation in critical systems.

## N.2 Requirements V&V

The practice of treating the AI/ML component of a system as a closed box has led to challenges in AI requirements and AI requirements engineering (explainable artificial intelligence (XAI) [B11]). One challenge is the lack of effective methods to specify both desired and undesired properties of the AI/ML component (towards verified artificial intelligence [B56]). As such, under-specification is possible, and thus it is important to provide assurance of the completeness of specifications of AI/ML correct behavior [B55]. One problem lies in the fact that the correctness of unsupervised AI/ML methods is embedded in the training data and the operational environment. Belani and colleagues outline a taxonomy of AI requirements engineering challenges [B2]. Requirements traceability for systems with AI elements is obscured by treating the AI element as a closed box.

Requirements review serves to determine the feasibility and testability of the requirements and to verify the consistency of requirements to satisfy user needs. A feasible requirement can be validated that it is satisfied by the technologies, methods and algorithms defined for the project. A requirement is testable if objective information supporting the requirement, demonstratable by testing, can be provided. If any specifications require the use of an adaptive or learning system, it must be verified that the type of AI/ML selected can meet the system and system/software requirements allocated to it. The appropriateness of a technique is primarily based on data availability, the task (e.g., regression, dimensionality reduction, clustering, or classification), whether a quantity or a category is being sought, and the availability of labeled data.<sup>12</sup> Verifying consistency of requirements for AI/ML must include information on how the AI/ML component acquires and acts on knowledge (control) and what that knowledge is [B54]).

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<sup>12</sup> An example technique selection map is provided at [https://scikit-learn.org/stable/tutorial/machine\\_learning\\_map/index.html](https://scikit-learn.org/stable/tutorial/machine_learning_map/index.html) (accessed 02/16/2024).

In verifying and validating AI/ML requirements, make sure data requirements are included and that the following considerations are taken into account:

- Relevance of the training and test data to the operational domain.
- What and how much data is required?
- What pre-processing of the data is required? An example is normalizing the data.
- Are the data to be treated equally during training, or will they be weighted? If weighted, what is the weighting scheme?
- Determine the dimensions of likely variance in the operational domain and require that data sufficiently cover that variance (robustness).
- Data completeness: consider the range of conditions or operating dimensions, e.g., for an autonomous vehicle’s vision system, data should cover different times of day, weather conditions, road conditions, traffic situations, etc.
- Label quality requirements.
- Bias and ethics requirements.
- Accuracy, relevance, completeness, and balance.

Provide assurance that V&V requirements address properties applicable to AI/ML systems, that is, AI/ML system accountability, controllability, explainability, interpretability, reliability, resilience, robustness, safety, and transparency. Terms often used in AI/ML requirements are described in [Table N.1](#).

**Table N.1—Machine learning terms used in requirements discussion**

| Term                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Accountability              | The degree to which the actions of an entity can be traced uniquely to the entity (IEEE Std 7002 <sup>TM</sup> -2022 <a href="#">[B29]</a> , IEEE Std 7007 <sup>TM</sup> -2021 <a href="#">[B30]</a> )                                                                                                                                                                                                                                                                                                                                                                                              |
| Classification              | A machine learning function that predicts the output class (signified by a label) for a given input                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Clustering                  | A machine learning process that groups a set of unlabeled objects such that objects in the same group (i.e., a cluster) are more similar to each other than to those in other clusters                                                                                                                                                                                                                                                                                                                                                                                                              |
| Controllability             | The degree to which there is (or can be) technical or organizational control over a system (IEEE Std 7000-2021 <a href="#">[B27]</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Dimension(ality) reduction  | In machine learning, the factors taken into consideration in a classification problem, are the basis of the classification. The number of those factors (or features) is based on the dimensions (attributes or variables) of the training dataset. When the number of dimensions is too high, then the classification problem is difficult to visualize, and many dimensions may be correlated or redundant. Thus, dimensionality reduction is the preprocessing of a dataset to reduce the number of its attributes while retaining as much of the variation of the original dataset as possible. |
| Explainability              | The extent to which the information made transparently available to a stakeholder can be readily interpreted and understood by a stakeholder. (IEEE Std 7001-2021 <a href="#">[B28]</a> )                                                                                                                                                                                                                                                                                                                                                                                                           |
| Interpretability            | The extent to which a human can understand the cause of, or algorithmic reasoning behind, a decision.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Labeled data                | Labeled data are data that have been tagged with one or more labels identifying certain properties, characteristics, or classifications.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Operational (design) domain | Operating conditions under which a given automated system or feature thereof is specifically designed to function, including, but not limited to, environmental, geographical, and time-of-day restrictions, and/or the requisite presence or absence of certain operational characteristics (IEEE Std 2846 <sup>TM</sup> -2022 <a href="#">[B26]</a> )                                                                                                                                                                                                                                             |
| Regression                  | Regression analysis is the process of estimating the relationship between a dependent variable and independent variables. In machine learning, regression is supported by a set of supervised learning methods that find a numerical or continuous value for a given input.                                                                                                                                                                                                                                                                                                                         |
| Reliability                 | The ability of a system or component to perform its required functions under stated conditions for a specified period of time (ISO/IEC/IEEE 24765:2017 <a href="#">[B43]</a> ) Note: Wear or                                                                                                                                                                                                                                                                                                                                                                                                        |

| Term                               | Description                                                                                                                                                                                                                                                                                     |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | aging does not occur in software. Limitations in reliability are due to faults in requirements, design, and implementation, or due to contextual changes.                                                                                                                                       |
| Resilience                         | Ability of a system or component to maintain an acceptable level of service in the face of disruption. (IEEE Std 2413™-2019 [B25])                                                                                                                                                              |
| Robustness                         | The degree to which a system or component can function correctly in the presence of invalid inputs or stressful environmental conditions (ISO/IEC/IEEE 24765:2017 [B43])                                                                                                                        |
| Safety                             | The expectation that a system does not, under defined conditions, lead to a state in which human life, health, property, or the environment is endangered (ISO/IEC/IEEE 12207:2017(E), [B40])                                                                                                   |
| Transparency                       | Characteristic of the transfer of information to a stakeholder, which is honest; contains information relevant to the causes of some action, decision, or behavior; and is presented at a level of technicality and in a form that is meaningful to the stakeholder. (IEEE Std 7000-2021 [B27]) |
| Variance in the operational domain | The degree to which the actual operational domain differs from the expected operational domain or environment.                                                                                                                                                                                  |

### N.3 Hazard analysis

Some hazards introduced by AI/ML are like those created by any software, e.g., incorrect results when correct results are expected. Pullum and colleagues ([B53], [B54]) and Pereira and Thomas [B50] provide lists of hazards that can be introduced during V&V processes and AI/ML life cycle phases, respectively. These have been combined to provide a comprehensive high-level list of hazards to consider for AI/ML (sub)systems, as shown in [Table N.2](#). Machine learning terms used in the discussion of hazard analysis are described in [Table N.3](#).

**Table N.2—Potential hazards introduced by AI/ML, organized by the IEEE Std 1012 software V&V process in which the hazard is introduced**

| IEEE Std 1012 software V&V process                                                                                                                                           | Hazard introduced                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Business or mission analysis V&V—conduct preliminary hazard analysis via hazards and operability studies (HAZOPS); potential hazards to consider for the HAZOPS are provided | AI/ML model does not adapt                                                                                                               |
|                                                                                                                                                                              | AI/ML model adapts, but is unable to reach convergence                                                                                   |
|                                                                                                                                                                              | AI/ML model adapts and converges, but converges to an incorrect output                                                                   |
|                                                                                                                                                                              | AI/ML model converges to a correct state, but cannot do so in the required time                                                          |
|                                                                                                                                                                              | AI/ML model grows beyond available system resources during adaptation (for online learning systems)                                      |
| Stakeholder needs and requirements definition V&V and system requirements definition V&V                                                                                     | Incomplete data definition                                                                                                               |
|                                                                                                                                                                              | Incorrect objective function                                                                                                             |
|                                                                                                                                                                              | Inadequate performance measure                                                                                                           |
|                                                                                                                                                                              | Incompleteness of testing and Verification                                                                                               |
|                                                                                                                                                                              | Inadequate safe operating values (if in a safety-critical system)                                                                        |
| Architecture definition V&V and design definition V&V—hazards introduced due to the design of AI/ML                                                                          | Lack of a run-time monitor that recognizes and reacts to out-of-tolerance AI/ML results (if a safety-critical system)                    |
|                                                                                                                                                                              | Inadequate distribution of training examples                                                                                             |
|                                                                                                                                                                              | Insufficient dataset size                                                                                                                |
|                                                                                                                                                                              | Bias                                                                                                                                     |
|                                                                                                                                                                              | Irrelevance of the training data                                                                                                         |
|                                                                                                                                                                              | Data quality deficiencies                                                                                                                |
|                                                                                                                                                                              | The training set does not model intended goals                                                                                           |
|                                                                                                                                                                              | The collection of training and test data sets is not tracked under configuration management and is not described in design documentation |
|                                                                                                                                                                              | Lack of adequate testing data to achieve component-level project-specific certification standards                                        |
|                                                                                                                                                                              | AI/ML architecture is not completely specified                                                                                           |

| IEEE Std 1012 software V&V process     | Hazard introduced                                                                                                                                                                                              |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation V&V                     | Implementation of the AI/ML prior to training is incorrect (e.g., for neural networks, the initial number of neurons, connection matrices, growing/learning functions, or activation functions contain errors) |
|                                        | Inappropriate learning algorithm is selected, causing suboptimal knowledge acquisition (training)                                                                                                              |
|                                        | Through learning, the AI/ML structure exceeds the computational limitations of the system                                                                                                                      |
|                                        | AI/ML model takes too long to adapt to new stimuli (online learning)                                                                                                                                           |
|                                        | AI/ML model never reaches stability and never achieves the intended performance                                                                                                                                |
|                                        | AI/ML model is over-generalized and cannot provide a suitable solution to the specific problem (underfitting)                                                                                                  |
|                                        | AI/ML model is too specialized and cannot provide a general solution for the problem domain (overfitting)                                                                                                      |
|                                        | Observable behavior of the AI/ML model is not predictable or repeatable                                                                                                                                        |
|                                        | Model mismatch                                                                                                                                                                                                 |
|                                        | Bias is introduced to the model                                                                                                                                                                                |
|                                        | (Hyper-) parameter mismatch                                                                                                                                                                                    |
|                                        | Performance measure mismatch                                                                                                                                                                                   |
|                                        | Error rate                                                                                                                                                                                                     |
|                                        | Lack of interpretability                                                                                                                                                                                       |
| Implementation V&V and integration V&V | Test techniques fail to adequately exercise the AI/ML model (Incompleteness)                                                                                                                                   |
|                                        | Failure to examine the training data for completeness and correctness                                                                                                                                          |
|                                        | Failure to review input selection (ensuring the input choices are correct, based on an acceptable technique, etc.)                                                                                             |
|                                        | Test procedures described in the test plans do not properly assess AI/ML performance criteria such as stability and convergence                                                                                |
|                                        | For software in the loop simulations (SILS), the test cases do not accurately reflect how the adaptive component performs within the entire system                                                             |
|                                        | For hardware in the loop simulations (HILS), near identical versions of the hardware do not accurately reflect the performance of the true hardware                                                            |
|                                        | Non-representative distribution                                                                                                                                                                                |
| Transition V&V                         | Differences in computational platforms                                                                                                                                                                         |
|                                        | Incompleteness of the delivery package, e.g., missing environment, check-cases, or operational monitors, if used                                                                                               |
|                                        | Incompatibility between development and target environment                                                                                                                                                     |
|                                        | Differences in hardware processing size and speeds may impact performance                                                                                                                                      |
|                                        | Incomplete or incompatible integration with the system may induce additional hazards                                                                                                                           |
|                                        | Differences between operational and train/test data                                                                                                                                                            |
|                                        | Non-detection of potentially incorrect outputs                                                                                                                                                                 |
|                                        | New data/continuous learning (for online learning systems)                                                                                                                                                     |
| Operation V&V                          | Operating procedures are inconsistent with user documentation or with system requirements                                                                                                                      |
|                                        | Operating conditions differ from those intended by system developers                                                                                                                                           |
|                                        | Input data is received from a source that was not originally intended                                                                                                                                          |

| IEEE Std 1012 software V&V process | Hazard introduced                                                                                 |
|------------------------------------|---------------------------------------------------------------------------------------------------|
| Maintenance V&V                    | Computational resource limits are approached or exceeded (possible with online learning)          |
|                                    | Operational monitors function inappropriately as either too restrictive or not restrictive enough |
|                                    | Consider the hazards identified under implementation V&V and potentially design V&V               |

**Table N.3—Machine learning terms used in hazard analysis discussion**

| Term                           | Description                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bias                           | Bias (inclination or prejudice for or against one person or group) occurs in machine learning when the algorithm produces systemically prejudiced results. This can occur due to inherent bias in the training data or can be introduced throughout the machine learning process. Types of bias include prejudicial, sampling, algorithmic, and confirmation bias. |
| Convergence                    | A machine learning model converges when additional training will not improve the model, that is, when it reaches a state during training in which loss settles to within a specified error range around the final value.                                                                                                                                           |
| Objective function             | A formula that relates a decision variable to either the cost or the revenue of an alternative (ISO/IEC/IEEE 24765:2017)                                                                                                                                                                                                                                           |
| Operational (run-time) monitor | A means to examine a component or system during operation. Typically used to identify safety-related operational faults and design assumption violations                                                                                                                                                                                                           |
| Overfitting                    | Overfitting occurs when the machine learning model matches too closely or exactly to a particular set of data, resulting in a model that cannot generalize (perform accurately against unseen data).                                                                                                                                                               |
| Stability                      | Refers to the changes in the output of a model when a) a different training set is used or b) there is variance in the inputs during operation.                                                                                                                                                                                                                    |
| Underfitting                   | Underfitting occurs when the machine learning model cannot capture the underlying trend of a particular set of data, resulting in a model with poor accuracy. The model may perform sufficiently on the training data, but does not perform well on test or operational data.                                                                                      |

## N.4 Testing V&V

In many aspects, testing V&V for AI/ML systems or components is like that for traditional systems or components. The user interface, data ingress, and visualization components typically used with AI/ML undergo traditional testing. This section addresses the differences in testing for the AI/ML component itself. When developing the software V&V plan, consider all the information provided in this annex, e.g., V&V addresses applicable hazards and does not introduce additional hazards. Testing may also include the use of special tools used in the Verification of neural networks (or other AI/ML techniques), such as development environments, visualization packages, test data generators, and others. In this section, we will examine AI/ML data and closed-box and open-box testing, including considerations with respect to testing V&V included throughout.

### N.4.1 Data

Data are intrinsically important to AI/ML because data are used to create, calibrate, and evaluate AI/ML. Different sets of data must be used for each of these purposes, and they are termed training, Validation, and test, respectively. The training dataset is what the model uses in learning, e.g., setting the weights and biases (parameters) of a neural network. The Validation dataset is used to evaluate the model's fit on the training data while tuning the model's hyperparameters (e.g., the activation function in a neural network or learning rate in a gradient descent optimization algorithm) and is sometimes called the calibration dataset for this reason. It is not used directly to affect the model, but indirectly based on the model's performance. The test dataset is used to evaluate the final model's performance (after the model has been trained and tuned). It is important that the datasets have adequate coverage of the operational space. If not, the behavior of the AI component can be questionable. The datasets must be large enough and must sufficiently cover the operational space. Generative adversarial networks (GANs) can be a good tool for generating test data sets.

Data quality attributes are important to the correct functioning of AI/ML and should be included in the V&V of an AI/ML component. These include accuracy, currency and timeliness, correctness, and consistency, in addition to usability, security and privacy, accessibility, accountability, scalability, and lack of bias, among others. [Table N.4](#) provides the description of ML terms used in data testing. Tests are recommended to provide assurance that the data are valid with respect to the requirements, specifically in the areas of accuracy, relevance, completeness, and balance. In addition, data Validation steps can include file Validation, import Validation, domain Validation, transformation Validation, aggregation rule, and business Validation ([\[B9\]](#)).

Another challenge with AI/ML data is that of adversarial inputs. Szegedy et al. [\[B59\]](#) identified that several ML models are vulnerable to adversarial examples. This has been shown many times in image classification software, though adversarial attacks can be made against other AI tasks (e.g., natural language processing) and against techniques other than neural networks (in image classification), such as reinforcement learning models. It is important to address robustness to adversarial attacks in the V&V of an AI/ML component.

**Table N.4—Machine learning terms used in data test V&V discussion**

| Term                  | Description                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Accessibility (data)  | The extent to which a data subject can use or have visibility of his/her/their data (IEEE Std 7000-2021 <a href="#">[B27]</a> )                                                                                                                                                                                                                                                        |
| Accountability (data) | This concerns who has a designated function/role/task. This differs from responsibility, where someone has a duty to fulfill a certain task/ function which they fail to fulfil or do so inadequately, for which there are legal sanctions. The person merely accounts for their actions (i.e., provides an explanation) and nothing more (IEEE Std 7000-2021 <a href="#">[B27]</a> ). |
| Accuracy (data)       | Data is free of errors (IEEE Std 7000-2021 <a href="#">[B27]</a> )                                                                                                                                                                                                                                                                                                                     |
| Balance (data)        | A balanced dataset has approximately the same number of examples of each class.                                                                                                                                                                                                                                                                                                        |
| Completeness (data)   | Expected comprehensiveness; lack of missing data elements; usable data; all requisite data available                                                                                                                                                                                                                                                                                   |
| Consistency (data)    | Data across all systems reflect the same information and are in sync with one another                                                                                                                                                                                                                                                                                                  |
| Correctness (data)    | The degree to which data accurately reflects an event or object described                                                                                                                                                                                                                                                                                                              |
| Currency (data)       | Data currency is a temporal, not a financial, concept. Currency measures the degree to which the data is current with the world it models and is important in, e.g., land use models, healthcare models, and other models in which time is a vital factor.                                                                                                                             |
| Relevance (data)      | Refers to ethical relevance or relevance to the user's information needs                                                                                                                                                                                                                                                                                                               |
| Scalability (data)    | The ability of a data system to accommodate the growth of data, in volume/size or amount of traffic                                                                                                                                                                                                                                                                                    |
| Timeliness (data)     | This refers to whether the data is available when it is expected and needed.                                                                                                                                                                                                                                                                                                           |
| Usability (data)      | The ability of a user to derive useful information from data                                                                                                                                                                                                                                                                                                                           |

#### N.4.2 Closed-box and open-box testing

Both closed-box and open-box testing can be used with AI/ML or traditional components. Here we provide a definition, then describe techniques or tools for the application of these techniques to AI/ML. Closed box testing is testing conducted without knowing implementation details and, as such, tests the functionality of the system as a whole. It is typically used to test the functionality (expected outputs); non-functional attributes such as scalability, usability, and performance; and in regression testing and can be used throughout the life cycle. Closed-box testing is sometimes referred to as behavioral testing. Closed-box testing techniques for AI/ML systems include combinatorial testing (pairwise testing), statistical testing, back-to-back, and A/B testing, metamorphic testing, and exploratory testing [\[B57\]](#). For testing AI/ML, it is vital to provide coverage of the dimensions of variability in the operational space. Testing requirements like explainability, interpretability, and transparency require a human-in-the-loop.

When evaluating model performance, consider not only the model performance in terms of, e.g., accuracy, but also in time to decision, training time (especially if frequent retraining is necessary), robustness, safety, area under the receiver operating characteristic (ROC) curve (AUC), false alarm rate, and other performance

measures that reflect the requirements [B54]. These and other terms used in open-box and closed-box testing are described in Table N.5.

In terms of completeness, this standard requires Verification that application-specific requirements such as functional diversity, fault detection, fault isolation, and diagnostic and error recovery satisfy user needs. In an AI/ML component, fault isolation may be identified at the AI/ML component level, but at this time, not inside the technique itself. Advances in AI/ML explainability and interpretability will have use here. In addition, the results of the component can be monitored and, if out of range, could be handled like any other system with the intent that no results should cause failure propagation or an adverse condition. Especially for safety-critical continuous learning components, operational monitoring is necessary.

Be aware of concept drift (statistically significant changes to operational environment or required capability) in online or streaming AI/ML applications, and the need to detect it if relevant to the operating environment. If concept drift is not handled effectively (detect and update the model as needed), the AI/ML model can be rendered ineffective.

*Closed-box testing* can be used to verify that the AI/ML model is robust to abnormal inputs and adversarial inputs. It is important to verify that AI/ML does not exhibit unintended behavior such as overfitting, underfitting, and sensitivity (e.g., exhibiting large changes to small differences in the input space).

*Open-box testing* tests the internal structures or internal operation of the software. Code coverage, path testing, and data flow testing are examples of open-box testing for traditional software. Test coverage metrics have been proposed in the research, namely neuron, threshold, sign change, value change, sign-sign, and layer coverage, but there is currently little data on the test effectiveness based on the current literature using these metrics [B71]. Several tools have been developed for open-box testing of AI/ML, including the following:

- DeepXplore (Deepxplore: Automated whitebox testing of deep learning systems [B49]) implements open-box testing of deep learning systems.
- DeepTest (Deeptest: Automated testing of deep-neural-network-driven autonomous cars [B59]) is used to test deep neural network-driven autonomous cars.
- DeepRoad (DeepRoad: GAN-Based Metamorphic Testing and Input Validation Framework for Autonomous Driving Systems [B64]) uses Generative Adversarial Network (GAN)-based techniques to create realistic and diverse driving scenarios to find inconsistencies in autonomous driving systems.
- DeepSafe (Deepsafe: A data-driven approach for checking adversarial robustness in neural networks [B10]) identifies safe regions in the input space for classification problems.
- DeepCover (Testing Deep Neural Networks [B58]) is a coverage-based bug finder for deep learning applications. When a neuron's state changes, DeepCover identifies the change and its immediate causes.

**Table N.5—Machine learning terms used in open- and closed-box testing discussion**

| Term                           | Description                                                                                                                                                                                           |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A/B testing                    | Technique to determine the effectiveness of minor changes in a product or design where A represents the original version and B represents the modified version (ISO/IEC/IEEE 26513:2017 [B44])        |
| Area under the ROC curve (AUC) | Performance measures for classification problems at different threshold levels. AUC is the measure of separability between classes. The higher the AUC, the better the performance of the classifier. |
| Back-to-back testing           | Testing in which two or more variants of a program are executed with the same inputs, the outputs are compared, and errors are analyzed in case of discrepancies (ISO/IEC/IEEE 24765:2017 [B43])      |
| Combinatorial testing          | Combinatorial testing is a software test design approach in which test cases are designed to execute multiple combinations of values of the input parameters.                                         |



| Term                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Explainability                           | The extent to which the information made transparently available to a stakeholder can be readily interpreted and understood by a stakeholder. (IEEE Std 7001-2021 [B28])                                                                                                                                                                                                                                                                  |
| Exploratory testing                      | Exploratory testing is a software test approach that is a combination of simultaneous learning, test design, and test execution.                                                                                                                                                                                                                                                                                                          |
| False alarm rate                         | Rate of observation of an effect when no effect exists. False alarm rate = (number of false alarms / (number of false alarms + number of correct rejections)) = Type I error rate                                                                                                                                                                                                                                                         |
| Interpretability                         | The degree to which a human can readily understand the reasoning behind predictions and decisions made by a model.                                                                                                                                                                                                                                                                                                                        |
| Metamorphic testing                      | Metamorphic testing is a property-based software test approach that consists of comparing test results of inputs that have a known relationship (a metamorphic relation). The tester conducts tests with input $a$ and results $f(a)$ . Then the tester transforms or morphs $a$ into $a'$ whose results $f(a')$ are related to the test results of $a$ by a metamorphic relation (MR). A correct result occurs when $f(a) = MR(f(a'))$ . |
| Operational monitoring                   | Refers to collecting performance metrics at periodic intervals over time.                                                                                                                                                                                                                                                                                                                                                                 |
| Receiver operating characteristics (ROC) | The receiver operating characteristics (ROC) is a probability curve plotting the True Positive Rate vs the False Positive Rate at different classification thresholds                                                                                                                                                                                                                                                                     |
| Transparency                             | Characteristic of the transfer of information to a stakeholder, which is honest; contains information relevant to the causes of some action, decision, or behavior; and is presented at a level of technicality and in a form that is meaningful to the stakeholder. (IEEE Std 7000-2021 [B27])                                                                                                                                           |

## N.5 Approaches to the V&V of AI/ML

There are several approaches to V&V AI/ML components, including formal methods (e.g., formal proofs, model checking, probabilistic Verification), software testing, simulation-based testing, and experiments. Some specific approaches are as follows:

- Metamorphic testing to test ML algorithms, addressing the oracle problem ([B63])
- Formal Verification techniques to verify some AI/ML properties; the properties must be formally specified. Formal results must be mapped to a context understandable by stakeholders (e.g., regulatory or those accepting the component/system/product).
- An ML test score consisting of tests for features and data, model development and ML infrastructure, and monitoring tests for ML ([B3])
- Checking for inconsistency with desired behavior and systematically searching for worst-case outcomes when testing consistency with specifications.
- Corroborative Verification (A corroborative approach to V&V of human-robot teams [B62]), in which several Verification methods, working at different levels of abstraction and applied to the same AI component, may prove useful for the Verification of AI components of systems.
- Testing against strong adversarial attacks ([B61]), researchers have found that models may show robustness to weak adversarial attacks and show little to no accuracy when faced with strong attacks ([B1], [B61], [B5]).
- Use of formal Verification to prove that models are consistent with specifications ([B12]).
- Assurance cases combining the results of V&V and other activities as evidence to support claims on the assurance of systems with AI components ([B47], [B51]).

## N.6 Model evaluation

One of the primary tasks for AI/ML components is classification. Many references suggest the use of accuracy for the evaluation of classification. Practitioners should note that using accuracy for evaluation of supervised classification algorithms is insufficient. It reveals the average-case performance of models ([B46]). Given this caveat, we describe the basic determination of performance of a binary classifier. To do so, we quantify the ability of the underlying model to correctly predict the class or outcome. Practical



applications of binary classification are often not symmetric. That is, the relative proportion of different types of errors are of more interest than the accuracy. The confusion matrix is a classic way to enumerate the correct and incorrect answers that a model provides. The matrix denotes the number of true positives, false positives (Type I error), false negatives (Type II error), and true negatives. Numerous performance metrics have been defined based on the contents of the confusion matrix, including true positive rate (TPR) or sensitivity, true negative rate (TNR), or specificity, false negative rate (FNR) or miss rate, false positive rate (FPR) or probability of false alarm, among others ([\[B8\]](#), [\[B48\]](#), [\[B52\]](#), [\[B4\]](#)). An extensive discourse on the various metrics and evaluation of classification algorithms is provided in [\[B46\]](#).

## N.7 Operation V&V

Most of the discussion above has been on V&V prior to the operational use of the AI/ML system. However, as in the discussion about hazard analysis (subclause [N.3](#)), there are AI/ML V&V considerations during operation and maintenance. Continuous assessment of AI/ML systems can be conducted on the inputs to the system, the results of the AI/ML operation, or both. Input checks are used to provide assurance that valid inputs enter the system. Other checks on the inputs include monitoring for changes in the operational environment. The system should be monitored to provide assurance it is not consuming more computational resources than allocated. The results of the AI/ML system can be monitored to check for concept drift, the results of adversarial attacks, and performance degradation, for example. As hinted in subclause [N.3](#), it should be verified that the operational monitor functions appropriately and is neither too restrictive nor not restrictive enough.

## Annex O

(informative)

### Vignettes to illustrate application of IEEE 1012

#### O.1 Introduction

A key aspect of IEEE 1012 is the determination of the integrity level to establish “the importance of the system to the user and acquirer based on complexity, criticality, risk, safety, security, desired performance, reliability, or other system unique characteristics.” Integrity levels are then used to select the V&V tasks, activities, rigor, and the level of intensity of the V&V to be performed. [Annex B](#) provides a risk-based integrity level schema that defines the four integrity levels. This is to help guide the determination of the integrity level of a system. However, such determination should be more involved than applying a given definition. The importance and complexity of the design should be part of the consideration in determining its integrity level. These vignettes provide illustrations of incorporating such consideration in assigning the integrity level for a design, using examples of systems with various integrity levels.

The five vignettes in this annex are as follows:

Integrity level assignment of nuclear power system ([O.2](#))

Determining the integrity level for a closed-circuit television (CCTV) monitoring system onboard a rail vehicle ([O.3](#))

Application of IEEE Std 1012 to a Non-defense government system ([O.4](#))

Determining integrity level for training simulators ([O.5](#))

V&V of field programmable gate array (FPGA) based systems ([O.6](#))

## O.2 Integrity level assignment of nuclear power system

### O.2.1 Introduction

This vignette provides the thought process behind the assignments of integrity levels to the software components and system of interests of a hypothesized instrumentation and control (I&C) system using a typical pressurized water reactor nuclear power plant as an example. Specifically, examples are provided for determining the integrity level 4. In addition, this vignette discusses certain V&V tasks, as an example, for increased rigor and intensity for integrity level 4. Considerations for the independence of the V&V team are also discussed.

NOTE—The criticality analyses for nuclear plant designs with different technologies may end up at a lower integrity level due to the plant design, fuel safety (accident tolerant fuel), I&C architecture, or other safety features.

### O.2.2 What is a nuclear power plant?

The basic function of a nuclear power plant (NPP) is to convert the energy of nuclear fission into electricity for the consumer. An NPP is not all that different from coal, oil, or gas-fueled plants. The difference between traditional energy methods and nuclear energy is that nuclear plants do not burn anything. At a nuclear energy facility, the heat used to make steam is produced by fission. Turbines, powered by steam, turn to produce electricity.

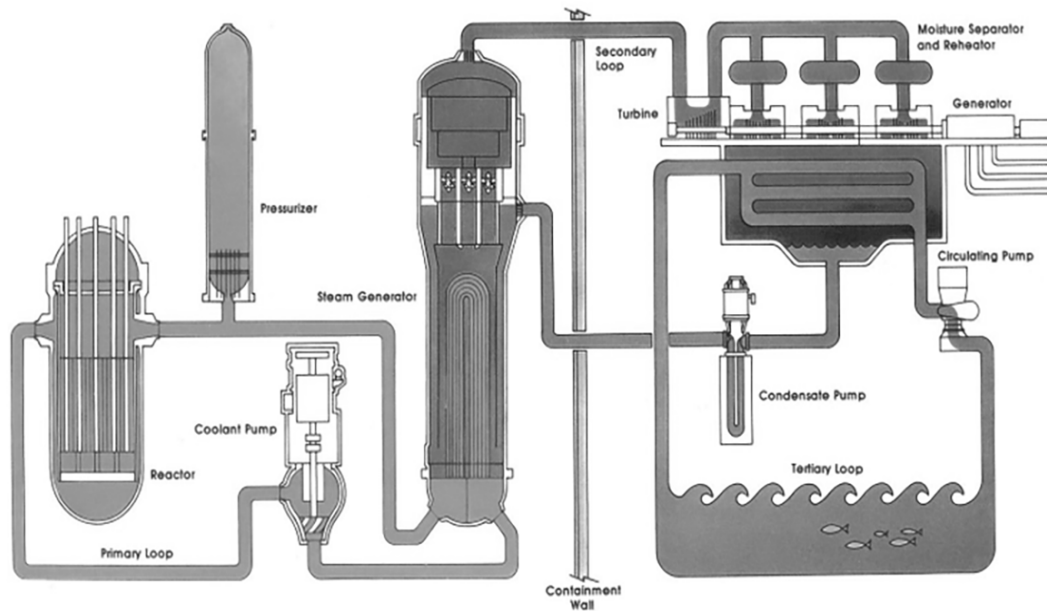
### O.2.3 Basic operation of a nuclear power plant

The basic principle of running a nuclear power plant with a pressurized water reactor (PWR) can be simplified in the following steps:

- a) Obtain thermal energy by nuclear fission of the nucleus of atoms of nuclear fuel.
- b) Generate steam by means of the thermal energy obtained previously.
- c) Operate a set of turbines using the steam obtained.
- d) Take advantage of the mechanical energy of the turbines to drive an electric generator. This electric generator will generate electricity.

A PWR power plant uses many separate systems to make this conversion, as shown in [Figure O.1](#). These systems are divided into the following categories:

- Primary systems
- Secondary systems
- Control systems



**Figure O.1—Systems of a nuclear power plant**

The primary systems convert fission energy into steam for the secondary systems. The secondary systems convert the steam into electricity. The control systems control key plant parameters in both the primary and the secondary systems, such as pressure, temperature, flow rate, and level. The control systems also allow the control room operators to monitor the operation of the key parameters and permit automatic controls. All automatic control systems can be shifted to manual control if necessary.

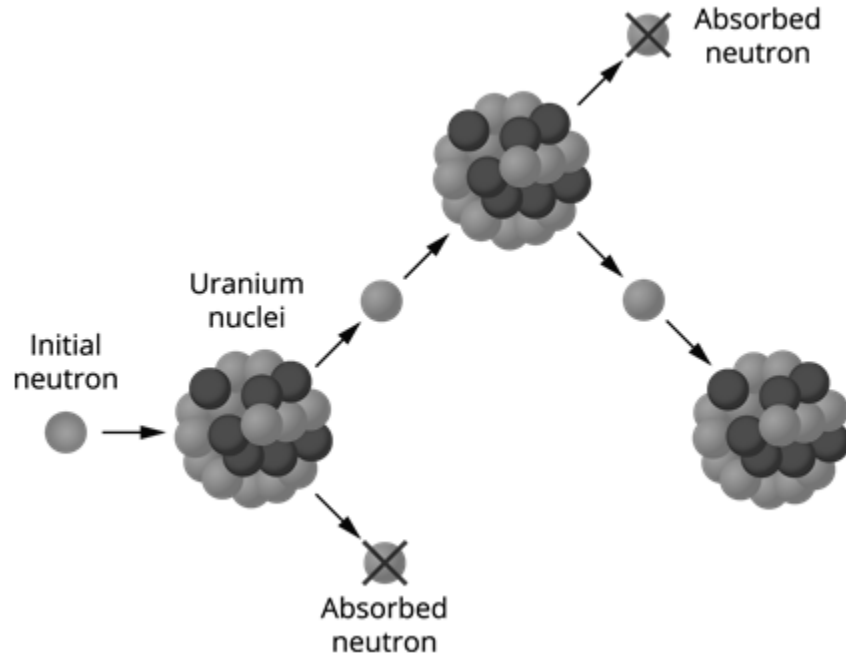
What makes the NPP different from other power plants is the way it produces heat via its primary systems.

#### **O.2.4 How is heat generated from nuclear fission?**

Fission is the splitting of atoms into smaller parts. Some atoms split when they are struck by neutrons. Each time this happens, more neutrons come out of the split atom and strike other atoms, causing a chain reaction as illustrated in [Figure O.2](#). The nuclear energy facility is able to control the chain reaction to keep it from releasing too much energy too fast. Through the chain reaction, fission becomes self-sustaining.

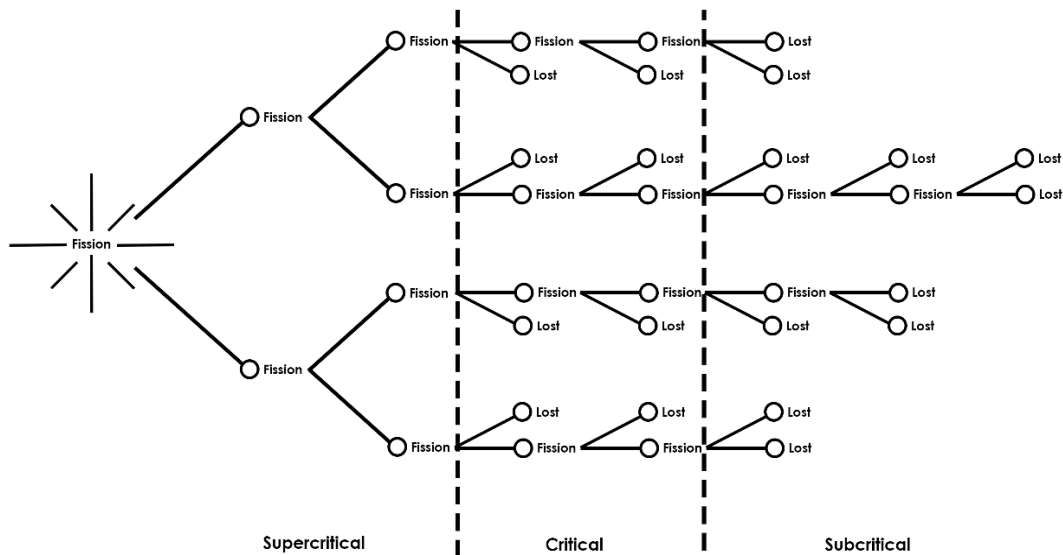
If the chain reaction is not controlled, it might lead to a catastrophic event.

Few natural elements have atoms that will split in a chain reaction; uranium is one of them. In commercial nuclear power plants, uranium is used as fuel. Fission reactions using uranium-235 as the fuel produce enormous amounts of heat capable of turning water into steam for use in generating electricity. The heat is produced from the collision of fissioned particles with other atoms, again converting kinetic energy into heat.



**Figure O.2—Controlled nuclear fission**

The release of two or three neutrons in fission is important. These neutrons can make other uranium-235 nuclei fission, releasing more neutrons, which are absorbed by the new atoms, causing them to fission, releasing more heat and neutrons. The repeating of this process over and over is a chain reaction and can continue as long as enough uranium-235 nuclei and neutrons are present. [Figure O.3](#) illustrates the amount and configuration of uranium-235 needed to sustain a chain reaction is called a critical mass.



**Figure O.3—Critical mass in a chain reaction**

Each fission produces heat, with the total number of fissions determining the amount of heat. The neutron controls the fission process. To maintain a sustained nuclear reaction, which is the objective of NPPs to

produce electricity steadily, the release of neutrons must be controlled. When the operator controls the neutron population in the fuel, the number of fissions, fuel heat production, and reactor power are controlled. Thus, controlling the neutron population in the fuel controls reactor power.

An absorber of neutrons must be present to control the number of free neutrons in the reaction space. Most reactors are controlled by control rods made of a strong material that absorbs neutrons (i.e., boron or cadmium).

## **O.2.5 Why is nuclear special?**

### **O.2.5.1 Chain reaction**

A major difference between a fossil fuel plant and a nuclear power plant is that the potential heat output of nuclear fuel is virtually unlimited. That is, it is possible to produce more heat in the fuel than can be effectively removed. This is due to the almost unlimited number of neutrons that can be made available to cause fission.

The chain reaction must be controlled.

### **O.2.5.2 Decay heat**

Each fission occurring in the reactor produces a great deal of energy. Most of this energy is released immediately after the fuel atom fissions. A small part of the energy produced by fission is released at a later time. Each fission produces fission fragments that are normally radioactive and are retained within the reactor fuel. These fragments, like all radioactive atoms, periodically give up some of their energy (or decay) in order to become stable, nonradioactive atoms. The release of energy from the radioactive fission fragments adds heat to the reactor fuel even after the reactor is shut down. Even though it decreases with time, the decay heat remains a significant enough value that removal of the heat must take place.

Decay heat must be removed even after the reactor is shut down.

## **O.2.6 Safety considerations**

Levels of physical barriers exist to contain radioactivity starting with the fuel assemblies to the reactor vessel to the containment building.

Numerous I&C systems help to prevent reactor accidents and reduce the effects if an accident were to occur due to equipment failure, human error, or a natural disaster. Those I&C systems that are specifically designed to shut down the reactor (to control the chain reaction) and monitor and control physical components to help remove decay heat are called “safety systems.”

For instance, to control the chain reaction, control rods in the reactor contain materials that absorb neutrons and prevent them from hitting atoms and causing more fission. The nuclear reaction can be increased or slowed down by varying (through control systems) how many of these control rods are withdrawn and how far. I&C systems exist to help the plant operator control the rods to adjust the amount of fission, thereby adjusting the power produced. If the control rods are inserted completely into the core, fission stops. Water flowing up through the fuel assemblies and control rods removes the heat of the chain reaction.

Safety systems automatically come on as soon as the plant operating parameters go outside of their normal levels. Critical safety functions have backups (e.g., redundancy) to increase reliability.

NPPs have systems that shut down the reactor when monitoring instruments find a problem. Leak detection equipment lets operators know any time the cooling system fails. In addition to the main cooling system, reactors have separate and independent emergency cooling systems, which automatically come on when safety sensors detect a loss of coolant. In case of a power failure, backup generators provide power to operate

cooling pumps and emergency equipment. The more recent generation of power plants utilizes natural forces (convection, gravity, etc.) to accomplish a safe shutdown without a need for external power.

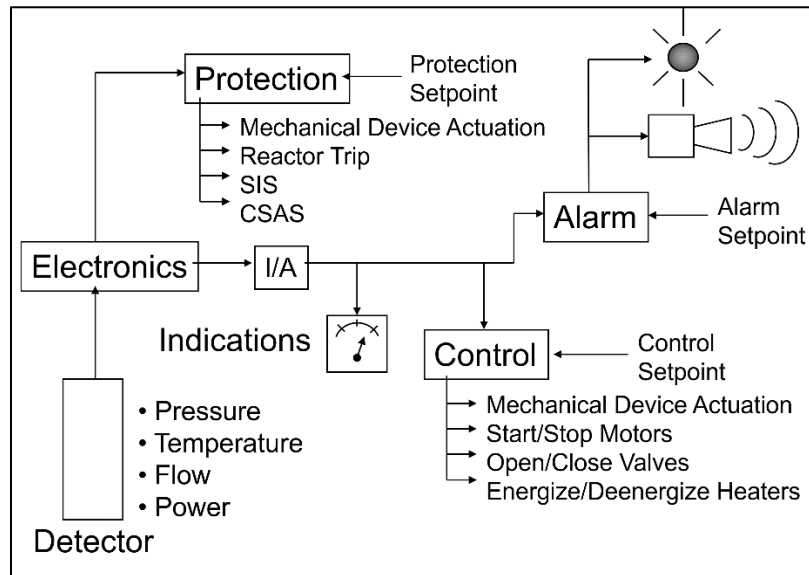
The automatic and manual controls of plant functions and equipment are accomplished through instrumentation and control (I&C), which includes hardware and software solutions. These safety systems need to be designed, developed, verified and validated at a higher scrutiny to provide assurance they will perform their functions correctly. This is where integrity levels play a role—to help determine the level of rigor and intensity necessary to apply to these control systems or components.

## 0.2.7 Instrumentation and control systems in nuclear power plants

Instrumentation and control systems utilize hardware and software to sense and monitor plant parameters, display parameters to operators, take input from operators, and send commands to actuate pumps, valves, and motors to safely operate and shut down the plant and protect the barriers of radioactivity.

Typically, an I&C system uses some electrical or mechanical detector to measure a plant parameter, such as pressure, temperature, flow, or power. That detector generates an electronic signal equivalent to the magnitude of the parameter. The electronic signal can be used to perform many functions, such as the following, as shown in [Figure O.4](#):

- Indications
- Control
- Alarms
- Plant protection



**Figure O.4—Components of a typical I&C System**

The operator uses the indication as information to control the parameter within a range of values. To aid the operator, the I&C system usually exerts some control over the parameters it monitors. To do this, the I&C system uses an electronic signal for the actual parameter and compares that to a second value, called the control setpoint signal. The set point can be a constant value, or it can vary within a specified range. Whenever the actual parameter signal differs from the setpoint signal, the I&C system causes a change in a plant system to make the actual match the setpoint. This relieves the operator from continuously monitoring and controlling hundreds of PWR systems, which is humanly impossible.

Usually, the parameter is allowed to deviate from the control setpoint within a specified range without any serious concerns for safe system operation. However, when the parameter exceeds the upper or lower range of control (i.e., exceeding an alarm setpoint), the I&C system produces an alarm. The alarm warns the operator of the parameter's abnormal behavior, and the operator is expected to take some action to return the parameter to within its normal range.

Usually, at the alarm setpoints, no harm can come to the system as long as the operator responds and corrects the condition. However, if the parameter goes beyond the alarm values and exceeds an "unsafe" protection setpoint, a separate safety I&C system generates an automatic protection signal.

### **O.2.8 Integrity levels**

Due to what makes nuclear energy separate from traditional fossil-fuel power plants, it should be designed to protect the public and the environment. The parts of the plant's I&C system that monitor, detect, and perform protection functions have to work. An integrity level of 4 is assigned to these functions so that required V&V tasks are applied to the design commensurate with the safety implications of the function, i.e., the failure of the function performed could cause catastrophic consequences.

Criticality analysis is intended to assign integrity levels to a hardware/software/system component based on the component's function. The identification of components that are required for the reactor protection functions determines which components are assigned integrity level 4. Usually, these functions are designed and implemented in a system that is separate and isolated from the remaining I&C systems (called plant control systems) to avoid creating new hazards to integrity level 4 system as a result of interfacing with the lower integrity level systems.

### **O.2.9 Reactor protection system**

Integrity level 4 is assigned to components where failure of such components may lead to catastrophic consequences. The I&C system that monitors, detects, and controls these critical components is usually called a reactor protection system (RPS).

One of the most important goals of a nuclear power plant is to prevent the release of radioactivity. The fuel offers the largest source of radiation and radioactivity. The fission of uranium-235 atoms produces immediate radiation and fission fragments. The fission fragments decay and produce additional radiation long after the fission occurs and even after plant shutdown. Because of this, a great deal of attention is paid to the physical condition of the fuel. The fuel rods encapsulate the fuel within the fuel assemblies, thus forming a barrier against radioactivity release. This is the first barrier.

If a fuel rod should break or crack, fission fragments would enter into the reactor coolant system (RCS). But as long as the RCS remains intact, the fission fragments would be restrained within the RCS. The RCS, therefore, acts as a backup, or second, barrier against the release of fission fragments.

If the RCS and the fuel rod fail, the fission fragments enter into the containment building. However, as long as the containment building remains intact, the fission fragments will be contained in the containment. The containment building, then, acts as a third barrier against release.

Because the fuel offers the largest potential release of radioactivity, the protection of the three barriers receives special attention. The term that encompasses this protection of the fission fragment barriers is engineered safety features (ESF). Any plant component (hardware or software) that initiates a reactor trip or ESF actuation in order to protect these barriers when the plant parameters reach unsafe conditions or in case of an accident is to be assigned integrity level 4.



### O.2.10 Criticality Analysis V&V task

To demonstrate how the Criticality Analysis task is performed to determine Integrity Level 4 assignment to necessary plant functions, consider the following safety analysis scenario:

One of the worst accidents that could occur is the loss of coolant accident (LOCA). During a LOCA, an RCS pipe breaks, releasing coolant into the containment. This loss of the second barrier is an assumption at the beginning of the LOCA.

The system designer determines what can be done to protect the other two barriers, fuel rods and containment. Both barriers are challenged during a LOCA. The loss of coolant accident (LOCA) causes coolant to leave the RCS and enter into the containment. The loss of coolant removes the medium that removes the heat from the fuel and transfers it to the steam generators. Without coolant, no transfer of heat can occur. This leads to the retention of heat within the fuel and fuel rods, causing their temperatures to increase. When the temperatures increase high enough, the fuel rods and the fuel break down, releasing fission fragments.

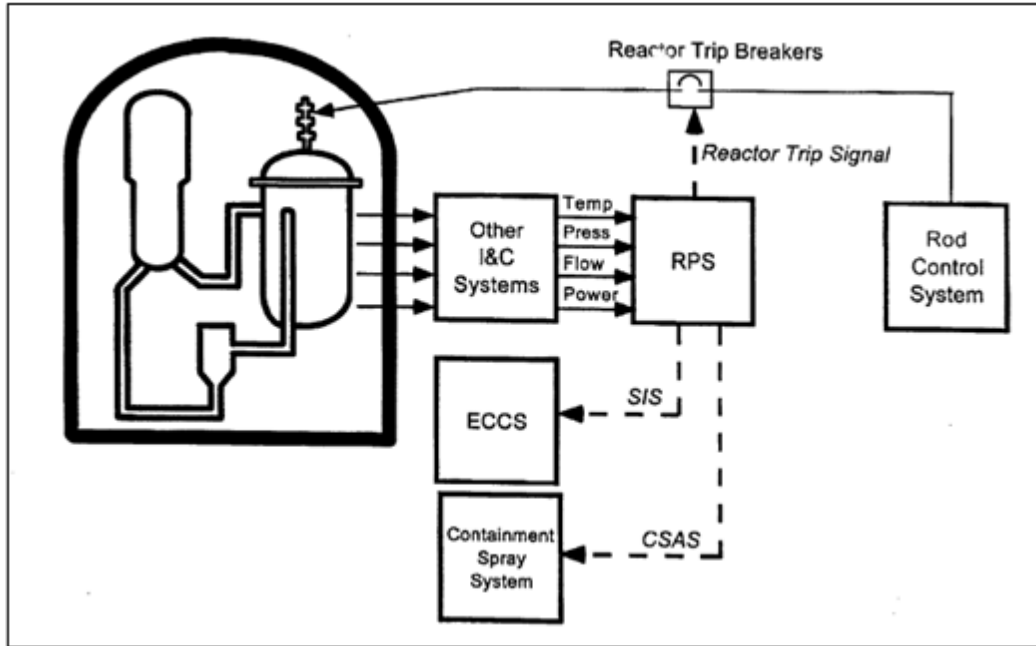
The reactor protection system (RPS part of the overall plant I&C system) monitors the following four key parameters in the plant to determine when an unsafe condition is achieved to warrant a reactor shutdown:

- Power
- RCS flow
- RCS temperatures
- RCS pressure

All of these create an envelope of protection. If reactor power production exceeds the analyzed value, the fuel's heat production exceeds the plant's heat removal capabilities, leading to fuel and fuel rod overheating and damage. If RCS flow decreases below the analyzed value, not enough coolant passes through the fuel. The coolant that does pass through the fuel increases in temperature, possibly boiling, which again leads to fuel and fuel rod overheating and damage. A similar sequence occurs for increasing RCS temperatures.

For the RCS pressure, a reactor trip occurs when the pressure decreases below the analyzed setpoint. With too low of an RCS pressure, the coolant boils, particularly near the highest heat area (the fuel rods). When that happens, less heat removal from the fuel occurs, and the fuel and fuel rods can again overheat and become damaged.

To protect the fuel and fuel rods against overheating and damage, the RPS initiates signals in three major categories shown in [Figure O.5](#): a reactor trip (shut down the reactor), a safety injection signal (SIS), and a containment spray actuation signal (CSAS).



**Figure O.5—Trips and actuations initiated by the reactor protection system**

To help protect the fuel, the fission heat production must stop. That requires a signal to cause the control rods to be inserted into the fuel assemblies. In the reactor trip subcategory, the RPS receives signals from other I&C systems. The RPS determines if the signals warrant a reactor trip, and if so, it sends a signal to the Rod Control System reactor trip breakers. This signal opens the reactor trip breakers, cutting off electrical power to the control rod drive mechanisms, thereby releasing the control rods and causing them to fall by gravity into the fuel assemblies. This action adds enough negative reactivity to stop the fission heat production.

The hardware and software components of the RPS that implement the reactor trip path (from sensors that produce indications of power and RCS flow/temperature/pressure all the way to the reactor trip breakers) are assigned integrity level 4 due to the critical protection nature of this function.

Remember that fission fragment decay continues to produce significant heat even after fission heat production stops, meaning fuel and fuel rod damage can still occur. One obvious resolution is to add a system to push an external source of water into the RCS and to pass the water over the fuel to remove the decay heat. The plant system that does it is called the emergency core cooling systems (ECCS). The signal that automatically starts the ECCS comes from RPS and is called a safety injection signal (SIS). Because the start of ECCS is vital to the protection of the fuel against overheating and damage, the I&C function that initiates SIS is also assigned integrity level 4.

The RPS (reactor trip) and ECCS can protect the first barrier, the fuel rods. However, the analyses also show that the third barrier, containment, needs protection too during a LOCA. The analyses show that if the RCS breaks, with its release of coolant, then the break causes containment pressure to increase. That is because the hot coolant immediately turns to steam when the pressure decreases. Thus, steam fills the containment, increasing pressure and temperature in the containment. With a high enough pressure, the containment can crack, causing a breach of the third barrier.

The method to protect the containment against high pressure is to cool the containment atmosphere—turning the steam back into water and lowering the pressure—through the containment spray system and the containment cooling system. This system sprays water into the containment building, cooling the air temperature and lowering the pressure. An automatic start signal for these systems is necessary. This signal also comes from the RPS and is called the containment spray actuation signal (CSAS). It is again assigned integrity level 4.

### **0.2.11 Improving reliability of I&C safety system**

Because these protection systems have radiation release connotations, the RPS of the I&C system has to be reliable. To increase reliability, redundancy is introduced in the system architecture. The number of complete (redundant) systems, or trains, depends on the reliability goals and can be anywhere between two to four.

Each train has to be fully capable (100 percent capacity) of protecting the barrier(s) without the other trains. To maintain operability, each train receives electrical power from independent sources. So that a problem in one train does not affect the other trains, they must be physically separated. This ensures that the loss of any single component affects the operation of only one train, leaving the other trains fully capable of barrier protection.

In the case of the RPS, that means the same logic is being implemented in separate trains, and these trains communicate with each other. That necessitates a voting function to be implemented in the RPS. This is called the local coincidence logic (LCL) subsystem, where each train receives input from other trains and votes on these signals before determining the outgoing trip or actuation signals. Voting should consider the quality of the signal (e.g., the sensor may be offline or a cable break may have been detected) or whether the train is under maintenance. When this new functionality is added to the existing subsystems, the criticality analysis dictates that the LCL subsystem (software) and the communication path between the trains (hardware) be integrity level 4.

### **0.2.12 Considerations for building the system when adding new functions**

Additional features may be added to the RPS for error detection (diagnostics), parameter update, or online or offline test injection features for maintenance. Each of these features is evaluated for their impact on the safety functions and, if they cause hazards to the integrity level 4 components, they are also assigned integrity level 4.

The integrity level of a component may be lower than integrity level 4 if it can be proven that it has no impact on the operation of the safety components. Even if the added feature is not to perform a safety function, the architectural design might necessitate the function be assigned integrity level 4. Examples of this include if the function is implemented in the same unit alongside the integrity level 4 software.

### **0.2.13 Rigor and intensity of V&V tasks based on integrity level**

This standard identifies the minimum required and optional V&V tasks for each integrity level. Many tasks are common across the integrity levels. However, the rigor and intensity applied to perform these tasks increases for the higher integrity level systems and components.

For instance, while generating test cases against the functional requirements of a system as part of qualification testing is an acceptable level of Validation effort for integrity levels 2 or 3. However, for integrity level 4 systems, the intensity is increased to include negative testing to provide assurance the system has implemented adequate safeguards against abnormal operating conditions.

Hazards analyses consider impacts from postulated failures of interfacing systems of lower integrity levels.

Additionally, the V&V activities are rigorously documented to help regulatory bodies follow and verify the objective evidence produced. Such strict documentation rules may not be necessary for integrity level 1 or 2 systems.

The use of automated tools should be considered to reduce the toil, reduce human errors, or cover more ground (e.g. more test cases) than what would be possible with limited resources. Where possible, simulation of the plant behavior based on approved models can provide additional reassurance or expand the test boundaries for scenarios that may be destructive to the equipment or for those scenarios that are too risky to exercise after installation at the site. If a simulator or a test tool is to be used as part of the Validation of the

system, it should also have been validated to the same rigor commensurate with the integrity level of the functions being tested. Any error in the simulation tool or other key analysis tools can cause a latent design error in the NPP system to remain undetected by the V&V analysis process.

Using an independent V&V team is also a consideration, and usually a requirement imposed by the regulators or clients, when performing V&V tasks. Classical or modified independent V&V should be strongly recommended for Integrity Level 4 systems and components.

Although a representative (surrogate) hardware may be used during the Validation of software functions, the target (installed) hardware is equally important to the performance of the integrity level 4 software in a system setting. Therefore, increased rigor on hardware V&V and qualification is necessary for systems at integrity level 4. Some activities that should be considered for to design, assembly, and qualification of hardware components and system include the following:

- Performing necessary V&V activities to give reasonable assurance that the COTS HW used in the system will perform its intended integrity level 4 functions,
- The hardware design to be in accordance with a set list of system/hardware requirements. Tracing each hardware requirement to a suitable fulfilment which includes testing, inspection, tracing to drawing or tech spec.
- Physically separate and electrically isolate safety assemblies (Integrity Level 3 or 4) from interfacing assemblies of lower integrity levels. This is mainly done through isolation devices and physical separation.
- Regardless if it is a first-of-a-kind or off-the-self, the hardware should be qualified in its intended environment. This includes temperature and humidity, electromagnetic compatibility (EMC), and seismic conditions. Another testing that is needed, such as isolation between safety and non-safety circuits, should be considered. Note these types of testing can be done individually or as part of an assembly.
- Validate performance requirements for the specific hardware used in the installed system (e.g., time response).
- Validate correct assembly (e.g. exercise all signal paths as part of the qualification testing) and other quality control/inspection activities.

The IV&V plan for the nuclear power plant (integrity level 4) would require all minimum V&V tasks (as identified in IEEE Std 1012) to be performed and because of the critical characteristics of the NPP, would strongly recommend that optional V&V tasks such as equations analysis, simulation tool V&V, full-spectrum testing V&V and failure mode and effects analysis (FMEA) be performed to provide assurance that a comprehensive, disciplined analysis of the NPP critical system requirements have been validated. The concept of applying greater rigor and intensity to V&V tasks is an additional vital and necessary method of achieving a comprehensive and disciplined analysis of the NPP system requirements.

To illustrate the concept of greater rigor and intensity as applied to V&V tasks, the following are some examples to illustrate how the formalism (rigor) and greater breadth and depth (intensity) to some sample V&V task analysis:

- Traceability analysis (minimum V&V task—greater rigor): The normal traceability analysis shows where each system/software/hardware requirement is being addressed in the design or implementation. The V&V analysis of the traceability mapping to design and implementation (requirements traceability matrix) provides evidence of the completeness of how each system/software/hardware requirement is being addressed. For critical systems such as the NPP system Integrity Level 4, additional rigor can be applied by conducting a formal analysis of how critical requirements are compartmentalized into containable design and implementation modules, thus isolating the functional criticality into known and measurable modules of the system/software/hardware. In addition, formal data coupling analysis methods can be applied to assess how critical data elements are isolated from interference or inadvertent modification by less

critical design and implementation modules, thus limiting any erroneous or extraneous changes to critical system integrity level 4 data.

- Equations analysis (optional V&V task—greater rigor): In cases where there are multiple choices in the selected algorithm or system performance calculations, the system integrity level 4 V&V analysis may be required to rederive the basic equations from formal engineering/mathematical principles to ensure that there are no solution singularities, unusual boundary conditions, and accuracy sensitivities in the selected equations or calculations that could cause the system to operate in a more hazardous condition. This additional analysis provides a rigorous examination of the system assumptions that affect the operations of the NPP system under both normal and potentially hazardous conditions.
- Full spectrum testing (minimum V&V task—greater intensity): For the critical system modules and requirements, the V&V testing can take on a greater intensity of effort by testing across the full spectrum of test conditions, such as stress conditions, abnormal events, boundary values, and comprehensive path coverage test patterns.

### **O.3 Determining the integrity level for a CCTV monitoring system onboard a rail vehicle**

#### **O.3.1 Introduction**

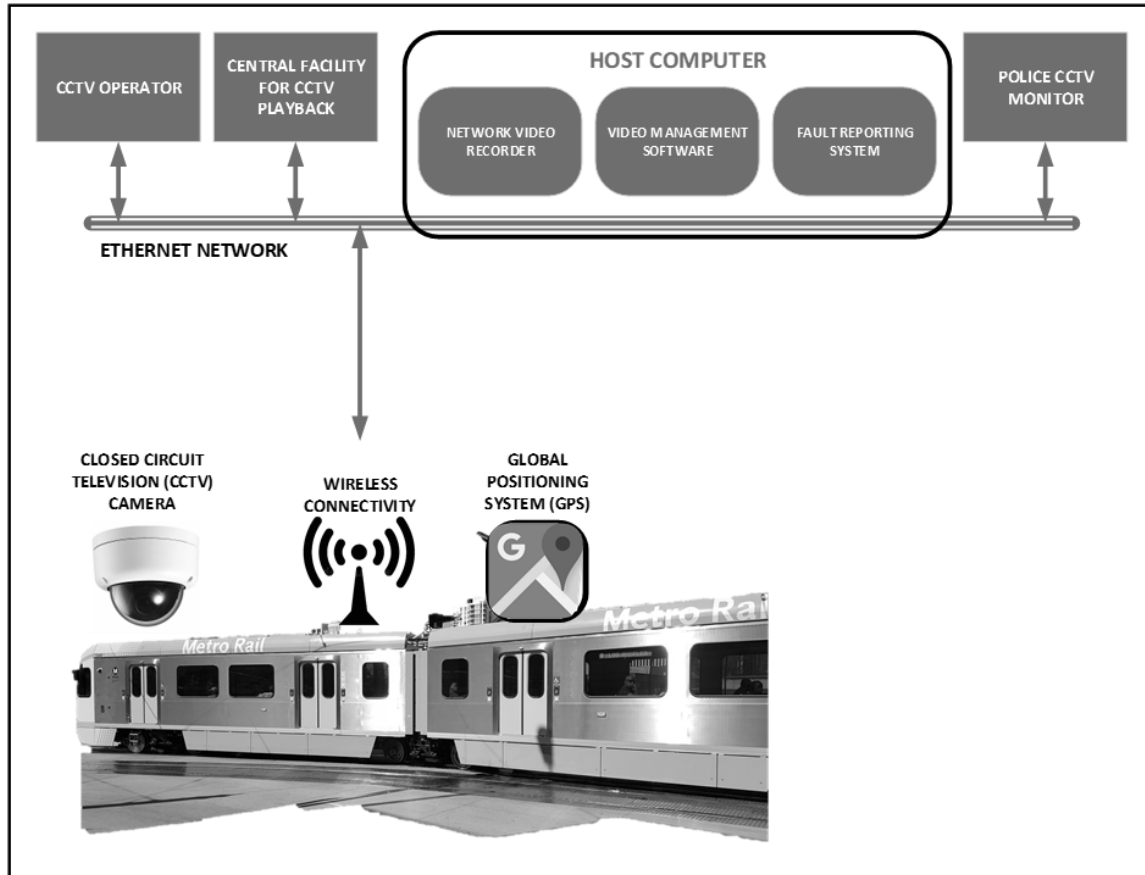
A key component of IEEE Std. 1012 is to use the determination of the integrity level to establish “the importance of the system to the user and acquirer based on complexity, criticality, risk, safety, security, desired performance, reliability, or other system unique characteristics.” Integrity levels are then used for the determination of the V&V tasks, activities, rigor, and the level of intensity of the V&V to be performed. [Annex B](#) provides a risk-based integrity level schema that defines four integrity levels. This is to help guide the determination of the integrity level of a system. However, such determination is more involved than applying the given definition. As such, the following vignette provides an example of the considerations one would contemplate to determine a given system’s integrity level and if the use of commercial off-the-shelf (COTS) software is appropriate. The importance and complexity of the design should also be part of the consideration in determining a system’s integrity level.

#### **O.3.2 Purpose**

The purpose of this vignette is to provide an example of how to approach a system that has a mix of integrity level 1 and integrity level 2 components where the system uses COTS components. This example pertains to a CCTV and video recording system of a railcar transporting people as a public service.

#### **O.3.3 System description**

Video surveillance is an important security tool to help protect passengers, employees, and equipment. The CCTV and recording system provides the ability to monitor situations inside railcars (vehicles) and access live and recorded video on demand from a police monitoring facility. [Figure O.6](#) illustrates the system as described.



**Figure O.6—Block diagram of a metro system’s CCTV monitoring and recording system**

The on-board CCTV provides a unified and secure method of video capture and storage, as well as the ability to rebroadcast videos for external use within a secure context e.g., to the police CCTV monitoring facility. further, the train operator has a CCTV monitor in the train cab where triggered events will play automatically. A triggered event is an event that signals the CCTV system to focus the camera and increase the frame rate, such as the opening of a door, a passenger initiating a call with the Operator, or other user-defined events that can potentially be defined as hazardous or unsafe. Triggered events do not cause changes to the train’s operation, such as a braking or emergency stopping event.

All CCTV on-board subsystems and components are connected to the CCTV ethernet network that provides communication between cameras and the host computer. In this manner, videos can be uploaded and played back at, for instance, the central facility. Videos can also be extracted from the vehicle and reviewed anywhere else. Further, the system uses a geographical positioning service (GPS) in the different cars of the train to identify the exact location of the recorded footage.

The CCTV and recording system consist of network video recorders (NVRs) and video management software (VMS). An NVR is a device that digitally records video data, audio data, and its associated metadata. NVRs are used to manage video equipment, multiple cameras, and vehicle activity through a single software solution. The VMS works in conjunction with the NVRs to manage video, audio, and associated metadata recordings to dynamically control the behavior of the recorder. The VMS allows operators to view live and recorded audio and video sequences, and provides alarm monitoring and post-event analysis while also supporting encryption of the audio and video data at the time of recording.

The NVR consists of the NVR platform, operating system, and application software. The NVR is a robust railway grade high-performance platform. The system includes self-diagnostics and fault reporting to monitor and alert of failures of any system component. When a failure is detected, the system will generate an alarm

that will be sent to the Fault Reporting system. These failures can be hardware, software, or connectivity failures.

The NVR selected was a COTS application used by other rail transportation companies. Using a COTS NVR enables users to configure computing performance based on their needs and allows the NVR to support a wide variety of cameras and utilize the latest generations of network video encoding and streaming protocols.

However, there are concerns when using COTS because their quality and reliability may not be demonstrated or documented throughout their development in accordance with industry standards or with a sufficient level of quality and reliability compared to a product specifically developed for the application. The V&V team relies on the use of industry standards to develop components to provide assurance that the system will work as required. If the use of industry standards is not confirmed, then the V&V team will need to perform testing and V&V to confirm that the system satisfies its intended requirements (not merely the engineering specifications) and that assessment of the component quality is performed throughout the system life cycle.

### **0.3.4 Critical characteristics—Determination of integrity level**

IEEE 1012 identifies the V&V tasks in relation to the organization's integrity level schema (see Table 2a through Table 2d). This standard uses integrity levels to describe the importance of the system, software, and hardware, varying from high integrity to low integrity, to the user. For this vignette, we use the risk-based integrity level schema described in [Annex B](#), which defines four integrity levels, levels 1 through 4.

To determine the integrity level of the train monitoring system, the V&V team considered the design and complexity of the system and the importance of its mission, i.e., to monitor and record activities in the vehicles to protect passengers, employees, and equipment.

The failure of the monitoring system will result in the following:

- Prevent operators from viewing CCTV footage or listening to audio from the cameras in case of emergency events occurring inside the railcars.
- Prevent the police monitoring facility from knowing if an event is occurring in a train, and from assessing the consequence in other trains or the entire rail network.
- Prevent incident investigators (police or rail) from knowing or understanding the incident and what led up to the incident. It can also prevent the determination of participants, co-occurring events, and timing of the incident.

### **0.3.5 Questions to ask and answer related to the CCTV and railcar system**

If the features of the CCTV system are unavailable, would the managers of the railway system consider the railcar with the non-functioning CCTV system out of service?

- Not likely: If another fully functioning railcar with an operational CCTV system were available, it would be used first. However, the absence of a functioning CCTV system would not render a railcar non-operational, simply as not fully functional but available for use.

Does the CCTV system contain any failure modes where if the system or one of its components were to fail via overheating would it pose a fire hazard to occupants of the railcar or the system operators?

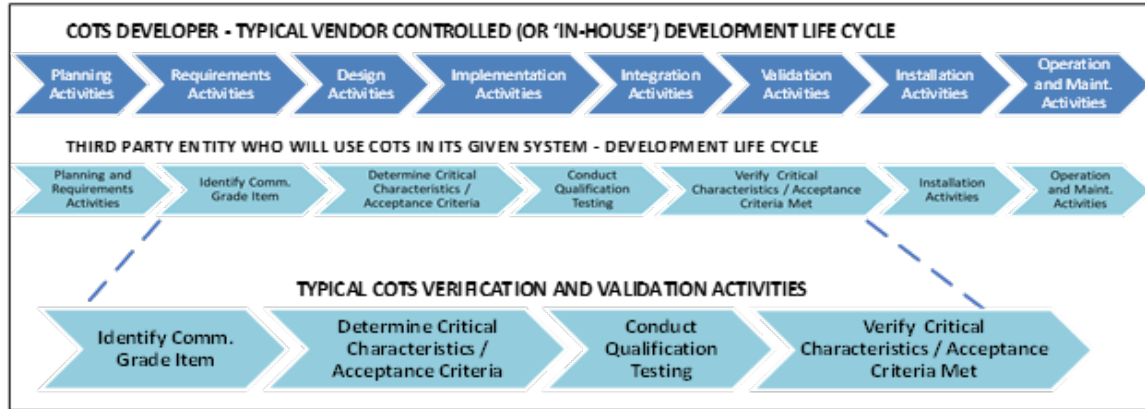
- Not likely: The components within the railcar do not pose any additional risk due to malfunction of new potential heat sources (e.g., new application or use of lithium-ion battery technology), as the power for the new CCTV system will tap into existing power supplies and/or the backup battery power.

Therefore, the CCTV system, including the NVR, is defined as an integrity level 2.

### 0.3.6 Verification & Validation

This vignette shows the process of applying V&V activities to the development of the application software. If the user has access to the source code, a traditional V&V process could be followed since modifications could be made to the code.

Figure O.7 depicts an example of the development life cycle for vendor-controlled development of a system, and how it interrelates with a third-party development life cycle and example COTS V&V activities.



**Figure O.7—Example of a development life cycle for “in-House,” third party, and associated COTS V&V activities**

Because we are using COTS and assuming no access to the source code, the assurance of proper functioning will require a combination of several approaches during all phases of development to assure that the system will perform when needed. This may include the following:

- Hazard analysis, especially hazard identification.
- Intensity and rigor of testing.
- Requirements Validation, including requirements/constraints (as a result of these V&V evaluations, it may be recommended to be independently verified at a higher integrity level).
- Architectural specification Verification.
- System detailed design Verification, including constraints.
- System implementation Verification, including coding and constraints.
- Software/system tool testing.
- Use of the given system, including information to support reasoning about residual uncertainty on the assurance that the functions will be performed as intended. The specific goal is to assure that the function is performed on demand and its performance is not degraded.

### 0.3.7 Hazard analysis

During the V&V process, the hazard analysis may produce cross-phase iterative cycles to validate system requirements. For example, a hazard analysis performed on the work product of a particular phase may lead to a change in the work product of the preceding phase.

For the NVR application software, a preliminary analysis should be performed to identify inherent hazards and contributory hazards and to formulate requirements and constraints to eliminate, prevent, or otherwise control the hazards. As the system development progresses, requirements are refined with additional safety constraints identified through the hazard analysis.



Starting from the initial hazard analysis performed on the functional concept, the hazard analysis is revised at every phase of the development life cycle, with change-control management and configuration management. This process is iterated for each added hazard control requirement (especially looking for any verifiability-decreasing side effect) until no new hazards are identified.

Since the monitoring system is a level 2, a hazard analysis is not required per the standard. However, although the defined system was determined to integrity level 2, which does not require a hazard analysis, the customer could request a hazard analysis regardless of the determined integrity level.

### **O.3.8 Intensity and rigor of testing**

The intensity and rigor applied to the V&V tasks vary according to the integrity level. Higher integrity levels require the application of greater intensity and rigor in the V&V tasks. Intensity includes a greater scope of analysis across all normal and abnormal system operating conditions.

The intensity and rigor applied to the V&V tasks for this system are appropriate for the system's integrity level (level 2). Regarding intensity, we will analyze normal and abnormal operating conditions as documented in industry operating guides. The amount of rigor of V&V for this integrity level does not include formal methods; however, all test results are recorded in the test results document.

### **O.3.9 Requirements specification and Validation**

#### **O.3.9.1 COTS definition**

Commercial off-the-shelf or commercially available off-the-shelf products that are packaged or 'canned' hardware or software, which are adapted aftermarket to the needs of the purchasing organization, rather than the commissioning of custom-made, or bespoke, solutions.

In this case, COTS includes adaptable software configured for the train CCTV cameras, Network Video Recorders, and interfaces.

#### **O.3.9.2 COTS consideration (when to use COTS)**

The following actions could be taken to determine when the use of COTS is appropriate:

- Determine if the software is in use elsewhere and ask for the experience of this (these) user(s).
- Make a system description of what you want the COTS to do, what are the main features, and where compromises can be evaluated.
- Perform a trade-off study of various applicable COTS products.
- Evaluate if COTS and its components could experience any failures so the potential failures are addressed at each stage of the life cycle.
- Analyze the Pros and Cons of the COTS software vs. a custom designed product.
- If a large number of modifications are required, study these and make an assessment of the modifications and amount of effort and compare to a custom development.
- Ask the COTS vendor for a demo and test period (V&V) to evaluate the product.
- Assess the feasibility of changes to the product if not completely satisfactory.
- Determine the feasibility and cost of transitioning from the old to the new system.
- Calculate the fixed and operational costs for the life of the product.
- Evaluate the cost related to a long-term single vendor.
- Determine the cost related to early termination of the COTS contract.
- Determine the cost associated with maintaining custom features.

- Identify the legal arrangements regarding proprietary data related to the COTS product.
- Analyze the source code if available.
- Analyze any test data from the software's product development if available.
- Review all documentation that has been developed.
- Any other concerns you may have.

Using a traditional development process to design and develop the application software and identify the part implemented in COTS, the first step is the concept phase, in which system requirements and desired system functionality are identified. Then, determine if these are available in COTS or can be developed in the application software. In this manner, the gap between what COTS provides and what is needed will help identify system/software requirements and design for the NVR. During requirement analysis, it is important to identify what can be handled with the COTS and how to use the COTS.

The next step would be to determine if additional functionality is required. This also requires determining how these requirements are translated to requirements for the subsystems, components, and elements that comprise the system. The use of COTS for the NVR imposes restrictions because the firmware (if any) cannot be modified or customized. Therefore, the requirements for the application software should identify any restrictions that the COTS may impose.

Then, the user should verify the system requirements with the objective of answering the question that the right system will be built (i.e., requirements are consistent and there are no conflicts across requirements). This information would be used to define items or artifacts that would be tested to demonstrate the correct operation of the system. Deficiencies in requirements are the highest contributing factor in the failure of a system to satisfy its intended goal, as experienced in all safety- and mission-critical application sectors. This confirmation also implies confirming that hazardous conditions (e.g., unspecified functionality) are not introduced. Once the hazards and contributing factors are identified, constraints can be formulated to prevent these hazards and their contributors.

The hazard analysis is independent of development and includes a change impact analysis and configuration control.

Using the requirements, the next step is to develop the detailed design, which is the specification to be satisfied by the implementation (e.g., for software, the program, code, or firmware). Also, requirements are used to identify test cases, which will help reduce uncertainties associated with the system design, detailed design, and implementation. Then, test procedures are generated from the optimized test cases based on the constraints defined in the test objectives.

### **O.3.10 Architectural specifications**

Using the requirements, the next step will be their translation into components and parts that will form the hardware of the COTS. This is important since COTS tends to include features and capabilities that may not be needed for the application and may increase the cost. Further, all these features or hardware components might not support the application's requirements or the required integrity level. Therefore, the user should consider what the system requires to operate and consider the features and accessories included to support or affect the different aspects of design. In this manner, the user would not incur additional costs and introduce unnecessary features and performance issues. For example, for the NVR, the COTS should include communication capabilities and provide information to rail personnel and/or police facilities.

### **O.3.11 Detailed design**

Once the system requirements and architectural specifications are established and verified, the developer proceeds to design the system. During this phase, the developer would specify the system details for implementation of the application software and interconnection with the system architecture. The developer should begin to address how the source code and application software will be developed using commercial software and then implemented in the system. The developer should consider the distribution and density of

signals and functions in the system, so the failure of a signal or module does not cause a total or partial shutdown of the system.

Then, the developer should evaluate the design approach against the results from the safety and hazard assessment to determine if deficiencies exist and their potential impact on safety. If deficiencies are identified, then the developer should identify additional requirements for the system to eliminate the impact on safety. The developer should re-evaluate the safety requirements and confirm that they are appropriate.

To assure proper functionality of the system, the developer should use a clear, comprehensive, and unambiguous set of functional requirements and design specifications against which the functions should be checked during design, manufacture, installation, and service, and will be used as a reference for any in-service modifications.

The system design should identify the COTS selected for the application, how to customize the COTS, and the hardware required [e.g., programmable logic controller (PLC) or digital computer system (DCS), local or remote input/output (I/O) racks, modbus, or fieldbus, etc.]. This would also help when identifying testing methods, testing tools, and acceptance criteria. This information can later be used for Validation of the system during factory acceptance testing to confirm that the system works as intended.

In addition, using the hazard analysis and system requirements, the designer should identify unwanted COTS functions or features and how to manage them so that the system performs as required.

The V&V activities are identified during the design specification to satisfy the requirements that flow down from the system specification such that the design specification satisfies the conditions to achieve the characteristics necessary for proper system operation. The system life cycle needs to be modified to address the use of COTS and the development of the software application for the rail system. Also, V&V activities would require modifications since there might be limited access to the COTS' original code (or operating system).

The reader is referred to [Annex D](#) for COTS considerations.

### **0.3.12 System implementation**

During implementation the design is transformed into a work product (e.g., a software component), ready for integration with other components of a system. Although the integration of components is also part of the implementation, deficiencies encountered during integration and integrated Verification are rooted in deficient upstream work products. Prevention of such deficiencies in implementation can reduce uncertainties associated with testing the implemented unit. The V&V team should determine whether implementation satisfies all system requirements. Then, the intent of the implementation is to satisfy the flowed-down requirements, including the flowed-down conditions. Although not required, an independent assurance could be used to confirm that the system performs its function when demanded.

Many people have seen that defect discovery occurs during implementation and integration testing, which has caused unexpected costs and delays. This issue resulted from inadequacies in earlier phases. Even after applying the right combination of best practices and state-of-the-art approaches, some uncertainties may remain in the preventative approaches and the resulting specifications and flow-down of requirements, architectural specifications, and detailed design specifications. Therefore, changes may be necessary after the system is configured. Nevertheless, these documents can be used to identify the effect a required change could have on the system design.

In addition, the use of COTS requires consideration of integration of the COTS with other systems since different vendors tend to use their own codes and protocols. Therefore, additional software might be necessary to bridge the gap. Verification during this phase should demonstrate proper operation of the software in the event of a hardware failure.

During system integration and Validation, the tester should emphasize evaluating the impact of the COTS on the rail system and repeat any tests as necessary to demonstrate that the system complies with its safety requirements and is appropriate for the intended function. If, based on the system's determined ILINTEGRITY LEVEL and/or the desires of the system developer, this testing should also evaluate that all hazards identified in the system hazard analysis (if one is determined to be desired or required) have been addressed.

### **O.3.13 Tool testing**

Once the system is designed and integrated, the question arises of how to test it to confirm that the system performs its functions when demanded. The first thing to consider is that a testing tool can be used with the system and with its interconnected devices, particularly with proprietary devices. For tool selection, the tester should consider versatility to support a wide variety of test cases, reliability, repeatability during each testing phase, compatibility with the system and its operating system, and efficiency in performing all test cases. The selection of the tool also depends on the necessary rigor of the process used for its development, Verification, and Validation. The test tool should be able to test all the test cases identified in the system test specification, so that all application features are tested. If commercially available tools are not feasible, then a custom tool might need to be developed.

### **O.3.14 Deployment/operation**

After demonstrating that COTS digital devices adhere to the functional, safety and environmental requirements, the system is ready for final site testing and use at the rail system.

## **O.4 Application of IEEE Std 1012 to a Non-defense government system**

### **O.4.1 Introduction**

A key component of IEEE 1012 is to use the determination of the integrity level to establish “the importance of the system to the user and acquirer based on complexity, criticality, risk, safety, security, desired performance, reliability, or other system unique characteristics.” Integrity levels are then used for the determination of the V&V tasks, activities, rigor, and the level of intensity of the V&V to be performed. [Annex B](#) provides a risk-based integrity level schema that defines the four integrity levels. This is to help guide the determination of the integrity level of a system. However, such determination should be more involved than applying the given definition. The importance and complexity of the design should be part of the consideration in determining its integrity level. This vignette provides an illustration of the incorporation of such consideration in assigning the integrity level for a design, using a non-defense government system, with issues such as management of large volumes of data, distributed data, identification, and privacy as an example.

### **O.4.2 Background**

Originally, government contracting was a labor-intensive process. Federal agencies created their contracts with lots of paper and pens and then typewriters. Each manually created contract and its related attachments were then placed in a folder, and the folder was placed in a file cabinet. The first upgrade came about when the agencies used contracts-creating software which allowed attachments to be affixed. The newly created contracts were then printed out, placed in a folder along with the attachments, and placed in a file cabinet. The storage and disposition of the paper files were done in accordance with the rules promulgated by the National Archives and Records Administration (NARA). The second upgrade came about years later, after the advent of cloud-based software known as Software as a Service (SaaS). This project was initiated as part of the second upgrade.

The objective of the Project is to implement an electronic contract filing system (ECFS) that has the look and feel of a paper filing system. A number of issues were identified, and these acted as the parameters for the selection of a new system. These included the following:

- Requirements used to develop the old system were now lost and could not be used to create the new system.
- New software would be cloud-based (SaaS).
- New software would be accessed through the Federal Risk and Authorization Management Program (FEDRAMP), and software would have to be already approved by FEDRAMP to enforce conformance to National Institute of Standards and Technology (NIST) security requirements.
- Created data (contracts, attachments, metadata) that would be stored on local (agency) servers.

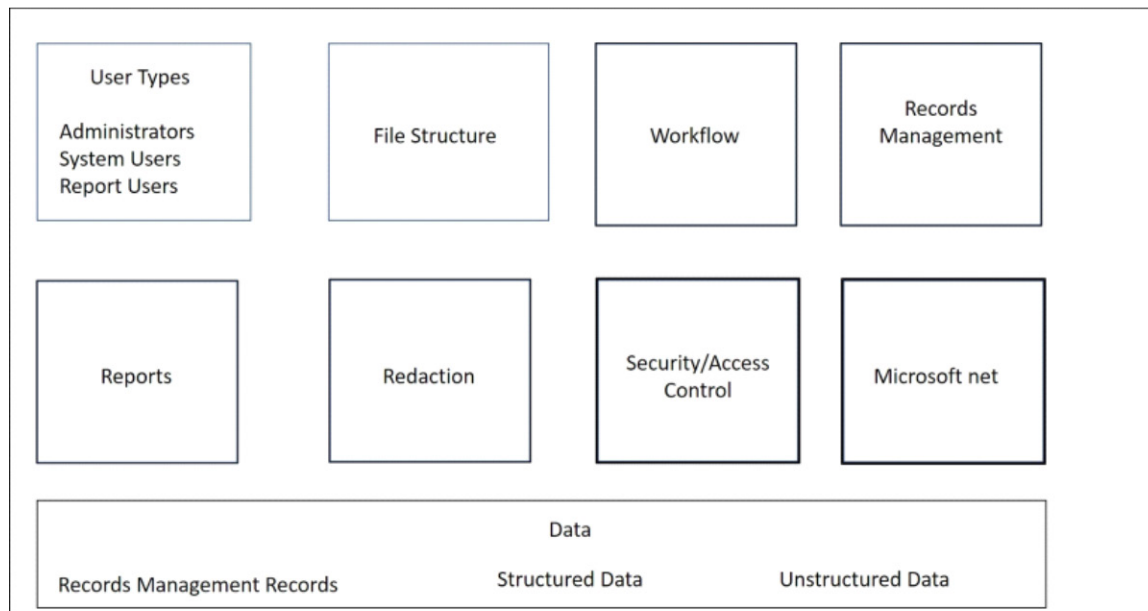
The software application selected was a commercial off-the-shelf (COTS) application currently being utilized by various federal agencies. The vendor assured the federal agency that minor modifications could be made to meet the needs of the federal agency. These mainly consisted of tab and folder names and the location of data fields on each of the screens. In addition, the data elements were kept consistent with the old database elements. Access to the ECFS is available to federal agency personnel through the agency's local network. However, ECFS will store personally identifiable information (PII) in a cloud environment, and there will be security considerations.

The next section of this document contains a high-level description of the ECFS; first, the textual description followed by functional diagrams. The second section of this document will describe the focus of the V&V effort.

The ECFS functions as a contract creation and storage system throughout the life cycle of a contract and will be the authoritative source of the contract file in an electronic format for federal government components as they move away from paper-based management. In addition to contract file storage, ECFS supports workflow, document management, and records management. A privacy impact assessment (PIA) is required because the ECFS stores PII on government personnel and vendor contacts.

### O.4.3 Diagrams

The features of the ECFS are shown in [Figure O.8](#).



**Figure O.8—Features of the ECFS**

#### O.4.4 Challenges to the V&V effort

IEEE Std 1012 is constructed so that an organization need only focus on the activities and tasks related to their particular project. A software project such as ECFS need only address the common activities of [Clause 7](#) and the Software activities of [Clause 9](#).

ECFS is SaaS. As with most federal agencies, V&V was not consulted during the Acquisition Process. The federal agency has accepted the software as meeting its requirements, and V&V will not have oversight of mission or business analysis, user needs and requirements, system/software requirements, architecture definition, design definition, implementation, or integration activities.

A V&V plan will be developed to address activities from the processes, as shown in the following [Table O.1](#). Some activities and tasks will be limited in scope as agreed upon with agency leadership and the SaaS provider.

**Table O.1—Relevant Activities**

| Clause 7                                                                                                                                                                                                                   | Clause 9                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Supply planning ( <a href="#">7.3</a> ),                                                                                                                                                                                   | Qualification testing <ul style="list-style-type: none"> <li>— System/software requirements definition (<a href="#">9.3</a>)</li> <li>— Architecture definition (<a href="#">9.4</a>)</li> <li>— Design definition (<a href="#">9.5</a>)</li> <li>— Implementation (<a href="#">9.7</a>)</li> <li>— Integration (<a href="#">9.8</a>)</li> </ul> |
| Project planning ( <a href="#">7.4</a> )                                                                                                                                                                                   | Acceptance testing <ul style="list-style-type: none"> <li>— System/software requirements definition (<a href="#">9.3</a>)</li> <li>— Architecture definition (<a href="#">9.4</a>)</li> <li>— Design definition (<a href="#">9.5</a>)</li> <li>— Implementation (<a href="#">9.7</a>)</li> <li>— Transition (<a href="#">9.10</a>)</li> </ul>    |
| Configuration management ( <a href="#">7.5</a> ) to the extent of tracking <ul style="list-style-type: none"> <li>— The agency requested changes to the system</li> <li>— Deliverables such as the V&amp;V plan</li> </ul> | Transition ( <a href="#">9.10</a> )                                                                                                                                                                                                                                                                                                              |
| —                                                                                                                                                                                                                          | Operation ( <a href="#">9.12</a> )                                                                                                                                                                                                                                                                                                               |
| —                                                                                                                                                                                                                          | Maintenance ( <a href="#">9.13</a> )                                                                                                                                                                                                                                                                                                             |
| —                                                                                                                                                                                                                          | Disposal ( <a href="#">9.14</a> )                                                                                                                                                                                                                                                                                                                |

#### O.4.5 Criticality

Although ECFS is SaaS, there is a criticality to the system. IEEE 1012 identifies the V&V tasks in relation to the organization's integrity level schema (see Table 2a through Table 2d). IEEE Standard Adoption of ISO/IEC 15026-3—Systems and Software Engineering—Systems and Software Assurance—Part 3: System Integrity Levels [\[B33\]](#), provides information on developing an integrity level schema. The standard defines Integrity Level 3 as follows:

The behavior of the system, in combination with its environment, causes the following:

- Catastrophic consequences for which the likelihood of the behavior occurring is at most infrequent
- Critical consequences for which the likelihood of the behavior occurring is at most occasional
- Marginal consequences for which the likelihood of the behavior occurring is at most probable

The federal agency has defined ECFS as a system integrity level 3 based on the following rationale:

- System failure results in loss of productivity and may trigger actions of the business continuity plan.

The business continuity plan outlines the actions an organization would take based on events that interrupt normal business operations. The business continuity plan may also be called the continuity of operations plan (COOP) or the disaster recovery plan (DRP).

- There are still legal and financial issues that include the high monetary value of contracts and PII implications.
- Data migration and interface issues with other/older contract systems which may cause loss of the data identified above.

#### **O.4.6 Focus of the V&V effort**

The ECFS will use a modified V&V approach as described in [Annex C](#) (C.2.3). ECFS is a SaaS product and the V&V effort will not have influence on the development of the solution. It will, however, have an unbiased evaluation of the solution therefore technical independence is maintained, and financial independence is maintained by a separate budget for V&V activities.

#### **O.4.7 Area of focus**

This vignette will not cover the application of all the V&V tasks listed above. Rather, it will show the process for one specific area of focus as a pattern that could be followed to apply V&V activities to federal, non-Department of Defense, software as a service (SaaS) systems. The V&V plan has already been developed, and traditional development and integration activities have been excluded. The chief concern of the agency at this point is acceptance testing and organizational change management. There are some related activities supporting this concern, so this vignette will focus on the following:

- User orientation and acceptance testing
- Organizational change management
- Assessment of the limited vendor documentation
- Criteria for assessment
- Issue—missing documents
- Records Management
- Business Continuity

##### **O.4.7.1 User orientation and acceptance:**

Because the system is SaaS and will be available through the cloud, the Federal agency client wants V&V to focus its direction on user acceptance testing (UAT). Strong user acceptance testing will require that agency personnel stationed in numerous departments around the country and who will be using this software are willing to devote time to either creating the test cases or participating in acceptance testing given time and resource constraints. User orientation will be required prior to any testing activity.

User acceptance tests will demonstrate that the software application will:

- Allow geographically dispersed agency personnel to accomplish everything they could do with the paper system and the old contracts creation system
- Allow agency personnel to download limited specific data items to the federal procurement data system (FPDS).  
The FPDS contains information from all agencies on current, past, and pending contracts. It allows contractors to know when proposals they are interested in will be coming up. ECFS uses an agency-specific data system to retrieve information about a particular vendor, e.g., the contract point of contact information, or the billing information.
- Allow agency personnel to upload limited specific data items from the internal vendor contact system.

Both before and during UAT, V&V will be engaged in the following tasks as appropriate to a project with a level 3 criticality:

- Many SaaS providers will limit or totally restrict access to their system documentation to delivering only user guides or system descriptions. To the extent possible, V&V will:
  - Verify the vendor developed system test plan if available
  - Verify the vendor developed system test cases if available
  - Verify the vendor developed (with agency input) user acceptance testing (UAT) plan
  - Verify the vendor developed (with agency input) UAT test cases (see example in attachment 1)
- Monitor user orientation on the new system
- Monitor UAT
- Verify the UAT test report
- Create a repository of regression tests
- Develop an approach to engage agency personnel in UAT
- Develop a value-added user orientation to UAT
- Create a UAT defect reporting system
- Support the development of UAT acceptance criteria

#### **O.4.7.2 Organizational change management:**

The federal agency must also consider that SaaS is a COTS-based, cloud-accessed system that will have an impact on organizational processes. In addition to system training, the agency will benefit from a strong organizational change management strategy. The following steps outline V&V considerations as an example of such a strategy. V&V would serve to inform the customer of what to expect from the developer or an organizational change management function:

- Clearly define the change and align it to business goals
  - Why do we need the change?
- Determine impacts and those affected
  - What are the impacts?
  - Who will the change affect the most?
  - How will the change be received?
- Develop communication strategy
  - How will the change be communicated?
  - How will feedback be managed?
- Provide effective training
  - What behaviors and skills are required to achieve business results?
  - How will training be delivered?
  - How will effectiveness be measured?
- Implement a support structure
- Where is support most required?
- What types of support will be offered?
- How will effectiveness be measured?
- Measure the change process
- Did the change assist in achieving business goals?
- Was the change management process successful?



- What could have been done differently?

#### **O.4.7.3 Records management:**

As cited in the section on criticality, the ECFS has the potential to process legal, financial, and PII data. Although a SaaS solution, the agency will still be required to establish a system of records and adhere to a records management process as established by NARA. Among the many records management activities are the filing of notices in the federal register and other forms required by NARA. V&V would review these notices and forms and assure all requirements and time periods have been met.<sup>13</sup>

#### **O.4.7.4 Business continuity:**

As discussed earlier, events that interrupt business practice require a business continuity plan. However, there is another type of business continuity to consider. The agency is also concerned about how to continue with its work after the expiration of the V&V contract. V&V documents developed in support of the contract are the property of the agency. To this end, V&V creates a document management capability within a collaborative system. The V&V staff creates numerous templates that are posted and thus are available to all agency personnel needing access to those templates.

To support the federal agency after the expiration of the contract, V&V creates a shared repository for the following:

- Defect reports
- Report templates based on defect reports
- Templates for
  - Test plans
  - Test cases
  - Project management plan
  - Risk management plan
  - Configuration management plan
  - Quality assurance plan
  - V&V plan

Training is provided for the agency on how to use the collaborative system to access the necessary templates or files when needed.

#### **O.4.7.5 Lessons learned**

Throughout the process of developing the V&V tasking for the acceptance phase of the ECFS project, the V&V team noted the following issues for agency consideration:

- The agency should periodically review the appropriate integrity level. level 3 was selected based on the processing of financial and PII data. There may be additional concerns as the system is used in production, which will raise the integrity level.
- The software was selected prior to V&V involvement. The agency will have many decision points to consider given their lack of oversight of:
  - Concept
  - Architecture definition
  - Design considerations

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<sup>13</sup> Details on records management activities may be found on the National Archives' website, [www.nara.gov](http://www.nara.gov).

- Requirements
  - Design
  - Construction
  - Integration activities
- Specific tasks (must have vs. nice to have) will require consideration given that the agency has accepted the software, for example:
    - Does the agency change its processes to adapt to the new software, or
    - Does the agency insist on modifications to the software to adapt to their processes?
  - The agency should consider identifying some acceptance criteria and review qualification test or acceptance test results in connection with accepting the software.
  - The agency should consider if there are risks involved with the following:
    - Downloading specific data items to the FPDS only takes place in the operational environment.
    - Uploading specific data items from the internal vendor contact system only takes place in the operation environment.
    - Vendor user guide does not match the procured software

Throughout the process of developing the V&V tasking for the acceptance phase of the ECFS project, the V&V team noted the following issues for V&V consideration:

- The V&V plan will be critical in defining the limitations, decisions, and expectations of the V&V team, given the following:
  - The procurement was made before V&V involvement
  - The areas of the standard listed above will not be in scope
  - System documentation may not be available
- V&V must stress the importance of user acceptance testing even though the software has been accepted. Customer involvement in assuring SaaS is meeting operational needs is essential.
- V&V will need to negotiate with the customer to define its role as a “trusted advisor.”

## O.4.8 Summary

With many federal agencies, and their components, relying on SaaS, it has become incumbent on V&V to become proficient in applying its skills and expertise to support a life-cycle process that may no longer begin with “development.” In this example, V&V actually began with Verification of user acceptance. Regardless of where in the life cycle V&V begins, it is vital to identify the criticality of the system or software in order to identify the desired scope and depth of the V&V effort. In this example, V&V deemed the software to be at a level 3 and thus required the identified V&V tasks. At the same time, there is another tension that must be anticipated. V&V must accede to the agency and its component’s direction as dictated by its V&V budget.

## O.5 Determining integrity level for training simulators

### O.5.1 Introduction

A nuclear power plant is comprised of multiple systems (see [Figure O.9](#)) to produce steam that generates electricity and to help ensure the safety of the plant and the public. A nuclear reactor generates heat via fission of the Uranium 235 in an assemble of multiple fuel rods that is known as the core. The core model (A1) water flows through the core to produce steam (A2), either directly as in a boiling water reactor (BWR), or indirectly via a steam generator as in a pressurized water reactor (PWR). The steam is then passed through a turbine generator (A3) to produce electricity with the condensate being recycled back to the reactor vessel. reactor protection system (RPS) (A4) and engineered safety features (ESF) (A5) are implemented to monitor the condition of the reactor, shut down the reactor, and provide emergency cooling in case of abnormal events. The containment (A6) and the containment isolation system are the controls of these multiple systems, and

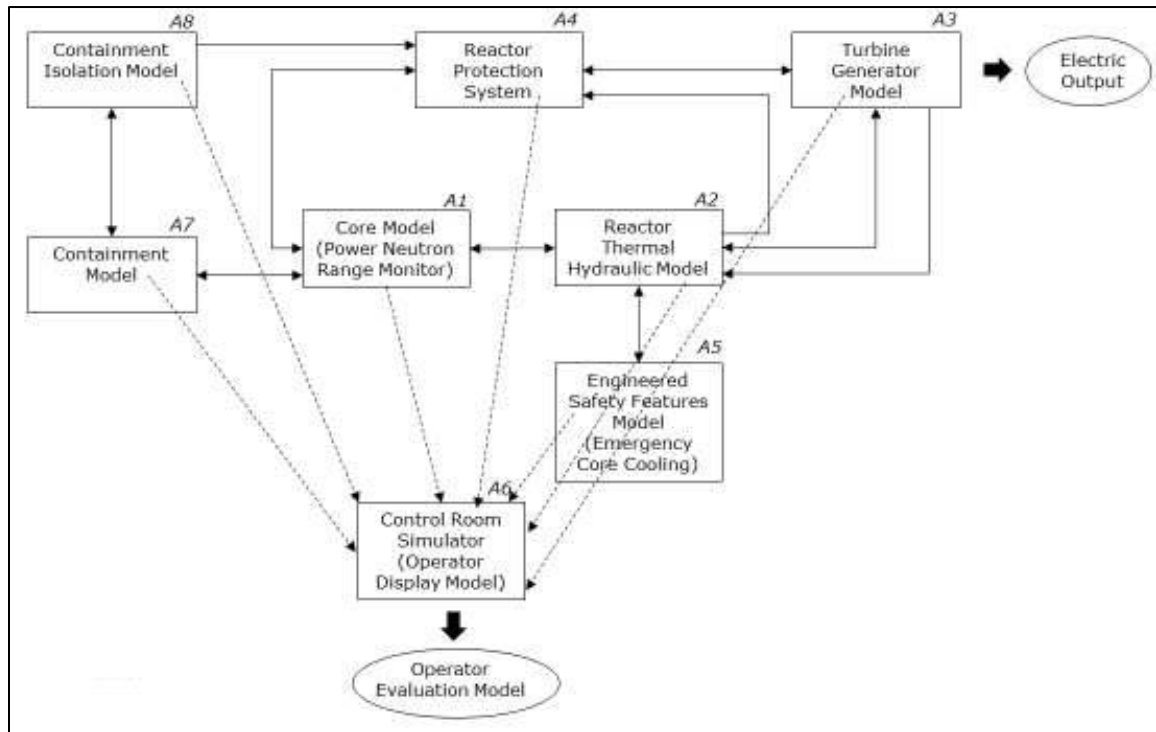
the ESF is in the control room simulator (A6), where the reactor operators play a major role in ensuring the safe operation of the nuclear power plant. Other power plant features, such as the plant containment model (A7) and plant isolation model (A8), are designed so that the potential release of radioactive materials in case of an accident are contained within the containment for public safety. Models are developed to evaluate the performance of these systems. These models are also used in simulators (A8) to train the reactor operators much the same way as the flight simulators for airplane pilots. The simulators provide effective hands-on training for normal and abnormal events so that the operators can respond correctly during plant operations. With today's technology, the training simulator can be a complicated computer-based system. The assignment of the integrity level of the training simulator will determine the necessary V&V tasks for developing the training simulator.

The definition of integrity level 1 is as follows:

The behavior of the system, in combination with its environment, causes the following:

- Critical consequences for which the likelihood of the behavior occurring is at most infrequent
- Marginal consequences for which the likelihood of the behavior occurring is at most occasional
- Negligible consequences for which the likelihood of the behavior occurring is at most probable

Negligible consequences are defined as resulting in “minor injury or illness, minor impact on system performance, or operator inconvenience.”



**Figure O.9—Nuclear power plant training simulator**

Training simulators are used to train reactor operators to be proficient in certain tasks or skills, e.g., startup, shutdown, and response to abnormal events. In theory, design failures of such training simulators would not result in “Minor injury or illness, minor impact on system performance, or operator inconvenience”. However, this does not mean that every training simulator should be integrity level 1 and that the V&V tasks should be minimal. The complexity and the potential impact of errors in the training simulator should also be considered, as illustrated in the following examples.

### **0.5.2 Example 1—Core model (power range neutron model) simulator for boiling water reactors (BWR)**

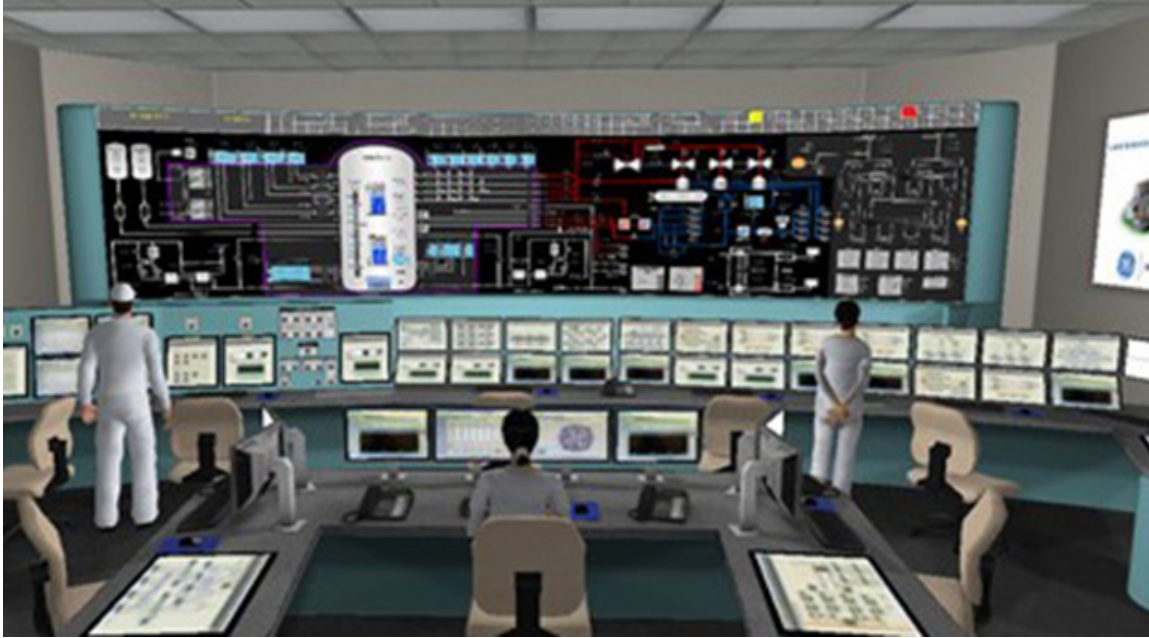
The power range neutron monitor (PRNM) is a safety-related neutron monitoring system that provides neutron flux information during power operation. The local neutron detectors in the core provide inputs for the PRNM to process to determine the flux (power) profile in the core. The core model (A1) provides the neutron flux information. PRNM provides the safety function of automatic trips to the reactor protection system (A4) should the power level exceed the predetermined limits. This function is a safety function, and therefore, the algorithms that perform the automatic trip would be considered integrity level 4. It also provides the operator with information regarding the power flux, individual detector information, and reactor stability. The PRNM Simulator for operator training should produce the same response as the plant system for various event scenarios based on the simulated inputs. The algorithms for the simulator would be the same as those for the plant system, and the algorithm that calculates the neutron flux and the automatic trip is considered integrity level 4. An additional consideration is whether the simulator software that drives the PRNM logic has any failure modes that would cause invalid training. The likelihood is very low, and the training instructor is a second check to make sure the scenarios are accurate. Given that the PRNM simulator only provides outputs and displays that mimic the plant system that would have been validated and that the likelihood of invalid training is low, the integrity level for the PRNM Simulator could be integrity level 1. However, one must verify and validate the implementation of the algorithms of the actual plant software/system and that the core abnormal or other event scenarios are correctly implemented in the training simulator. If the actual core performance model and other event behaviors are different in the simulation model than the actual BWR plant system, the operator may be trained to react incorrectly or incompletely. The consequences of improperly trained operators due to simulation errors could require an initial assignment of system integrity level 1 to the core model to be elevated to integrity level 2 or even level 3, depending on the degree of serious consequences of incorrect operation-trained actions.

### **0.5.3 Example 2—Control room simulator (A8)**

The control room of a nuclear power plant, as depicted in [Figure O.10](#), provides the controls for the operation of the plant and the status of all the safety and power generation systems in the plant. It is the central location where all the operating information is collected and displayed to the operator. The status of all safety and plant systems is on display. The manual operation of the systems is also provided in the control room. In accordance with the NRC regulations, the power generation systems, e.g., turbine generators would be considered as non-safety systems, whereas the ESF would be safety systems. The integrity level for the power generation system may be 2 or 3, depending on its complexity and importance to plant operation. The integrity level for the safety systems would be 4.

A control room simulator is a full-size mockup of the control room of a nuclear power plant as illustrated in [Figure O.11](#). It is used to provide operator training so that the operators can perform the necessary actions for various operating and accident scenarios. Such scenarios may include the following:

- Unit startup, shutdown, and steady-state operations
- Normal responses to small or moderate changes in demand
- Retraining due to unit equipment modifications/additions
- Loss of large pump function
- Loss of feedwater heater(s)
- Loss of feedwater flow
- Loss of coolant accident (LOCA)
- Anticipated transient without scram (ATWS)



**Figure O.10—Control room of a nuclear power plant**



**Figure O.11—Control room simulator**

The assignment of integrity level for control room simulators should be commensurate with the importance of its mission, i.e., correct operator response during plant operations, especially during emergency events. The consequences of improper operator response during emergency events would have grave consequences, such as damages to plant equipment or injury to plant personnel. Therefore, the simulator model must be validated to help ensure proper training. For example, loss of the recirculation pump flow in a BWR will result in a large water level transient with two possible scenarios. One scenario is that the large water level transient would cause a high reactor water level trip with subsequent reactor scram and loss of the feedwater pumps. For this scenario, the operator should be ready to bring the plant to safe shutdown conditions. Another scenario is that the water level transient is insufficient to cause a high reactor water level trip, and the reactor would not scram. The loss of recirculation pump flow would leave the reactor in the instability region of low recirculation flow and high power. The training objective for this event is for the operator to understand the possible consequences following the recirculation pump trip, including loss of the feedwater pumps, be able to initiate procedures to provide emergency core cooling or operation in the instability region, and be able to initiate procedures to exit the region. Errors in simulation may result in only a feedwater pump trip and no reactor instability response, resulting in improper operator training. The V&V tasks for the simulator may be based on an assigned integrity level of 3 to provide assurance that the simulator would provide the same response as the actual system. The scope of the V&V would include the simulation of direct response and subsequent responses to an abnormal event.

### **0.5.4 Example 3—Smart control room simulator—Future design consideration**

With the advancement in computer simulation and artificial intelligence, it is conceivable that one can develop a smart control room simulator to provide more functions than basic operator training. For example, the smart control room simulator may provide automatic scoring or warning to the operator, should the operator perform an inappropriate response during a training scenario. The smart control room simulator may also be used for the following:

- Review of existing control logic and process responses
- Replays of plant problem scenarios to improve controls and responses
- New control logic testing and debugging
- Alarm management studies and corrective actions

With such applications, any error in the smart control room simulator may cause wrong behavior during plant operation and wrong response from the operator during emergency conditions. The increase in complexity and importance would require a high Integrity level, e.g., 4, to be assigned to such a smart control room simulator.

### **0.5.5 V&V of tools/simulators/special test drivers**

As illustrated in previous examples, V&V tools/simulators require special attention and scrutiny as an error in the tool could cause an error in the target system (e.g., the nuclear instrumentation and control system that is the system of interest for V&V) from being detected by the V&V analyst. In the case of the simulation/test driver, an error in the simulation/test driver model could cause the simulator/test driver to produce the wrong inputs or incomplete set events to the target system (e.g., nuclear power instrumentation and control system) and thus not allow thorough and accurate testing to the target system (e.g., nuclear instrumentation and control system). The degree of V&V required of any tool or simulation/test driver will be dictated by the importance and criticality of the target system undergoing V&V. For high integrity systems (e.g., integrity level 4), V&V tools and simulations/test drivers should be subjected to a comprehensive review, analysis, and testing (e.g., sample testing with known error inputs, coverage testing of all conditions/events). The full extent of V&V tools/simulations/test drivers should be coordinated with the procurement organization and any regulatory policy requirements.

### **0.5.6 Summary**

This vignette shows that the assignment of integrity level to a training simulator should include the following considerations, which are necessary for the assignment of the integrity level so that the appropriate V&V tasks may then be defined:

- The importance of the system, e.g., if errors in the training simulator would result in grave consequences, V&V tasks should be performed to provide assurance that the training simulator would perform the same as the simulated system.
- The complexity of the system, e.g., if the system provides automatic scoring or critique of the operator's actions, the accuracy of the scoring and critique would require additional consideration by V&V.

## 0.6 V&V of field programmable gate array (FPGA) based systems

### 0.6.1 Introduction

The field programmable gate array (FPGA) based systems are not all that different from microprocessor-based systems from the system perspective. Therefore, the System Technical Processes summarized in [Figure 5](#) (Relationship of system, software, and hardware processes) applies at the high level.

However, an FPGA is a type of integrated circuit (IC) that can be programmed after manufacturing, whereas a microprocessor-based controller has its own circuitry and instructions set beforehand. Therefore, the specific implementation of an FPGA-based system is unique and has characteristics that are also dictated by the FPGA manufacturers. The implementation activities of the FPGA-based system requirements and design involve FPGA hardware description language (HDL) programming, simulation, and bitstream file (FPGA downloadable file) generation processes. These activities are an amalgamation of both software and hardware technical processes. As such, the required V&V activities are dictated by the fusion of software and hardware aspects of the FPGA.

This vignette presents an example of a V&V process of FPGA-based systems. First, the system life cycle of the FPGA-based system is described. Then the FPGA-based system specific Implementation Process that triggers both FPGA software and hardware development is presented. Next, the V&V of the FPGA implementation from hardware and software perspective is illustrated. And the FPGA-based system V&V is discussed. Finally, the process is demonstrated using an implementation case study in a digital instrumentation & control (I&C) example replacing microprocessor with FPGA for solving component obsolete issue.

### 0.6.2 FPGA-based system life cycle development and V&V

#### 0.6.2.1 FPGA-based system life cycle development process

The FPGA-based system, as a system of interest element (FPGA device or FPGA-based printed circuit board (PCB) when used in specific applications, its development follows the technical processes of ISO/IEC/IEEE 15288(E) [\[B41\]](#) and ISO/IEC/IEEE 12207(E) [\[B40\]](#). The FPGA application development is consistent with International Atomic Energy Agency (IAEA) (see [\[B13\]](#)).

Within the implementation phase of its system technical process, both hardware and software technical processes are invoked. FPGAs are used to implement the system using both software and hardware development life cycle (concept, requirements, design, implementation, integration, and test).

However, the technical processes for software and hardware overlap with the architecture definition, design definition, system analysis, integration, and transition processes at the system level. At the system element level, the FPGA implementation will be realized within the Software Implementation Process and the Hardware Implementation Process.

The following approach regarding application of [Clause 8](#) (system), [Clause 9](#) (software), and [Clause 10](#) (hardware) to V&V of FPGA-based systems is utilized:

- An HDL program (prior to integration with hardware) can be considered a software system element (i.e., developed using software life cycle processes); therefore, [Clause 9](#), Software V&V processes, applies
- A “not-yet-programmed” FPGA can be considered a hardware system element (i.e., developed using hardware life cycle processes); therefore, [Clause 10](#), hardware V&V processes, applies
- The smallest entity that contains the fully integrated FPGA (i.e., the FPGA has been programmed using the software) can be considered a system (or subsystem of the system-of-interest); therefore, [Clause 8](#), system V&V processes, applies



### O.6.2.2 The Implementation Process and V&V of FPGAs

The most common FPGA architecture consists of an array of logic blocks (also known as configurable logic block or logic array block, depending on a vendor), I/O pads, and routing channels. Generally, all the routing channels have the same width (number of wires). Multiple I/O pads may fit into the height of one row or the width of one column in the array. The specific implementation of the FPGA chip fulfilling both software and hardware requirements and design consists of the following steps:

Step 1: To implement FPGA design, designers use a HDL or a schematic design. The HDL form is well suited to work with large structures because it is possible to specify them numerically rather than drawing every piece by hand.

This step is translating the application logic for use in the FPGA. Designers use functional simulation to verify that the output of the logic is the intended result.

The development of the application logic (see Design entry—Application logic box in [Figure O.12](#)) is the software part of the FPGA implementation. The V&V effort follows the software life cycle processes defined in the standard to provide assurance that the software performs its specific requirements and implements the design. Therefore, the V&V effort performs functional simulation to verify the output of the logic. V&V testing ranges from software component (library) to integrated logic testing to provide assurance that the application logic produces satisfactory results, as illustrated in [Figure O.12](#).

Step 2: A technology-mapped netlist is generated using an electronic design automation tool. This process is called synthesis, in which the HDL or schematic design is translated into logic gates, memory units, registers, and connections.

During the synthesis process, an FPGA manufacturer's synthesis engine compiles the design to transform HDL sources into an architecture-specific design netlist.

Step 3: The netlist can then be implemented by the FPGA manufacturer's proprietary software to fit the actual FPGA architecture. This includes translation, map, and place-and-route processes. The designer verifies the map, and place-and-route results via timing analysis, simulation, and other Verification methodologies.

Step 4: Once software design and implementation are complete, the bitstream programming file (a binary file that contains the configuration information for an FPGA) is generated (also using the FPGA manufacturer's proprietary software). This file is transferred to the FPGA.

Steps 3 and 4 above are the so-called FPGA post-synthesis and will lead to the completed logic being realized in the FPGA chip. Although this process is primarily hardware-driven, we still treat it as software because the end result of the process is the bitstream file to be downloaded into the FPGA chip. Therefore, during the process, the designers will perform the required development testing, including timing simulation, to provide assurance that the circuit will work at the required speed.

The V&V effort also performs synthesis and post-synthesis analysis, Verification, and timing simulation/testing. The V&V effort performs the programmed FPGA chip qualification and acceptance testing after the FPGA bitstream file is downloaded into the device (or FPGA PCB). [Figure O.12](#) illustrates the above FPGA-specific implementation steps.



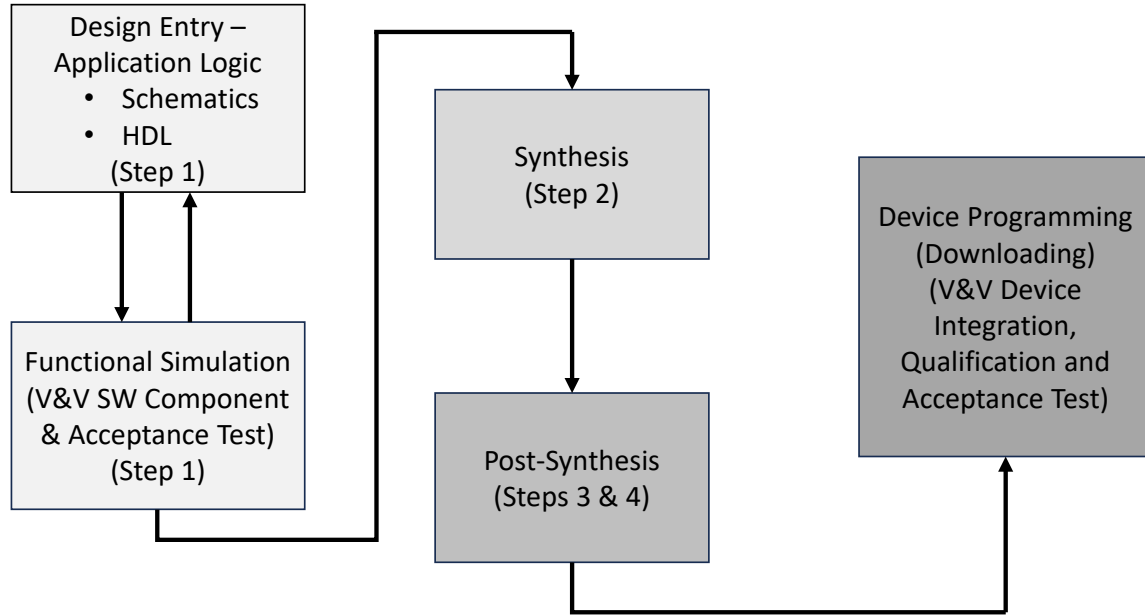


Figure O.12—FPGA implementation steps

### O.6.3 An example demonstration

#### O.6.3.1 FPGA emulation of obsolete microprocessor chip for nuclear safety applications

The first generation of single board controllers (SBC) used in an NPP was a microprocessor-based controller with a now obsolete chip. One of the applications for the SBC was a safety shutdown function in a single-loop system. Because of its obsolescence, an upgrade to replace this chip was imminent for the continued safe operation of the power plant.

The proposal for adopting an FPGA chip to replace the obsolete chip was accepted by the plant owner. The key element within the proposal was to meet the plant requirements, which included those of functional, mechanical and environmental, electrical, memory capability, and licensing constraints. Specifically, the following functional and mechanical requirements were to be met:

##### Functional requirements

- The updated FPGA-based controller card shall be capable of operating in a mode that is functionally equivalent to the operation of the existing microprocessor-based controller.
- The selected FPGA for the updated controller shall be code-compatible with the obsolete chip.
- The updated controller shall be capable of using the firmware erasable programmable read-only memory (EPROM) and patch panel EPROM from the existing controller card so that a control function executing on an updated SBC operates identically to the same control function executing on the existing controller.

##### Mechanical requirements

- The updated controller card shall conform to the standard form factor for all existing cards as specified in their original design.
- The updated controller shall have mounting hardware, which enables the card to be mounted and secured in the same system chassis slot as the existing controller card.
- The physical structure of the updated controller card shall be designed to meet environmental requirements.

To meet these requirements, design and implementation activities were performed using new FPGA components to replace the obsolete chip and other logic circuits on the previous SBC. The FPGA-based controller would then emulate the obsolete chip's execution of the assembly code (which was stored in onboard EPROM), control of external/onboard IO channels, and communication with the loop master. Additionally, the FPGA would provide an access point that a tester could use to monitor and evaluate the SBC functionality and performance. Therefore, the updated FPGA controller's architecture, comprised of several independent functional blocks, implemented functions for generating the following:

- System clock
- The obsolete chip IP core
- Memory address decoder
- Light-emitting diode control
- Complementary error check
- Onboard and external I/O channels controls
- Implementation of the intercommunication bus interface
- Miscellaneous control features including dual-in-line package switch selection.

The design and implementation were carried out through disciplined specification and rigorous V&V for both the correctness of the technical implementation, to provide assurance that safety requirements were met, and the simplicity of the licensing process to provide regulatory compliance. Testing and requirements traceability activities were performed, which are critical for providing these assurances.

### **0.6.3.2 Comprehensive functional/structural, and simulation testing**

#### **0.6.3.2.1 FPGA functional and structural testing**

In a similar fashion to microprocessor-based systems, FPGA test activities consisted of test planning (component test plan, integration test plan, system test plan, and acceptance test plan generation and Verification), test plan implementation (test design, test case, and test procedure generation and Verification), and test execution, as well as reporting (component, integration, system, and acceptance test execution, Verification and reporting). These activities were conducted in accordance with nuclear regulatory guidance and IEEE 1012.

These testing activities aimed to verify and validate the HDL code used to program the FPGA on the new controller card. Thus, they proved that the redesigned controller card executed the same test cases, and produced identical results/outputs as the original controller card.

Complete functional and structural tests were required to provide comprehensive test coverage, in which a given test or set of tests addressed all specified requirements for a given system or component.

Requirements traceability analysis was also necessary for ensuring complete test coverage of system and software requirements.

#### **0.6.3.2.2 FPGA simulation testing**

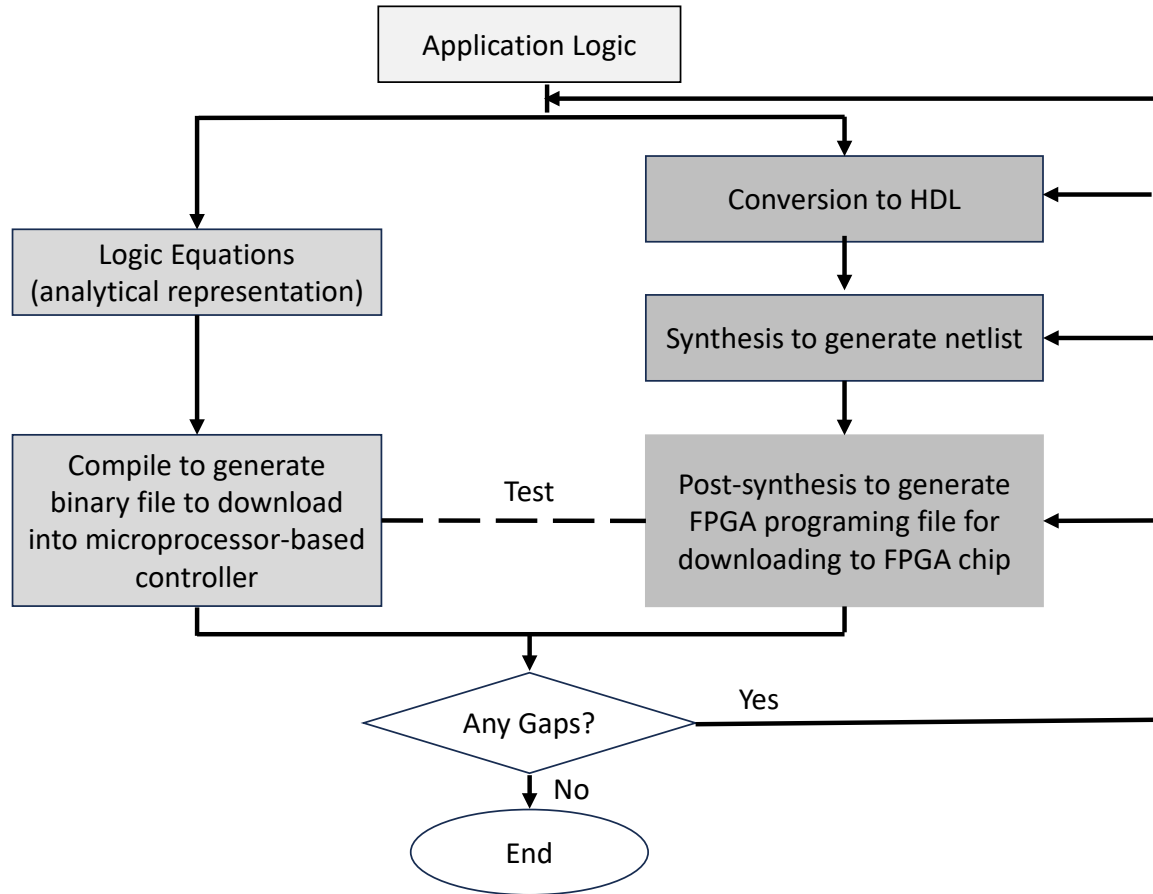
Simulation is the most commonly used test technique for modeling the behaviors or operations of a given system when a set of controlled inputs is provided.

Simulations can be performed in any phase of the software and system development life cycle. However, it is most effective before the integrated system and software are completed since, through simulation activities, errors can be identified as early as possible so that the project risk is minimized.

### O.6.3.3 Final Verification using proven reference

Before the updated controller was integrated into the safe shutdown system at the power plant, a final Verification was necessary to provide assurance that the application logic performed its intended function with the programmed FPGA chip and met the specific requirements.

This was realized using the proven microprocessor-based system as a reference. [Figure O.13](#) shows the final Verification to identify possible gaps between the FPGA-based implementation and the proven microprocessor-based implementation. This final check ensured that the downloaded FPGA files were the correct applications.



**Figure O.13—Equivalence check for final assurance**

It is important to know that during the process, gaps introduced by a malfunctioning FPGA chip, including its damaged gates or defects, can mask the intended software evaluation results. These results should be fully investigated, analyzed, and resolved.

The application logic, representing the safe shutdown function, served as a starting point for the equivalence check. For microprocessor-based execution, the verified logic was converted into the logic equations (an analytical representation) and then compiled into a binary file to be downloaded into the controller to be executed. Predesigned test cases were executed, and results were recorded.

On the other hand, the same verified logic (converted to HDL and synthesized to netlist) was programmed into the FPGA bitstream file via the post-synthesis process. The FPGA file was then downloaded into the FPGA chip for test execution of the safe shutdown function. The FPGA chip executed the same test cases in the microprocessor-based controller. The results were then recorded.

Both results were compared to identify gaps, which were back-traced and analyzed to find resolutions. In the end, the consistency of both results was achieved, which provided further assurance that the downloaded FPGA bitstream file contained the intended logic to execute the required safe shutdown function.

#### **O.6.4 Summary**

For FPGA-based system applications, its system development life cycle is essentially the same as the microprocessor-based systems. However, the specific implementation of the FPGA-based systems is unique and has its own characteristics that are also dictated by the FPGA manufacturers. The implementation activities of the FPGA-based system requirements and design encompass the FPGA HDL code programming, synthesis, simulation, and downloadable file generation processes. These activities are a combination of software and hardware technical processes, and the required V&V activities are dictated by both aspects of the FPGA-based system.

## Annex P

(informative)

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